CHROMATIC VARIANTS OF GEOMETRIC SATAKE

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ABSTRACT. The goal of this document is to explain an extension of the derived geometric Satake equivalence of Bezrukavnikov-Finkelberg to the setting of sheaves of k-modules on the affine Grassmannian of a connected complex reductive group G, where k is a commutative ring spectrum. We describe the "spectral side" in terms of the Langlands dual group $\check{\mathrm{G}}$ and the 1-dimensional formal group associated to k via chromatic homotopy theory. One of the key computations is that of the equivariant k-homology of the affine Grassmannian, which is a generalization of results of Bezrukavnikov-Finkelberg-Mirkovic and Yun-Zhu.

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1. Introduction

1.1. **Main result.** The goal of this document is to study a variant of the derived geometric Satake equivalence of [BF08] with coefficients in an $\mathbf{E}_\infty$-ring. Recall the statement of the geometric Satake equivalence from [MV07, Gin95]: let G be a connected reductive group over $\mathbf{C}$, let k be a noetherian ring of finite global dimension, and let $\check{\mathrm{G}}$ be the split form of the Langlands dual group of G viewed as an algebraic group over k. If $\mathrm{G}_{\mathcal{O}} = \mathrm{G}(\mathbf{C}[\![t]\!])$, $\mathrm{G}_{\mathcal{K}} = \mathrm{G}(\mathbf{C}(\!(t)\!))$, and $\mathcal{G}\mathrm{r}_{\mathrm{G}} = \mathrm{G}_{\mathcal{K}}/\mathrm{G}_{\mathcal{O}}$ is the affine Grassmannian, then there is a symmetric monoidal k-linear equivalence of abelian categories $\mathrm{Perv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \simeq \mathrm{Rep}(\check{\mathrm{G}})^\heartsuit$. Here, $\mathrm{Perv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ is the abelian category of perverse sheaves on $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ which are $\mathrm{G}_{\mathcal{O}}$-equivariant. It is equipped with the symmetric monoidal structure coming from convolution on the affine Grassmannian.

At least when $k = \mathbf{Q}$, this equivalence admits an extension to the derived category $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ as shown in [BF08]. Namely, there is a $\mathbf{Q}$-linear equivalence $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathbf{Q}) \simeq \mathrm{IndPerf}(\check{\mathfrak{g}}^*[2]/\check{\mathrm{G}})$, where $\check{\mathrm{G}}$ lives over $\mathbf{Q}$ and $\check{\mathfrak{g}}^*[2]$ denotes the shifted affine space $\mathrm{Spec}\,\mathrm{Sym}_{\mathbf{Q}}(\check{\mathfrak{g}}[-2])$.[1] This equivalence is monoidal, and is furthermore t-exact for the perverse t-structure on $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathbf{Q})$ and a certain t-structure on $\mathrm{IndPerf}(\check{\mathfrak{g}}^*[2]/\check{\mathrm{G}})$ whose heart is $\mathrm{Rep}(\check{\mathrm{G}})^\heartsuit$. For reference later in the introduction, let me note here that if G is simply-laced and simply-connected, then $\check{\mathfrak{g}}^*$ is isomorphic to the simply-connected form $\check{\mathfrak{g}}^{\mathrm{sc}}$ via the minimal bilinear form, and that this isomorphism in fact holds over $\mathbf{Z}$ itself and not just over $\mathbf{Q}$; so in this case the stack $\check{\mathfrak{g}}^*[2]/\check{\mathrm{G}}$ appearing on the Langlands dual side can be rewritten as $\check{\mathfrak{g}}^{\mathrm{sc}}[2]/\check{\mathrm{G}}$.

Our goal in this document is to prove a similar result for $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ where k is a general $\mathbf{E}_\infty$-ring. However, there are two important (related) difficulties. First, there is no perverse t-structure on $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ for a general $\mathbf{E}_\infty$-ring k. This is not special to homotopically interesting k: there is no such t-structure already for $k = \mathbf{Q}[u^{\pm 1}]$ with u in homological degree 2. In particular, any variant of [BF08] for $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ cannot rely on [MV07]. Second, the category $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ is k-linear, so in order for it to be equivalent to $\mathrm{IndPerf}(\mathcal{X})$, the stack $\mathcal{X}$ has to live over k. In particular, it must be a spectral stack (e.g., in the sense of [Lur17]). However, I do not know how to construct a lift of $\check{\mathrm{G}}$ to a general $\mathbf{E}_\infty$-ring (see Appendix B for some discussion).

The first difficulty is easy to fix: we simply give up on using perversity. Instead, we rely on a spectral version of the theory of parity sheaves [JMW14], building on ideas from [Pst23a] (the latter can be thought of as studying the case of parity sheaves of spectra on a point). To resolve the second difficulty, we produce a filtered lift of $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ which is $\tau_{\geq\star}(k)$-linear, whose associated graded category $\mathrm{Shv}^{\mathrm{gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ (which is now $\pi_*(k)$-linear) is equivalent to $\mathrm{IndPerf}^{\mathrm{gr}}(\mathcal{X}) := \mathrm{IndPerf}(\mathcal{X}/\mathbf{G}_m)$ for a certain stack $\mathcal{X}$ with $\mathbf{G}_m$-action which is

[1] In [CR23, Theorem 6.6.1], it was shown that the above equivalence can be refined to an equivalence of monoidal factorization stable ∞-categories. We will not need this refinement here, and only mention it for the sake of completeness.

equipped with a $\mathbf{G}_m$-equivariant map $\mathcal{X} \to \mathrm{Spec}(\pi_*(k))$. This does *not* require producing a lift of $\check{\mathrm{G}}$ to a group scheme over k; instead, the ordinary group scheme $\check{\mathrm{G}}$ over $\pi_*(k)$ suffices. If X is a topological space and one views $\mathrm{Shv}(\mathrm{X}; k)$ as a categorification of the *cochains* $\mathrm{C}^*(\mathrm{X}; k)$, then the filtered lift of $\mathrm{Shv}(\mathrm{X}; k)$ mentioned above should be viewed as a categorification of the Whitehead filtration $\tau_{\geq\star}\mathrm{C}^*(\mathrm{X}; k)$, so that its associated graded $\mathrm{Shv}^{\mathrm{gr}}(\mathrm{X}; k)$ is a categorification of the *cohomology* $\mathrm{H}^{-*}(\mathrm{X}; k)$. (This filtered lift appears in the body of the text as Construction 2.4.2; it unfortunately looks somewhat clunky as I have presented it, but I plan to give a more "principled" definition following [Pst23a] in future work.)

To state our main result, we need to set up some notation. For simplicity, I will assume that k is an *even* $\mathbf{E}_\infty$-ring, meaning that its homotopy groups are concentrated in even degrees. There is a canonical 1-dimensional graded formal group over $\pi_*(k)$, defined as $\mathrm{F} := \mathrm{Spf}(\mathrm{H}^{-*}(\mathbf{C}\mathrm{P}^\infty; k))$. Let $\mathrm{F}^\vee$ denote its Cartier dual, so that $\mathrm{F}^\vee = \mathrm{Spec}(\mathrm{H}_*(\mathbf{C}\mathrm{P}^\infty; k))$. Let G be the Chevalley split form of a connected reductive group, viewed as a group scheme over $\mathbf{Z}$, and let ${}^{2\check{\rho}}\mathrm{G}$ denote the graded group scheme over $\mathbf{Z}$ given by the $\mathbf{G}_m$-action on G via $\mathrm{Ad}_{2\check{\rho}}$; here, $2\check{\rho} : \mathbf{G}_m \to \mathrm{G}$ is the sum of positive coroots of G. We will use the same notation to denote the base-change of ${}^{2\check{\rho}}\mathrm{G}$ to $\pi_*(k)$. Note that ${}^{2\check{\rho}}\mathrm{G}$ is essentially the C-group of [Ber20, BG14]. Define ${}^{2\check{\rho}}\mathrm{G}_\mathrm{F}$ to be the (ind-)scheme $\mathrm{Hom}(\mathrm{F}^\vee, {}^{2\check{\rho}}\mathrm{G})$, so that it admits an action of ${}^{2\check{\rho}}\mathrm{G}$ by conjugation. (This object was introduced in [Mou21].)

Below, we will need to assume Conjecture 3.1.14, which says essentially that the fibers of the convolution map on $\mathcal{G}\mathrm{r}_\mathrm{G}$ are affinely paved; in the minuscule case, this was shown in [Hai06]. Recall that the $\mathrm{G}_\mathcal{O}$-orbits on $\mathcal{G}\mathrm{r}_\mathrm{G}$ are parametrized by dominant weights of $\check{\mathrm{G}}$. For every dominant weight μ of $\check{\mathrm{G}}$ which is 0, minuscule, or quasi-minuscule, we define an object $\widetilde{\mathrm{IC}}_\mu \in \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ in Section 5.1, such that when $k = \mathbf{Q}$ it is isomorphic to a direct sum of δ-sheaves at the basepoint of $\mathcal{G}\mathrm{r}_\mathrm{G}$ with (a shift of) the IC-sheaf of the corresponding Schubert subvariety in $\mathcal{G}\mathrm{r}_\mathrm{G}$. We then study the subcategory $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ of $\mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ generated by convolutions of the objects $\widetilde{\mathrm{IC}}_\mu$ as μ ranges over dominant weights of $\check{\mathrm{G}}$ which are 0, minuscule, or quasi-minuscule. On the dual side, if X is a graded stack with an action of ${}^{2\rho}\check{\mathrm{G}}$, we study the subcategory $\mathrm{IndPerf}^{\text{q-min,gr}}(\mathrm{X}/{}^{2\rho}\check{\mathrm{G}})$ of $\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{X}/{}^{2\rho}\check{\mathrm{G}})$ generated by tensor products of the vector bundles $\mathcal{O}_\mathrm{X} \otimes \mathrm{T}_\mu$, where T_μ is the tilting representation of $\check{\mathrm{G}}$ (see [Jan03]) with highest weight μ which is 0, minuscule, or quasi-minuscule.

Theorem 1.1.1 (Theorem 5.1.16 and Theorem 5.2.9)**.** *Suppose* G *is simply-laced and simply-connected, and suppose that the bad primes (see Table 1) of* G *are inverted in* $\pi_0(k)$*. Assume Conjecture 3.1.14 for every sequence* $(\mu_1, \cdots, \mu_n)$ *of dominant coweights of* G^{ad} *such that each* μ_i *is* 0*, minuscule, or quasi-minuscule. If* $\check{\mathrm{G}}^{\mathrm{sc}}$ *denotes the simply-connected form of* $\check{\mathrm{G}}$*, then there is an equivalence of graded* $\pi_*(k)$*-linear categories*

$$\mathcal{S} : \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \xrightarrow{\simeq} \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}}).$$

The composite of $\mathcal{S}$ *with the "cohomology" functor*

$$\underline{\tau}_\star : \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \to \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$$

sends $\widetilde{\mathrm{IC}}_\mu$ *to* $\widetilde{\mathrm{T}}_\mu \otimes_\mathbf{Z} \mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}}}$ *for every dominant weight* μ *of* $\check{\mathrm{G}}$ *which is either* 0*, minuscule, or quasi-minuscule; here,* $\widetilde{\mathrm{T}}_\mu$ *is a direct sum of the tilting*

$\check{\mathrm{G}}$*-representation with highest weight* μ *and trivial representations. In particular,* $\mathcal{S} \circ \underline{\mathcal{T}}_\star$ *sends* IC_0 *(the skyscraper sheaf at the basepoint of* $\mathcal{G}\mathrm{r}_\mathrm{G}$*) to the structure sheaf* $\mathcal{O}_{^{2\rho}\check{\mathrm{G}}_\mathrm{F}^\mathrm{sc}/^{2\rho}\check{\mathrm{G}}}$. *Moreover, when restricted to the full subcategory of those* $\mathcal{F} \in \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ *whose !-restriction to each Iwahori orbit is "perfect even" (in the sense of Definition 2.3.1),* $\mathcal{S} \circ \underline{\mathcal{T}}_\star$ *is monoidal for the convolution monoidal structure on* $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ *and the pointwise tensor product on* $\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\check{\mathrm{G}}_\mathrm{F}^\mathrm{sc}/{}^{2\rho}\check{\mathrm{G}})$.

The assumption that G is simply-laced and simply-connected is only for convenience in stating the result; Theorem 5.1.16 and Theorem 5.2.9 extend Theorem 1.1.1 to general connected reductive G, possibly with multiply-laced derived subgroup. In the multiply-laced case, ${}^{2\rho}\check{\mathrm{G}}_\mathrm{F}^\mathrm{sc}$ must be replaced by ${}^{2\rho}\check{\mathrm{G}}_\mathrm{F}$.

Remark 1.1.2. I expect Theorem 1.1.1 to hold verbatim even if the bad primes of G are not inverted. Along the way of proving the above result, we prove several results with weaker assumptions than those of Theorem 1.1.1 which support this expectation. In the multiply-laced case, I only expect the resulting analogue of Theorem 1.1.1 to hold verbatim if the lacing number of G is inverted in $\pi_0(k)$; a more "canonical" object replacing ${}^{2\rho}\check{\mathrm{G}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}}$ is therefore warranted. I also expect that $\mathcal{S} \circ \underline{\mathcal{T}}_\star$ is in fact $\mathbf{E}_3$-*monoidal* when restricted to the full subcategory of those $\mathcal{F} \in \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ whose !-restriction to each Iwahori orbit is perfect even, but giving a precise proof of this is beyond the scope of this document.

Example 1.1.3. Assume G is simply-laced and simply-connected. When k is an even $\mathbf{E}_\infty$-$\mathbf{Z}$-algebra, $\mathrm{F} = \hat{\mathbf{G}}_a(2)$, and ${}^{2\rho}\check{\mathrm{G}}_\mathrm{F}^\mathrm{sc} \cong ({}^{2\rho}\check{\mathfrak{g}}^\mathrm{sc})^\wedge_\mathcal{N}(2)$ where $\mathcal{N}$ is the nilpotent cone. As mentioned earlier in the introduction, this is isomorphic (even over $\mathbf{Z}$, without inverting any primes) to $({}^{2\rho}\check{\mathfrak{g}}^*)^\wedge_\mathcal{N}(2)$, so that Theorem 1.1.1 states that if the bad primes of G are inverted in $\pi_0(k)$, there is an equivalence

$$\mathcal{S} : \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \xrightarrow{\simeq} \mathrm{IndPerf}^{\text{q-min,gr}}(({}^{2\rho}\check{\mathfrak{g}}^*)^\wedge_\mathcal{N}(2)/{}^{2\rho}\check{\mathrm{G}}).$$

This equivalence can be viewed as a "graded" variant of the (renormalized) derived geometric Satake equivalence [BF08, AG15].

Example 1.1.4. Assume G is simply-laced and simply-connected. When k is an even $\mathbf{E}_\infty$-KU-algebra, $\mathrm{F} = \hat{\mathbf{G}}_m$, and ${}^{2\rho}\check{\mathrm{G}}_\mathrm{F}^\mathrm{sc} \cong ({}^{2\rho}\check{\mathrm{G}}^\mathrm{sc})^\wedge_\mathcal{U}$ where $\mathcal{U}$ is the unipotent cone. Theorem 1.1.1 states that if the bad primes of G are inverted in $\pi_0(k)$, there is an equivalence

$$\mathcal{S} : \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \xrightarrow{\simeq} \mathrm{IndPerf}^{\text{q-min}}((\check{\mathrm{G}}^\mathrm{sc})^\wedge_\mathcal{U}/\check{\mathrm{G}}).$$

This is related to [CK18, Conjecture 1.4], but differs from it in two important respects: first, we allow quasi-minuscule representations of $\check{\mathrm{G}}$, while [CK18] restrict to minuscule representations; and second, the stack appearing on the Langlands dual side in Theorem 1.1.1 is a completion of $\check{\mathrm{G}}^\mathrm{sc}/\check{\mathrm{G}}$, as opposed to $\check{\mathrm{G}}/\check{\mathrm{G}}$.

1.2. **Methods.** The proof of Theorem 1.1.1 follows the argument of [BF08], but numerous modifications to the argument in *loc. cit.* are needed. Many of our arguments below work in arbitrary characteristic, while the arguments of [BF08] rely on results of Kostant's [Kos63, Kos78] which tend to work only away from some small primes [Ric17]. To explain, we give a linear overview of this document. If k is an $\mathbf{E}_\infty$-ring and H is a group object in anima/homotopy types, we will use the notation $k^{h\mathrm{H}}$ and $\mathrm{Mod}_k^{h\mathrm{H}}$ to denote the homotopy fixed points of the *trivial* H-action on k and Mod_k (unless otherwise specified), i.e., to denote $\mathrm{C}^*(\mathrm{BH}; k)$

and $\mathrm{Fun}(\mathrm{BH}, \mathrm{Mod}_k)$. This is (perhaps unfortunately) common notation in stable homotopy theory.

We begin in Section 2.1 with a definition of the category of sheaves on (suitable) ind-compact Hausdorff spaces, like the affine Grassmannian. This is mostly a straightforward adaptation of [Ras15]. Working with coefficients in a general $\mathbf{E}_\infty$-ring k carries some possibly interesting subtleties which have to do with the fact that manifolds may not be k-orientable. The only effect of this is that "dimension theories" on ind-(pro-)spaces are no longer specified by integers, but rather by (compatible families of) line bundles.

Next, in Section 2.2, we describe an adaptation of Goresky-Kottwitz-MacPherson theory ([GKM98]), which describes torus-equivariant cohomology on suitable equivariant cell complexes, to the setting of ring spectra. This was already done in [HHH05], and this section is essentially a review of their arguments (with emphasis placed on homology and its *co*algebra structure). We then use this theory to provide a description of the T-equivariant k-cohomology of the flag variety G/B, where G is a connected complex reductive group, $\mathrm{B} \subseteq \mathrm{G}$ is a Borel subgroup, $\mathrm{T} \subseteq \mathrm{B}$ is a maximal torus, and k is an even $\mathbf{E}_\infty$-ring. This calculation showcases one of the utilities of working with general ring spectra, in particular of complex cobordism MU. While ordinary GKM theory cannot describe $\mathrm{H}_\mathrm{T}^{-*}(\mathrm{G/B}; \mathbf{F}_p)$ for all p (because some roots of G may end up becoming equal in $\mathfrak{t}_{\mathbf{F}_p} = \mathrm{Spec}\, \mathrm{H}^{-*}(\mathrm{BT}; \mathbf{F}_p)$), one can nevertheless give a GKM description of $\mathrm{H}_\mathrm{T}^{-*}(\mathrm{G/B}; \mathrm{MU})$, whose base-change along the map $\pi_*(\mathrm{MU}) \to \mathbf{F}_p$ computes $\mathrm{H}_\mathrm{T}^{-*}(\mathrm{G/B}; \mathbf{F}_p)$. This lets us define the *Weyl groupoid* $\mathcal{W} = \mathrm{Spf}\, \mathrm{H}_{\mathrm{T}\times\mathrm{T}}^{-*}(\mathrm{G}; k)$, which is a groupoid scheme (i.e., Hopf algebroid) over $\mathrm{T}_\mathrm{F} := \mathrm{Spf}\, \pi_*(k^{h\mathrm{T}})$; when $\mathrm{F} = \hat{\mathbf{G}}_a(2)$ and the order of the Weyl group is invertible in $\pi_0(k)$, then $\mathcal{W}$ is just $\hat{\mathfrak{t}}(2) \times_{\hat{\mathfrak{t}}(2)/\!\!/\mathrm{W}} \hat{\mathfrak{t}}(2)$. If X is any G-space, then there is a canonical $\mathcal{W}$-action on $\mathrm{H}_\mathrm{T}^{-*}(\mathrm{X}; k)$ (by "Hecke operators"). By writing $k^{h\mathrm{G}}$ as the totalization of $\mathrm{C}^*_{\mathrm{T}^{\times \bullet+1}}(\mathrm{G}^{\times\bullet}; k)$ and taking levelwise Whitehead filtrations, one obtains a filtration denoted $\mathrm{fil}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$ on $k^{h\mathrm{G}}$ whose associated graded is isomorphic to $\mathrm{R}\Gamma(\mathrm{T}_\mathrm{F}/\mathcal{W}; \mathcal{O})$. This filtration is described in Appendix A (motivated by [HRW25]) without reference to a maximal torus of G; more generally, if $\mathrm{M} \in \mathrm{Mod}_k^{h\mathrm{G}}$, there is a filtration $\mathrm{fil}^\star_{\text{G-ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$ given by the totalization of the levelwise Whitehead filtration of $(\mathrm{C}^*(\mathrm{G}^{\times\bullet}; k) \otimes \mathrm{M})^{h\mathrm{T}^{\times\bullet+1}}$.

In Section 2.3, we adapt an ingenious argument of Ginzburg's [Gin25] (which was preceded by [Gin91]) to show "full faithfulness of cohomology". More precisely, suppose X is a (connected) locally compact Hausdorff topological space equipped with a T-action and a finite stratification indexed by a poset Λ, where each stratum $j_\lambda : \mathrm{X}_\lambda \hookrightarrow \mathrm{X}$ is locally closed and T-stable, and is T-equivariantly isomorphic to a complex vector space with a *linear* T-action which fixes only the origin. Following [JMW14, Pst23a], say that a sheaf $\mathcal{F} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X}; k)$ is *!-even* if its !-restriction to each X_λ is built under extensions and retracts by even shifts $\Sigma^\mathrm{V} \underline{k}_{\mathrm{X}_\lambda}$ of the constant sheaf by complex representations of T, and similarly for *$*$-even.* If k is even, $\mathcal{F} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X}; k)$ is $*$-even, and $\mathcal{G} \in \mathrm{Shv}_\mathrm{T}(\mathrm{Y}; k)$ is !-even, then there is an isomorphism of graded $\mathrm{H}^*_\mathrm{T}(\mathrm{X}; k)$-modules:

$$\mathrm{Ext}^\star_{\mathrm{Shv}_\mathrm{T}(\mathrm{X};k)}(\mathcal{F}, \mathcal{G}) \xrightarrow{\cong} \mathrm{Hom}^\star_{\mathrm{H}^*_\mathrm{T}(\mathrm{X};k)}(\mathrm{H}^*_\mathrm{T}(\mathrm{X}; \mathcal{F}), \mathrm{H}^*_\mathrm{T}(\mathrm{X}; \mathcal{G})).$$

Here, the right-hand side is given by Hom in the abelian 1-category (and, crucially, *not* the derived category!) of graded $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}; k)$-modules. This isomorphism is extremely powerful, and provides the key transition from topology to algebra.

Finally, in Section 2.4, we define the "degeneration" of $\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}; k)$. This is a slight variant of the following: if $\mathcal{C}$ is a presentable stable category and $x \in \mathcal{C}$ is compact, then the localizing subcategory $\langle x \rangle$ of $\mathcal{C}$ generated by x is equivalent to $\mathrm{RMod}_{\mathrm{End}_{\mathcal{C}}(x)}$, so there is a filtered lift $\langle x \rangle_{\mathrm{fil}}$ of $\langle x \rangle$ given by $\mathrm{RMod}^{\mathrm{fil}}_{\tau_{\geq \star} \mathrm{End}_{\mathcal{C}}(x)}$. Its associated graded category $\langle x \rangle_{\mathrm{gr}}$ is equivalent to $\mathrm{RMod}^{\mathrm{gr}}_{\pi_* \mathrm{End}_{\mathcal{C}}(x)}$. The degeneration in question is obtained by applying this to $\mathcal{C} = \mathrm{Shv}_{\mathrm{T}}(\mathrm{X}; k)$ and some chosen object $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X}; k)$. When X is as in the preceding paragraph and $\mathcal{F}$ is both !- and $*$-even, Section 2.3 gives a description of the associated graded category $\langle x \rangle_{\mathrm{gr}}$ as modules over $\mathrm{End}^{\star}_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}; k)}(\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}; \mathcal{F}))$, where we remind the reader that these endomorphisms are taken in the abelian 1-category (and not the derived category!) of graded $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}; k)$-modules. When X is ind-compact Hausdorff, one instead needs to consider the algebra of endomorphisms of $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}; \mathcal{F})$ in the category of graded $\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathrm{X}; k)$-comodules. These results are easily extended to $\mathrm{Shv}_{\mathrm{G}}(\mathrm{X}; k)$ using the $\mathcal{W}$-action on T-equivariant cohomology.

In Section 3, we finally turn to the case $\mathrm{X} = \mathcal{G}\mathrm{r}_{\mathrm{G}}$. In order to apply the results of the previous sections, we need to compute $\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$. This is one of the main tasks of Section 3.1; using Section 2.2, we generalize the results of [BFM05, YZ11] to compute $\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ (with its $\mathcal{W}$-action) for arbitrary even $\mathbf{E}_\infty$-rings k in terms of an affine blowup (denoted $\mathrm{Spec}(\mathcal{A}_{\mathrm{G}})$) of $\check{\mathrm{T}} \times \mathrm{T}_{\mathrm{F}}$, where $\check{\mathrm{T}}$ is the Langlands dual torus to $\mathrm{T} \subseteq \mathrm{G}$. Again, it is important to work with $k = \mathrm{MU}$ here in order to apply GKM theory. (See also [LSS10, KK90, Zho25].) In Section 3.2, we use the results of Section 2.2 to describe the equivariant cohomology $\mathrm{H}^*_{\mathrm{G}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ (and its loop-rotation equivariant version $\mathrm{H}^*_{\mathrm{G} \times \mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$) in terms of the Hochschild homology of $k^{h\mathrm{G}}$. Using this, and a very crude form of the Hochschild-Kostant-Rosenberg theorem, we show how this perspective leads to an alternative proof of [BF08, Theorem 1] which describes $\mathrm{H}^*_{\mathrm{G} \times \mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathbf{Q})$ as the ring of functions on the deformation to the normal cone of the diagonal $\mathfrak{t}/\!\!/\mathrm{W} \hookrightarrow \mathfrak{t}/\!\!/\mathrm{W} \times \mathfrak{t}/\!\!/\mathrm{W}$. Using this perspective and [Ste60], we also prove the somewhat surprising result that the equivariant homology $\mathrm{H}^{\mathrm{T}}_*(\Omega(\mathrm{G}/\mathrm{T}); k)$ is concentrated in even degrees, at least if the bad primes of G are inverted in $\pi_0(k)$. Finally, in Section 3.3, we define "by hand" a Poisson structure on the W-quotient of the affine blowup $\mathrm{Spec}(\mathcal{A}_{\mathrm{G}})$ of $\check{\mathrm{T}} \times \mathrm{T}_{\mathrm{F}}$, and show that it agrees with the natural Poisson structure on $\mathrm{H}^{\mathrm{G},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ defined using its $\mathbf{E}_3$-algebra structure.

Most of our work takes place on the spectral side; let us assume for simplicity that G is simply-laced and simply-connected, as in Theorem 1.1.1. We need to show that many classical results about $\check{\mathfrak{g}}^* \cong \check{\mathfrak{g}}^{\mathrm{sc}}$ continue to hold for ${}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_{\mathrm{F}}$. The strategy is to study the F-analogue ${}^{2\rho}\widetilde{\check{\mathrm{G}}}{}^{\mathrm{sc}}_{\mathrm{F}} := {}^{2\rho}\check{\mathrm{G}} \times^{{}^{2\rho}\check{\mathrm{B}}} {}^{2\rho}\check{\mathrm{B}}^{\mathrm{sc}}_{\mathrm{F}}$ of the Grothendieck-Springer resolution ${}^{2\rho}\widetilde{\check{\mathfrak{g}}} = {}^{2\rho}\check{\mathrm{G}} \times^{{}^{2\rho}\check{\mathrm{B}}} {}^{2\rho}\check{\mathfrak{u}}^{\perp}$ of $\check{\mathfrak{g}}^*$, which is much more accessible. Note that our assumption on G allows us to write ${}^{2\rho}\widetilde{\check{\mathfrak{g}}} = {}^{2\rho}\check{\mathrm{G}} \times^{{}^{2\rho}\check{\mathrm{B}}} {}^{2\rho}\check{\mathfrak{u}}^{\mathrm{sc}}$. In Section 4.1, we define an F-analogue of the Kostant slice $\kappa : \mathfrak{t} \cong \check{\mathfrak{t}}^* \to \widetilde{\check{\mathfrak{g}}}$, which is given by a map $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}} \to {}^{2\rho}\widetilde{\check{\mathrm{G}}}{}^{\mathrm{sc}}_{\mathrm{F}}$. Although this map depends on a choice (just like κ depends on the choice of a regular $f \in {}^{2\rho}\check{\mathfrak{u}}^{\mathrm{sc}}$), the resulting map $\mathrm{T}_{\mathrm{F}} \to {}^{2\rho}\widetilde{\check{\mathrm{G}}}{}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}} \cong {}^{2\rho}\check{\mathrm{B}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}}$ is independent of this choice, at least if the bad primes of G are inverted in $\pi_0(k)$.

In Section 4.2, we study the centralizer of $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}} \to {}^{2\rho}\check{\mathrm{B}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}}$; this is a subgroup scheme of ${}^{2\rho}\check{\mathrm{B}} \times \mathrm{T}_{\mathrm{F}}$ over T_{F} which is denoted $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$. We show that if the bad primes of G are inverted in $\pi_0(k)$, then $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ is flat over T_{F}, and furthermore that there is an isomorphism $\operatorname{Spec} \mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \cong \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ of group schemes over T_{F}. This generalizes the main results of [BFM05, YZ11]. The proof is like in *loc. cit.*: one reduces (essentially using Hartogs' lemma) to the case when G has semisimple rank 1, where the calculation can be done explicitly. Along the way, we observe that for arbitrary even $\mathbf{E}_\infty$-$\mathbf{Z}$-algebras k and connected reductive G, possibly with multiply-laced derived subgroup, $\operatorname{Spec} \mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ is in fact isomorphic to the centralizer of the Kostant slice $\kappa : \check{\mathfrak{t}}^* \to \widetilde{\check{\mathfrak{g}}}/\check{\mathrm{G}} \cong \check{\mathfrak{u}}^\perp/\check{\mathrm{B}}$. Since we do not require any primes to be invertible in $\pi_0(k)$, it is important here to work with the *coadjoint*, and not the adjoint, representation of $\check{\mathrm{G}}$.

To understand the quotient stack ${}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ itself, we need to understand the descent along ${}^{2\rho}\widetilde{\check{\mathrm{G}}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}} \to {}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$. This is done in Section 4.3. We show that the map $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}} \to {}^{2\rho}\widetilde{\check{\mathrm{G}}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ descends to a map $\mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$. When k is an even $\mathbf{E}_\infty$-$\mathbf{Q}$-algebra, so that $\mathrm{F} = \hat{\mathbf{G}}_a(2)$, this is the composite $\check{\mathfrak{t}}/\!\!/\mathrm{W} \to \check{\mathfrak{g}}^* \to \check{\mathfrak{g}}^*/\check{\mathrm{G}}$ where the first map is the Kostant slice (see [Kos78], as well as [Ric17] away from bad characteristic). However, the construction of the map $\mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ in Section 4.3 does not proceed by constructing a map $\mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_{\mathrm{F}}$; instead, it proceeds by producing a map $\mathcal{W} \to {}^{2\rho}\widetilde{\check{\mathrm{G}}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}} \times_{{}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}} {}^{2\rho}\widetilde{\check{\mathrm{G}}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ of groupoids over the map κ_{F}. Our work in the preceding section shows that $\operatorname{Spec} \mathrm{gr}^\star_{\mathrm{G}\text{-ev}/k^{h\mathrm{G}}}(\mathrm{C}^{\mathrm{G},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$ is isomorphic as a group scheme over $\mathrm{T}_{\mathrm{F}}/\mathcal{W}$ to the centralizer $\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}$ of the map $\mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$; again, this can be viewed as a generalization of the main results of [BFM05, YZ11]. This computation gives a homomorphism $\operatorname{Spec} \mathrm{gr}^\star_{\mathrm{G}\text{-ev}/k^{h\mathrm{G}}}(\mathrm{C}^{\mathrm{G},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \to {}^{2\rho}\check{\mathrm{G}} \times \operatorname{Spec} \mathrm{gr}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}})$. If $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, then $\mathrm{gr}^\star_{\mathrm{G}\text{-ev}/k^{h\mathrm{G}}}(\mathrm{R}\Gamma_{c,\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F}))$ is a $\mathrm{gr}^\star_{\mathrm{G}\text{-ev}/k^{h\mathrm{G}}}(\mathrm{C}^{\mathrm{G},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$-comodule, so this homomorphism defines a functor $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \to \mathrm{IndPerf}(\mathrm{B}^{2\rho}\check{\mathrm{G}} \times \operatorname{Spec} \mathrm{gr}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}}))$, which can be viewed as a version of the fiber functor of [MV07].

Along the way, we observe (essentially using Hartogs' lemma) that for arbitrary even $\mathbf{E}_\infty$-$\mathbf{Z}$-algebras k and connected reductive G, possibly with multiply-laced derived subgroup, the stack $\mathfrak{t}/\mathcal{W}$ is in fact isomorphic to the Whittaker quotient $(\psi + \check{\mathfrak{u}}^\perp_-)/\check{\mathrm{U}}_-$. This is motivated by discussions with Akshay Venkatesh. It in particular shows that the cohomology of the structure sheaf of $(\psi + \check{\mathfrak{u}}^\perp_-)/\check{\mathrm{U}}_-$ is isomorphic to $\mathrm{gr}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}})$. We illustrate some computations of this stack in Appendix C. It follows from our discussion that $\operatorname{Spec} \mathrm{gr}^\star_{\mathrm{G}\text{-ev}/k^{h\mathrm{G}}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$ is isomorphic to the biWhittaker quotient $\mathrm{T}^*(\check{\mathrm{U}}_\psi\backslash\check{\mathrm{G}}/_\psi\check{\mathrm{U}})$, over an arbitrary even $\mathbf{E}_\infty$-$\mathbf{Z}$-algebra. Perhaps one interesting remark about Section 4.2 and Section 4.3 is that we only ever study the stacks ${}^{2\rho}\widetilde{\check{\mathrm{G}}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ and $\check{\mathfrak{g}}^*(2)/\check{\mathrm{G}}$ through the Grothendieck-Springer resolution (which on occasion requires inverting bad primes), since the latter behaves uniformly in all characteristics because of structural properties of Borel subalgebras; for instance, as mentioned above, we construct a Kostant slice $\mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ without needing to show that it lifts to a map $\mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_{\mathrm{F}}$.

We begin collating these results in Section 5. In Section 5.1, we prove Theorem 1.1.1. Since we do not have an *a priori* analogue of the abelian geometric

Satake equivalence of [MV07], we need a replacement of Mirkovic-Vilonen's calculation that the cohomology of the IC-sheaf of a Schubert stratum in $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ is isomorphic to the corresponding irreducible representation of $\check{\mathrm{G}}$. We do this by studying convolutions of even sheaves (i.e., parity sheaves in the sense of [JMW14]) on minuscule and quasi-minuscule Schubert strata. The minuscule case is not very difficult to generalize, because the corresponding IC-sheaf is smooth: we show that the cohomology of the pushforward of the constant sheaf along a minuscule Schubert stratum into $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ is given by the restriction along the homomorphism $\operatorname{Spec} \mathrm{gr}^\star_{\mathrm{G}\text{-ev}/k^{h\mathrm{G}}}(\mathrm{C}^{\mathrm{G},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \to {}^{2\rho}\check{\mathrm{G}} \times \operatorname{Spec} \mathrm{gr}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}})$ of the corresponding minuscule ${}^{2\rho}\check{\mathrm{G}}$-representation. The quasi-minuscule case is more interesting; taking a hint from [NP01], we consider the pushforward $\widetilde{\mathrm{IC}}_\theta$ of the constant sheaf along a smooth resolution of the corresponding Schubert stratum. Rationally, this contains the corresponding IC-sheaf as a direct summand (by the decomposition theorem), and in [JMW14, JMW16] it is shown that this pushforward has a unique indecomposable parity sheaf summand. We do not show that $\widetilde{\mathrm{IC}}_\theta$ splits, but instead directly compute that its cohomology is given by the restriction along the homomorphism $\operatorname{Spec} \mathrm{gr}^\star_{\mathrm{G}\text{-ev}/k^{h\mathrm{G}}}(\mathrm{C}^{\mathrm{G},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \to {}^{2\rho}\check{\mathrm{G}} \times \operatorname{Spec} \mathrm{gr}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}})$ of a direct sum of trivial representations with the indecomposable tilting ${}^{2\rho}\check{\mathrm{G}}$-module with quasi-minuscule highest weight. Using these calculations and some cohomological vanishing guaranteed by the existence of good filtrations, we obtain Theorem 1.1.1. In Section 5.2, we extend these computations to the multiply-laced case. (As mentioned above, some of the calculations of equivariant (co)homology in terms of ${}^{2\rho}\check{\mathrm{G}}$ hold for arbitrary even $\mathbf{E}_\infty$-$\mathbf{Z}$-algebras k without inverting any primes. This suggests that at least for such k, Theorem 1.1.1 holds as stated without inverting any primes. The primary obstacle to proving this given the calculations we have already done is the lack of good filtrations in the sense of [Don81], but I hope this can be overcome in the near future.)

One interesting aspect of Theorem 1.1.1 is that the category $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ admits a 1-parameter deformation given by the loop-rotation equivariant category $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}} \rtimes \mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, and hence the same must be true of the category $\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$. When k is an even $\mathbf{E}_\infty$-$\mathbf{Q}$-algebra, it follows from [BF08] that $\mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_{\mathcal{O}} \rtimes \mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ is equivalent to the category $\mathrm{DMod}_\hbar({}^{2\rho}\check{\mathrm{G}})^{({}^{2\rho}\check{\mathrm{G}} \times {}^{2\rho}\check{\mathrm{G}}, \mathrm{wk})}$ of (graded) Harish-Chandra bimodules for $\check{\mathrm{G}}$. We do not address the analogue of this result for arbitrary G and $\mathbf{E}_\infty$-rings k in this document (this will be in future work), but in Section 5.3, we study the case when G is a torus. The key definition is that of "F-D-modules" on $\check{\mathrm{T}}$, which was introduced already in [Dev26, DM23], and recovers ordinary D-modules (resp. q-Weyl algebra) when k is an even $\mathbf{E}_\infty$-$\mathbf{Z}$-algebra (resp. $\mathbf{E}_\infty$-KU-algebra).

The equivalence Theorem 1.1.1 is compatible with restriction to Levi subgroups, as we show in Section 6.1. More precisely, we show that restriction along $\mathcal{G}\mathrm{r}_{\mathrm{L}} \subseteq \mathcal{G}\mathrm{r}_{\mathrm{G}}$ corresponds to pull-push along the affinization of an F-analogue of the parabolic Grothendieck-Springer resolution.

Section 6.2 discusses an approximation to the abelian geometric Satake equivalence of [MV07]. While the notion of perverse t-structure seems poorly behaved for coefficients in general $\mathbf{E}_\infty$-rings, we take motivation from [MV07, Section 3] to study those $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ whose cohomology on the semi-infinite orbit indexed by $\lambda \in \mathbb{X}_*(\mathrm{T})$ is a retract of direct sum of copies of $\Sigma^{-\langle 2\rho, \lambda \rangle} k$; we call such $\mathcal{F}$ "projectively perverse". If k is an ordinary commutative ring, then projectively perverse sheaves are precisely those perverse sheaves whose cohomology over semi-infinite orbits are projective k-modules. We use some of the calculations of the previous sections to construct variants of IC-sheaves which are projectively perverse and compute their cohomologies over semi-infinite orbits.

In Section 6.3, we study some simple applications of our calculations to describe the effect of Steenrod operations on k in terms of the Langlands dual side of Theorem 1.1.1. This allows us to reprove some (simple) results about the restricted power operation $x \mapsto x^{[p]}$ on $\check{\mathfrak{g}}$. We also show a generalized version of the Springer isomorphism [Spr69], namely that the "F-nilpotent cone" ${}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}^{\mathrm{sc}} \times_{\mathrm{T_F}/\mathcal{W}} \{0\}$ is isomorphic to the usual nilpotent cone away from bad primes, at least for G of classical type and G_2.

In Appendix A, we review some background on gradings and filtrations (following [HP25, HP24]) and even $\mathbf{E}_\infty$-rings (following [HRW25]). Appendix A.2 may be of independent interest: we suggest a definition of even filtration in the setting of Borel-equivariant $\mathbf{E}_\infty$-rings for nonabelian groups. Appendix B describes a conjectural story which motivated the present work, concerning lifting the Chevalley split form of ${}^{2\rho}\check{\mathrm{G}}$ over $\mathbf{Z}$ to the sphere spectrum. (This seems to have been first considered in [Lur10, Remark 10.3].) In Appendix C, we illustrate some examples of the isomorphism $\mathrm{gr}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}}) \cong \mathrm{R}\Gamma((\psi + \check{\mathfrak{u}}_-^\perp)/\check{\mathrm{U}}_-; \mathcal{O})$ by explicit calculation.

1.2.1. *Notation.* We end this introduction with some notation that will be used throughout this document. The symbol k will always denote an $\mathbf{E}_\infty$-ring, often assumed to be even. (The assumption that k is $\mathbf{E}_\infty$ is essentially never used, but it does help simplify some of our discussion, so we assume this structure throughout.) Examples include complex cobordism MU, complex K-theory KU, connective complex K-theory ku, the integers $\mathbf{Z}$, and the finite field $\mathbf{F}_p$. If f is a morphism between objects of a category (like spectra, sheaves of spectra on a topological space, quasicoherent sheaves on a stack, etc.), I will say that f is an "isomorphism" if f is invertible, and generally reserve the word "equivalence" for an invertible functor between categories. Except in Section 2.3, we will always work with *homological* grading. In particular, the k-cohomology of a space X will be denoted $\mathrm{H}^{-*}(\mathrm{X}; k)$. If k is an $\mathbf{E}_\infty$-ring and X is a homotopy type, we will write $k[\mathrm{X}]$ or $\mathrm{C}_*(\mathrm{X}; k)$ to denote $k \otimes \Sigma^\infty_+ \mathrm{X}$ (and refer to it as "k-chains"), and $\mathrm{C}^*(\mathrm{X}; k)$ to denote its k-linear dual (and refer to it as "k-cochains"). If k is an $\mathbf{E}_\infty$-ring and H is a group object in anima/homotopy types, we will use the notation $k^{h\mathrm{H}}$ and $\mathrm{Mod}_k^{h\mathrm{H}}$ to denote the homotopy fixed points of the *trivial* H-action on k and Mod_k (unless otherwise specified), i.e., to denote $\mathrm{C}^*(\mathrm{BH}; k)$ and $\mathrm{Fun}(\mathrm{BH}, \mathrm{Mod}_k)$. This is (perhaps unfortunately) common notation in stable homotopy theory.

The symbol G will always mean a connected reductive group, unless mentioned otherwise. (The case when G is possibly disconnected shows up only in Appendix A.2.) We will also use the same notation G to denote the corresponding Chevalley group scheme over $\mathbf{Z}$. This will only be used when discussing the algebraic geometry of G; on the topological side, we will only use the $\mathbf{C}$-points of G as a complex Lie group. We will often assume the derived subgroup of G to be simply-laced, and will specify when we do so. The utility of this assumption is that the minimal bilinear form on $\mathfrak{g}$ defines an equivariant isomorphism $\check{\mathfrak{g}}^* \cong \mathfrak{g}$ over $\mathbf{Z}$ (see Remark 4.2.2). If $\mathrm{B} \subseteq \mathrm{G}$ is a Borel subgroup, there is an associated Iwahori subgroup $\mathrm{I} \subseteq \mathrm{G}_\mathcal{O}$ given by $\mathrm{G}_\mathcal{O} \times_\mathrm{G} \mathrm{B}$. Note that if T is the abstract Cartan of G, there is a canonical map $\mathrm{B} \to \mathrm{T}$ which is a homotopy equivalence. We will also fix a maximal torus contained in B (which we will continue to denote by T).We will write $\mathbb{X}_*$ to denote the cocharacter lattice of T, and $\mathbb{X}^*$ to denote its character lattice. Let $\Phi \subseteq \mathbb{X}^*$ (resp. $\check{\Phi} \subseteq \mathbb{X}_*$) be the set of roots (resp. coroots) of G, and let $2\rho \in \mathbb{X}^*$ be the sum of all positive roots. Let $\Delta \subseteq \Phi$ denote a chosen base of simple roots. Let W be the Weyl group of G, so that W acts on T. The quotient $\mathbb{X}_*/\mathrm{W}$ is isomorphic to the subset $\mathbb{X}_*^+$ of dominant coweights; let $\Phi^+ \subseteq \Phi$ denote the associated set of positive roots. The Bruhat order on $\mathbb{X}_*$ declares $\lambda \leq \mu$ for $\lambda, \mu \in \mathbb{X}_*$ if $\mu - \lambda$ is a non-negative integral linear combinations of simple coroots. A dominant coweight $\mu \in \mathbb{X}_*^+$ is minuscule if $\mu \neq 0$ and $(\alpha, \mu) \leq 1$ for every positive root α. Similarly, $\mu \in \mathbb{X}_*^+$ is called quasi-minuscule if $\mu \neq 0$, it is not minuscule, and $(\alpha, \mu) \leq 2$ for every positive root α. If the derived subgroup of G is simple, then there is a unique quasi-minuscule dominant coweight, given by the highest short coroot. Let δ be the null root, let $\hat{\Phi} = \{\alpha + n\delta | \alpha \in \Phi, n \in \mathbf{Z}\}$ denote the set of real roots, and let $\hat{\Phi}^+ \subseteq \hat{\Phi}$ denote the set of positive real roots (so $\hat{\Phi}^+ = \{\alpha + n\delta | \alpha \in \Phi, n > 0\} \cup \Phi^+$).

We write ${}^{2\check{\rho}}\mathrm{G}$ denote the graded group scheme over $\mathbf{Z}$ given by the $\mathbf{G}_m$-action on G via $\mathrm{Ad}_{2\check{\rho}}$, where $2\check{\rho} \in \mathbb{X}_*(\mathrm{T})$ is the sum of positive coroots of G. Of course, ${}^{2\check{\rho}}\mathrm{G}$ is closely related to the C-group (see [Ber20] and [BG14, Section 5]): the classifying stack over $\mathrm{B}\mathbf{G}_m$ of ${}^{2\check{\rho}}\mathrm{G}/\mathbf{G}_m$ is isomorphic, as a stack over $\mathrm{B}\mathbf{G}_m$, to the classifying stack of the C-group ${}^{\mathrm{C}}\mathrm{G}$ of G, which lies over $\mathrm{B}\mathbf{G}_m$ thanks to the homomorphism ${}^{\mathrm{C}}\mathrm{G} \to \mathbf{G}_m$. (Note that if G is simply-laced, then $2\check{\rho} = 2\rho$ under the isomorphism $\mathbb{X}_*(\mathrm{T}) \cong \mathbb{X}^*(\mathrm{T})$ coming from the minimal bilinear form; so ${}^{2\check{\rho}}\mathrm{G} \cong {}^{2\rho}\mathrm{G}$.)

All our stacks are to be understood as *fpqc* stacks. Similarly, all categories are implicitly assumed to be ∞-categories, and we will make it clear when we discuss 1-categories (often via the symbol $\heartsuit$). The symbols Shv, QCoh, Mod, etc. (and therefore functors between these categories, like pushforward and pullback) are all to be understood in the derived sense, as are all fiber and tensor products; we will make it clear if any of these operations are to be understood in their classical sense. In particular, if X is a scheme or (fpqc) stack, we will write $\mathrm{QCoh}(\mathrm{X})^\heartsuit$ to denote the abelian 1-category of quasicoherent sheaves on X. We will also write $\otimes$ to denote the smash product of spectra. One important place where our terminology does *not* adhere to the "everything is derived" convention is that if X is a stack, we will refer to its *affinization* as $\mathrm{X}^{\mathrm{aff}} := \mathrm{Spec}\,\mathrm{H}^0(\mathrm{X}; \mathcal{O}_\mathrm{X})$, and refer to $\mathrm{Spec}\,\mathrm{R}\Gamma(\mathrm{X}; \mathcal{O}_\mathrm{X})$ as its *derived affinization*. If G acts on X, I will often abusively write $\mathcal{O}_\mathrm{X}$ to denote the structure sheaf of X/G, viewed as $\mathcal{O}_\mathrm{X} \in \mathrm{Perf}(\mathrm{X})$ equipped with its G-action. If k is an $\mathbf{E}_\infty$-ring, $\mathcal{C}$ is a k-linear ∞-category (that is, a Mod_k-module in the ∞-category of presentable stable ∞-categories), and $k \to \mathrm{A}$ is a map of $\mathbf{E}_\infty$-rings, then $\mathcal{C} \otimes_k \mathrm{A}$ will denote the A-linear ∞-category $\mathcal{C} \otimes_{\mathrm{Mod}_k} \mathrm{Mod}_\mathrm{A}$; similarly if k is a classical commutative ring and $\mathcal{C}$ is an abelian k-linear category. If X is a stack over a commutative ring R and $\mathrm{R} \to \mathrm{S}$ is a map of commutative rings, I will write X_S or $\mathrm{X} \otimes_\mathrm{R} \mathrm{S}$ to denote the base change $\mathrm{X} \times_{\mathrm{Spec(R)}} \mathrm{Spec(S)}$.

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[CG10], and also for introducing me to [BFM05], which led me down the beautiful road to geometric representation theory. Since the first version of this article, my understanding of the subject has been greatly influenced by [BZSV23]. I would also like to thank some anonymous referees for helpful suggestions on improving an old version of this article. I'm also very grateful to the PD Soros Fellowship, the Simons Collaboration on Perfection in Algebra and Geometry, and the National Science Foundation for funding my work (the latter under grant DGE-2140743 and DMS-2502909). Last, but certainly far from least, the influence, support, advice, and encouragement of my advisors Dennis Gaitsgory and Mike Hopkins is evident throughout this project; I cannot thank them enough.

2. Sheaf theory

2.1. **Definition of sheaf category.**

In this section, we review some of the setup of sheaf theory in the setting of topological stacks. There are several sources for this material; I refer the reader to [Kha25] for some of the constructions below.

Recall that a topological stack is a sheaf of ∞-groupoids (also known as "anima") on the category Top of locally compact Hausdorff topological spaces which satisfies descent with respect to surjective local homeomorphisms. A representable map $\mathrm{X} \to \mathrm{Y}$ of topological stacks will be called a topological submersion if the pullback along any $\mathrm{Z} \to \mathrm{X}$ from a topological space Z defines a topological submersion $\mathrm{X} \times_{\mathrm{Y}} \mathrm{Z} \to \mathrm{Z}$. A topological stack X is called Artin if the diagonal $\mathrm{X} \to \mathrm{X} \times \mathrm{X}$ is a representable morphism, and there is a topological space Y with a map $\mathrm{Y} \to \mathrm{X}$ which is representable by a surjective topological submersion. (In the language of [Kha25], this is a "1-Artin" stack.)

Definition 2.1.1. Let k be an $\mathbf{E}_\infty$-ring and let X be a topological stack. Then the category $\mathrm{Shv}(\mathrm{X}; k)$ of sheaves of k-modules on X is defined as

$$\mathrm{Shv}(\mathrm{X}; k) = \lim_{\mathrm{Y} \to \mathrm{X}} \mathrm{Shv}(\mathrm{Y}; k),$$

where the limit runs over maps $\mathrm{Y} \to \mathrm{X}$ where Y is a locally compact Hausdorff space. If X is an Artin stack, it suffices to restrict the limit diagram to topological submersions $\mathrm{Y} \to \mathrm{X}$.

By [Kha25, Theorem 1.2], there is a six-functor formalism associated to the functor $\mathrm{X} \mapsto \mathrm{Shv}(\mathrm{X}; k)$ on topological stacks. There are a few important objects of $\mathrm{Shv}(\mathrm{X}; k)$ which we will use below. If $q : \mathrm{X} \to *$ is the canonical map to a point, the constant sheaf $\underline{k}_{\mathrm{X}}$ (resp. the dualizing sheaf ω_{X}) is the pullback $q^* k$ (resp. $q^! k$). If $q : \mathrm{X} \to *$ is the canonical map to a point and $\mathcal{F} \in \mathrm{Shv}(\mathrm{X}; k)$, we write $\mathrm{R}\Gamma(\mathrm{X}; \mathcal{F})$ for the global sections and $\mathrm{R}\Gamma_c(\mathrm{X}; \mathcal{F}) = q_! \mathcal{F}$ for the compactly supported global sections. When $\mathcal{F} = \underline{k}_{\mathrm{X}}$, we write $\mathrm{C}^*(\mathrm{X}; k) = \mathrm{R}\Gamma(\mathrm{X}; \underline{k}_{\mathrm{X}})$; $\mathrm{C}^*_c(\mathrm{X}; k) = \mathrm{R}\Gamma_c(\mathrm{X}; \underline{k}_{\mathrm{X}})$; $\mathrm{C}^{\mathrm{BM}}_*(\mathrm{X}; k) = \mathrm{R}\Gamma(\mathrm{X}; \omega_{\mathrm{X}})$; and $\mathrm{C}_*(\mathrm{X}; k) = \mathrm{R}\Gamma_c(\mathrm{X}; \omega_{\mathrm{X}})$. Note that Verdier duality identifies the k-linear dual of $\mathrm{C}^*_c(\mathrm{X}; k)$ with $\mathrm{C}^{\mathrm{BM}}_*(\mathrm{X}; k)$. Also, if the natural map from $\underline{k}_{\mathrm{X}}$ to its Verdier bidual is an isomorphism, then the k-linear dual of $\mathrm{C}_*(\mathrm{X}; k)$ is $\mathrm{C}^*(\mathrm{X}; k)$. If $\mathcal{F}, \mathcal{G} \in \mathrm{Shv}(\mathrm{X}; k)$, we will also write $\mathrm{Ext}^*_{\mathrm{Shv}(\mathrm{X}; k)}(\mathcal{F}, \mathcal{G})$ to mean $\pi_{-*} \mathrm{RHom}_{\mathrm{Shv}(\mathrm{X}; k)}(\mathcal{F}, \mathcal{G})$.

Remark 2.1.2. The pushforward functor q_* is lax symmetric monoidal (since it is right adjoint to q^*, which is symmetric monoidal). Since $\underline{k}_{\mathrm{X}}$ is the unit object in the standard symmetric monoidal structure on $\mathrm{Shv}(\mathrm{X}; k)$, it follows that $\mathrm{C}^*(\mathrm{X}; k)$

is an $\mathbf{E}_\infty$-k-algebra. Moreover, if $\mathcal{F} \in \mathrm{Shv}(\mathrm{X}; k)$, then $\mathrm{R}\Gamma(\mathrm{X}; \mathcal{F})$ is canonically a $\mathrm{C}^*(\mathrm{X}; k)$-module.

The category $\mathrm{Shv}(\mathrm{X}; k)$ admits a different symmetric monoidal structure given by !-tensor product. Then $q^!$ is !-symmetric monoidal, so its left adjoint $q_!$ is oplax !-symmetric monoidal. Since ω_X is the unit object for the !-tensor product (and the !-tensor product is the usual tensor product on Mod_k when $\mathrm{X} = *$), it follows that $\mathrm{C}_*(\mathrm{X}; k)$ is an $\mathbf{E}_\infty$-k-coalgebra. Moreover, if $\mathcal{F} \in \mathrm{Shv}(\mathrm{X}; k)$, then $\mathrm{R}\Gamma_c(\mathrm{X}; \mathcal{F})$ is canonically a $\mathrm{C}_*(\mathrm{X}; k)$-comodule.

Remark 2.1.3. Let $f : \mathrm{Y} \to \mathrm{X}$ be a map of topological stacks which is representable by Artin stacks, and let $\mathrm{Y}^\bullet$ denote its Čech nerve. If f is a topological submersion, then by [Kha25, Proposition 1.3], there is an equivalence

$$\mathrm{Shv}(\mathrm{X}; k) \xrightarrow{\simeq} \mathrm{Tot}(\mathrm{Shv}(\mathrm{Y}^\bullet; k)),$$

where the transition maps are given by $*$-pullback. Using !-pullback instead also produces an equivalence.

Remark 2.1.4. By [Lur16, Theorem A.4.19], if X is a locally compact Hausdorff space which is locally of singular shape in the sense of [Lur16, Definition A.4.15] (e.g., X is locally weakly contractible), then there is a fully faithful functor $\mathrm{Loc}(\mathrm{X}; k) := \mathrm{Fun}(\mathrm{X}, \mathrm{Mod}_k) \hookrightarrow \mathrm{Shv}(\mathrm{X}; k)$ whose essential image consists of locally constant sheaves. Remark 2.1.3 implies the same claim when X is an Artin stack which admits a topological submersion from a locally compact Hausdorff space which is locally of singular shape.

Example 2.1.5. For instance, let G be a locally contractible locally compact topological group. Then the map $* \to \mathrm{BG}$ is a topological submersion, and so Remark 2.1.3 gives an equivalence

$$\mathrm{Shv}(\mathrm{BG}; k) \xrightarrow{\simeq} \mathrm{Tot}(\mathrm{Shv}(\mathrm{G}^{\times\bullet}; k)).$$

However, since the $*$-pullback functor $\mathrm{Shv}(*; k) \to \mathrm{Shv}(\mathrm{G}; k)$ lands in the subcategory of (locally) constant sheaves on G, and the functor $\mathrm{Loc}(\mathrm{G}; k) \to \mathrm{Shv}(\mathrm{G}; k)$ is fully faithful by Remark 2.1.4, this implies that there is an equivalence

$$\mathrm{Shv}(\mathrm{BG}; k) \simeq \mathrm{Loc}(\mathrm{BG}; k) = \mathrm{Fun}(\mathrm{BG}, \mathrm{Mod}_k).$$

An object of $\mathrm{Shv}(\mathrm{BG}; k)$ can therefore be viewed as a k-module with a naive G-action[2]. If $q : \mathrm{BG} \to *$ is the canonical map, the global sections functor $q_* : \mathrm{Shv}(\mathrm{BG}; k) \to \mathrm{Mod}_k$ sends a k-module M with naive G-action to its homotopy fixed points $\mathrm{M}^{h\mathrm{G}}$. Similarly $q_!$ sends M to the homotopy orbits $\mathrm{M}_{h\mathrm{G}}$. For notational simplicity, I will also write $k^{h\mathrm{G}} = \mathrm{C}^*(\mathrm{BG}; k)$. This means one takes the homotopy fixed points for the trivial G-action on k.

Remark 2.1.6. Following [MNN17], let us briefly summarize some facts about the category $\mathrm{Fun}(\mathrm{BG}, \mathrm{Mod}_k)$. There is a functor $\mathrm{Fun}(\mathrm{BG}, \mathrm{Mod}_k) \to \mathrm{Mod}_{k^{h\mathrm{G}}}$ given by homotopy fixed points, and it is often close to being an equivalence in the following sense. Suppose that the natural map $k \otimes_{k^{h\mathrm{G}}} k \to \mathrm{C}^*(\mathrm{G}; k)$ is an isomorphism (for instance, k could be even and G could be a torus; or k could be any $\mathbf{E}_\infty$-$\mathbf{Z}$-algebra).

[2] In the homotopy-theory literature, this is called "Borel-equivariant". I will not use this terminology here since it might confuse with equivariance for the Borel subgroup of a complex reductive group.

By [MNN17, Theorem 7.29], this implies that the homotopy fixed points functor $\mathrm{Fun}(\mathrm{BG}, \mathrm{Mod}_k) \to \mathrm{Mod}_{k^{h\mathrm{G}}}$ is an equivalence onto the subcategory $\mathrm{Mod}_{k^{h\mathrm{G}}}^{\hat{k}} \subseteq \mathrm{Mod}_{k^{h\mathrm{G}}}$ of k-complete $k^{h\mathrm{G}}$-modules. The homotopy fixed points functor is lax symmetric monoidal (since it is right adjoint to the functor $\mathrm{Mod}_k \to \mathrm{Fun}(\mathrm{BG}, \mathrm{Mod}_k)$ given by restriction along $\mathrm{BG} \to *$), and when $k \otimes_{k^{h\mathrm{G}}} k \xrightarrow{\cong} \mathrm{C}^*(\mathrm{G}; k)$, it is symmetric monoidal. The equivalence $\mathrm{Fun}(\mathrm{BG}, \mathrm{Mod}_k) \xrightarrow{\simeq} \mathrm{Mod}_{k^{h\mathrm{G}}}^{\hat{k}}$ sends the tensor product on $\mathrm{Fun}(\mathrm{BG}, \mathrm{Mod}_k)$ to the k-completed tensor product $-\widehat{\otimes}_{k^{h\mathrm{G}}}-$.

Note that the natural map $k \otimes_{k^{h\mathrm{G}}} k \to \mathrm{C}^*(\mathrm{G}; k)$ is an isomorphism (and so $\mathrm{Fun}(\mathrm{BG}, \mathrm{Mod}_k) \simeq \mathrm{Mod}_{k^{h\mathrm{G}}}^{\hat{k}}$) if k is even and G is a torus; we will often use this fact below without further reference to this remark. In fact, the map is also an isomorphism if G is connected and k is a connective $\mathbf{E}_\infty$-ring such that $\pi_0(k)$ is a perfect k-module; this can be viewed as a regularity condition on k (see [BL14]). Indeed, since G is assumed to be connected, Lemma 3.2.2 reduces us to the case when k is discrete, where it follows from [MNN17, Proposition 7.33]. However, the map $k \otimes_{k^{h\mathrm{G}}} k \to \mathrm{C}^*(\mathrm{G}; k)$ need not be an isomorphism in general; for instance, it fails if $k = \mathrm{KU}$ and $\mathrm{G} = \mathrm{SO}_3$ by [MNN17, Example 7.38].

If $\mathrm{Y} = \mathrm{X}/\mathrm{G}$, we will write $\mathrm{Shv}_\mathrm{G}(\mathrm{X}; k)$ to denote $\mathrm{Shv}(\mathrm{Y}; k)$. In this case, one has a refinement of the (Borel-Moore) (co)homology groups of Y, defined as follows.

Definition 2.1.7. Let X be a locally compact Hausdorff space equipped with a continuous action of a locally contractible locally compact Hausdorff topological group G. There is a map $q : \mathrm{X}/\mathrm{G} \to \mathrm{BG}$. Let $\omega_\mathrm{X} \in \mathrm{Shv}_\mathrm{G}(\mathrm{X}; k)$ denote $q^! \underline{k}_{\mathrm{BG}}$. Define

$$\begin{aligned} \mathrm{C}^*_\mathrm{G}(\mathrm{X}; k) &:= \mathrm{R}\Gamma(\mathrm{BG}; q_* \underline{k}_{\mathrm{X}/\mathrm{G}}) \cong \mathrm{C}^*(\mathrm{X}; k)^{h\mathrm{G}}, \\ \mathrm{C}^*_{\mathrm{G},c}(\mathrm{X}; k) &:= \mathrm{R}\Gamma(\mathrm{BG}; q_! \underline{k}_{\mathrm{X}/\mathrm{G}}) \cong \mathrm{C}^*_c(\mathrm{X}; k)^{h\mathrm{G}}, \\ \mathrm{C}^{\mathrm{G},\mathrm{BM}}_*(\mathrm{X}; k) &:= \mathrm{R}\Gamma(\mathrm{BG}; q_* \omega_\mathrm{X}) \cong \mathrm{C}^{\mathrm{BM}}_*(\mathrm{X}; k)^{h\mathrm{G}}, \\ \mathrm{C}^\mathrm{G}_*(\mathrm{X}; k) &:= \mathrm{R}\Gamma(\mathrm{BG}; q_! \omega_\mathrm{X}) \cong \mathrm{C}_*(\mathrm{X}; k)^{h\mathrm{G}}. \end{aligned}$$

Note that $\mathrm{C}^*_\mathrm{G}(\mathrm{X}; k) = \mathrm{C}^*(\mathrm{X}/\mathrm{G}; k)$, but the other constructions differ from $\mathrm{C}^*_c(\mathrm{X}/\mathrm{G}; k)$, $\mathrm{C}^{\mathrm{BM}}_*(\mathrm{X}/\mathrm{G}; k)$, and $\mathrm{C}_*(\mathrm{X}/\mathrm{G}; k)$. If $\mathcal{F} \in \mathrm{Shv}_\mathrm{G}(\mathrm{X}; k)$, we will similarly define $\mathrm{R}\Gamma_\mathrm{G}(\mathrm{X}; \mathcal{F}) = \mathrm{R}\Gamma(\mathrm{BG}; q_* \mathcal{F}) \cong \mathrm{C}^*(\mathrm{X}; \mathcal{F})^{h\mathrm{G}}$ and $\mathrm{R}\Gamma_{\mathrm{G},c}(\mathrm{X}; \mathcal{F}) = \mathrm{R}\Gamma(\mathrm{BG}; q_! \mathcal{F}) \cong \mathrm{C}^*_c(\mathrm{X}; \mathcal{F})^{h\mathrm{G}}$.

Just like in Remark 2.1.2, $\mathrm{C}^*_\mathrm{G}(\mathrm{X}; k)$ is an $\mathbf{E}_\infty$-$k^{h\mathrm{G}}$-algebra, and if $\mathcal{F} \in \mathrm{Shv}_\mathrm{G}(\mathrm{X}; k)$, then $\mathrm{R}\Gamma_\mathrm{G}(\mathrm{X}; \mathcal{F})$ is canonically a $\mathrm{C}^*_\mathrm{G}(\mathrm{X}; k)$-module. Similarly, $q_! \omega_\mathrm{X}$ canonically acquires the structure of an $\mathbf{E}_\infty$-coalgebra in $\mathrm{Shv}(\mathrm{BG}; k) = \mathrm{Fun}(\mathrm{BG}, \mathrm{Mod}_k)$; in other words, $\mathrm{C}_*(\mathrm{X}; k)$ is a naive G-equivariant $\mathbf{E}_\infty$-k-coalgebra. Moreover, if $\mathcal{F} \in \mathrm{Shv}_\mathrm{G}(\mathrm{X}; k)$, then $q_! \mathcal{F}$ is an $\mathbf{E}_\infty$-$q_! \omega_\mathrm{X}$-coalgebra.

Remark 2.1.8. Note that $\mathrm{C}^\mathrm{G}_*(\mathrm{X}; k)$ may not be an $\mathbf{E}_\infty$-$k^{h\mathrm{G}}$-coalgebra, since the functor of homotopy fixed points is lax symmetric monoidal and not oplax symmetric monoidal. However, if $k \otimes_{k^{h\mathrm{G}}} k \xrightarrow{\cong} \mathrm{C}^*(\mathrm{G}; k)$, then Remark 2.1.6 implies that $\mathrm{C}^\mathrm{G}_*(\mathrm{X}; k)$ is an $\mathbf{E}_\infty$-$k^{h\mathrm{G}}$-coalgebra. Moreover, if $\mathcal{F} \in \mathrm{Shv}_\mathrm{G}(\mathrm{X}; k)$, then $\mathrm{R}\Gamma_{\mathrm{G},c}(\mathrm{X}; \mathcal{F})$ is canonically a $\mathrm{C}^\mathrm{G}_*(\mathrm{X}; k)$-comodule in $\mathrm{Mod}_{k^{h\mathrm{G}}}$. Note that in this case, Verdier duality identifies the $k^{h\mathrm{G}}$-linear dual of $\mathrm{C}^*_{\mathrm{G},c}(\mathrm{X}; k)$ with $\mathrm{C}^{\mathrm{G},\mathrm{BM}}_*(\mathrm{X}; k)$. Also, if the natural map from $\underline{k}_\mathrm{X}$ to its Verdier bidual is an isomorphism, then the $k^{h\mathrm{G}}$-linear dual of $\mathrm{C}^\mathrm{G}_*(\mathrm{X}; k)$ is $\mathrm{C}^*_\mathrm{G}(\mathrm{X}; k)$.

Remark 2.1.9. The definition of $\mathrm{Shv}_\mathrm{G}(\mathrm{X}; k)$ (as well as the various objects therein, like ω_X and $\underline{k}_\mathrm{X}$) can be extended to the case when k itself admits a naive G-action.

Namely, suppose $k \in \mathrm{CAlg}(\mathrm{Fun}(\mathrm{BG}, \mathrm{Sp}))$. If $q : \mathrm{X} \to *$ denotes the canonical map to a point, then $q^*k \in \mathrm{Shv_G}(\mathrm{X}; \mathbb{S})$ is an $\mathbf{E}_\infty$-algebra; define $\mathrm{Shv_G}(\mathrm{X}; k) = \mathrm{Mod}_{q^*k}(\mathrm{Shv_G}(\mathrm{X}; \mathbb{S}))$. When $\mathrm{X} = *$, the category $\mathrm{Shv_G}(*; k)$ is just $\mathrm{Mod}_k(\mathrm{CAlg}(\mathrm{Fun}(\mathrm{BG}, \mathrm{Sp})))$, which we will denote by $\mathrm{Mod}_k^{h\mathrm{G}}$.

Below, we will need to consider $\mathrm{Shv_G}(\mathrm{X}; k)$ when G is the complex points of a pro-algebraic group and X is the complex points of an ind-G-scheme. This necessitates discussion of sheaf theory on spaces of "infinite type". I will follow Raskin [Ras15], who described this picture in the setting of D-modules (see also [Nad05, Section 2.2]).

Setup 2.1.10. Let H be a topological group of the form $\lim_j \mathrm{H}_j$ where each H_j is a locally contractible locally compact Hausdorff topological group, and where the transition maps $\mathrm{H}_i \to \mathrm{H}_j$ for $i > j$ are surjections with contractible kernel. Let $\mathrm{Y} = \lim_\alpha \mathrm{Y}_\alpha$ be a pro-locally compact Hausdorff space where the transition maps $f_{\alpha\beta} : \mathrm{Y}_\alpha \to \mathrm{Y}_\beta$ for $\alpha > \beta$ are topological submersion with contractible fibers. Suppose H acts continuously on the pro-system Y_α, where for each α, the H-action on Y_α factors through the surjection $\mathrm{H} \twoheadrightarrow \mathrm{H}_j$ for some j. We will refer to the above setup by simply saying that Y is a *good* H-space.

Construction 2.1.11. Let Y be a good H-space as in Setup 2.1.10. Fix α, and suppose j is such that the H-action on Y_α factors through $\mathrm{H} \twoheadrightarrow \mathrm{H}_j$. Since the maps $\mathrm{H}_i \to \mathrm{H}_j$ are surjections with contractible kernels, pullback along $\mathrm{Y}_\alpha/\mathrm{H}_i \to \mathrm{Y}_\alpha/\mathrm{H}_j$ induces an equivalence on sheaf categories. Thus $\mathrm{Shv_H}(\mathrm{Y}_\alpha; k) := \mathrm{Shv}_{\mathrm{H}_j}(\mathrm{Y}_\alpha; k)$ is well-defined and independent of j.

Define $\mathrm{Shv_H}(\mathrm{Y}; k)$ as the direct limit $\mathrm{colim}_\alpha \mathrm{Shv_H}(\mathrm{Y}_\alpha; k)$ along the !-pullback functors $f^!_{\alpha\beta} : \mathrm{Shv_H}(\mathrm{Y}_\beta; k) \to \mathrm{Shv_H}(\mathrm{Y}_\alpha; k)$. There is an object $\omega_\mathrm{Y} \in \mathrm{Shv_H}(\mathrm{Y}; k)$ defined via the natural identifications $\omega_{\mathrm{Y}_\alpha} \cong f^!_{\alpha\beta}\omega_{\mathrm{Y}_\beta}$. There is also a symmetric monoidal structure on $\mathrm{Shv_H}(\mathrm{Y}; k)$ given by !-tensor product, and if $f : \mathrm{Y} \to \mathrm{Y}'$ is a map of good H-spaces, there is a pullback functor $f^! : \mathrm{Shv_H}(\mathrm{Y}'; k) \to \mathrm{Shv_H}(\mathrm{Y}; k)$. The projection formula also implies that if $f : \mathrm{Y} \to \mathrm{Y}'$ is a map of good H-spaces, there is a pushforward functor $f_{*,!} : \mathrm{Shv_H}(\mathrm{Y}; k) \to \mathrm{Shv_H}(\mathrm{Y}'; k)$.

Similarly, define $\mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}; k)$ as the direct limit $\mathrm{colim}_\alpha \mathrm{Shv_H}(\mathrm{Y}_\alpha; k)$ along the $*$-pullback functors $f^*_{\alpha\beta} : \mathrm{Shv_H}(\mathrm{Y}_\beta; k) \to \mathrm{Shv_H}(\mathrm{Y}_\alpha; k)$. Equivalently, since $*$-pushforward is right adjoint to $*$-pullback, $\mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}; k)$ is equivalent to the inverse limit $\lim_\alpha \mathrm{Shv_H}(\mathrm{Y}_\alpha; k)$ along $*$-pushforward. There is an object $\underline{k}_\mathrm{Y} \in \mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}; k)$ defined via the natural identifications $\underline{k}_{\mathrm{Y}_\alpha} \cong f^*_{\alpha\beta}\underline{k}_{\mathrm{Y}_\beta}$. There is also a symmetric monoidal structure on $\mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}; k)$ given by $*$-tensor product, and if $f : \mathrm{Y} \to \mathrm{Y}'$ is a map of good H-spaces, there is a pushforward functor $f_* : \mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}; k) \to \mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}'; k)$ as well as a pullback functor $f^* : \mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}'; k) \to \mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}; k)$. Moreover, f^* is left adjoint to f_*. In particular, when $\mathrm{Y}' = *$, we obtain a global sections functor $\mathrm{R}\Gamma_\mathrm{H}(\mathrm{Y}; -) : \mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}; k) \to \mathrm{Shv}^*_\mathrm{H}(*; k) \simeq \mathrm{Fun}(\mathrm{BH}_j, \mathrm{Mod}_k)$. The projection formula also implies that if $f : \mathrm{Y} \to \mathrm{Y}'$ is a map of good H-spaces, there is a pullback functor $f^{!,*} : \mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}; k) \to \mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}'; k)$.

The projection formula furthermore implies that there is an action of $\mathrm{Shv_H}(\mathrm{Y}; k)$ (equipped with the !-tensor product) on $\mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}; k)$: if $\mathcal{F} \in \mathrm{Shv_H}(\mathrm{Y}_\alpha; k)$, $\mathcal{G} \in \mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}; k)$, and $f_\alpha : \mathrm{Y} \to \mathrm{Y}_\alpha$ is the canonical map, then $\mathcal{F} \otimes f_{\alpha,*}(\mathcal{G}) \cong f_{\alpha,*}(f^!_\alpha(\mathcal{F}) \otimes^! \mathcal{G})$, so that $f^!_\alpha(\mathcal{F}) \otimes^! \mathcal{G}$ is a well-defined object of $\mathrm{Shv}^*_\mathrm{H}(\mathrm{Y}; k)$.

Construction 2.1.12. Suppose Y is a good H-space. A *local dimension theory* $\mathcal{L}$ for Y is an assignment $\alpha \mapsto \mathcal{L}_\alpha \in \mathrm{Shv_H}(\mathrm{Y}_\alpha; k)$, where each $\mathcal{L}_\alpha$ is $\otimes^*$-invertible, along with the data of compatible isomorphisms $f_{\alpha\beta}^!(\mathcal{L}_\beta) \cong \mathcal{L}_\alpha$ for each $\alpha > \beta$. Since $f_{\alpha\beta}$ is a topological submersion, this implies that there are compatible isomorphisms

$$\begin{aligned} f_{\alpha\beta}^*(\omega_{\mathrm{Y}_\beta} \otimes \mathcal{L}_\beta^{-1}) &\cong f_{\alpha\beta}^!(\omega_{\mathrm{Y}_\beta}) \otimes f_{\alpha\beta}^*(\mathcal{L}_\beta)^{-1} \otimes f_{\alpha\beta}^!(\underline{k}_{\mathrm{Y}_\beta})^{-1} \\ &\cong f_{\alpha\beta}^!(\omega_{\mathrm{Y}_\beta}) \otimes f_{\alpha\beta}^!(\mathcal{L}_\beta)^{-1} \cong \omega_{\mathrm{Y}_\alpha} \otimes \mathcal{L}_\alpha^{-1}. \end{aligned}$$

Therefore, the $*$-pullback of $\omega_{\mathrm{Y}_\alpha}^{\mathrm{ren}} := \omega_{\mathrm{Y}_\alpha} \otimes \mathcal{L}_\alpha^{-1}$ along the map $\mathrm{Y} \to \mathrm{Y}_\alpha$ (for any α) defines an object $\omega_{\mathrm{Y}}^{\mathrm{ren}} \in \mathrm{Shv}_{\mathrm{H}}^*(\mathrm{Y}; k)$. This will be called the *renormalized dualizing sheaf* of Y. The global sections $\mathrm{R}\Gamma_{\mathrm{H}}(\mathrm{Y}; \omega_{\mathrm{Y}}^{\mathrm{ren}})$ will sometimes be denoted $\mathrm{C}_{*+\mathcal{L}}^{\mathrm{H,BM}}(\mathrm{Y}; k)$ to denote emphasis on $\mathcal{L}$.

Example 2.1.13. If Y is a locally compact Hausdorff space, then $\mathrm{Y} = \lim_\alpha \mathrm{Y}_\alpha$ along the constant diagram $\mathrm{Y}_\alpha = \mathrm{Y}$. It follows that $\mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}; k) \simeq \mathrm{Shv}_{\mathrm{H}}^*(\mathrm{Y}; k)$, although the monoidal structures differ. In this case, there is a canonical local dimension theory given by $\alpha \mapsto \mathcal{L}_\alpha = \underline{k}_{\mathrm{Y}}$, and $\omega_{\mathrm{Y}}^{\mathrm{ren}} \cong \omega_{\mathrm{Y}}$, so that $\mathrm{R}\Gamma_{\mathrm{H}}(\mathrm{Y}; \omega_{\mathrm{Y}}^{\mathrm{ren}}) \cong \mathrm{C}_*^{\mathrm{H,BM}}(\mathrm{Y}; k) \cong \mathrm{C}_*^{\mathrm{BM}}(\mathrm{Y}; k)^{h\mathrm{H}}$.

Example 2.1.14. If each Y_α is a smooth manifold, then $\alpha \mapsto \mathcal{L}_\alpha = \omega_{\mathrm{Y}_\alpha}$ is a local dimension theory. Then $\omega_{\mathrm{Y}_\alpha}^{\mathrm{ren}} \cong \underline{k}_{\mathrm{Y}_\alpha}$, so that $\omega_{\mathrm{Y}}^{\mathrm{ren}} = \underline{k}_{\mathrm{Y}}$. In particular, $\mathrm{R}\Gamma_{\mathrm{H}}(\mathrm{Y}; \omega_{\mathrm{Y}}^{\mathrm{ren}}) \cong \mathrm{C}_{\mathrm{H}}^*(\mathrm{Y}; k)$.

Example 2.1.15. Suppose Y' is a locally compact Hausdorff space equipped with trivial H-action, and $g : \mathrm{Y} \to \mathrm{Y}'$ is a map of good H-spaces. Equip Y' with the local dimension theory of Example 2.1.13, so that $\omega_{\mathrm{Y}'}^{\mathrm{ren}} \cong \omega_{\mathrm{Y}'}$. If $\mathcal{L}$ is a local dimension theory on Y, then $\mathcal{L}_g \in \mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}; k)$ is just $\mathcal{L}$.

Assume further that H is an inverse limit of Lie groups H_j along smooth surjections with contractible kernel, and that $\mathrm{Y} = \lim_\alpha \mathrm{Y}_\alpha$ is an H-torsor over Y'. In other words, for each α, there is some j for which the H-action on Y_α descends to a H_j-action, and a map $g_\alpha : \mathrm{Y}_\alpha \to \mathrm{Y}'$ which is an H_j-torsor. Then, there is a natural choice of local dimension theory on Y, given by $\mathcal{L}_\alpha = g_\alpha^! \underline{k}_{\mathrm{Y}'} = \omega_{g_\alpha}$. (Clearly $f_{\alpha\beta}^! \mathcal{L}_\beta \cong \mathcal{L}_\alpha$, so it only remains to check that $\mathcal{L}_\alpha$ is $\otimes^*$-invertible. Since the map g_α is an H_j-torsor (for some j depending on α), this follows from the assumption that each H_j is a Lie group.) In particular, $\omega_{\mathrm{Y}}^{\mathrm{ren}}$ is the $*$-pullback of $\omega_{\mathrm{Y}_\alpha} \otimes \omega_{g_\alpha}^{-1}$ (for any α) along the map $\mathrm{Y} \to \mathrm{Y}_\alpha$. But the natural map $g_\alpha^*(\omega_{\mathrm{Y}'}) \to \omega_{\mathrm{Y}_\alpha} \otimes \omega_{g_\alpha}^{-1}$ is an isomorphism, so we conclude that $\omega_{\mathrm{Y}}^{\mathrm{ren}}$ is the $*$-pullback of $g_\alpha^*(\omega_{\mathrm{Y}'})$ along the map $\mathrm{Y} \to \mathrm{Y}_\alpha$. In other words, $\omega_{\mathrm{Y}}^{\mathrm{ren}} \cong g^*(\omega_{\mathrm{Y}'})$.

Definition 2.1.16. Suppose Y is a good H-space, and let $\mathcal{L}$ be a local dimension theory. Then the functor $\eta_{\mathrm{Y}} : \mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}; k) \to \mathrm{Shv}_{\mathrm{H}}^*(\mathrm{Y}; k)$ given by acting on $\omega_{\mathrm{Y}}^{\mathrm{ren}}$ is an equivalence. Indeed it is the colimit of the functors $\mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}_\alpha; k) \to \mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}_\alpha; k)$ sending $\mathcal{F} \mapsto \mathcal{F} \otimes^! \omega_{\mathrm{Y}_\alpha}^{\mathrm{ren}} \cong \mathcal{F} \otimes \mathcal{L}_\alpha^{-1}$, which are all equivalences.

If $f : \mathrm{Y} \to \mathrm{Y}'$ is a map of good H-spaces, each equipped with a local dimension theory $\mathcal{L}$ and $\mathcal{L}'$, then there are two functors

$$f_*^{\mathrm{ren}} : \mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}; k) \xrightarrow{\eta_{\mathrm{Y}}} \mathrm{Shv}_{\mathrm{H}}^*(\mathrm{Y}; k) \xrightarrow{f_*} \mathrm{Shv}_{\mathrm{H}}^*(\mathrm{Y}'; k) \xrightarrow{\eta_{\mathrm{Y}'}^{-1}} \mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}'; k),$$

and similarly

$$f^{!,\mathrm{ren}} : \mathrm{Shv}_{\mathrm{H}}^*(\mathrm{Y}'; k) \xrightarrow{\eta_{\mathrm{Y}'}^{-1}} \mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}'; k) \xrightarrow{f^!} \mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}; k) \xrightarrow{\eta_{\mathrm{Y}}} \mathrm{Shv}_{\mathrm{H}}^*(\mathrm{Y}; k).$$

Note that $f^{!,\mathrm{ren}}(\omega_{\mathrm{Y}'}^{\mathrm{ren}}) = \omega_{\mathrm{Y}}^{\mathrm{ren}}$.

Remark 2.1.17. Suppose Y and Y' are locally compact Hausdorff H-spaces (viewed as good H-spaces via the constant limit diagram) with local dimension theories $\mathcal{L}$ and $\mathcal{L}'$, so these are just invertible objects of $\mathrm{Shv}_{\mathrm{H}}(\mathrm{Y};k)$ and $\mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}';k)$. Then $\mathrm{Shv}_{\mathrm{H}}(\mathrm{Y};k)$ is canonically $\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{Y};k)$. If $f:\mathrm{Y}\to\mathrm{Y}'$ is a map of H-spaces, there are two functors $f^!, f^{!,\mathrm{ren}}:\mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}';k)\to\mathrm{Shv}_{\mathrm{H}}(\mathrm{Y};k)$. Let $\mathcal{L}_f\in\mathrm{Shv}_{\mathrm{H}}(\mathrm{Y};k)$ denote $f^*(\mathcal{L}')^{-1}\otimes\mathcal{L}$. If $\mathcal{F}\in\mathrm{Shv}_{\mathrm{H}}(\mathrm{Y};k)$, then

$$f^{!,\mathrm{ren}}(\mathcal{F})\cong f^!(\mathcal{F}\otimes\mathcal{L}')\otimes\mathcal{L}^{-1}\cong f^!(\mathcal{F})\otimes f^*(\mathcal{L}')\otimes\mathcal{L}^{-1}\cong f^!(\mathcal{F})\otimes\mathcal{L}_f^{-1},$$

so that $f^!\cong f^{!,\mathrm{ren}}\otimes\mathcal{L}_f$. Since $f^!(\omega_{\mathrm{Y}'})=\omega_{\mathrm{Y}}$, it follows that $f^{!,\mathrm{ren}}(\omega_{\mathrm{Y}'})=\omega_{\mathrm{Y}}\otimes\mathcal{L}_f^{-1}$.

Now suppose that Y and Y' are general good H-spaces, and $g:\mathrm{Y}\to\mathrm{Y}'$ is a map of good H-spaces such that there is some α for which f is pulled back from a map $g_\alpha:\mathrm{Y}_\alpha\to\mathrm{Y}'_\alpha$, i.e., such that there is a Cartesian square

$$\begin{tikzcd} \mathrm{Y} \arrow[r, "g"] \arrow[d, "f_\alpha"'] & \mathrm{Y}' \arrow[d, "f'_\alpha"] \\ \mathrm{Y}_\alpha \arrow[r, "g_\alpha"'] & \mathrm{Y}'_\alpha. \end{tikzcd}$$

Let $\mathcal{F}\in\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{Y};k)$. Then

$$g^{!,*}(\mathcal{F})=\mathrm{colim}_\alpha\, f^*_\alpha f_{\alpha,*}g^{!,*}(\mathcal{F})\cong\mathrm{colim}_\alpha\, f^*_\alpha g^!_\alpha f'_{\alpha,*}(\mathcal{F}).$$

Our discussion above identifies this with

$$\mathrm{colim}_\alpha\, f^*_\alpha(\mathcal{L}_{g_\alpha}\otimes g^{!,\mathrm{ren}}_\alpha f'_{\alpha,*}(\mathcal{F}))\cong\mathrm{colim}_\alpha\, f^*_\alpha(\mathcal{L}_{g_\alpha})\otimes f^*_\alpha(g^{!,\mathrm{ren}}_\alpha f'_{\alpha,*}(\mathcal{F})).$$

The definition of a local dimension theory implies that $f^*_\alpha(\mathcal{L}_{g_\alpha})$ is independent of α, so it defines an object $\mathcal{L}_g\in\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{Y};k)$. Pulling $\mathcal{L}_g$ out of the colimit, it follows that $g^{!,*}(\mathcal{F})\cong\mathcal{L}_g\otimes g^{!,\mathrm{ren}}(\mathcal{F})$ for $\mathcal{F}\in\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{Y}';k)$.

Construction 2.1.18. Suppose $\mathrm{X}=\mathrm{colim}_n\,\mathrm{X}^n$ is a filtered colimit of good H-spaces along closed embeddings $i_{nm}:\mathrm{X}^n\hookrightarrow\mathrm{X}^m$; we will say that X is an *ind-good* H-space. Define $\mathrm{Shv}_{\mathrm{H}}(\mathrm{X};k)$ as the inverse limit $\lim_n\mathrm{Shv}_{\mathrm{H}}(\mathrm{X}^n;k)$, where the inverse limit is taken along !-pullbacks. Since the left adjoint to !-pullback is !-pushforward (and the transition maps are closed embeddings), one can alternatively write $\mathrm{Shv}_{\mathrm{H}}(\mathrm{X};k)$ as $\mathrm{colim}_n\,\mathrm{Shv}_{\mathrm{H}}(\mathrm{X}^n;k)$, where the colimit is over $(*,!)$-pushforwards. Similarly, define $\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{X};k)$ as the direct limit $\mathrm{colim}_n\,\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{X}^n;k)$, where the direct limit is taken along $*$-pushforwards. Equivalently, it is $\lim_n\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{X}^n;k)$, where the limit is over $(!,*)$-pullbacks.

If $f:\mathrm{X}\to\mathrm{Y}$ is a map of ind-good H-spaces, then the maps $f_{*,!}:\mathrm{Shv}_{\mathrm{H}}(\mathrm{X}^n;k)\to\mathrm{Shv}_{\mathrm{H}}(\mathrm{Y}^n;k)$ give a pushforward functor $f_{*,!}:\mathrm{Shv}_{\mathrm{H}}(\mathrm{X};k)\to\mathrm{Shv}_{\mathrm{H}}(\mathrm{Y};k)$. The maps $f_*:\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{X}^n;k)\to\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{Y}^n;k)$ give a pushforward functor $f_*:\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{X};k)\to\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{Y};k)$.

Construction 2.1.19. Suppose $\mathrm{X}=\mathrm{colim}_n\,\mathrm{X}^n$ is an ind-good H-space. Equip each X^n with a local dimension theory $\mathcal{L}_n$, and suppose that for each $n\leq m$, there is some α such that the closed embedding $i:\mathrm{X}^n\to\mathrm{X}^m$ is pulled back from a closed embedding $i_\alpha:\mathrm{X}^n_\alpha\to\mathrm{X}^m_\alpha$. It is not true that $i^{!,*}\omega^{\mathrm{ren}}_{\mathrm{X}_m}\cong\omega^{\mathrm{ren}}_{\mathrm{X}^n}$; so the sheaves $\omega^{\mathrm{ren}}_{\mathrm{X}^n}\in\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{X}^n;k)$ do not assemble to an object of $\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{X};k)$.

Instead, we must ask for additional data. An *($\mathcal{L}$-compatible) ind-dimension theory* τ on X is an assignment $n\mapsto\tau^n\in\mathrm{Shv}^*_{\mathrm{H}}(\mathrm{X}^n;k)$ where each τ^n is dualizable, along with compatible isomorphisms $\tau^n \cong \mathcal{L}_i \otimes i^*(\tau^m)$. Since $\omega_{X^n}^{\mathrm{ren}} \cong i^{!,\mathrm{ren}}(\omega_{X_m}^{\mathrm{ren}})$, there are isomorphisms

$$\omega_{X^n}^{\mathrm{ren}} \otimes \tau^n \cong i^{!,\mathrm{ren}}(\omega_{X_m}^{\mathrm{ren}}) \otimes \mathcal{L}_i \otimes i^*(\tau^m) \cong i^{!,*}(\omega_{X_m}^{\mathrm{ren}}) \otimes i^*(\tau^m) \cong i^{!,*}(\omega_{X_m}^{\mathrm{ren}} \otimes \tau^m).$$

Here, we have used Remark 2.1.17 in the third isomorphism, and the final isomorphism uses that τ^m is dualizable. It follows that the objects $\omega_{X^n}^{\mathrm{ren}} \otimes \tau^n$ assemble to an object $\omega_X^{\mathrm{ren}} \in \mathrm{Shv}_H^*(X; k)$. (I will suppress τ from the notation for readability; if emphasis is needed, I will write it as $\omega_X^{\mathrm{ren}}[\tau]$.) This will be called the *(τ-)renormalized dualizing sheaf* of X. The global sections $\mathrm{R}\Gamma_H(X; \omega_X^{\mathrm{ren}})$ will sometimes be denoted $C_{*+\mathcal{L}}^{H,BM}(X; k[\tau])$ to denote emphasis on $\mathcal{L}$ and τ.

Example 2.1.20. If X is a good H-space viewed as an ind-good H-space via the constant colimit diagram, and $\mathcal{L}$ is a local dimension theory on X, then $\tau = \underline{k}_X$ is an ind-dimension theory on X. It follows that $\omega_X^{\mathrm{ren}}[\tau] = \omega_X^{\mathrm{ren}}$.

Example 2.1.21. Suppose X is a filtered colimit of locally compact Hausdorff spaces X^n (each viewed as a good H-space via the constant limit diagram); then there is a canonical equivalence $\mathrm{Shv}_H(X; k) \simeq \mathrm{Shv}_H^*(X; k)$, since there is a canonical equivalence $\mathrm{Shv}_H(X^n; k) \simeq \mathrm{Shv}_H^*(X^n; k)$ for each n. Let $\mathcal{L} = \underline{k}_{X^n}$ be the local dimension theory of Example 2.1.13; note that the equivalence $\eta_X : \mathrm{Shv}_H(X^n; k) \xrightarrow{\cong} \mathrm{Shv}_H^*(X^n; k)$ is just the canonical equivalence mentioned above. Then $\tau^n = \mathcal{L}_n$ is an ind-dimension theory on X. Note that $\omega_{X^n}^{\mathrm{ren}} \otimes \tau^n = \omega_{X^n}$, so that

$$\mathrm{R}\Gamma_H(X; \omega_X^{\mathrm{ren}}) = \mathrm{colim}_n \, \mathrm{R}\Gamma_H(X^n; \omega_{X^n}) \cong \mathrm{colim}_n(C_*^{H,BM}(X^n; k)).$$

Note that ω_X^{ren} is independent of the choice of local dimension theories $\mathcal{L}_n$. Moreover, if $k \otimes_{k^{hH}} k \xrightarrow{\cong} C^*(H; k)$, then by Remark 2.1.8, $\mathrm{R}\Gamma_H(X; \omega_X^{\mathrm{ren}})$ is an $\mathbf{E}_\infty$-k^{hH}-coalgebra, and if $\mathcal{F} \in \mathrm{Shv}_H(X; k)$, then $\mathrm{R}\Gamma_{H,c}(X; \mathcal{F})$ is canonically a $\mathrm{R}\Gamma_H(X; \omega_X^{\mathrm{ren}})$-comodule in $\mathrm{Mod}_{k^{hH}}$. I will abusively write $C_*^{H,BM}(X; k)$ to denote $\mathrm{R}\Gamma_H(X; \omega_X^{\mathrm{ren}})$.

2.2. **GKM theory.** The goal of this section is to prove an analogue of Goresky-Kottwitz-MacPherson (GKM) theory [GKM98] for even $\mathbf{E}_\infty$-rings. A generalization of GKM theory to ring spectra was already studied in [HHH05], and our exposition here closely follows their presentation. For the remainder of this section, fix a torus T and an even $\mathbf{E}_\infty$-ring k.

Construction 2.2.1. Let G be a locally compact Hausdorff topological group, and let $k \in \mathrm{Fun}(BG, Sp)$, so that $\mathrm{Mod}_k^{hG} = \mathrm{Mod}_k(\mathrm{Fun}(BG, Sp))$. Let $\mathcal{P}$ be a set of proper closed subgroups of G which is closed under taking subgroups, and let $\langle \mathcal{P} \rangle \subseteq \mathrm{Mod}_k^{hG}$ denote the smallest thick $\otimes$-ideal which contains the essential image of the inflation functor $\mathrm{Mod}_k^{hG'} \to \mathrm{Mod}_k^{hG}$ for all $G' \in \mathcal{P}$. The left Kan extension of the homotopy fixed points functor $\mathrm{Mod}_k^{hG} \to \mathrm{Mod}_{k^{hG}}$ along the Verdier quotient $\mathrm{Mod}_k^{hG} \to \mathrm{Mod}_k^{hG}/\langle \mathcal{P} \rangle$ defines a lax symmetric monoidal functor $-^{t_{\mathcal{P}}G}$: $\mathrm{Mod}_k^{hG}/\langle \mathcal{P} \rangle \to \mathrm{Mod}_k$. The image of k (with trivial G-action) under this functor will be denoted $k^{t_{\mathcal{P}}G}$. The composite

$$\mathrm{Mod}_k^{hG} \to \mathrm{Mod}_k^{hG}/\langle \mathcal{P} \rangle \xrightarrow{-^{t_{\mathcal{P}}G}} \mathrm{Mod}_k$$

refines to a lax symmetric monoidal functor $\mathrm{Mod}_k^{hG} \to \mathrm{Mod}_{k^{t_{\mathcal{P}}G}}$, which we will continue to denote by $-^{t_{\mathcal{P}}G}$.

Lemma 2.2.2. *Let* X *be a locally compact Hausdorff topological space equipped with a* G*-action such that* X *admits a* G*-invariant cell decomposition, with only finitely many cells in each dimension. Let* $i : \mathrm{X}^{\mathrm{G}} \hookrightarrow \mathrm{X}$ *denote the inclusion. Suppose* $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}}(\mathrm{X}; k)$ *is such that for each cell* $j : \mathrm{U} \to \mathrm{X}$*, the pullback* $j^*\mathcal{F}$ *is a* G*-equivariant locally constant sheaf on* U*. Let* $\mathcal{P}$ *be the set of proper closed subgroups of* G *which are stabilizers of points of* X*. Then the maps* $\mathrm{R}\Gamma(\mathrm{X}; \mathcal{F}) \to \mathrm{R}\Gamma(\mathrm{X}^{\mathrm{G}}; i^*\mathcal{F})$ *and* $\mathrm{R}\Gamma_c(\mathrm{X}^{\mathrm{G}}; i^!\mathcal{F}) \to \mathrm{R}\Gamma_c(\mathrm{X}; \mathcal{F})$ *induce isomorphisms upon applying* $-^{t_{\mathcal{P}}\mathrm{G}}$*.*

Proof. The statement is clear if $\mathrm{X}^{\mathrm{G}} = \mathrm{X}$. In general, the desired statement follows by induction on the cell structure of X. Suppose $i : \mathrm{X}' \hookrightarrow \mathrm{X}$ is a subcomplex of X for which the maps $\mathrm{R}\Gamma(\mathrm{X}'; \mathcal{F}) \to \mathrm{R}\Gamma((\mathrm{X}')^{\mathrm{G}}; i^*\mathcal{F})$ and $\mathrm{R}\Gamma_c((\mathrm{X}')^{\mathrm{G}}; \mathcal{F}) \to \mathrm{R}\Gamma_c(\mathrm{X}'; \mathcal{F})$ induce isomorphisms upon applying $-^{t_{\mathcal{P}}\mathrm{G}}$. Let $\mathrm{G}' \subsetneq \mathrm{G}$ be a proper closed subgroup along with a G-equivariant map $\alpha : \mathrm{S}^n \times \mathrm{G}/\mathrm{G}' \to \mathrm{X}'$. If $\mathrm{X} = \mathrm{X}' \sqcup_{\mathrm{S}^n \times \mathrm{G}/\mathrm{G}'} (\mathrm{D}^{n+1} \times \mathrm{G}/\mathrm{G}')$ is obtained by attaching an equivariant G-cell along α, then the inclusion $(\mathrm{X}')^{\mathrm{G}} \subseteq \mathrm{X}^{\mathrm{G}}$ is a bijection since G' is a proper subgroup. The complement of $\mathrm{X}' \subseteq \mathrm{X}$ is G-equivariantly isomorphic to $\mathbf{R}^{n+1} \times \mathrm{G}/\mathrm{G}'$. By excision, the cofiber of the map $\mathrm{R}\Gamma_c(\mathrm{X}'; i^!\mathcal{F}) \to \mathrm{R}\Gamma_c(\mathrm{X}; \mathcal{F})$ is $\mathrm{R}\Gamma(\mathbf{R}^{n+1} \times \mathrm{G}/\mathrm{G}'; j^*\mathcal{F})$, and similarly the fiber of the map $\mathrm{R}\Gamma(\mathrm{X}; \mathcal{F}) \to \mathrm{R}\Gamma(\mathrm{X}'; i^*\mathcal{F})$ is $\mathrm{R}\Gamma_c(\mathbf{R}^{n+1} \times \mathrm{G}/\mathrm{G}'; j^*\mathcal{F})$.

Recall now that $j^*\mathcal{F}$ was assumed to be G-equivariantly locally constant. Since $\mathbf{R}^{n+1}$ is contractible, and G acts transitively on G/G', the functor $\mathrm{Loc}_{\mathrm{G}}(\mathbf{R}^{n+1} \times \mathrm{G}/\mathrm{G}') \to \mathrm{Mod}_k^{h\mathrm{G}'}$ is an equivalence, and under this equivalence, the functor $\mathrm{R}\Gamma(\mathbf{R}^{n+1} \times \mathrm{G}/\mathrm{G}'; -)$ (resp. $\mathrm{R}\Gamma_c(\mathbf{R}^{n+1} \times \mathrm{G}/\mathrm{G}'; -)$) identifies with the functor $\mathrm{Mod}_k^{h\mathrm{G}'} \to \mathrm{Mod}_{k^{h\mathrm{G}'}}^{\hat{k}}$ sending $\mathrm{M} \mapsto \mathrm{M}^{h\mathrm{G}'}$ (resp. $\mathrm{M} \mapsto \Sigma^{n+1}\mathrm{M}^{h\mathrm{G}'}$). It therefore suffices to show that if $\mathrm{M} \in \mathrm{Mod}_k^{h\mathrm{G}'}$, then $\mathrm{C}_*^{\mathrm{BM}}(\mathrm{G}/\mathrm{G}'; \mathcal{F})^{t_{\mathcal{P}}\mathrm{G}} = 0$. Since $\mathrm{G}' \in \mathcal{P}$, this follows from the definition of $-^{t_{\mathcal{P}}\mathrm{G}}$. □

For the remainder of this section, fix a torus T and an $\mathbf{E}_\infty$-ring $k \in \mathrm{Fun}(\mathrm{BT}, \mathrm{Sp})$ whose underlying nonequivariant spectrum is even.

Lemma 2.2.3. *Let* $\mathcal{P}$ *be a set of proper closed subgroups of* T *which is closed under taking subgroups. Let* $\mathrm{V}_{\mathcal{P}}$ *be a representation of* T *such that for any* $\mathrm{T}' \in \mathcal{P}$*, the weights of* $\mathrm{V}_{\mathcal{P}}$ *are primitive in* $\ker(\mathbb{X}^*(\mathrm{T}) \to \mathbb{X}^*(\mathrm{T}'))$ *and form a* $\mathbf{Z}$*-basis thereof. If* $\mathrm{M} \in \mathrm{Fun}(\mathrm{BT}, \mathrm{Mod}_k)$*, then the map* $\mathrm{M}^{h\mathrm{T}} \to \mathrm{M}^{t_{\mathcal{P}}\mathrm{T}}$ *is localization at the Euler class* $u_{\mathcal{P}} := e(\mathrm{V}_{\mathcal{P}}) \in \pi_*(k^{h\mathrm{T}})$*. In particular, the map* $\mathrm{M}^{h\mathrm{T}} \otimes_{k^{h\mathrm{T}}} k^{t_{\mathcal{P}}\mathrm{T}} \to \mathrm{M}^{t_{\mathcal{P}}\mathrm{T}}$ *is an isomorphism.*

Proof. Recall that $\mathrm{C}_*^{\mathrm{BM}}(\mathrm{T}/\mathrm{T}'; k)$ is the k-linear dual of $\mathrm{C}_c^*(\mathrm{T}/\mathrm{T}'; k)$. Since $\mathrm{C}_c^*(\mathrm{T}/\mathrm{T}'; k)$ is a perfect k-module, we may equivalently view $\langle \mathcal{P} \rangle \subseteq \mathrm{Fun}(\mathrm{BT}, \mathrm{Mod}_k)$ as the thick $\otimes$-ideal which contains all naive T-spectra of the form $\mathrm{C}_c^*(\mathrm{T}/\mathrm{T}'; k)$ for $\mathrm{T}' \in \mathcal{P}$. If $\mathrm{M} = \mathrm{C}_c^*(\mathrm{T}/\mathrm{T}'; k)$, then $\mathrm{M}^{h\mathrm{T}} \cong k^{h\mathrm{T}'}$, which we claim vanishes upon inverting $u_{\mathcal{P}}$. In fact, $k^{h\mathrm{T}'}$ will vanish upon inverting $e(\mathrm{V}')$ for any T-representation V' whose weights are primitive in $\ker(\mathbb{X}^*(\mathrm{T}) \to \mathbb{X}^*(\mathrm{T}'))$ and form a $\mathbf{Z}$-basis thereof. Since $k^{h\mathrm{T}} \cong (k^{h\mathrm{T}'})^{h(\mathrm{T}/\mathrm{T}')}$, we may assume that in fact T' is trivial. We therefore need to show that if V' is a T-representation whose weights are primitive in $\mathbb{X}^*(\mathrm{T})$ and form a $\mathbf{Z}$-basis thereof (for instance, if we choose an isomorphism $\mathrm{T} \cong (\mathbf{C}^\times)^n$, then one can take $\mathrm{V}' = \mathbf{C}^n$), then $k[e(\mathrm{V}')^{-1}] \cong 0$. In fact, there is even an isomorphism $k^{h\mathrm{T}}/e(\mathrm{V}') \xrightarrow{\cong} k$. To see this, we may reduce to the case when T has rank 1. In this case, we may take V' to be the standard 1-dimensional complex representation λ of T. Then there is a cofiber sequence of T-equivariant spectra $\mathrm{T}_+ \to \mathrm{S} \to \mathrm{S}^\lambda$, which induces the map $(\Sigma^{-\lambda}k)^{h\mathrm{T}} \to k^{h\mathrm{T}} \to k$ on homotopy fixed points. Under a Thom isomorphism $\Sigma^{-\lambda}k \cong \Sigma^{-2}k$ (which uses the evenness of k), the map $(\Sigma^{-\lambda}k)^{h\mathrm{T}} \to k^{h\mathrm{T}}$ is simply multiplication by the Euler class of λ, so we conclude that $k^{h\mathrm{T}}/e(\lambda) \cong k$ as desired.

It follows that there is a natural map $\mathrm{M}^{h\mathrm{T}}[\frac{1}{u_{\mathcal{P}}}] \to \mathrm{M}^{t_{\mathcal{P}}\mathrm{T}}$. To check that it is an equivalence, [NS18, Theorem I.4.1] reduces to checking that the functor sending M to the fiber of the map $\mathrm{M}^{h\mathrm{T}} \to \mathrm{M}^{h\mathrm{T}}[\frac{1}{u_{\mathcal{P}}}]$ preserves all colimits. Since this fiber is a direct limit of the functors $\mathrm{M} \mapsto \mathrm{M}^{h\mathrm{T}}/u_{\mathcal{P}}^n$, it suffices to check that each of the latter preserve colimits. By induction on n, it suffices to prove the claim when $n = 1$. Again arguing as above, we may reduce to the case when $\mathcal{P}$ is the set of all proper closed subgroups of T, and then further to the case when T has rank 1. In this case, we need to check that if $t \in \pi_{-2}(k^{h\mathrm{T}})$ is the Euler class of the standard 1-dimensional complex representation of T, the functor $\mathrm{M} \mapsto \mathrm{M}^{h\mathrm{T}}/t$ commutes with all colimits. But this is just the forgetful functor $\mathrm{Fun}(\mathrm{BT}, \mathrm{Mod}_k) \to \mathrm{Mod}_k$, since it identifies with the composite $\mathrm{Fun}(\mathrm{BT}, \mathrm{Mod}_k) \simeq \mathrm{Mod}_{k^{h\mathrm{T}}}^{\hat{k}} \to \mathrm{Mod}_{k^{h\mathrm{T}}} \to \mathrm{Mod}_k$, where the final functor is base-change along $k^{h\mathrm{T}} \mapsto k$. $\square$

For instance, if $\mathrm{T} \cong \mathbf{C}^\times$ is rank 1 and $\mathcal{P}$ is the set of subgroups of the cyclic group μ_n, then $\mathrm{V}_{\mathcal{P}}$ can be the 1-dimensional complex representation of T of weight n. If $t \in \pi_{-2}(k^{h\mathrm{T}})$ is the Euler class of the standard 1-dimensional complex representation of T, then $u_{\mathcal{P}}$ is the n-series $[n](t)$, and $\mathrm{M}^{t_{\mathcal{P}}\mathrm{T}} \cong \mathrm{M}^{h\mathrm{T}}[\frac{1}{[n](t)}]$.

Combining Lemma 2.2.2 and Lemma 2.2.3, we obtain:

Proposition 2.2.4. *Let* X *be a locally compact Hausdorff topological space equipped with a* T*-action such that* X *admits a* T*-invariant cell decomposition, with only finitely many cells in each dimension. Let* $i : \mathrm{X}^\mathrm{T} \hookrightarrow \mathrm{X}$ *denote the inclusion. Suppose* $\mathcal{F} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X}; k)$ *is such that for each cell* $j : \mathrm{U} \to \mathrm{X}$*, the pullback* $j^*\mathcal{F}$ *is a* T*-equivariant locally constant sheaf on* U*. Let* $\mathcal{P}$ *be the set of proper closed subgroups of* T *which are stabilizers of points of* X*. Suppose that* $\mathrm{V}_{\mathcal{P}}$ *is a representation of* T *such that for any* $\mathrm{T}' \in \mathcal{P}$*, the weights of* $\mathrm{V}_{\mathcal{P}}$ *are primitive in* $\ker(\mathbb{X}^*(\mathrm{T}) \to \mathbb{X}^*(\mathrm{T}'))$ *and form a* **Z***-basis thereof. Then the maps* $\mathcal{F} \to i_*i^*\mathcal{F}$ *and* $i_*i^!\mathcal{F} \to \mathcal{F}$ *induce isomorphisms upon applying* $\mathrm{R}\Gamma_\mathrm{T}(\mathrm{X}; -)$ *and* $\mathrm{R}\Gamma_{\mathrm{T},c}(\mathrm{X}; -)$ *and inverting* $u_{\mathcal{P}} = e(\mathrm{V}_{\mathcal{P}})$*, i.e., upon base-change along the map* $k^{h\mathrm{T}} \to k^{t_{\mathcal{P}}\mathrm{T}}$*.*

Remark 2.2.5. Fix notation as in Proposition 2.2.4. Suppose that the stabilizers of points in X are connected, so that if $\mathrm{T}' \in \mathcal{P}$, then $\mathbb{X}^*(\mathrm{T}')$ is itself a **Z**-lattice (i.e., T' has no torsion characters). Since the map $\mathbb{X}^*(\mathrm{T}) \to \mathbb{X}^*(\mathrm{T}')$ is a surjection, there exists a subtorus $\mathrm{T}'' \subseteq \mathrm{T}$ such that $\mathrm{T} \cong \mathrm{T}' \times \mathrm{T}''$. If V'' is a representation of T'' whose weights are primitive in $\mathbb{X}^*(\mathrm{T}'')$ and form a **Z**-basis thereof, and V'' is viewed as a T-representation via the projection $\mathrm{T} \to \mathrm{T}''$, then $u_{\mathcal{P}} = e(\mathrm{V}'')$.

If V'' is in fact a representation of T whose weights are primitive in $\mathbb{X}^*(\mathrm{T})$ and form a **Z**-basis thereof (corresponding to the case when $\mathcal{P}$ is the family of proper connected closed subgroups of T), we will write $u := u_{\mathcal{P}}$ and $k^{t\mathrm{T}} := k^{t_{\mathcal{P}}\mathrm{T}} \cong k^{h\mathrm{T}}[u^{-1}]$. Proposition 2.2.4 implies that if X is a finite T-CW-complex such that the stabilizers of points in X are connected, then the map $\mathrm{C}_*^{\mathrm{T},\mathrm{BM}}(\mathrm{X}^\mathrm{T}; k) \to \mathrm{C}_*^{\mathrm{T},\mathrm{BM}}(\mathrm{X}; k)$ of $\mathbf{E}_\infty$-$k^{h\mathrm{T}}$-coalgebras is an isomorphism upon inverting u, i.e., upon base-changing along $k^{h\mathrm{T}} \to k^{t\mathrm{T}}$.

Lemma 2.2.6. *Fix notation as in Proposition 2.2.4. Suppose that* $u_{\mathcal{P}} \in \pi_*(k^{h\mathrm{T}})$ *is not a zero divisor. If* $\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathrm{X}; k)$ *is a finitely generated projective* $\pi_*(k^{h\mathrm{T}})$*-module,* *then the following map of cocommutative $\pi_*(k^{h\mathrm{T}})$-coalgebras is injective*

$$\mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X};k) \hookrightarrow \mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X};k)[u_{\mathcal{P}}^{-1}] \xleftarrow{\cong} \mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X}^{\mathrm{T}};k)[u_{\mathcal{P}}^{-1}].$$

Under the canonical isomorphism $\mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X}^{\mathrm{T}};k)[u_{\mathcal{P}}^{-1}] \cong \pi_* k^{t_{\mathcal{P}}\mathrm{T}}[\mathrm{X}^{\mathrm{T}}]$*, the* $\pi_* k^{t_{\mathcal{P}}\mathrm{T}}$*-linear coproduct on* $\mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X}^{\mathrm{T}};k)[u_{\mathcal{P}}^{-1}]$ *is given by the diagonal on* X^{T}*.*

Proof. The isomorphism $\mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X}^{\mathrm{T}};k)[u_{\mathcal{P}}^{-1}] \to \mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X};k)[u_{\mathcal{P}}^{-1}]$ follows from Proposition 2.2.4 since homotopy groups commute with filtered colimits. It only remains to see that $\mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X};k) \to \mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X};k)[u_{\mathcal{P}}^{-1}]$ is injective. But this follows from the conditions that $u_{\mathcal{P}} \in \pi_*(k^{h\mathrm{T}})$ is not a zero divisor and that $\mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X};k)$ is a finitely generated projective $\pi_*(k^{h\mathrm{T}})$-module. □

Notation 2.2.7. If $\chi : \mathrm{T} \to \mathbf{C}^\times$ is a character, then the standard action of $\mathbf{C}^\times$ on S^2 defines an action of T on S^2. To emphasize the dependence on the character, this sphere will be denoted S^χ. The Euler class of the 1-dimensional complex representation of T corresponding to χ defines a class $e(\chi) \in \pi_{-2} k^{h\mathrm{T}}$.

Definition 2.2.8. Let X be a locally compact Hausdorff topological space equipped with a T-action such that X is a finite T-CW-complex. Let X^{T} denote the fixed point set, and let $\mathrm{X}^{(1)}$ denote the union of those points in X whose stabilizer is a subgroup of T of codimension ≤ 1. Say that X is a *k-GKM* T*-space* if it satisfies the following conditions:

(1) X admits a T-invariant cell decomposition with only (real) even-dimensional cells, and there are only finitely many in each dimension. This implies that $\mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X};k)$ is a finitely generated projective (in fact, free) $\pi_*(k^{h\mathrm{T}})$-module.
(2) The subspace $\mathrm{X}^{(1)}$ consists of a finite number of spheres S^χ for characters $\chi \in \mathbb{X}^*(\mathrm{T})$ which meet only at points in X^{T}; let E denote the multiset of characters χ such that $\mathrm{S}^\chi \subseteq \mathrm{X}^{(1)}$. We require that if $\chi \in \mathrm{E}$, then $e(\chi) \in \pi_*(k^{h\mathrm{T}})$ is a nontrivial nonzerodivisor (i.e., is nonzero, and is a nonzerodivisor). Furthermore, for any tuple $\chi_1, \cdots, \chi_j \in \mathrm{E}$ of distinct elements, the elements $e(\chi_1), \cdots, e(\chi_j) \in \pi_*(k^{h\mathrm{T}})$ are coprime.[3]

The second condition above implies that X^{T} is finite. There are two maps $\mathrm{E} \rightrightarrows \mathrm{X}^{\mathrm{T}}$ sending χ to the images of $0, \infty \in \mathrm{S}^\chi$ in X^{T}; the diagram $\mathrm{E} \rightrightarrows \mathrm{X}^{\mathrm{T}}$ of finite sets will be denoted Γ_{X}. These images will sometimes be denoted $x_0(\chi), x_\infty(\chi)$. The two maps $\mathrm{E} \rightrightarrows \mathrm{X}^{\mathrm{T}}$ can be viewed as a single map $\mathrm{E} \sqcup \mathrm{E} \to \mathrm{X}^{\mathrm{T}}$. If $x \in \mathrm{X}^{\mathrm{T}}$, let $\mathrm{E}_x \subseteq \mathrm{E} \sqcup \mathrm{E}$ denote the preimage of x, so $\chi \in \mathrm{E}_x$ if and only if $x_\epsilon(\chi) = x$ with $\epsilon \in \{0, \infty\}$.

Let $\mathcal{P}$ be the set of proper closed subgroups of T which are stabilizers of points of X. Fix a representation $\mathrm{V}_{\mathcal{P}}$ of T such that for any $\mathrm{T}' \in \mathcal{P}$, the weights of $\mathrm{V}_{\mathcal{P}}$ are primitive in $\ker(\mathbb{X}^*(\mathrm{T}) \to \mathbb{X}^*(\mathrm{T}'))$ and form a **Z**-basis thereof. Let $u_{\mathcal{P}} = e(\mathrm{V}_{\mathcal{P}}) \in \pi_*(k^{h\mathrm{T}})$. (The choice of $\mathrm{V}_{\mathcal{P}}$ is immaterial.)

Theorem 2.2.9. *Let* X *be a k-GKM* T*-space. Under the injection* $\mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X};k) \hookrightarrow \mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X}^{\mathrm{T}};k)[u_{\mathcal{P}}^{-1}] \cong \pi_* k^{t_{\mathcal{P}}\mathrm{T}}[\mathrm{X}^{\mathrm{T}}]$ *of Lemma 2.2.6,* $\mathrm{H}_*^{\mathrm{T,BM}}(\mathrm{X};k)$ *is isomorphic as a cocommutative* $\pi_*(k^{h\mathrm{T}})$*-coalgebra to the subset of those* $\sum_{x\in\mathrm{X}^{\mathrm{T}}} f_x[x]$ *with* $f_x \in \pi_* k^{t_{\mathcal{P}}\mathrm{T}}$ *satisfying the following conditions:*

[3] If R is a commutative ring and $x_1, \cdots, x_n \in \mathrm{R}$ are elements, we say that they are coprime if the map $\mathrm{R}/x_1 \cdots x_n \to \mathrm{R}/x_1 \times \cdots \times \mathrm{R}/x_n$ is injective, i.e., $(x_1) \cap \cdots \cap (x_n) = (x_1 \cdots x_n)$.

(1) *For each* $x \in \mathrm{X}^{\mathrm{T}}$, $f_x \prod_{\chi \in \mathrm{E}_x} e(\chi) \in \pi_*(k^{h\mathrm{T}})$. *In other words, all poles of* f_x *are simple, and lie on the divisors cut out by* $e(\chi)$ *for* $\chi \in \mathrm{E}_x$.
(2) *If* $\chi \in \mathrm{E}$, *then the element* $e(\chi)(f_{x_0(\chi)} - f_{x_\infty(\chi)}) \in \pi_*(k^{h\mathrm{T}})[\prod_{\chi' \in \mathrm{E}_x - \{\chi\}} e(\chi')^{-1}]$ *is zero modulo* $e(\chi)$.

Implicit in Theorem 2.2.9 is the claim that the subset of $\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathrm{X}^{\mathrm{T}}; k)[u_{\mathcal{P}}^{-1}]$ defined above is in fact a cocommutative $\pi_*(k^{h\mathrm{T}})$-coalgebra.

Remark 2.2.10. Suppose $k \to k'$ is a map of $\mathbf{E}_\infty$-rings. If X is a k-GKM T-space, it may not be the case that X is a k'-GKM T-space (for example, the image of $e(\chi) \in \pi_*(k^{h\mathrm{T}})$ in $\pi_*((k')^{h\mathrm{T}})$ could vanish). However, the natural map $\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathrm{X}; k) \hat{\otimes}_{\pi_*(k^{h\mathrm{T}})} \pi_*((k')^{h\mathrm{T}}) \to \mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathrm{X}; k')$ will be an isomorphism. To apply Theorem 2.2.9 to k', it therefore suffices to find some k with a map $k \to k'$ for which X is a k-GKM T-space.

Definition 2.2.11. Let $\mathrm{E} \rightrightarrows \mathrm{F}$ be a diagram of finite sets, denoted Γ for simplicity. Denote by $x_0(\chi), x_\infty(\chi) \in \mathrm{F}$ the images of an element $\chi \in \mathrm{E}$. Let R be a commutative ring, and let $e : \mathrm{E} \to \mathrm{R}$ be a map of sets such that $e(\chi)$ is a nontrivial nonzerodivisor for every $\chi \in \mathrm{E}$, and such that for any tuple $\chi_1, \cdots, \chi_j \in \mathrm{E}$ of distinct elements, the elements $e(\chi_1), \cdots, e(\chi_j) \in \mathrm{R}$ are coprime. Set $\mathrm{M}_\Gamma \subseteq \mathrm{R}^{\mathrm{F}}$ to be the subset of maps $g : \mathrm{F} \to \mathrm{R}$ such that $g(x_0(\chi)) - g(x_\infty(\chi)) \equiv 0 \pmod{e(\chi)}$ for each $\chi \in \mathrm{E}$, so that M_Γ is a commutative R-subalgebra of R^{F}. It can be checked that M_Γ is a free R-module of rank equal to $|\mathrm{F}|$.

The following is the more standard formulation of GKM theory.

Theorem 2.2.12. *Let* X *be a* k*-GKM* T*-space. Let* $\mathrm{E} \rightrightarrows \mathrm{X}^{\mathrm{T}}$ *denote the diagram of finite sets associated to* X*, which we will denote by* Γ*. Then the map* $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathrm{X}; k) \to \mathrm{H}_{\mathrm{T},c}^{-*}(\mathrm{X}^{\mathrm{T}}; k) \cong \pi_*(k^{h\mathrm{T}})^{\mathrm{X}^{\mathrm{T}}}$ *is injective, and is isomorphic to the subset* $\mathrm{M}_\Gamma \subseteq \pi_*(k^{h\mathrm{T}})^{\mathrm{X}^{\mathrm{T}}}$ *of Definition 2.2.11.*

Again, one has an analogue of Remark 2.2.10.

Recall that $\mathrm{C}_*^{\mathrm{T},\mathrm{BM}}(\mathrm{X}; k)$ is the $k^{h\mathrm{T}}$-linear dual of $\mathrm{C}_{\mathrm{T},c}^*(\mathrm{X}; k)$. If X is a k-GKM T-space, then $\mathrm{C}_{\mathrm{T},c}^*(\mathrm{X}; k)$ is a free $k^{h\mathrm{T}}$-module, so $\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathrm{X}; k)$ is the $\pi_*(k^{h\mathrm{T}})$-linear dual of $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathrm{X}; k)$. Theorem 2.2.9 therefore follows from Theorem 2.2.12 along with the following elementary algebraic lemma:

Lemma 2.2.13. *In Definition 2.2.11,* M_Γ *is a free* R*-module of rank* $|\mathrm{F}|$*, and the* R*-linear dual* $\mathrm{M}_\Gamma^\vee$ *of* M_Γ *is isomorphic to the subset of* $\mathrm{R}[\prod_{\chi \in \mathrm{E}} e(\chi)^{-1}][\mathrm{F}]$ *of those elements* $\sum_{x \in \mathrm{F}} f_x[x]$ *such that* $t f_v \in \mathrm{R}$ *for* $v \in \mathrm{F}$ *and such that* $t(f_0 - f_\infty) \equiv 0 \pmod{t}$*. Furthermore, the inclusion* $\mathrm{M}_\Gamma^\vee \subseteq \mathrm{R}[\prod_{\chi \in \mathrm{E}} e(\chi)^{-1}][\mathrm{F}]$ *is a map of cocommutative* R*-coalgebras.*

Let us now prove Theorem 2.2.12.

Example 2.2.14. Let $\chi : \mathrm{T} \to \mathbf{C}^\times$ be a character. We will verify Theorem 2.2.9 in the case $\mathrm{X} = \mathrm{S}^\chi$. The conditions that X be a k-GKM T-space are that $e(\chi)$ is a nontrivial nonzerodivisor in $\pi_*(k^{h\mathrm{T}})$. In this case, $\mathrm{E} = \{\chi\}$, $\mathrm{X}^{\mathrm{T}} = \{0, \infty\}$, $x_0(\chi) = 0$, and $x_\infty(\chi) = \infty$. Since $e(\chi)$ is a nontrivial nonzerodivisor, the map $\pi_* k^{h\mathbf{C}^\times} \to \pi_*(k^{h\mathrm{T}})$ sending $t \mapsto e(\chi)$ is injective. We may therefore reduce to the case $\mathrm{T} = \mathbf{C}^\times$ itself, and χ is the standard 1-dimensional representation of T.

There is a cofiber sequence

$$\mathrm{T}_+ \to \mathrm{S}^0 = \mathrm{X}^{\mathrm{T}} \to \mathrm{S}^\chi = \mathrm{X},$$

which induces a fiber sequence

$$\mathrm{C}^*_{\mathrm{T},c}(\mathrm{X}; k) \to \mathrm{C}^*_{\mathrm{T},c}(\mathrm{X}^{\mathrm{T}}; k) \cong (k^{h\mathrm{T}})^{\mathrm{X}^{\mathrm{T}}} \to \mathrm{C}^*_{\mathrm{T},c}(\mathrm{T}; k) \cong k,$$

where the map $k^{h\mathrm{T}}[\mathrm{X}^{\mathrm{T}}] \cong k^{h\mathrm{T}} \times k^{h\mathrm{T}} \to k$ is the sum of the canonical map $k^{h\mathrm{T}} \to k$ with itself. The map $k^{h\mathrm{T}} \to k$ induces a π_*-surjection, where it is given by killing t. It follows that $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}; k) \cong \pi_*(k^{h\mathrm{T}})^{\mathrm{X}^{\mathrm{T}}} \times_{\pi_*(k)} 0$, i.e., it is the subalgebra of $\pi_*(k^{h\mathrm{T}})^{\mathrm{X}^{\mathrm{T}}}$ of those maps $f : \mathrm{X}^{\mathrm{T}} \to \pi_*(k^{h\mathrm{T}})$ such that $f(0) - f(\infty) \equiv 0 \pmod t$. Explicitly, if $\pi_*(k^{h\mathrm{T}})^{\mathrm{X}^{\mathrm{T}}} = \pi_*(k^{h\mathrm{T}})[a]/a(a-1)$, where a is the δ-function at 0 (so $1-a$ is the δ-function at ∞), then $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}; k) \subseteq \pi_*(k^{h\mathrm{T}})^{\mathrm{X}^{\mathrm{T}}}$ is isomorphic to the subset $\pi_*(k^{h\mathrm{T}})[b]/(b^2 - bt)$ where $b = at$.

Example 2.2.15. Let V be a complex T-representation. If E is the multiset of weights of V, the sphere $\mathrm{X} = \mathrm{S}^{\mathrm{V}}$ is a k-GKM T-space exactly when $e(\chi) \in \pi_*(k^{h\mathrm{T}})$ is a nontrivial nonzerodivisor for every $\chi \in \mathrm{E}$, and that the elements $e(\chi), e(\lambda) \in \pi_*(k^{h\mathrm{T}})$ are pairwise coprime for any distinct $\chi, \lambda \in \mathrm{E}$. The latter implies in particular that all the weights of V are distinct, so E is in fact a set (and not just a multiset). In this case, $\mathrm{X}^{\mathrm{T}} = \{0, \infty\}$, and $x_\epsilon(\chi) = \epsilon$ for each $\chi \in \mathrm{E}$ and $\epsilon \in \{0, \infty\}$. The assumptions on $e(\chi)$ guarantee that the map $\pi_* k^{h(\mathbf{C}^\times)^{\mathrm{E}}} \to \pi_*(k^{h\mathrm{T}})$ induced by the action map $\mathrm{T} \to (\mathbf{C}^\times)^{\mathrm{E}}$ is injective. We may therefore reduce to the case $\mathrm{T} = (\mathbf{C}^\times)^{\mathrm{E}}$ itself, and V is the standard representation of T.

In this case, one can argue exactly as in Example 2.2.14. Namely, there is a cube $2^{\mathrm{E}} \to \mathrm{Sp}$ sending a subset $\mathrm{I} \subseteq \mathrm{E}$ to $\Sigma^\infty_+ (\mathbf{C}^\times)^{\mathrm{I}}$. The total cofiber of this cube is precisely S^{V}. It follows that $\mathrm{C}^*_{\mathrm{T},c}(\mathrm{X}; k)$ is the total fiber of the cube $(2^{\mathrm{E}})^{\mathrm{op}} \to \mathrm{Mod}_{k^{h\mathrm{T}}}$ sending $\mathrm{I} \subseteq \mathrm{E} \mapsto k^{h(\mathbf{C}^\times)^{\mathrm{I}}}$. Writing the total fiber of the cube as an iterated fiber, it follows that $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}; k)$ is the subalgebra of $\pi_*(k^{h\mathrm{T}})^{\mathrm{X}^{\mathrm{T}}}$ of those maps $f : \mathrm{X}^{\mathrm{T}} \to \pi_*(k^{h\mathrm{T}})$ such that $f(0) - f(\infty) \equiv 0 \pmod{e(\chi)}$ for each $\chi \in \mathrm{E}$. Explicitly, if $\pi_*(k^{h\mathrm{T}})^{\mathrm{X}^{\mathrm{T}}} = \pi_*(k^{h\mathrm{T}})[a]/a(a-1)$, where a is the δ-function at 0 (so $1-a$ is the δ-function at ∞), then $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}; k) \subseteq \pi_*(k^{h\mathrm{T}})^{\mathrm{X}^{\mathrm{T}}}$ is isomorphic to the subset $\pi_*(k^{h\mathrm{T}})[b]/(b^2 - b\prod_{\chi \in \mathrm{E}} e(\chi))$ where $b = a\prod_{\chi \in \mathrm{E}} e(\chi)$.

Proof of Theorem 2.2.12. Let us first show that the map $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}; k) \to \mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^{\mathrm{T}}; k)$ is injective. Let $\mathcal{P}_{\mathrm{X}}$ be the set of proper closed subgroups of T which are stabilizers of points of X. Fix a representation $\mathrm{V}_{\mathcal{P}_{\mathrm{X}}}$ of T such that for any $\mathrm{T}' \in \mathcal{P}_{\mathrm{X}}$, the weights of $\mathrm{V}_{\mathcal{P}_{\mathrm{X}}}$ are primitive in $\ker(\mathbb{X}^*(\mathrm{T}) \to \mathbb{X}^*(\mathrm{T}'))$ and form a $\mathbf{Z}$-basis thereof. Let $u_{\mathrm{X}} = e(\mathrm{V}_{\mathcal{P}_{\mathrm{X}}}) \in \pi_*(k^{h\mathrm{T}})$. (Of course, this depends on the choice of $\mathrm{V}_{\mathcal{P}_{\mathrm{X}}}$; but $\mathrm{V}_{\mathcal{P}_{\mathrm{X}}}$ will soon be immaterial.) Recall from Proposition 2.2.4 that the map $\mathrm{C}^*_{\mathrm{T},c}(\mathrm{X}; k) \to \mathrm{C}^*_{\mathrm{T},c}(\mathrm{X}^{\mathrm{T}}; k)$ induces an isomorphism upon inverting u_{X}. The composite map

$$\mathrm{C}^*_{\mathrm{T},c}(\mathrm{X}; k) \to \mathrm{C}^*_{\mathrm{T},c}(\mathrm{X}^{\mathrm{T}}; k) \to \mathrm{C}^*_{\mathrm{T},c}(\mathrm{X}^{\mathrm{T}}; k)[u_{\mathrm{X}}^{-1}] \xleftarrow{\cong} \mathrm{C}^*_{\mathrm{T},c}(\mathrm{X}; k)[u_{\mathrm{X}}^{-1}]$$

is injective on π_* since $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}; k)$ is a finitely generated projective $\pi_*(k^{h\mathrm{T}})$-module. The first map is therefore also an injection.

Observe that in fact, $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}; k) \subseteq \mathrm{M}_\Gamma$. Indeed, the map $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}; k) \to \mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^{\mathrm{T}}; k)$ factors through the map $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^{(1)}; k) \to \mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}; k)$. The calculation of Example 2.2.14 implies that the image of the latter is already contained in M_Γ.

To prove that the inclusion $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X};k) \subseteq \mathrm{M}_\Gamma$ is an isomorphism, we will induct on the cell structure on X. Let $\mathrm{X}^n \subseteq \mathrm{X}$ denote the union of cells of real dimension $\leq 2n$. Then X^n is still a k-GKM T-space, so let Γ_n denote the subdiagram of Γ given by $\mathrm{E}_n \rightrightarrows (\mathrm{X}^n)^\mathrm{T}$, where $\mathrm{E}_n \subseteq \mathrm{E}$ consists of those $\chi \in \mathrm{E}$ such that $x_0(\chi), x_\infty(\chi) \in (\mathrm{X}^n)^\mathrm{T} \subseteq \mathrm{X}^\mathrm{T}$. Note that the inclusion $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^n;k) \subseteq \mathrm{M}_{\Gamma_n}$ is clearly an isomorphism when $n = 0$, since $\mathrm{X}^0 \subseteq \mathrm{X}^\mathrm{T}$.

Assume for the sake of induction that the inclusion $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^n;k) \subseteq \mathrm{M}_{\Gamma_n}$ is an isomorphism. The map $\Gamma_n \to \Gamma_{n+1}$ of diagrams induces a map $\mathrm{M}_{\Gamma_{n+1}} \to \mathrm{M}_{\Gamma_n}$. This map is surjective. Indeed, the composite

$$\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^{n+1};k) \subseteq \mathrm{M}_{\Gamma_{n+1}} \to \mathrm{M}_{\Gamma_n}$$

identifies with the map $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^{n+1};k) \to \mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^n;k)$ composed with the isomorphism given by the inductive hypothesis. It therefore suffices to observe that the map $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^{n+1};k) \to \mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^n;k)$ is surjective, but this is clear since the boundary map $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^n;k) \to \mathrm{H}^{-*+1}_{\mathrm{T},c}(\mathrm{X}^{n+1},\mathrm{X}^n;k)$ in the long exact sequence in cohomology vanishes for parity reasons.

For the same parity reasons, the kernel of the map $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^{n+1};k) \to \mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^n;k)$ is isomorphic to $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^{n+1},\mathrm{X}^n;k)$. Since the map $\mathrm{M}_{\Gamma_{n+1}} \to \mathrm{M}_{\Gamma_n}$ is surjective as shown above, it suffices to show that $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}^{n+1},\mathrm{X}^n;k)$ is isomorphic to the kernel of $\mathrm{M}_{\Gamma_{n+1}} \to \mathrm{M}_{\Gamma_n}$. This kernel consists of those functions $g : \Gamma_{n+1} \to \pi_*(k^{h\mathrm{T}})$ satisfying the relevant condition to lie in $\mathrm{M}_{\Gamma_{n+1}}$, with the additional condition that $g(x) = 0$ for $x \in \Gamma_n \subseteq \Gamma_{n+1}$. In other words, g may be viewed as a function supported on $\Gamma_n \subseteq \Gamma_{n+1}$. The existence of the cell structure of X^{n+1} reduces us to checking that the map $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X};k) \subseteq \mathrm{M}_\Gamma$ is an isomorphism when $\mathrm{X} = \mathrm{S}^\mathrm{V}$ for a general complex T-representation V which is a k-GKM T-space. This is exactly Example 2.2.15. □

Remark 2.2.16. Theorem 2.2.9 admits an extension to ind-T-spaces. Namely, suppose that $\mathrm{X} = \mathrm{colim}_n \mathrm{X}^n$ is an ind-T-space, where each X^n is a k-GKM T-space. Say that X is a k-GKM ind T-space if X has isolated T-fixed points, and for any $x_0, x_\infty \in \mathrm{X}^\mathrm{T}$, there are finitely many 1-dimensional orbits whose boundary is $\{x_0, x_\infty\}$. In this case, the statement of Theorem 2.2.9 continues to hold verbatim (because the functors of homotopy groups and Borel-Moore homology preserve filtered colimits).

Suppose that G is a complex reductive group. Let $\mathrm{B} \subseteq \mathrm{G}$ be a Borel subgroup with unipotent radical U, and let T be a maximal torus contained in $\mathrm{B} \subseteq \mathrm{G}$. Let W be the Weyl group of G, let $\Phi \subseteq \mathbb{X}^*$ (resp. $\check{\Phi} \subseteq \mathbb{X}_*$) be the set of roots (resp. coroots), and let $\Phi^+ \subseteq \Phi$ be the subset of positive roots. Assume that the T-action on k extends to a G-action.

Definition 2.2.17. Say that k is *good for* G if the following conditions are satisfied:

1. if $e(\alpha) \in \pi_*(k^{h\mathrm{T}})$ is a nontrivial nonzerodivisor for each $\alpha \in \Phi^+$;
2. for any tuple $\alpha_1, \cdots, \alpha_j \in \Phi^+$ of distinct positive roots, the elements $e(\alpha_1), \cdots, e(\alpha_j) \in \pi_*(k^{h\mathrm{T}})$ are coprime.

More generally, if R is a graded commutative ring with a graded 1-dimensional formal group law F, and $\mathrm{T}_\mathrm{F} = \mathrm{Hom}(\mathbb{X}^*(\mathrm{T}), \mathrm{F})$, say that R is *good for* G if the above conditions are satisfied for $e(\alpha) \in \mathcal{O}_{\mathrm{T}_\mathrm{F}}$ given by the image of $e(\alpha) \in \pi_*(\mathrm{MU}^{h\mathrm{T}})$ under the map $\pi_*(\mathrm{MU}^{h\mathrm{T}}) \to \mathcal{O}_{\mathrm{T}_\mathrm{F}}$ induced by the map $\pi_*(\mathrm{MU}) \to \mathrm{R}$ classifying F.

Example 2.2.18. The field $k = \mathbf{F}_p$ is good for G if and only if each $\alpha \in \mathrm{Sym}(\mathfrak{t}^*)$ is a nontrivial nonzerodivisor for each $\alpha \in \Phi^+$ and each pair α, β of distinct positive roots, α and β are not scalar multiples of each other. If for simplicity we assume that G is semisimple, then these conditions hold exactly when p is not a bad prime and p does not divide the order of the center of G, i.e., p is a "very good" prime for G. Indeed, note that when G is of adjoint type, $\mathbb{X}^*(\mathrm{T}) = \mathbf{Z}\Phi$, and if $\alpha, \beta \in \Phi^+$, then the saturation of their span in $\mathbf{Z}\Phi$ is the root lattice of the rank-2 closed subsystem Σ they generate; so if α and β are proportional in $\mathbb{X}^*(\mathrm{T})$, then $\mathbf{Z}\Phi/\mathbf{Z}\Sigma$ has p-torsion, i.e., p is a bad prime of G.

Lemma 2.2.19. *Let* A *be a compact abelian group, and let* $\beta \in \mathbb{X}^*(\mathrm{A})$ *be a non-torsion character of* A*. Then* $e(\beta) \in \pi_*\mathrm{MU}^{h\mathrm{A}}$ *is a nonzerodivisor.*

Proof. Observe first that $\mathrm{MU}^{h\mathrm{A}}$ is concentrated in even degrees (in fact, $\pi_*(\mathrm{MU}^{h\mathrm{A}})$ is also flat over $\pi_*(\mathrm{MU})$). This was shown in [Lan70]. Let A_β denote the kernel of the homomorphism $\beta : \mathrm{A} \to \mathbf{C}^\times$. Since β is non-torsion, the map $\beta : \mathrm{A} \to \mathbf{C}^\times$ is surjective, so that the map $\mathrm{BA}_\beta \to \mathrm{BA}$ is a principal S^1-bundle. This implies that $\mathrm{MU}^{h\mathrm{A}}/e(\beta) \cong \mathrm{MU}^{h\mathrm{A}_\beta}$. Since both $\mathrm{MU}^{h\mathrm{A}}$ and $\mathrm{MU}^{h\mathrm{A}_\beta}$ are concentrated in even degrees, the long exact sequence in homotopy (i.e., the Gysin sequence for the S^1-bundle $\mathrm{BA}_\beta \to \mathrm{BA}$) implies that multiplication by $e(\beta)$ is injective on $\pi_*(\mathrm{MU}^{h\mathrm{A}})$, as desired. $\square$

Lemma 2.2.20. *The* $\mathbf{E}_\infty$*-ring* MU *is good for* G*.*

Proof. Since $\pi_*(\mathrm{MU}^{h\mathrm{T}})$ is a domain, $e(\alpha) \in \pi_*(\mathrm{MU}^{h\mathrm{T}})$ is a nonzerodivisor for each $\alpha \in \Phi^+$. It is also nonzero, so it is a nontrivial nonzerodivisor. We need to check that for any tuple $\alpha_1, \cdots, \alpha_j \in \Phi^+$ of distinct positive roots, the elements $e(\alpha_1), \cdots, e(\alpha_j) \in \pi_*(k^{h\mathrm{T}})$ are coprime. Assume by induction that we have proved the claim for any tuple of $(j-1)$ distinct positive roots. Suppose $x \in \pi_*(\mathrm{MU}^{h\mathrm{T}})$ vanishes modulo $e(\alpha_i)$ for each $1 \leq i \leq j$. By the inductive hypothesis, $x = y\prod_{i=1}^{j-1} e(\alpha_i)$ for some $y \in \pi_*(\mathrm{MU}^{h\mathrm{T}})$. To show that x is divisible by $\prod_{i=1}^{j} e(\alpha_i)$, it suffices to show that y is zero modulo $e(\alpha_j)$. For this, it in turn suffices to show that $\prod_{i=1}^{j-1} e(\alpha_i)$ (and stronger, that each $e(\alpha_i)$ for $1 \leq i \leq j-1$) is a nontrivial nonzerodivisor in $\pi_*(\mathrm{MU}^{h\mathrm{T}})/e(\alpha_j)$. Let T_{α_j} denote the kernel of $\alpha_j : \mathrm{T} \to \mathbf{C}^\times$, so that $\mathrm{MU}^{h\mathrm{T}}/e(\alpha_j) \cong \mathrm{MU}^{h\mathrm{T}_{\alpha_j}}$. In particular, $\mathrm{MU}^{h\mathrm{T}}/e(\alpha_j)$ is concentrated in even degrees, so that $\pi_*(\mathrm{MU}^{h\mathrm{T}}/e(\alpha_j)) \cong \pi_*(\mathrm{MU}^{h\mathrm{T}})/e(\alpha_j)$. Since the root system of G is reduced, no α_i is a scalar multiple of α_j, so $\alpha_i \in \mathbb{X}^*(\mathrm{T}_{\alpha_j}) = \mathbb{X}^*(\mathrm{T})/\mathbf{Z}\alpha_j$ is non-torsion for $1 \leq i \leq j-1$. Lemma 2.2.19 implies that $e(\alpha_i) \in \pi_*(\mathrm{MU}^{h\mathrm{T}_{\alpha_j}})$ is a nontrivial nonzerodivisor for each $1 \leq i \leq j-1$, as desired. $\square$

Example 2.2.21. Let $n \geq 0$, and let $\mathrm{X} = \mathcal{B} = \mathrm{G}/\mathrm{B}$ be the flag variety. Then $\mathrm{X}^\mathrm{T} = \mathrm{W}$, and $\mathrm{E} = \mathrm{W} \times \Phi^+$; the two maps $\mathrm{E} \rightrightarrows \mathrm{W}$ send $(w, \alpha) \mapsto w, s_\alpha w$. Moreover, the map $e : \mathrm{E} \to \pi_*(k^{h\mathrm{T}})$ sends $(w, \alpha) \mapsto e(\alpha)$. The flag variety is a k-GKM T-space if and only if k is good for G. In particular, by Lemma 2.2.20, the flag variety is always an MU-GKM T-space. Remark 2.2.10 implies that $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; k) \cong \mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; \mathrm{MU})\hat{\otimes}_{\pi_*(\mathrm{MU}^{h\mathrm{T}})}\pi_*(k^{h\mathrm{T}})$ for any T-equivariant MU-algebra k. We will use this observation several times below.

Let $\Gamma_\mathcal{B}$ denote the diagram $\mathrm{E} \rightrightarrows \mathrm{W}$ of finite sets, and let $\mathrm{M}_{\Gamma_\mathcal{B}}$ denote the $\pi_*(k^{h\mathrm{T}})$-algebra defined via Definition 2.2.11. Note that this is a combinatorially-defined $\pi_*(k^{h\mathrm{T}})$-algebra associated to G which can be defined with no reference to the flag variety. In this case, Theorem 2.2.12 and the preceding paragraph yield an isomorphism $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; k) \cong \mathrm{M}_{\Gamma_{\mathcal{B}}}$ of $\pi_*(k^{h\mathrm{T}})$-algebras; note that this isomorphism holds for any T-equivariant MU-algebra k.

Proposition 2.2.22. *If k is any even $\mathbf{E}_\infty$-ring, then $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{B}; k)$ is a complete intersection over $\pi_*(k^{h\mathrm{T}})$.*

Proof. Unfortunately, I do not have a direct combinatorial argument for this; instead, we will rely on [Tod75, TW74]. Since $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; k) \cong \mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; \mathrm{MU})\hat{\otimes}_{\pi_*(\mathrm{MU}^{h\mathrm{T}})}\pi_*(k^{h\mathrm{T}})$, and $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; \mathrm{MU})$ is flat (in fact, free) over $\pi_*(\mathrm{MU}^{h\mathrm{T}})$, it suffices to prove that $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{B}; \mathrm{MU})$ is a complete intersection over $\pi_*(\mathrm{MU}^{h\mathrm{T}})$. There is a homotopy fixed points spectral sequence

$$\mathrm{E}_2 \cong \mathrm{H}^{-*}(\mathrm{BT}; \mathbf{Z}) \otimes_{\mathbf{Z}} \mathrm{H}^{-*}(\mathcal{B}; \mathrm{MU}) \Rightarrow \mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{B}; \mathrm{MU}),$$

which collapses because the E_2-term is concentrated in even degrees. Since $\mathrm{H}^{-*}(\mathrm{BT}; \mathbf{Z})$ is a polynomial $\mathbf{Z}$-algebra, it suffices to show that $\mathrm{H}^{-*}(\mathcal{B}; \mathrm{MU})$ is a complete intersection over $\pi_*(\mathrm{MU})$. The Atiyah-Hirzebruch spectral sequence runs

$$\mathrm{E}_2 \cong \mathrm{H}^{-*}(\mathcal{B}; \mathbf{Z}) \otimes_{\mathbf{Z}} \pi_*(\mathrm{MU}) \Rightarrow \mathrm{H}^{-*}(\mathcal{B}; \mathrm{MU}),$$

and it too collapses because the E_2-term is concentrated in even degrees. Since $\pi_*(\mathrm{MU})$ is a polynomial $\mathbf{Z}$-algebra, it suffices to show that $\mathrm{H}^{-*}(\mathcal{B}; \mathbf{Z})$ is a complete intersection over $\mathbf{Z}$. This follows from the explicit calculations of [Tod75, Proposition 3.1 and Proposition 3.2] and [TW74, Theorem 2.1 and Corollary 2.2] for each simple G. □

Definition 2.2.23. The map $\mathcal{B} \to \mathrm{BB}$ defines a map $\eta_{\mathrm{R}} : \pi_*(k^{h\mathrm{B}}) \cong \pi_*(k^{h\mathrm{T}}) \to \mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; k) \cong \mathrm{M}_{\Gamma_{\mathcal{B}}}$ called the *right unit*; note that this map is $\pi_*(k)$-linear, but *not* $\pi_*(k^{h\mathrm{T}})$-linear! In terms of $\mathrm{M}_{\Gamma_{\mathcal{B}}}$, the composite

$$\pi_*(k^{h\mathrm{T}}) \xrightarrow{\eta_{\mathrm{R}}} \mathrm{M}_{\Gamma_{\mathcal{B}}} \subseteq \mathrm{Map}(\mathrm{W}, \pi_*(k^{h\mathrm{T}}))$$

is adjoint to the natural W-action on $\pi_*(k^{h\mathrm{T}})$. To check that this is well-defined, we only need to verify that the image of the map $\pi_*(k^{h\mathrm{T}}) \to \mathrm{Map}(\mathrm{W}, \pi_*(k^{h\mathrm{T}}))$ in fact lies in $\mathrm{M}_{\Gamma_{\mathcal{B}}}$. That this is true reduces to the observation that if $\chi \in \mathbb{X}^*(\mathrm{T})$ and $\alpha \in \Phi^+$, then $s_\alpha\chi - \chi$ is divisible by α.

The double coset formula $\mathrm{B}\backslash\mathcal{B} \cong \mathrm{G}\backslash(\mathcal{B}\times\mathcal{B})$ defines an involution c on $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; k) \cong \mathrm{M}_{\Gamma_{\mathcal{B}}}$ by swapping the two factors of $\mathcal{B}$. In terms of $\mathrm{M}_{\Gamma_{\mathcal{B}}}$, this is induced by the involution of $\Gamma_{\mathcal{B}}$ induced by the involution on $\mathrm{E} = \mathrm{W}\times\Phi^+$ sending $(w, \alpha) \mapsto (w^{-1}, w\alpha)$. That this is well-defined (i.e., preserves $\mathrm{M}_{\Gamma_{\mathcal{B}}} \subseteq \mathrm{Map}(\mathrm{W}, \pi_*(k^{h\mathrm{T}}))$) reduces to the identity $s_\alpha w^{-1} = w^{-1}s_{w\alpha}$.

There is a map $\mathrm{B}\backslash(\mathcal{B} \times \mathcal{B}) \to \mathrm{B}\backslash\mathcal{B}$ given by $(g_1\mathrm{B}, g_2\mathrm{B}) \mapsto g_1g_2\mathrm{B}$ defines a $\pi_*(k^{h\mathrm{T}})$-linear coproduct Δ on $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; k) \cong \mathrm{M}_{\Gamma_{\mathcal{B}}}$. In terms of $\mathrm{M}_{\Gamma_{\mathcal{B}}}$, this is induced by restriction $m^* : \mathrm{Map}(\mathrm{W}, \pi_*(k^{h\mathrm{T}})) \to \mathrm{Map}(\mathrm{W} \times \mathrm{W}, \pi_*(k^{h\mathrm{T}}))$ along the multiplication map $m : \mathrm{W} \times \mathrm{W} \to \mathrm{W}$. That this sends $\mathrm{M}_{\Gamma_{\mathcal{B}}}$ to $\mathrm{M}_{\Gamma_{\mathcal{B}}} \otimes_{\pi_*(k^{h\mathrm{T}})} \mathrm{M}_{\Gamma_{\mathcal{B}}}$ reduces to the identity $w's_\alpha w = w'ws_{w^{-1}\alpha}$.

Together, these data define the structure of a graded commutative Hopf algebroid $(\pi_*(k^{h\mathrm{T}}), \mathrm{M}_{\Gamma_{\mathcal{B}}})$, which we will call the *Weyl algebroid*. Following Construction A.1.4, this is the Hopf algebroid associated to the cosimplicial diagram $\mathrm{C}^*_{\mathrm{T}}(\mathcal{B}^{\times\bullet}; k)$ of $\mathbf{E}_\infty$-rings. We will sometimes write $\mathcal{W}$ to denote $\mathrm{Spf}(\mathrm{M}_{\Gamma_{\mathcal{B}}})$, so that $\mathcal{O}_{\mathcal{W}} = \mathrm{M}_{\Gamma_{\mathcal{B}}} = \mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; k)$, and $\mathcal{W}$ is a graded groupoid scheme over $\mathrm{Spf}(\pi_*(k^{h\mathrm{T}}))$.

By construction, there is a dominant map $\underline{\mathrm{W}} \to \mathcal{W}$ of groupoid schemes over $\mathrm{Spf}(\pi_*(k^{h\mathrm{T}}))$, where $\underline{\mathrm{W}}$ is the constant group scheme on the Weyl group.

Note that $\mathcal{W}$ depends on the formal group over $\pi_*(k)$ coming from $\mathrm{Spf}(\pi_*(k^{h\mathrm{S}^1}))$, so it might be better denoted $\mathcal{W}_{\mathrm{F}}$; but we will omit the subscript F from the notation.

Example 2.2.24. For simplicity, let k be an ordinary commutative ring. Let us describe $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$ when G is SL_2 or PGL_2, so that $\pi_*(k^{h\mathrm{T}}) \cong k[t]$ with $|t| = -2$. If $\mathrm{G} = \mathrm{SL}_2$, then $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; k) \cong k[t,b]/b(b-2t)$ with $|b| = -2$. The right unit is the map $k[t] \to k[t,b]/b(b-2t)$ sending $t \mapsto t - b$. The coproduct on b is given by $\Delta(b) = b \otimes 1 + 1 \otimes b$. The cohomology of this Hopf algebroid is isomorphic to $k[t^2]$ with $|t^2| = -4$. Since $k^{h\mathrm{SL}_2}$ is concentrated in even degrees, this of course is precisely $\pi_*\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{SL}_2})$, where t^2 represents the second Chern class.

If $\mathrm{G} = \mathrm{PGL}_2$, then $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; k) \cong k[t,b]/b(b-t)$ with $|b| = -2$. The right unit η_{R} is the map $k[t] \to k[t,b]/b(b-t)$ sending $t \mapsto t - 2b$. The coproduct on b is given by $\Delta(b) = b \otimes 1 + 1 \otimes b$. There is a 2-torsion 1-cocycle τ_0 in the cohomology of this Hopf algebroid sending $b \mapsto b$. Using this, a simple calculation shows that if 2 is not a zero divisor in k, then the cohomology of this Hopf algebroid is given by $\pi_*\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{PGL}_2}) \cong k[t^2, \tau]/2\tau$ where $|t^2| = -4$, and τ lives in weight -2 and degree -1. On the other hand, if $2 = 0 \in k$, then $\pi_*\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{PGL}_2}) \cong k[t,\tau]$. The class t represents the Stiefel-Whitney class w_2, and τ represents the class w_3.[4] See also Example C.1.

Remark 2.2.25. The map $\pi_*(k^{h\mathrm{T}}) \hat{\otimes}_{\pi_*(k)} \pi_*(k^{h\mathrm{T}}) \to \mathrm{Map}(\mathrm{W}, \pi_*(k^{h\mathrm{T}}))$ sending $x \otimes y$ to the function $f(w) = x + wy$ induces a map $\pi_*(k^{h\mathrm{T}}) \hat{\otimes}_{\pi_*(k^{h\mathrm{T}})^{\mathrm{W}}} \pi_*(k^{h\mathrm{T}}) \to \mathrm{M}_{\Gamma_{\mathcal{B}}}$. This map is an isomorphism if $|\mathrm{W}|$ is invertible in $\pi_0(k)$ (and this recovers Borel's computation [Bor53] of $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{B}; \mathbf{Q})$); however, to get isomorphy, it is not sufficient to invert the torsion primes of G (see Table 1). We will often consider the quotient stack $\mathrm{Spf}(\pi_*(k^{h\mathrm{T}}))/\mathcal{W}$ below. It follows, for instance, that if $|\mathrm{W}|$ is invertible in $\pi_0(k)$, then the stack $\mathrm{Spf}(\pi_*(k^{h\mathrm{T}}))/\mathcal{W}$ is isomorphic to $\mathrm{Spf}(\pi_*(k^{h\mathrm{T}}))/\!\!/\mathrm{W}$, and hence is in fact a scheme. However, even if $\mathrm{Spf}(\pi_*(k^{h\mathrm{T}}))/\mathcal{W}$ turns out to be a scheme, we will always treat it as a stack below.

Note that this fails badly for arbitrary k; for instance, if $k = \mathbf{F}_2$ and G has semisimple rank 1 (so $\mathrm{W} = \mathbf{Z}/2$), then $\pi_*(k^{h\mathrm{T}})^{\mathrm{W}} \cong \pi_*(k^{h\mathrm{T}})$, so $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{B}; k)$ is not isomorphic to $\pi_*(k^{h\mathrm{T}}) \hat{\otimes}_{\pi_*(k^{h\mathrm{T}})^{\mathrm{W}}} \pi_*(k^{h\mathrm{T}})$. The issue in this case is that the W-action on $\pi_*(k^{h\mathrm{T}})$ is not faithful. (Note that $p = 2$ is not a torsion prime for type A_1.) Even if one replaces the invariants $\pi_*(k^{h\mathrm{T}})^{\mathrm{W}}$ by the *derived* invariants $\pi_*(k^{h\mathrm{T}})^{h\mathrm{W}}$ (meaning the derived W-invariants of $\pi_*(k^{h\mathrm{T}})$, not π_* of the derived W-invariants of $k^{h\mathrm{T}}$; the latter would instead compute $\pi_*(k^{h\mathrm{N}_{\mathrm{G}}(\mathrm{T})})$), it is still not true that $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{B}; k) \cong \pi_*(k^{h\mathrm{T}}) \otimes_{\pi_*(k^{h\mathrm{T}})^{h\mathrm{W}}} \pi_*(k^{h\mathrm{T}})$. The same counterexample works: if

[4] Let us illustrate the calculation in the case $k = \mathbf{F}_2$. Then the left and right units agree, so $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; \mathbf{F}_2)$ is a Hopf algebra over $\mathbf{F}_2[t]$. The group scheme $\mathrm{Spec}(\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; \mathbf{F}_2))$ is the kernel of the homomorphism $(\mathbf{G}_a)_{\mathbf{F}_2[t]} \to (\mathbf{G}_a)_{\mathbf{F}_2[t]}$ sending $b \mapsto b^2 - bt$. The cohomology $\mathrm{H}^*(B\mathbf{G}_a; \mathcal{O})$ is isomorphic to $\mathbf{F}_2[\tau_0, \tau_1, \cdots]$ where each τ_i lives in $\mathrm{H}^1(B\mathbf{G}_a; \mathcal{O})$ (and in weight -2^{i+1}) and represents the homomorphism $b \mapsto b^{2^i}$. The homomorphism $(\mathbf{G}_a)_{\mathbf{F}_2[t]} \to (\mathbf{G}_a)_{\mathbf{F}_2[t]}$ induces the map $\mathbf{F}_2[t, \tau_0, \tau_1, \cdots] \to \mathbf{F}_2[t, \tau_0, \tau_1, \cdots]$ sending $\tau_i \mapsto \tau_{i+1} + t^{2^i}\tau_i$, and this implies that the cohomology of the group scheme $\mathrm{Spec}(\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; \mathbf{F}_2))$ is given by $\mathbf{F}_2[t, \tau_0, \tau_1, \cdots]/(\tau_{i+1} + t^{2^i}\tau_i) \cong \mathbf{F}_2[t, \tau_0]$, as desired.

TABLE 1. Various numbers associated to simple groups. Bad primes p are those such that p divides a coefficient of the highest root of $\mathfrak{g}$, or equivalently if there is a closed subsystem Σ such that $\mathbf{Z}\Phi/\mathbf{Z}\Sigma$ has p-torsion. Similarly, torsion primes p are those such that p divides a coefficient of the coroot dual to the highest root of $\mathfrak{g}$, equivalently if p divides a coefficient of the highest short root of the Langlands dual $\check{\mathfrak{g}}$, or equivalently if there is a closed subsystem Σ such that $\mathbf{Z}\check{\Phi}/\mathbf{Z}\check{\Sigma}$ has p-torsion (see [Ste75]).

Type	A_n	B_n	C_n	D_n	E_6	E_7	E_8	F_4	G_2
$\lvert\mathrm{Z}(\mathrm{G})\rvert$	$n+1$	2	2	2	3	2	1	1	1
Bad primes	none	2	2	2	$2,3$	$2,3$	$2,3,5$	$2,3$	$2,3$
Torsion primes	$p\mid(n+1)$	2 $(n\geq 3)$	none	2	$2,3$	$2,3$	$2,3,5$	$2,3$	2
(Dual) Coxeter	$n+1$	$2n$ $(2n-1)$	$2n$ $(n+1)$	$2n-2$	12	18	30	12 (9)	6 (4)

$k=\mathbf{F}_2$ and G has semisimple rank 1 (so $\mathrm{W}=\mathbf{Z}/2$), then the above tensor product is $k[x,a]/(a^2-a)$ with $|x|=-2$ and $|a|=0$.

Remark 2.2.26. There is a map $\underline{\mathrm{W}}\to\mathcal{W}$ of groupoid schemes over T_{F}. If M is a $\mathcal{W}$-equivariant $\pi_*(k^{h\mathrm{T}})$-module (which we assume is concentrated in degree zero, for simplicity), then there is a map $\mathrm{M}^{h\mathcal{W}}\to\mathrm{R}\Gamma(\mathrm{W};\mathrm{M})$. If the map $\mathcal{O}_{\mathcal{W}}\to\mathrm{Map}(\mathrm{W},\pi_*(k^{h\mathrm{T}}))$ is injective (e.g., if k is good for G) and M is flat over $\pi_*(k^{h\mathrm{T}})$, then the map $\pi_0(\mathrm{M}^{h\mathcal{W}})\to\mathrm{H}^0(\mathrm{W};\mathrm{M})=\mathrm{M}^{\mathrm{W}}$ is an isomorphism.

Let $\mathrm{M}\in\mathrm{Mod}_k^{h\mathrm{B}}$; since U is contractible, the map $\mathrm{M}^{h\mathrm{U}}\to\mathrm{M}$ is a T-equivariant equivalence, so in particular, the map $\mathrm{M}^{h\mathrm{B}}\to\mathrm{M}^{h\mathrm{T}}$ is an isomorphism. There is a cosimplicial diagram

$$k\longrightarrow \mathrm{C}^*(\mathcal{B};k) \rightrightarrows \mathrm{C}^*(\mathcal{B}^2;k) \rightrightarrows\!\!\!\to \cdots$$

in $\mathrm{CAlg}(\mathrm{Mod}_k^{h\mathrm{G}})$.

Lemma 2.2.27. *Let k be an $\mathbf{E}_\infty$-ring (possibly with* G*-action), and let* $\mathrm{M}\in\mathrm{Mod}_k^{h\mathrm{G}}$*. Then the natural map* $\mathrm{M}\to\mathrm{Tot}(\mathrm{M}\otimes_k\mathrm{C}^*(\mathcal{B}^{\times\bullet+1};k))$ *is a* G*-equivariant isomorphism.*

Proof. By [Mat16b, Theorem 3.38], it suffices to show that $\mathrm{C}^*(\mathcal{B};k)$ is dualizable and faithful in $\mathrm{Mod}_k^{h\mathrm{G}}$. Dualizability follows from the fact that $\mathcal{B}$ is a finite G-CW-complex. For faithfulness, suppose that $\mathrm{M}\in\mathrm{Mod}_k^{h\mathrm{G}}$ and $\mathrm{M}\otimes_k\mathrm{C}^*(\mathcal{B};k)=0$; we need to show that $\mathrm{M}=0\in\mathrm{Mod}_k^{h\mathrm{G}}$. The forgetful functor $\mathrm{Mod}_k^{h\mathrm{G}}\to\mathrm{Mod}_k$ is conservative, so it suffices to show that the underlying k-module of M is zero. Nonequivariantly, k is a retract of $\mathrm{C}^*(\mathcal{B};k)$, so M is retract of $\mathrm{M}\otimes_k\mathrm{C}^*(\mathcal{B};k)$, and therefore must be zero. □

Definition 2.2.28. Let $\mathrm{M}\in\mathrm{Mod}_k^{h\mathrm{G}}$. Define a filtration on $\mathrm{M}^{h\mathrm{G}}$ via

$$\mathrm{fil}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}}):=\mathrm{Tot}(\tau_{\geq\star}((\mathrm{M}\otimes_k\mathrm{C}^*(\mathcal{B}^{\times\bullet+1};k))^{h\mathrm{G}}))\cong\mathrm{Tot}(\tau_{\geq\star}((\mathrm{M}\otimes_k\mathrm{C}^*(\mathcal{B}^{\times\bullet};k))^{h\mathrm{T}})).$$

This filtration is very nontrivial already when $\mathrm{M}=k$; by Example A.2.12, there is an isomorphism $\mathrm{fil}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}})\cong\mathrm{fil}^\star_{\mathrm{G}}(k^{h\mathrm{G}})$. More generally, there is an isomorphism $\mathrm{fil}^\star_{\mathrm{G}\text{-ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})\cong\mathrm{fil}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}})$; as such, we will often write $\mathrm{fil}^\star_{\mathrm{G}\text{-ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$ or $\mathrm{fil}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}})$ to denote the same object. The notation $\mathrm{fil}^\star_{\mathrm{G}\text{-ev}}$ used here is from

Definition A.2.7, where it is defined intrinsically to G (i.e., without reference to a choice of maximal torus in G). It follows from Lemma 2.2.27 that the underlying $k^{h\mathrm{G}}$-module of $\mathrm{fil}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}})$ is $\mathrm{M}^{h\mathrm{G}}$. We define $\mathrm{gr}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}})$ as the shearing of the associated graded of $\mathrm{fil}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}})$, so that

$$\mathrm{gr}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}}) := \mathrm{Tot}(\pi_\star((\mathrm{M} \otimes_k \mathrm{C}^*(\mathcal{B}^{\times \bullet+1}; k))^{h\mathrm{G}})) \cong \mathrm{Tot}(\pi_\star((\mathrm{M} \otimes_k \mathrm{C}^*(\mathcal{B}^{\times \bullet}; k))^{h\mathrm{T}})).$$

The symmetric monoidality of the homotopy fixed points equivalence $\mathrm{Mod}_k^{h\mathrm{T}} \xrightarrow{\simeq} \mathrm{Mod}_{k^{h\mathrm{T}}}^{\hat{k}}$ implies that

$$\mathrm{M}^{h\mathrm{T}} \otimes_{k^{h\mathrm{T}}} \mathrm{C}^*(\mathcal{B}^{\times \bullet}; k)^{h\mathrm{T}} \xrightarrow{\cong} (\mathrm{M} \otimes_k \mathrm{C}^*(\mathcal{B}^{\times \bullet}; k))^{h\mathrm{T}}.$$

The natural map

$$\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{M}^{h\mathrm{T}}) \otimes_{\pi_*(k^{h\mathrm{T}})} \mathrm{H}_{\mathrm{T}}^{-*}(\mathcal{B}^{\times \bullet}; k) \xrightarrow{\cong} \mathrm{gr}^\star_{\mathrm{ev}}((\mathrm{M} \otimes_k \mathrm{C}^*(\mathcal{B}^{\times \bullet}; k))^{h\mathrm{T}})$$

is an isomorphism. In particular, it follows that if $\mathrm{M} \in \mathrm{Mod}_k^{h\mathrm{G}}$, then $\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{M}^{h\mathrm{T}})$ admits the structure of a graded module over the Weyl algebroid $\mathcal{W}$, and that $\mathrm{gr}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}})$ is isomorphic to the derived invariants $\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{M}^{h\mathrm{T}})^{h\mathcal{W}}$ of $\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{M}^{h\mathrm{T}})$. These identifications are moreover natural (and lax symmetric monoidal) in M. In particular, $\mathrm{gr}^\star_{\mathrm{G}}(k^{h\mathrm{G}}) = \mathrm{gr}^\star_{\mathrm{ev}}(k^{h\mathrm{G}})$ is isomorphic to $\mathcal{O}_{\mathrm{T}_{\mathrm{F}}}^{h\mathcal{W}} \cong \mathrm{R}\Gamma(\mathrm{T}_{\mathrm{F}}/\mathcal{W}; \mathcal{O})$, and the isomorphism $\mathrm{gr}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}}) \cong \mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{M}^{h\mathrm{T}})^{h\mathcal{W}}$ is one of $\mathrm{gr}^\star_{\mathrm{ev}}(k^{h\mathrm{G}})$-modules.

Note that Remark 2.2.26 implies that k is good for G and $\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{M}^{h\mathrm{T}})$ is a flat $\pi_*(k^{h\mathrm{T}})$-module (concentrated in degree zero, say), then $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}}) \cong \pi_0(\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{M}^{h\mathrm{T}})^{h\mathcal{W}})$ is just the W-invariants $\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{M}^{h\mathrm{T}})^{\mathrm{W}}$. Moreover, if the order of W is invertible in $\pi_0(k)$, then in fact $\mathrm{gr}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}}) \cong \mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{M}^{h\mathrm{T}})^{\mathrm{W}}$.

Construction 2.2.29. Let Γ be a diagram of finite sets $x_0, x_\infty : \mathrm{E} \rightrightarrows \mathrm{F}$ along with a map $\mathrm{E} \to \mathbb{X}^*(\mathrm{T})$, and let $e : \mathrm{E} \to \pi_*(\mathrm{MU}^{h\mathrm{T}})$ be the composite $\mathrm{E} \to \mathbb{X}^*(\mathrm{T}) \to \pi_*(\mathrm{MU}^{h\mathrm{T}})$, where the final map is given by the Euler class. Suppose that this data satisfies the conditions of Definition 2.2.11, i.e., that $e(\chi)$ is a nontrivial nonzerodivisor for every $\chi \in \mathrm{E}$, and such that if $\chi \neq \lambda \in \mathrm{E}$, then $e(\chi)$ and $e(\lambda)$ are coprime in $\pi_*(\mathrm{MU}^{h\mathrm{T}})$. This defines an $\pi_*(\mathrm{MU}^{h\mathrm{T}})$-algebra M_Γ; if k is an MU-algebra, we will abusively continue to write M_Γ to denote the completed base-change of M_Γ along $\pi_*(\mathrm{MU}^{h\mathrm{T}}) \to \pi_*(k^{h\mathrm{T}})$.

Suppose that Γ is a diagram of finite sets with W-action, where the map $\mathrm{E} \to \mathbb{X}^*(\mathrm{T})$ is W-equivariant, such that if $v \in \mathrm{F}$ and $\alpha \in \Phi^+$ are such that $s_\alpha v \neq v$, then $\alpha \in \mathbb{X}^*(\mathrm{T})$ is the image of some $\chi \in \mathrm{E}$ (so $e(\chi) = e(\alpha)$) with $x_0(\chi) = v$ and $x_\infty(\chi) = s_\alpha v$. Then M_Γ acquires the structure of a $\mathcal{W}$-module in the following way. The coaction map $\rho : \mathrm{M}_\Gamma \to \mathrm{M}_\Gamma \otimes_{\pi_*(k^{h\mathrm{T}})} \mathrm{M}_{\Gamma_{\mathcal{B}}}$ is the unique map such that the following diagram commutes:

$$\begin{tikzcd} \mathrm{M}_\Gamma \arrow[r, dashed, "\rho"] \arrow[d, hook] & \mathrm{M}_\Gamma \otimes_{\pi_*(k^{h\mathrm{T}})} \mathrm{M}_{\Gamma_{\mathcal{B}}} \arrow[d, hook] \\ \pi_*(k^{h\mathrm{T}})^{\mathrm{F}} \arrow[r] & \pi_*(k^{h\mathrm{T}})^{\mathrm{F}} \otimes_{\pi_*(k^{h\mathrm{T}})} \mathrm{Map}(\mathrm{W}, \pi_*(k^{h\mathrm{T}})), \end{tikzcd}$$

where the bottom horizontal map is induced by the W-action on F. To show this, note that uniqueness of ρ is clear since the vertical maps are injective; so it suffices to prove that ρ is well-defined.

Note that a function $g : \mathrm{W} \times \mathrm{F} \to \pi_*(k^{h\mathrm{T}})$ lies in $\mathrm{M}_\Gamma \otimes_{\pi_*(k^{h\mathrm{T}})} \mathrm{M}_{\Gamma_{\mathcal{B}}}$ if and only if the following conditions are satisfied:

- If $\chi \in \mathrm{E}$ and $x_\epsilon := x_\epsilon(\chi)$ for $\epsilon \in \{0, \infty\}$, then $g(w, x_0) - g(w, x_\infty) \equiv 0 \pmod{we(\chi)}$;
- If $\alpha \in \Phi^+$ and $x \in \mathrm{F}$, then $g(w, x) - g(s_\alpha w, x) \equiv 0 \pmod{we(\alpha)}$.

Suppose $g : \mathrm{F} \to \pi_*(k^{h\mathrm{T}})$ lies in M_Γ; we need to show that the map $\rho_g : \mathrm{W} \times \mathrm{F} \to \pi_*(k^{h\mathrm{T}})$ given by $\rho_g(w, x) = g(wx)$ lies in $\mathrm{M}_\Gamma \otimes_{\pi_*(k^{h\mathrm{T}})} \mathrm{M}_{\Gamma_{\mathscr{B}}}$. Then:

- Let $\chi \in \mathrm{E}$ and $x_\epsilon := x_\epsilon(\chi)$ for $\epsilon \in \{0, \infty\}$. Then
$$g(w, x_0) - g(w, x_\infty) = g(wx_0) - g(wx_\infty) = w(g(x_0) - g(x_\infty)).$$
Since $g \in \mathrm{M}_\Gamma$, this vanishes modulo $we(\chi)$.
- Let $\alpha \in \Phi^+$ and $x \in \mathrm{F}$. Note that if $w \in \mathrm{W}$, then $s_\alpha wx = ws_\alpha w^{-1}wx = s_{w\alpha}wx$. It follows that
$$g(w, x) - g(s_\alpha w, x) = g(wx) - g(s_\alpha wx) = g(wx) - g(s_{w\alpha}wx).$$
This is tautologically zero if $s_{w\alpha}$ fixes wx. Otherwise, $w\alpha \in \mathbb{X}^*(\mathrm{T})$ is the image of some $\chi \in \mathrm{E}$ (so $e(\chi) = e(w\alpha)$); since $g \in \mathrm{M}_\Gamma$, it follows that $g(wx) - g(s_{w\alpha}wx)$ vanishes modulo $e(w\alpha) = we(\alpha)$, as desired.

Example 2.2.30. Let X be a k-GKM T-space such that the T-action on X extends to a continuous G-action, and let Γ_X be the diagram of finite sets from Theorem 2.2.12. Then $\mathrm{W} = \mathrm{N_G(T)/T}$ acts on $\mathrm{X^T}$ since $\mathrm{N_G(T)}$ normalizes T, and similarly W acts on $\chi \in \mathrm{E}$ via the natural W-action on $\mathbb{X}^*(\mathrm{T})$. Moreover, the maps $\mathrm{E} \rightrightarrows \mathrm{X^T}$ are W-equivariant. If $x \in \mathrm{X^T}$ and $\alpha \in \Phi^+$ are such that $s_\alpha x \neq x$, then $\alpha \in \mathbb{X}^*(\mathrm{T})$ is the image of some $\chi \in \mathrm{E}$ (so $e(\chi) = e(\alpha)$) with $x_0(\chi) = x$ and $x_\infty(\chi) = s_\alpha x$. Indeed, if $\alpha \in \Phi^+$ defines an associated homomorphism $\mathrm{SL}_2 \to \mathrm{G}$, then the SL_2-orbit of x is either 0-dimensional (in which case $s_\alpha x = x$) or is isomorphic to $\mathbf{C}\mathrm{P}^1 \cong \mathrm{S}^2$ for the standard action of SL_2 on $\mathbf{C}\mathrm{P}^1$. Since this 2-sphere is contained in $\mathrm{X}^{(1)}$, it defines the desired $\chi \in \mathrm{E}$. It follows that the W-action on $\mathrm{Map}(\mathrm{W}, \pi_*(k^{h\mathrm{T}}))$ defines a $\mathcal{W}$-action on $\mathrm{M}_{\Gamma_\mathrm{X}}$.

The action map $\mathrm{G} \times^{\mathrm{B}} \mathrm{X} \to \mathrm{X}$ can also be used to construct a $\mathcal{W}$-action on $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}; k)$ (via Definition 2.2.28), and it is straightforward (using Proposition 2.2.4) to verify that the isomorphism $\mathrm{M}_{\Gamma_\mathrm{X}} \cong \mathrm{H}^{-*}_{\mathrm{T},c}(\mathrm{X}; k)$ of Theorem 2.2.12 is a $\mathcal{W}$-equivariant isomorphism of commutative $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$-algebras.

Combining Definition 2.2.28 with Example 2.2.30, we find:

Corollary 2.2.31. *Let* X *be a k-GKM* T*-space such that the* T*-action on* X *extends to a continuous* G*-action. Then there is an isomorphism* $\mathrm{gr}^\star_\mathrm{G}(\mathrm{C}^*_{\mathrm{G},c}(\mathrm{X}; k)) \cong \mathrm{M}^{h\mathcal{W}}_{\Gamma_\mathrm{X}}$ *of commutative* $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$*-algebras. Dualizing, there is an isomorphism* $\mathrm{gr}^\star_\mathrm{G}(\mathrm{C}^{\mathrm{G,BM}}_*(\mathrm{X}; k)) \cong (\mathrm{M}^\vee_{\Gamma_\mathrm{X}})^{h\mathcal{W}}$ *of cocommutative* $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$*-coalgebras.*

In particular, if k is good for G, then $\mathrm{gr}^0_\mathrm{G}(\mathrm{C}^*_{\mathrm{G},c}(\mathrm{X}; k)) \cong (\mathrm{M}_{\Gamma_\mathrm{X}})^\mathrm{W}$ and $\mathrm{gr}^0_\mathrm{G}(\mathrm{C}^{\mathrm{G,BM}}_*(\mathrm{X}; k)) \cong (\mathrm{M}^\vee_{\Gamma_\mathrm{X}})^\mathrm{W}$. Like in Remark 2.2.16, Corollary 2.2.31 generalizes easily to ind-G-spaces whose underlying ind-T-space is k-GKM.

2.3. **Full faithfulness.**

We will now review (a variant of) an ingenious argument of Ginzburg's [Gin25], which proves that taking (equivariant) cohomology is "fully faithful".

Definition 2.3.1. Let k be an $\mathbf{E}_\infty$-ring (which is not necessarily even), let T be a torus, let $k^{h\mathrm{T}}$ denote the homotopy fixed points for the trivial T-action on k, and let $k^{t\mathrm{T}}$ denote the Tate construction for the trivial T-action on k. If M is a $k^{h\mathrm{T}}$-module, we will denote its base-change along the map $k^{h\mathrm{T}} \to k^{t\mathrm{T}}$ by $\mathrm{M}_\circ$. Let X be a (connected) locally compact Hausdorff topological space equipped with a T-action and a finite stratification indexed by a poset Λ, where each stratum $j_\lambda : \mathrm{X}_\lambda \hookrightarrow \mathrm{X}$ is locally closed, smooth, and T-stable.

Let $\mathrm{Shv}_\mathrm{T}(\mathrm{X}; k)_\mathrm{pev}$ denote the smallest subcategory of $\mathrm{Shv}_\mathrm{T}(\mathrm{X}; k)$ which contains all shifts $\Sigma^\mathrm{V} \underline{k}_\mathrm{X}$ of the constant sheaf by complex representations of T, and which is closed under extensions and retracts. An object of $\mathrm{Shv}_\mathrm{T}(\mathrm{X}; k)_\mathrm{pev}$ will be called *perfect even* (this definition is motivated by [Pst23a]). Say that $\mathcal{F} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X}; k)$ is $*$*-even* (resp. *!-even*) if for all $\lambda \in \Lambda$, the pullback $j_\lambda^* \mathcal{F}$ (resp. $j_\lambda^! \mathcal{F}$) lies in $\mathrm{Shv}_\mathrm{T}(\mathrm{X}_\lambda; k)_\mathrm{pev}$. Say that $\mathcal{F}$ is *even* if it is both $*$- and !-even. See [JMW14], where this definition is motivated from.

For $\mathcal{F} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X}; k)$, we will often just write $\mathrm{H}^*_\mathrm{T}(\mathcal{F})$ to denote $\mathrm{H}^*_\mathrm{T}(\mathrm{X}; \mathcal{F})$. (In this section, and only in this section, we use cohomological indexing, to decrease clutter.)

Theorem 2.3.2. *Let $f : \mathrm{Y} \subseteq \mathrm{X}$ be a closed embedding of* T*-spaces, and let $\mathrm{Y}_\lambda = \mathrm{X}_\lambda \cap \mathrm{Y}$. Let Σ denote the set of those $\lambda \in \Lambda$ for which Y_λ is nonempty, and assume that the stabilizer in* T *of any point of* X *is connected. Following Ginzburg, assume:*

1. *For each $\lambda \in \Lambda$, the stratum X_λ is* T*-equivariantly isomorphic to a complex vector space with a linear* T*-action; and moreover, $\mathrm{Y}_\lambda \subseteq \mathrm{X}_\lambda$ is a vector subspace. I will write $0_\lambda \subseteq \mathrm{X}_\lambda$ to denote the origin, and denote this inclusion also by 0_λ^*.*
2. *For each $\lambda \in \Sigma$, the* T*-invariant set $\mathrm{X}_\lambda^\mathrm{T}$ is just the origin 0_λ of the vector space X_λ.*

Let k be an even $\mathbf{E}_\infty$-ring. Let $\mathcal{F} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X}; k)$ be $*$-even, and let $\mathcal{G} \in \mathrm{Shv}_\mathrm{T}(\mathrm{Y}; k)$ be !-even. Then there is an isomorphism of graded $\mathrm{H}^*_\mathrm{T}(\mathrm{X}; k)$-modules:

$$\mathrm{Ext}^\star_{\mathrm{Shv}_\mathrm{T}(\mathrm{X};k)}(\mathcal{F}, f_*\mathcal{G}) \xrightarrow{\cong} \mathrm{Hom}^\star_{\mathrm{H}^*_\mathrm{T}(\mathrm{X};k)}(\mathrm{H}^*_\mathrm{T}(\mathrm{X}; \mathcal{F}), \mathrm{H}^*_\mathrm{T}(\mathrm{X}; f_*\mathcal{G})).$$

The left-hand side can be rewritten as $\pi_{-\star}\mathrm{R}\Gamma_\mathrm{T}(\mathrm{X}; \mathcal{F}^\vee \otimes^! f_*(\mathcal{G}))$.

Remark 2.3.3. The assumptions on k and X imply that $\mathrm{H}^*_\mathrm{T}(\mathrm{X}; k)$ is concentrated in even weights. In the proof, we will see that $\mathrm{H}^*_\mathrm{T}(\mathrm{X}; \mathcal{F})$ and $\mathrm{H}^*_\mathrm{T}(\mathrm{X}; f_*\mathcal{G}) = \mathrm{H}^*_\mathrm{T}(\mathrm{Y}; \mathcal{G})$ are also concentrated in even weights. It follows that $\mathrm{Hom}^\star_{\mathrm{H}^*_\mathrm{T}(\mathrm{X};k)}(\mathrm{H}^*_\mathrm{T}(\mathrm{X}; \mathcal{F}), \mathrm{H}^*_\mathrm{T}(\mathrm{X}; f_*\mathcal{G}))$ is *also* concentrated in even weights, so in particular $\mathrm{RHom}_{\mathrm{Shv}_\mathrm{T}(\mathrm{X};k)}(\mathcal{F}, f_*\mathcal{G})$ is concentrated in even degrees. Moreover, the proof shows that it is a projective $\pi_{-*}(k^{h\mathrm{T}})$.

Remark 2.3.4. Suppose X is in fact compact. Then $\mathrm{H}^*_\mathrm{T}(\mathrm{X}; k)$ is a finitely generated $\pi_{-*}(k^{h\mathrm{T}})$-module; in fact, it is also flat and concentrated in even weights, since X is assumed to have a stratification by complex vector spaces. The same is true of $\mathrm{H}^\mathrm{T}_{-*}(\mathrm{X}; k) = \pi_{-*}\mathrm{R}\Gamma_\mathrm{T}(\mathrm{X}; \omega_\mathrm{X})$. Note that $\mathrm{R}\Gamma_\mathrm{T}(\mathrm{X}; \omega_\mathrm{X})$ is an $\mathbf{E}_\infty$-$k^{h\mathrm{T}}$-coalgebra by Remark 2.1.8. The Künneth formula (which applies since $\mathrm{H}^\mathrm{T}_*(\mathrm{X}; k)$ is a flat $\pi_*(k^{h\mathrm{T}})$-module) implies that $\mathrm{H}^\mathrm{T}_*(\mathrm{X}; k)$ is a flat cocommutative $\pi_*(k^{h\mathrm{T}})$-coalgebra.

Since $\mathrm{H}^*_\mathrm{T}(\mathrm{X}; k)$ is a finitely generated $\pi_{-*}(k^{h\mathrm{T}})$-module, $\mathrm{Mod}^\heartsuit_{\mathrm{H}^*_\mathrm{T}(\mathrm{X};k)}$ is equivalent to the (1-)category of $\mathrm{H}^\mathrm{T}_{-*}(\mathrm{X}; k)$-comodules in $\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})}$. (If $\mathcal{F} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X}; k)$, then $\mathrm{H}^*_\mathrm{T}(\mathrm{X}; \mathcal{F})$ acquires the structure of a $\mathrm{H}^\mathrm{T}_{-*}(\mathrm{X}; k)$-comodule. If $\mathrm{C}^*_c(\mathrm{X}; \mathcal{F}) \cong \mathrm{C}^*(\mathrm{X}; \mathcal{F})$ is a compact object of $\mathrm{Fun}(\mathrm{BT}, \mathrm{Mod}_k)$, then the $\mathrm{H}^\mathrm{T}_{-*}(\mathrm{X}; k)$-comodule structure on $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X};\mathcal{F}) = \mathrm{H}^*_{\mathrm{T},c}(\mathrm{X};\mathcal{F})$ is obtained via Remark 2.1.8 and the Künneth formula.) The conclusion of Theorem 2.3.2 can therefore be restated as an isomorphism

$$\mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}, f_*\mathcal{G}) \xrightarrow{\cong} \mathrm{Hom}^\star_{\mathrm{coMod}_{\mathrm{H}^{\mathrm{T}}_{-*}(\mathrm{X};k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})})}(\mathrm{H}^*_{\mathrm{T}}(\mathrm{X};\mathcal{F}), \mathrm{H}^*_{\mathrm{T}}(\mathrm{Y};\mathcal{G})).$$

Before proving Theorem 2.3.2, let us discuss its extension to the case when X is ind-compact Hausdorff (Construction 2.1.18).

Setup 2.3.5. Suppose $\mathrm{X} = \mathrm{colim}_n \mathrm{X}^n$ is an ind-good T-space, where each X^n is locally compact Hausdorff (viewed as a good T-space via the constant limit diagram) and is equipped a finite stratification indexed by a poset Λ^n, where each stratum $j^n_\lambda : \mathrm{X}^n_\lambda \hookrightarrow \mathrm{X}^n$ is locally closed, smooth, and T-stable. Let $i^n : \mathrm{X}^n \to \mathrm{X}$ denote the closed inclusion. An object $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ will be called *!-even* if $i^{n,!}(\mathcal{F})$ is !-even for each n. Note that the notion of $*$-evenness is not well-defined for an arbitrary object $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$. However, if $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ is pushed forward along i^n for some n, it does make sense to say that $\mathcal{F}$ is $*$-even, namely if $(i^n)^!(\mathcal{F})$ is $*$-even in the sense of Definition 2.3.1.

Remark 2.3.6. Suppose $\mathrm{X} = \mathrm{colim}_n \mathrm{X}^n$ is an ind-good T-space, where each X^n is compact Hausdorff (viewed as a good T-space via the constant limit diagram). Suppose also that $f : \mathrm{Y} \subseteq \mathrm{X}$ is a closed embedding of ind-good T-spaces such that each inclusion $f^n : \mathrm{Y}^n \subseteq \mathrm{X}^n$ satisfies the hypotheses of Theorem 2.3.2. We will assume that the maps $i_* : \mathrm{H}^{\mathrm{T}}_{-*}(\mathrm{X}^m;k) \to \mathrm{H}^{\mathrm{T}}_{-*}(\mathrm{X}^n;k)$ of cocommutative $\pi_{-*}(k^{h\mathrm{T}})$-coalgebras induced by the maps $i : \mathrm{X}^m \to \mathrm{X}^n$ are all injective. Then an analogue of Theorem 2.3.2 continues to hold; see (2.3.1).

If $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X}^n;k)$ is $*$-even and $\mathcal{G} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{Y}^n;k)$ is !-even, then Theorem 2.3.2 gives an isomorphism of graded $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}^n;k)$-modules:

$$\mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}^n;k)}(\mathcal{F}, f^n_*\mathcal{G}) \xrightarrow{\cong} \mathrm{Hom}^\star_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}^n;k)}(\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}^n;\mathcal{F}), \mathrm{H}^*_{\mathrm{T}}(\mathrm{Y}^n;\mathcal{G})).$$

Recall that $\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k) = \mathrm{colim}_n \mathrm{Shv}_{\mathrm{T}}(\mathrm{X}^n;k)$, where the colimit is taken along ($*$-)pushforwards, and similarly for $\mathrm{Shv}_{\mathrm{T}}(\mathrm{Y};k)$. Suppose that $\mathcal{G} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{Y};k)$ is !-even, and that $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ is $*$-even and pushed forward along the inclusion $i^n : \mathrm{X}^n \hookrightarrow \mathrm{X}$ for some n. Since the colimit defining $\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ is filtered, and $f_*\mathcal{G} = \mathrm{colim}_n f^n_* f^{n,!}\mathcal{G}$, it follows that

$$\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}, f_*\mathcal{G}) \cong \mathrm{colim}_n \mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}^n;k)}(\mathcal{F}, f^n_* f^{n,!}\mathcal{G}).$$

Since $\pi_\star$ commutes with filtered colimits, it follows from Remark 2.3.4 that

$$\begin{aligned}\mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}, f_*\mathcal{G}) &\cong \mathrm{colim}_n \mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}^n;k)}(\mathcal{F}, f^n_* f^{n,!}\mathcal{G}) \\ &\xrightarrow{\cong} \mathrm{colim}_n \mathrm{Hom}^\star_{\mathrm{coMod}_{\mathrm{H}^{\mathrm{T}}_{-*}(\mathrm{X}^n;k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})})}(\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}^n;\mathcal{F}), \mathrm{H}^*_{\mathrm{T}}(\mathrm{Y}^n; f^{n,!}\mathcal{G})).\end{aligned}$$

The maps $i_* : \mathrm{H}^{\mathrm{T}}_{-*}(\mathrm{X}^m;k) \to \mathrm{H}^{\mathrm{T}}_{-*}(\mathrm{X}^n;k)$ of cocommutative $\pi_{-*}(k^{h\mathrm{T}})$-coalgebras define a functor

$$\mathrm{coMod}_{\mathrm{H}^{\mathrm{T}}_{-*}(\mathrm{X}^m;k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})}) \to \mathrm{coMod}_{\mathrm{H}^{\mathrm{T}}_{-*}(\mathrm{X}^n;k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})}).$$

Recall from Example 2.1.21 that $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};\omega^{\mathrm{ren}}_{\mathrm{X}}) = \mathrm{colim}_n \mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}^n;\omega_{\mathrm{X}^n})$. This is an $\mathbf{E}_\infty$-$k^{h\mathrm{T}}$-coalgebra. Since $\tau_{\geq\star}$ is a lax symmetric monoidal functor, there is a natural map

$$\tau_{\geq\star}\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};\omega^{\mathrm{ren}}_{\mathrm{X}}) \otimes_{\tau_{\geq\star}k^{h\mathrm{T}}} \tau_{\geq\star}\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};\omega^{\mathrm{ren}}_{\mathrm{X}}) \to \tau_{\geq\star}(\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};\omega^{\mathrm{ren}}_{\mathrm{X}}) \otimes_{k^{h\mathrm{T}}} \mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};\omega^{\mathrm{ren}}_{\mathrm{X}})),$$

which is an isomorphism since $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}; \omega_{\mathrm{X}}^{\mathrm{ren}})$ is a colimit of the perfect free $k^{h\mathrm{T}}$-modules $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}^n; \omega_{\mathrm{X}^n})$. In particular, it follows that $\tau_{\geq\star}\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}; \omega_{\mathrm{X}}^{\mathrm{ren}})$ is a filtered $\mathbf{E}_\infty$-$\tau_{\geq\star} k^{h\mathrm{T}}$-coalgebra. Since π_* commutes with filtered colimits, it follows that $\mathrm{H}_{-*}^{\mathrm{T,BM}}(\mathrm{X}; k) \cong \mathrm{colim}_n \mathrm{H}_{-*}^{\mathrm{T}}(\mathrm{X}^n; k)$, and this is a graded cocommutative $\pi_{-*}(k^{h\mathrm{T}})$-coalgebra. In particular, since the maps $\mathrm{H}_{-*}^{\mathrm{T}}(\mathrm{X}^m; k) \to \mathrm{H}_{-*}^{\mathrm{T}}(\mathrm{X}^n; k)$ are assumed to be injective, there are isomorphisms

$$\mathrm{colim}_n \mathrm{Hom}^\star_{\mathrm{coMod}_{\mathrm{H}_{-*}^{\mathrm{T}}(\mathrm{X}^n;k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})})}(\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}^n; \mathcal{F}), \mathrm{H}^*_{\mathrm{T}}(\mathrm{Y}^n; f^{n,!}\mathcal{G})) \cong \\ \mathrm{colim}_n \mathrm{Hom}^\star_{\mathrm{coMod}_{\mathrm{H}_{-*}^{\mathrm{T,BM}}(\mathrm{X};k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})})}(\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}; \mathcal{F}), \mathrm{H}^*_{\mathrm{T}}(\mathrm{Y}^n; f^{n,!}\mathcal{G})),$$

and we conclude that there is an isomorphism
(2.3.1)

$$\mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}, f_*\mathcal{G}) \xrightarrow{\cong} \mathrm{colim}_n \mathrm{Hom}^\star_{\mathrm{coMod}_{\mathrm{H}_{-*}^{\mathrm{T,BM}}(\mathrm{X};k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})})}(\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}; \mathcal{F}), \mathrm{H}^*_{\mathrm{T}}(\mathrm{Y}^n; f^{n,!}\mathcal{G})).$$

Suppose that $\mathcal{G} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{Y}; k)$ is !-even, and more generally that $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X}; k)$ is a direct limit of $*$-even sheaves $\mathcal{F}^m$ such that the support of $\mathcal{F}^m$ is contained in X^m. Then there are isomorphisms

$$\mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}, f_*\mathcal{G}) \cong \lim_m \mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}^m, f_*\mathcal{G}) \xrightarrow{\cong} \\ \lim_m \mathrm{colim}_n \mathrm{Hom}^\star_{\mathrm{coMod}_{\mathrm{H}_{-*}^{\mathrm{T,BM}}(\mathrm{X};k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})})}(\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}; \mathcal{F}^m), \mathrm{H}^*_{\mathrm{T}}(\mathrm{Y}^n; f^{n,!}\mathcal{G})).$$

Remark 2.3.7. The map $\mathrm{X}^m \to \mathrm{X}^n$ induces a functor

$$\mathrm{coMod}_{\mathrm{H}_{-*}^{\mathrm{T}}(\mathrm{X}^m;k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})}) \to \mathrm{coMod}_{\mathrm{H}_{-*}^{\mathrm{T}}(\mathrm{X}^n;k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})}),$$

which is isomorphic to the functor $\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}^m;k)} \to \mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}^n;k)}$ given by restriction of scalars along the map $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}^n; k) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}^m; k)$. One could therefore replace the category $\mathrm{coMod}_{\mathrm{H}_{-*}^{\mathrm{T,BM}}(\mathrm{X};k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_{-*}(k^{h\mathrm{T}})})$ in Remark 2.3.6 with $\mathrm{colim}_n \mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}^n;k)}$. This can be viewed as the category of complete graded $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}; k)$-modules, where $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}; k)$ is equipped with the topology coming from the inverse limit $\lim_n \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}^n; k)$. Since it is technically more involved to work with the category of complete modules over a topological ring, we will instead work with the category of $\mathrm{H}_{-*}^{\mathrm{T,BM}}(\mathrm{X}; k)$-comodules.

Let us now turn to the proof of Theorem 2.3.2. We first need to understand the case when X is a (complex) vector space V with a linear T-action.

Lemma 2.3.8. *Let* V *be a complex vector space with a linear* T*-action, and let* $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{V}; k)_{\mathrm{pev}}$*. Then there is an isomorphism* $\mathrm{H}^*_{\mathrm{T},c}(\mathrm{V}; \mathcal{F}) \cong \mathrm{H}^*_{\mathrm{T}}(\mathrm{V}; \mathcal{F})(-2\dim(\mathrm{V}))$*. (Here,* $\dim = \dim_{\mathbf{C}}$*.)*

Proof. This reduces to the case when $\mathcal{F}$ is the constant sheaf. If $q : \mathrm{V} \to *$ denotes the quotient, there is a T-equivariant equivalence $q_! \underline{k} \cong \Sigma^{-\mathrm{V}} k$, so that $\mathrm{R}\Gamma_{\mathrm{T},c}(\mathrm{V}; k) \cong (\Sigma^{-\mathrm{V}} k)^{h\mathrm{T}}$. (Indeed, recall that V is the complement of $\mathbf{P}(\mathrm{V}) \subseteq \mathbf{P}(\mathrm{V} \oplus \mathbf{C})$. Excision implies that $q_! \underline{k} \cong \Sigma^{-\mathrm{V}} k$.) Since k is even, it is complex-orientable, so there is a T-equivariant equivalence $\Sigma^{-\mathrm{V}} k \simeq \Sigma^{-2\dim(\mathrm{V})} k$, so that $\mathrm{H}^*_{\mathrm{T},c}(\mathrm{V}; k) \cong \pi_{*+2\dim(\mathrm{V})}(k^{h\mathrm{T}})$, as desired. □

Lemma 2.3.9. *Let* $i : \mathrm{X}' \subseteq \mathrm{X}$ *be a closed subset which is a union of strata in* X*. Suppose* $j_\lambda : \mathrm{X}_\lambda \subseteq \mathrm{X}'$ *is open, and let* $i_\lambda : \mathrm{X}'_{<\lambda} = \mathrm{X}' - \mathrm{X}_\lambda \subseteq \mathrm{X}'$ *be its complement.*

- *If* $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X}'; k)$ *is* !*-even, then there is a short exact sequence of* $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; k)$*-modules*
$$0 \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (i_\lambda)_! i^!_\lambda \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_* j^*_\lambda \mathcal{F}) \to 0,$$
and moreover each term is even and projective as a $\pi_{-*}(k^{h\mathrm{T}})$*-module.*
- *If* $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X}'; k)$ *is* $*$*-even, then there is a short exact sequence of* $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; k)$*-modules*
$$0 \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_! j^!_\lambda \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (i_\lambda)_* i^*_\lambda \mathcal{F}) \to 0,$$
and moreover each term is even and projective as a $\pi_{-*}(k^{h\mathrm{T}})$*-module.*

Proof. The two statements are proved similarly, so let us show the first one. Since $\mathcal{F}$ is !-even, $j^!_\lambda \mathcal{F}$ is perfect even (and since $\mathrm{X}_\lambda \subseteq \mathrm{X}'$ is open, $j^!_\lambda \mathcal{F} \cong j^*_\lambda \mathcal{F}$). By Lemma 2.3.8, it follows that $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_! j^!_\lambda \mathcal{F})$ is isomorphic to $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}_\lambda; j^*_\lambda \mathcal{F})(-2\dim(\mathrm{X}_\lambda))$, which is just a (graded) shift of $\pi_* k^{h\mathrm{T}}$. By induction on the stratification of X', one also knows that $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (i_\lambda)_! i^!_\lambda \mathcal{F})$ is even and projective as a $\pi_* k^{h\mathrm{T}}$-module. The cofiber sequence
$$(i_\lambda)_! i^!_\lambda \mathcal{F} \to \mathcal{F} \to (j_\lambda)_* j^*_\lambda \mathcal{F}$$
defines a cofiber sequence
$$\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; (i_\lambda)_! i^!_\lambda \mathcal{F}) \to \mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; \mathcal{F}) \to \mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_* j^*_\lambda \mathcal{F})$$
of $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}; k)$-modules. Since it is an exact sequence of even projective $k^{h\mathrm{T}}$-modules, it follows that the extension splits (in $k^{h\mathrm{T}}$-modules), so we obtain an exact sequence on homotopy groups, as desired. □

Lemma 2.3.10. *Let* $i : \mathrm{X}' \subseteq \mathrm{X}$ *be a closed subset which is a union of strata in* X*. Suppose* $j_\lambda : \mathrm{X}_\lambda \subseteq \mathrm{X}'$ *is open, and let* $i_\lambda : \mathrm{X}'_{<\lambda} = \mathrm{X}' - \mathrm{X}_\lambda \subseteq \mathrm{X}'$ *be its complement.*

Let $\mathcal{F}$ *be a* $*$*-even sheaf on* X'*. Then, the map* $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_* j^*_\lambda \mathcal{F})$ *is surjective, and there is a short exact sequence of* $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; k)$*-modules*
$$0 \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (i_\lambda)_! i^!_\lambda \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_* j^*_\lambda \mathcal{F}) \to 0.$$

Proof. To show that the map $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_* j^*_\lambda \mathcal{F})$ is surjective, observe that the map
$$\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_* j^*_\lambda \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (0_\lambda)_* 0^*_\lambda \mathcal{F})$$
is an isomorphism, since X_λ is an affine space on which T acts linearly, and $j^*_\lambda \mathcal{F}$ is perfect even. The inclusion $0_\lambda \subseteq \mathrm{X}_\lambda \subseteq \mathrm{X}'$ is closed. Let $a_\lambda : \mathrm{U}_\lambda \subseteq \mathrm{X}'$ denote its complement, so a_λ is an open embedding. Then recollement gives a cofiber sequence
$$\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; (a_\lambda)_! a^*_\lambda \mathcal{F}) \to \mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; \mathcal{F}) \to \mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; (0_\lambda)_* 0^*_\lambda \mathcal{F}).$$
Note that $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; (0_\lambda)_* 0^*_\lambda \mathcal{F})$ is an even projective $k^{h\mathrm{T}}$-module, so to show that the map $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; \mathcal{F}) \to \mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; (0_\lambda)_* 0^*_\lambda \mathcal{F})$ is surjective on homotopy, it suffices to show that $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; (a_\lambda)_! a^*_\lambda \mathcal{F})$ is also an even projective $k^{h\mathrm{T}}$-module. But $(a_\lambda)_! a^*_\lambda \mathcal{F}$ is $*$-even, so the claim follows from the second part of Lemma 2.3.9.

Recollement gives a cofiber sequence
$$\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; (i_\lambda)_! i^!_\lambda \mathcal{F}) \to \mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; \mathcal{F}) \to \mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_* j^*_\lambda \mathcal{F})$$
of $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}; k)$-modules, which induces a long exact sequence
$$\cdots \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (i_\lambda)_! i^!_\lambda \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_* j^*_\lambda \mathcal{F}) \xrightarrow{\partial} \mathrm{H}^{*+1}_{\mathrm{T}}(\mathrm{X}'; (i_\lambda)_! i^!_\lambda \mathcal{F}) \to \cdots.$$

To get the claimed short exact sequence, it suffices to show that the boundary map ∂ vanishes, or equivalently that the map $\mathrm{H}^*_\mathrm{T}(\mathrm{X}';\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}(\mathrm{X}';(j_\lambda)_* j_\lambda^* \mathcal{F})$ is surjective. But this was proved above. $\square$

Induction on strata then shows:

Corollary 2.3.11. *Let $i : \mathrm{X}' \to \mathrm{X}$ be a closed inclusion where X' is a union of strata, and let $j : \mathrm{X} - \mathrm{X}' \to \mathrm{X}$ denote the complement. Suppose $\mathcal{F} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X};k)$ is !-even, then $\mathrm{H}^*_\mathrm{T}(\mathrm{X}; i_! i^! \mathcal{F})$ is even and projective over $\pi_{-*}(k^{h\mathrm{T}})$. Also, there is a short exact sequence of $\mathrm{H}^*_\mathrm{T}(\mathrm{X};k)$-modules*

$$0 \to \mathrm{H}^*_\mathrm{T}(\mathrm{X}; i_! i^! \mathcal{F}) \to \mathrm{H}^*_\mathrm{T}(\mathrm{X}; \mathcal{F}) \to \mathrm{H}^*_\mathrm{T}(\mathrm{X}; j_* j^* \mathcal{F}) \to 0.$$

Similarly, if $\mathcal{F} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X};k)$ is $*$-even, then $\mathrm{H}^*_\mathrm{T}(\mathrm{X}; i_* i^* \mathcal{F})$ is even and projective over $\pi_{-*}(k^{h\mathrm{T}})$. Also, there is a short exact sequence of $\mathrm{H}^*_\mathrm{T}(\mathrm{X};k)$-modules

$$0 \to \mathrm{H}^*_\mathrm{T}(\mathrm{X}; j_! j^! \mathcal{F}) \to \mathrm{H}^*_\mathrm{T}(\mathrm{X}; \mathcal{F}) \to \mathrm{H}^*_\mathrm{T}(\mathrm{X}; i_* i^* \mathcal{F}) \to 0.$$

We will now show that certain maps between equivariant cohomology groups are automatically zero for reasons of localization. Let $\mathrm{X}^\mathrm{T} = \coprod_{v \in \mathrm{V}} \mathrm{X}_v$ where V is a finite set and each X_v is connected. Let $\epsilon_v : \mathrm{X}_v \to \mathrm{X}$ denote the inclusion. Note that Remark 2.2.5 says that the pullback map $\mathrm{H}^*_\mathrm{T}(\mathrm{X};k) \to \mathrm{H}^*_\mathrm{T}(\mathrm{X}^\mathrm{T};k)$ becomes an isomorphism upon base-change along the map $k^{h\mathrm{T}} \to k^{t\mathrm{T}}$ (i.e., upon inverting all Euler classes). The assumption that the stabilizers in T of points in X are connected is used here. There is an isomorphism $\mathrm{H}^*_\mathrm{T}(\mathrm{X}^\mathrm{T};k) \cong \bigoplus_{v \in \mathrm{V}} \mathrm{H}^*_\mathrm{T}(\mathrm{X}_v;k)$ of rings, so let e_v denote the unit of $\mathrm{H}^*_\mathrm{T}(\mathrm{X}_v;k)$ inside $\mathrm{H}^*_\mathrm{T}(\mathrm{X}^\mathrm{T};k)$. Then $1 = \sum_{v \in \mathrm{V}} e_v$ in $\mathrm{H}^*_\mathrm{T}(\mathrm{X};k)_\circ$. If $\mathcal{F} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X};k)$, then there are maps $(\epsilon_v)_! \epsilon_v^! \mathcal{F} \to \mathcal{F}$ and $\mathcal{F} \to (\epsilon_v)_* \epsilon_v^* \mathcal{F}$ which induce isomorphisms

$$e_v \mathrm{H}^*_\mathrm{T}((\epsilon_v)_! \epsilon_v^! \mathcal{F})_\circ \xrightarrow{\cong} e_v \mathrm{H}^*_\mathrm{T}(\mathcal{F})_\circ,$$
$$e_v \mathrm{H}^*_\mathrm{T}(\mathcal{F})_\circ \xrightarrow{\cong} e_v \mathrm{H}^*_\mathrm{T}((\epsilon_v)_* \epsilon_v^* \mathcal{F})_\circ.$$

Lemma 2.3.12. *Stick with the setup of Lemma 2.3.9. Assume also that X_λ intersects Y, so $\mathrm{Y}_\lambda \subseteq \mathrm{Y} \cap \mathrm{X}'$ is open and $0_\lambda \in \mathrm{Y}_\lambda$. Let X_v denote the connected component of X^T containing 0_λ. Then $\mathrm{X}_v \cap \mathrm{X}'_{<\lambda}$ is empty.*

Proof. Indeed, $\mathrm{X}_v \cap \mathrm{X}'$ is a union $\sqcup_{\mu \in \mathrm{S}} \mathrm{X}_\mu^\mathrm{T}$ for some $\mathrm{S} \subseteq \Lambda$. By assumption (b), $\mathrm{X}_v \cap \mathrm{X}_\lambda = \mathrm{X}_\lambda^\mathrm{T}$ is just $\{0_\lambda\}$. Because X_v is connected, 0_λ must lie in the closure of $\mathrm{X}_\mu^\mathrm{T}$ for each $\mu \in \mathrm{S}$. Since X_λ is open in X', $\{0_\lambda\}$ is open in $\mathrm{X}_v \cap \mathrm{X}'$, so 0_λ is not contained in the closure of any $\mathrm{X}_\mu^\mathrm{T}$ (unless $\mu = \lambda$). It follows that $\mathrm{S} = \{\lambda\}$, as desired. $\square$

Corollary 2.3.13. *Let $\mathcal{F}_\lambda \in \mathrm{Shv}_\mathrm{T}(\mathrm{X}_\lambda;k)$, and let $\mathcal{F}_{<\lambda} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X}'_{<\lambda})$. Let H_λ denote either $\mathrm{H}^*_\mathrm{T}(\mathrm{X}';(j_\lambda)_* \mathcal{F}_\lambda)$ or $\mathrm{H}^*_\mathrm{T}(\mathrm{X}';(j_\lambda)_! \mathcal{F}_\lambda)$, and similarly let $\mathrm{H}_{<\lambda}$ denote $\mathrm{H}^*_\mathrm{T}(\mathrm{X}';(i_\lambda)_* \mathcal{F}_{<\lambda})$. Let $\mathrm{V}_{<\lambda}$ denote the set of those $v \in \mathrm{V}$ such that $\mathrm{X}_v \cap \mathrm{X}'_\lambda$ is nonempty. Then:*

- *The element $e_{<\lambda} = \sum_{v \in \mathrm{V}_{<\lambda}} e_v \in \mathrm{H}^*_\mathrm{T}(\mathrm{X}^\mathrm{T};k)$ acts by zero on $(\mathrm{H}_\lambda)_\circ$, and by the identity on $(\mathrm{H}_{<\lambda})_\circ$.*
- *Suppose $\mathrm{H}_{<\lambda}$ is a flat $\pi_{-*}(k^{h\mathrm{T}})$-module. Then $\mathrm{Hom}_{\mathrm{H}^*_\mathrm{T}(\mathrm{X};k)}(\mathrm{H}_\lambda, \mathrm{H}_{<\lambda}) = 0$. Similarly, if H_λ is a flat $\pi_{-*}(k^{h\mathrm{T}})$-module, then $\mathrm{Hom}_{\mathrm{H}^*_\mathrm{T}(\mathrm{X};k)}(\mathrm{H}_{<\lambda}, \mathrm{H}_\lambda) = 0$.*

Proof. That $e_{<\lambda}$ acts by the identity on $(\mathrm{H}_{<\lambda})_\circ$ is clear. To show that $e_{<\lambda}$ acts by zero on $(\mathrm{H}_\lambda)_\circ$, it suffices to show the same for e_v for any $v \in \mathrm{V}_{<\lambda}$. By Lemma 2.3.12, $\mathrm{X}_v \cap \mathrm{X}' \subseteq \mathrm{X}_v \cap \mathrm{X}'_{<\lambda}$, so the inclusion of $\mathrm{X}_v \cap \mathrm{X}'$ into X' factors as

$$\epsilon'_v : \mathrm{X}_v \cap \mathrm{X}' \to \mathrm{X}'_{<\lambda} \xrightarrow{i_\lambda} \mathrm{X}'.$$

Recall that $i_\lambda^*(j_\lambda)_! = 0$, so $(\epsilon'_v)^*(j_\lambda)_! = 0$. Similarly, $(i_\lambda)^!(j_\lambda)_* = 0$, so $(\epsilon'_v)^!(j_\lambda)_* = 0$. Therefore

$$0 \cong e_v \mathrm{H}^*_{\mathrm{T}}((\epsilon'_v)_!(\epsilon'_v)^!(j_\lambda)_*\mathcal{F}_\lambda)_\circ \xrightarrow{\cong} e_v \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_*\mathcal{F}_\lambda)_\circ,$$

and similarly

$$e_v \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}'; (j_\lambda)_!\mathcal{F}_\lambda)_\circ \xrightarrow{\cong} e_v \mathrm{H}^*_{\mathrm{T}}((\epsilon'_v)_*(\epsilon'_v)^*(j_\lambda)_!\mathcal{F}_\lambda)_\circ \cong 0,$$

as desired.

Now suppose $f : \mathrm{H}_\lambda \to \mathrm{H}_{<\lambda}$ is a $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}; k)$-module map. If $\mathrm{H}_{<\lambda}$ is a flat $\pi_{-*}(k^{h\mathrm{T}})$-module, the map $\mathrm{H}_{<\lambda} \to (\mathrm{H}_{<\lambda})_\circ$ is injective; but $e_{<\lambda}$ acts by zero on H_λ and by the identity on $\mathrm{H}_{<\lambda}$, so f must be zero. Similarly for maps $\mathrm{H}_{<\lambda} \to \mathrm{H}_\lambda$ if H_λ is a flat $\pi_{-*}(k^{h\mathrm{T}})$-module. ☐

Proposition 2.3.14. *Stick with the setup of Lemma 2.3.9. Consider the diagram*

$$\begin{tikzcd}
\mathrm{Y} \cap \mathrm{X}'_{<\lambda} \arrow[r, "i'_\lambda"] \arrow[d, "f_{<\lambda}"'] & \mathrm{Y} \cap \mathrm{X}' \arrow[d, "f_{\leq\lambda}"'] & \mathrm{Y}_\lambda \arrow[l, "j'_\lambda"'] \arrow[d, "f_\lambda"] \\
\mathrm{X}'_{<\lambda} \arrow[r, "i_\lambda"'] & \mathrm{X}' & \mathrm{X}_\lambda. \arrow[l, "j_\lambda"']
\end{tikzcd}$$

Note that each square is Cartesian. Let $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X}'; k)$ be $*$-even, and let $\mathcal{G} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{Y} \cap \mathrm{X}'; k)$ be !-even. Then there is a commutative diagram

$$\begin{tikzcd}
0 \arrow[d] & 0 \arrow[d] \\
\mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}_{<\lambda};k)}(i_\lambda^*\mathcal{F}, (f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}) \arrow[r, "h_1"] \arrow[d, "\overline{\alpha}"'] & \mathrm{Hom}^\star_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X};k)}(\mathrm{H}^*_{\mathrm{T}}((i_\lambda)_*i_\lambda^*\mathcal{F}), \mathrm{H}^*_{\mathrm{T}}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G})) \arrow[d, "\alpha"] \\
\mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}';k)}(\mathcal{F}, (f_{\leq\lambda})_*\mathcal{G}) \arrow[r, "h_2"] \arrow[d, "\overline{\beta}"'] & \mathrm{Hom}^\star_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X};k)}(\mathrm{H}^*_{\mathrm{T}}(\mathcal{F}), \mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G})) \arrow[d, "\beta"] \\
\mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}_\lambda;k)}(j_\lambda^*\mathcal{F}, (f_\lambda)_*(j'_\lambda)^*\mathcal{G}) \arrow[r, "h_3"'] \arrow[d] & \mathrm{Hom}^\star_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X};k)}(\mathrm{H}^*_{\mathrm{T}}((j_\lambda)_*j_\lambda^*\mathcal{F}), \mathrm{H}^*_{\mathrm{T}}((j_\lambda \circ f_\lambda)_*(j'_\lambda)^*\mathcal{G})) \\
0 &
\end{tikzcd}$$

where the left column is short exact and the right column is exact on the left and the middle.

Proof. Let $\mathcal{G}' = (f_{\leq\lambda})_*\mathcal{G}$. Then $i_\lambda^!\mathcal{G}' \cong (f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}$, so that

$$\mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}_{<\lambda};k)}(i_\lambda^*\mathcal{F}, (f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}) \cong \mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}_{<\lambda};k)}(i_\lambda^*\mathcal{F}, i_\lambda^!\mathcal{G}') \cong \mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}';k)}((i_\lambda)_*i_\lambda^*\mathcal{F}, \mathcal{G}').$$

Similarly, there is an isomorphism $(f_\lambda)_*(j'_\lambda)^*\mathcal{G} \cong j_\lambda^*\mathcal{G}'$, so that

$$\mathrm{Ext}^\star_{\mathrm{Shv_T(X_\lambda};k)}(j_\lambda^*\mathcal{F}, (f_\lambda)_*(j'_\lambda)^*\mathcal{G}) \cong \mathrm{Ext}^\star_{\mathrm{Shv_T(X_\lambda};k)}(j_\lambda^*\mathcal{F}, j_\lambda^*\mathcal{G}') \cong \mathrm{Ext}^\star_{\mathrm{Shv_T(X'};k)}((j_\lambda)_!j_\lambda^!\mathcal{F}, \mathcal{G}').$$

The cofiber sequence

$$(j_\lambda)_!j_\lambda^!\mathcal{F} \to \mathcal{F} \to (i_\lambda)_*i_\lambda^*\mathcal{F}$$

defines a cofiber sequence

$$\mathrm{Map}_{\mathrm{Shv_T(X'};k)}((i_\lambda)_*i_\lambda^*\mathcal{F}, \mathcal{G}') \to \mathrm{Map}_{\mathrm{Shv_T(X'};k)}(\mathcal{F}, \mathcal{G}') \to \mathrm{Map}_{\mathrm{Shv_T(X'};k)}((j_\lambda)_!j_\lambda^!\mathcal{F}, \mathcal{G}')$$

of $k^{h\mathrm{T}}$-modules. We claim that flanking terms are perfect even $k^{h\mathrm{T}}$-modules; then the boundary map is null, and one obtains the short exact sequence in the left column.

To prove that the flanking terms are perfect even $k^{h\mathrm{T}}$-modules, observe that they can be rewritten alternatively as

$$\mathrm{Map}_{\mathrm{Shv_T(X_{<\lambda}};k)}(i_\lambda^*\mathcal{F}, (f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}) \cong (f_{<\lambda}^*i_\lambda^*\mathcal{F})^\vee \otimes^! (i'_\lambda)^!\mathcal{G} \cong (i'_\lambda)^!(f_{\leq\lambda}^!(\mathcal{F}^\vee) \otimes^! \mathcal{G}),$$

$$\mathrm{Map}_{\mathrm{Shv_T(X_\lambda};k)}(j_\lambda^*\mathcal{F}, (f_\lambda)_*(j'_\lambda)^*\mathcal{G}) \cong (f_\lambda^*j_\lambda^*\mathcal{F})^\vee \otimes^! (j'_\lambda)^!\mathcal{G} \cong (j'_\lambda)^!(f_{\leq\lambda}^!(\mathcal{F}^\vee) \otimes^! \mathcal{G}).$$

Now, observe that $f_{\leq\lambda}^!(\mathcal{F}^\vee) \otimes^! \mathcal{G}$ is !-even: indeed, $\mathcal{F}^\vee$ is !-even, and the !-tensor product of two !-even sheaves remains !-even. It follows from Corollary 2.3.11 that the cohomology of $(i'_\lambda)^!(f_{\leq\lambda}^!(\mathcal{F}^\vee) \otimes^! \mathcal{G})$ and $(j'_\lambda)^!(f_{\leq\lambda}^!(\mathcal{F}^\vee) \otimes^! \mathcal{G})$ are both even and projective over $\pi_{-*}(k^{h\mathrm{T}})$.

Let us now define the maps α and β.

- To define α, let $f : \mathrm{H}^*_\mathrm{T}((i_\lambda)_*i_\lambda^*\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G})$. Observe that
$$(i_\lambda)_*(f_{<\lambda})_*(i'_\lambda)^!\mathcal{G} \cong (i_\lambda)_*i_\lambda^!(f_{\leq\lambda})_*\mathcal{G},$$
so that there is a canonical map to $(f_{\leq\lambda})_*\mathcal{G}$. Then α_F is the composite
$$\mathrm{H}^*_\mathrm{T}(\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}((i_\lambda)_*i_\lambda^*\mathcal{F}) \xrightarrow{f} \mathrm{H}^*_\mathrm{T}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}) \to \mathrm{H}^*_\mathrm{T}((f_{\leq\lambda})_*\mathcal{G}).$$
- To define β, let $f : \mathrm{H}^*_\mathrm{T}(\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}((f_{\leq\lambda})_*\mathcal{G})$. Without loss of generality assume that $\mathrm{Y}_\lambda \neq \emptyset$ (if $\mathrm{Y}_\lambda = \emptyset$, then $\mathrm{H}^*_\mathrm{T}((j_\lambda \circ f_\lambda)_*(j'_\lambda)^*\mathcal{G}) = 0$, so $\beta = 0$). By Lemma 2.3.10, it follows that there is a short exact sequence of $\mathrm{H}^*_\mathrm{T}(\mathrm{X}'; k)$-modules
$$0 \to \mathrm{H}^*_\mathrm{T}((i_\lambda)_!i_\lambda^!\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}(\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}((j_\lambda)_*j_\lambda^*\mathcal{F}) \to 0.$$
Given the map $f : \mathrm{H}^*_\mathrm{T}(\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}((f_{\leq\lambda})_*\mathcal{G})$, notice that the following composite is zero by Corollary 2.3.13:
$$\mathrm{H}^*_\mathrm{T}((i_\lambda)_!i_\lambda^!\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}(\mathcal{F}) \xrightarrow{f} \mathrm{H}^*_\mathrm{T}((f_{\leq\lambda})_*\mathcal{G}) \to \mathrm{H}^*_\mathrm{T}((j_\lambda)_*(j_\lambda)^*(f_{\leq\lambda})_*\mathcal{G}) \cong \mathrm{H}^*_\mathrm{T}((j_\lambda \circ f_\lambda)_*(j'_\lambda)^*\mathcal{G}).$$
One therefore obtains a map
$$\mathrm{H}^*_\mathrm{T}(\mathcal{F})/\mathrm{H}^*_\mathrm{T}((i_\lambda)_!i_\lambda^!\mathcal{F}) \cong \mathrm{H}^*_\mathrm{T}((j_\lambda)_*j_\lambda^*\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}((j_\lambda \circ f_\lambda)_*(j'_\lambda)^*\mathcal{G}).$$
This is our definition of β_F.

To check that α is injective, note that the map $\mathrm{H}^*_\mathrm{T}(\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}((i_\lambda)_*i_\lambda^*\mathcal{F})$ is surjective by Lemma 2.3.9 since $\mathcal{F}$ is $*$-even, and that the map $\mathrm{H}^*_\mathrm{T}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}) \to \mathrm{H}^*_\mathrm{T}((f_{\leq\lambda})_*\mathcal{G})$ is injective by Lemma 2.3.9 since $\mathcal{G}$ is !-even.

We now check that $\mathrm{im}(\alpha) = \ker(\beta)$. First, let us show that $\beta\alpha = 0$. If $f : \mathrm{H}^*_\mathrm{T}((i_\lambda)_*i_\lambda^*\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G})$, then $\beta\alpha(f)$ is induced by the following composite:

$$\mathrm{H}^*_\mathrm{T}(\mathcal{F}) \to \mathrm{H}^*_\mathrm{T}((i_\lambda)_*i_\lambda^*\mathcal{F}) \xrightarrow{f} \mathrm{H}^*_\mathrm{T}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}) \to \mathrm{H}^*_\mathrm{T}((f_{\leq\lambda})_*\mathcal{G}) \to \mathrm{H}^*_\mathrm{T}((j_\lambda \circ f_\lambda)_*(j'_\lambda)^*\mathcal{G}).$$

But the latter two maps compose to zero (by recollement), so the composite is zero, and $\beta\alpha = 0$.

It remains to show that $\ker(\beta) \subseteq \mathrm{im}(\alpha)$. So, suppose $f : \mathrm{H}^*_{\mathrm{T}}(\mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G})$ is such that $\beta_{\mathrm{F}} = 0$, i.e., that in the commutative diagram

$$\begin{tikzcd}
\mathrm{H}^*_{\mathrm{T}}(\mathcal{F}) \arrow[r, two heads] \arrow[d, "f"'] & \mathrm{H}^*_{\mathrm{T}}((j_\lambda)_* j_\lambda^* \mathcal{F}) \arrow[d, "\beta_{\mathrm{F}}"] \\
\mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G}) \arrow[r] & \mathrm{H}^*_{\mathrm{T}}((j_\lambda \circ f_\lambda)_*(j'_\lambda)^*\mathcal{G}),
\end{tikzcd}$$

the left vertical arrow is zero. The natural map $(j_\lambda)_! j_\lambda^* \mathcal{F} \to (j_\lambda)_* j_\lambda^* \mathcal{F}$ factors through $\mathcal{F}$, so the composite map

$$\mathrm{H}^*_{\mathrm{T}}((j_\lambda)_! j_\lambda^* \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathcal{F}) \xrightarrow{f} \mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G}) \to \mathrm{H}^*_{\mathrm{T}}((j_\lambda \circ f_\lambda)_*(j'_\lambda)^*\mathcal{G})$$

is zero. By the first point in Lemma 2.3.9, the kernel of the map $\mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G}) \to \mathrm{H}^*_{\mathrm{T}}((j_\lambda \circ f_\lambda)_*(j'_\lambda)^*\mathcal{G})$ is $\mathrm{H}^*_{\mathrm{T}}((i_\lambda)_! i^!_\lambda (f_{\leq\lambda})_*\mathcal{G}) \cong \mathrm{H}^*_{\mathrm{T}}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G})$. It follows that there is a dotted map making the following diagram commute:

$$\begin{tikzcd}
\mathrm{H}^*_{\mathrm{T}}((j_\lambda)_! j_\lambda^* \mathcal{F}) \arrow[r] \arrow[d, dashed] & \mathrm{H}^*_{\mathrm{T}}(\mathcal{F}) \arrow[d, "f"] \\
\mathrm{H}^*_{\mathrm{T}}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}) \arrow[r] & \mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G}).
\end{tikzcd}$$

But $\mathrm{H}^*_{\mathrm{T}}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G})$ is a flat $\pi_{-*}(k^{h\mathrm{T}})$-module (here, we used that $\mathcal{G}$ is !-even and induction on the stratification), so Corollary 2.3.13 implies that the dotted map is in fact zero. In particular, the composite map

$$\mathrm{H}^*_{\mathrm{T}}((j_\lambda)_! j_\lambda^* \mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathcal{F}) \xrightarrow{f} \mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G})$$

is zero. The second point of Lemma 2.3.9 implies that f factors as a map

$$f : \mathrm{H}^*_{\mathrm{T}}(\mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}(\mathcal{F})/\mathrm{H}^*_{\mathrm{T}}((j_\lambda)_! j_\lambda^* \mathcal{F}) \cong \mathrm{H}^*_{\mathrm{T}}((i_\lambda)_* i_\lambda^* \mathcal{F}) \xrightarrow{f'} \mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G})$$

for some map f'. The composite

$$\mathrm{H}^*_{\mathrm{T}}((i_\lambda)_* i_\lambda^* \mathcal{F}) \xrightarrow{f'} \mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G}) \to \mathrm{H}^*_{\mathrm{T}}((j_\lambda \circ f_\lambda)_*(j'_\lambda)^*\mathcal{G})$$

is also zero by Corollary 2.3.13, because $\mathrm{H}^*_{\mathrm{T}}((j_\lambda \circ f_\lambda)_*(j'_\lambda)^*\mathcal{G})$ is a flat $\pi_{-*}(k^{h\mathrm{T}})$-module. (Here, we use that $j'_\lambda : \mathrm{Y}_\lambda \to \mathrm{Y} \cap \mathrm{X}'$ is an open embedding since $j_\lambda : \mathrm{X}_\lambda \to \mathrm{X}'$ is an open embedding, so that $(j'_\lambda)^* = (j'_\lambda)^!$; since $\mathcal{G}$ is assumed to be !-even, $(j'_\lambda)^*\mathcal{G}$ is perfect even.) By the first point of Lemma 2.3.9, the kernel of the map $\mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G}) \to \mathrm{H}^*_{\mathrm{T}}((j_\lambda \circ f_\lambda)_*(j'_\lambda)^*\mathcal{G})$ is $\mathrm{H}^*_{\mathrm{T}}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G})$; so we find that f' factors as a map

$$f' : \mathrm{H}^*_{\mathrm{T}}((i_\lambda)_* i_\lambda^* \mathcal{F}) \xrightarrow{f''} \mathrm{H}^*_{\mathrm{T}}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}) \to \mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G}).$$

It follows then that f is the composite

$$\mathrm{H}^*_{\mathrm{T}}(\mathcal{F}) \to \mathrm{H}^*_{\mathrm{T}}((i_\lambda)_* i_\lambda^* \mathcal{F}) \xrightarrow{f''} \mathrm{H}^*_{\mathrm{T}}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}) \to \mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G}),$$

but this is exactly $\alpha(f'')$. $\square$

Lemma 2.3.15. *Suppose* A *is an even* $\mathbf{E}_\infty$*-ring and that* $\mathrm{M}, \mathrm{N} \in (\mathrm{Mod}_\mathrm{A})_\mathrm{pev}$ *(i.e., they lie in the smallest subcategory containing all even shifts of* A *and which is closed under extensions and retracts). Then the natural map*

$$\mathrm{Ext}^\star_\mathrm{A}(\mathrm{M}, \mathrm{N}) \xrightarrow{\cong} \mathrm{Hom}^\star_{\pi_*(\mathrm{A})}(\pi_*(\mathrm{M}), \pi_*(\mathrm{N}))$$

is an isomorphism.

Proof. There is a spectral sequence

$$\mathrm{E}_1 = \mathrm{Ext}^\star_{\pi_*(\mathrm{A})}(\pi_*(\mathrm{M}), \pi_*(\mathrm{N})) \Rightarrow \mathrm{Ext}^\star_\mathrm{A}(\mathrm{M}, \mathrm{N}),$$

so to prove the isomorphism, it suffices to show that $\pi_*(\mathrm{M})$ is a projective $\pi_*(\mathrm{A})$-module. Since this is clearly true when M is a shift of A, it suffices to show that the property that $\pi_*(\mathrm{M})$ be a projective $\pi_*(\mathrm{A})$-module is stable under extensions and retracts. This is clear for retracts, since a retract of a projective module remains projective. Observe that if $\mathrm{M}' \to \mathrm{M} \to \mathrm{M}''$ is an extension of A-modules where M' and M'' are perfect even, then the boundary map $\mathrm{M}'' \to \Sigma\mathrm{M}'$ is null. This is because $\mathrm{Hom}_\mathrm{A}(\mathrm{M}'', \mathrm{M}')$ is concentrated in even degrees; to see this, note that the property that $\mathrm{Hom}_\mathrm{A}(\mathrm{M}'', \mathrm{M}')$ is even is closed under retracts, extensions, and even suspensions separately in M'' and M'. Since the boundary map $\mathrm{M}'' \to \Sigma\mathrm{M}'$ is null, there is a short exact sequence

$$0 \to \pi_*(\mathrm{M}') \to \pi_*(\mathrm{M}) \to \pi_*(\mathrm{M}'') \to 0.$$

By induction, we may assume that $\pi_*(\mathrm{M}')$ and $\pi_*(\mathrm{M}'')$ are both projective $\pi_*(\mathrm{A})$-modules; but then $\pi_*(\mathrm{M})$ is also projective, as desired. $\square$

Proof of Theorem 2.3.2. When X has a single stratum, the claim is clear. Indeed, it suffices to show that if V is a complex vector space with T-action and $\mathcal{F}, \mathcal{G} \in \mathrm{Shv}_\mathrm{T}(\mathrm{V}; k)_\mathrm{pev}$ are perfect even, then there is an isomorphism of graded $\mathrm{H}^*_\mathrm{T}(\mathrm{V}; k) \cong \pi_{-*}(k^{h\mathrm{T}})$-modules:

$$\mathrm{Ext}^\star_{\mathrm{Shv}_\mathrm{T}(\mathrm{V};k)}(\mathcal{F}, \mathcal{G}) \xrightarrow{\cong} \mathrm{Hom}^\star_{\mathrm{H}^*_\mathrm{T}(\mathrm{V};k)}(\mathrm{H}^*_\mathrm{T}(\mathrm{V}; \mathcal{F}), \mathrm{H}^*_\mathrm{T}(\mathrm{V}; f_*\mathcal{G})).$$

Any object of $\mathrm{Shv}_\mathrm{T}(\mathrm{V}; k)_\mathrm{pev}$ is locally constant, so $\mathcal{F}, \mathcal{G} \in \mathrm{Loc}_\mathrm{T}(\mathrm{V}; k)$. Since V is contractible, pullback along the canonical map $\mathrm{V} \to *$ defines an equivalence $\mathrm{Fun}(\mathrm{BT}, \mathrm{Mod}_k) \xrightarrow{\simeq} \mathrm{Loc}_\mathrm{T}(\mathrm{V}; k)$. Since k is even, the map $k \otimes_{k^{h\mathrm{T}}} k \to \mathrm{C}^*(\mathrm{T}; k)$ is an isomorphism, so Remark 2.1.8 gives a symmetric monoidal equivalence $\mathrm{Fun}(\mathrm{BT}, \mathrm{Mod}_k) \xrightarrow{\simeq} \mathrm{Mod}^\wedge_{k^{h\mathrm{T}}}$. Under the resulting equivalence $\mathrm{Loc}_\mathrm{T}(\mathrm{V}; k) \simeq \mathrm{Mod}^\wedge_{k^{h\mathrm{T}}}$, $\mathcal{F} \in \mathrm{Loc}_\mathrm{T}(\mathrm{V}; k)$ is perfect even if and only if the corresponding $\mathrm{R}\Gamma_\mathrm{T}(\mathrm{V}; \mathcal{F}) \in \mathrm{Mod}^\wedge_{k^{h\mathrm{T}}}$ lies in the smallest subcategory containing all even shifts of $k^{h\mathrm{T}}$ and which is closed under extensions and retracts. The desired isomorphism is therefore a consequence of Lemma 2.3.15.

If X has more than one stratum, let $\mathrm{X}' = \mathrm{X}_{\leq\lambda} = \mathrm{X}_{<\lambda} \sqcup \mathrm{X}_\lambda$, and let $i : \mathrm{X}' \to \mathrm{X}$ and $i' : \mathrm{Y} \cap \mathrm{X}' \to \mathrm{Y}$ denote the closed embeddings. Let $\mathcal{F}' = i^*\mathcal{F}$, and let $\mathcal{G}' = (i')^!\mathcal{G}$. Then $\mathcal{F} \in \mathrm{Shv}_\mathrm{T}(\mathrm{X}'; k)$ is $*$-even, and the map $\mathrm{H}^*_\mathrm{T}(\mathrm{X}'; \mathcal{F}) \to \mathrm{H}^*_\mathrm{T}(\mathrm{X}'; (j_\lambda)_* j_\lambda^* \mathcal{F})$ is surjective if Y_λ is nonempty by assumption. Also, $\mathcal{G}' \in \mathrm{Shv}_\mathrm{T}(\mathrm{Y} \cap \mathrm{X}'; k)$ is $!$-even.

By Proposition 2.3.14, there is a commutative diagram

$$\begin{tikzcd}
0 \arrow[d] & 0 \arrow[d] \\
\mathrm{Ext}^{\star}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}_{<\lambda};k)}(i_\lambda^*\mathcal{F}', (f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}') \arrow[r, "h_1"] \arrow[d, "\overline{\alpha}"'] & \mathrm{Hom}_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X};k)}(\mathrm{H}^*_{\mathrm{T}}((i_\lambda)_* i_\lambda^*\mathcal{F}'), \mathrm{H}^*_{\mathrm{T}}((i_\lambda \circ f_{<\lambda})_*(i'_\lambda)^!\mathcal{G}')) \arrow[d, "\alpha"] \\
\mathrm{Ext}^{\star}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}';k)}(\mathcal{F}', (f_{\leq\lambda})_*\mathcal{G}') \arrow[r, "h_2"] \arrow[d, "\overline{\beta}"'] & \mathrm{Hom}^{\star}_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X};k)}(\mathrm{H}^*_{\mathrm{T}}(\mathcal{F}'), \mathrm{H}^*_{\mathrm{T}}((f_{\leq\lambda})_*\mathcal{G}')) \arrow[d, "\beta"] \\
\mathrm{Ext}^{\star}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}_{\lambda};k)}(j_\lambda^*\mathcal{F}', (f_{\lambda})_*(j'_\lambda)^*\mathcal{G}') \arrow[r, "h_3"'] \arrow[d] & \mathrm{Hom}_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X};k)}(\mathrm{H}^*_{\mathrm{T}}((j_\lambda)_* j_\lambda^*\mathcal{F}'), \mathrm{H}^*_{\mathrm{T}}((j_\lambda \circ f_{\lambda})_*(j'_\lambda)^*\mathcal{G}')) \\
0 &
\end{tikzcd}$$

where the left column is short exact and the right column is exact on the left and the middle. We need to show that h_2 is an isomorphism. Induction gives that h_1 is an isomorphism, so it suffices to check that h_3 is an isomorphism. When $\mathrm{Y}_\lambda = \emptyset$, clearly the source and target of h_3 are zero, so h_3 is an isomorphism. So, suppose Y_λ is nonempty. Then $j_\lambda^*\mathcal{F}'$ and $(j'_\lambda)^*\mathcal{G}'$ are perfect even, so that

$$\mathrm{Ext}^{\star}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}_{\lambda};k)}(j_\lambda^*\mathcal{F}', (f_{\lambda})_*(j'_\lambda)^*\mathcal{G}') \xrightarrow{\cong} \mathrm{Hom}_{\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}_\lambda;k)}(\mathrm{H}^*_{\mathrm{T}}((j_\lambda)_* j_\lambda^*\mathcal{F}'), \mathrm{H}^*_{\mathrm{T}}((j_\lambda \circ f_{\lambda})_*(j'_\lambda)^*\mathcal{G}')).$$

To show that h_3 is an isomorphism, it suffices to show that Hom in $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X}_\lambda;k)$-modules agrees with Hom in $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X};k)$-modules. But this is true because the map $\mathrm{H}^*_{\mathrm{T}}(\mathrm{X};k) \to \mathrm{H}^*_{\mathrm{T}}(\mathrm{X}_\lambda;k)$ is surjective. □

2.4. **Degenerations.**

Setup 2.4.1. Let T be a torus, and let X be a (connected) locally compact Hausdorff topological space equipped with a T-action. Fix an even $\mathbf{E}_\infty$-ring k. Recall from Remark 2.1.6 that homotopy fixed points defines a symmetric monoidal equivalence $\mathrm{Fun}(\mathrm{BT}, \mathrm{Mod}_k) \xrightarrow{\simeq} \mathrm{Mod}^{\hat{k}}_{k^{h\mathrm{T}}}$. If A is an $\mathbf{E}_1$-algebra in $\mathrm{Fun}(\mathrm{BT}, \mathrm{Mod}_k)$, we will write $\mathrm{RMod}^{\hat{k}}_{\mathrm{A}^{h\mathrm{T}}}$ to denote $\mathrm{RMod}_{\mathrm{A}^{h\mathrm{T}}}(\mathrm{Mod}^{\hat{k}}_{k^{h\mathrm{T}}})$.

Construction 2.4.2. Let $q : \mathrm{X} \to *$ denote the canonical map to a point. Fix an object $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ such that the functor $\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}, -) : \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k) \to \mathrm{Mod}^{h\mathrm{T}}_k$ preserves filtered colimits; we will say that $\mathcal{F}$ is $\mathrm{Mod}^{h\mathrm{T}}_k$-*compact*. By [Lur16, Theorem 7.1.2.1], the $\mathrm{Mod}^{h\mathrm{T}}_k$-linear localizing subcategory $\langle\mathcal{F}\rangle$ of $\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ generated by $\mathcal{F}$ is equivalent to the $k^{h\mathrm{T}}$-linear category $\mathrm{RMod}^{\hat{k}}_{\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F})}$. This category admits a filtered lift (in the sense of Remark A.1.2) given by $\mathrm{RMod}^{\mathrm{fil},\hat{k}}_{\tau_{\geq\star}\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F})}$, which will be denoted by $\langle\mathcal{F}\rangle_{\mathrm{fil}}$. It is a $\tau_{\geq\star}(k^{h\mathrm{T}})$-linear category. The associated graded of $\langle\mathcal{F}\rangle_{\mathrm{fil}}$ (in the sense of Remark A.1.3) is given by $\mathrm{RMod}^{\mathrm{gr},\hat{k}}_{\mathrm{gr}(\tau_{\geq\star}\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}))[-\star]}$. This category will be denoted by $\langle\mathcal{F}\rangle_{\mathrm{gr}}$. It is a $\pi_*(k^{h\mathrm{T}})$-linear category, and there is a 1-parameter degeneration

$$\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k) \supseteq \langle\mathcal{F}\rangle \rightsquigarrow \langle\mathcal{F}\rangle_{\mathrm{gr}}.$$

Remark 2.4.3. Construction 2.4.2 is rather *ad hoc*; a more "principled" approach, following [Pst23a, Pst23b], will be given in future work. There, we will also prove a "relative" version of the comparison between synthetic spectra and cellular motives over $\mathrm{Spec}(\mathbf{C})$ from [Pst23a, GIKR18]. As we will explain in future work, Construction 2.4.2 is only well-behaved if $\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F})$ is concentrated in even degrees.

Example 2.4.4. Suppose that the constant sheaf $\underline{k}_{\mathrm{X}}$ is $\mathrm{Mod}_k^{h\mathrm{T}}$-compact (e.g., X is proper). The full subcategory $\langle \underline{k}_{\mathrm{X}} \rangle \subseteq \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ is equivalent to the category $\mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k)$ of locally constant sheaves. Under the equivalence $\mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k) \simeq \mathrm{RMod}_{\mathrm{R}\Gamma(\mathrm{X};k)}(\mathrm{Fun}(\mathrm{BT}, \mathrm{Mod}_k))$, the global sections functor $\mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k) \to \mathrm{Loc}_{\mathrm{T}}(*;k)$ is given by restriction of scalars along the unit map $k \to \mathrm{R}\Gamma(\mathrm{X};k)$. Note that $\langle \underline{k}_{\mathrm{X}} \rangle_{\mathrm{fil}} = \mathrm{RMod}^{\mathrm{fil},\hat{k}}_{\tau_{\geq\star}\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};k)}$. This category will also be denoted $\mathrm{Loc}^{\mathrm{fil}}_{\mathrm{T}}(\mathrm{X};k)$. Since each stratum X_λ is an affine space and k is assumed to be even, $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}_\lambda;k)$ is even. It follows by induction on strata that $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};k)$ is also even. In particular, $\langle \underline{k}_{\mathrm{X}} \rangle_{\mathrm{gr}} = \mathrm{RMod}^{\mathrm{gr},\hat{k}}_{\mathrm{H}_{\mathrm{T}}^{-\star}(\mathrm{X};k)}$. This category will also be denoted $\mathrm{Loc}^{\mathrm{gr}}_{\mathrm{T}}(\mathrm{X};k)$.

Remark 2.4.5. There is a functor $\eta : \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k) \to \mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k)$ given by $\underline{\mathrm{Hom}}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\underline{k}_{\mathrm{X}}, -)$; it is right adjoint to the fully faithful inclusion $\mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k) \hookrightarrow \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$. The global sections functor $\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k) \to \mathrm{Shv}_{\mathrm{T}}(*;k)$ factors through η.

If $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ is $\mathrm{Mod}_k^{h\mathrm{T}}$-compact, there is a $k^{h\mathrm{T}}$-linear functor

$$\langle \mathcal{F} \rangle \hookrightarrow \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k) \xrightarrow{\eta} \mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k). \tag{2.4.1}$$

Under the identification $\langle \mathcal{F} \rangle \simeq \mathrm{RMod}_{\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F})}$, this functor is given by the $\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F})$-$\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};k)$-bimodule $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};\mathcal{F}) = \mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\underline{k}_{\mathrm{X}}, \mathcal{F})$. Since $\tau_{\geq\star}$ is a lax symmetric monoidal functor, $\tau_{\geq\star}\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};\mathcal{F})$ acquires the structure of a filtered $\tau_{\geq\star}\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F})$-$\tau_{\geq\star}\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};k)$-bimodule. The functor (2.4.1) therefore refines to a $\tau_{\geq\star}(k^{h\mathrm{T}})$-linear functor $\eta^{\mathrm{fil},\hat{k}} : \langle \mathcal{F} \rangle_{\mathrm{fil}} \to \mathrm{Loc}^{\mathrm{fil}}_{\mathrm{T}}(\mathrm{X};k)$, and hence defines a $\pi_*(k^{h\mathrm{T}})$-linear functor $\eta^{\mathrm{gr}} : \langle \mathcal{F} \rangle_{\mathrm{gr}} \to \mathrm{Loc}^{\mathrm{gr}}_{\mathrm{T}}(\mathrm{X};k)$.

More generally, suppose $f : \mathrm{X} \to \mathrm{Y}$ is a continuous T-equivariant map of locally compact Hausdorff spaces, and let $\mathcal{F}_{\mathrm{X}} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ and $\mathcal{F}_{\mathrm{Y}} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ be two $\mathrm{Mod}_k^{h\mathrm{T}}$-compact objects. Then there is a functor

$$\langle \mathcal{F}_{\mathrm{X}} \rangle \hookrightarrow \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k) \xrightarrow{f_*} \mathrm{Shv}_{\mathrm{T}}(\mathrm{Y};k) \to \langle \mathcal{F}_{\mathrm{Y}} \rangle,$$

where the final functor is the right adjoint of the fully faithful functor $\langle \mathcal{F}_{\mathrm{Y}} \rangle \to \mathrm{Shv}_{\mathrm{T}}(\mathrm{Y};k)$. Just as above, $\tau_{\geq\star}\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{Y};k)}(\mathcal{F}_{\mathrm{Y}}, f_*\mathcal{F}_{\mathrm{X}})$ acquires the structure of a filtered $\tau_{\geq\star}\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}_{\mathrm{X}})$-$\tau_{\geq\star}\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{Y};k)}(\mathcal{F}_{\mathrm{X}})$-bimodule. The above functor therefore refines to a $\tau_{\geq\star}(k^{h\mathrm{T}})$-linear functor $\eta^{\mathrm{fil},\hat{k}} : \langle \mathcal{F}_{\mathrm{X}} \rangle_{\mathrm{fil}} \to \langle \mathcal{F}_{\mathrm{Y}} \rangle_{\mathrm{fil}}$, and hence defines a $\pi_*(k^{h\mathrm{T}})$-linear functor $\eta^{\mathrm{gr}} : \langle \mathcal{F}_{\mathrm{X}} \rangle_{\mathrm{gr}} \to \langle \mathcal{F}_{\mathrm{X}} \rangle_{\mathrm{gr}}$.

Remark 2.4.6. The equivalence $\langle \mathcal{F} \rangle \simeq \mathrm{RMod}_{\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F})}$ sends an object $\mathcal{G} \in \langle \mathcal{F} \rangle$ to $\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}, \mathcal{G})$. Since $\tau_{\geq\star}$ is a lax symmetric monoidal functor, it follows that $\tau_{\geq\star}\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}, \mathcal{G})$ acquires the structure of a filtered right $\tau_{\geq\star}\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F})$-module. In particular, there is a functor

$$\underline{\tau}_{\geq\star} : \langle \mathcal{F} \rangle \to \langle \mathcal{F} \rangle_{\mathrm{fil}},\ \mathcal{G} \mapsto \tau_{\geq\star}\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}, \mathcal{G}).$$

Composing with the functor $\langle \mathcal{F}\rangle_{\mathrm{fil}} \to \langle \mathcal{F}\rangle_{\mathrm{gr}} \simeq \mathrm{RMod}^{\mathrm{gr},\hat{k}}_{\mathrm{gr}(\tau_{\geq\star}\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}))[-\star]}$, we obtain a functor

$$\underline{\tau}_\star : \langle \mathcal{F}\rangle \to \langle \mathcal{F}\rangle_{\mathrm{gr}},\ \mathcal{G} \mapsto \mathrm{gr}(\tau_{\geq\star}\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F},\mathcal{G}))[-\star].$$

For example, if $\mathcal{F}$ is the constant sheaf as in Example 2.4.4, then $\underline{\tau}_{\geq\star}$ is the functor $\mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k) \to \mathrm{RMod}^{\mathrm{fil},\hat{k}}_{\tau_{\geq\star}\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};k)}$ sending $\mathcal{G} \mapsto \tau_{\geq\star}\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};\mathcal{G})$.

Setup 2.4.7. Assume that X has a finite stratification indexed by Λ, where each stratum $j_\lambda : \mathrm{X}_\lambda \hookrightarrow \mathrm{X}$ is locally closed, smooth, and T-stable. Assume that for each $\lambda \in \Lambda$, the stratum X_λ is T-equivariantly isomorphic to a complex vector space with a linear T-action whose T-invariant set $\mathrm{X}_\lambda^{\mathrm{T}}$ is just the origin 0_λ of the vector space X_λ.

Proposition 2.4.8. *Assume Setup 2.4.7, and suppose that $\mathcal{F}$ is even. The graded $\pi_*(k^{h\mathrm{T}})$-linear category $\langle \mathcal{F}\rangle_{\mathrm{gr}}$ is equivalent to the category of graded right $\mathrm{End}^\star_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}(\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}))$-module spectra, where we remind that $\mathrm{End}^\star_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}(\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}))$ denotes endomorphisms in the* abelian *category $\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}$. Under this equivalence, the $\pi_*(k^{h\mathrm{T}})$-linear functor $\eta^{\mathrm{gr}} : \langle \mathcal{F}\rangle_{\mathrm{gr}} \to \mathrm{Loc}^{\mathrm{gr}}_{\mathrm{T}}(\mathrm{X};k)$ identifies with the functor on derived categories induced by the functor $\mathrm{RMod}^{\mathrm{gr},\heartsuit}_{\mathrm{End}^\star_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}(\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}))} \to \mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}$ given by the bimodule $\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F})$. Moreover, the functor $\underline{\tau}_\star : \langle \mathcal{F}\rangle \to \langle \mathcal{F}\rangle_{\mathrm{gr}}$ sends*

$$\langle \mathcal{F}\rangle \ni \mathcal{G} \mapsto \mathrm{Hom}^\star_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}(\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}),\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{G})) \in \mathrm{RMod}^{\mathrm{gr}}_{\mathrm{End}^\star_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}(\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}))}$$

if $\mathcal{G}$ is !-even.

Proof. Since each stratum X_λ is an affine space and k is assumed to be even, $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}_\lambda;k)$ is even. It follows that $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}_\lambda;\mathcal{G})$ is even for any perfect even $\mathcal{G} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X}_\lambda;k)_{\mathrm{pev}}$. Induction on strata implies that $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}_\lambda;\mathcal{F})$ is also even for any even $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$. In particular, $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X};k)$ is even. Theorem 2.3.2 says that there is an isomorphism of graded $\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)$-algebras $\pi_\star\,\mathrm{REnd}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}) \xrightarrow{\cong} \mathrm{End}^\star_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}(\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}))$. Since $\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)$ and $\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F})$ are concentrated in even weights, the right-hand side is also concentrated in even weights.

Recall that $\eta^{\mathrm{gr}} : \langle \mathcal{F}\rangle_{\mathrm{gr}} \to \mathrm{Loc}^{\mathrm{gr}}_{\mathrm{T}}(\mathrm{X};k) \simeq \mathrm{Mod}^{\mathrm{gr},\hat{k}}_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}$ is given by the $\mathrm{End}^\star_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}(\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}))$-$\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)$-bimodule $\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F})$. It follows that η^{gr} is isomorphic to the functor on derived categories induced by $\langle \mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F})\rangle^\heartsuit \hookrightarrow \mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}$.

Finally, recall that $\underline{\tau}_\star$ sends $\mathcal{G} \mapsto \mathrm{gr}(\tau_{\geq\star}\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F},\mathcal{G}))[-\star]$. Theorem 2.3.2 gives an isomorphism

$$\mathrm{gr}(\tau_{\geq\star}\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F},\mathcal{G}))[-\star] \cong \mathrm{Hom}^\star_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}(\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}),\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{G}))$$

if $\mathcal{G}$ is !-even, as desired. □

Remark 2.4.9. The above discussion can be extended to the case when X is ind-compact Hausdorff. Suppose $\mathrm{X} = \mathrm{colim}_n\,\mathrm{X}^n$ is an ind-good T-space, where each X^n is compact Hausdorff (viewed as a good T-space via the constant limit diagram) and satisfies Setup 2.4.7. Let $i^n : \mathrm{X}^n \hookrightarrow \mathrm{X}$ denote the closed embedding of the n-stratum. Let $\mathcal{F} \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$. Suppose that $\mathcal{F} = \mathrm{colim}_n\,\mathcal{F}^n$ where $\mathcal{F}^n$ is supported on X^n, and that the transition maps $\mathcal{F}^m \to \mathcal{F}^n$ admit retracts. Then there is a fully faithful functor $\langle \mathcal{F}^m \rangle \to \langle \mathcal{F}^n \rangle$, and we set $\langle \mathcal{F} \rangle = \mathrm{colim}_n \langle \mathcal{F}^n \rangle$. Similarly, set $\langle \mathcal{F} \rangle_{\mathrm{fil}} = \mathrm{colim}_n \langle \mathcal{F}^n \rangle_{\mathrm{fil}}$, and analogously for $\langle \mathcal{F} \rangle_{\mathrm{gr}}$.

Suppose that each $\mathcal{F}^n$ is even. Then Remark 2.3.6 implies that $\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}^n)$ is concentrated in even degrees, and there is an isomorphism

$$\pi_\star \mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)}(\mathcal{F}^n) \xrightarrow{\cong} \mathrm{End}^\star_{\mathrm{coMod}_{\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathrm{X};k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_*(k^{h\mathrm{T}})})}(\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}^n)).$$

For notational simplicity, let us denote this algebra by $\mathcal{A}_n$. The map $\mathcal{F}^m \to \mathcal{F}^n$ induces a map $\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}^m) \to \mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}^n)$. Since $\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}^m)$ is a retract of $\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}^n)$, there is a fully faithful functor $\langle \mathcal{F}^m \rangle_{\mathrm{gr}} \to \langle \mathcal{F}^n \rangle_{\mathrm{gr}}$. In particular, taking the colimit, we find that

$$\langle \mathcal{F} \rangle_{\mathrm{gr}} \simeq \mathrm{colim}_n \, \mathrm{RMod}^{\mathrm{gr}}_{\mathcal{A}_n}. \tag{2.4.2}$$

Remark 2.4.10. If $\mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k) = \mathrm{colim}_n \mathrm{Loc}_{\mathrm{T}}(\mathrm{X}^n;k)$, where the colimit is taken along pushforwards (i.e., along right Kan extension along the maps $\mathrm{X}^m/\mathrm{T} \to \mathrm{X}^n/\mathrm{T}$), then there is a functor $\eta : \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k) \to \mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k)$. Since $\mathrm{Loc}_{\mathrm{T}}(\mathrm{X}^n;k) \simeq \mathrm{RMod}^{\hat{k}}_{\mathrm{R\Gamma}_{\mathrm{T}}(\mathrm{X}^n;k)}$, and $\mathrm{R\Gamma}_{\mathrm{T}}(\mathrm{X}^n;k)$ is a perfect $k^{h\mathrm{T}}$-module whose k-linear dual is $\mathrm{R\Gamma}_{\mathrm{T}}(\mathrm{X}^n;\omega_{\mathrm{X}^n})$ by Remark 2.1.8, there is an equivalence

$$\mathrm{Loc}_{\mathrm{T}}(\mathrm{X}^n;k) \simeq \mathrm{coMod}_{\mathrm{R\Gamma}_{\mathrm{T}}(\mathrm{X}^n;\omega_{\mathrm{X}^n})}(\mathrm{Mod}^{\hat{k}}_{k^{h\mathrm{T}}}).$$

It follows that there is an equivalence

$$\mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k) \simeq \mathrm{coMod}_{\mathrm{R\Gamma}_{\mathrm{T}}(\mathrm{X};\omega^{\mathrm{ren}}_{\mathrm{X}})}(\mathrm{Mod}^{\hat{k}}_{k^{h\mathrm{T}}}).$$

There is a filtered refinement of $\mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k)$, given by

$$\mathrm{Loc}^{\mathrm{fil}}_{\mathrm{T}}(\mathrm{X};k) \simeq \mathrm{coMod}_{\tau_{\geq\star}\mathrm{R\Gamma}_{\mathrm{T}}(\mathrm{X};\omega^{\mathrm{ren}}_{\mathrm{X}})}(\mathrm{Mod}^{\mathrm{fil},\hat{k}}_{\tau_{\geq\star}(k^{h\mathrm{T}})}).$$

Recall that our assumptions on X guarantee that $\tau_{\geq\star}\mathrm{R\Gamma}_{\mathrm{T}}(\mathrm{X};\omega^{\mathrm{ren}}_{\mathrm{X}})$ is a filtered $\mathbf{E}_\infty$-$\tau_{\geq\star}(k^{h\mathrm{T}})$-coalgebra, so $\mathrm{Loc}^{\mathrm{fil}}_{\mathrm{T}}(\mathrm{X};k)$ is indeed well-defined. In particular,

$$\mathrm{Loc}^{\mathrm{gr}}_{\mathrm{T}}(\mathrm{X};k) \simeq \mathrm{coMod}_{\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathrm{X};k)}(\mathrm{Mod}^{\mathrm{gr},\hat{k}}_{\pi_*(k^{h\mathrm{T}})}).$$

As with (2.4.1), there is a functor

$$\langle \mathcal{F} \rangle \hookrightarrow \mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k) \xrightarrow{\eta} \mathrm{Loc}_{\mathrm{T}}(\mathrm{X};k).$$

It admits a filtered refinement, and hence defines a functor $\eta^{\mathrm{fil}} : \langle \mathcal{F} \rangle \to \mathrm{Loc}^{\mathrm{fil}}_{\mathrm{T}}(\mathrm{X};k)$. Under the equivalence (2.4.2), the $\pi_*(k^{h\mathrm{T}})$-linear functor $\eta^{\mathrm{gr}} : \langle \mathcal{F} \rangle_{\mathrm{gr}} \to \mathrm{Loc}^{\mathrm{gr}}_{\mathrm{T}}(\mathrm{X};k)$ identifies with the functor on derived categories induced by the functor

$$\mathrm{colim}_n \, \mathrm{RMod}^{\mathrm{gr},\heartsuit}_{\mathcal{A}_n} \to \mathrm{coMod}_{\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathrm{X};k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_*(k^{h\mathrm{T}})})$$

coming from the bimodules $\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}^n)$. Moreover, there is a functor $\underline{\tau}_\star : \langle \mathcal{F} \rangle \to \langle \mathcal{F} \rangle_{\mathrm{gr}}$ like in Remark 2.4.6, which sends a !-even $\mathcal{G} \in \langle \mathcal{F} \rangle$ to $\mathrm{Hom}^\star_{\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};k)}(\mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{F}), \mathrm{H}_{\mathrm{T}}^{-*}(\mathrm{X};\mathcal{G})) \in \mathrm{colim}_n \, \mathrm{RMod}^{\mathrm{gr},\heartsuit}_{\mathcal{A}_n}$.

Construction 2.4.11. Let G be a complex reductive group, let $\mathrm{B} \subseteq \mathrm{G}$ be a Borel subgroup with unipotent radical U, and let T be a maximal torus contained in $\mathrm{B} \subseteq \mathrm{G}$. Write $\mathcal{B} = \mathrm{G}/\mathrm{B}$ denote the flag variety. Recall the filtration $\mathrm{fil}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}})$ for $\mathrm{M} \in \mathrm{Mod}^{h\mathrm{G}}_k$ from Definition 2.2.28. Suppose that X is a (connected) locally compact Hausdorff topological space equipped with a G-action, such that the underlying T-space satisfies Setup 2.4.7. One can then define an analogue of $\mathrm{Shv}^{\mathrm{fil}}_{\mathrm{T}}(\mathrm{X};k)$ for $\mathrm{Shv}_{\mathrm{G}}(\mathrm{X};k)$ as follows.

Fix an object $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}}(\mathrm{X}; k)$, and let $q_n : \mathrm{X} \times \mathcal{B}^n \to \mathrm{X}$ denote the projection. For $n \geq 1$, let $\mathcal{F}_n$ denote the object $q_n^* \mathcal{F}$ in $\mathrm{Shv}_{\mathrm{G}}(\mathrm{X} \times \mathcal{B}^n; k) \simeq \mathrm{Shv}_{\mathrm{B}}(\mathrm{X} \times \mathcal{B}^{n-1}; k)$. Then the categories $\langle \mathcal{F}_n \rangle$, which are equivalent to $\mathrm{RMod}^{\hat{k}}_{\mathrm{End}_{\mathrm{Shv}_{\mathrm{B}}(\mathrm{X} \times \mathcal{B}^{n-1}; k)}(\mathcal{F}_n)}$, assemble into a cosimplicial category. Define $\langle \mathcal{F} \rangle_{\mathrm{fil}} = \mathrm{Tot}(\langle \mathcal{F}_\bullet \rangle_{\mathrm{fil}})$. Since $\langle \mathcal{F}_\bullet \rangle_{\mathrm{fil}}$ is a filtered $\tau_{\geq \star} \mathrm{C}_{\mathrm{T}}(\mathcal{B}^\bullet; k)$-linear category, $\langle \mathcal{F} \rangle_{\mathrm{fil}}$ is a $\mathrm{fil}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$-linear category. The underlying $k^{h\mathrm{G}}$-linear category of $\langle \mathcal{F} \rangle_{\mathrm{fil}}$ is $\langle \mathcal{F} \rangle := \mathrm{Tot}(\langle \mathcal{F}_\bullet \rangle)$, and also the associated graded $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$-linear category $\langle \mathcal{F} \rangle_{\mathrm{gr}}$ is $\mathrm{Tot}(\langle \mathcal{F}_\bullet \rangle_{\mathrm{gr}})$.

Note that $\mathcal{F}_2 = q_1^*(\mathcal{F}_1)$, so that

$$\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X} \times \mathcal{B}; k)}(\mathcal{F}_2) \cong \mathrm{Map}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}; k)}(\mathcal{F}_1, (q_1)_* q_1^*(\mathcal{F}_1)) \cong \mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}; k)}(\mathcal{F}_1) \otimes_{k^{h\mathrm{T}}} \mathrm{C}^*_{\mathrm{T}}(\mathcal{B}; k).$$

Since $\mathrm{C}^*_{\mathrm{T}}(\mathcal{B}; k)$ is a direct sum of even shifts of $k^{h\mathrm{T}}$, it follows that

$$\pi_* \mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X} \times \mathcal{B}; k)}(\mathcal{F}_2) \cong \pi_* \mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X}; k)}(\mathcal{F}_1) \otimes_{\pi_*(k^{h\mathrm{T}})} \mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{B}; k).$$

Unwinding the definition of $\langle \mathcal{F} \rangle_{\mathrm{gr}}$, it follows that $\langle \mathcal{F} \rangle_{\mathrm{gr}}$ is equivalent to the category of $\mathcal{W}$-equivariant objects in the $\pi_*(k^{h\mathrm{T}})$-linear category $\langle \mathcal{F}_1 \rangle_{\mathrm{gr}}$, where $\mathcal{F}_1 = q_1^*(\mathcal{F}) \in \mathrm{Shv}_{\mathrm{T}}(\mathrm{X}; k)$.

The equivalence $\langle \mathcal{F} \rangle \simeq \mathrm{Tot}(\mathrm{RMod}_{\mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X} \times \mathcal{B}^\bullet; k)}(q_\bullet^* \mathcal{F})})$ sends

$$\langle \mathcal{F} \rangle \in \mathcal{G} \mapsto \mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X} \times \mathcal{B}^\bullet; k)}(q_\bullet^* \mathcal{F}, q_\bullet^* \mathcal{G}).$$

Since $\tau_{\geq \star}$ is a lax symmetric monoidal functor, it follows that $\tau_{\geq \star} \mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X} \times \mathcal{B}^\bullet; k)}(q_\bullet^* \mathcal{F}, q_\bullet^* \mathcal{G})$ acquires the structure of a filtered right $\tau_{\geq \star} \mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X} \times \mathcal{B}^\bullet; k)}(q_\bullet^* \mathcal{F})$-module. In particular, there is a functor

$$\underline{\tau}_{\geq \star} : \langle \mathcal{F} \rangle \to \langle \mathcal{F} \rangle_{\mathrm{fil}}, \ \mathcal{G} \mapsto \mathrm{Tot}(\tau_{\geq \star} \mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X} \times \mathcal{B}^\bullet; k)}(q_\bullet^* \mathcal{F}, q_\bullet^* \mathcal{G})).$$

Composing with the functor $\langle \mathcal{F} \rangle_{\mathrm{fil}} \to \langle \mathcal{F} \rangle_{\mathrm{gr}}$, we obtain a functor

$$\underline{\tau}_{\star} : \langle \mathcal{F} \rangle \to \langle \mathcal{F} \rangle_{\mathrm{gr}}, \ \mathcal{G} \mapsto \mathrm{Tot}(\mathrm{gr}(\tau_{\geq \star} \mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathrm{X} \times \mathcal{B}^\bullet; k)}(q_\bullet^* \mathcal{F}, q_\bullet^* \mathcal{G}))[-\star]).$$

Remark 2.4.12. Construction 2.4.11 roughly amounts to replacing $\mathrm{Shv}_{\mathrm{G}}(\mathrm{X}; k)$ by $\mathrm{Tot}(\mathrm{Shv}_{\mathrm{G}}(\mathrm{X} \times \mathcal{B}^\bullet; k))$, i.e., by $\mathrm{Tot}(\mathrm{Shv}_{\mathrm{T}}(\mathrm{X} \times \mathcal{B}^\bullet; k))$. Already when X is a point, this amounts to replacing $\mathrm{Mod}_k^{h\mathrm{G}}$ by the totalization of $\mathrm{Mod}^{h\mathrm{T}}_{\mathrm{C}^*(\mathcal{B}^\bullet; k)} \simeq \mathrm{Mod}^{\hat{k}}_{\mathrm{C}^*_{\mathrm{T}}(\mathcal{B}^\bullet; k)}$. The natural functor $\mathrm{Shv}_{\mathrm{G}}(\mathrm{X}; k) \to \mathrm{Tot}(\mathrm{Shv}_{\mathrm{T}}(\mathrm{X} \times \mathcal{B}^\bullet; k))$ may not be an equivalence; to emphasize this, we will write $\mathrm{Shv}^{\mathrm{toral}}_{\mathrm{G}}(\mathrm{X}; k)$ to denote $\mathrm{Tot}(\mathrm{Shv}_{\mathrm{T}}(\mathrm{X} \times \mathcal{B}^\bullet; k))$. If $\mathcal{G}$ is a topological group acting continuously on X and $\mathcal{G} \to \mathrm{G}$ is a surjective homomorphism with contractible kernel K, so that $\mathrm{Shv}_{\mathcal{G}}(\mathrm{X}; k)$ is equivalent to the full subcategory of $\mathrm{Shv}_{\mathrm{G}}(\mathrm{X}; k)$ of sheaves which are locally constant along K-orbits, then we will write $\mathrm{Shv}^{\mathrm{toral}}_{\mathcal{G}}(\mathrm{X}; k)$ to denote $\mathrm{Tot}(\mathrm{Shv}_{\mathcal{G} \times_{\mathrm{G}} \mathrm{T}}(\mathrm{X} \times \mathcal{B}^\bullet; k))$.

Example 2.4.13. The full subcategory $\langle \underline{k}_{\mathrm{X}} \rangle \subseteq \mathrm{Shv}_{\mathrm{G}}(\mathrm{X}; k)$ compactly generated by $\underline{k}_{\mathrm{X}}$ is equivalent to the category $\mathrm{Loc}_{\mathrm{G}}(\mathrm{X}; k)$ of locally constant sheaves. Using Example 2.4.4, it follows that $\langle \underline{k}_{\mathrm{X}} \rangle_{\mathrm{gr}} = (\mathrm{RMod}^{\mathrm{gr}, \hat{k}}_{\mathrm{H}^{-\star}_{\mathrm{T}}(\mathrm{X}; k)})^{h\mathcal{W}}$. Note that $\mathrm{H}^{-\star}_{\mathrm{T}}(\mathrm{X}; k)^{h\mathcal{W}} = \mathrm{gr}^\star_{\mathrm{G}}(\mathrm{C}^*_{\mathrm{G}}(\mathrm{X}; k))$ in the notation of Definition 2.2.28. The category $\langle \underline{k}_{\mathrm{X}} \rangle_{\mathrm{gr}}$ is in turn equivalent to $\mathrm{RMod}^{\mathrm{gr}, \hat{k}}_{\mathrm{gr}^\star_{\mathrm{G}}(\mathrm{C}^*_{\mathrm{G}}(\mathrm{X}; k))}$. This category will also be denoted $\mathrm{Loc}^{\mathrm{gr}}_{\mathrm{G}}(\mathrm{X}; k)$. Just as in Remark 2.4.5, there is also a $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$-linear functor $\eta^{\mathrm{gr}} : \langle \mathcal{F} \rangle_{\mathrm{gr}} \to \mathrm{Loc}^{\mathrm{gr}}_{\mathrm{G}}(\mathrm{X}; k)$.

Proposition 2.4.8 has an evident G-equivariant analogue:

Proposition 2.4.14. *Assume we are in the setup of Construction 2.4.11, and suppose that the pullback of $\mathcal{F}$ to $\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ is even. Then the graded $\mathrm{gr}^{\star}_{\text{G-ev}}(k^{h\mathrm{G}})$-linear category $\langle\mathcal{F}\rangle_{\mathrm{gr}}$ is equivalent to the category $(\mathrm{RMod}^{\mathrm{gr}}_{\mathrm{End}^{\star}_{\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};k)}(\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F}))})^{h\mathcal{W}}$ of $\mathcal{W}$-equivariant objects, where $\mathcal{W}$ acts on $\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F})$ as in Definition 2.2.28. Under this equivalence, the $\mathrm{gr}^{\star}_{\text{G-ev}}(k^{h\mathrm{G}})$-linear functor $\eta^{\mathrm{gr}} : \langle\mathcal{F}\rangle_{\mathrm{gr}} \to \mathrm{Loc}^{\mathrm{gr}}_{\mathrm{G}}(\mathrm{X};k)$ identifies with the functor on $\mathcal{W}$-equivariant derived categories induced by the functor $\mathrm{RMod}^{\mathrm{gr},\heartsuit}_{\mathrm{End}^{\star}_{\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};k)}(\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F}))} \hookrightarrow \mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};k)}$ given by the bimodule $\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F})$. Moreover, the functor $\underline{\tau}_{\star} : \langle\mathcal{F}\rangle \to \langle\mathcal{F}\rangle_{\mathrm{gr}}$ sends an object $\mathcal{G} \in \langle\mathcal{F}\rangle$ whose pullback to $\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ is !-even to $\mathrm{Hom}^{\star}_{\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};k)}(\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F}), \mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{G}))$ in $\mathrm{RMod}^{\mathrm{gr},\heartsuit}_{\mathrm{End}^{\star}_{\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};k)}(\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F}))}$ equipped with its $\mathcal{W}$-action from Definition 2.2.28.*

Just as in Remark 2.4.9, Proposition 2.4.14 extends to the case when X is ind-compact Hausdorff; namely, if $\mathrm{X} = \mathrm{colim}_n \mathrm{X}^n$ is a colimit of compact Hausdorff G-spaces whose underlying T-spaces satisfy Setup 2.4.7, and $\mathcal{F} = \mathrm{colim}_n \mathcal{F}^n$ where the pullback of $\mathcal{F}^n$ to $\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ is supported on X^n, and the transition maps $\mathcal{F}^m \to \mathcal{F}^n$ exhibit $\mathcal{F}^m$ as a retract of $\mathcal{F}^n$, then there is an equivalence

$$\langle\mathcal{F}\rangle_{\mathrm{gr}} := \mathrm{colim}_n \langle\mathcal{F}^n\rangle_{\mathrm{gr}} \simeq \mathrm{colim}_n (\mathrm{RMod}^{\mathrm{gr},\heartsuit}_{\mathcal{A}_n})^{h\mathcal{W}},$$

where $\mathcal{A}_n$ is the algebra of endomorphisms of $\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F}^n)$ in $\mathrm{coMod}_{\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathrm{X};k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_*(k^{h\mathrm{T}})})$.

Finally, let us describe the analogue of the above story when k is not assumed to be even.

Definition 2.4.15. Let k be an $\mathbf{E}_\infty$-ring. Let G be a connected complex reductive group, let $\mathrm{B} \subseteq \mathrm{G}$ be a Borel subgroup with unipotent radical U, and let T be a maximal torus contained in $\mathrm{B} \subseteq \mathrm{G}$. Write $\mathcal{B} = \mathrm{G}/\mathrm{B}$ denote the flag variety. Suppose that X is a (connected) locally compact Hausdorff topological space equipped with a G-action, such that the underlying T-space satisfies Setup 2.4.7. Fix an object $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}}(\mathrm{X};k)$. If A is any $\mathbf{E}_\infty$-k-algebra, let $\mathcal{F}_{\mathrm{A}} = \mathcal{F} \otimes_k \mathrm{A} \in \mathrm{Shv}_{\mathrm{G}}(\mathrm{X};\mathrm{A})$. We define $\langle\mathcal{F}\rangle_{\mathrm{fil}} := \lim_{k\to\mathrm{A}} \langle\mathcal{F}_{\mathrm{A}}\rangle_{\mathrm{fil}}$, where the limit ranges over all even $\mathbf{E}_\infty$-k-algebras A. Note that $\langle\mathcal{F}\rangle_{\mathrm{fil}}$ is $\lim_{k\to\mathrm{A}} \mathrm{fil}^{\star}_{\text{G-ev}}(\mathrm{A}^{h\mathrm{G}})$-linear. By Example A.2.12, $\lim_{k\to\mathrm{A}} \mathrm{fil}^{\star}_{\text{G-ev}}(\mathrm{A}^{h\mathrm{G}}) = \mathrm{fil}^{\star}_{\text{G-ev}}(k^{h\mathrm{G}})$, so that $\langle\mathcal{F}\rangle_{\mathrm{fil}}$ is in fact $\mathrm{fil}^{\star}_{\text{G-ev}}(k^{h\mathrm{G}})$-linear. We write $\langle\mathcal{F}\rangle_{\mathrm{gr}}$ to denote the associated graded category of $\langle\mathcal{F}\rangle_{\mathrm{fil}}$, so that it is $\mathrm{gr}^{\star}_{\text{G-ev}}(k^{h\mathrm{G}})$-linear (and hence, in particular, $\mathrm{gr}^{\star}_{\mathrm{ev}}(k)$-linear).

One can similarly define $\mathrm{Loc}^{\mathrm{gr}}_{\mathrm{G}}(\mathrm{X};k)$, so that it is $\lim_{k\to\mathrm{A}} \mathrm{Loc}^{\mathrm{gr}}_{\mathrm{G}}(\mathrm{X};\mathrm{A})$, where the limit ranges over all even $\mathbf{E}_\infty$-k-algebras A.

The obvious analogue of Proposition 2.4.14 continues to hold: namely, suppose that the pullback of $\mathcal{F}$ to $\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ is even. Then the graded $\mathrm{gr}^{\star}_{\text{G-ev}}(k^{h\mathrm{G}})$-linear category $\langle\mathcal{F}\rangle_{\mathrm{gr}}$ is equivalent to the category $\lim_{k\to\mathrm{A}}(\mathrm{RMod}^{\mathrm{gr}}_{\mathrm{End}^{\star}_{\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathrm{A})}(\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F}_{\mathrm{A}}))})^{h\mathcal{W}}$, where the limit ranges over all even $\mathbf{E}_\infty$-k-algebras A. Under this equivalence, the $\mathrm{gr}^{\star}_{\text{G-ev}}(k^{h\mathrm{G}})$-linear functor $\eta^{\mathrm{gr}} : \langle\mathcal{F}\rangle_{\mathrm{gr}} \to \mathrm{Loc}^{\mathrm{gr}}_{\mathrm{G}}(\mathrm{X};k)$ identifies with the functor on $\mathcal{W}$-equivariant derived categories induced by the functor $\mathrm{RMod}^{\mathrm{gr},\heartsuit}_{\mathrm{End}^{\star}_{\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathrm{A})}(\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F}_{\mathrm{A}}))} \hookrightarrow \mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathrm{A})}$ given by the bimodule $\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F}_{\mathrm{A}})$. Moreover, the functor $\underline{\tau}_{\star} : \langle\mathcal{F}\rangle \to \langle\mathcal{F}\rangle_{\mathrm{gr}}$ sends an object $\mathcal{G} \in \langle\mathcal{F}\rangle$ whose pullback to $\mathrm{Shv}_{\mathrm{T}}(\mathrm{X};k)$ is !-even to $\lim_{k\to\mathrm{A}} \mathrm{Hom}^{\star}_{\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathrm{A})}(\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F}_{\mathrm{A}}), \mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{G}_{\mathrm{A}}))$ in $\lim_{k\to\mathrm{A}} \mathrm{RMod}^{\mathrm{gr},\heartsuit}_{\mathrm{End}^{\star}_{\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathrm{A})}(\mathrm{H}^{-*}_{\mathrm{T}}(\mathrm{X};\mathcal{F}_{\mathrm{A}}))}$

equipped with its $\mathcal{W}$-action from Definition 2.2.28. These statements also extend to the case when X is ind-compact Hausdorff.

3. Equivariant (co)homology of $\mathcal{G}\mathrm{r}_{\mathrm{G}}$

3.1. **Background on $\mathcal{G}\mathrm{r}_{\mathrm{G}}$.**

Recollection 3.1.1. Let $\mathrm{G}_{\mathcal{K}} = \mathrm{G}(\mathbf{C}(\!(t)\!))$, and let $\mathcal{G}\mathrm{r}_{\mathrm{G}} = \mathrm{G}_{\mathcal{K}}/\mathrm{G}_{\mathcal{O}}$ be the affine Grassmannian of G. (Note that $\mathrm{G}_{\mathcal{K}}$ is the complex points of a group object in ind-schemes, but it is not an ind-group scheme.) Each $\mu \in \mathbb{X}_*$ defines a point $t^\mu = \mu(t) \in \mathrm{T}_{\mathcal{K}}$, whose image in $\mathrm{G}_{\mathcal{K}}$ and in $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ will continue to be denoted t^μ. There is a left action of $\mathrm{G}_{\mathcal{O}}$ on $\mathcal{G}\mathrm{r}_{\mathrm{G}}$, and the Cartan decomposition identifies $\mathrm{G}_{\mathcal{K}} \cong \coprod_{\mu\in\mathbb{X}_*^+} \mathrm{G}_{\mathcal{O}} t^\mu \mathrm{G}_{\mathcal{O}}$. Therefore, the $\mathrm{G}_{\mathcal{O}}$-orbits on $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ are parametrized by $\mathbb{X}_*^+$; we will write $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ (resp. $\mathrm{Shv}_{(\mathrm{G}_{\mathcal{O}})}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$) to denote the category of $\mathrm{G}_{\mathcal{O}}$-equivariant sheaves of k-modules on $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ (resp. sheaves of k-modules on $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ which are locally constant on $\mathrm{G}_{\mathcal{O}}$-orbits) in the sense of Section 2, so that $\mathrm{Shv}_{(\mathrm{G}_{\mathcal{O}})}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \simeq \mathrm{Mod}_k \otimes_{\mathrm{Mod}_k^{h\mathrm{G}}} \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$. Let $\mathcal{G}\mathrm{r}_{\mathrm{G}}^\mu$ denote the associated $\mathrm{G}_{\mathcal{O}}$-orbit, and let $\mathcal{G}\mathrm{r}_{\mathrm{G}}^{\leq\mu}$ denote its closure. Write $i_\mu : \mathcal{G}\mathrm{r}_{\mathrm{G}}^{\leq\mu} \hookrightarrow \mathcal{G}\mathrm{r}_{\mathrm{G}}$ to denote the closed inclusion. Then $\mathcal{G}\mathrm{r}_{\mathrm{G}}^\mu$ is the $\mathbf{C}$-points of a smooth quasiprojective variety, and has complex dimension $(2\rho, \mu)$; thus $\mathcal{G}\mathrm{r}_{\mathrm{G}}^{\leq\mu}$ is the $\mathbf{C}$-points of a (generally singular) projective variety of the same dimension. Let P_μ be the parabolic subgroup of G generated by the root subgroups corresponding to those roots α for which $(\alpha, \mu) \leq 0$. Then there is a map $\mathcal{G}\mathrm{r}_{\mathrm{G}}^\mu \to \mathrm{G}/\mathrm{P}_\mu$ whose fibers are isomorphic to affine spaces. If μ is minuscule, then $\mathcal{G}\mathrm{r}_{\mathrm{G}}^\mu$ is already closed, and in fact $\mathcal{G}\mathrm{r}_{\mathrm{G}}^\mu \cong \mathrm{G}/\mathrm{P}_\mu$.

If $\mathrm{I} \subseteq \mathrm{G}_{\mathcal{O}}$ is the Iwahori subgroup associated to $\mathrm{B} \subseteq \mathrm{G}$, then there is an Iwasawa decomposition $\mathrm{G}_{\mathcal{K}} \cong \coprod_{\mu\in\mathbb{X}_*} \mathrm{I}t^\mu \mathrm{G}_{\mathcal{O}}$, so that the I-orbits on $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ are parametrized by $\mathbb{X}_*$. If I^0 denotes the prounipotent radical of I and $\mathrm{I}^0_\mu \subseteq \mathrm{I}^0$ denotes the stabilizer of $t^\mu \in \mathcal{G}\mathrm{r}_{\mathrm{G}}$, then the action map $\mathrm{I}^0 \to \mathrm{I}t^\mu$ induces an isomorphism $\mathrm{I}^0/\mathrm{I}^0_\mu \xrightarrow{\cong} \mathrm{I}t^\mu$. The quotient $\mathrm{I}^0/\mathrm{I}^0_\mu$ is isomorphic (by the exponential on the Lie algebra of I^0) to a finite-dimensional complex T-representation. We will write $j_\mu : \mathrm{I}t^\mu \hookrightarrow \mathcal{G}\mathrm{r}_{\mathrm{G}}$ to denote the inclusion.

Let $\mathrm{U} \subseteq \mathrm{B}$ denote the unipotent radical of B. For each $\lambda \in \mathbb{X}_*(\mathrm{T})$, there is a $\mathrm{U}_{\mathcal{K}}$-orbit $\mathrm{S}_\lambda \subseteq \mathcal{G}\mathrm{r}_{\mathrm{G}}$ (called a semi-infinite orbit) given by $i(q^{-1}(\lambda))$, where $i : \mathcal{G}\mathrm{r}_{\mathrm{B}} \to \mathcal{G}\mathrm{r}_{\mathrm{G}}$ and $q : \mathcal{G}\mathrm{r}_{\mathrm{B}} \to \mathcal{G}\mathrm{r}_{\mathrm{T}} = \mathbb{X}_*(\mathrm{T})$. Equivalently, $\mathrm{S}_\lambda = t^\lambda \mathcal{G}\mathrm{r}_{\mathrm{U}}$, so that $s_\lambda : \mathrm{S}_\lambda \hookrightarrow \mathcal{G}\mathrm{r}_{\mathrm{G}}$ is a locally closed embedding. If $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, then the fiber over $\lambda \in \mathcal{G}\mathrm{r}_{\mathrm{T}}$ of $q_! i^*(\mathcal{F})$ is precisely $\mathrm{R}\Gamma_c(\mathrm{S}_\lambda; s_\lambda^*\mathcal{F})$.

Let $\mathrm{Shv}_{\mathrm{I}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{\mathrm{ev}}$ (resp. $\mathrm{Shv}_{\mathrm{I}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{!\text{-ev}}$; resp. $\mathrm{Shv}_{\mathrm{I}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{*\text{-ev}}$) denote the full subcategory of $\mathrm{Shv}_{\mathrm{I}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ spanned by those sheaves which are even (resp. !-even; resp. $*$-even) in the sense of Definition 2.3.1 with respect to the I-orbit stratification of $\mathcal{G}\mathrm{r}_{\mathrm{G}}$. Similarly, let $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{\mathrm{ev}}$ (resp. $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{!\text{-ev}}$; resp. $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{*\text{-ev}}$) denote the full subcategory of $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ spanned by those sheaves whose restriction to $\mathrm{Shv}_{\mathrm{I}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ lies in $\mathrm{Shv}_{\mathrm{I}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{\mathrm{ev}}$ (resp. $\mathrm{Shv}_{\mathrm{I}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{!\text{-ev}}$; resp. $\mathrm{Shv}_{\mathrm{I}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{*\text{-ev}}$). For $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, we will use the same symbol $\mathcal{F}$ to denote its restriction to $\mathrm{Shv}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, where $\mathrm{T} \subseteq \mathrm{G}$ is a maximal torus.

Let G_c be a maximal compact subgroup of G, and let $\Omega_{\mathrm{poly}}\mathrm{G}_c \subseteq \mathrm{G}_{\mathcal{K}}$ denote the subspace of *polynomial* maps. Then the composite map $\Omega_{\mathrm{poly}}\mathrm{G}_c \subseteq \mathrm{G}_{\mathcal{K}} \to \mathcal{G}\mathrm{r}_{\mathrm{G}}$ is a homeomorphism. The canonical inclusion $\Omega_{\mathrm{poly}}\mathrm{G}_c \subseteq \Omega\mathrm{G}_c$ is also a homotopy equivalence. In fact, since the inclusion $\mathrm{G}_c \subseteq \mathrm{G}$ is also a homotopy equivalence, it follows that $\Omega\mathrm{G}_c \subseteq \Omega\mathrm{G}$ (and hence $\Omega_{\mathrm{poly}}\mathrm{G}_c \subseteq \Omega\mathrm{G}$) is a homotopy equivalence.

Construction 3.1.2. By [Noc20, Theorem 1.4], the category $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ admits a k-linear $\mathbf{E}_{3,\mathrm{BS}^1}$-monoidal structure $\star$ called *convolution.* The underlying $\mathbf{A}_2$-monoidal structure of $\star$, i.e., the underlying k-linear functor $\star : \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k) \otimes_k \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k) \to \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$, can be described as follows. There is a cospan

$$\begin{tikzcd}
(\mathcal{G}r_{\mathrm{G}} \times \mathcal{G}r_{\mathrm{G}})/\mathrm{G}_{\mathcal{O}}^{\mathrm{diag}} \arrow[r, "\cong"] \arrow[d, "p"'] & (\mathrm{G}_{\mathcal{K}} \times^{\mathrm{G}_{\mathcal{O}}} \mathcal{G}r_{\mathrm{G}})/\mathrm{G}_{\mathcal{O}} \arrow[d, "m"] \\
(\mathcal{G}r_{\mathrm{G}} \times \mathcal{G}r_{\mathrm{G}})/(\mathrm{G}_{\mathcal{O}} \times \mathrm{G}_{\mathcal{O}}) & \mathcal{G}r_{\mathrm{G}}/\mathrm{G}_{\mathcal{O}}.
\end{tikzcd}$$

One can rewrite $(\mathcal{G}r_{\mathrm{G}} \times \mathcal{G}r_{\mathrm{G}})/\mathrm{G}_{\mathcal{O}}^{\mathrm{diag}}$ as $(\mathrm{G}_{\mathcal{K}} \times \mathcal{G}r_{\mathrm{G}})/(\mathrm{G}_{\mathcal{O}} \times \mathrm{G}_{\mathcal{O}})$. Note that while $(\mathcal{G}r_{\mathrm{G}} \times \mathcal{G}r_{\mathrm{G}})/(\mathrm{G}_{\mathcal{O}} \times \mathrm{G}_{\mathcal{O}})$ and $\mathcal{G}r_{\mathrm{G}}/\mathrm{G}_{\mathcal{O}}$ are quotients of ind-(locally) compact Hausdorff spaces by pro-locally compact Hausdorff groups, $\mathrm{G}_{\mathcal{K}} \times \mathcal{G}r_{\mathrm{G}}$ is only an ind-good $\mathrm{G}_{\mathcal{O}} \times \mathrm{G}_{\mathcal{O}}$-space. (To see this, it suffices to show that $\mathrm{G}_{\mathcal{K}}$ is an ind-good $\mathrm{G}_{\mathcal{O}} \times \mathrm{G}_{\mathcal{O}}$-space via left and right multiplication. For this, let $\mathrm{G}_{\mathcal{K}}^{\leq \lambda}$ denote the preimage of $\mathcal{G}r_{\mathrm{G}}^{\leq \lambda}$ under the surjection $\mathrm{G}_{\mathcal{K}} \to \mathcal{G}r_{\mathrm{G}}$. If $(\mathrm{G}_{\mathcal{K}}^{\leq \lambda})^{(n)} := \ker(\mathrm{G}_{\mathcal{O}} \to \mathrm{G}_{\mathcal{O}}^{(n)})\backslash \mathrm{G}_{\mathcal{K}}^{\leq \lambda}$, then $\mathrm{G}_{\mathcal{K}}^{\leq \lambda} \cong \lim_n (\mathrm{G}_{\mathcal{K}}^{\leq \lambda})^{(n)}$, and the $\mathrm{G}_{\mathcal{O}} \times \mathrm{G}_{\mathcal{O}}$-action on $(\mathrm{G}_{\mathcal{K}}^{\leq \lambda})^{(n)}$ factors through $\mathrm{G}_{\mathcal{O}}^{(n)} \times \mathrm{G}_{\mathcal{O}}^{(m)}$ for some $m \gg n$. Using this, the conditions of being an ind-good space are easily verified.) Nevertheless, the categories $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}} \times \mathrm{G}_{\mathcal{O}}}(\mathrm{G}_{\mathcal{K}} \times \mathcal{G}r_{\mathrm{G}}; k)$ and $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}} \times \mathrm{G}_{\mathcal{O}}}(\mathrm{G}_{\mathcal{K}} \times \mathcal{G}r_{\mathrm{G}}; k)$ are well-defined. The functor $\star$ is now given by

$$\mathcal{F} \star \mathcal{G} := m_! p^*(\mathcal{F} \boxtimes \mathcal{G}).$$

Since m is ind-proper, the natural map $m_! \to m_*$ is an isomorphism. The unit for $\star$ is given by $\mathrm{IC}_0 := i_* \underline{k}$, where $i : * \to \mathcal{G}r_{\mathrm{G}}$ is the basepoint of $\mathcal{G}r_{\mathrm{G}}$.

There is an isomorphism

$$\mathrm{R\Gamma}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; \mathcal{F} \star \mathcal{G}) \xrightarrow{\cong} \mathrm{R\Gamma}_{\mathrm{G}_{\mathcal{O}}}(\mathrm{G}_{\mathcal{K}} \times^{\mathrm{G}_{\mathcal{O}}} \mathcal{G}r_{\mathrm{G}}; p^*(\mathcal{F} \boxtimes \mathcal{G})) \cong \mathrm{R\Gamma}_{\mathrm{G}_{\mathcal{O}}^{\mathrm{diag}}}(\mathcal{G}r_{\mathrm{G}} \times \mathcal{G}r_{\mathrm{G}}; \mathcal{F} \boxtimes \mathcal{G}).$$

Of course, there is a similar isomorphism with I-equivariance. If the map $k \otimes_{k^{h\mathrm{T}}} k \to \mathrm{C}^*(\mathrm{T}; k)$ is an isomorphism, then Remark 2.1.6 implies that there is an isomorphism

$$\mathrm{R\Gamma}_{\mathrm{I}}(\mathcal{G}r_{\mathrm{G}}; \mathcal{F}) \otimes_{k^{h\mathrm{T}}} \mathrm{R\Gamma}_{\mathrm{I}}(\mathcal{G}r_{\mathrm{G}}; \mathcal{G}) \xrightarrow{\cong} \mathrm{R\Gamma}_{\mathrm{I}^{\mathrm{diag}}}(\mathcal{G}r_{\mathrm{G}} \times \mathcal{G}r_{\mathrm{G}}; \mathcal{F} \boxtimes \mathcal{G}).$$

It follows that there is a natural isomorphism

$$\mathrm{R\Gamma}_{\mathrm{I}}(\mathcal{G}r_{\mathrm{G}}; \mathcal{F} \star \mathcal{G}) \cong \mathrm{R\Gamma}_{\mathrm{I}}(\mathcal{G}r_{\mathrm{G}}; \mathcal{F}) \otimes_{k^{h\mathrm{T}}} \mathrm{R\Gamma}_{\mathrm{I}}(\mathcal{G}r_{\mathrm{G}}; \mathcal{G}). \tag{3.1.1}$$

Remark 3.1.3. In Construction 3.1.2, we used I-equivariant cohomology. The statements therein hold verbatim with I-equivariance replaced by $\mathrm{G}_{\mathcal{O}}$-equivariance if the map $k \otimes_{k^{h\mathrm{G}}} k \to \mathrm{C}^*(\mathrm{G}; k)$ is an isomorphism. By Remark 2.1.6, this is true if k is even and G is a torus, or for arbitrary connected G if k is a connective $\mathbf{E}_\infty$-ring such that $\pi_0(k)$ is a perfect k-module. In fact, it can be shown that if the map $k \otimes_{k^{h\mathrm{G}}} k \to \mathrm{C}^*(\mathrm{G}; k)$ is an isomorphism, then the functor $\mathrm{R\Gamma}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; -) : \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k) \to \mathrm{Mod}_{k^{h\mathrm{G}}}$ is $\mathbf{E}_{2,\mathrm{BS}^1}$-monoidal, where $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ is equipped with the convolution $\mathbf{E}_{3,\mathrm{BS}^1}$-monoidal structure; but we will not use this here.

Notation 3.1.4. Let $\mu \in \mathbb{X}_*^+$, and let X be a $\mathrm{G}_{\mathcal{O}}$-space. Define $\mathcal{G}r_{\mathrm{G}}^{\leq \mu} \widetilde{\times} \mathrm{X}$ to be the quotient $\mathrm{G}_{\mathcal{K}}^{\leq \mu} \times^{\mathrm{G}_{\mathcal{O}}} \mathrm{X}$, so that there is a projection map $\mathcal{G}r_{\mathrm{G}}^{\leq \mu} \widetilde{\times} \mathrm{X} \to \mathcal{G}r_{\mathrm{G}}^{\leq \mu}$. If $\mu_\bullet = (\mu_1, \cdots, \mu_n)$ is a sequence of dominant coweights, define $\mathcal{G}r_{\mathrm{G}}^{\leq \mu_\bullet}$ to be $\mathcal{G}r_{\mathrm{G}}^{\leq \mu_1} \widetilde{\times} \cdots \widetilde{\times} \mathcal{G}r_{\mathrm{G}}^{\leq \mu_n}$. The action map $\mathrm{G}_{\mathcal{K}} \times^{\mathrm{G}_{\mathcal{O}}} \mathcal{G}r_{\mathrm{G}} \to \mathcal{G}r_{\mathrm{G}}$ defines a map m :

$\mathcal{G}r_{\mathrm{G}}^{\leq \mu_\bullet} \to \mathcal{G}r_{\mathrm{G}}^{\leq |\mu_\bullet|} \subseteq \mathcal{G}r_{\mathrm{G}}$, where $|\mu_\bullet| = \sum_i \mu_i$. Note that the map $m : \mathcal{G}r_{\mathrm{G}}^{\leq \mu_\bullet} \to \mathcal{G}r_{\mathrm{G}}^{\leq |\mu_\bullet|}$ is proper and surjective, and is moreover an isomorphism over $\mathcal{G}r_{\mathrm{G}}^{|\mu_\bullet|}$.

Construction 3.1.5. In general, $\mathcal{G}r_{\mathrm{G}} = \mathrm{colim}_\mu\, \mathcal{G}r_{\mathrm{G}}^{\leq \mu}$ where μ ranges over dominant coweights. Moreover, for each μ, there is some i such that the action of $\mathrm{G}_{\mathcal{O}}$ on $\mathcal{G}r_{\mathrm{G}}^{\leq \mu}$ factors through the quotient $\mathrm{G}_{\mathcal{O}} \to \mathrm{G}_{\mathcal{O}}^{(i)}$. It follows that $\mathcal{G}r_{\mathrm{G}}$ is an ind-good $\mathrm{G}_{\mathcal{O}}$-space in the terminology of Construction 2.1.18, so the category $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ is well-defined (and is canonically equivalent to $\mathrm{Shv}^*_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$).

As in Example 2.1.21, $\mathcal{L}_\mu = \underline{k}_{\mathcal{G}r_{\mathrm{G}}^{\leq \mu}}$ is a local dimension theory, $\tau^\mu = \mathcal{L}_\mu$ is an ind-dimension theory on $\mathcal{G}r_{\mathrm{G}}$, and $\omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}} = \mathrm{colim}_\mu\, i_{\mu,*}\omega_{\mathcal{G}r_{\mathrm{G}}^{\leq \mu}}$. In particular,

$$\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; \omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}}) = \mathrm{colim}_\mu(\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}^{\leq \mu}; \omega_{\mathcal{G}r_{\mathrm{G}}^{\leq \mu}})) = \mathrm{colim}_\mu\, \mathrm{C}^{\mathrm{G}}_*(\mathcal{G}r_{\mathrm{G}}^{\leq \mu}; k).$$

Note that if the map $k \otimes_{k^{h\mathrm{T}}} k \to \mathrm{C}^*(\mathrm{T}; k)$ is an isomorphism, then $\mathrm{R}\Gamma_{\mathrm{I}}(\mathcal{G}r_{\mathrm{G}}; \omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}})$ is an $\mathbf{E}_\infty$-$k^{h\mathrm{T}}$-coalgebra by Example 2.1.21. In fact, it is an $\mathbf{E}_\infty$-$k^{h\mathrm{T}}$-coalgebra equipped with a coaction of $\mathrm{C}^*_{\mathrm{T}}(\mathrm{G}/\mathrm{B}; k)$. It follows that $\mathrm{R}\Gamma_{\mathrm{I}}(\mathcal{G}r_{\mathrm{G}}; -)$ refines to a functor $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k) \to \mathrm{coMod}_{\mathrm{R}\Gamma_{\mathrm{I}}(\mathcal{G}r_{\mathrm{G}}; \omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}})}(\mathrm{Mod}_{k^{h\mathrm{T}}})$.

Remark 3.1.6. The object $\omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ is !-even. Indeed, it suffices to show that $\omega_{\mathcal{G}r_{\mathrm{G}}^{\leq \mu}}$ is !-even. If $j_\lambda : \mathrm{I}t^\lambda \hookrightarrow \mathcal{G}r_{\mathrm{G}}^{\leq \mu}$, then $j^!_\lambda \omega_{\mathcal{G}r_{\mathrm{G}}^{\leq \mu}} \cong \omega_{\mathrm{I}t^\lambda}$, which is indeed !-even since $\mathrm{I}t^\lambda$ is isomorphic (by the exponential on the Lie algebra of I^0) to a finite-dimensional complex T-representation.

Construction 3.1.7. Let λ be a dominant coweight. Then there is a local dimension theory $\mathcal{L}_\lambda$ on $\mathrm{G}_{\mathcal{K}}^{\leq \lambda} = \lim_n (\mathrm{G}_{\mathcal{K}}^{\leq \lambda})^{(n)}$ defined as follows. There is a map $q^\lambda_n : (\mathrm{G}_{\mathcal{K}}^{\leq \lambda})^{(n)} \to \mathcal{G}r_{\mathrm{G}}^{\leq \lambda}$ which is a $\mathrm{G}_{\mathcal{O}}^{(n)}$-bundle. The local dimension theory $\mathcal{L}_\lambda$ sends $n \mapsto \omega_{q^\lambda_n} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}} \times \mathrm{G}_{\mathcal{O}}}((\mathrm{G}_{\mathcal{K}}^{\leq \lambda})^{(n)}; k)$.

If $m : \mathrm{G}_{\mathcal{K}} \times^{\mathrm{G}_{\mathcal{O}}} \mathcal{G}r_{\mathrm{G}} \to \mathcal{G}r_{\mathrm{G}}$ is the action map, then $m^{!,\mathrm{ren}}\omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}} \cong \omega^{\mathrm{ren}}_{\mathrm{G}_{\mathcal{K}} \times^{\mathrm{G}_{\mathcal{O}}} \mathcal{G}r_{\mathrm{G}}}$. Since $\omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}} \boxtimes \omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}} = \omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}} \times \mathcal{G}r_{\mathrm{G}}}$, it follows from Example 2.1.15 that there is also an isomorphism $p^*(\omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}} \boxtimes \omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}}) \cong \omega^{\mathrm{ren}}_{\mathrm{G}_{\mathcal{K}} \times^{\mathrm{G}_{\mathcal{O}}} \mathcal{G}r_{\mathrm{G}}}$. In particular, there is an isomorphism

$$p^*(\omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}} \boxtimes \omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}}) \cong m^{!,\mathrm{ren}}\omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}}. \tag{3.1.2}$$

The counit $m_! m^{!,\mathrm{ren}} \to \mathrm{id}$ therefore defines a map

$$\omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}} \star \omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}} = m_! p^*(\omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}} \boxtimes \omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}}) \to \omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}}.$$

Arguing in this way using the isomorphism (3.1.2), the proof of [Noc20, Theorem 1.4] can be adapted to show that $\omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}}$ admits the structure of an $\mathbf{E}_{3,\mathrm{BS}^1}$-algebra in $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ under $\star$. It would then follow that $\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; \omega^{\mathrm{ren}}_{\mathcal{G}r_{\mathrm{G}}})$ is an $\mathbf{E}_{2,\mathrm{BS}^1}$-$k^{h\mathrm{G}}$-algebra in $\mathbf{E}_\infty$-$k^{h\mathrm{G}}$-coalgebras, and in fact it is an $\mathbf{E}_{3,\mathrm{BS}^1}$-$k$-algebra in $\mathbf{E}_\infty$-k-coalgebras. We will not need this below.

Recall the *extended* torus $\widetilde{\mathrm{T}} = \mathrm{T} \times \mathbf{C}^\times$; this acts on $\mathcal{G}r_{\mathrm{G}}$, where $\mathbf{C}^\times$ acts on $\mathcal{G}r_{\mathrm{G}}$ by loop-rotation and T acts in the standard way.

Definition 3.1.8. Let k be an even $\mathbf{E}_\infty$-ring. Say that k is *good for* $\mathrm{G}_{\mathcal{K}}$ if it is good for $\hat{\Phi}^+$ in the sense that $e(\alpha) \in \pi_*(k^{h\widetilde{\mathrm{T}}})$ is a nontrivial nonzero divisor for each $\alpha \in \hat{\Phi}^+$, and for distinct $\alpha, \beta \in \hat{\Phi}^+$, the elements $e(\alpha), e(\beta) \in \pi_*(k^{h\widetilde{\mathrm{T}}})$ are coprime.

Like in Lemma 2.2.20, MU is good for $\mathrm{G}_{\mathcal{K}}$.

Lemma 3.1.9. *The affine Grassmannian* $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ *is an* MU*-GKM ind-*$\widetilde{\mathrm{T}}$*-space. The associated diagram* $\Gamma_{\mathcal{G}\mathrm{r}_{\mathrm{G}}}$ *is given by* $\mathrm{E} \rightrightarrows \mathbb{X}_*$ *where* $\mathrm{E} = \mathbb{X}_* \times \Phi^+ \times \mathbf{Z} \setminus \{0\}$*, and one has* $x_0(\mu, \beta, n) = \mu$ *and* $x_\infty(\mu, \beta, n) = \mu + n\check{\beta}$*. Moreover, the map* $e : \mathrm{E} \to \pi_*(k^{h\widetilde{\mathrm{T}}})$ *sends* $(\mu, \beta, n) \mapsto e(\beta + (n + \langle \beta, \mu \rangle)\delta)$*.*

Proof. The fixed locus $\mathcal{G}\mathrm{r}_{\mathrm{G}}^{\widetilde{\mathrm{T}}} = \mathbb{X}_*$ via the map $\mu \mapsto t^\mu$. The weights of the $\widetilde{\mathrm{T}}$ action on the tangent space to t^μ show that for distinct $\mu, \lambda \in \mathbb{X}_*(\mathrm{T})$, then there is some $\chi \in \mathrm{E}$ for which $x_0(\chi) = \mu$ and $x_\infty(\chi) = \lambda$ if and only if there is some $\alpha \in \hat{\Phi}^+$ such that $s_\alpha(\mu) = \lambda$. This in turn happens if and only if $\lambda - \mu$ is a nonzero multiple $n\check{\beta}$ of a coroot $\check{\beta} \in \check{\Phi}$; in that case, $\alpha = \beta + (n + \langle \beta, \mu \rangle)\delta$. The map $e : \mathrm{E} \to \pi_*(\mathrm{MU}^{h\widetilde{\mathrm{T}}})$ sends (λ, β, n) to the Euler class $e(\alpha) = e(\beta) +_{\mathrm{F}} [n + \langle \beta, \mu \rangle]\, e(\delta)$. Recall that MU is good for $\mathrm{G}_{\mathcal{K}}$. To show that $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ is an MU-GKM ind-$\widetilde{\mathrm{T}}$-space, we only need to check that for any $\mu_0, \mu_\infty \in \mathbb{X}_*$, there are only finitely many 1-dimensional orbits whose boundary is $\{\mu_0, \mu_\infty\}$, but this is clear. $\square$

Definition 3.1.10. For any subset $\mathrm{S} \subseteq \Phi$, let $e(\mathrm{S}) = \prod_{\alpha \in \mathrm{S}} e(\alpha) \in \pi_*(\mathrm{MU}^{h\mathrm{T}})$. Let $\mathcal{A}_{\mathrm{G}}$ denote the $\mathcal{O}_{\check{\mathrm{T}} \times \mathrm{T}_{\mathrm{F}}} \cong \pi_*(\mathrm{MU}^{h\mathrm{T}})[\mathbb{X}_*]$-algebra given by the intersection

$$\bigcap_{\alpha \in \Phi} \pi_*(\mathrm{MU}^{h\mathrm{T}})[\mathbb{X}_*]\left[\tfrac{e^{\check{\alpha}} - 1}{e(\alpha)}, e(\Phi - \{\alpha\})^{-1}\right] \subseteq \pi_*(\mathrm{MU}^{h\mathrm{T}})[\mathbb{X}_*][e(\Phi)^{-1}].$$

Since MU is good for G, $\mathcal{A}_{\mathrm{G}}$ is a flat $\pi_*(\mathrm{MU}^{h\mathrm{T}})$-algebra. There are inclusions

$$\pi_*(\mathrm{MU}^{h\mathrm{T}})[\mathbb{X}_*] \cong \mathcal{A}_{\mathrm{T}} \subseteq \mathcal{A}_{\mathrm{G}} \subseteq \pi_*(\mathrm{MU}^{h\mathrm{T}})[\mathbb{X}_*][e(\Phi)^{-1}] \cong \mathcal{A}_{\mathrm{T}}[e(\Phi)^{-1}].$$

Both $\mathcal{A}_{\mathrm{T}}$ and $\mathcal{A}_{\mathrm{T}}[e(\Phi)^{-1}]$ are Hopf algebras over $\pi_*(\mathrm{MU}^{h\mathrm{T}})$ (and all the maps above are maps of Hopf algebras), and the coproduct on $\mathcal{A}_{\mathrm{T}}[e(\Phi)^{-1}]$ restricts to define the structure of a Hopf $\pi_*(\mathrm{MU}^{h\mathrm{T}})$-algebra on $\mathcal{A}_{\mathrm{G}}$. Indeed, it suffices to observe that since $\Delta(e^{\check{\alpha}}) = e^{\check{\alpha}} \otimes e^{\check{\alpha}}$, the identity $xy - 1 = x(y-1) + (x-1)$ implies that

$$\Delta(\tfrac{e^{\check{\alpha}} - 1}{e(\alpha)}) = e^{\check{\alpha}} \otimes \tfrac{e^{\check{\alpha}} - 1}{e(\alpha)} + \tfrac{e^{\check{\alpha}} - 1}{e(\alpha)} \otimes 1.$$

This is contained in $\mathcal{A}_{\mathrm{G}} \otimes_{\pi_*(\mathrm{MU}^{h\mathrm{T}})} \mathcal{A}_{\mathrm{G}} \subseteq \mathcal{A}_{\mathrm{T}} \otimes_{\pi_*(\mathrm{MU}^{h\mathrm{T}})} \mathcal{A}_{\mathrm{T}}[e(\Phi)^{-1}]$, as desired.

Let $\mathcal{W}$ denote the Weyl algebroid of G. Then the W-action on $\mathcal{A}_{\mathrm{G}}[e(\Phi)^{-1}] \cong \mathcal{A}_{\mathrm{T}}[e(\Phi)^{-1}]$ uniquely restricts to a $\mathcal{W}$-comodule structure $\mathcal{A}_{\mathrm{G}}$ (via Hopf algebra maps), i.e., there is a unique dashed arrow making the following diagram commute:

$$\begin{tikzcd}
\mathcal{A}_{\mathrm{G}} \arrow[r, dashed] \arrow[d] & \mathcal{A}_{\mathrm{G}} \otimes_{\pi_*(\mathrm{MU}^{h\mathrm{T}})} \mathcal{W} \arrow[d] \\
\mathcal{A}_{\mathrm{T}}[e(\Phi)^{-1}] \arrow[r, "\rho"'] & (\mathcal{A}_{\mathrm{T}} \otimes_{\pi_*(\mathrm{MU}^{h\mathrm{T}})} \mathrm{Map}(\mathrm{W}, \pi_*(\mathrm{MU}^{h\mathrm{T}})))[e(\Phi)^{-1}]
\end{tikzcd}$$

where both vertical arrows are injections. There is a W-action on $\mathcal{A}_{\mathrm{G}}$, because if $\alpha \in \Phi$, then $\rho(\frac{e^{\check{\alpha}} - 1}{e(\alpha)})$ sends w to $\frac{e^{w\check{\alpha}} - 1}{e(w\alpha)}$, which lies in $\mathcal{A}_{\mathrm{G}} \subseteq \mathcal{A}_{\mathrm{T}}[e(\Phi)^{-1}]$. Arguing as in Construction 2.2.29 shows that this W-action on $\mathcal{A}_{\mathrm{G}}$ is restricted from a $\mathcal{W}$-action on $\mathcal{A}_{\mathrm{G}}$, giving the dotted arrow as indicated.

If R is a graded commutative ring with a graded 1-dimensional formal group law F, and $\mathrm{T}_{\mathrm{F}} = \mathrm{Hom}(\mathbb{X}^*(\mathrm{T}), \mathrm{F})$, we will continue to write $\mathcal{A}_{\mathrm{G}}$ to denote the Hopf $\mathcal{O}_{\mathrm{T}_{\mathrm{F}}}$-algebra given by base-changing the Hopf $\pi_*(\mathrm{MU}^{h\mathrm{T}})$-algebra $\mathcal{A}_{\mathrm{G}}$ defined above along the map $\pi_*(\mathrm{MU}^{h\mathrm{T}}) \to \mathcal{O}_{\mathrm{T}_{\mathrm{F}}}$ induced by the map $\pi_*(\mathrm{MU}) \to \mathrm{R}$ which classifies F.

Theorem 3.1.11. *Let k be an even $\mathbf{E}_\infty$-ring. Then $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ is a commutative and cocommutative Hopf $\pi_*(k^{h\mathrm{T}})$-algebra which is flat over $\pi_*(k^{h\mathrm{T}})$. Moreover, if $\mathcal{W}$ denotes the Weyl algebroid of G, there is an isomorphism $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \xrightarrow{\cong} \mathcal{A}_\mathrm{G}$ of commutative and cocommutative Hopf $\pi_*(k^{h\mathrm{T}})$-algebras in $\mathcal{W}$-comodules, such that there is a commutative diagram*

$$\begin{tikzcd}
\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{T}; k) \arrow[r] \arrow[d, "\cong"] & \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \arrow[r] \arrow[d, "\cong"] & \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{T}; k)[e(\Phi)^{-1}] \arrow[d, "\cong"] \\
\mathcal{A}_\mathrm{T} \arrow[r] & \mathcal{A}_\mathrm{G} \arrow[r] & \mathcal{A}_\mathrm{T}[e(\Phi)^{-1}],
\end{tikzcd}$$

where all horizontal arrows are injections. By Definition 2.2.28, there is an isomorphism $\mathrm{gr}^\star(\mathrm{C}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)) \cong \mathcal{A}_\mathrm{G}^{h\mathcal{W}}$ of $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$-algebras.

If $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$, then $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathcal{F})$ is canonically a graded $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$-comodule in $\mathcal{W}$-comodules. Moreover, if $i : \mathcal{G}\mathrm{r}_\mathrm{T} \hookrightarrow \mathcal{G}\mathrm{r}_\mathrm{G}$ is the inclusion, there is a commutative diagram of $\mathcal{W}$-comodules

$$\begin{tikzcd}
\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathcal{F})[e(\Phi)^{-1}] \arrow[r] \arrow[d, "\cong"'] & \mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathcal{F}) \otimes_{\pi_*(k^{h\mathrm{T}})} \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)[e(\Phi)^{-1}] \arrow[d, "\cong"] \\
\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_\mathrm{T}; i^!\mathcal{F})[e(\Phi)^{-1}] \arrow[r] & \mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_\mathrm{T}; i^!\mathcal{F}) \otimes_{\pi_*(k^{h\mathrm{T}})} \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{T}; k)[e(\Phi)^{-1}],
\end{tikzcd}$$

where the bottom horizontal arrow is induced by the $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{T}; k)$-coaction on $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_\mathrm{T}; i^!\mathcal{F})$. The vertical arrows are isomorphisms by Proposition 2.2.4.

Proof. The case of general k follows from the case of MU using the trick from Example 2.2.21 of (completed) base-changing along the map $\pi_*(\mathrm{MU}^{h\mathrm{T}}) \to \pi_*(k^{h\mathrm{T}})$ induced by the T-equivariant MU-algebra structure on k. We will therefore assume that $k = \mathrm{MU}$ below.

Note first that Lemma 3.1.9 implies that $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ is an ind-projective $\pi_*(k^{h\mathrm{T}})$-module. It follows from the Künneth formula that if $\mathrm{M} \in \mathrm{Ind}(\mathrm{Perf}_k^{h\mathrm{T}})$, then the natural map

$$\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \otimes_{\pi_*(k^{h\mathrm{T}})} \pi_*\mathrm{M}^{h\mathrm{T}} \to \pi_*(\mathrm{C}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \otimes_k \mathrm{M})^{h\mathrm{T}} \tag{3.1.3}$$

is an isomorphism.

There is a homeomorphism $\Omega_{\mathrm{poly}}\mathrm{G} \times \Omega_{\mathrm{poly}}\mathrm{G} \xrightarrow{\cong} \mathrm{G}_\mathcal{K} \times^{\mathrm{G}_\mathcal{O}} \mathcal{G}\mathrm{r}_\mathrm{G}$ which fits into a commutative diagram

$$\begin{tikzcd}
\Omega_{\mathrm{poly}}\mathrm{G} \times \Omega_{\mathrm{poly}}\mathrm{G} \arrow[r, "\cong"] \arrow[d, "m"'] & \mathrm{G}_\mathcal{K} \times^{\mathrm{G}_\mathcal{O}} \mathcal{G}\mathrm{r}_\mathrm{G} \arrow[d, "m"] \\
\Omega_{\mathrm{poly}}\mathrm{G} \arrow[r] & \mathcal{G}\mathrm{r}_\mathrm{G}.
\end{tikzcd}$$

Under the homotopy equivalence $\Omega_{\mathrm{poly}}\mathrm{G} \xrightarrow{\sim} \Omega\mathrm{G}$, the T-equivariant multiplication map $\Omega\mathrm{G} \times \Omega\mathrm{G} \to \Omega\mathrm{G}$ induces the map $\mathrm{C}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \otimes_{k^{h\mathrm{T}}} \mathrm{C}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \to \mathrm{C}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$. This is a map of $\mathbf{E}_\infty$-$k^{h\mathrm{T}}$-coalgebras.[5] Using the isomorphism

[5]Note that the map $\Omega\mathrm{G} \times \Omega\mathrm{G} \to \Omega\mathrm{G}$ is *not* $\widetilde{\mathrm{T}}$-equivariant (even though it is T-equivariant)! In particular, $\mathrm{H}_*^{\widetilde{\mathrm{T}},\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ is *not* a $\pi_* k^{h\widetilde{\mathrm{T}}}$-algebra. In fact, it is not even an algebra!

(3.1.3), this induces a map $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)\otimes_{\pi_*(k^{h\mathrm{T}})}\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)\to \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$ which equips $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$ with the structure of an associative algebra in cocommutative $\pi_*(k^{h\mathrm{T}})$-algebras. Using (3.1.3), it also follows that $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{F})$ is a graded $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$-comodule for any $\mathcal{F}\in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}:k)$.

The map $\mathcal{G}\mathrm{r}_{\mathrm{T}}\to\mathcal{G}\mathrm{r}_{\mathrm{G}}$ is a T-equivariant map of $\mathbf{E}_2$-spaces, so the map $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)\to \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$ is one of Hopf $\pi_*(k^{h\mathrm{T}})$-algebras. Theorem 2.2.9 gives an injection

$$\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)\subseteq \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)[u_{\mathcal{P}}^{-1}]\xrightarrow{\cong}\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)[u_{\mathcal{P}}^{-1}].$$

Since $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)$ is a *commutative* and cocommutative Hopf $\pi_*(k^{h\mathrm{T}})$-algebra, the same is true of $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)[u_{\mathcal{P}}^{-1}]$, and hence of $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$.

Using Theorem 2.2.9 and base-changing along $\pi_*(k^{h\widetilde{\mathrm{T}}})\twoheadrightarrow\pi_*(k^{h\mathrm{T}})$, we find that $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$ is isomorphic to the the subset of $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)[e(\Phi)^{-1}]\cong \pi_*(k^{h\mathrm{T}})[\mathbb{X}_*][e(\Phi)^{-1}]$ of those $\sum_{\mu\in\mathbb{X}_*}f_\mu[\mu]$ with $f_\mu\in\pi_*(k^{h\mathrm{T}})[e(\Phi)^{-1}]$ satisfying the following conditions:

(1) For each $\mu\in\mathbb{X}_*$, $f_\mu\prod_{\chi\in\mathrm{E}_\mu}e(\chi)\in\pi_*(k^{h\mathrm{T}})$. In other words, all poles of f_μ are simple, and lie on the divisors cut out by $e(\chi)$ for $\chi\in\mathrm{E}_\mu$.
(2) If $\chi\in\mathrm{E}$, then $e(\chi)(f_{x_0(\chi)}-f_{x_\infty(\chi)})=0 \bmod e(\chi)$.

It follows that $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$ is in turn isomorphic to the $\pi_*(k^{h\mathrm{T}})[\mathbb{X}_*]$-submodule of $\pi_*(k^{h\mathrm{T}})[\mathbb{X}_*][e(\Phi)^{-1}]$ generated by expressions of the form $\frac{e^{\mu+n\check{\alpha}}-e^{\mu}}{e(\alpha)}$ for $\mu\in\mathbb{X}_*$, $\alpha\in\Phi$, and $n\in\mathbf{Z}$. Observe that $\frac{e^{\mu+n\check{\alpha}}-e^{\mu}}{e(\alpha)}=e^{\mu}\frac{e^{n\check{\alpha}}-1}{e(\alpha)}$. The identity $xy-1=x(y-1)+(x-1)$ implies that

$$\frac{e^{n\check{\alpha}}-1}{e(\alpha)}=e^{\check{\alpha}}\frac{e^{(n-1)\check{\alpha}}-1}{e(\alpha)}+\frac{e^{\check{\alpha}}-1}{e(\alpha)};$$

inductively, it follows that $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$ is isomorphic to the $\pi_*(k^{h\mathrm{T}})[\mathbb{X}_*]$-submodule of $\pi_*(k^{h\mathrm{T}})[\mathbb{X}_*][e(\Phi)^{-1}]$ generated by expressions of the form $\frac{e^{\check{\alpha}}-1}{e(\alpha)}$. But this is exactly the ring $\mathcal{A}_{\mathrm{G}}$. The commutativity of the diagrams in the statement of Theorem 3.1.11 follow from naturality. ☐

The following says that the functor

$$\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}};-):\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)\to\mathrm{coMod}_{\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_*(k^{h\mathrm{T}})})$$

is monoidal when restricted to a suitable subcategory of the source, where $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$ is equipped with the convolution monoidal structure and $\mathrm{coMod}_{\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_*(k^{h\mathrm{T}})})$ is equipped with the monoidal structure coming from the Hopf $\pi_*(k^{h\mathrm{T}})$-algebra structure on $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$.

Corollary 3.1.12. *Let k be an even $\mathbf{E}_\infty$-ring. For $?\in\{!,*\}$, let $\mathcal{F}$ be an object of $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)^{?\mathrm{ev}}\subseteq\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$, and let $\mathcal{G}\in\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$. Then the functor*

$$\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}};-):\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)\to\mathrm{coMod}_{\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_*(k^{h\mathrm{T}})})$$

coming from Theorem 3.1.11 induces an isomorphism

$$\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{F})\otimes_{\pi_*(k^{h\mathrm{T}})}\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{G})\xrightarrow{\cong}\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{F}\star\mathcal{G}).$$

Proof. By Theorem 3.1.11, if $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, then $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F})$ is canonically a graded $\mathrm{H}^{\mathrm{T,BM}}_{*}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$-comodule. If $\mathcal{F}, \mathcal{G} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, (3.1.1) gives a natural isomorphism

$$\mathrm{R}\Gamma_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F}) \otimes_{k^{h\mathrm{T}}} \mathrm{R}\Gamma_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{G}) \xrightarrow{\cong} \mathrm{R}\Gamma_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F} \star \mathcal{G}).$$

Since $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{?\mathrm{ev}}$, it follows by induction on I-orbit strata that $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F})$ is a projective $\pi_*(k^{h\mathrm{T}})$-module. In particular, it is a flat $\pi_*(k^{h\mathrm{T}})$-module, so the Künneth formula therefore implies that there is a natural isomorphism

$$\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F}) \otimes_{\pi_*(k^{h\mathrm{T}})} \mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{G}) \xrightarrow{\cong} \mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F} \star \mathcal{G}),$$

which implies the desired monoidal structure on $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; -)$. □

Construction 3.1.13. Recall that the $\mathrm{I} \times \mathrm{I}$-orbits on $\mathrm{G}_{\mathcal{K}}$ are indexed by $w \in \mathrm{W}^{\mathrm{aff}}$, and that each $\mathrm{I}w\mathrm{I}/\mathrm{I}$ is isomorphic to $\mathbf{C}^{\ell(w)}$ on which T acts linearly. Similarly, the I-orbits on $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ are parametrized by $\mathbb{X}_*$, and each $\mathrm{I}t^\mu \mathrm{G}_{\mathcal{O}}/\mathrm{G}_{\mathcal{O}}$ is isomorphic to a finite-dimensional complex T-representation. This induces a stratification on $\mathrm{G}_{\mathcal{K}} \times^{\mathrm{I}} \mathcal{G}\mathrm{r}_{\mathrm{G}}$ by $\coprod_{w \in \mathrm{W}^{\mathrm{aff}}, \mu \in \mathbb{X}_*} \mathrm{I}w\mathrm{I} \times^{\mathrm{I}} \mathrm{I}t^\mu \mathrm{G}_{\mathcal{O}}/\mathrm{G}_{\mathcal{O}}$. Note that each stratum $\mathrm{I}w\mathrm{I} \times^{\mathrm{I}} \mathrm{I}t^\mu \mathrm{G}_{\mathcal{O}}/\mathrm{G}_{\mathcal{O}}$ is an affine space on which T acts linearly.

Conjecture 3.1.14. *The fibers of the action/convolution map* $m_{\mathrm{I}} : \mathrm{G}_{\mathcal{K}} \times^{\mathrm{I}} \mathcal{G}\mathrm{r}_{\mathrm{G}} \to \mathcal{G}\mathrm{r}_{\mathrm{G}}$ *are* T*-equivariantly paved by affine spaces with linear* T*-action. Moreover, if* $\mathrm{G}_{\mathcal{K}} \times^{\mathrm{I}} \mathcal{G}\mathrm{r}_{\mathrm{G}}$ *is equipped with the affine paving from Construction 3.1.13, the inclusions of the fibers of* m_{I} *into* $\mathrm{G}_{\mathcal{K}} \times^{\mathrm{I}} \mathcal{G}\mathrm{r}_{\mathrm{G}}$ *respect this affine paving, and are given on each stratum by a disjoint union of linear* T*-equivariant embeddings into the strata of* $\mathrm{G}_{\mathcal{K}} \times^{\mathrm{I}} \mathcal{G}\mathrm{r}_{\mathrm{G}}$ *(whose underlying* T*-spaces are vector spaces on which* T *acts linearly, see Recollection 3.1.1).*

Since we rely on perfect evenness (in the sense of Definition 2.3.1) in Theorem 2.3.2 (instead of mere evenness of cohomology), we will unfortunately need to assume Conjecture 3.1.14 in some of our statements below.

Remark 3.1.15. If $\mu_\bullet = (\mu_1, \cdots, \mu_n)$ is a sequence of dominant coweights, we may ask that the map $\mathrm{G}_{\mathcal{K}}^{\leq \mu_1} \times^{\mathrm{I}} \cdots \times^{\mathrm{I}} \mathcal{G}\mathrm{r}_{\mathrm{G}}^{\leq \mu_n} \to \mathcal{G}\mathrm{r}_{\mathrm{G}}^{\leq |\mu_\bullet|}$ satisfy the conclusions of Conjecture 3.1.14; we will refer to this as "Conjecture 3.1.14 for $\mu_\bullet$". For example, [Hai06, Corollary 1.2] implies that the fibers of the map $\mathcal{G}\mathrm{r}_{\mathrm{G}}^{\leq \mu_\bullet} = \mathrm{G}_{\mathcal{K}}^{\leq \mu_1} \times^{\mathrm{G}_{\mathcal{O}}} \cdots \times^{\mathrm{G}_{\mathcal{O}}} \mathcal{G}\mathrm{r}_{\mathrm{G}}^{\leq \mu_n} \to \mathcal{G}\mathrm{r}_{\mathrm{G}}^{\leq |\mu_\bullet|}$ admit affine pavings. From this and the fact that the map $\mathrm{BI} \to \mathrm{BG}_{\mathcal{O}}$ of classifying stacks is a locally trivial fibration with fibers G/B which admit a paving by affine spaces (via the Bruhat decomposition), it is easy to show that Conjecture 3.1.14 holds for sequences of dominant minuscule coweights.

Proposition 3.1.16. *Assume Conjecture 3.1.14. Let* $? \in \{!, *\}$*. Then the subcategory* $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{?\mathrm{ev}} \subseteq \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ *is closed under convolution* $\star$ *(and hence similarly for* $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{\mathrm{ev}}$*).*

Proof. Let $\lambda \in \mathbb{X}_*$, and let $j_\lambda : \mathrm{I}t^\lambda \hookrightarrow \mathcal{G}\mathrm{r}_{\mathrm{G}}$ denote the inclusion of the corresponding I-orbit. An object $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ is ?-even if $j_\lambda^?(\mathcal{F})$ is perfect even in $\mathrm{Shv}_{\mathrm{I}}(\mathrm{I}t^\lambda; k)$ (in the sense of Definition 2.3.1) for each $\lambda \in \mathbb{X}_*$. As mentioned in Recollection 3.1.1, $\mathrm{I}t^\lambda$ is a vector space on which T acts linearly. In particular, $\mathrm{Shv}_{\mathrm{I}}(\mathrm{I}t^\lambda; k) \simeq \mathrm{Mod}_k^{h\mathrm{T}}$ (via restriction to the basepoint $t^\lambda \in \mathrm{I}t^\lambda$). It follows that if $i_\lambda : t^\lambda \hookrightarrow \mathcal{G}\mathrm{r}_{\mathrm{G}}$ denotes the inclusion of the T-fixed point, then $\mathcal{F}$ is ?-even if $i_\lambda^?(\mathcal{F})$ is perfect even in $\mathrm{Mod}_k^{h\mathrm{T}} \simeq \mathrm{Mod}_{k^{h\mathrm{T}}}^{\hat{k}}$ for each $\lambda \in \mathbb{X}_*$.

Denote the quotient map

$$\mathrm{G}_{\mathcal{K}} \times^{\mathrm{I}} \mathcal{G}\mathrm{r}_{\mathrm{G}} \to \mathrm{G}_{\mathcal{K}} \times^{\mathrm{G}_{\mathcal{O}}} \mathcal{G}\mathrm{r}_{\mathrm{G}} = \mathcal{G}\mathrm{r}_{\mathrm{G}} \widetilde{\times} \mathcal{G}\mathrm{r}_{\mathrm{G}}$$

by q, so that $m_{\mathrm{I}} = m \circ q$, where $m : \mathcal{G}\mathrm{r}_{\mathrm{G}} \widetilde{\times} \mathcal{G}\mathrm{r}_{\mathrm{G}} \to \mathcal{G}\mathrm{r}_{\mathrm{G}}$ is the convolution map. Let $\mathcal{F}, \mathcal{G} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$. Then $q^*(\mathcal{F} \widetilde{\boxtimes} \mathcal{G}) \cong q^*(\mathcal{F}) \widetilde{\boxtimes} \mathcal{G}$; its pushforward along m_{I} will be denoted $q^*(\mathcal{F}) \star_{\mathrm{I}} \mathcal{G}$. Observe that $q^*(\mathcal{F}) \star_{\mathrm{I}} \mathcal{G} \cong m_* q_* q^*(\mathcal{F} \widetilde{\boxtimes} \mathcal{G})$. Since q is a G/B-bundle, the inclusion of the basepoint of G/B defines a splitting of $m_*(\mathcal{F} \widetilde{\boxtimes} \mathcal{G}) = \mathcal{F} \star \mathcal{G}$ off $q^*(\mathcal{F}) \star_{\mathrm{I}} \mathcal{G}$. (This splitting only exists in $\mathrm{Shv}_{\mathrm{I}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, since the basepoint of G/B is only B-equivariantly fixed, and not G-equivariantly fixed.) Since a retract of a ?-even sheaf is ?-even, it suffices to show that if $\mathcal{F}$ and $\mathcal{G}$ are ?-even, then $q^*(\mathcal{F}) \star_{\mathrm{I}} \mathcal{G}$ is also ?-even. Let $\lambda \in \mathbb{X}_*(\mathrm{T})$, and define X by the fiber product

$$\begin{tikzcd}
\mathrm{X} \arrow[r, "i_\lambda"] \arrow[d, "m_{\mathrm{I}}"'] & \mathrm{G}_{\mathcal{K}} \times^{\mathrm{I}} \mathcal{G}\mathrm{r}_{\mathrm{G}} \arrow[d, "m_{\mathrm{I}}"] \\
t^\lambda \arrow[r, "i_\lambda"'] & \mathcal{G}\mathrm{r}_{\mathrm{G}}.
\end{tikzcd}$$

Then proper base-change identifies $i_\lambda^?(q^*(\mathcal{F}) \star_{\mathrm{I}} \mathcal{G}) \cong \mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}; i_\lambda^?(q^*(\mathcal{F}) \widetilde{\boxtimes} \mathcal{G}))$. Because $\mathcal{F}$ and $\mathcal{G}$ are both ?-even, the ?-restriction of $q^*(\mathcal{F}) \widetilde{\boxtimes} \mathcal{G}$ to any stratum of $\mathrm{G}_{\mathcal{K}} \times^{\mathrm{I}} \mathcal{G}\mathrm{r}_{\mathrm{G}}$ (via the stratification of Construction 3.1.13) is perfect even in the sense of Definition 2.3.1 (i.e., is built via extensions and retracts from shifts of the constant sheaf by complex representations of T). By Conjecture 3.1.14, this implies that the ?-restriction of $q^*(\mathcal{F}) \widetilde{\boxtimes} \mathcal{G}$ to each stratum $\mathrm{X}_i - \mathrm{X}_{i-1}$ of X is perfect even: this is clear when $? = *$; and when ? is !, it follows from the observation that the !-restriction of the constant sheaf along a linear inclusion $\mathrm{V}' \hookrightarrow \mathrm{V}$ of complex vector spaces is isomorphic to $\Sigma^{\mathrm{V}/\mathrm{V}'} k$. Since each $\mathrm{X}_i - \mathrm{X}_{i-1}$ is a disjoint union of affine spaces on which T acts linearly, it follows that $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}_i - \mathrm{X}_{i-1}; i_\lambda^?(q^*(\mathcal{F}) \widetilde{\boxtimes} \mathcal{G}))$ and $\mathrm{R}\Gamma_{\mathrm{T},c}(\mathrm{X}_i - \mathrm{X}_{i-1}; i_\lambda^?(q^*(\mathcal{F}) \widetilde{\boxtimes} \mathcal{G}))$ are perfect even in $\mathrm{Mod}_{k^{h\mathrm{T}}}^{\hat{k}}$. Using recollement, this implies that $\mathrm{R}\Gamma_{\mathrm{T}}(\mathrm{X}; i_\lambda^?(q^*(\mathcal{F}) \widetilde{\boxtimes} \mathcal{G}))$ is itself perfect even in $\mathrm{Mod}_{k^{h\mathrm{T}}}^{\hat{k}}$. □

Corollary 3.1.12 and Proposition 3.1.16 therefore imply that the functor

$$\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; -) : \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \to \mathrm{coMod}_{\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\pi_*(k^{h\mathrm{T}})})$$

coming from Theorem 3.1.11 admits a monoidal structure when restricted to $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{?\mathrm{ev}}$.

3.2. **Cohomology of $\mathcal{G}\mathrm{r}_{\mathrm{G}}$.** Our goal in this section is to describe a calculation of $\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, which we define as the $k^{h\mathrm{G}}$-linear dual to $\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \omega^{\mathrm{ren}}_{\mathcal{G}\mathrm{r}_{\mathrm{G}}})$, using Hochschild cohomology. Let $\mathbf{T} \cong \mathbf{C}^\times$, acting on $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ by loop-rotation.

Lemma 3.2.1. *Let* G *be a topological group, and let* G *act on itself by conjugation. Then the homotopy quotient* $(\mathrm{BG})_{h\mathrm{G}}$ *is homotopy equivalent to* $\mathrm{BG} \times \mathrm{BG}$*, and under this equivalence, the natural map* $\mathrm{BG} \to (\mathrm{BG})_{h\mathrm{G}}$ *is homotopic to the diagonal on* BG*.*

Proof. There are several ways to see this: one is to note that $(\mathrm{BG})_{h\mathrm{G}} \cong \mathrm{B}(\mathrm{G} \rtimes \mathrm{G})$. But there is an isomorphism $\mathrm{G} \rtimes \mathrm{G} \cong \mathrm{G} \times \mathrm{G}$ by $(g, h) \mapsto (gh, h)$, so $(\mathrm{BG})_{h\mathrm{G}} \cong \mathrm{B}(\mathrm{G} \times \mathrm{G})$. One could alternatively think about the homotopy triviality of the G-action on BG as the observation that conjugating the G-action on a G-representation V produces a G-representation which is canonically isomorphic to V. □

Lemma 3.2.2. *Let* X *be a simply-connected space, and assume that* k *is a connective* $\mathbf{E}_\infty$*-ring such that* $\pi_0(k)$ *is a perfect* k*-module. Then the map* $k \otimes_{\mathrm{C}^*(\mathrm{X};k)} k \to \mathrm{C}^*(\Omega\mathrm{X};k)$ *is an isomorphism upon* $\pi_0(k)$*-completion.*

Proof. It suffices to check that the map $k \otimes_{\mathrm{C}^*(\mathrm{X};k)} k \to \mathrm{C}^*(\Omega\mathrm{X};k)$ is an isomorphism upon base-changing along the map $k \to \pi_0(k)$. Since $\pi_0(k)$ is a perfect k-module, $\mathrm{C}^*(\mathrm{Y};k) \otimes_k \pi_0(k) \cong \mathrm{C}^*(\mathrm{Y};\pi_0(k))$ for any anima Y. We are therefore reduced to showing that the map $k \otimes_{\mathrm{C}^*(\mathrm{X};k)} k \to \mathrm{C}^*(\Omega\mathrm{X};k)$ is an isomorphism when k is concentrated in degree zero. This is the classical statement of Eilenberg-Moore, which uses that X is simply-connected (see [Dwy74]). □

As mentioned in Remark 2.1.6, the condition on k in Lemma 3.2.2 can be viewed as a regularity condition on k (see [BL14]).

Proposition 3.2.3. *Let* k *be a connective* $\mathbf{E}_\infty$*-ring such that* $\pi_0(k)$ *is a perfect* k*-module. Suppose that* G *is a connected complex reductive group. Let* $\widetilde{\mathrm{G}}$ *be the universal cover of* G*, so that* $\pi_1(\mathrm{G})$ *acts on* $\widetilde{\mathrm{G}}$ *with quotient* G*. There is a* $\mathbf{T}$*-equivariant map* $\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k) \to \mathrm{R}\Gamma_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G};k)$ *of* $\mathbf{E}_\infty$*-*k*-algebras, which exhibits a* $\mathbf{T}$*-equivariant isomorphism upon* $\pi_0(k)$*-completion*

$$\mathrm{R}\Gamma_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G};k) \cong \mathrm{Map}(\pi_1(\mathrm{G}), \mathrm{HH}_{\mathbf{E}_2}(k^{h\widetilde{\mathrm{G}}}/k)^{h\pi_1(\mathrm{G})}).$$

Proof. Recall that $\mathrm{R}\Gamma_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G};k) \cong \mathrm{C}^*_\mathrm{G}(\Omega\mathrm{G};k) \cong \mathrm{C}^*(\Omega\mathrm{G};k)^{h\mathrm{G}}$, where G acts on itself by conjugation, and that

$$\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k) = k^{h\mathrm{G}} \otimes_{k^{h\mathrm{G}} \otimes_{k^{h\mathrm{G}} \otimes_k k^{h\mathrm{G}}} k^{h\mathrm{G}}} k^{h\mathrm{G}}.$$

The natural map $k \otimes_{k^{h\mathrm{G}}} k \to \mathrm{C}^*(\mathrm{G};k)$ is G-equivariant for the G-action on itself by conjugation. The assumption on k implies that the natural map $k \otimes_{k^{h\mathrm{G}}} k \to \mathrm{C}^*(\mathrm{G};k)$ is an isomorphism: by Lemma 3.2.2, this reduces to the claim when k is discrete, where it follows from [MNN17, Proposition 7.33]. Taking homotopy G-fixed points and using the equivalence $\mathrm{Mod}_k^{h\mathrm{G}} \to \mathrm{Mod}_{k^{h\mathrm{G}}}^{\hat{k}}$ from Remark 2.1.6, it then follows that

$$k^{h\mathrm{G}} \otimes_{(k^{h\mathrm{G}})^{h\mathrm{G}}} k^{h\mathrm{G}} \xrightarrow{\cong} (k \otimes_{k^{h\mathrm{G}}} k)^{h\mathrm{G}} \xrightarrow{\cong} \mathrm{C}^*_\mathrm{G}(\mathrm{G};k).$$

We claim that $(k^{h\mathrm{G}})^{h\mathrm{G}} \cong k^{h(\mathrm{G}\times\mathrm{G})}$, where G acts on itself, hence on BG, by conjugation; moreover, under this identification, the map $(k^{h\mathrm{G}})^{h\mathrm{G}} \to k^{h\mathrm{G}}$ induced by taking G-fixed points of the canonical map $k^{h\mathrm{G}} \to k$ is given by the map $k^{h(\mathrm{G}\times\mathrm{G})} \to k^{h\mathrm{G}}$ induced by restriction along the diagonal of G. This follows from Lemma 3.2.1.

The preceding discussion implies that

$$k^{h\mathrm{G}} \otimes_{(k^{h\mathrm{G}})^{h\mathrm{G}}} k^{h\mathrm{G}} \cong k^{h\mathrm{G}} \otimes_{k^{h(\mathrm{G}\times\mathrm{G})}} k^{h\mathrm{G}},$$

and it is easy to check that this is in turn just $k^{h\mathrm{G}} \otimes_{k^{h\mathrm{G}} \otimes_k k^{h\mathrm{G}}} k^{h\mathrm{G}}$. It follows that

$$\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k) \cong k^{h\mathrm{G}} \otimes_{\mathrm{C}^*_\mathrm{G}(\mathrm{G};k)} k^{h\mathrm{G}} \cong (k \otimes_{\mathrm{C}^*(\mathrm{G};k)} k)^{h\mathrm{G}}.$$

There is a G-equivariant map $k \otimes_{\mathrm{C}^*(\mathrm{G};k)} k \to \mathrm{C}^*(\Omega\mathrm{G};k)$, which is an equivalence upon $\pi_0(k)$-completion by Lemma 3.2.2 since G is simply-connected. In particular, when G is simply-connected, there is a $\mathbf{T}$-equivariant isomorphism $\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k) \cong \mathrm{C}^*_\mathrm{G}(\Omega\mathrm{G};k)$. In general, if $\widetilde{\mathrm{G}}$ denotes the universal cover of G, then the fibration $\widetilde{\mathrm{G}} \to \mathrm{G} \to \mathrm{B}\pi_1(\mathrm{G})$ splits upon looping. Therefore, $\mathrm{C}^*(\Omega\mathrm{G};k) \cong \mathrm{Map}(\pi_1(\mathrm{G}), \mathrm{C}^*(\Omega\widetilde{\mathrm{G}};k))$. Since $\mathrm{HH}_{\mathbf{E}_2}(k^{h\widetilde{\mathrm{G}}}/k) \cong \mathrm{C}^*_{\widetilde{\mathrm{G}}}(\Omega\widetilde{\mathrm{G}};k)$ upon $\pi_0(k)$-completion as discussed above, it follows that $\mathrm{HH}_{\mathbf{E}_2}(k^{h\widetilde{\mathrm{G}}}/k)^{h\pi_1(\mathrm{G})} \cong \mathrm{C}^*_{\mathrm{G}}(\Omega\widetilde{\mathrm{G}}; k)$ upon $\pi_0(k)$-completion, so there is a $\mathbf{T}$-equivariant equivalence upon $\pi_0(k)$-completion

$$\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \cong \mathrm{Map}(\pi_1(\mathrm{G}), \mathrm{HH}_{\mathbf{E}_2}(k^{h\widetilde{\mathrm{G}}}/k)^{h\pi_1(\mathrm{G})}),$$

as desired. □

Remark 3.2.4. The proof of Proposition 3.2.3 shows that if k is an $\mathbf{E}_\infty$-ring such that the natural map $k \otimes_{k^{h\mathrm{G}}} k \to \mathrm{C}^*(\mathrm{G}; k)$ is an isomorphism, then $\mathrm{HH}(k^{h\mathrm{G}}/k) \cong \mathrm{C}^*_{\mathrm{G}}(\mathrm{G}; k)$, where G acts on itself by conjugation.

Corollary 3.2.5. *Let k be a connective $\mathbf{E}_\infty$-ring such that $\pi_0(k)$ is a perfect k-module. Suppose that* G *is a simply-connected complex reductive group. Then the double $k^{h\mathrm{G}}$-linear dual of $\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ is $\mathbf{T}$-equivariantly isomorphic to the $\mathbf{E}_2$-center $\mathfrak{Z}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)$.*

Proof. Recall that $\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \cong \mathrm{C}^{\mathrm{G}}_*(\Omega\mathrm{G}; k)$ is the colimit $\mathrm{colim}_\lambda(\mathrm{C}^{\mathrm{G}}_*(\mathcal{G}\mathrm{r}^{\leq\lambda}_{\mathrm{G}}; k))$. The $k^{h\mathrm{G}}$-linear dual of $\mathrm{C}^{\mathrm{G}}_*(\mathcal{G}\mathrm{r}^{\leq\lambda}_{\mathrm{G}}; k)$ is equivalent to $\mathrm{C}^*_{\mathrm{G}}(\mathcal{G}\mathrm{r}^{\leq\lambda}_{\mathrm{G}}; k)$. It follows that the $k^{h\mathrm{G}}$-linear dual of $\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ is equivalent to $\lim_\lambda \mathrm{C}^*_{\mathrm{G}}(\mathcal{G}\mathrm{r}^{\leq\lambda}_{\mathrm{G}}; k)$, which is $\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \cong \mathrm{C}^*_{\mathrm{G}}(\Omega\mathrm{G}; k)$. Proposition 3.2.3 identifies this with the $\pi_0(k)$-completion of $\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)$. Now the claim follows from the observation that the $k^{h\mathrm{G}}$-linear dual of $\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)$ is $\mathfrak{Z}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)$. □

Remark 3.2.6. An analogous observation (see [Tel14]) is that the completion $\mathrm{C}_*(\Omega\mathrm{G}; k)^{h\mathrm{G}} \cong (\mathrm{colim}_\lambda \mathrm{C}_*(\mathcal{G}\mathrm{r}^{\leq\lambda}_{\mathrm{G}}; k))^{h\mathrm{G}}$ of $\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \cong \mathrm{colim}_\lambda(\mathrm{C}^{\mathrm{G}}_*(\mathcal{G}\mathrm{r}^{\leq\lambda}_{\mathrm{G}}; k))$ is isomorphic to the $\mathbf{E}_2$-center $\mathfrak{Z}_{\mathbf{E}_2}(\mathrm{C}_*(\Omega\mathrm{G}; k)/k)$ of the $\mathbf{E}_2$-k-algebra $\mathrm{C}_*(\Omega\mathrm{G}; k)$. This can be proved exactly as in Proposition 3.2.3, except it is easier, in that one only needs to use *homological* Eilenberg-Moore, as opposed to the statement of cohomological Eilenberg-Moore from Lemma 3.2.2. Explicitly, $k \otimes_{\mathrm{C}_*(\Omega\mathrm{G};k)} k \cong \mathrm{C}_*(\mathrm{G}; k)$, so that

$$\mathrm{HH}_{\mathbf{E}_2}(\mathrm{C}_*(\Omega\mathrm{G}; k)/k) \cong \mathrm{End}_{\mathrm{C}_*(\mathrm{G}\times\Omega\mathrm{G};k)}(\mathrm{C}_*(\Omega\mathrm{G}; k)) \cong \mathrm{C}_*(\Omega\mathrm{G}; k)^{h\mathrm{G}}.$$

Remark 3.2.7. Assume k is an even $\mathbf{E}_\infty$-ring. Since $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ admits a T-equivariant cell decomposition (see Recollection 3.1.1), $\mathrm{R}\Gamma_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ is concentrated in even degrees. Recall from Definition 2.2.28 that $\mathrm{gr}^\star_{\mathrm{G}}(\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}\rtimes\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \cong \mathrm{H}^{-*}_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{h\mathcal{W}}$. If k is good for $\mathrm{G}_{\mathcal{K}}$ (in the sense of Definition 3.1.8), then Theorem 2.2.12 and Lemma 3.1.9 give a complete description of $\mathrm{H}^{-*}_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ (along with its $\mathcal{W}$-action by Example 2.2.30). From this perspective, the results discussed below are rather weak, in the sense that they describe the associated graded of a filtration on $\mathrm{H}^{-*}_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ in terms of simpler objects, but there are very interesting multiplicative extensions in assembling $\mathrm{H}^{-*}_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ from its graded pieces. To illustrate, let us describe $\mathrm{H}^{-*}_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ when G is $\mathbf{G}_m$ and SL_2.

Example 3.2.8. Assume k is an even $\mathbf{E}_\infty$-ring, and write $\pi_*(k^{h(\mathrm{T}\times\mathbf{T})}) \cong \pi_*(k)[x, \hbar]^\wedge$, where x and $\hbar$ live in weight -2. When $\mathrm{G} = \mathrm{T} = \mathbf{G}_m$, there is an isomorphism

$$\mathrm{H}^{-*}_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; k) \cong \pi_*(k)[x, \hbar]^\wedge[\tbinom{b}{n}]^\wedge_{n\geq 0},$$

where b lives in weight 0. Here, the first completion is taken in the sense of Remark A.1.6, and the second completion is taken at the ideal generated by the sequence of elements $\binom{b}{n}$. Indeed, this follows from the isomorphism $\mathcal{G}\mathrm{r}_{\mathrm{G}} \cong \mathbf{Z}$ and Newton's theorem that the binomial functions form a basis for integer-valued polynomials. However, it is convenient, especially for the purpose of using Theorem 2.2.12 and Lemma 3.1.9, to observe that one can rewrite $\mathrm{H}^{-*}_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)$ in terms of a basis which is adapted to the formal group F over $\pi_*(k)$. Define a formal group law over $\pi_*(k)[\hbar]^\wedge$ by $a +_{\widetilde{\mathrm{F}}} a' = \frac{(a\hbar)+_{\mathrm{F}}(a'\hbar)}{\hbar}$, where a and a' live in weight 0. Let $\langle n\rangle = \frac{[n](\hbar)}{\hbar}$, so that $\langle n\rangle$ is the n-fold sum of 1 under $\widetilde{\mathrm{F}}$, and define $\langle n\rangle! = \langle n\rangle\langle n-1\rangle\cdots\langle 1\rangle$ for $n\geq 1$. Finally, define $\binom{a}{n}_{\widetilde{\mathrm{F}}} = \frac{1}{\langle n\rangle!}\prod_{i=0}^{n-1}(a -_{\widetilde{\mathrm{F}}} \langle i\rangle)$. Note that, just like with the usual binomial coefficients, if $m\in\mathbf{Z}$, then $\binom{\langle m\rangle}{n}_{\widetilde{\mathrm{F}}}$ is well-defined in $\pi_*(k)[\hbar]^\wedge$, i.e., does not have any poles in $\hbar$ (see [DM23, Proposition 2.1.3 and Proposition 4.3.4]). Then, there is an isomorphism

$$\mathrm{H}^{-*}_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)\cong\pi_*(k)[x,\hbar]^\wedge\left[\binom{a}{n}_{\widetilde{\mathrm{F}}}\right]^\wedge_{n\geq 0},$$

where a lives in weight 0. This was already observed in [Dev26, Remark 5.8] from the perspective of Proposition 3.2.3; namely, this algebra is $\pi_*\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{T}}/k)^{h\mathbf{T}}$. For instance, if $k=\mathrm{KU}$ so that $\pi_*(k)\cong\mathbf{Z}[\beta^{\pm1}]$ and we write $q-1=\beta\hbar$, then $\langle n\rangle = [n]_q = \frac{q^n-1}{q-1}$, and $\langle n\rangle! = [n]_q! = [n]_q[n-1]_q\cdots[1]_q$. Since $x+_{\widetilde{\mathrm{F}}}y = x+y+(q-1)xy$, we have

$$a -_{\widetilde{\mathrm{F}}} [n]_q = a +_{\widetilde{\mathrm{F}}} [-n]_q = q^{-n}a + [-n]_q = q^{-n}(a-[n]_q),$$

so that

$$\binom{a}{n}_{\widetilde{\mathrm{F}}} = \frac{a(a-[1]_q)\cdots(a-[n]_q)}{q^{\binom{n}{2}}[n]_q!}.$$

In particular, $\mathrm{H}^0_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};\mathrm{KU})$ is isomorphic to a completion of the ring of quantum integer-valued polynomials from [HH17].

Example 3.2.9. Assume k is an even $\mathbf{E}_\infty$-ring. When $\mathrm{G}=\mathrm{SL}_2$ and k is good for $\mathrm{G}_{\mathcal{K}}$, Theorem 2.2.12 and Lemma 3.1.9 imply that there is an isomorphism

$$\mathrm{H}^{-*}_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)\cong\pi_*(k)[x,\hbar]^\wedge\left[\frac{1}{\langle n\rangle!\hbar^n}\prod_{i=0}^{n-1}(y-_{\mathrm{F}}x-_{\mathrm{F}}[i](\hbar))(y+_{\mathrm{F}}x+_{\mathrm{F}}[i](\hbar))\right]^\wedge_{n\geq 0},$$

where $\hbar$, x, and y all live in weight -2. For instance, if $k=\mathbf{Z}$, then

$$\mathrm{H}^{-*}_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathbf{Z})\cong\mathbf{Z}[x,\hbar]\left[\frac{1}{n!\hbar^n}\prod_{i=0}^{n-1}(y^2-(x+i\hbar)^2)\right]^\wedge_{n\geq 0}.$$

If $k=\mathrm{KU}$, so that $\pi_*(k)=\mathbf{Z}[\beta^{\pm1}]$ with β in weight 2, then we may replace $\hbar$, x, and y by $q-1=\hbar\beta$, $u-1=x\beta$, and $v-1=y\beta$, where q, u, and v live in weight 0. Then,

$$\mathrm{H}^{-*}_{\mathrm{T}\times\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathrm{KU})\cong\mathbf{Z}[\![q-1,u-1]\!][\beta^{\pm1}]\left[\frac{1}{[n]_q!(q-1)^n}\prod_{i=0}^{n-1}(v^2-v(q^iu+q^{-i}u^{-1})+1)\right]^\wedge_{n\geq 0}.$$

Theorem 4.2.12 gives an alternative approach to describing $\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$ and its $\mathbf{T}$-equivariant version $\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}\rtimes\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)$ through more "differential" invariants. We now describe a very crude approximation to these rich invariants.

Construction 3.2.10. Let $\mathrm{A}\to\mathrm{B}$ be a map of $\mathbf{E}_\infty$-rings, and let $n\geq 1$. Then there is a $\mathbf{T}$-equivariant filtration on $\mathrm{HH}_{\mathbf{E}_n}(\mathrm{B}/\mathrm{A})$ defined by $\mathrm{HH}_{\mathbf{E}_n}(\mathrm{fil}^\star_{\mathrm{ev}}(\mathrm{B})/\mathrm{fil}^\star_{\mathrm{ev}}(\mathrm{A}))$; we will denote it by $\mathrm{fil}^\star\mathrm{HH}_{\mathbf{E}_n}(\mathrm{B}/\mathrm{A})$. Let $\mathrm{Y}=\mathrm{Spec}(\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{A}))$ and $\mathrm{X}=\mathrm{Spec}(\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{B}))$, so that there is a map $\mathrm{X}\to\mathrm{Y}$. The associated graded $\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_n}(\mathrm{B}/\mathrm{A})$ is isomorphic upon inverse shearing to $\mathrm{HH}_{\mathbf{E}_n}(\mathrm{X}/\mathrm{Y})$. Note that $\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{A})$ (and hence

$\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{B}))$ is a graded $\mathbf{Z}$-algebra, so $\mathrm{sh}^{-1}(\mathrm{gr}^\star \mathrm{HH}_{\mathbf{E}_n}(\mathrm{B}/\mathrm{A})) \cong \mathrm{HH}_{\mathbf{E}_n}(\mathrm{X}/\mathrm{Y})$ in turn admits a Hochschild-Kostant-Rosenberg filtration with associated graded given by $\mathrm{LSym}^*(\Sigma^n \mathrm{L}_{\mathrm{X}/\mathrm{Y}})$. In other words,

$$\mathrm{gr}^*_{\mathrm{HKR}}\mathrm{sh}^{-1}(\mathrm{gr}^\star \mathrm{HH}_{\mathbf{E}_n}(\mathrm{B}/\mathrm{A})) \cong \mathrm{LSym}^*(\Sigma^n \mathrm{L}_{\mathrm{X}/\mathrm{Y}}).$$

When $n = 1$, Illusie's [Lur17, Proposition 25.2.4.2] implies that

$$\mathrm{gr}^*_{\mathrm{HKR}}\mathrm{sh}^{-1}(\mathrm{gr}^\star \mathrm{HH}(\mathrm{B}/\mathrm{A})) \cong \mathrm{L}\Omega^*_{\mathrm{X}/\mathrm{Y}}[*];$$

when $n = 2$, Illusie's [Lur17, Proposition 25.2.4.2] implies that

$$\mathrm{gr}^*_{\mathrm{HKR}}\mathrm{sh}^{-1}(\mathrm{gr}^\star \mathrm{HH}_{\mathbf{E}_2}(\mathrm{B}/\mathrm{A})) \cong \mathrm{L}\Gamma^*_{\mathcal{O}_\mathrm{X}}(\mathrm{L}_{\mathrm{X}/\mathrm{Y}})[2*].$$

Note that the case $n = 2$ is in fact a special case of the case $n = 1$, because $\mathrm{HH}_{\mathbf{E}_2}(\mathrm{X}/\mathrm{Y}) \cong \mathrm{HH}(\mathrm{X}/(\mathrm{X} \times_\mathrm{Y} \mathrm{X}))$ and $\mathrm{L}_{\mathrm{X}/(\mathrm{X}\times_\mathrm{Y}\mathrm{X})} \cong \mathrm{L}_{\mathrm{X}/\mathrm{Y}}[1]$.

There is also a loop-rotation equivariant version of the above discussion.

Definition 3.2.11. Fix a derived stack Y. Let $(\mathbf{G}_a^{\hat{\mathrm{dR}},+})_\mathrm{Y}$ denote the ring stack over $(\mathbf{A}^1/\mathbf{G}_m)_\mathrm{Y}$ defined as the cofiber of the map $\mathcal{O}(-1)^{\hat\sharp} \to \mathcal{O}$, where $\mathcal{O}(-1)^{\hat\sharp}$. (The closely related object given by the cofiber of the map $\mathcal{O}(-1)^\sharp \to \mathcal{O}$ was defined in [Bha24, Definition 2.5.1] when Y is p-nilpotent. This object is studied in general in forthcoming work of Bhatt-Mathew-Vologodsky-Zhang under the name "sheared filtered de Rham stack"; one should not confuse the word "shearing" here with our use of it in this document.) If $f : \mathrm{X} \to \mathrm{Y}$ is a morphism of derived stacks, then transmuting the ring stack $(\mathbf{G}_a^{\hat{\mathrm{dR}},+})_\mathrm{Y}$ (in the sense of [Bha24, Remark 2.3.8]) defines a ring stack $\mathrm{Def}^\sharp_f$ over $(\mathbf{A}^1/\mathbf{G}_m)_\mathrm{Y}$. Explicitly, if f is a map of stacks over $\mathbf{Z}$, and $\mathrm{X}^{\hat{\mathrm{dR}},+}$ and $\mathrm{Y}^{\hat{\mathrm{dR}},+}$ denote the transmutations of X and Y given by the filtered stack $(\mathbf{G}_a^{\hat{\mathrm{dR}},+})_\mathbf{Z}$ over $\mathbf{Z}$, then $\mathrm{Def}^\sharp_f \cong \mathrm{X}^{\hat{\mathrm{dR}},+} \times_{\mathrm{Y}^{\hat{\mathrm{dR}},+}} (\mathbf{A}^1/\mathbf{G}_m)_\mathrm{Y}$. This can be viewed as a divided power version of the deformation to the normal cone. We will often continue to write $\mathrm{Def}^\sharp_f$ to denote the pullback of $\mathrm{Def}^\sharp_f$ along $\mathbf{A}^1_\mathrm{Y} \to (\mathbf{A}^1/\mathbf{G}_m)_\mathrm{Y}$; this should hopefully not cause confusion.

Example 3.2.12. Suppose $\mathrm{X} \to \mathrm{Y}$ is the diagonal map $\mathbf{A}^1 \to \mathbf{A}^1 \times_\mathbf{Z} \mathbf{A}^1$, and equip $\mathbf{A}^1$ with coordinate x. Then $\mathrm{Def}^\sharp_f \cong \mathrm{Spf}\, \mathbf{Z}[t][x,y]\langle \frac{x-y}{t} \rangle^\wedge_{(x-y)}$, where $\langle - \rangle$ denotes the divided power envelope.

Example 3.2.13. If Y is a $\mathbf{Q}$-scheme, then $\mathbf{G}_a^{\hat{\mathrm{dR}},+}$ is the stack $\mathbf{G}_a^{\mathrm{dR},+}$ from [Bha24, Construction 2.3.4], so $\mathrm{Def}^\sharp_f$ is the relative filtered de Rham stack $(\mathrm{X}/\mathrm{Y})^{\mathrm{dR},+} = \mathrm{X}^{\mathrm{dR},+} \times_{\mathrm{Y}^{\mathrm{dR},+}} (\mathbf{A}^1/\mathbf{G}_m)_\mathrm{Y}$. For instance, if $\mathrm{X} \to \mathrm{Y}$ is the diagonal map (relative to some base commutative ring R), then $(\mathrm{X}/\mathrm{Y})^{\mathrm{dR},+} \cong (\mathbf{A}^1/\mathbf{G}_m)_\mathrm{X} \times_{\mathrm{X}^{\mathrm{dR},+}} (\mathbf{A}^1/\mathbf{G}_m)_\mathrm{X}$. The fiber over $1 \in \mathbf{A}^1/\mathbf{G}_m$ is just $\mathrm{X} \times_{\mathrm{X}^{\mathrm{dR}}} \mathrm{X} \cong (\mathrm{X} \times_\mathrm{R} \mathrm{X})^\wedge_\mathrm{X}$, and the fiber over $B\mathbf{G}_m \subseteq \mathbf{A}^1/\mathbf{G}_m$ is $\mathrm{X} \times_{B\hat{\mathrm{T}}_{\mathrm{X}/\mathrm{R}}(-1)} \mathrm{X} \cong \hat{\mathrm{T}}_{\mathrm{X}/\mathrm{R}}(-1)$.

Proposition 3.2.14. *Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism of derived stacks, which we assume to be affine for simplicity. Then the HKR filtration on the negative cyclic homology $\mathrm{HH}(\mathrm{X}/\mathrm{Y})^{h\mathbf{T}}$, defined as in* [Ant19]*, has associated graded isomorphic to the shearing of $\mathrm{R}\Gamma(\mathrm{Def}^\sharp_f; \mathcal{O})$ as algebras over $\mathrm{gr}(\tau_{\geq\star}(\mathcal{O}_\mathrm{Y}^{h\mathbf{T}})) \cong \mathrm{sh}(\mathcal{O}_{(\mathbf{A}^1)_\mathrm{Y}})$.*

Proof sketch. This can be proved as in [Ant19]. An alternative approach is as follows: using that $\mathrm{Def}^\sharp_f$ is defined by the ring stack $\mathbf{G}_a^{\hat{\mathrm{dR}},+}$, we are reduced to checking the key case when f is the map $\mathbf{A}^1_\mathbf{Z} \to \mathrm{Spec}(\mathbf{Z})$. In this case, the

HKR filtration on $\mathrm{HH}(\mathbf{A}^1_{\mathbf{Z}}/\mathbf{Z})^{h\mathbf{T}}$ is defined as the totalization of the HKR filtration on the cosimplicial $\mathbf{E}_\infty$-ring $\mathrm{HH}(\mathbf{A}^1_{\mathbf{Z}}/\mathbf{A}^{\bullet+1}_{\mathbf{Z}})^{h\mathbf{T}}$, where $\mathbf{A}^{\bullet+1}_{\mathbf{Z}} = (\mathbf{A}^1_{\mathbf{Z}})^{\times_{\mathbf{Z}}\bullet+1}$. Since $\mathrm{Def}^\sharp_f$ is computed similarly, we are reduced to showing that the HKR filtration on $\mathrm{HH}(\mathbf{A}^1_{\mathbf{Z}}/\mathbf{A}^{\bullet+1}_{\mathbf{Z}})^{h\mathbf{T}}$ has associated graded isomorphic to the shearing of $\mathrm{R\Gamma}(\mathrm{Def}^\sharp_{f^\bullet}; \mathcal{O})$, where $f^\bullet : \mathbf{A}^{\bullet+1}_{\mathbf{Z}} \to \mathbf{A}^1_{\mathbf{Z}}$ is the multiplication map. This can be further reduced to the case $\bullet = 1$, i.e., to showing that the HKR filtration on $\mathrm{HH}(\mathbf{A}^1_{\mathbf{Z}}/\mathbf{A}^2_{\mathbf{Z}})^{h\mathbf{T}}$ has associated graded isomorphic to the shearing of $\mathrm{R\Gamma}(\mathrm{Def}^\sharp_\Delta; \mathcal{O})$. This, in turn, can be checked by direct calculation: by Example 3.2.12, we need to show that

$$\pi_*\mathrm{HH}(\mathbf{Z}[x]/\mathbf{Z}[x,y])^{h\mathbf{T}} \cong \mathbf{Z}[t][x,y]\langle \tfrac{x-y}{t}\rangle^\wedge_{(x-y)}.$$

It is easy to see that $\mathrm{HH}(\mathbf{Z}[x]/\mathbf{Z}[x,y])$ is a free $\mathbf{Z}[x]$-module; so its homotopy injects into its x-localization. We are therefore reduced to showing that

$$\pi_*\mathrm{HH}(\mathbf{Z}[x^{\pm 1}]/\mathbf{Z}[x^{\pm 1}, y^{\pm 1}])^{h\mathbf{T}} \cong \mathbf{Z}[t][x^{\pm 1}, y^{\pm 1}]\langle \tfrac{x-y}{t}\rangle^\wedge_{(x-y)}.$$

Upon replacing y with $z := y/x$, the computation reduces to showing that

$$\pi_*\mathrm{HH}(\mathbf{Z}/\mathbf{Z}[z^{\pm 1}])^{h\mathbf{T}} \cong \mathbf{Z}[t][z^{\pm 1}]\langle \tfrac{z-1}{t}\rangle^\wedge_{(z-1)}.$$

There is a $\mathbf{T}$-equivariant assembly map $\mathbf{Z}[\mathrm{B}\mathbf{Z}] \to \mathrm{HH}(\mathbf{Z}[z^{\pm 1}]/\mathbf{Z})$ which detects the class $d\log(z) = \frac{dz}{z}$ on π_1, and an isomorphism

$$\mathbf{Z} \otimes_{\mathbf{Z}[\mathrm{B}\mathbf{Z}]} \mathrm{HH}(\mathbf{Z}[z^{\pm 1}]/\mathbf{Z}) \cong \mathbf{Z}[z^{\pm 1}].$$

This implies that there is a $\mathbf{T}$-equivariant isomorphism $\mathrm{HH}(\mathbf{Z}/\mathbf{Z}[z^{\pm 1}]) \cong \mathbf{Z}[\mathrm{B}^2\mathbf{Z}]$, where $\mathbf{T}$ acts trivially on $\mathbf{Z}[\mathrm{B}^2\mathbf{Z}]$, and the generator $u \in \pi_2\mathrm{HH}(\mathbf{Z}/\mathbf{Z}[z^{\pm 1}])$ is given by the suspension of the class $d\log(z) \in \pi_1\mathrm{HH}(\mathbf{Z}[z^{\pm 1}]/\mathbf{Z})$. Since $\pi_*\mathbf{Z}[\mathrm{B}^2\mathbf{Z}] \cong \mathbf{Z}\langle u\rangle$, it follows that $\pi_*\mathrm{HH}(\mathbf{Z}/\mathbf{Z}[z^{\pm 1}])^{h\mathbf{T}}$ is isomorphic to $\mathbf{Z}[t]\langle u\rangle^\wedge_{tu}$. It follows from the construction of the assembly map that the class tu in weight 0 is in fact the power series $\log(z)$. Now the identity $\sum_{n\geq 0} \frac{\log(z)^n}{n!} w^n = \sum_{n\geq 0} w(w-1)\cdots(w-(n-1))\frac{(z-1)^n}{n!}$ implies that in fact $\mathbf{Z}[t]\langle u\rangle^\wedge_{tu} \cong \mathbf{Z}[t]\langle \frac{z-1}{t}\rangle^\wedge_{z-1}$. □

Remark 3.2.15. The strategy of proof above was also exposited in [Dev26, Section 4], where the calculation of $\pi_*\mathrm{HH}(k[x]/k[x,y])^{h\mathbf{T}}$ was generalized to the case when k is any even $\mathbf{E}_\infty$-ring.

Construction 3.2.16. Extending Construction 3.2.10, let $\mathrm{A} \to \mathrm{B}$ be a map of $\mathbf{E}_\infty$-rings, and let $n \geq 1$. Then there is a filtration on $\mathrm{HH}_{\mathbf{E}_n}(\mathrm{B}/\mathrm{A})^{h\mathbf{T}}$ defined by $\mathrm{HH}_{\mathbf{E}_n}(\mathrm{fil}^\star_{\mathrm{ev}}(\mathrm{B})/\mathrm{fil}^\star_{\mathrm{ev}}(\mathrm{A}))^{h\mathbf{T}}$; we will denote it by $\mathrm{fil}^\star\mathrm{HH}_{\mathbf{E}_n}(\mathrm{B}/\mathrm{A})^{h\mathbf{T}}$. Let $\mathrm{Y} = \mathrm{Spec}(\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{A}))$ and $\mathrm{X} = \mathrm{Spec}(\mathrm{gr}^\star_{\mathrm{ev}}(\mathrm{B}))$, so that there is a map $f : \mathrm{X} \to \mathrm{Y}$. Let $\Delta^{(n)}_f$ denote the n-fold diagonal, defined inductively as the pullback of $\Delta^{(n-1)}_f$ along itself, where $\Delta^{(0)}_f = f$. The associated graded $\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_n}(\mathrm{B}/\mathrm{A})^{h\mathbf{T}}$ is isomorphic upon inverse shearing to $\mathrm{HH}_{\mathbf{E}_n}(\mathrm{X}/\mathrm{Y})^{h\mathbf{T}}$. This in turn admits a HKR filtration with associated graded given by

$$\mathrm{gr}^*_{\mathrm{HKR}}\mathrm{sh}^{-1}(\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_n}(\mathrm{B}/\mathrm{A})^{h\mathbf{T}}) \cong \mathrm{sh}(\mathrm{R\Gamma}(\mathrm{Def}^\sharp_{\Delta^{(n-1)}_f}; \mathcal{O})).$$

It follows in particular that

$$\mathrm{gr}^*_{\mathrm{HKR}}\mathrm{sh}^{-1}(\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}}) \cong \mathrm{sh}(\mathrm{R\Gamma}(\mathrm{Def}^\sharp_{\Delta_f}; \mathcal{O})),$$

where $f : \mathrm{Spec}(\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})) \to \mathrm{Spec}(\mathrm{gr}^\star_{\mathrm{ev}}(k))$. We have described $\mathrm{Spec}(\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}}))$ in Definition 2.2.28 as $\mathrm{T}_{\mathrm{F}}/\mathcal{W}$, so it follows that:

Proposition 3.2.17. *Suppose k is an even $\mathbf{E}_\infty$-ring, and let $\mathrm{T_F}/\mathcal{W}$ be as in Definition 2.2.23. Then there is an isomorphism*

$$\mathrm{gr}^*_{\mathrm{HKR}}\mathrm{sh}^{-1}(\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}}) \cong \mathrm{sh}(\mathrm{R\Gamma}(\mathrm{Def}^\sharp_{\Delta:\mathrm{T_F}/\mathcal{W}\to\mathrm{T_F}/\mathcal{W}\times_{\pi_*(k)}\mathrm{T_F}/\mathcal{W}};\mathcal{O})).$$

By Proposition 3.2.3, this in turn gives a description of the associated graded of a filtration on the associated graded of a filtration (!) on $\mathrm{R\Gamma}_{\mathrm{G}_\mathcal{O}\rtimes\mathbf{T}}(\mathcal{G}r_\mathrm{G};k)$ in terms of the divided power deformation to the normal cone of the diagonal of $\mathrm{T_F}/\mathcal{W}$.

Remark 3.2.18. It follows similarly that if $f:\mathrm{T_F}/\mathcal{W}\to\mathrm{Spec}(\pi_*(k))$ is the canonical map, then

$$\mathrm{gr}^*_{\mathrm{HKR}}\mathrm{sh}^{-1}(\mathrm{gr}^\star\mathrm{HH}(k^{h\mathrm{G}}/k)^{h\mathbf{T}}) \cong \mathrm{sh}(\mathrm{R\Gamma}(\mathrm{Def}^\sharp_f;\mathcal{O})).$$

Remark 3.2.4 identifies the left-hand side with the associated graded of a filtration on the associated graded of a filtration on $\mathrm{C}^*_\mathrm{G}(\mathrm{G};k)^{h\mathbf{T}}$, i.e., on the $\mathbf{T}$-equivariant cohomology of the free loop space of G. It follows from the definition that $\mathrm{Def}^\sharp_f\cong((\mathrm{T_F}/\mathcal{W})/\pi_*(k))^{\mathrm{dR},+}$, so that $\mathrm{gr}^*_{\mathrm{HKR}}\mathrm{sh}^{-1}(\mathrm{gr}^\star\mathrm{HH}(k^{h\mathrm{G}}/k)^{h\mathbf{T}})$ is isomorphic to the shearing of the Hodge filtration on the derived de Rham cohomology of $\mathrm{T_F}/\mathcal{W}$ relative to $\pi_*(k)$. This is analogous to (and can be viewed as a generalization of) a result of Brylinski-Zhang [BZ00].

Example 3.2.19. Suppose k is an even $\mathbf{Z}$-algebra in which all the torsion primes of G are inverted. Let W be the Weyl group of G, and let $d_1,\cdots,d_n$ denote the fundamental degrees of W. Then $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$ is a completed polynomial $\pi_*(k)$-algebra on generators in weights $-2d_1,\cdots,-2d_n$ (where the completion is in the sense of Remark A.1.6). It follows that $\mathrm{L}_{\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})/\pi_*(k)}$ is a free $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$-module on generators in weights $-2d_1,\cdots,-2d_n$. Proposition 3.2.17 implies that

$$\mathrm{Spec}\,\mathrm{sh}^{-1}(\mathrm{gr}^*_{\mathrm{HKR}}\mathrm{sh}^{-1}(\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}}))\cong\mathrm{Def}^\sharp_{\Delta:\hat{\mathfrak{t}}(2)/\mathcal{W}\to\hat{\mathfrak{t}}(2)/\mathcal{W}\times_{\pi_*(k)}\hat{\mathfrak{t}}(2)/\mathcal{W}}.$$

If $k=\mathbf{Q}$, for instance, this is just the deformation to the normal cone of the diagonal of $\hat{\mathfrak{t}}(2)/\mathcal{W}\cong\hat{\mathfrak{t}}(2)/\!\!/\mathrm{W}$. Both the filtrations $\mathrm{fil}^\star_{\mathrm{HKR}}$ and $\mathrm{fil}^\star$ split in this case, so we find that $\mathrm{Spec}\,\pi_*\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}}$ is isomorphic to the deformation to the normal cone of the diagonal of $\hat{\mathfrak{t}}(2)/\mathcal{W}\cong\hat{\mathfrak{t}}(2)/\!\!/\mathrm{W}$, which recovers [BF08, Theorem 1]. Similarly, when $k=\mathbf{Q}$, Remark 3.2.18 specializes to the statement that $\mathrm{H}^*_\mathrm{G}(\mathrm{G};k)^{h\mathbf{T}}$, i.e., the $\mathbf{T}$-equivariant cohomology of the free loop space of G, is isomorphic to the Rees construction with respect to the variable t of the Hodge-filtered de Rham cohomology of $\hat{\mathfrak{t}}(2)/\!\!/\mathrm{W}$.

Example 3.2.20. While the HKR filtration on $\mathrm{sh}(\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k))$ might split, the filtration $\mathrm{fil}^\star\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)$ itself rarely splits (except if k is a $\mathbf{Q}$-algebra). For example, suppose $\mathrm{G}=\mathbf{C}^\times$ and $k=\mathbf{Z}$. Then $k^{h\mathrm{G}}$ is the free binomial ring $\mathrm{LBin}_\mathbf{Z}(\mathbf{Z}[-2])$ in the sense of [KSZ23], so that $\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)\cong\mathrm{LBin}_\mathbf{Z}(\mathbf{Z}[-2]\oplus\mathbf{Z})$. In particular, $\pi_0\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)$ is isomorphic to $\mathbf{Z}[\binom{x}{n}]_{n\geq 0}$; this also follows from Example 3.2.8. However, $\pi_0\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)\cong\Gamma_\mathbf{Z}(x)$, which is not isomorphic to $\mathbf{Z}[\binom{x}{n}]_{n\geq 0}$. In other words, there are many interesting multiplicative extensions in assembling $\pi_*\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)$ from $\pi_*\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)$.

Remark 3.2.21. Proposition 3.2.17 is helpful in getting a handle on the "size" of $\pi_*\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)$, but it is a very poor approximation to the algebraic richness of $\pi_*\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)$. We have already seen this in Example 3.2.20 when G is a torus. Another example is the case $\mathrm{G}=\mathrm{SL}_2$. In this case, $k^{h\mathrm{G}}$ is concentrated in even degrees for any even $\mathbf{E}_\infty$-ring k, and $\pi_*(k^{h\mathrm{G}})\cong\pi_*(k)[y^2]^\wedge$ with y^2 in weight -4. The algebra $\mathrm{sh}^{-1}\mathrm{gr}^*_{\mathrm{HKR}}\mathrm{sh}^{-1}(\mathrm{gr}^\star \mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}})$ is just $\pi_*(k)[x, \hbar, \frac{y^2-x^2}{\hbar}]^\wedge$. However, if k is not a $\mathbf{Q}$-algebra, then $\pi_*\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}} \cong \mathrm{H}^{-*}_{\mathrm{G}\times\mathbf{T}}(\mathcal{G}r_\mathrm{G}; k)$ is much more complicated, as seen from Example 3.2.9. Again, there are many interesting multiplicative extensions in assembling $\pi_*\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}}$ from $\pi_*\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}}$.

Our discussion in this section only describes a way of computing $\pi_*\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}}$, and unfortunately does not describe $\pi_*\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}}$ itself. We will return to the question of assembling $\mathrm{gr}^\star\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}}$ into $\mathrm{HH}_{\mathbf{E}_2}(k^{h\mathrm{G}}/k)^{h\mathbf{T}}$ in future work (related to Section 5.3).

We will now set $\mathbf{T}$-equivariance aside, and give a direct proof of the following result (which will not be used in our later discussions, but is rather interesting nonetheless).

Proposition 3.2.22. *Let k be an even $\mathbf{E}_\infty$-ring, and assume that 2, the bad primes (see Table 1) of G, and the order of the center of G are invertible in $\pi_0(k)$.*[6] *Then the $\mathbf{E}_2$-$k^{h\mathrm{G}}$-algebra $\mathfrak{Z}(k^{h\mathrm{T}}/k^{h\mathrm{G}})$ is concentrated in even degrees.*

Proof. For notational simplicity, let $\mathrm{A} = \mathrm{gr}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}})$ and $\mathrm{B} = \pi_*(k^{h\mathrm{T}})$. The filtered $\mathbf{E}_2$-$\mathrm{fil}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}})$-algebra $\mathfrak{Z}(\mathrm{fil}^\star_{\mathrm{T}\text{-ev}}(k^{h\mathrm{T}})/\mathrm{fil}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}}))$ on $\mathfrak{Z}(k^{h\mathrm{T}}/k^{h\mathrm{G}})$ has associated graded $\mathfrak{Z}(\mathrm{B}/\mathrm{A})$, so it suffices to prove that $\mathfrak{Z}(\mathrm{B}/\mathrm{A})$ is concentrated in even degrees. By definition of the center, $\mathfrak{Z}(\mathrm{B}/\mathrm{A}) \cong \mathrm{REnd}_{\mathrm{B}\otimes_\mathrm{A}\mathrm{B}}(\mathrm{B})$. It follows from Definition 2.2.28 that $\mathrm{B}\otimes_\mathrm{A}\mathrm{B} \cong \mathrm{H}^{-*}_\mathrm{T}(\mathcal{B}; k)$, so we need to show that $\mathrm{REnd}_{\mathrm{H}^{-*}_\mathrm{T}(\mathcal{B};k)}(\pi_*(k^{h\mathrm{T}}))$ is concentrated in even degrees. Here, $\mathrm{H}^{-*}_\mathrm{T}(\mathcal{B}; k)$ is an augmented $\pi_*(k^{h\mathrm{T}})$-algebra via the inclusion of the basepoint in $\mathcal{B}$. Let $\mathbf{Z}'$ denote the localization of $\mathbf{Z}$ at the primes in the statement of Proposition 3.2.22. We will argue that $\mathrm{REnd}_{\mathrm{H}^{-*}_\mathrm{T}(\mathcal{B};\mathbf{Z}')}(\pi_*({\mathbf{Z}'}^{h\mathrm{T}}))$ is in fact a free $\mathbf{Z}'$-module concentrated in even degrees; for this, it suffices to show that if p is odd and not a torsion prime, then $\mathrm{REnd}_{\mathrm{H}^{-*}_\mathrm{T}(\mathcal{B};\mathbf{F}_p)}(\pi_*(\mathbf{F}_p^{h\mathrm{T}}))$ is concentrated in even degrees. Using the Atiyah-Hirzebruch spectral sequence, this in turn implies that $\mathrm{REnd}_{\mathrm{H}^{-*}_\mathrm{T}(\mathcal{B};k)}(\pi_*(k^{h\mathrm{T}}))$ is a free $\pi_*(k)$-algebra concentrated in even degrees, as desired.

For this, consider the following more general setup. Let R be a commutative ring, and let S be an augmented commutative R-algebra which is a complete intersection over R. The ring $\mathrm{REnd}_\mathrm{S}(\mathrm{R})$ is the R-linear dual of $\mathrm{R}\otimes_\mathrm{S}\mathrm{R}$. If I denotes the fiber of the map $\mathrm{R}\otimes_\mathrm{S}\mathrm{R}\to\mathrm{R}$, then the I-adic filtration on $\mathrm{R}\otimes_\mathrm{S}\mathrm{R}$ defines a filtration on $\mathrm{REnd}_\mathrm{S}(\mathrm{R})$ whose associated graded is isomorphic to $\mathrm{L}\Gamma_\mathrm{R}(\mathrm{T}_{\mathrm{R}/\mathrm{S}})$, so if the latter is even, then $\mathrm{REnd}_\mathrm{S}(\mathrm{R})$ is also even. The tangent complex $\mathrm{T}_{\mathrm{R}/\mathrm{S}}$ is isomorphic to $\mathrm{T}_{\mathrm{S}/\mathrm{R}}\otimes_\mathrm{S}\mathrm{R}[-1]$, which can be computed directly as follows. Write $\mathrm{S} = \mathrm{R}[x_1,\cdots,x_n]/(f_1,\cdots,f_n)$ so that the map $\mathrm{S}\to\mathrm{R}$ sends $x_i\mapsto 0$ (in particular, $f_i(0)=0$). Let $\mathrm{J} = (\partial_{x_j}f_i)(0)$ denote the Jacobian matrix evaluated at zero. Then $\mathrm{L}_{\mathrm{S}/\mathrm{R}}\otimes_\mathrm{S}\mathrm{R}$ is given by the two-term complex $\mathrm{J}:\mathrm{R}^n\to\mathrm{R}^n$, so that $\mathrm{L}\Gamma_\mathrm{R}(\mathrm{T}_{\mathrm{R}/\mathrm{S}}) \cong \mathrm{R}[y_1,\cdots,y_n]/\mathrm{J}^\mathrm{T}\vec{y}$ where each y_i lives in homological degree -2, J^T is the transpose of J, and $\vec{y} = (y_1,\cdots,y_n)$. In order for $\mathrm{L}\Gamma_\mathrm{R}(\mathrm{T}_{\mathrm{R}/\mathrm{S}})$ to be even, the sequence of linear forms $\mathrm{J}^\mathrm{T}\vec{y}$ must be regular in $\mathrm{R}[y_1,\cdots,y_n]$. This in turn happens if for every $1\le j\le n$, the depth of the ideal $\mathrm{I}_j(\mathrm{J}^\mathrm{T})$ in R generated by the $j\times j$-minors of J^T is at least $n-j+1$. Let us call this the "depth condition".

Returning to our task, recall from Proposition 2.2.22 that $\mathrm{H}^{-*}_\mathrm{T}(\mathcal{B};\mathbf{F}_p)$ is a complete intersection over $\pi_*(\mathbf{F}_p^{h\mathrm{T}}) = \mathcal{O}_\mathfrak{t}$. (We will ignore the grading on $\mathfrak{t}$ below.) In

[6] I expect the conclusion to be true more generally if k is good for G (for instance, for $k = \mathrm{MU}$ by Lemma 2.2.20 without inverting any primes).

fact, since p is not a bad prime for G, $\mathbf{F}_p^{h\mathrm{G}}$ is concentrated in even degrees, where it agrees with $\mathcal{O}_{\mathfrak{g}}^{\mathrm{G}}$. This implies that there is an isomorphism $\mathrm{H}_{\mathrm{T}}^{-*}(\mathcal{B}; \mathbf{F}_p) \cong \mathcal{O}_{\mathfrak{t}} \otimes_{\mathcal{O}_{\mathfrak{g}}^{\mathrm{G}}} \mathcal{O}_{\mathfrak{t}}$, where as usual the tensor product is *a priori* derived (but it is in fact a classical tensor product because $\mathrm{H}_{\mathrm{T}}^{-*}(\mathcal{B}; \mathbf{F}_p)$ is a complete intersection). Choose a basis $x_1, \cdots, x_n$ for $\mathfrak{t}$ over $\mathbf{F}_p$, and write $\mathcal{O}_{\mathfrak{t}}^{\mathrm{W}} = \mathbf{F}_p[f_1, \cdots, f_n]$; then $\mathrm{J} = (\partial_{x_j} f_i)$. To check the depth condition, it suffices to show that the set $\mathrm{V}(\mathrm{I}_j(\mathrm{J}^{\mathrm{T}})) \subseteq \mathfrak{t}$ of points $x \in \mathfrak{t}$ where $\mathrm{J}^{\mathrm{T}}|_x$ has rank $\leq j-1$ is of codimension $n-j+1$. We will momentarily review the proof of the main theorem of [Ste60], which states that $\dim(\ker(\mathrm{J}|_x))$ is equal to the maximum number m_x of root hyperplanes which contain x. In fact, we will only need that $\dim(\ker(\mathrm{J}|_x)) \leq m_x$. Since $\mathrm{rank}(\mathrm{J}^{\mathrm{T}}|_x) = n - \dim(\ker(\mathrm{J}|_x))$, the condition $\mathrm{rank}(\mathrm{J}^{\mathrm{T}}|_x) \leq j-1$ means that $n-j+1 \leq \dim(\ker(\mathrm{J}|_x)) \leq m_x$. Taking the union over all x such that $\mathrm{rank}(\mathrm{J}^{\mathrm{T}}|_x) \leq j-1$ shows that $\mathrm{V}(\mathrm{I}_j(\mathrm{J}^{\mathrm{T}}))$ has codimension at least $n-j+1$, as desired.

It remains to observe that the relevant part of the main theorem of [Ste60] continues to hold away from the bad primes and $p=2$. Let α be a positive root, and assume that the basis $x_1, \cdots, x_n$ for $\mathfrak{t}$ has $x_1 = \alpha$. We first note that if $g \in \mathcal{O}_{\mathfrak{t}}^{\mathrm{W}}$, then α divides $\partial_\alpha g$. Since g is W-invariant, $s_\alpha g = g$. Since $p \neq 2$, this implies that $g \in \mathbf{F}_p[\alpha^2, x_2, \cdots, x_n]$; it follows that 2α divides $\partial_\alpha g$, and so α divides $\partial_\alpha g$ since $p \neq 2$. Since α divides $\partial_\alpha g$ for an arbitrary invariant g, it follows that for any sequence $g_1, \cdots, g_n \in \mathcal{O}_{\mathfrak{t}}^{\mathrm{W}}$, the associated Jacobian matrix $\mathrm{J}_g = (\partial_{x_j} g_i)$ has $\det(\mathrm{J}_g)$ divisible by α for every positive root α. Because p is not a bad prime and does not divide the order of the center of G, $\mathbf{F}_p$ is good for G by Example 2.2.18 (in the sense of Definition 2.2.17), so the positive roots of G are coprime in $\mathcal{O}_{\mathfrak{t}}$, and therefore $\det(\mathrm{J}_g)$ is divisible by $\prod_{\alpha \in \Phi^+} \alpha$. Note also that if $g_1, \cdots, g_n$ is a basic set of W-invariants (so that $\mathrm{J}_g = \mathrm{J}$), then $\det(\mathrm{J})$ is a polynomial of degree $\sum_{i=1}^n (d_i - 1) = |\Phi^+|$, where $d_1, \cdots, d_n$ are the fundamental degrees of W. It follows that $\det(\mathrm{J}) = c \prod_{\alpha \in \Phi^+} \alpha$ for some $c \in \mathbf{F}_p$. Note that $c \neq 0$ because $\det(\mathrm{J}) \neq 0$ (since p is not a bad prime). Now we can run Steinberg's argument for the implication "(c) implies (a)" (called "$m \leq k$" in [Ste60]) in the main theorem of [Ste60]. □

Remark 3.2.23. Arguing as in Proposition 3.2.3 shows that if k is a connective even $\mathbf{E}_\infty$-ring such that $\pi_0(k)$ is a perfect k-module, the $\pi_0(k)$-completion of $\mathfrak{Z}(k^{h\mathrm{T}}/k^{h\mathrm{G}})$ is isomorphic to $\mathrm{C}_*^{\mathrm{T},\mathrm{BM}}(\Omega(\mathrm{G}/\mathrm{T}); k)$. This can be used to explain why Proposition 3.2.22 is somewhat surprising from both an algebraic and topological point of view. First, $\mathrm{C}_*^{\mathrm{T},\mathrm{BM}}(\Omega(\mathrm{G}/\mathrm{T}); k) \cong \mathrm{C}_*^{\mathrm{BM}}(\Omega(\mathrm{G}/\mathrm{T}); k)^{h\mathrm{T}}$ is typically computed from a homotopy fixed point spectral sequence

$$\mathrm{E}_2 \cong \mathrm{H}^{-*}(\mathrm{BT}; \mathrm{H}_*^{\mathrm{BM}}(\Omega(\mathrm{G}/\mathrm{T}); k)) \Rightarrow \mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\Omega(\mathrm{G}/\mathrm{T}); k).$$

However, $\mathrm{H}_*^{\mathrm{BM}}(\Omega(\mathrm{G}/\mathrm{T}); k)$ is *not* concentrated in even degrees (unless G is a torus), so the E_2-page is not concentrated in even degrees. In order for $\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\Omega(\mathrm{G}/\mathrm{T}); k)$ to be concentrated in even degrees, there must therefore be several differentials in this spectral sequence, whose existence is not obvious at all.

Alternatively, Proposition 3.2.22 is surprising from an algebraic point of view for the following reason. The equivariant homology $\mathrm{C}_*^{\mathrm{T},\mathrm{BM}}(\Omega(\mathrm{G}/\mathrm{T}); k)$ is isomorphic to $\mathrm{C}_*^{\mathrm{T},\mathrm{BM}}(\Omega\mathrm{G}; k) \otimes_{\mathrm{C}_*^{\mathrm{T},\mathrm{BM}}(\Omega\mathrm{T}; k)} k^{h\mathrm{T}}$, so that there is a Tor-spectral sequence

$$\mathrm{E}^1 \cong \mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\Omega\mathrm{G}; k) \otimes_{\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\Omega\mathrm{T}; k)} \pi_*(k^{h\mathrm{T}}) \Rightarrow \mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\Omega(\mathrm{G}/\mathrm{T}); k).$$

The E^1-page can be computed using Theorem 4.2.12 below as the ring of functions on the derived fiber product $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \times_{\check{\mathrm{T}}\times\mathrm{T}_\mathrm{F}} \mathrm{T}_\mathrm{F}$ (see Section 4.2, and in particular Definition 4.2.4, for unexplained notation). The E^1-page is not obviously concentrated in degree zero, because the map $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \to \check{\mathrm{T}} \times \mathrm{T}_\mathrm{F}$ from Definition 4.2.4 is not flat. (Nevertheless, it turns out that the E^1-page is indeed concentrated in degree zero, and this can be argued similarly to Proposition 3.2.22.)

Remark 3.2.24. One might be tempted to argue that Proposition 3.2.22 continues to hold if merely the torsion primes of G were inverted. The only cases of torsion primes which are not bad primes are (G, p) being $(\mathrm{Sp}_{2n}, 2)$ and $(\mathrm{G}_2, 3)$. However, in these cases, the desired evenness fails when $k = \mathbf{F}_p$. For instance, for $(\mathrm{G}, p) = (\mathrm{SL}_2, 2)$, $\mathrm{H}_\mathrm{T}^{-*}(\mathcal{B}; \mathbf{F}_p) \cong \mathcal{O}_\mathfrak{t}[x]/x^2$ where the map $\mathrm{H}_\mathrm{T}^{-*}(\mathrm{G}/\mathrm{B}; \mathbf{F}_p) \to \pi_*(\mathbf{F}_p^{h\mathrm{T}})$ sends $x \mapsto 0$. This implies that $\mathrm{REnd}_{\mathrm{H}_\mathrm{T}^{-*}(\mathcal{B};\mathbf{F}_p)}(\mathcal{O}_\mathfrak{t}) \cong \mathbf{F}_2[x,y]/2xy$ (where this means the derived cofiber). But $2y = 0$, and it follows from this that $\mathrm{H}_*^\mathrm{T}(\Omega(\mathrm{SL}_2/\mathrm{T}); \mathbf{F}_2)$ is not concentrated in even degrees; note, on the other hand, that $\mathrm{H}_*^\mathrm{T}(\Omega(\mathrm{PGL}_2/\mathrm{T}); \mathbf{F}_2)$ *is* concentrated in even degrees. A similar argument works in type C_n at $p = 2$ for arbitrary n. Similarly, for $(\mathrm{G}, p) = (\mathrm{G}_2, 3)$, $\pi_*(\mathbf{F}_3^{h\mathrm{G}_2}) \cong \mathbf{F}_3[x_1^2+x_2^2+x_1x_2, x_1^2x_2^2(x_1+x_2)^2]$ as a subalgebra of $\mathcal{O}_\mathfrak{t} \cong \mathbf{F}_3[x_1, x_2]$. The Jacobian matrix is therefore

$$\mathrm{J} = \begin{pmatrix} 2x_1+x_2 & 4x_1^3x_2^2+2x_1x_2^4+6x_1^2x_2^3 \\ 2x_2+x_1 & 4x_1^2x_2^3+2x_1^4x_2+6x_1^3x_2^2 \end{pmatrix} \equiv \begin{pmatrix} x_2-x_1 & x_1x_2^2(x_1^2-x_2^2) \\ x_1-x_2 & x_1^2x_2(x_2^2-x_1^2) \end{pmatrix} \pmod 3.$$

It is now clear that $\mathrm{J}^\mathrm{T}\vec{y} = ((x_1-x_2)(y_2-y_1), x_1x_2(x_1^2-x_2^2)(x_2y_1-x_1y_2))$ is not a regular sequence in $\mathbf{F}_3[x_1, x_2, y_1, y_2]$, since they share the common factor $x_1 - x_2$. This implies that $\mathrm{REnd}_{\mathrm{H}_\mathrm{T}^{-*}(\mathcal{B};\mathbf{F}_p)}(\mathcal{O}_\mathfrak{t})$ is not concentrated in even degrees, and in fact $\mathfrak{Z}(\mathbf{F}_3^{h\mathrm{T}}/\mathbf{F}_3^{h\mathrm{G}_2})$ is itself not concentrated in even degrees.

3.3. **Poisson structures.**

The equivariant homology $\mathrm{H}_*^{\mathrm{G}_\mathcal{O},\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Q})$ admits the structure of a graded Poisson $\mathbf{Q}$-algebra, where the Poisson bracket has weight 2. There are two equivalent ways of constructing this Poisson structure. The first, more traditional approach in geometric representation theory, is by noting that $\mathrm{H}_*^{\mathrm{G}\times\mathbf{T},\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Q})$ is an associative deformation of $\mathrm{H}_*^{\mathrm{G}_\mathcal{O},\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Q})$ to $\mathbf{Q}[\hbar]$, so the reduction of the commutator modulo $\hbar$ produces a Poisson bracket on $\mathrm{H}_*^{\mathrm{G}_\mathcal{O},\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Q})$; here, $\mathbf{T} \cong \mathbf{C}^\times$, acting by loop-rotation. The second, more homotopy-theoretic approach, is by observing that $\mathrm{C}_*^{\mathrm{G}_\mathcal{O},\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Q})$ is an $\mathbf{E}_3$-$\mathbf{Q}$-algebra, and the homotopy groups of any such object acquires the structure of a graded Poisson $\mathbf{Q}$-algebra where the Poisson bracket has weight 2. The equivariant k-homology of $\mathcal{G}\mathrm{r}_\mathrm{G}$ for an arbitrary (even) $\mathbf{E}_\infty$-ring k still acquires a Poisson bracket of weight 2, and we will give a description of it via Theorem 3.1.11.

Construction 3.3.1. Let R be a graded commutative ring with a 1-dimensional formal group law F, and let $\mathrm{T}_\mathrm{F} = \mathrm{Hom}(\mathbb{X}^*(\mathrm{T}), \mathrm{F})$. Let θ denote a generating invariant 1-form on F; concretely, if x is a coordinate on F, then $\theta = u(x)dx$ for some power series $u(x)$ with invertible constant term. Then θ defines a 1-form $\sum_\mu \theta_\mu$ on T_F. There is a symplectic form ω_F on $\mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_\mathrm{F}} = \mathcal{O}_{\mathrm{T}_\mathrm{F}}[\mathbb{X}_*(\mathrm{T})]$ given by $\sum_\mu \theta_\mu \wedge d\log(e^\mu) \in \Omega^2_{(\check{\mathrm{T}}\times\mathrm{T}_\mathrm{F})/\mathrm{R}}$. This in turn defines a Poisson bracket of weight 2 on $\mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_\mathrm{F}}$, determined by $\{x_\mu, e^\lambda\} = \delta_{\mu\lambda}e^\lambda$.

Remark 3.3.2. In Definition 5.3.1, we define a graded associative $\pi_*(k^{h\mathbf{T}})$-algebra $\mathrm{D}_{\check{\mathrm{T}}}^\mathrm{F}$. If we write $\pi_*(k^{h\mathbf{T}}) \cong \pi_*(k)[\hbar]^\wedge$, where the completion is taken in the sense of Remark A.1.6, then $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}/\hbar \cong \mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}}$. Moreover, the reduction modulo $\hbar$ of the commutator in $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$ defines a Poisson bracket on $\mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}}$. This is precisely the Poisson bracket from Construction 3.3.1.

Recall the $\mathcal{W}$-equivariant Hopf R-algebra $\mathcal{A}_{\mathrm{G}}$ from Definition 3.1.10.

Proposition 3.3.3. *Suppose* R *is good for* G *in the sense of Definition 2.2.17, and if* G *has a simple factor of adjoint type* B_n $(n \geq 1)$*, assume further that* 2 *is a unit in* R*. Then the* 2*-form* ω_{F} *on* $\mathcal{A}_{\mathrm{T}} = \pi_*(k^{h\mathrm{T}})[\mathbb{X}_*(\mathrm{T})]$ *from Construction 3.3.1 extends to a* 2*-form on* $\mathcal{A}_{\mathrm{G}}$*, and the cyclic* $\mathcal{A}_{\mathrm{G}}$*-submodule of* $\Omega^2_{\mathcal{A}_{\mathrm{G}}/\mathrm{R}}$ *generated by* ω_{F} *is isomorphic to* $e(\Phi)\Omega^2_{\mathcal{A}_{\mathrm{G}}/\mathrm{R}} \subseteq \Omega^2_{\mathcal{A}_{\mathrm{G}}/\mathrm{R}}$*. Moreover, the Poisson bracket associated to* ω_{F} *is regular (i.e., has no poles) on the* W*-invariant subring* $(\mathcal{A}_{\mathrm{G}})^{\mathrm{W}}$*.*

The proof will require a calculation with formal group laws.

Lemma 3.3.4. *Let* x *be a coordinate on* F*, let* $\overline{x} := [-1](x)$ *be the inverse of* x *in the group law* F*, and let* $q(x) = \partial_y(x +_{\mathrm{F}} y)|_{y=0}$*. If we write* $\theta = u(x)dx$*, then* $u(x) = \frac{u(0)}{q(x)}$*. Moreover, the fraction* $\frac{1+\partial_x(\overline{x})}{x-\overline{x}}$ *does not have poles in* x*, i.e., is well-defined in* $\mathrm{R}[x]^\wedge$*.*

Proof. Since $u(x)dx$ is an invariant 1-form,

$$u(x +_{\mathrm{F}} y)(\partial_x(x +_{\mathrm{F}} y)dx + \partial_y(x +_{\mathrm{F}} y)dy) = u(x)dx + u(y)dy.$$

Comparing coefficients of dy, we find that $u(y) = u(x +_{\mathrm{F}} y)\partial_y(x +_{\mathrm{F}} y)$. Taking $y = 0$, it follows that $u(0) = u(x)\partial_y(x +_{\mathrm{F}} y)|_{y=0} = u(x)q(x)$. But $q(x)$ has constant term 1, so it is a unit, and $u(x) = \frac{u(0)}{q(x)}$ as desired.

We now show that $\frac{1+\partial_x(\overline{x})}{x-\overline{x}} \in \mathrm{R}[x]^\wedge$. Observe that translation invariance of $u(x)dx$ implies that $q(x +_{\mathrm{F}} y)\partial_x(x +_{\mathrm{F}} y) = q(x)$; by symmetry, $q(x +_{\mathrm{F}} y)\partial_y(x +_{\mathrm{F}} y) = q(y)$. Taking $y = \overline{x}$ and using $q(0) = 1$, we find that $\partial_x(x +_{\mathrm{F}} y)|_{y=\overline{x}} = q(x)$ and $\partial_y(x +_{\mathrm{F}} y)|_{y=\overline{x}} = q(\overline{x})$. Since $x +_{\mathrm{F}} \overline{x} = 0$, the chain rule implies that

$$\partial_x(x +_{\mathrm{F}} y)|_{y=\overline{x}} + \partial_y(x +_{\mathrm{F}} y)|_{y=\overline{x}}\partial_x(\overline{x}) = 0.$$

Rearranging, we find that $\partial_x(\overline{x}) = -\frac{q(x)}{q(\overline{x})}$. Now it follows that

$$\tfrac{1+\partial_x(\overline{x})}{x-\overline{x}} = q(\overline{x})^{-1}\tfrac{q(\overline{x})-q(x)}{x-\overline{x}},$$

but this lies in $\mathrm{R}[x]^\wedge$ because $q(\overline{x})$ is a unit and $\frac{q(\overline{x})-q(x)}{x-\overline{x}}$ lies in $\mathrm{R}[x]^\wedge$ (this is true for any power series). $\square$

Proof of Proposition 3.3.3. Since $e(\Phi) = \prod_{\alpha\in\Phi^+} e(\alpha)$ and R is good for G by assumption (so that the $e(\alpha)$ are coprime), it suffices to show that for each $\alpha \in \Phi^+$, ω_{F} extends to a 2-form on $\mathcal{A}_{\mathrm{G}}[e(\Phi - \{\alpha\})^{-1}]$, and the cyclic $\mathcal{A}_{\mathrm{G}}[e(\Phi - \{\alpha\})^{-1}]$-submodule of $\Omega^2_{\mathcal{A}_{\mathrm{G}}[e(\Phi-\{\alpha\})^{-1}]/\mathrm{R}}$ generated by ω_{F} is isomorphic to the submodule of $\Omega^2_{\mathcal{A}_{\mathrm{G}}[e(\Phi-\{\alpha\})^{-1}]/\mathrm{R}}$ generated by $e(\alpha)$. It follows from the definition of $\mathcal{A}_{\mathrm{G}}$ that we may replace G by a Levi subgroup L with semisimple rank 1; in fact, we may further replace L by its derived subgroup, which is isomorphic either to SL_2 or PGL_2. We may therefore assume G is one of these groups. First consider the case $\mathrm{G} = \mathrm{SL}_2$. Upon choosing a coordinate on F, we obtain an isomorphism $\mathcal{A}_{\mathrm{G}} \cong \mathrm{R}[x]^\wedge[a^{\pm 1}, \frac{a-1}{[2](x)}]$. If we write $b = \frac{a-1}{[2](x)}$, then $\omega = \theta \wedge d\log(a) = \frac{u(x)[2](x)}{a}dx \wedge db$, where we recall that $\theta = u(x)dx$ for some unit power series $u(x)$. Since $\frac{u(x)}{a}$ is a unit, ω is a unit multiple of $[2](x)dx \wedge db$, as desired.

Similarly, suppose $\mathrm{G} = \mathrm{PGL}_2$; note that this case is possible if and only if the original group G has a simple factor of adjoint type B_n and α is a short root (see [KW76, Proposition 2.1]), so that our assumptions require 2 to be a unit in R. Then $\mathcal{A}_{\mathrm{G}} \cong \mathrm{R}[x]^\wedge[a^{\pm 1}, \frac{a^2-1}{x}]$. If we write $b = \frac{a^2-1}{x}$, then $da = \frac{bdx+xdb}{2a}$. It follows that $\omega = \theta \wedge d\log(a) = \frac{u(x)x}{2a^2} dx \wedge db$. Again, this is a unit multiple of $xdx \wedge db$, as desired.

It remains to check that the Poisson bracket associated to ω_{F} is regular (i.e., has no poles) on the W-invariant subring $(\mathcal{A}_{\mathrm{G}})^{\mathrm{W}}$. Since W is generated by simple reflections, and $\mathcal{A}_{\mathrm{G}}$ is defined as the intersection of the subrings $\mathcal{A}_{\mathrm{G}}[e(\Phi - \{\alpha\})^{-1}] \subseteq \mathcal{A}_{\mathrm{G}}[e(\Phi)^{-1}]$, it suffices to to show that the Poisson bracket associated to ω_{F} is regular (i.e., has no poles in $e(\alpha)$) on $\mathcal{A}_{\mathrm{G}}[e(\Phi - \{\alpha\})^{-1}]^{\langle s_\alpha \rangle}$. As before, this reduces us to the cases $\mathrm{G} = \mathrm{SL}_2, \mathrm{PGL}_2$. First consider the case $\mathrm{G} = \mathrm{SL}_2$, so that $\mathcal{A}_{\mathrm{G}} \cong \mathrm{R}[x]^\wedge[a^{\pm 1}, \frac{a-1}{[2](x)}]$ and the $\mathbf{Z}/2$-action sends $a \mapsto a^{-1}$ and $x \mapsto \overline{x}$. Note that $x - \overline{x}$ is a unit multiple of $[2](x)$, since both reduce modulo x^2 to $2x$. The invariant subring $\mathcal{A}_{\mathrm{G}}^{\mathrm{W}}$ is then generated by

$$\mathrm{U} = a + a^{-1},\ y = -x\overline{x},\ \mathrm{V} = \tfrac{a-a^{-1}}{x-\overline{x}}.$$

These satisfy a somewhat complicated relation, but to prove the regularity of the Poisson bracket, it suffices to check that the Poisson bracket between these generators has no poles. This is a direct calculation, starting from the bracket $\{x, a\} = au(x)$, where we recall that $\theta = u(x)dx$ for some unit power series $u(x)$. Indeed, this bracket implies that $\{x, a^{-1}\} = -a^{-1}u(x)$, and more interestingly that $\{\overline{x}, a\} = \partial_x(\overline{x})u(x)a$ and $\{\overline{x}, a^{-1}\} = -\partial_x(\overline{x})u(x)a^{-1}$. Since $\{x - \overline{x}, a\} = au(x)(1 - \partial_x(\overline{x}))$, it follows that $\{(x-\overline{x})^{-1}, a\} = -(x-\overline{x})^{-2}au(x)(1-\partial_x(\overline{x}))$ and similarly $\{(x-\overline{x})^{-1}, a^{-1}\} = a^{-1}(x-\overline{x})^{-2}u(x)(1-\partial_x(\overline{x}))$. It follows that

$$\{y, \mathrm{V}\} = -\{x\overline{x}, \tfrac{a-a^{-1}}{x-\overline{x}}\} = -u(x)\tfrac{\overline{x}+x\partial_x(\overline{x})}{x-\overline{x}}\mathrm{U}.$$

Note that $\frac{\overline{x}+x\partial_x(\overline{x})}{x-\overline{x}} = x\frac{1+\partial_x(\overline{x})}{x-\overline{x}} - 1$, so Lemma 3.3.4 implies that the fraction $\frac{\overline{x}+x\partial_x(\overline{x})}{x-\overline{x}}$ is defined in $\mathrm{R}[x]^\wedge$, as desired. Similarly,

$$\{\mathrm{U}, \mathrm{V}\} = \mathrm{V}^2 u(x)(1 - \partial_x(\overline{x})),$$

which indeed does not have any poles. Finally, $\{\mathrm{U}, y\}$ clearly has no poles; for completeness, we remark that

$$\{\mathrm{U}, y\} = -u(x)(\overline{x} + x\partial_x(\overline{x}))(x - \overline{x})\mathrm{V}.$$

Now consider the case $\mathrm{G} = \mathrm{PGL}_2$, so that $\mathcal{A}_{\mathrm{G}} \cong \mathrm{R}[x]^\wedge[a^{\pm 1}, \frac{a^2-1}{x}]$, and the $\mathbf{Z}/2$-action sends $a \mapsto a^{-1}$ and $x \mapsto \overline{x}$. The invariant subring $\mathcal{A}_{\mathrm{G}}^{\mathrm{W}}$ is generated by

$$\mathrm{U} = a + a^{-1},\ y = -x\overline{x},\ \mathrm{V} = a^{-1}\tfrac{a^2-1}{x}(1 - \tfrac{x}{\overline{x}}) = (a - a^{-1})(\tfrac{1}{x} - \tfrac{1}{\overline{x}}),\ \mathrm{V}' = \tfrac{(a-a^{-1})^2}{x\overline{x}}.$$

The assumption that 2 is a unit in R implies that the generator V' is in fact redundant. Indeed,

$$\mathrm{V}^2 = \tfrac{(a-a^{-1})^2}{y^2}(\overline{x} - x)^2.$$

By $\mathbf{Z}/2$-invariance, $(\overline{x} - x)^2$ is a power series in $y = -x\overline{x}$. It has constant term 2, because $\overline{x} - x \equiv -2x$ modulo x^2. Because 2 is a unit in R, $(\overline{x} - x)^2$ is a unit multiple of y, so that V^2 is a unit multiple of $\frac{(a-a^{-1})^2}{y} = \mathrm{V}'$, as desired.

Again, the generators U, y, and V satisfy a somewhat complicated relation, but to prove the regularity of the Poisson bracket, it suffices to check that the Poisson bracket between these generators has no poles. The bracket $\{\mathrm{U}, y\}$ is exactly as in the case $\mathrm{G} = \mathrm{SL}_2$, so again we only need to compute $\{y, \mathrm{V}\}$ and $\{\mathrm{U}, \mathrm{V}\}$. One finds that

$$\{y, \mathrm{V}\} = (x - \overline{x})\tfrac{\overline{x} + x\partial_x(\overline{x})}{x\overline{x}} u(x)\mathrm{U}.$$

Write $\overline{x} = x\iota(x)$ for some power series $\iota(x)$ with constant term -1 (so $\iota(x)$ is a unit); then $x\overline{x}$ is a unit multiple of x^2. Since both $(x - \overline{x})$ and $\overline{x} + x\partial_x(\overline{x})$ are divisible by x, their product is divisible by x^2, and hence by $x\overline{x}$. In other words, the fraction $(x - \overline{x})\frac{\overline{x} + x\partial_x(\overline{x})}{x\overline{x}}$ is defined in $\mathrm{R}[x]^\wedge$, and there are no poles in $\{y, \mathrm{V}\}$. Similarly,

$$\{\mathrm{V}, \mathrm{U}\} = -\mathrm{V}^2 u(x)\tfrac{\overline{x}^2 - x^2\partial_x(\overline{x})}{(\overline{x} - x)^2}.$$

It is easy to check that the fraction $\frac{\overline{x}^2 - x^2\partial_x(\overline{x})}{(\overline{x} - x)^2}$ lies in $\mathrm{R}[x]^\wedge$: since 2 is a unit in R, the fraction $\frac{(\overline{x}-x)^2}{x^2}$ is a unit in $\mathrm{R}[x]^\wedge$, so $\frac{\overline{x}^2 - x^2\partial_x(\overline{x})}{(\overline{x} - x)^2}$ is a unit multiple of the element $\frac{\overline{x}^2 - x^2\partial_x(\overline{x})}{x^2} = \iota(x)^2 - \partial_x(\overline{x}) \in \mathrm{R}[x]^\wedge$, as desired. □

Lemma 3.3.5. *Suppose k is an even $\mathbf{E}_\infty$-ring which is good for* G *(in the sense of Definition 2.2.17). There is an isomorphism $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \cong (\mathcal{A}_{\mathrm{G}})^{\mathrm{W}}$ of graded commutative $\pi_*(k)$-algebras.*

Proof. Recall from Definition 2.2.28 that if k is an even $\mathbf{E}_\infty$-ring which is good for G and X is a G-space with even-dimensional cells, then $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G},\mathrm{BM}}_*(\mathrm{X}; k)) \cong \mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathrm{X}; k)^{\mathrm{W}}$. The claim now follows from Theorem 3.1.11. □

Construction 3.3.6. There is a Poisson bracket of weight 2 on $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$ defined as follows. Recall that $\mathbf{T} = \mathbf{C}^\times$ acts by loop rotation on $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ and on $\mathrm{G}_{\mathcal{O}}$, so one can form $\mathrm{C}^{\mathrm{G}_{\mathcal{O}} \rtimes \mathbf{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$. Construction 3.1.2 implies that this is an $\mathbf{E}_1$-$k^{h\mathbf{T}}$-algebra; but for our purposes it suffices to note that the standard convolution diagram produces an $\mathbf{A}_3$-$k^{h\mathbf{T}}$-algebra structure on $\mathrm{C}^{\mathrm{G}_{\mathcal{O}} \rtimes \mathbf{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, so that $\mathrm{gr}^\star_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}} \rtimes \mathbf{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$ is a graded $\mathbf{E}_1$-$\pi_*(k^{h\mathbf{T}})$-algebra. This implies that $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}} \rtimes \mathbf{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$ is an associative graded $\pi_*(k^{h\mathbf{T}})$-algebra. Moreover, if we write $\pi_*(k^{h\mathbf{T}}) = \pi_*(k)[\hbar]^\wedge$, where the completion is taken in the sense of Remark A.1.6, then $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}} \rtimes \mathbf{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))/\hbar \cong \mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$. This is a commutative graded $\pi_*(k)$-algebra, so the reduction of the commutator in $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}} \rtimes \mathbf{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$ modulo $\hbar$ defines a Poisson bracket on $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$. This Poisson bracket has weight 2, because $\hbar$ lives in weight -2.

Theorem 3.3.7. *Suppose k is an even $\mathbf{E}_\infty$-ring which is good for* G, *and if* G *has a simple factor of adjoint type B_n ($n \geq 1$), assume further that 2 is a unit in $\pi_0(k)$. Then the isomorphism $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \cong (\mathcal{A}_{\mathrm{G}})^{\mathrm{W}}$ of Lemma 3.3.5 is one of graded Poisson $\pi_*(k)$-algebras, where $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$ is equipped with the Poisson structure of Construction 3.3.6, and $(\mathcal{A}_{\mathrm{G}})^{\mathrm{W}}$ is equipped with the Poisson structure of Proposition 3.3.3.*

Proof. There are injections

$$(\mathcal{A}_{\mathrm{G}})^{\mathrm{W}} \hookrightarrow \mathcal{A}_{\mathrm{G}} \hookrightarrow \mathcal{A}_{\mathrm{G}}[e(\Phi)^{-1}] \xleftarrow{\cong} \mathcal{A}_{\mathrm{T}}[e(\Phi)^{-1}].$$

Although $\mathcal{A}_{\mathrm{G}}$ does not have a Poisson bracket, Proposition 3.3.3 implies that $\mathcal{A}_{\mathrm{G}}[e(\Phi)^{-1}]$ does acquire a Poisson bracket (inherited from $\mathcal{A}_{\mathrm{T}}$), and the map

$(\mathcal{A}_{\mathrm{G}})^{\mathrm{W}} \hookrightarrow \mathcal{A}_{\mathrm{G}}[e(\Phi)^{-1}]$ is a Poisson algebra map. Similarly, there are injections

$$\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \hookrightarrow \mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \hookrightarrow \mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)[e(\Phi)^{-1}] \xleftarrow{\cong} \mathrm{H}^{\mathrm{T}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}}; k)[e(\Phi)^{-1}].$$

Moreover, the map $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \to \mathrm{H}^{\mathrm{T}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}}; k)[e(\Phi)^{-1}]$ is one of graded Poisson $\pi_*(k)$-algebras. To check that the isomorphism $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \cong (\mathcal{A}_{\mathrm{G}})^{\mathrm{W}}$ of Lemma 3.3.5 is one of graded Poisson $\pi_*(k)$-algebras, it therefore suffices to check that the isomorphism $\mathrm{H}^{\mathrm{T}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}}; k) \cong \mathcal{A}_{\mathrm{T}} = \mathcal{O}_{\check{\mathrm{T}} \times \mathrm{T}_{\mathrm{F}}}$ is one of graded Poisson $\pi_*(k)$-algebras. By Remark 3.3.2, this follows from Proposition 5.3.2, which gives an isomorphism $\mathrm{H}^{\mathrm{T}_{\mathcal{O}} \rtimes \mathbf{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}}; k) \cong \mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$ of graded associative $\pi_*(k^{h\mathbf{T}})$-algebras. □

Remark 3.3.8. There are other structures on $\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, like analogues of restricted Poisson structures in the sense of [BK08]. When $k = \mathbf{F}_p$, these structures were studied in [Lon18]. For general (even) $\mathbf{E}_\infty$-rings k, the interaction between the Poisson structure above and the "Frobenius" coming from Dyer-Lashof operations is more subtle and depends on the formal group F in a highly nontrivial way. We will study this in future work, where we will also discuss the full loop-rotation equivariant $\mathrm{C}^{\mathrm{G}_{\mathcal{O}} \rtimes \mathbf{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$.

4. Spectral side

4.1. **F-adaptations.** Recall that the Lie algebra $\mathfrak{g}$ of a connected reductive group G splits as $\mathfrak{t} \oplus \bigoplus_{\alpha\in\Phi} \mathfrak{g}_\alpha$, where $\mathfrak{g}_\alpha$ are the root subspaces and $\mathfrak{t}$ is a Cartan subalgebra. We will need a variant of $\mathfrak{g}$ which is "adapted" to a 1-dimensional formal group F over a commutative ring R, where the semisimple part $\mathfrak{t}$ is replaced by the formal scheme $\mathrm{Hom}(\mathbb{X}^*(\mathrm{T}), \mathrm{F})$, and where the root subspaces $\mathfrak{g}_\alpha \cong \mathbf{A}^1$ are replaced by the line bundle Lie(F) over R. The required "adaptation" of $\mathfrak{g}$ is provided by the following, which was originally defined in the context of "F-loop spaces" in [MRT22, Mou21].

Definition 4.1.1. Let F be a graded 1-dimensional formal group over a graded commutative ring R. Let H be a graded group scheme over R; I will always assume that H is evenly graded. Define a new graded group scheme H_{F} over R by $\mathrm{H}_{\mathrm{F}} = \mathrm{Hom}(\mathrm{F}^\vee, \mathrm{H}) \cong \mathrm{Map}_*(\mathrm{BF}^\vee, \mathrm{BH})$. Note that H_{F} is an ind-scheme. If H is flat over R, then H_{F} is also flat over R. There is an action of H on H_{F} given by conjugation; note that if H is commutative, then H_{F} acquires the structure of a commutative graded group scheme over R.

Lemma 4.1.2. *Suppose* R *is a graded commutative ring, let* $\mathrm{F} = \hat{\mathbf{G}}_a(2)$, *and let* H *be a graded group scheme over* R *with Lie algebra* $\mathfrak{h}$. *Let* $\mathfrak{h}(2)^\wedge_{\mathcal{N}}$ *denote the functor* $\mathrm{CAlg}^{\mathrm{gr},\heartsuit}_{\mathrm{R}} \to \mathrm{Set}$ *sending*

$$\mathrm{R} \mapsto \{x \in \mathfrak{h}(2)(\mathrm{R}) | \mathrm{ad}^n_x = 0 \textit{ for some } n \geq 0\}.$$

Then there is an H*-equivariant isomorphism* $\mathrm{H}_{\mathrm{F}} \cong \mathfrak{h}(2)^\wedge_{\mathcal{N}}$.

Proof sketch. This is [Mun24, Example 4.2.13]; let us sketch the argument. Recall that $\hat{\mathbf{G}}_a(2)^\vee \cong \mathrm{Spec}\,\Gamma_{\mathrm{R}}(x)$, where $\Gamma_{\mathrm{R}}(x)$ is the divided power R-algebra on a class x (in weight 2). A homomorphism $f : \hat{\mathbf{G}}_a(2)^\vee \to \mathrm{H}$ is the data of a (graded) map $f : \mathcal{O}_{\mathrm{H}} \to \mathcal{O}_{\hat{\mathbf{G}}_a(2)^\vee} \cong \Gamma_{\mathrm{R}}(x)$ of Hopf R-algebras. That is, for each $h \in \mathcal{O}_{\mathrm{H}}$, it is a *finite* sum $\sum_{n\geq 0} f_n(h)\gamma_n(x)$ such that f_0 is the augmentation $\mathcal{O}_{\mathrm{H}} \to \mathrm{R}$, $f(h_1h_2) = f(h_1)f(h_2)$, and $(f \otimes f) \circ \Delta_{\mathcal{O}_{\mathrm{H}}} = \Delta_{\Gamma_{\mathrm{R}}(x)} \circ f$. Unwinding, this implies that f is determined by $f_1(h) \in \mathrm{R}$ which satisfies the Leibniz rule $f_1(h_1 h_2) = f(h_1)\epsilon(h_2) + \epsilon(h_1) f(h_2)$. In other words, f is determined by $f_1 \in \mathfrak{h}(2)$. The expression $f(h) = \sum_{n\geq 0} f_n(h)\gamma_n(x)$ is a finite sum if $\mathrm{ad}_{f_1}^n = 0$ for some $n \geq 0$. $\square$

Remark 4.1.3. Let $\mathrm{R} \to \mathrm{S}$ be a map of graded commutative rings. If H is a graded group scheme over R, then the natural map $(\mathrm{H} \otimes_{\mathrm{R}} \mathrm{S})_{\mathrm{F}} \xrightarrow{\cong} \mathrm{H}_{\mathrm{F}} \otimes_{\mathrm{R}} \mathrm{S}$ is an isomorphism. In particular, when $\mathrm{S} = \mathrm{R}_{\mathbf{Q}}$ is the rationalization of R, a choice of isomorphism $\ell_{\mathrm{F}} : \mathrm{F} \xrightarrow{\cong} \hat{\mathbf{G}}_a(2)$ (which always exists) defines an isomorphism $\ell_{\mathrm{F}} : (\mathrm{H}_{\mathrm{F}})_{\mathbf{Q}} \xrightarrow{\cong} \mathfrak{h}_{\mathbf{Q}}(2)^\wedge_{\mathcal{N}}$, whose inverse will be denoted ℓ_{F}^{-1}. If R is torsion-free and H is flat over R, it follows that the map $\mathcal{O}_{\mathrm{H}_{\mathrm{F}}} \hookrightarrow \mathcal{O}_{\mathfrak{h}_{\mathbf{Q}}(2)^\wedge_{\mathcal{N}}}$ is injective. Note, however, that the natural map $(\mathcal{O}_{\mathrm{H}_{\mathrm{F}}})_{\mathbf{Q}} \to \mathcal{O}_{\mathfrak{h}_{\mathbf{Q}}(2)^\wedge_{\mathcal{N}}}$ is *not* an isomorphism; for instance, when $\mathrm{H} = \mathbf{G}_m$, this amounts to the failure of the map $\mathrm{R}[\![t]\!] \otimes \mathbf{Q} \to (\mathrm{R} \otimes \mathbf{Q})[\![t]\!]$ to be an isomorphism.

Example 4.1.4. One can similarly show that if $\mathrm{F} = \hat{\mathbf{G}}_m$, and $\mathrm{H}^\wedge_{\mathcal{U}}$ denote the functor $\mathrm{CAlg}_{\mathrm{R}}^{\mathrm{gr},\heartsuit} \to \mathrm{Set}$ sending

$$\mathrm{R} \mapsto \{x \in \mathrm{H}(\mathrm{R}) | (x-1)^n = 0 \text{ on } \mathcal{O}_{\mathrm{H}} \text{ for some } n \geq 0\},$$

then there is an H-equivariant isomorphism $\mathrm{H}_{\mathrm{F}} \cong \mathrm{H}^\wedge_{\mathcal{U}}$. (Here, $x-1$ acts on $\mathcal{O}_{\mathrm{H}}$ by right translation.) This follows as in Lemma 4.1.2 from the isomorphism $\hat{\mathbf{G}}_m^\vee \cong \mathrm{Spec}(\mathrm{Bin}_{\mathrm{R}}(x))$, where $\mathrm{Bin}_{\mathrm{R}}(x)$ is the free binomial R-algebra on a class x in weight 2. Formally, $\mathrm{Bin}_{\mathrm{R}}(x) = \mathrm{R} \otimes_{\mathbf{Z}} \mathrm{Bin}_{\mathbf{Z}}(x)$ where $\mathrm{Bin}_{\mathbf{Z}}(x)$ is the $\mathbf{Z}$-subalgebra of $\mathbf{Q}[x]$ generated by the polynomials $\binom{x}{n} = \frac{x(x-1)\cdots(x-(n-1))}{n!}$. (The scheme $\hat{\mathbf{G}}_m^\vee$ is equivalently the affinization of the constant group scheme $\underline{\mathbf{Z}}$ over R.)

More generally, suppose R is a graded commutative ring, and let β be a graded variable in weight 2. Let $\mathrm{F} = \hat{\mathbf{G}}_\beta$ denote the graded formal group over $\mathrm{R}[\beta]$ given by the 1-parameter degeneration of $\hat{\mathbf{G}}_m$ to $\hat{\mathbf{G}}_a(2)$, i.e., with group law $x+y+uxy$. Let $(\mathrm{H}_\beta)^\wedge_{\mathcal{U}}$ denote the functor $\mathrm{CAlg}_{\mathrm{R}}^{\mathrm{gr},\heartsuit} \to \mathrm{Set}$ sending

$$\mathrm{R} \mapsto \{x \in \mathrm{H}(\mathrm{R}) | (\tfrac{x-1}{\beta})^n = 0 \text{ for } n \geq 0\}.$$

Then there is an H-equivariant isomorphism $\mathrm{H}_{\mathrm{F}} \cong (\mathrm{H}_\beta)^\wedge_{\mathcal{U}}$. This follows as in Lemma 4.1.2 from the isomorphism $\hat{\mathbf{G}}_m^\vee \cong \mathrm{Spec}(\mathrm{R} \otimes_{\mathbf{Z}[\beta]} \mathbf{Z}[\beta, \frac{x(x-\beta)\cdots(x-(n-1)\beta)}{n!}])$.

Remark 4.1.5. One can also prove non-completed analogues of Lemma 4.1.2 and Example 4.1.4. For instance, if $\mathrm{F} = \mathbf{G}_a(2)$ or $\mathbf{G}_m$, then $\mathrm{Hom}(\mathrm{F}^\vee, \mathrm{H})$ is isomorphic to $\mathrm{Lie}(\mathrm{H})$ or H, respectively. One can also define a decompletion of $\hat{\mathbf{G}}_\beta$ over $\mathrm{R}[\beta]$, given by the group scheme $\mathbf{G}_\beta = \mathrm{Spec}\,\mathrm{R}[\beta][x^{\pm 1}, \frac{x-1}{\beta}]$ with group law determined by $\Delta(x) = x \otimes x$. Then $\mathrm{Hom}(\mathbf{G}_\beta^\vee, \mathrm{H} \times \mathrm{Spec}(\mathrm{R}[\beta]))$ is isomorphic to the deformation to the normal cone of the diagonal $\mathrm{H} \hookrightarrow \mathrm{H} \times_{\mathrm{R}} \mathrm{H}$. If $\mathrm{H} = \mathrm{SL}_n$, for instance, then $\mathrm{Hom}(\mathbf{G}_\beta^\vee, \mathrm{SL}_n \times \mathrm{Spec}(\mathrm{R}[\beta]))$ is isomorphic to the subscheme of $\mathbf{A}^{n^2} \times \mathrm{Spec}(\mathrm{R}[\beta])$ of matrices A such that $\frac{\det(\mathrm{id}_n + \beta \mathrm{A}) - 1}{\beta} = 0$.

Example 4.1.6. If E is an elliptic curve over R and F is its formal completion at the identity section, then $\mathrm{BF}^\vee$ is isomorphic to the derived affinization of $\mathrm{E}^\vee \cong \mathrm{E}$. The quotient stack $\mathrm{H}_{\mathrm{F}}/\mathrm{H}$ is isomorphic to the formal completion of the moduli stack of semistable H-bundles over E of degree 0 at the locus of "unipotent" bundles; see [FGL22] for more.

Lemma 4.1.7. *Let* $1 \to \mathrm{V} \to \mathrm{G} \to \mathrm{H} \to 1$ *be a short exact sequence of flat affine group schemes over* R*, and assume that* V *is a vector group which is central in* G*. Then there is an isomorphism* $\mathrm{V_F} \cong \mathrm{Lie(F)} \otimes_{\mathbf{G}_a} \mathrm{V}$ *of group schemes over* R*. Moreover,* $\mathrm{G_F} \to \mathrm{H_F}$ *is a* $\mathrm{V_F}$*-torsor.*

Proof. That $\mathrm{V_F}$ is a group scheme over R is clear because V is commutative. It follows from Cartier duality that $\mathrm{Hom}(\mathrm{F}^\vee, \mathbf{G}_a) \cong \mathrm{Lie(F)}$, so $\mathrm{V_F} \cong \mathrm{Lie(F)} \otimes_{\mathbf{G}_a} \mathrm{V}$. Now, the translation action of V on G defines an action of $\mathrm{V_F}$ on $\mathrm{G_F}$, and the action map induces an isomorphism $\mathrm{V_F} \times \mathrm{G_F} \to \mathrm{G_F} \times_{\mathrm{H_F}} \mathrm{G_F}$. To show that $\mathrm{G_F} \to \mathrm{H_F}$ is a $\mathrm{V_F}$-torsor, it suffices to show that the map $\mathrm{G_F} \to \mathrm{H_F}$ is surjective. Let $x \in \mathrm{H_F}$, so that x can be viewed as a pointed map $\mathrm{BF}^\vee \to \mathrm{BH}$. This defines a V-gerbe on $\mathrm{BF}^\vee$, which we need to show is trivial. Since F is a 1-dimensional formal group scheme over R, $\mathrm{H}^*\mathrm{R}\Gamma(\mathrm{BF}^\vee; \mathcal{O})$ is an exterior R-algebra on a class in cohomological degree 1. It follows that $\mathrm{H}^i(\mathrm{BF}^\vee; \mathrm{V}) = 0$ for $i > 1$, so all V-gerbes are trivial (albeit not canonically so), and x lifts to a map $\mathrm{BF}^\vee \to \mathrm{BG}$. $\square$

Remark 4.1.8. Note that $\mathrm{Hom}(\mathrm{F}^\vee, \mathbf{G}_a) \cong \mathrm{Map}_*(\mathrm{BF}^\vee, \mathrm{B}\mathbf{G}_a)$. There is an isomorphism $\mathrm{Map}(\mathrm{BF}^\vee, \mathrm{B}\mathbf{G}_a) \cong \mathrm{R}\Gamma(\mathrm{BF}^\vee; \mathcal{O})[1]$. The isomorphism $\mathrm{Hom}(\mathrm{F}^\vee, \mathbf{G}_a) \cong \mathrm{Lie(F)}$ therefore gives a map $\mathrm{Lie(F)} \to \mathrm{R}\Gamma(\mathrm{BF}^\vee; \mathcal{O})[1]$. This defines a rank 2 vector bundle $\mathcal{V}$ over $\mathrm{BF}^\vee$ which is a nonsplit extension of $\mathcal{O}_{\mathrm{BF}^\vee}$ by $\mathrm{Lie(F)}^{-1}$. For instance, when $\mathrm{F} = \hat{\mathrm{E}}$ is the formal group of an elliptic curve over R, the stack $\mathrm{BF}^\vee$ can be identified with the derived affinization of E, and then (choosing a trivialization of Lie(F)) the pullback of $\mathcal{V}$ along the natural map $\mathrm{E} \to \mathrm{BF}^\vee$ identifies with the Atiyah bundle [Ati57].

Lemma 4.1.9. *Let* U *be a connected unipotent graded group scheme over* R*. Then* $\mathrm{U_F}$ *is a bundle of affine spaces over* R*, and it has relative dimension* dim(U)*.*

Proof. Denote the lower central series of U by

$$1 \subseteq \mathrm{U}_n \subseteq \cdots \subseteq \mathrm{U}_1 = \mathrm{U}.$$

We will show that $\mathrm{U_F}$ is a bundle of affine spaces over R by descending induction on n. There is a central extension $\mathrm{U}_n \to \mathrm{U} \to \mathrm{U}/\mathrm{U}_n$ and U_n is a vector group, so $\mathrm{U_F}$ is a $(\mathrm{U}_n)_\mathrm{F} \cong \mathrm{Lie(F)} \otimes_{\mathbf{G}_a} \mathrm{U}_n$-torsor over $(\mathrm{U}/\mathrm{U}_n)_\mathrm{F}$ by Lemma 4.1.7. But $(\mathrm{U}/\mathrm{U}_n)_\mathrm{F}$ is a bundle of affine spaces over R, so it follows that $\mathrm{U_F}$ is also a bundle of affine spaces over R. $\square$

We will apply Definition 4.1.1 to the case when H is a split reductive group.

Definition 4.1.10. Let G be the Chevalley split form of a connected reductive group, viewed as a group scheme over $\mathbf{Z}$. Let $\mathrm{B} \subseteq \mathrm{G}$ be a fixed Borel subgroup, with unipotent radical U. We write ${}^{2\check{\rho}}\mathrm{G}$ denote the graded group scheme over $\mathbf{Z}$ given by the $\mathbf{G}_m$-action on G via $\mathrm{Ad}_{2\check{\rho}}$, where $2\check{\rho} \in \mathbb{X}_*(\mathrm{T})$ is the sum of positive coroots of G. Of course, ${}^{2\check{\rho}}\mathrm{G}$ is closely related to the C-group (see [Ber20] and [BG14, Section 5]): the classifying stack over $\mathrm{B}\mathbf{G}_m$ of ${}^{2\check{\rho}}\mathrm{G}/\mathbf{G}_m$ is isomorphic, as a stack over $\mathrm{B}\mathbf{G}_m$, to the classifying stack of the C-group ${}^{\mathrm{C}}\mathrm{G}$ of G, which lies over $\mathrm{B}\mathbf{G}_m$ thanks to the homomorphism ${}^{\mathrm{C}}\mathrm{G} \to \mathbf{G}_m$. (Note that if the derived subgroup of G is simply-laced, then $2\check{\rho} = 2\rho$ under the isomorphism $\mathbb{X}_*(\mathrm{T}) \cong \mathbb{X}^*(\mathrm{T})$ coming from the minimal bilinear form; so we will write ${}^{2\rho}\mathrm{G}$ to denote ${}^{2\check{\rho}}\mathrm{G}$.) As we will see in Theorem 4.2.12, the graded lift ${}^{2\rho}\check{\mathrm{G}}$ of the group scheme $\check{\mathrm{G}}$ is the natural "home" for the grading on equivariant (co)homology (this was observed already in [Ber20]).

If R is a graded commutative ring, the base-change of G and ${}^{2\check{\rho}}\mathrm{G}$ along the map $\mathbf{Z} \to \mathrm{R}$ (i.e., the pullback along the map $\mathrm{Spec}(\mathrm{R})/\mathbf{G}_m \to \mathrm{B}\mathbf{G}_m$) will continue to be denoted G and ${}^{2\check{\rho}}\mathrm{G}$ respectively. (If we wish to emphasize the base R, these will instead be denoted G_R and ${}^{2\check{\rho}}\mathrm{G}_\mathrm{R}$.) Let ${}^{2\check{\rho}}\mathfrak{g}$ denote the Lie algebra of ${}^{2\check{\rho}}\mathrm{G}$. If F is a 1-dimensional graded formal group over R, then Definition 4.1.1 defines an ind-scheme ${}^{2\check{\rho}}\mathrm{G}_\mathrm{F}$ over R. There is a graded action of ${}^{2\check{\rho}}\mathrm{G}$ on ${}^{2\check{\rho}}\mathrm{G}_\mathrm{F}$.

Lemma 4.1.11. *The map* $\mathrm{B}_\mathrm{F} \to (\mathrm{B}/[\mathrm{U},\mathrm{U}])_\mathrm{F}$ *is surjective.*

Proof. Consider the filtration on U given by root height, denoted

$$1 = \mathrm{U}_n \subseteq \cdots \subseteq \mathrm{U}_2 \subseteq \mathrm{U}_1 = \mathrm{U}.$$

We will see more generally that $(\mathrm{B}/\mathrm{U}_{i+1})_\mathrm{F} \to (\mathrm{B}/\mathrm{U}_i)_\mathrm{F}$ is surjective. The map $\mathrm{B}/\mathrm{U}_{i+1} \to \mathrm{B}/\mathrm{U}_i$ is surjective with kernel $\mathrm{U}_{i+1}/\mathrm{U}_i$. Let $x \in (\mathrm{B}/\mathrm{U}_i)_\mathrm{F}$, viewed as a B/U_i-bundle over $\mathrm{BF}^\vee$, and let $\mathcal{P}_x$ denote the associated T-bundle over $\mathrm{BF}^\vee$ (induced by the map $\mathrm{B}/\mathrm{U}_i \to \mathrm{B}/\mathrm{U} \cong \mathrm{T}$). The obstruction to lifting x to $(\mathrm{B}/\mathrm{U}_{i+1})_\mathrm{F}$ lies in $\mathrm{H}^2(\mathrm{BF}^\vee; \mathcal{P}_x \times^\mathrm{T} \mathrm{U}_i/\mathrm{U}_{i+1})$. But $\mathcal{P}_x \times^\mathrm{T} \mathrm{U}_i/\mathrm{U}_{i+1} \cong \bigoplus_{\mathrm{height}(\beta)=i} \mathcal{L}_\beta$, where $\mathcal{L}_\beta$ is the line bundle associated to $\mathcal{P}_x$ through $\beta \in \mathbb{X}^*(\mathrm{T})$. Since $\mathrm{H}^i(\mathrm{BF}^\vee; \mathrm{V}) = 0$ for $i > 1$ and any vector bundle V, it follows that x lifts to $(\mathrm{B}/\mathrm{U}_{i+1})_\mathrm{F}$, as desired. □

In the case $\mathrm{F} = \hat{\mathbf{G}}_a(2)$, many of the results below were proved in [Ric17], as well as [Jan04] (and over a $\mathbf{Q}$-algebra, these results are all well-known and follow from the work of Kostant [Kos63, Kos78]). In fact, as we will see below, many results below reduce to the case $\mathrm{F} = \hat{\mathbf{G}}_a(2)$, using the canonical degeneration from F to its Lie algebra Lie(F).

Construction 4.1.12. Let $[\mathrm{U},\mathrm{U}] \subseteq \mathrm{U}$ denote the derived subgroup of U, so that $\mathrm{U}/[\mathrm{U},\mathrm{U}] \cong \mathbf{G}_a^{\oplus\Delta}$, where Δ is the set of simple roots of G. Since there is an extension $1 \to \mathbf{G}_a^{\oplus\Delta} \to \mathrm{B}/[\mathrm{U},\mathrm{U}] \to \mathrm{T} \to 1$, it follows from Lemma 4.1.7 that the map $({}^{2\check{\rho}}\mathrm{B}/[{}^{2\check{\rho}}\mathrm{U}, {}^{2\check{\rho}}\mathrm{U}])_\mathrm{F} \to \mathrm{T}_\mathrm{F}$ is a torsor for $({}^{2\check{\rho}}\mathrm{U}/[{}^{2\check{\rho}}\mathrm{U}, {}^{2\check{\rho}}\mathrm{U}])_\mathrm{F} \cong \mathrm{Lie}(\mathrm{F})^{\oplus\Delta}$. The section $\mathrm{T} \to \mathrm{B}$ of the quotient map $\mathrm{B} \to \mathrm{T}$ splits this torsor, so that there is an isomorphism $({}^{2\check{\rho}}\mathrm{B}/[{}^{2\check{\rho}}\mathrm{U}, {}^{2\check{\rho}}\mathrm{U}])_\mathrm{F} \cong \mathrm{T}_\mathrm{F} \times_{\mathrm{Spec}(\mathrm{R})} \mathrm{Lie}(\mathrm{F})^{\oplus\Delta}$. The map $\Delta \to *$ of finite sets defines a diagonal map $\mathrm{Lie}(\mathrm{F}) \to \mathrm{Lie}(\mathrm{F})^{\oplus\Delta}$. A choice of trivialization of Lie(F) therefore yields a point $f \in \mathrm{Lie}(\mathrm{F})^{\oplus\Delta}$ (via the image of $1 \in \mathrm{R} \cong \mathrm{Lie}(\mathrm{F})$). In particular, there is a map $\overline{\kappa}_\mathrm{F} : \mathrm{T}_\mathrm{F} \to ({}^{2\check{\rho}}\mathrm{B}/[{}^{2\check{\rho}}\mathrm{U}, {}^{2\check{\rho}}\mathrm{U}])_\mathrm{F}$ sending $x \mapsto (x, f)$. Using Lemma 4.1.11, $\overline{\kappa}_\mathrm{F}$ admits a lift to a map $\kappa_\mathrm{F} : \mathrm{T}_\mathrm{F} \to {}^{2\check{\rho}}\mathrm{B}_\mathrm{F}$; such a lift of $\overline{\kappa}_\mathrm{F}$ is, of course, far from canonical, but we will fix one such lift once and for all. Let us abusively write $f \in {}^{2\check{\rho}}\mathrm{U}_\mathrm{F} \subseteq {}^{2\check{\rho}}\mathrm{B}_\mathrm{F}$ to denote the resulting lift of $f \in \mathrm{Lie}(\mathrm{F})^{\oplus\Delta}$.

Example 4.1.13. Let $n \in \mathbf{Z}$, and consider the weight n action of $\mathbf{G}_m$ on $\mathbf{G}_a$. This defines an extension $\mathbf{G}_a \to \mathrm{B}_n \to \mathbf{G}_m$, where B_n can be viewed as the group of matrices of the form $\left(\begin{smallmatrix} a & b \\ 0 & a^{1-n} \end{smallmatrix}\right)$. Note that the center of B_n is isomorphic to μ_n. Equip B_n with the grading where a lives in weight 0 and b lives in weight 2; let us denote the resulting graded group by ${}^{2\check{\rho}}\mathrm{B}_n$. Recall that $(\mathbf{G}_m)_\mathrm{F} = \mathrm{F}$. If S is a graded commutative R-algebra, then each S-point $x \in \mathrm{F}$ defines a line bundle $\mathcal{L}_x$ over $(\mathrm{BF}^\vee)_\mathrm{S}$. Since $\mathrm{F} \cong \hat{\mathbf{A}}^1(2)$, the S-point x defines an element $x \in \mathrm{S}_{-2}$ which is nilpotent. Cartier duality implies that $\mathrm{R}\Gamma(\mathrm{BF}^\vee; \mathcal{L}_x) \cong \mathrm{S}/x[-1]$, where this means the derived cofiber. In particular, $\mathrm{R}\Gamma(\mathrm{BF}^\vee; \mathcal{L}_x)$ is concentrated in two degrees, $\mathrm{H}^0(\mathrm{BF}^\vee; \mathcal{L}_x) \cong \mathrm{S}^{x\text{-tors}}$ is the x-torsion submodule of S, and $\mathrm{H}^1(\mathrm{BF}^\vee; \mathcal{L}_x)$ is the underived quotient of S by x.

Let us view the stack $({}^{2\check{\rho}}\mathrm{B}_n)_{\mathrm{F}}/{}^{2\check{\rho}}\mathrm{B}_n = \mathrm{Hom}(\mathrm{BF}^\vee, \mathrm{B}^{2\check{\rho}}\mathrm{B}_n)$ as the moduli stack of ${}^{2\check{\rho}}\mathrm{B}_n$-bundles on $\mathrm{BF}^\vee \otimes_{\mathrm{R}} \mathrm{S}$. Then the map $\kappa_{\mathrm{F}} : \mathrm{F} \to ({}^{2\check{\rho}}\mathrm{B}_n)_{\mathrm{F}}$ sends an S-point $x \in \mathrm{F}$, viewed as a line bundle $\mathcal{L}_x$ over $\mathrm{BF}^\vee$, to the extension of $\mathcal{L}_x$ by $\mathcal{L}_x^{\otimes(1-n)}$ classified by the unique S-module generator of $\mathrm{Ext}^1(\mathcal{L}_x^{\otimes(1-n)}, \mathcal{L}_x) \cong \mathrm{H}^1(\mathrm{BF}^\vee; \mathcal{L}_x^{\otimes n})$. (For instance, if S is a field and $\mathrm{F} = \hat{\mathrm{E}}$ is the formal group of an elliptic curve over R, so that $\mathrm{BF}^\vee$ is isomorphic to the derived affinization of E, then $\kappa_{\mathrm{F}} : \mathrm{F} \to (\mathrm{B}_n)_{\mathrm{F}}$ is the derived affinization of the map $\kappa_{\mathrm{E}} : \mathrm{E} \to \mathrm{Bun}_{\mathrm{B}_n}(\mathrm{E}) \times_{\mathrm{BB}_n} \mathrm{Spec}(\mathrm{S})$ which can informally be described as follows. The map κ_{E} sends x to the split extension $\mathcal{O}_{x-e} \oplus \mathcal{O}_{(1-n)x-e}$ if x is not an n-torsion point of E, and to the tensor product of $\mathcal{O}_{x-e}$ with the Atiyah bundle [Ati57] if x is an n-torsion point of E.)

By Construction 4.1.12, there is an isomorphism $({}^{2\check{\rho}}\mathrm{B}_n)_{\mathrm{F}} \cong \mathrm{F} \times_{\mathrm{Spec(R)}} \mathrm{Lie(F)}$. The group ${}^{2\check{\rho}}\mathrm{B}_n$ acts on $({}^{2\check{\rho}}\mathrm{B}_n)_{\mathrm{F}}$ as follows: if $g = \left(\begin{smallmatrix} a & b \\ 0 & a^{1-n} \end{smallmatrix}\right) \in {}^{2\check{\rho}}\mathrm{B}_n$ with $|a| = 0$ and $|b| = 2$, then unwinding definitions shows that g acts on $(x, y) \in ({}^{2\check{\rho}}\mathrm{B}_n)_{\mathrm{F}} \cong \mathrm{F} \times_{\mathrm{Spec(R)}} \mathrm{Lie(F)}$ by

$$(x, y) \mapsto (x, a^n y - a^{n-1} b[n](x)).$$

Here, $[n](x)$ denotes the n-fold sum of x in the formal group F. Assume for the moment that R is the Lazard ring L and that F is the universal formal group law over L equipped with a coordinate. Then $[n](x) \in \mathcal{O}_{\mathrm{F}}$ is a nonzerodivisor. Since $\kappa_{\mathrm{F}} : \mathrm{F} \to ({}^{2\check{\rho}}\mathrm{B}_n)_{\mathrm{F}}$ sends $x \mapsto (x, 1)$, it follows that g stabilizes (x, y) if and only if $b = \frac{a - a^{1-n}}{[n](x)} \in \mathcal{O}_{\mathrm{F} \times {}^{2\check{\rho}}\mathrm{B}_n}[\frac{1}{[n](x)}]$. If $d|n$, then the conjugation action of ${}^{2\check{\rho}}\mathrm{B}_n$ on $({}^{2\check{\rho}}\mathrm{B}_n)_{\mathrm{F}}$ descends to an action of ${}^{2\check{\rho}}\mathrm{B}_n/\mu_d$ on $({}^{2\check{\rho}}\mathrm{B}_n)_{\mathrm{F}}$. The calculation above shows that the subgroup of $\mathrm{F} \times {}^{2\check{\rho}}\mathrm{B}_n/\mu_d$ given by the stabilizer of κ_{F} is isomorphic (as a group scheme over F) to

$$\mathrm{Spec}\, \mathcal{O}_{\mathrm{F}}[a^{\pm d}, b]/(a^n - 1 = [n](x)b) \hookrightarrow \mathrm{F} \times {}^{2\check{\rho}}\mathrm{B}_n/\mu_d.$$

The group law is given by $\Delta(a^d) = a^d \otimes a^d$ and $\Delta(b) = b \otimes 1 + 1 \otimes b + [n](x) b \otimes b$. We have only proved this isomorphism when R is the Lazard ring, but this implies the claim for arbitrary R by base-changing along the map $\mathrm{L} \to \mathrm{R}$ which determines F with its chosen coordinate.

Proposition 4.1.14. *Suppose that the bad primes (see Table 1) of* G *are inverted in* R. *Let* $\widetilde{\mathrm{J}}_{\mathrm{F}} \hookrightarrow {}^{2\check{\rho}}\mathrm{B} \times \mathrm{T}_{\mathrm{F}}$ *denote the stabilizer of* $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}} \to {}^{2\check{\rho}}\mathrm{B}_{\mathrm{F}}$ *(we will use similar notation for a slightly different object below in Definition 4.2.4). If* V *is a graded* ${}^{2\check{\rho}}\mathrm{B}$*-representation in finitely generated projective* R*-modules, then the* $\mathcal{O}_{\mathrm{T}_{\mathrm{F}}}$*-linear map* $(\mathrm{V} \otimes_{\mathrm{R}} \mathcal{O}_{{}^{2\check{\rho}}\mathrm{B}_{\mathrm{F}}})^{{}^{2\check{\rho}}\mathrm{B}} \to (\mathrm{V} \otimes_{\mathrm{R}} \mathcal{O}_{\mathrm{T}_{\mathrm{F}}})^{\widetilde{\mathrm{J}}_{\mathrm{F}}}$ *on invariants is an isomorphism.*

Proof. The closed subgroup $\widetilde{\mathrm{J}}_{\mathrm{F}} \hookrightarrow {}^{2\check{\rho}}\mathrm{B} \times \mathrm{T}_{\mathrm{F}}$ defines a $\widetilde{\mathrm{J}}_{\mathrm{F}}$-action on $\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B} \times \mathrm{T}_{\mathrm{F}}}$ via right translation. View $\mathrm{V} \otimes_{\mathrm{R}} \mathcal{O}_{{}^{2\check{\rho}}\mathrm{B} \times \mathrm{T}_{\mathrm{F}}}$ as a $\widetilde{\mathrm{J}}_{\mathrm{F}}$-module via the coaction on the second tensor factor. There is a commuting action of ${}^{2\check{\rho}}\mathrm{B}$ on $\mathrm{V} \otimes_{\mathrm{R}} \mathcal{O}_{{}^{2\check{\rho}}\mathrm{B} \times \mathrm{T}_{\mathrm{F}}}$ via the diagonal ${}^{2\check{\rho}}\mathrm{B}$-action on V and $\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B} \times \mathrm{T}_{\mathrm{F}}}$. There is a ${}^{2\check{\rho}}\mathrm{B}$-equivariant isomorphism $\mathrm{V} \otimes_{\mathrm{R}} (\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B} \times \mathrm{T}_{\mathrm{F}}})^{\widetilde{\mathrm{J}}_{\mathrm{F}}} \xrightarrow{\cong} (\mathrm{V} \otimes_{\mathrm{R}} \mathcal{O}_{{}^{2\check{\rho}}\mathrm{B} \times \mathrm{T}_{\mathrm{F}}})^{\widetilde{\mathrm{J}}_{\mathrm{F}}}$, and taking ${}^{2\check{\rho}}\mathrm{B}$-invariants produces exactly $(\mathrm{V} \otimes_{\mathrm{R}} \mathcal{O}_{\mathrm{T}_{\mathrm{F}}})^{\widetilde{\mathrm{J}}_{\mathrm{F}}}$. It therefore suffices to show that the ${}^{2\check{\rho}}\mathrm{B}$-equivariant restriction map $\mathrm{V} \otimes_{\mathrm{R}} \mathcal{O}_{{}^{2\check{\rho}}\mathrm{B}_{\mathrm{F}}} \to \mathrm{V} \otimes_{\mathrm{R}} (\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B} \times \mathrm{T}_{\mathrm{F}}})^{\widetilde{\mathrm{J}}_{\mathrm{F}}}$ is an isomorphism. In turn, it suffices to check this when $\mathrm{V} = \mathrm{R}$, i.e., to show that the ${}^{2\check{\rho}}\mathrm{B}$-equivariant restriction map $\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B}_{\mathrm{F}}} \to (\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B} \times \mathrm{T}_{\mathrm{F}}})^{\widetilde{\mathrm{J}}_{\mathrm{F}}}$ is an isomorphism.

There is a complete filtration on $\mathcal{O}_{\mathrm{F}}$ given by powers of the augmentation ideal $\ker(\mathcal{O}_{\mathrm{F}} \to \mathrm{R})$, with associated graded $\mathcal{O}_{\hat{\mathbf{G}}_a(2)}$. This in turn induces a complete filtration on $\mathcal{O}_{\mathrm{T}_{\mathrm{F}}}$ and $\mathcal{O}_{^{2\check{\rho}}\mathrm{B}_{\mathrm{F}}}$, and defines a filtered lift of the map $\mathcal{O}_{^{2\check{\rho}}\mathrm{B}_{\mathrm{F}}} \to (\mathcal{O}_{^{2\check{\rho}}\mathrm{B}\times\mathrm{T}_{\mathrm{F}}})^{\widetilde{\mathrm{J}}_{\mathrm{F}}}$. Since both sides are complete for this filtration, it suffices to show that the induced map on associated graded pieces is an isomorphism. Since ${}^{2\check{\rho}}\mathrm{B}_{\hat{\mathbf{G}}_a(2)} \cong {}^{2\check{\rho}}\mathfrak{b}(2)^{\wedge}_{\mathcal{N}}$ by Lemma 4.1.2 (and similarly $\mathrm{T}_{\hat{\mathbf{G}}_a(2)} \cong \hat{\mathfrak{t}}(2)$), we are reduced to showing that the ${}^{2\check{\rho}}\mathrm{B}$-equivariant restriction map $\mathcal{O}_{^{2\check{\rho}}\mathfrak{b}(2)^{\wedge}_{\mathcal{N}}} \to (\mathcal{O}_{^{2\check{\rho}}\mathrm{B}\times\hat{\mathfrak{t}}(2)})^{\widetilde{\mathrm{J}}_{\hat{\mathbf{G}}_a(2)}}$ is an isomorphism. It suffices to show that the underlying ungraded map is an isomorphism. This is in turn follows if we show that the (uncompleted and ungraded) B-equivariant restriction map $\mathcal{O}_{\mathfrak{b}} \to (\mathcal{O}_{\mathrm{B}\times\mathfrak{t}})^{\widetilde{\mathrm{J}}}$ is an isomorphism, where now $\widetilde{\mathrm{J}}$ denotes the stabilizer of the map $\kappa : \mathfrak{t} \to \mathfrak{b}$ sending $x \mapsto f + x$.

At this point, we are in the realm of ordinary (i.e., non-formal) algebraic geometry. By the algebraic Hartogs lemma, it suffices to show that the map $\mathfrak{b}^{\mathrm{reg}} := (\mathrm{B}\times\mathfrak{t})/\widetilde{\mathrm{J}} \to \mathfrak{b}$ is an open immersion with complement of codimension 2. (By definition, $\mathfrak{b}^{\mathrm{reg}}$ is the B-orbit of $\kappa : \mathfrak{t} \to \mathfrak{b}$.) This is well-known, but we review a proof for completeness. Below, we work over $\mathbf{Z}$, and will invert primes only as needed.

(1) Suppose the centralizer of $x \in \mathfrak{b}$ has dimension equal to $\mathrm{rank}(\mathrm{G})$, and write $x = x_{\mathrm{ss}} + x_n$ with $x_{\mathrm{ss}} \in \mathfrak{t}$ and $x_n = \sum_{\alpha\in\Phi^+} x_\alpha e_\alpha$. We claim that there is no simple root $\alpha \in \Delta$ with both $\alpha(x_{\mathrm{ss}}) = 0$ and $x_\alpha = 0$. Suppose otherwise, and let P_α denote the corresponding minimal parabolic subgroup, with unipotent radical $\mathrm{U}_{\mathrm{P}_\alpha}$ and Levi quotient L_α. Then $x_n \in \mathfrak{u}_{\mathrm{P}_\alpha}$. One also has $x_{\mathrm{ss}} \in \mathrm{Z}(\mathfrak{l}_\alpha)$, because $\mathfrak{l}_\alpha = \mathfrak{g}_{-\alpha} \oplus \mathfrak{t} \oplus \mathfrak{g}_\alpha$, and $\mathrm{ad}_{x_{\mathrm{ss}}}$ vanishes on $\mathfrak{g}_{-\alpha}\oplus\mathfrak{g}_\alpha$ since $\alpha(x_{\mathrm{ss}}) = 0$. Since $x_{\mathrm{ss}} \in \mathrm{Z}(\mathfrak{l}_\alpha)$, it follows that $\mathrm{ad}_x(\mathfrak{p}_\alpha) \subseteq \mathfrak{u}_\alpha$. But then the kernel of $\mathrm{ad}_x|_{\mathfrak{p}_\alpha}$ has dimension at least that of $\mathfrak{l}_\alpha$, which is strictly larger than $\mathrm{rank}(\mathrm{G})$. This contradicts the assumption that the centralizer of x has dimension $\mathrm{rank}(\mathrm{G})$.

(2) Let $\mathfrak{b}^{\mathrm{reg}'}$ denote the open subvariety of $\mathfrak{b}$ of those $x \in \mathfrak{b}$ whose centralizer has dimension equal to $\mathrm{rank}(\mathrm{G})$. We now claim that $\mathfrak{b} \setminus \mathfrak{b}^{\mathrm{reg}'} \hookrightarrow \mathfrak{b}$ has codimension 2. Let $x \in \mathfrak{b}$, and write $x = x_{\mathrm{ss}} + x_n$ with $x_{\mathrm{ss}} \in \mathfrak{t}$ and $x_n \in \mathfrak{u}$. Let $\mathfrak{b}^{(i)}$ denote the closed subset of $\mathfrak{b}$ of those x such that x_{ss} lies on the intersection of i root hyperplanes. Since $\mathfrak{b}^{(2)} \subseteq \mathfrak{b}$ has codimension 2, it suffices to check that $(\mathfrak{b}\setminus\mathfrak{b}^{\mathrm{reg}'}) \cap (\mathfrak{b}\setminus\mathfrak{b}^{(2)}) \subseteq \mathfrak{b}\setminus\mathfrak{b}^{(2)}$ has codimension 2. Suppose first that $x \in \mathfrak{b}\setminus\mathfrak{b}^{(1)}$, i.e., that x_{ss} does not lie on any root hyperplane. Then (the proof of) Proposition 4.2.8 implies that x is conjugate to x_{ss} by some element of $\mathrm{U} \subseteq \mathrm{B}$. In this case, the neutral connected component of the centralizer of x_{ss} is a maximal torus, so x is regular. In other words, $(\mathfrak{b}\setminus\mathfrak{b}^{\mathrm{reg}'}) \cap (\mathfrak{b}\setminus\mathfrak{b}^{(1)})$ is empty. Next, suppose that $x \in \mathfrak{b}^{(1)}\setminus\mathfrak{b}^{(2)}$, i.e., there is some $\alpha \in \Phi^+$ with $\alpha(x_{\mathrm{ss}}) = 0$. (Conjugating by some element of $\mathrm{N}_{\mathrm{G}}(\mathrm{T})$, we may assume that $\alpha \in \Delta$.) By Item 1, if x is regular, then $x_\alpha \neq 0$. More is true: in order for the centralizer of x to have dimension equal to $\mathrm{rank}(\mathrm{G})$, we need $x_n \in \mathfrak{l}_\alpha = \mathfrak{g}_{-\alpha}\oplus\mathfrak{t}\oplus\mathfrak{g}_\alpha$ to have 1-dimensional centralizer. Therefore, if $\alpha(x_{\mathrm{ss}}) = 0$, then $x \in \mathfrak{b}\setminus\mathfrak{b}^{\mathrm{reg}'}$ iff $x_n = 0 \in \mathfrak{g}_\alpha$. In particular, $(\mathfrak{b}\setminus\mathfrak{b}^{\mathrm{reg}'}) \cap (\mathfrak{b}^{(1)}\setminus\mathfrak{b}^{(2)})$ has codimension 2 inside $\mathfrak{b}\setminus\mathfrak{b}^{(2)}$, as desired.

(3) We claim that $\mathfrak{b}^{\mathrm{reg}} \hookrightarrow \mathfrak{b}$ is isomorphic to $\mathfrak{b}^{\mathrm{reg}'}$, i.e., to the open subvariety of $\mathfrak{b}$ of those $x \in \mathfrak{b}$ whose centralizer has dimension equal to $\mathrm{rank}(\mathrm{G})$. (In particular, for $x \in \mathfrak{b}^{\mathrm{reg}'}$, there is some element $g \in \mathrm{B}$ with $\mathrm{Ad}_g(x) \in f + \mathfrak{t}$;

this element of B will be unique up to translation by the centralizer $\widetilde{\mathrm{J}}$.) Write $x = x_{\mathrm{ss}} + \sum_{\alpha\in\Delta} x_\alpha e_\alpha + y$, where $x_{\mathrm{ss}} \in \mathfrak{t}$ and $y \in [\mathfrak{u},\mathfrak{u}]$. If $\alpha \in \Delta$ has $x_\alpha \neq 0$, then acting by an element of T allows us to rescale and assume $x_\alpha = 1$; if $x_\alpha = 0$, then $\alpha(x_{\mathrm{ss}})$ must be nonzero by our preceding discussion, and then conjugating x by $-\alpha(x_{\mathrm{ss}})^{-1} \in \mathrm{U}_\alpha \cong \mathbf{G}_a$ replaces the coefficient of e_α by 1 (instead of 0). In other words, by conjugating x by some element of B, we may assume that $x_\alpha = 1$ for each $\alpha \in \Delta$, so that $x = x_{\mathrm{ss}} + f + y$. We need to show that there is some $g \in \mathrm{B}$ with $\mathrm{Ad}_g(x) = x_{\mathrm{ss}} + f$; equivalently (by replacing x with $x - x_{\mathrm{ss}}$), we need to show that $f + [\mathfrak{u},\mathfrak{u}]$ forms a single B-orbit.

Let $f_0 = \sum_{\alpha\in\Delta} f_\alpha$. Consider the filtration on U given by root height, denoted

$$1 = \mathrm{U}_n \subseteq \cdots \subseteq \mathrm{U}_2 \subseteq \mathrm{U}_1 = \mathrm{U}.$$

Suppose $y \in \mathfrak{u}_2$; we will show that there is $g \in \mathrm{B}$ with $\mathrm{Ad}_g(f_0 + y) = f_0$. (Since any other choice of f, which for us was equal to f_0 modulo $\mathfrak{u}_2$ by Construction 4.1.12, can be written as $f_0 + y'$ for some $y' \in \mathfrak{u}_2$, the elements f and f_0 are conjugate by B. This also implies that for each $y \in \mathfrak{u}_2$, there is some $g \in \mathrm{B}$ with $\mathrm{Ad}_g(f + y) = f$.) Assume we have shown that $f_0 + y$ can be conjugated to $f_0 \pmod{\mathfrak{u}_i}$; then replacing $f_0 + y$ with this conjugate, we may assume that $y \in \mathfrak{u}_i$, and we need to show that $f_0 + y$ can be conjugated to $f_0 \pmod{\mathfrak{u}_{i+1}}$. Let $g \in \mathrm{U}_{i-1}/\mathrm{U}_i \cong \mathfrak{u}_{i-1}/\mathfrak{u}_i \cong \prod_{\mathrm{ht}(\alpha)=i-1} \mathbf{G}_a$. Conjugating $f+y$ by g produces $f_0+y+[g, f_0] \pmod{\mathfrak{u}_{i+1}}$. To finish, it therefore suffices that $\mathrm{ad}_{f_0} : \mathfrak{u}_{i-1} \to \mathfrak{u}_i$ be surjective for $i \geq 2$.

In [Spr66, Proposition 2.2], it is shown that ad_{f_0} has rank equal to the dimension of $\mathfrak{u}_i$. Moreover, in [Spr66, Theorem 2.6], it is shown that $\mathrm{coker}(\mathrm{ad}_{f_0})$ has p-torsion exactly if p is a bad prime for G. It follows that $\mathrm{ad}_{f_0} : \mathfrak{u}_{i-1} \to \mathfrak{u}_i$ is surjective for $i \geq 2$ as soon as the bad primes of G are inverted. (This is the only place where any primes need to be inverted.) □

Informally, Proposition 4.1.14 says that ${}^{2\check\rho}\mathrm{B}_{\mathrm{F}}^{\mathrm{reg}} := {}^{2\check\rho}\mathrm{B} \times^{\widetilde{\mathrm{J}}} \mathrm{T}_{\mathrm{F}} \to {}^{2\check\rho}\mathrm{B}_{\mathrm{F}}$ is open with complement of codimension 2, and ${}^{2\check\rho}\mathrm{B}_{\mathrm{F}}$ is the affine closure of ${}^{2\check\rho}\mathrm{B}_{\mathrm{F}}^{\mathrm{reg}}$.

Remark 4.1.15. The argument in Item 3 in the proof of Proposition 4.1.14 generalizes to show that if the bad primes (see Table 1) of G are inverted in R, then any two choices of lifts κ_{F} of $\overline{\kappa}_{\mathrm{F}}$ are conjugate under ${}^{2\check\rho}\mathrm{U}$. Note that the maps $\overline{\kappa}_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}} \to ({}^{2\check\rho}\mathrm{B}/[{}^{2\check\rho}\mathrm{U}, {}^{2\check\rho}\mathrm{U}])_{\mathrm{F}}$ associated to any two choices of trivializations of Lie(F) are conjugate under ${}^{2\check\rho}\mathrm{B}$. Indeed, two trivializations of Lie(F) differ by a unit $u \in \mathrm{R}^\times$, which defines an element of ${}^{2\check\rho}\mathrm{B}$ that conjugates the two resulting maps $\overline{\kappa}_{\mathrm{F}}$. In fact, this shows that $\overline{\kappa}_{\mathrm{F}}$ itself is independent of the choice of map $\mathrm{Lie}(\mathrm{F}) \to \mathrm{Lie}(\mathrm{F})^{\oplus\Delta}$ used in Construction 4.1.12 (i.e., given a trivialization of Lie(F), of the choice of $f \in \mathrm{Lie}(\mathrm{F})^{\oplus\Delta}$), as long as it is nondegenerate in the following sense: the induced composite $\mathrm{Lie}(\mathrm{F}) \to \mathrm{Lie}(\mathrm{F})^{\oplus\Delta} \xrightarrow{\mathrm{pr}_\alpha} \mathrm{Lie}(\mathrm{F})$ is an isomorphism for each $\alpha \in \Delta$.

Together, these statements imply that if the bad primes (see Table 1) of G are inverted in R, then the composite

$$\mathrm{T}_{\mathrm{F}} \xrightarrow{\kappa_{\mathrm{F}}} {}^{2\check\rho}\mathrm{B}_{\mathrm{F}} \to {}^{2\check\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\check\rho}\mathrm{B}$$

is independent of the choice of trivialization of Lie(F), the choice of $f \in \mathrm{Lie}(\mathrm{F})^{\oplus\Delta}$ used to define $\overline{\kappa}_{\mathrm{F}}$ (as long as f is nondegenerate in the above sense), and the choice of lift κ_{F} of $\overline{\kappa}_{\mathrm{F}}$. In particular, the above composite map $\mathrm{T_F} \to {}^{2\check{\rho}}\mathrm{B_F}/{}^{2\check{\rho}}\mathrm{B}$ depends only on the choice of Borel $\mathrm{B} \subseteq \mathrm{G}$.

Corollary 4.1.16. *Suppose that the bad primes (see Table 1) of* G *are inverted in* R. *Let* $\widetilde{\mathrm{J}}_{\mathrm{F}}$ *be as in Proposition 4.1.14, and let* V *and* V′ *be projective graded* R*-module representations of* ${}^{2\check{\rho}}\mathrm{B}$. *Then pullback along the natural map* $\mathrm{T_F}/\widetilde{\mathrm{J}}_{\mathrm{F}} \to {}^{2\check{\rho}}\mathrm{B_F}/{}^{2\check{\rho}}\mathrm{B}$ *of stacks defines an isomorphism*

$$\mathrm{Hom}_{\mathrm{Perf}^{\mathrm{gr}}({}^{2\check{\rho}}\mathrm{B_F}/{}^{2\check{\rho}}\mathrm{B})^{\heartsuit}}(\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B_F}}\otimes_{\mathrm{R}}\mathrm{V}, \mathcal{O}_{{}^{2\check{\rho}}\mathrm{B_F}}\otimes_{\mathrm{R}}\mathrm{V}') \xrightarrow{\cong} \mathrm{Hom}_{\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T_F}}\widetilde{\mathrm{J}}_{\mathrm{F}})^{\heartsuit}}(\mathcal{O}_{\mathrm{T_F}}\otimes_{\mathrm{R}}\mathrm{V}, \mathcal{O}_{\mathrm{T_F}}\otimes_{\mathrm{R}}\mathrm{V}').$$

Proof. Indeed,

$$\mathrm{Hom}_{\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T_F}}\widetilde{\mathrm{J}}_{\mathrm{F}})^{\heartsuit}}(\mathcal{O}_{\mathrm{T_F}}\otimes_{\mathrm{R}}\mathrm{V}, \mathcal{O}_{\mathrm{T_F}}\otimes_{\mathrm{R}}\mathrm{V}') \cong (\mathcal{O}_{\mathrm{T_F}}\otimes_{\mathrm{R}}(\mathrm{V}^{\vee}\otimes_{\mathrm{R}}\mathrm{V}'))^{\widetilde{\mathrm{J}}_{\mathrm{F}}}.$$

By Proposition 4.1.14, this is isomorphic to

$$(\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B_F}}\otimes_{\mathrm{R}}(\mathrm{V}^{\vee}\otimes_{\mathrm{R}}\mathrm{V}'))^{{}^{2\check{\rho}}\mathrm{B}} \cong \mathrm{Hom}_{\mathrm{Perf}^{\mathrm{gr}}({}^{2\check{\rho}}\mathrm{B_F}/{}^{2\check{\rho}}\mathrm{B})^{\heartsuit}}(\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B_F}}\otimes_{\mathrm{R}}\mathrm{V}, \mathcal{O}_{{}^{2\check{\rho}}\mathrm{B_F}}\otimes_{\mathrm{R}}\mathrm{V}'),$$

as desired. □

Lemma 4.1.17. *The composite* ${}^{2\check{\rho}}\mathrm{B_F}/{}^{2\check{\rho}}\mathrm{B} \to \mathrm{T_F}/\mathrm{T} \to \mathrm{T_F}$ *defines an isomorphism* ${}^{2\check{\rho}}\mathrm{B_F}/\!\!/{}^{2\check{\rho}}\mathrm{B} \xrightarrow{\cong} \mathrm{T_F}$, *where* ${}^{2\check{\rho}}\mathrm{B_F}/\!\!/{}^{2\check{\rho}}\mathrm{B}$ *is the derived invariant-theoretic quotient* $\mathrm{Spec}(\mathrm{R}\Gamma(\mathrm{B}^{2\check{\rho}}\mathrm{B};\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B_F}}))$.

Proof. We need to prove that the natural map $\mathcal{O}_{\mathrm{T_F}} \to \mathrm{R}\Gamma(\mathrm{B}^{2\check{\rho}}\mathrm{B};\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B_F}})$ is an isomorphism. As in the proof of Proposition 4.1.14, there is a complete filtration on $\mathcal{O}_{\mathrm{F}}$ given by powers of the augmentation ideal $\ker(\mathcal{O}_{\mathrm{F}} \to \mathrm{R})$, with associated graded $\mathcal{O}_{\hat{\mathbf{G}}_a(2)}$. This in turn induces a complete filtration on $\mathcal{O}_{\mathrm{T_F}}$ and $\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B_F}}$, and hence a complete filtration on $\mathrm{R}\Gamma(\mathrm{B}^{2\check{\rho}}\mathrm{B};\mathcal{O}_{{}^{2\check{\rho}}\mathrm{B_F}})$. Since both sides are complete for this filtration, it suffices to show that the induced map on associated graded pieces is an isomorphism. Since ${}^{2\check{\rho}}\mathrm{B}_{\hat{\mathbf{G}}_a(2)} \cong {}^{2\check{\rho}}\mathfrak{b}(2)^{\wedge}_{\mathcal{N}}$ by Lemma 4.1.2 (and similarly $\mathrm{T}_{\hat{\mathbf{G}}_a(2)} \cong \hat{\mathfrak{t}}(2)$), we are reduced to showing that the map $\mathcal{O}_{\hat{\mathfrak{t}}(2)} \to \mathrm{R}\Gamma(\mathrm{B}^{2\check{\rho}}\mathrm{B};\mathcal{O}_{{}^{2\check{\rho}}\mathfrak{b}(2)^{\wedge}_{\mathcal{N}}})$ is an isomorphism. In turn, it suffices to show that the underlying ungraded and uncompleted map $\mathcal{O}_{\mathfrak{t}} \to \mathrm{R}\Gamma(\mathrm{BB};\mathcal{O}_{\mathfrak{b}})$ is an isomorphism. This follows from weight considerations; see [Tot18, Theorem 6.1]. □

4.2. **F-regular centralizers.** Recall that the action of G on itself by conjugation factors through its adjoint quotient $\mathrm{G_{ad}}$. We will need the following:

Lemma 4.2.1. *Let* G *be the Chevalley split form over* $\mathbf{Z}$ *of a connected reductive group whose derived subgroup is* simply-laced. *Then the minimal bilinear form on* $\mathbb{X}^*(\mathrm{T})$ *defines an isomorphism between the adjoint quotients of* G *and* $\check{\mathrm{G}}$*; in particular, the action of* G *on itself by conjugation defines an action of* $\check{\mathrm{G}}$ *on* G *by conjugation.*

Proof. It suffices to take G semisimple. Let $\mathrm{T} \subseteq \mathrm{G^{sc}}$ denote a maximal torus of the simply-connected cover $\mathrm{G^{sc}}$, and let $\mathrm{T_{ad}}$ denote the maximal torus of the adjoint form of $\mathrm{G^{sc}}$. Write $\mathrm{G} = \mathrm{G^{sc}}/\pi_1^{\mathrm{alg}}(\mathrm{G})$ with $\pi_1^{\mathrm{alg}}(\mathrm{G}) \subseteq \mathrm{Z}(\mathrm{G^{sc}})$, so that $\mathbb{X}^*(\mathrm{Z}(\mathrm{G^{sc}})) \twoheadrightarrow \mathbb{X}^*(\pi_1^{\mathrm{alg}}(\mathrm{G}))$. Let $\langle -,- \rangle : \mathbb{X}_*(\mathrm{T}) \otimes_{\mathbf{Z}} \mathbb{X}_*(\mathrm{T}) \to \mathbf{Z}$ be the minimal bilinear form, so it is the unique symmetric W-invariant bilinear form such that $(\check{\alpha}, \check{\alpha}) = 2$ for each coroot $\check{\alpha}$. Then the dual map $\mathbb{X}^*(\check{\mathrm{T}}) \to \mathbb{X}^*(\mathrm{T})$ is injective with cokernel $\mathbb{X}^*(\mathrm{Z}(\mathrm{G^{sc}}))$, so the minimal bilinear form defines a surjection $\mathrm{T} \to \check{\mathrm{T}}$ whose kernel is $\mathrm{Z}(\mathrm{G^{sc}})$. The minimal bilinear form gives a perfect pairing $\mathbb{X}^*(\mathrm{Z}(\mathrm{G^{sc}})) \otimes \mathbb{X}^*(\mathrm{Z}(\mathrm{G^{sc}})) \to \mathbf{Q}/\mathbf{Z}$, and the orthogonal complement to the kernel of $\mathbb{X}^*(\mathrm{Z}(\mathrm{G}^{\mathrm{sc}})) \twoheadrightarrow \mathbb{X}^*(\pi_1^{\mathrm{alg}}(\mathrm{G}))$ under this perfect pairing defines a quotient $\mathbb{X}^*(\mu)$ of $\mathbb{X}^*(\mathrm{Z}(\mathrm{G}^{\mathrm{sc}}))$. Let μ denote the Cartier dual of $\mathbb{X}^*(\mu)$; then, there is an induced map from the root data of G to the root data of $\check{\mathrm{G}}$ which defines an isomorphism $\check{\mathrm{G}} \cong \mathrm{G}^{\mathrm{sc}}/\mu$. The adjoint forms of G and $\check{\mathrm{G}}$ are isomorphic. □

Remark 4.2.2. If G is semisimple and simply-laced, there is a $\check{\mathrm{G}} \cong \mathrm{G}/\mu$-equivariant isomorphism $\check{\mathfrak{g}} \cong \mathfrak{g}/\mathrm{Lie}(\mu)$ over $\mathbf{Z}$, where μ is as in the proof of Lemma 4.2.1. In fact, this implies that there is a $\check{\mathrm{G}} \cong \mathrm{G}/\mu$-equivariant isomorphism $\check{\mathfrak{g}}^* \cong \mathfrak{g}$. (Note that this isomorphism exists over $\mathbf{Z}$ itself, and does not require inverting any primes.)

To see this, we may (and will) assume again for simplicity that G is semisimple and simply-connected. The restriction of the Killing form $(-,-) : \mathfrak{g} \otimes_{\mathbf{Z}} \mathfrak{g} \to \mathbf{Z}$ to $\mathfrak{t}$ is a W-equivariant even bilinear form; moreover, by [SS70, Item 1, Page E-14], if α is a simple root and $h_\alpha \in \mathfrak{t}$ is the corresponding Chevalley element, then $(h_\alpha, h_\alpha) = 4h^\vee$. Therefore, $(-,-)|_{\mathfrak{t}} = 2h^\vee\langle -,-\rangle$, where $h^\vee$ is the dual Coxeter number. In [SS70, Item 3, Page E-15], it is also shown that if e_α is a $\mathbf{Z}$-module generator of the root space $\mathfrak{g}_\alpha$ and $h_\alpha \in \mathfrak{t}$ is the corresponding Chevalley element, then $(e_\alpha, e_\alpha) = \frac{(h_\alpha, h_\alpha)}{2} = h^\vee\langle h_\alpha, h_\alpha\rangle$. It follows that the Killing form on $\mathfrak{g}$ is divisible by $2h^\vee$, i.e., there is a pairing $\langle -,-\rangle = \frac{(-,-)}{2h^\vee} : \mathfrak{g} \otimes_{\mathbf{Z}} \mathfrak{g} \to \mathbf{Z}$. The adjoint map $\mathfrak{g} \to \mathfrak{g}^*$ is injective, with $\mathfrak{g}^*/\mathfrak{g} \cong \mathbb{X}^*(\mathrm{Z}(\mathrm{G}))$. Together with the isomorphism $\check{\mathfrak{g}} \cong \mathfrak{g}/\mathrm{Lie}(\mathrm{Z}(\mathrm{G}))$ from Lemma 4.2.1, this implies that $\check{\mathfrak{g}}^* \cong (\mathfrak{g}/\mathrm{Lie}(\mathrm{Z}(\mathrm{G})))^* \cong \mathfrak{g}$.

Let us note that the map $\langle -,-\rangle : \mathfrak{g} \otimes_{\mathbf{Z}} \mathfrak{g} \to \mathbf{Z}$ still exists even when $\mathfrak{g}$ is not simply-laced, where $\langle -,-\rangle|_{\mathfrak{t}}$ is given by $\langle h_\alpha, h_\alpha\rangle = 2$ for long roots α. The map $\mathfrak{g} \to \mathfrak{g}^*$ is an isomorphism upon inverting the lacing number of G (see [GN04] and [GGN17, Section 8]). In fact, if G is simple, the determinant of $\langle -,-\rangle$ is divisible by exactly one prime, namely the lacing number of G.

Below, we will often use the isomorphism $\check{\mathfrak{g}}^* \cong \mathfrak{g}$ described in Remark 4.2.2, but we will not use an isomorphism $\check{\mathfrak{g}}^* \cong \check{\mathfrak{g}}$ (except in Proposition 4.2.15, and that too only in a rather soft way).

Setup 4.2.3. For the remainder of this section, fix a graded commutative ring R, and let F be a 1-dimensional graded formal group over R equipped with a coordinate. Let G be the base-change to R of the Chevalley split form of a connected reductive group whose derived subgroup is *simply-laced*. (We will make it clear if we ever consider G with non-simply-laced derived subgroup.) Recall that since $2\check{\rho} = 2\rho$ under the isomorphism $\mathbb{X}_*(\mathrm{T}) \cong \mathbb{X}^*(\mathrm{T})$ coming from the minimal bilinear form, we will often write ${}^{2\rho}\mathrm{G}$ to denote ${}^{2\check{\rho}}\mathrm{G}$. Lemma 4.2.1 defines an action of ${}^{2\rho}\check{\mathrm{G}}$ on ${}^{2\rho}\mathrm{G}$, and hence on ${}^{2\rho}\mathrm{G}_{\mathrm{F}}$, by conjugation. If ${}^{2\rho}\check{\mathrm{B}} \subseteq {}^{2\rho}\check{\mathrm{G}}$ is the Borel subgroup, then the action of ${}^{2\rho}\check{\mathrm{G}}$ on ${}^{2\rho}\mathrm{G}_{\mathrm{F}}$ by conjugation defines an action of ${}^{2\rho}\check{\mathrm{B}}$ on ${}^{2\rho}\mathrm{B}_{\mathrm{F}}$ by conjugation. Define the F-*Grothendieck-Springer resolution* as ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}} := {}^{2\rho}\check{\mathrm{G}} \times^{{}^{2\rho}\check{\mathrm{B}}} {}^{2\rho}\mathrm{B}_{\mathrm{F}}$, so that the ${}^{2\rho}\check{\mathrm{G}}$-action on ${}^{2\rho}\mathrm{G}_{\mathrm{F}}$ defines a map $\mu : {}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}$.

In many places below, we will assume that the bad primes of G are invertible in R. It seems very likely that this can be removed with a more careful analysis. To hopefully assist with future analyses, we will specify in what (typically minimal) way this assumption is used in the argument. (We will not use this assumption to choose a $\check{\mathrm{G}}$-equivariant isomorphism $\check{\mathfrak{g}}^* \cong \check{\mathfrak{g}}$, for instance.)

Definition 4.2.4. Let $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ denote the groupoid scheme over T_{F} given by the (*a priori* derived) fiber product

$$\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} := \mathrm{T}_{\mathrm{F}} \times_{{}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}}} \mathrm{T}_{\mathrm{F}},$$

where both maps $\mathrm{T}_{\mathrm{F}} \to {}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}}$ are given by composing $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}} \to {}^{2\rho}\mathrm{B}_{\mathrm{F}}$ with the quotient map ${}^{2\rho}\mathrm{B}_{\mathrm{F}} \to {}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}}$. Since the invariant-theoretic quotient ${}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}}$ is isomorphic to T_{F} and κ is a section of the natural map ${}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}} \to \mathrm{T}_{\mathrm{F}}$, both projections $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \to \mathrm{T}_{\mathrm{F}}$ agree, and there is an (*a priori* derived) Cartesian square

$$\begin{tikzcd} \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \arrow[r] \arrow[d] & \mathrm{T}_{\mathrm{F}} \times {}^{2\rho}\check{\mathrm{B}} \arrow[d, "x{,}b \mapsto \mathrm{Ad}_b(\kappa_{\mathrm{F}}(x))"] \\ \mathrm{T}_{\mathrm{F}} \arrow[r, "\kappa_{\mathrm{F}}"'] & {}^{2\rho}\mathrm{B}_{\mathrm{F}}. \end{tikzcd} \tag{4.2.1}$$

This equips $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ with the structure of a graded affine group scheme over T_{F}. We will show in Proposition 4.2.7 that $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ is in fact a classical scheme which is flat over T_{F}, so it is in particular a closed subgroup scheme of the constant graded group scheme $\mathrm{T}_{\mathrm{F}} \times {}^{2\rho}\check{\mathrm{B}}$ over T_{F}.

Note that the map ${}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}} \to \mathrm{T}_{\mathrm{F}}/\check{\mathrm{T}} \cong \mathrm{T}_{\mathrm{F}} \times \mathrm{B}\check{\mathrm{T}}$ induced by the quotient map $\mathrm{B} \to \mathrm{T}$ induces a homomorphism

$$\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \to \widetilde{\mathrm{J}}_{\check{\mathrm{T}},\mathrm{F}} \cong \mathrm{T}_{\mathrm{F}} \times \check{\mathrm{T}}$$

of affine group schemes over T_{F}. This homomorphism fits into a commutative diagram

$$\begin{tikzcd} \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \arrow[r] \arrow[d] & \mathrm{T}_{\mathrm{F}} \times {}^{2\rho}\check{\mathrm{B}} \arrow[d] \\ \widetilde{\mathrm{J}}_{\check{\mathrm{T}},\mathrm{F}} \arrow[r, "\cong"'] & \mathrm{T}_{\mathrm{F}} \times \check{\mathrm{T}} \end{tikzcd}$$

of affine group schemes over T_{F}, where the right vertical map is induced by the quotient map ${}^{2\rho}\check{\mathrm{B}} \to \check{\mathrm{T}}$.

Remark 4.2.5. Suppose G is semisimple (and simply-laced). The $\check{\mathrm{G}} \cong \mathrm{G}/\mu$-equivariant isomorphism $\check{\mathfrak{g}}^* \cong \mathfrak{g}$ of Remark 4.2.2 restricts to a $\check{\mathrm{B}} \cong \mathrm{B}/\mu$-equivariant isomorphism $\check{\mathfrak{u}}^\perp \cong \mathfrak{b}$, where $\check{\mathfrak{u}} \subseteq \check{\mathfrak{g}}$ is the unipotent radical of $\check{\mathfrak{b}}$. When $\mathrm{F} = \hat{\mathbf{G}}_a(2)$, the map $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}} \to {}^{2\rho}\mathrm{B}_{\mathrm{F}}$ is the usual Kostant slice $\hat{\mathfrak{t}}(2) \hookrightarrow {}^{2\rho}\mathfrak{b}(2)^\wedge_{\mathcal{N}}$. Under the preceding isomorphisms, this can be identified with the map $\hat{\check{\mathfrak{t}}}^*(2) \hookrightarrow {}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}$ sending $x \mapsto \psi + x$, where ψ is a nondegenerate (additive) character of $\check{\mathfrak{u}}$. It follows that $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\hat{\mathbf{G}}_a(2)}$ is isomorphic to $\hat{\check{\mathfrak{t}}}^*(2) \times_{{}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{B}}} \hat{\check{\mathfrak{t}}}^*(2)$. Moreover, the underlying ungraded scheme of ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}} \cong {}^{2\rho}\check{\mathrm{G}} \times^{{}^{2\rho}\check{\mathrm{B}}} {}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}$ is isomorphic to the completion of $\widetilde{\check{\mathfrak{g}}} = \mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{U}})/\check{\mathrm{T}}$ at $\mathcal{N}$.

Remark 4.2.6. The statements of both Proposition 4.1.14 and Corollary 4.1.16 continue to hold with $\widetilde{\mathrm{J}}$ replaced by $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$. Namely, suppose that the bad primes (see Table 1) of G are inverted in R. If V is a graded ${}^{2\rho}\mathrm{B}$-representation in finitely generated projective R-modules, then the $\mathcal{O}_{\mathrm{T}_{\mathrm{F}}}$-linear map $(\mathrm{V} \otimes_{\mathrm{R}} \mathcal{O}_{{}^{2\rho}\mathrm{B}_{\mathrm{F}}})^{{}^{2\rho}\check{\mathrm{B}}} \to$

$(\mathrm{V} \otimes_{\mathrm{R}} \mathcal{O}_{\mathrm{T_F}})^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$ on invariants is an isomorphism. Moreover, if $\mathcal{V}$ is a set of projective graded R-module representations of ${}^{2\rho}\check{\mathrm{B}}$, then pullback along the natural map $\mathrm{T_F}/\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \to {}^{2\rho}\mathrm{B_F}/{}^{2\rho}\check{\mathrm{B}}$ of stacks defines an equivalence between the full subcategory $\mathcal{C}$ of $\mathrm{Perf}({}^{2\rho}\mathrm{B_F}/{}^{2\rho}\check{\mathrm{B}})^{\mathrm{gr},\heartsuit}$ projectively generated by the ${}^{2\rho}\check{\mathrm{B}}$-equivariant $\mathcal{O}_{{}^{2\rho}\mathrm{B_F}}$-modules $\mathcal{O}_{{}^{2\rho}\mathrm{B_F}} \otimes_{\mathrm{R}} \mathrm{V}$ for $\mathrm{V} \in \mathcal{V}$ and the full subcategory of $\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T_F}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})^{\heartsuit}$ projectively generated by the $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$-modules $\mathcal{O}_{\mathrm{T_F}} \otimes_{\mathrm{R}} \mathrm{V}$ for $\mathrm{V} \in \mathcal{V}$.

If $\mathcal{V}$ is a set of projective graded R-module representations of ${}^{2\rho}\check{\mathrm{G}}$ (and not just of ${}^{2\rho}\check{\mathrm{B}}$), then the category $\mathcal{C}$ described above is equivalent to the full subcategory of $\mathrm{Perf}({}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})^{\mathrm{gr},\heartsuit}$ projectively generated by the ${}^{2\rho}\check{\mathrm{G}}$-equivariant $\mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}}$-modules $\mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}} \otimes_{\mathrm{R}} \mathrm{V}$ for $\mathrm{V} \in \mathcal{V}$. If ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}^{\mathrm{aff}}$ denotes the affinization of ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}$, then $\mathcal{C}$ is in turn equivalent to the full subcategory of $\mathrm{Perf}({}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}^{\mathrm{aff}}/{}^{2\rho}\check{\mathrm{G}})^{\mathrm{gr},\heartsuit}$ projectively generated by the ${}^{2\rho}\check{\mathrm{G}}$-equivariant $\mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}^{\mathrm{aff}}}$-modules $\mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}^{\mathrm{aff}}} \otimes_{\mathrm{R}} \mathrm{V}$ for $\mathrm{V} \in \mathcal{V}$.

Proposition 4.2.7. *Suppose that the bad primes (see Table 1) of* G *are inverted in* R.[7] *The projection map* $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \to \mathrm{T_F}$ *is flat (in particular,* $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ *is a classical scheme).*

Proof. As in the proof of Proposition 4.1.14, there is a complete filtration on $\mathcal{O}_{\mathrm{F}}$ given by powers of the augmentation ideal $\ker(\mathcal{O}_{\mathrm{F}} \to \mathrm{R})$, with associated graded $\mathcal{O}_{\hat{\mathbf{G}}_a(2)}$. This in turn induces a complete filtration on $\mathrm{T_F}$ and $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$, and defines a filtered lift of the map $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \to \mathrm{T_F}$. Since both sides are complete for this filtration, it suffices to show that the induced map on associated graded pieces is flat. Since ${}^{2\rho}\mathrm{B}_{\hat{\mathbf{G}}_a(2)} \cong {}^{2\rho}\mathfrak{b}(2)^{\wedge}_{\mathcal{N}}$ by Lemma 4.1.2 (and similarly $\mathrm{T}_{\hat{\mathbf{G}}_a(2)} \cong \hat{\mathfrak{t}}(2)$), we are reduced to showing that the map $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\hat{\mathbf{G}}_a(2)} \to \hat{\mathfrak{t}}(2)$ is flat over L. Since $\hat{\mathbf{G}}_a(2)$ over L is base-changed along the unit map $\mathbf{Z} \to \mathrm{L}$, it suffices to show that the map $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\hat{\mathbf{G}}_a(2)} \to \hat{\mathfrak{t}}(2)$ is flat over $\mathbf{Z}$. The map κ_{F} is the completion of the usual Kostant slice $\check{\mathfrak{t}}^*(2) \cong \mathfrak{t}(2) \to {}^{2\rho}\check{\mathfrak{u}}^{\perp}(2) \cong {}^{2\rho}\mathfrak{b}(2)$ sending $x \mapsto f + x$; here, we have used Remark 4.2.5 to identify $\check{\mathfrak{u}}^{\perp} \cong \mathfrak{b}$. Also, by Item 3 in the proof of Proposition 4.1.14, we may conjugate f to $f_0 = \sum_{\alpha \in \Delta} f_\alpha$, where $f_\alpha \in \mathfrak{g}_\alpha$ are generators of the root space. (This is the only place in the argument where we need to assume that the bad primes of G are inverted, and could likely be removed by proving the results of [Spr66, Section 2] with f used in place of f_0. If we assume that $f = f_0$ from the beginning, then the argument below works over $\mathbf{Z}$ itself, instead of $\mathbf{Z}'$.)

Following [YZ11, Step II of Theorem 6.1], let us argue that the (*a priori* derived) fiber product

$$\mathrm{Y} := \mathfrak{t}(2) \times_{{}^{2\rho}\mathfrak{b}(2)/{}^{2\rho}\mathrm{B}} \mathfrak{t}(2) \cong \mathfrak{t}(2) \times_{{}^{2\rho}\mathfrak{b}(2)} ({}^{2\rho}\mathrm{B} \times \mathfrak{t}(2))$$

is flat over $\mathfrak{t}(2)$. This fiber product is a subgroup scheme of $\mathfrak{t}(2) \times {}^{2\rho}\mathrm{B}$, and its quotient by μ is isomorphic to $\mathfrak{t}(2) \times_{{}^{2\rho}\mathfrak{b}(2)/{}^{2\rho}\check{\mathrm{B}}} \mathfrak{t}(2)$. In particular, the latter remains flat, as desired.

If ${}^{2\rho}\mathrm{B} \times \mathfrak{t}(2) \to {}^{2\rho}\mathfrak{u}(2)$ is the map sending $(g, x) \mapsto \mathrm{Ad}_g(f_0) + \mathrm{Ad}_g(x) - x$, then there is an isomorphism $\mathrm{Y} \cong \{f_0\} \times_{{}^{2\rho}\mathfrak{u}(2)} ({}^{2\rho}\mathrm{B} \times \mathfrak{t}(2))$, where the fiber product is

[7]This assumption is used in a very minor way below (namely, to conjugate f to $f_0 = \sum_{\alpha \in \Delta} f_\alpha$ when $\mathrm{F} = \hat{\mathbf{G}}_a(2)$), and is probably unnecessary.

again derived. Let Y_{cl} denote the underlying classical scheme of Y, so that

$$\dim_{\mathfrak{t}(2)}(\mathrm{Y}_{\mathrm{cl}}) \geq \dim({}^{2\rho}\mathrm{B}) - \dim({}^{2\rho}\mathfrak{u}(2)) = \mathrm{rank}(\mathrm{G}).$$

If we show that the fibers of $\mathrm{Y}_{\mathrm{cl}} \to \mathfrak{t}(2)$ have dimension equal to the rank of G, it would follow that $\mathrm{Y} \cong \mathrm{Y}_{\mathrm{cl}}$, and that Y is a local complete intersection (in particular, Cohen-Macaulay) over $\mathfrak{t}(2)$. Since $\mathfrak{t}(2)$ is an affine space, and all the fibers of $\mathrm{Y} \to \mathfrak{t}(2)$ would be equidimensional, miracle flatness implies that Y is flat, as desired.

There is a contracting $\mathbf{G}_m$-action on $\mathfrak{t}(2)$ (with limit point being the origin), and the map $\mathrm{Y}_{\mathrm{cl}} \to \mathfrak{t}(2)$ is $\mathbf{G}_m$-equivariant. By upper semicontinuity, the dimensions of the fibers of $\mathrm{Y}_{\mathrm{cl}} \to \mathfrak{t}(2)$ are bounded above by the dimension of the fiber of $\mathrm{Y}_{\mathrm{cl}} \to \mathfrak{t}(2)$ over the origin. Let $\mathrm{Y}_{\mathrm{cl},0}$ denote the (underived) pullback $\mathrm{Y}_{\mathrm{cl}} \times_{\mathfrak{t}(2)} \mathrm{Spec}(\mathrm{R})$, where the map $\mathrm{Spec}(\mathrm{R}) \to \mathfrak{t}(2)$ is the inclusion of the origin. Then $\mathrm{Y}_{\mathrm{cl},0}$ is the centralizer $\mathrm{Z}_{{}^{2\rho}\mathrm{B}}(f_0) \subseteq {}^{2\rho}\mathrm{B}$ of $f_0 \in {}^{2\rho}\mathfrak{u}(2)$. If $s \in \mathrm{Spec}(\mathrm{R})$ is a closed point, we need to show that $\mathrm{Z}_{{}^{2\rho}\mathrm{B}}(f_0)|_s \to s$ has dimension rank(G). This follows from the splitting $\mathrm{Z}_{{}^{2\rho}\mathrm{B}}(f_0)|_s \cong \mathrm{Z}(\mathrm{G})|_s \times \mathrm{Z}_{{}^{2\rho}\mathrm{U}}(f_0)|_s$ and [Ken87, Theorem] (where f_0 is denoted X). □

Proposition 4.2.8. *Let $\mathrm{I} \subseteq \Phi^+$ be a subset, and let $\mathrm{U}_\mathrm{I} \subseteq \mathrm{U}$ be a closed subgroup such that the root subgroup $\mathrm{U}_\alpha \subseteq \mathrm{U}_\mathrm{I}$ for each $\alpha \in \mathrm{I}$, and such that U_I is closed under conjugation by T. Let $\mathrm{B}_\mathrm{I} = \mathrm{T}\mathrm{U}_\mathrm{I} \subseteq \mathrm{B}$. Suppose that $e(\alpha) \in \mathcal{O}_{\mathrm{T}_\mathrm{F}}$ is a nontrivial nonzerodivisor for each $\alpha \in \Phi^+ \setminus \mathrm{I}$. The $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-algebra map on rings of functions induced by the action map ${}^{2\rho}\mathrm{B} \times^{{}^{2\rho}\mathrm{B}_\mathrm{I}} ({}^{2\rho}\mathrm{B}_\mathrm{I})_\mathrm{F} \to {}^{2\rho}\mathrm{B}_\mathrm{F}$ is an isomorphism upon base-change along the map $\mathcal{O}_{\mathrm{T}_\mathrm{F}} \to \mathcal{O}_{\mathrm{T}_\mathrm{F}}[e(\Phi^+ \setminus \mathrm{I})^{-1}]$.*

For instance, if $\mathrm{I} = \emptyset$ and $\mathrm{U}_\mathrm{I} = 1 \subseteq \mathrm{U}$, Proposition 4.2.8 gives an isomorphism $\mathcal{O}_{{}^{2\rho}\mathrm{B}_\mathrm{F}}[e(\Phi)^{-1}] \xrightarrow{\cong} \mathcal{O}_{{}^{2\rho}\mathrm{U} \times \mathrm{T}_\mathrm{F}}[e(\Phi)^{-1}]$.

Proof. Let $\mathrm{X} = {}^{2\rho}\mathrm{B} \times^{{}^{2\rho}\mathrm{B}_\mathrm{I}} ({}^{2\rho}\mathrm{B}_\mathrm{I})_\mathrm{F}$, so we need to show that $\mathcal{O}_{{}^{2\rho}\mathrm{B}_\mathrm{F}} \to \mathcal{O}_\mathrm{X}$ is an isomorphism of $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-algebras upon inverting $e(\Phi^+ \setminus \mathrm{I})^{-1}$. As in the proof of Proposition 4.1.14, there is a complete filtration on $\mathcal{O}_\mathrm{F}$ given by powers of the augmentation ideal $\ker(\mathcal{O}_\mathrm{F} \to \mathrm{R})$, with associated graded $\mathcal{O}_{\hat{\mathbf{G}}_a(2)}$. This in turn induces a complete filtration on ${}^{2\rho}\mathrm{B}_\mathrm{F}$ and X, and defines a filtered lift of these objects. Since both $\mathcal{O}_{{}^{2\rho}\mathrm{B}_\mathrm{F}}$ and $\mathcal{O}_\mathrm{X}$ are complete for this filtration, it suffices to show that the induced map on associated graded pieces is an isomorphism. Recall that ${}^{2\rho}\mathrm{B}_{\hat{\mathbf{G}}_a(2)} \cong {}^{2\rho}\mathfrak{b}(2)^\wedge_\mathcal{N}$ by Lemma 4.1.2. When $\mathrm{F} = \hat{\mathbf{G}}_a(2)$, the map $\mathrm{X} \to {}^{2\rho}\mathrm{B}_{\hat{\mathbf{G}}_a(2)} \cong {}^{2\rho}\mathfrak{b}(2)^\wedge_\mathcal{N}$ is the completion of the action map ${}^{2\rho}\mathrm{B} \times^{{}^{2\rho}\mathrm{B}_\mathrm{I}} {}^{2\rho}\mathfrak{b}_\mathrm{I} \to {}^{2\rho}\mathfrak{b}$, so it suffices to show that this map induces an equivalence upon restriction to $\mathfrak{t}_\mathrm{I} := \mathfrak{t} - \bigcup_{\alpha \in \Phi^+ \setminus \mathrm{I}} \mathfrak{t}_\alpha \hookrightarrow \mathfrak{t}$, where $\mathfrak{t}_\alpha$ is the hyperplane in $\mathfrak{t}$ corresponding to the root α. It also suffices to prove that the action map is an isomorphism after forgetting the grading, i.e., that the action map $\mathrm{U} \times^{\mathrm{U}_\mathrm{I}} \mathfrak{b}_\mathrm{I} \to \mathrm{B} \times^{\mathrm{B}_\mathrm{I}} \mathfrak{b}_\mathrm{I} \to \mathfrak{b}$ is an isomorphism upon restriction to $\mathfrak{t}_\mathrm{I} \hookrightarrow \mathfrak{t}$. This result is well-known (for instance, in the case $\mathrm{I} = \emptyset$ and $\mathrm{U}_\mathrm{I} = 1 \subseteq \mathrm{U}$, this is [CG10, Lemma 3.1.44]), but we give a proof for completeness.

Below, we will write $\chi : \mathfrak{b} \to \mathfrak{t}$ denote the natural quotient map. Let

$$1 = \mathrm{U}_n \subseteq \cdots \subseteq \mathrm{U}_2 \subseteq \mathrm{U}_1 = \mathrm{U}.$$

be the filtration on U by root height, and let $\mathrm{U}_{\mathrm{I},i} = \mathrm{U}_\mathrm{I} \cap \mathrm{U}_i$. Let $\mathrm{U}^i = \mathrm{U}/\mathrm{U}_i$, and similarly let $\mathrm{U}_\mathrm{I}^i = \mathrm{U}_\mathrm{I}/\mathrm{U}_{\mathrm{I},i}$. Set $\mathrm{gr}^i(\mathfrak{u}) = \mathfrak{u}_i/\mathfrak{u}_{i+1}$, $\mathfrak{b}^i = \mathfrak{b}/\mathfrak{u}_i$, and $\mathfrak{b}_\mathrm{I}^i = \mathfrak{b}_\mathrm{I}/\mathfrak{u}_\mathrm{I}^i$. Note that $\mathrm{gr}^i(\mathfrak{u}) \cong \mathrm{U}_i/\mathrm{U}_{i+1} \subseteq \mathrm{U}^{i+1}$ is central. In particular, if $v \in \mathrm{gr}^i(\mathfrak{u})$ and $x \in \mathfrak{u}^{i+1}$, then $\mathrm{Ad}_v(x) = x$. More generally, if $x \in \mathfrak{b}^{i+1}$, then $\mathrm{Ad}_v(x) = x + [v, \chi(x)]$. The action map $\mathrm{U} \times^{\mathrm{U_I}} \mathfrak{b}_{\mathrm{I}} \to \mathfrak{b}$ induces a map $q_i : \mathrm{U}^i \times^{\mathrm{U}^i_{\mathrm{I}}} \mathfrak{b}^i_{\mathrm{I}} \to \mathfrak{b}^i$. We will prove by induction on i that q_i induces an isomorphism upon restriction to $\mathfrak{t}_{\mathrm{I}}$. The base case $i = 1$ is clear since $\mathrm{U}^1 = \mathrm{U}$, $\mathrm{U}^1_{\mathrm{I}} = \mathrm{U_I}$, and $\mathfrak{b}^1 \cong \mathfrak{b}^1_{\mathrm{I}} \cong \mathfrak{t}$.

Suppose that q_i is an isomorphism upon restriction to $\mathfrak{t}_{\mathrm{I}}$; we need to show that q_{i+1} is an isomorphism. Let $x \in \mathfrak{b}^{i+1}$ be an element whose image under $\chi : \mathfrak{b}^{i+1} \to \mathfrak{t}$ lies in $\mathfrak{t}_{\mathrm{I}} \subseteq \mathfrak{b}$, and let $\overline{x} \in \mathfrak{b}^i$ denote its image under the quotient map $\mathfrak{b}^{i+1} \to \mathfrak{b}^i$. Since q_i is an isomorphism, there is some $\overline{u} \in \mathrm{U}^i$ and $\overline{y} \in \mathfrak{b}^i_{\mathrm{I}}$ with $\mathrm{Ad}_{\overline{u}}(\overline{y}) = \overline{x}$. Choose lifts $u \in \mathrm{U}^{i+1}$ and $y \in \mathfrak{b}^{i+1}_{\mathrm{I}}$ of $\overline{u}$ and $\overline{y}$; then $n := \mathrm{Ad}_u(y) - x$ lies in the subspace $\mathfrak{u}_i/\mathfrak{u}_{i+1} = \mathrm{gr}^i(\mathfrak{u}) \subseteq \mathfrak{b}^{i+1}$.

If $u' \in \mathrm{U}^{i+1}$ and $y' \in \mathfrak{b}^{i+1}_{\mathrm{I}}$ are other lifts of $\overline{u}$ and $\overline{y}$, then there are some $v \in \mathrm{gr}^i(\mathfrak{u}) \cong \mathrm{U}_i/\mathrm{U}_{i+1} \subseteq \mathrm{U}^{i+1}$ (which is central) and $z \in \mathfrak{b}^{i+1}_{\mathrm{I}} \cap \mathrm{gr}^i(\mathfrak{u}) \subseteq \mathfrak{b}^{i+1}_{\mathrm{I}}$ such that $u' = uv$ and $y' = y + z$. We need to show that there are choices of z and v so that $\mathrm{Ad}_{u'}(y') = x$. Note that $\chi(y') = \chi(y) = \chi(x) \in \mathfrak{t}$. Since

$$\mathrm{Ad}_v(y') = y' + [v, \chi(y')] = y + z + [v, \chi(x)],$$

it follows that

$$\mathrm{Ad}_{u'}(y') = \mathrm{Ad}_u \mathrm{Ad}_v(y') = \mathrm{Ad}_u(y) + z + [v, \chi(x)].$$

Recall the element $n = \mathrm{Ad}_u(y) - x \in \mathrm{gr}^i(\mathfrak{u})$; we need to show that there are choices of z and v so that $[v, \chi(x)] + z = n$. But this is clear: if we write $n = \sum_{\mathrm{ht}(\gamma)=i} n_\gamma e_\gamma$, then the elements

$$v = \sum_{\beta \in \Phi^+ \setminus \mathrm{I}, \mathrm{ht}(\beta)=i} \tfrac{-n_\beta}{\beta(\chi(x))} e_\beta, \; z = \sum_{\alpha \in \mathrm{I}, \mathrm{ht}(\alpha)=i} n_\alpha e_\alpha$$

solve the desired linear system, as desired (indeed, $[v, \chi(x)] = \sum_{\beta \in \Phi^+ \setminus \mathrm{I}, \mathrm{ht}(\beta)=i} n_\beta e_\beta$). Note that the fraction $\frac{n_\beta}{\beta(\chi(x))}$ is well-defined because $\chi(x) \in \mathfrak{t}_{\mathrm{I}}$, so $\beta(\chi(x))$ is a unit when $\beta \in \Phi^+ \setminus \mathrm{I}$. □

Informally, Proposition 4.2.8 says that the map $({}^{2\rho}\mathrm{B_I})_{\mathrm{F}}/{}^{2\rho}\mathrm{B_I} \to {}^{2\rho}\mathrm{B_F}/{}^{2\rho}\mathrm{B}$ induces an isomorphism upon restriction to "$(\mathrm{T_F})_{\mathrm{I}} := \mathrm{T_F} - \bigcup_{\alpha \in \Phi^+ \setminus \mathrm{I}} (\mathrm{T}_\alpha)_{\mathrm{F}}$" $\subseteq \mathrm{T_F}$. (This localization does not make sense, at least in the setting of classical algebraic geometry, because $\mathrm{T_F}$ is a formal scheme.) There are two Kostant slices $\kappa^{\mathrm{I}}_{\mathrm{F}} : \mathrm{T_F} \to ({}^{2\rho}\mathrm{B_I})_{\mathrm{F}}$ and $\kappa_{\mathrm{F}} : \mathrm{T_F} \to {}^{2\rho}\mathrm{B_F}$, and while the composite

$$\mathrm{T_F} \xrightarrow{\kappa^{\mathrm{I}}_{\mathrm{F}}} ({}^{2\rho}\mathrm{B_I})_{\mathrm{F}} \to {}^{2\rho}\mathrm{B_F}$$

need not identify with κ_{F}, Proposition 4.2.8 implies that the restriction of the above composite to "$(\mathrm{T_F})_{\mathrm{I}}$" $\subseteq \mathrm{T_F}$ must be conjugate by an element of ${}^{2\rho}\mathrm{B}$ to the restriction of κ_{F} to "$(\mathrm{T_F})_{\mathrm{I}}$". In particular, there is an isomorphism

$$\text{“}(\mathrm{T_F})_{\mathrm{I}}\text{”} \times_{({}^{2\rho}\mathrm{B_I})_{\mathrm{F}}|_{\text{“}(\mathrm{T_F})_{\mathrm{I}}\text{”}}/{}^{2\rho}\mathrm{B_I}} \text{“}(\mathrm{T_F})_{\mathrm{I}}\text{”} \xrightarrow{\cong} \text{“}(\mathrm{T_F})_{\mathrm{I}}\text{”} \times_{({}^{2\rho}\mathrm{B})_{\mathrm{F}}|_{\text{“}(\mathrm{T_F})_{\mathrm{I}}\text{”}}/{}^{2\rho}\mathrm{B_I}} \text{“}(\mathrm{T_F})_{\mathrm{I}}\text{”}.$$

As mentioned above, this argument is not precise since "$(\mathrm{T_F})_{\mathrm{I}}$" is not well-defined; nevertheless, it is straightforward to translate the above argument to the level of rings of functions (where $\mathcal{O}_{\text{“}(\mathrm{T_F})_{\mathrm{I}}\text{”}}$ is interpreted to mean $\mathcal{O}_{\mathrm{T_F}}[e(\Phi^+ \setminus \mathrm{I})^{-1}]$), where it gives:

Corollary 4.2.9. *Fix notation as in Proposition 4.2.8, and let $\check{\mathrm{G}}_{\mathrm{I}} \subseteq \check{\mathrm{G}}$ denote the neutral connected component of the centralizer of a point of $\mathfrak{t}$ which lies in*

$\mathfrak{t}_{\mathrm{I}} = \mathfrak{t} - \bigcup_{\alpha\in\Phi^+\setminus\mathrm{I}} \mathfrak{t}_\alpha \hookrightarrow \mathfrak{t}$. *(Note that the derived subgroup of* $\check{\mathrm{G}}_{\mathrm{I}}$ *is also simply-laced.) Then the inclusion* $\mathrm{B}_{\mathrm{I}} \subseteq \mathrm{B}$ *induces an isomorphism* $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}[e(\Phi^+\setminus\mathrm{I})^{-1}] \to \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}}_{\mathrm{I}},\mathrm{F}}}[e(\Phi^+\setminus\mathrm{I})^{-1}]$ *of Hopf* $\mathcal{O}_{\mathrm{T}_{\mathrm{F}}}$*-algebras.*

Example 4.2.10. For instance, when $\mathrm{I} = \emptyset$ and $\mathrm{U}_{\mathrm{I}} = 1 \subseteq \mathrm{U}$, Corollary 4.2.9 says that if $e(\alpha) \in \mathcal{O}_{\mathrm{T}_{\mathrm{F}}}$ is a nontrivial nonzerodivisor for each $\alpha \in \Phi^+$, then there is an isomorphism $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}[e(\Phi)^{-1}] \xrightarrow{\cong} \mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}}[e(\Phi)^{-1}]$. Composition with the map $q : \mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}} \to \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$ induced by the homomorphism $q : \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \to \check{\mathrm{T}} \times \mathrm{T}_{\mathrm{F}}$ from Definition 4.2.4 results in the obvious localization map; in other words, the composite

$$\mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}} \to \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \to \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}[e(\Phi)^{-1}] \xrightarrow{\cong} \mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}}[e(\Phi)^{-1}]$$

is the localization map. In particular, the map $\mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}} \to \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$ is an isomorphism upon inverting $e(\Phi)^{-1}$. If the bad primes of G are inverted in R, then $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ is flat over T_{F} by Proposition 4.2.7, and so there is an injection $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \hookrightarrow \mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}}[e(\Phi)^{-1}]$.

Warning 4.2.11. Note that there is a commutative diagram

$$\begin{tikzcd}
 & \mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}} \arrow[dl, "q"'] \arrow[d, "q"] \arrow[dr, "\mathrm{id}"] & \\
\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} & \mathcal{O}_{{}^{2\rho}\check{\mathrm{B}}\times\mathrm{T}_{\mathrm{F}}} \arrow[l] \arrow[r] & \mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}},
\end{tikzcd}$$

which comes from the commutative diagram

$$\begin{tikzcd}
\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \arrow[r, hook] \arrow[dr, "q"'] & {}^{2\rho}\check{\mathrm{B}} \times \mathrm{T}_{\mathrm{F}} \arrow[d, "q"] & \check{\mathrm{T}} \times \mathrm{T}_{\mathrm{F}} \arrow[l, hook'] \arrow[dl, "\mathrm{id}"] \\
 & \check{\mathrm{T}} \times \mathrm{T}_{\mathrm{F}} &
\end{tikzcd}$$

of group schemes over T_{F} (see Definition 4.2.4). The map q induces an isomorphism upon inverting $e(\Phi)$. However, the composite

$$\mathcal{O}_{{}^{2\rho}\check{\mathrm{B}}\times\mathrm{T}_{\mathrm{F}}}[e(\Phi)^{-1}] \to \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}[e(\Phi)^{-1}] \xleftarrow[q]{\cong} \mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}}[e(\Phi)^{-1}]$$

is *not* induced by the inclusion $\check{\mathrm{T}} \hookrightarrow {}^{2\rho}\check{\mathrm{B}}$.

Theorem 4.2.12. *Suppose that* k *is an even* $\mathbf{E}_\infty$*-ring, and that the bad primes of* G *(see Table 1) are invertible in* $\pi_0(k)$.[8] *Continue to assume that* G *is the base-change to* $\pi_*(k)$ *of the Chevalley split form of a connected reductive group whose derived subgroup is simply-laced. Then there is an isomorphism* $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \cong \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ *of Hopf* $\pi_*(k^{h\mathrm{T}})$*-algebras such that the following diagram commutes:*

$$\begin{tikzcd}
\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{T}},\mathrm{F}}} \arrow[r, hook] \arrow[d, "\cong"] & \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \arrow[r, hook] \arrow[d, "\cong"] & \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{T}},\mathrm{F}}}[e(\Phi)^{-1}] \arrow[d, "\cong"] \\
\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; k) \arrow[r, hook] & \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \arrow[r, hook] & \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; k)[e(\Phi)^{-1}].
\end{tikzcd}$$

[8]This assumption is only used to appeal to Proposition 4.2.7; as remarked there, it is used only in a very minor way, and is probably unnecessary.

Proof. We may assume $k = \mathrm{MU}'$, where MU' denotes the localization of MU at the bad primes of G; the case of general k follows by (completed) base-change along the map $\pi_*(\mathrm{MU}'^{h\mathrm{T}}) \to \pi_*(k^{h\mathrm{T}})$. We will apply the preceding results to the case $\mathrm{R} = \pi_*(k)$, i.e., to the Lazard ring L' localized at the bad primes of G, with F being the universal formal group law with a coordinate. Recall from Theorem 3.1.11 that $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ is isomorphic to the Hopf algebra $\mathcal{A}_\mathrm{G}$ from Definition 3.1.10. It therefore suffices to show that $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \cong \mathcal{A}_\mathrm{G}$. Since $\pi_*(k) \cong \mathrm{L}'$ is a normal domain, the same is true of $\pi_*(k^{h\mathrm{T}}) \cong \mathcal{O}_{\mathrm{T}_\mathrm{F}}$. Because $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$ is flat and finitely presented over $\pi_*(k^{h\mathrm{T}})$ by Proposition 4.2.7, the natural map

$$\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \to \bigcap_{\alpha \in \Phi} \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \left[e(\Phi \setminus \alpha)^{-1}\right]$$

is an isomorphism. Since $\mathcal{A}_\mathrm{G}$ is defined similarly, it suffices to show that $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \left[e(\Phi \setminus \alpha)^{-1}\right]$ isomorphic to

$$\mathcal{A}_\mathrm{G}\left[e(\Phi \setminus \alpha)^{-1}\right] \cong \pi_*(k^{h\mathrm{T}})[\mathbb{X}_*]\left[\tfrac{e^{\check{\alpha}}-1}{e(\alpha)}, e(\Phi \setminus \alpha)^{-1}\right]$$

as Hopf subalgebras of $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{T}},\mathrm{F}}}[e(\Phi)^{-1}] \cong \mathcal{A}_\mathrm{T}[e(\Phi)^{-1}]$.

For this, we apply Corollary 4.2.9 to the case $\mathrm{I} = \{\alpha\}$ (note that since our base is L', the element $e(\alpha) \in \mathcal{O}_{\mathrm{T}_\mathrm{F}}$ is a nontrivial nonzerodivisor by Lemma 2.2.20; so Corollary 4.2.9 is applicable). Then $\check{\mathrm{G}}_\mathrm{I} \subseteq \check{\mathrm{G}}$ has semisimple rank 1, so we are reduced to checking that there is an isomorphism $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \cong \mathcal{A}_\mathrm{G}$ of Hopf $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-algebras when G has semisimple rank 1. (We will check this without inverting any primes!) Since G is then a finite central quotient of $\mathrm{T}' \times \mathrm{SL}_2$ for a torus T', it suffices to consider the cases $\mathrm{G} = \mathrm{PGL}_2, \mathrm{SL}_2$. In these cases, the desired isomorphism was proved in Example 4.1.13 (corresponding to $n = 1$ and $n = 2$, respectively). □

Remark 4.2.13. Let L' be the localization of the Lazard ring at the bad primes of G, where G is the Chevalley split form of a connected reductive group whose derived subgroup is simply-laced. The argument for Theorem 4.2.12 shows that $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \cong \mathcal{A}_\mathrm{G}$, and we have already seen in Definition 3.1.10 that $\mathcal{W}$ acts on $\mathcal{A}_\mathrm{G} \subseteq \mathcal{O}_{\check{\mathrm{T}} \times \mathrm{T}_\mathrm{F}}[e(\Phi)^{-1}]$. (By base-changing along the map $\mathrm{L}' \to \mathrm{R}$ classifying a 1-dimensional formal group over R with a coordinate, it follows that $\mathcal{W}$ acts on $\mathcal{A}_\mathrm{G}$, even though $\mathcal{A}_\mathrm{G}$ may not inject into $\mathcal{O}_{\check{\mathrm{T}} \times \mathrm{T}_\mathrm{F}}[e(\Phi)^{-1}]$.) Theorem 3.1.11 implies that the isomorphism $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \cong \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ of Theorem 4.2.12 is $\mathcal{W}$-equivariant.

Remark 4.2.14. Recall from Theorem 3.1.11 that if $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$, then $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathcal{F})$ is canonically a graded $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$-comodule (in $\mathcal{W}$-comodules). Using Theorem 4.2.12, we obtain a functor $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathcal{G}\mathrm{r}_\mathrm{G}; -) : \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \to \mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T}_\mathrm{F}} \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})$. There is also a functor $\mathrm{Rep}^{\mathrm{gr}}({}^{2\rho}\check{\mathrm{G}} \times \mathrm{T}_\mathrm{F}) \to \mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T}_\mathrm{F}} \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})$ which comes from the homomorphisms $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \hookrightarrow {}^{2\rho}\check{\mathrm{B}} \times \mathrm{T}_\mathrm{F} \hookrightarrow {}^{2\rho}\check{\mathrm{G}} \times \mathrm{T}_\mathrm{F}$. It follows from Corollary 5.1.11 that there is a certain full subcategory $\mathcal{C} \subseteq \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ (which is the entirety of $\mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ rationally, for instance) for which the functor $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathcal{G}\mathrm{r}_\mathrm{G}; -) : \mathcal{C} \to \mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T}_\mathrm{F}} \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})$ lifts to a functor $\mathcal{C} \to \mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}} \times \mathrm{T}_\mathrm{F})$. This should be viewed as an analogue for sheaves with k-coefficients of the equivalence $\mathrm{Perv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Z}) \xrightarrow{\simeq} \mathrm{Rep}(\check{\mathrm{G}}_\mathbf{Z})$ from [MV07].

Suppose $\mathrm{F} = \hat{\mathbf{G}}_a(2)$. The assumption that the bad primes of G are inverted in $\pi_0(k)$ was only used in Proposition 4.2.7 to conjugate f to $f_0 = \sum_{\alpha\in\Delta} f_\alpha$. If we worked with f_0 itself (instead of f), then Theorem 4.2.12 can therefore be refined to show that if k is an even $\mathbf{Z}$-algebra (with no primes inverted!) and G is the Chevalley split form of a connected reductive group whose derived subgroup is simply-laced, there is an isomorphism

$$\operatorname{Spec} \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \cong \hat{\mathfrak{t}}(2) \times_{{}^{2\rho}\mathfrak{b}(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{B}}} \hat{\mathfrak{t}}(2)$$

of group schemes over $\pi_*(k^{h\mathrm{T}}) \cong \hat{\mathfrak{t}}(2)$, where the map $\hat{\mathfrak{t}}(2) \to {}^{2\rho}\mathfrak{b}(2)^\wedge_{\mathcal{N}}$ sends $x \mapsto f_0 + x$. In fact, an essentially identical argument shows the following (which generalizes [YZ11, Theorem 1.2]); note that the argument is very nearly classification-free.

Proposition 4.2.15. *Let k be an even $\mathbf{Z}$-algebra (with no primes inverted!). If G is the Chevalley split form of a connected reductive group (not necessarily with simply-laced derived subgroup), there is an isomorphism*

$$\operatorname{Spec} \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \cong \hat{\check{\mathfrak{t}}}^*(2) \times_{{}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{B}}} \hat{\check{\mathfrak{t}}}^*(2) \tag{4.2.2}$$

of group schemes over $\pi_*(k^{h\mathrm{T}}) \cong \hat{\check{\mathfrak{t}}}^*(2)$, where the map $\hat{\check{\mathfrak{t}}}^*(2) \to {}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}$ sends $x \mapsto \psi_0 + x$, where ψ_0 is the map ${}^{2\rho}\check{\mathfrak{u}}^-(2) \to \mathbf{A}^1$ given by

$${}^{2\rho}\check{\mathfrak{u}}^-(2) \to {}^{2\rho}\check{\mathfrak{u}}^-(2)/[{}^{2\rho}\check{\mathfrak{u}}^-(2), {}^{2\rho}\check{\mathfrak{u}}^-(2)] \cong \prod_{\alpha\in\Delta} \check{\mathfrak{g}}^*_{-\alpha} \xrightarrow{\sum} \mathbf{A}^1.$$

Proof sketch. This can be shown by arguing exactly as in Theorem 4.2.12, with some care needed in the analogue of Proposition 4.2.7 in showing that $\hat{\check{\mathfrak{t}}}^*(2) \times_{{}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{B}}} \hat{\check{\mathfrak{t}}}^*(2)$ is flat over $\hat{\check{\mathfrak{t}}}^*(2)$. Let us sketch the argument. Arguing as in the proof of Proposition 4.2.7, we are reduced to showing that over an algebraically closed field k of characteristic p, the centralizer $\mathrm{Z}_{\check{\mathrm{B}}}(\psi_0)$ has dimension equal to rank(G). If p does not divide the lacing number ℓ of $\check{\mathrm{G}}$, then the minimal bilinear form defines an isomorphism $\check{\mathfrak{g}} \cong \check{\mathfrak{g}}^*$ by Remark 4.2.2. Under this isomorphism, the element ψ_0 is sent to $\ell \sum_{\alpha\in\Delta^s} f_\alpha + \sum_{\alpha\in\Delta^l} f_\alpha$, where Δ^s (resp. Δ^l) is the set of short (resp. long) simple roots of $\check{\mathrm{G}}$. Scaling f_α for short simple roots α by ℓ^{-1}, we may assume that ψ_0 is sent to $f_0 = \sum_{\alpha\in\Delta} f_\alpha$, in which case the centralizer $\mathrm{Z}_{\check{\mathrm{B}}}(\psi_0)$ was shown to have dimension equal to rank(G) in [Ken87]. (Said differently, the isomorphism $\check{\mathfrak{g}} \cong \check{\mathfrak{g}}^*$ essentially reduces the flatness to [YZ11, Step II of Theorem 6.1].)

It remains only to verify the case when $p|\ell$, in which case $\check{\mathfrak{g}}$ and $\check{\mathfrak{g}}^*$ are *not* isomorphic. For this, we may assume $\check{\mathrm{G}}$ is simple, and then argue case-by-case. The relevant cases to consider are $(\check{\mathrm{G}}, p)$ equal to $(\mathrm{Sp}_{2n}, 2)$, $(\mathrm{SO}_{2n+1}, 2)$, $(\mathrm{G}_2, 3)$, and $(\mathrm{F}_4, 2)$.

- When $(\check{\mathrm{G}}, p) = (\mathrm{Sp}_{2n}, 2)$ with standard $2n$-dimensional representation V, the coadjoint representation $\mathfrak{sp}_{2n}^*$ is isomorphic to $\mathrm{Sym}^2(\mathrm{V})$. Choose a symplectic basis for V, denoted $\{x_1, \cdots, x_n, y_1, \cdots, y_n\}$. Under the isomorphism $\mathfrak{sp}_{2n}^* \cong \mathrm{Sym}^2(\mathrm{V})$, ψ_0 corresponds to $x_n^2 + \sum_{i=1}^{n-1} x_i y_{i+1}$. A direct calculation shows that the stabilizer of this element under $\check{\mathrm{B}} \subseteq \check{\mathrm{G}}$ is n-dimensional. For instance, when $n = 2$, the stabilizer is isomorphic to $\mu_2 \times \begin{pmatrix} 1 & a & a^2 & b \\ 0 & 1 & 0 & a^2 \\ 0 & 0 & 1 & a \\ 0 & 0 & 0 & 1 \end{pmatrix}$.

- When $(\check{\mathrm{G}}, p) = (\mathrm{SO}_{2n+1}, 2)$ one can argue exactly as in the case $(\mathrm{Sp}_{2n}, 2)$, except now using the isomorphism $\mathfrak{so}_{2n+1} \cong \wedge^2(\mathrm{W})$, where W is the standard $(2n+1)$-dimensional representation of $\check{\mathrm{G}}$. If we choose a basis for W^* denoted $x_1, \cdots, x_n, y_1, \cdots, y_n, z$, where z is dual to a vector which spans the 1-dimensional radical of the bilinear form associated to the quadratic form on W, then ψ_0 corresponds to $x_n \wedge z + \sum_{i=1}^{n-1} x_i \wedge y_{i+1}$. Again, a direct calculation shows that the stabilizer of this element under $\check{\mathrm{B}} \subseteq \check{\mathrm{G}}$ is n-dimensional.
- The cases $(\check{\mathrm{G}}, p) = (\mathrm{G}_2, 3)$ and $(\mathrm{F}_4, 2)$ can also be checked by direct calculation: see, e.g., [Xue14, Tables 1 and 2]. □

Example 4.2.16. As a very explicit illustration of Proposition 4.2.15, consider the case $\mathrm{G} = \mathrm{G}_2$. By [Bot58], there is an isomorphism

$$\mathrm{H}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}_2}; \mathbf{Z}) \cong \mathbf{Z}[u, v, w]/(2v = u^2),$$

where u, v, and w live in weights 2, 4, and 10. The (cocommutative) coalgebra structure on $\mathrm{H}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}_2}; \mathbf{Z})$ is given by

$$\begin{aligned}
\Delta(u) &= u \otimes 1 + 1 \otimes u,\\
\Delta(v) &= v \otimes 1 + u \otimes u + 1 \otimes v\\
\Delta(w) &= w \otimes 1 + 3v^2 \otimes u + 6uv \otimes v + 6v \otimes uv + 3u \otimes v^2 + 1 \otimes w.
\end{aligned}$$

(Note that Bott's formulas in [Bot58, Equation 11.6], as well as in the calculation of $\mathrm{H}^*(\mathrm{G}_2/\mathrm{GL}_2^{\mathrm{short}}; \mathbf{Z})$, have minor typos.) On the other hand, if α is a short simple root of G_2 and β is a long simple root, then [Ken87] shows that the centralizer $\mathrm{Z}_{\check{\mathrm{B}}}(f_0) \subseteq \check{\mathrm{G}} \cong \mathrm{G}_2$ of G_2 acting on $f_0 = f_\alpha + f_\beta \in \mathfrak{g}_2$ (not $\psi_0 \in \mathfrak{g}_2^*$!) is

$$\mathrm{Z}_{\check{\mathrm{B}}}(f_0) \cong \operatorname{Spec} \mathbf{Z}[x_\alpha, x_\beta, x_{\alpha+\beta}, x_{2\alpha+\beta}, x_{3\alpha+\beta}, x_{3\alpha+2\beta}]/\mathrm{I},$$

where the ideal I is generated by the functions

$$x_\beta - x_\alpha,\ 2x_{\alpha+\beta} - 2x_\alpha x_\beta + x_\alpha^2,\ 3x_\alpha^2 x_\beta - 6x_\alpha x_{\alpha+\beta} + 3x_{2\alpha+\beta} - x_\alpha^3,\ 3(x_{\alpha+\beta}^2 - x_\beta x_{2\alpha+\beta}) + x_{3\alpha+\beta}.$$

The element x_μ lies in weight given by twice the height of the root μ. Of course, this simplifies to

$$\mathrm{Z}_{\mathrm{B}}(f_0) \cong \operatorname{Spec} \mathbf{Z}[x_\alpha, x_{\alpha+\beta}, x_{2\alpha+\beta}, x_{3\alpha+2\beta}]/(2x_{\alpha+\beta} - x_\alpha^2,\ 2x_\alpha^3 - 6x_\alpha x_{\alpha+\beta} + 3x_{2\alpha+\beta}).$$

In particular, as already observed in [YZ11], $\mathrm{Z}_{\check{\mathrm{B}}}(f_0)_{\mathbf{F}_3}$ is not isomorphic (even as a scheme) to $\operatorname{Spec} \mathrm{H}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}_2}; \mathbf{F}_3)$ (since the latter does not admit a function whose cube is zero). However, following Proposition 4.2.15, if one instead considers the centralizer $\mathrm{Z}_{\check{\mathrm{B}}}(\psi_0) \subseteq \check{\mathrm{G}} \cong \mathrm{G}_2$ of G_2 acting on $\psi_0 \in \mathfrak{g}_2^*$, then

$$\mathrm{Z}_{\check{\mathrm{B}}}(\psi_0) \cong \operatorname{Spec} \mathbf{Z}[y_\alpha, y_\beta, y_{\alpha+\beta}, y_{2\alpha+\beta}, y_{3\alpha+\beta}, y_{3\alpha+2\beta}]/\mathrm{J},$$

where the ideal J is generated by the functions

$$3y_\beta - y_\alpha,\ 2y_{\alpha+\beta} - 6y_\alpha y_\beta + y_\alpha^2,\ 3y_\alpha^2 y_\beta - 2y_\alpha y_{\alpha+\beta} + y_{2\alpha+\beta} - y_\alpha^3/3,\ y_{\alpha+\beta}^2 - 3y_\beta y_{2\alpha+\beta} + y_{3\alpha+\beta}.$$

Again, the element y_μ lies in weight given by twice the height of the root μ. (These relations can be deduced by setting $y_\mu = 3x_\mu$ for long positive roots μ in the generating relations for the ideal I, and dividing any relation which vanishes modulo 3 by 3.) This simplifies to

$$\mathrm{Z}_{\check{\mathrm{B}}}(\psi_0) \cong \operatorname{Spec} \mathbf{Z}[y_\beta, y_{\alpha+\beta}, y_{3\alpha+2\beta}]/(2y_{\alpha+\beta} + 3y_\beta^2),$$

which is indeed isomorphic to $\mathrm{Spec}\,\mathrm{H}_*(\mathcal{G}r_{\mathrm{G}_2};\mathbf{Z})$ before inverting any primes. They are even isomorphic as group schemes.

Let us note, for the sake of completeness, that the relations in $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}r_{\mathrm{G}};\mathbf{Z})$ sometimes deform nontrivially to $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}r_{\mathrm{G}};\mathrm{MU})$; for instance, the arguments of [Cla74, Proposition 7.1] using Bott's generating variety $\mathrm{G}_2/\mathrm{GL}_2^{\mathrm{long}} \to \Omega\mathrm{G}_2$ [Bot58] can be used to show that there is an isomorphism

$$\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}r_{\mathrm{G}_2};\mathrm{MU}) \cong \pi_*(\mathrm{MU})[u,v,w]/(u^2 = 2v + 3x_1 u),$$

where $\pi_*(\mathrm{MU}) \cong \mathbf{Z}[x_1, x_2, \cdots]$, and x_1 is represented by the cobordism class of $\mathbf{C}\mathrm{P}^1$.

4.3. $\mathcal{W}$**-descent.** We now discuss the effect of taking $\mathcal{W}$-quotients in Proposition 4.2.7. If k is a $\mathbf{Q}$-algebra, then $\mathcal{W}$ is isomorphic to the groupoid $\hat{\check{\mathfrak{t}}}^*(2) \times_{\hat{\check{\mathfrak{t}}}^*(2)/\!\!/\mathrm{W}} \hat{\check{\mathfrak{t}}}^*(2)$, and there is a Cartesian square

$$\begin{tikzcd}
\hat{\check{\mathfrak{t}}}^*(2) \arrow[r] \arrow[d] & {}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2)/{}^{2\rho}\check{\mathrm{G}} \cong {}^{2\rho}\check{\mathfrak{u}}^\perp(2)/{}^{2\rho}\check{\mathrm{B}} \arrow[d] \\
\hat{\check{\mathfrak{t}}}^*(2)/\!\!/\mathrm{W} \arrow[r] & {}^{2\rho}\check{\mathfrak{g}}(2)/{}^{2\rho}\check{\mathrm{G}} \cong {}^{2\rho}\check{\mathfrak{g}}^*(2)/{}^{2\rho}\check{\mathrm{G}}.
\end{tikzcd}$$

This follows from its uncompleted and ungraded version, where it follows from the Cartesian square

$$\begin{tikzcd}
\widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}} \arrow[r] \arrow[d] & \check{\mathfrak{t}}^* \arrow[d] \\
\check{\mathfrak{g}}^{\mathrm{reg}} \arrow[r] & \check{\mathfrak{t}}^*/\!\!/\mathrm{W}
\end{tikzcd}$$

proved by Kostant [Kos78] by pulling back along the Kostant slice $\check{\mathfrak{t}}^*/\!\!/\mathrm{W} \to \check{\mathfrak{g}}^{\mathrm{reg}}$. This implies that

$$\mathrm{Spec}\,\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}r_{\mathrm{G}};k) \cong \hat{\check{\mathfrak{t}}}^*(2) \times_{{}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{B}}} \hat{\check{\mathfrak{t}}}^*(2) \cong \hat{\check{\mathfrak{t}}}^*(2) \times_{{}^{2\rho}\check{\mathfrak{g}}^*(2)/{}^{2\rho}\check{\mathrm{G}}} \hat{\check{\mathfrak{t}}}^*(2)/\!\!/\mathrm{W},$$

and taking $\mathcal{W}$-quotients (equivalently, taking the invariant-theoretic quotient by W) produces an isomorphism

$$\mathrm{Spec}\,\mathrm{H}_*^{\mathrm{G}_\mathcal{O},\mathrm{BM}}(\mathcal{G}r_{\mathrm{G}};k) \cong \hat{\check{\mathfrak{t}}}^*(2)/\!\!/\mathrm{W} \times_{{}^{2\rho}\check{\mathfrak{g}}^*(2)/{}^{2\rho}\check{\mathrm{G}}} \hat{\check{\mathfrak{t}}}^*(2)/\!\!/\mathrm{W}$$

of group schemes over $\mathrm{Spec}(\mathrm{gr}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}})) \cong \mathrm{Spf}(\pi_*(k^{h\mathrm{G}})) \cong \hat{\check{\mathfrak{t}}}^*(2)/\!\!/\mathrm{W}$. (The right-hand side above is a completion of the regular centralizer group scheme of $\check{\mathrm{G}}$ acting on $\check{\mathfrak{g}}^*$.) These statements do not hold as stated for more general k, but variants nevertheless hold, as we now discuss.

Lemma 4.3.1. *Suppose that the bad primes (see Table 1) of* G *and the order of the fundamental group of its derived subgroup are inverted in* R*. Then the centralizer inside* ${}^{2\rho}\check{\mathrm{U}}^- \times \mathfrak{t}(2)$ *of* $\mathfrak{t}(2) \xrightarrow{\kappa_{\mathrm{F}}} {}^{2\rho}\mathfrak{b}(2) \subseteq {}^{2\rho}\mathfrak{g}(2)$ *is trivial.*

Proof. For $x \in \mathfrak{t}$, we need to show that $\mathrm{Z}_{\check{\mathrm{U}}^-}(f+x) = \{1\}$. This is classical, but we review an argument for completeness. (See also the proof of [Cot22, Theorem 3.10], where our claim follows from the regularity of $f + x$ and the isomorphism $\mathrm{Z}_{\check{\mathrm{B}}}(f+x) \xrightarrow{\cong} \mathrm{Z}_{\check{\mathrm{G}}}(f+x)$.) Using the $\mathbf{G}_m$-action on $\mathfrak{t}$ scaling it to the origin and semicontinuity, it suffices to show that $\mathrm{Z}_{\check{\mathrm{U}}^-}(f) = \{1\}$. This follows from the isomorphism $\mathrm{Z}_{\mathrm{B}}(f) \xrightarrow{\cong} \mathrm{Z}_{\mathrm{G}}(f)$ from (the proof of) [Cot22, Theorem 3.10]. $\square$

Proposition 4.3.2. *Let* R *be a graded commutative ring, and let* F *be a* 1*-dimensional graded formal group over* R *equipped with a coordinate. Suppose* G *is the base-change to* R *of the Chevalley split form of a connected reductive group whose derived subgroup is simply-laced. The composite map*

$$\mathrm{T}_{\mathrm{F}} \xrightarrow{\kappa_{\mathrm{F}}} {}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$$

admits a factorization through the stacky quotient $\mathrm{T}_{\mathrm{F}}/\mathcal{W}$*; I will continue to denote the resulting map* $\mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ *by* κ_{F}*.*

Proof. For this, we first produce a map $\sigma : \underline{\mathrm{W}} \to {}^{2\rho}\check{\mathrm{G}} \times {}^{2\rho}\mathrm{G}_{\mathrm{F}}$ of groupoid schemes which lives over the map $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}} \to {}^{2\rho}\mathrm{B}_{\mathrm{F}} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}$. In fact, we will produce a map $\sigma : \underline{\mathrm{W}} \to {}^{2\rho}\check{\mathrm{U}}^- \times {}^{2\rho}\mathrm{G}_{\mathrm{F}}$, which will make the following diagram commute:

$$\begin{tikzcd}
\underline{\mathrm{W}} \arrow[r, "\sigma"] \arrow[d, "\eta"'] & {}^{2\rho}\check{\mathrm{U}}^- \times {}^{2\rho}\mathrm{G}_{\mathrm{F}} \arrow[d, "\eta"] \\
\mathrm{T}_{\mathrm{F}} \arrow[r, "\kappa_{\mathrm{F}}"'] & {}^{2\rho}\mathrm{G}_{\mathrm{F}},
\end{tikzcd}$$

where η denotes the action map.

Let us begin by defining $\sigma_{s_\alpha,x}$ such that $\mathrm{Ad}_{\sigma_{s_\alpha,x}}(\kappa_{\mathrm{F}}(x)) = \kappa_{\mathrm{F}}(s_\alpha x)$. Using Remark 4.1.15, this reduces to the case when G has semisimple rank 1; since the claim itself depends only on the derived subgroup of G, we may further reduce to G being SL_2 or PGL_2. For simplicity, let us take $\mathrm{G} = \mathrm{SL}_2$ (a similar argument works for PGL_2 with minor modifications). In this case, $\kappa_{\mathrm{F}}(x) \in \mathrm{B}_{\mathrm{F}}$ is described in Example 4.1.13: if $\mathcal{L}_x$ is the line bundle on $\mathrm{BF}^\vee$ associated to an S-point $x \in \mathrm{F}$, then $\kappa_{\mathrm{F}}(x)$ is the extension of $\mathcal{L}_x$ by $\mathcal{L}_x^{-1}$ determined by the unique S-module generator of $\mathrm{H}^1(\mathrm{BF}^\vee; \mathcal{L}_x^{\otimes 2})$. The underlying $\check{\mathrm{G}}$-bundles of $\kappa_{\mathrm{F}}(x)$ and $\kappa_{\mathrm{F}}(s_\alpha x)$ are isomorphic, even though the underlying $\check{\mathrm{B}}$-bundles of $\kappa_{\mathrm{F}}(x)$ and $\kappa_{\mathrm{F}}(s_\alpha x)$ are not isomorphic (since the line subbundles $\mathcal{L}_x$ and $\mathcal{L}_{s_\alpha(x)} = \mathcal{L}_x^{-1}$ are not isomorphic). Since $\kappa_{\mathrm{F}}(x), \kappa_{\mathrm{F}}(s_\alpha x) \in \mathrm{G}_{\mathrm{F}}$ are chosen trivializations of the underlying $\check{\mathrm{G}}$-bundles of $\kappa_{\mathrm{F}}(x)$ and $\kappa_{\mathrm{F}}(s_\alpha x)$ at the basepoint of $\mathrm{BF}^\vee$, this implies that there exists a unique S-point $\sigma_{s_\alpha,x} \in \check{\mathrm{G}}$ such that $\mathrm{Ad}_{\sigma_{s_\alpha,x}}(\kappa_{\mathrm{F}}(x)) = \kappa_{\mathrm{F}}(s_\alpha x)$. In fact, upon choosing a coordinate on F, we may write $\sigma_{s_\alpha,x} = \left(\begin{smallmatrix} 1 & 0 \\ \alpha(x) & 1 \end{smallmatrix}\right) \in \mathbf{G}_a \cong \check{\mathrm{U}}^- \subseteq \check{\mathrm{G}}$. (Upon choosing a coordinate on F, the element $\sigma_{s_\alpha,x}$ for general $\check{\mathrm{G}}$ can be written as the element $\alpha(x) \in \mathbf{G}_a \cong \check{\mathrm{U}}_{-\alpha} \subseteq \check{\mathrm{U}}^-$.)

We now return to defining σ in general. Concretely, suppose $w \in \mathrm{W}$ and $x \in \mathrm{T}_{\mathrm{F}}$. We need to construct $\sigma_{w,x} \in {}^{2\rho}\check{\mathrm{U}}^- \subseteq {}^{2\rho}\check{\mathrm{G}}$ such that $\mathrm{Ad}_{\sigma_{w,x}}(\kappa_{\mathrm{F}}(x)) = \kappa_{\mathrm{F}}(wx)$. Write $w = s_{\alpha_1} \cdots s_{\alpha_n}$ for some sequence $\alpha_1, \cdots, \alpha_n$ of simple roots, and define $\sigma_{w,x} = \sigma_{s_{\alpha_1}, s_{\alpha_2} \cdots s_{\alpha_n} x} \cdots \sigma_{s_{\alpha_n},x} \in {}^{2\rho}\check{\mathrm{U}}^-$. This will satisfy the identity $\mathrm{Ad}_{\sigma_{w,x}}(\kappa_{\mathrm{F}}(x)) = \kappa_{\mathrm{F}}(wx)$, so we need to show that $\sigma_{w,x}$ will be independent of the choice of presentation of w as a product of simple reflections, and that the assignment $(w, x) \mapsto (\sigma_{w,x}, x)$ defines a homomorphism $\underline{\mathrm{W}} \to {}^{2\rho}\check{\mathrm{G}} \times {}^{2\rho}\mathrm{G}_{\mathrm{F}}$ of groupoid schemes which lives over the map $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}} \to {}^{2\rho}\mathrm{B}_{\mathrm{F}} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}$ (i.e., $\sigma_{w',wx}\sigma_{w,x} = \sigma_{w'w,x}$). Let us check independence of presentation; a similar argument proves that σ is a homomorphism of groupoids. If $\sigma'_{w,x} \in {}^{2\rho}\check{\mathrm{U}}^-$ is obtained by another choice of presentation of w, we need to show that $\sigma_{w,x} = \sigma'_{w,x}$. It suffices to check this in the initial case when L is the Lazard ring and F is the universal formal group law with a coordinate. The assignments $x \mapsto \sigma_{w,x}, \sigma'_{w,x}$ define two maps $\mathrm{T}_\mathrm{F} \to {}^{2\rho}\check{\mathrm{U}}^- \times \mathrm{T}_\mathrm{F}$, i.e., two maps $\mathcal{O}_{{}^{2\rho}\check{\mathrm{U}}^-} \otimes \mathcal{O}_{\mathrm{T}_\mathrm{F}} \to \mathcal{O}_{\mathrm{T}_\mathrm{F}}$. Since ${}^{2\rho}\check{\mathrm{U}}^-$ is just an affine space, it suffices to check that these two maps agree upon composition with the injective map $\mathcal{O}_{\mathrm{T}_\mathrm{F}} \hookrightarrow \mathcal{O}_{(\mathrm{T}_\mathrm{F})_\mathbf{Q}}$. (This map is injective because L, and hence $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$, is flat over $\mathbf{Z}$.) The isomorphism $\mathrm{F}_\mathbf{Q} \cong \hat{\mathbf{G}}_a(2)_\mathbf{Q}$ over $\mathrm{L}_\mathbf{Q}$ defines an isomorphism $(\mathrm{T}_\mathrm{F})_\mathbf{Q} \cong \hat{\mathfrak{t}}(2)_\mathbf{Q}$, and so we are reduced to the case $\mathrm{F} = \hat{\mathbf{G}}_a(2)$ over $\mathbf{Q}$. Since $\sigma_{w,x}^{-1}\sigma'_{w,x}$ centralizes $\kappa_\mathrm{F}(x)$, Lemma 4.3.1 implies that it must be the identity.

The preceding discussion defines a map $\sigma : \underline{\mathrm{W}} \to {}^{2\rho}\check{\mathrm{G}} \times {}^{2\rho}\mathrm{G}_\mathrm{F}$ of groupoid schemes which lives over the map $\kappa_\mathrm{F} : \mathrm{T}_\mathrm{F} \to {}^{2\rho}\mathrm{B}_\mathrm{F} \to {}^{2\rho}\mathrm{G}_\mathrm{F}$. To obtain the desired map $\mathcal{W} \to {}^{2\rho}\check{\mathrm{G}} \times {}^{2\rho}\mathrm{G}_\mathrm{F}$ of groupoids which lives over the map $\kappa_\mathrm{F} : \mathrm{T}_\mathrm{F} \to {}^{2\rho}\mathrm{G}_\mathrm{F}$, we need to check that the map $\sigma^* : \mathcal{O}_{{}^{2\rho}\check{\mathrm{G}} \times {}^{2\rho}\mathrm{G}_\mathrm{F}} \to \mathrm{Map}(\mathrm{W}, \mathcal{O}_{\mathrm{T}_\mathrm{F}})$ factors through the map $\mathcal{O}_\mathcal{W} \to \mathrm{Map}(\mathrm{W}, \mathcal{O}_{\mathrm{T}_\mathrm{F}})$. For this, we may consider the universal case when R is the Lazard ring. In this case, the map $\mathcal{O}_\mathcal{W} \to \mathrm{Map}(\mathrm{W}, \mathcal{O}_{\mathrm{T}_\mathrm{F}})$ is injective by Example 2.2.21. More precisely, $\mathcal{O}_\mathcal{W}$ is the subalgebra of $\mathrm{Map}(\mathrm{W}, \mathcal{O}_{\mathrm{T}_\mathrm{F}})$ of those functions $h : \mathrm{W} \to \mathcal{O}_{\mathrm{T}_\mathrm{F}}$ such that $h(w) \equiv h(s_\alpha w) \pmod{e(\alpha)}$. That σ^* factors through $\mathcal{O}_\mathcal{W}$ now follows from the construction of σ and the identity $x \equiv s_\alpha(x) \pmod{e(\alpha)}$ for $x \in \mathcal{O}_{\mathrm{T}_\mathrm{F}}$. □

Remark 4.3.3. In characteristic 0 (or large enough characteristic), Kostant's slice [Kos78] is a map $\mathfrak{t}/\mathcal{W} \cong \mathfrak{t}/\!\!/\mathrm{W} \to \mathfrak{g}$. Note that Proposition 4.3.2 does *not* produce such a map; instead, it only produces a map $\mathfrak{t}/\!\!/\mathrm{W} \to \mathfrak{g}/\check{\mathrm{G}}$ of which Kostant's slice is a lift. This is particularly important in small characteristics: we will see in Proposition 4.3.9 below that there is always an isomorphism $\mathfrak{t}/\mathcal{W} \cong (f + \mathfrak{b})/\check{\mathrm{U}}^-$, under which the map $\mathfrak{t}/\mathcal{W} \to \mathfrak{g}/\check{\mathrm{G}}$ from Proposition 4.3.2 is the canonical map $(f + \mathfrak{b})/\check{\mathrm{U}}^- \to \mathfrak{g}/\check{\mathrm{G}}$. However, unlike in large characteristic, the $\check{\mathrm{U}}^-$-action on $f + \mathfrak{b}$ may fail to be free, so there is no slice of $(f + \mathfrak{b})/\check{\mathrm{U}}^-$, and one cannot lift the map $(f + \mathfrak{b})/\check{\mathrm{U}}^- \to \mathfrak{g}/\check{\mathrm{G}}$ along $\mathfrak{g} \to \mathfrak{g}/\check{\mathrm{G}}$. (This happens for $\mathrm{G} = \mathrm{PGL}_n$ in characteristic p if $p|n$; see, e.g., Appendix C for some calculations.)

Lemma 4.3.4. *Let* Z_F *denote the "*F*-Steinberg variety", defined as the (a priori) derived fiber product* ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F} \times_{{}^{2\rho}\mathrm{G}_\mathrm{F}} {}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}$*. Then* Z_F *is a flat locally complete intersection over* R*. In particular, it is a classical scheme.*

Proof. It suffices to prove the claim in the universal case, when R is the Lazard ring L and F is the universal formal group with a coordinate. Arguing as in Proposition 4.2.7, we may further reduce to the case $\mathrm{R} = \mathbf{Z}$ and $\mathrm{F} = \hat{\mathbf{G}}_a(2)$, in which case it suffices to show that the derived fiber product $\mathrm{Z} := {}^{2\rho}\widetilde{\check{\mathfrak{g}}} \times_{{}^{2\rho}\mathfrak{g}} {}^{2\rho}\widetilde{\check{\mathfrak{g}}}$ is flat over $\mathbf{Z}$.

There is a derived pullback square

$$\begin{tikzcd} \mathrm{Z} \arrow[r] \arrow[d] & {}^{2\rho}\widetilde{\check{\mathfrak{g}}} \times {}^{2\rho}\widetilde{\check{\mathfrak{g}}} \arrow[d] \\ \mathfrak{g} \arrow[r, "\Delta"'] & \mathfrak{g} \times \mathfrak{g}. \end{tikzcd}$$

Let Z_cl denote the underlying classical scheme of Z. Since ${}^{2\rho}\widetilde{\check{\mathfrak{g}}}$ is flat of dimension $\dim(\mathrm{G})$ over $\mathbf{Z}$, it follows that $\dim(\mathrm{Z}_\mathrm{cl}) \geq \dim(\mathrm{G})$. If we show that the fibers of the map $\mathrm{Z} \to \mathrm{Spec}(\mathbf{Z})$ have dimension exactly $\dim(\mathrm{G})$, it would follow that $\mathrm{Z} \cong \mathrm{Z}_\mathrm{cl}$, and that Z is a local complete intersection (in particular, Cohen-Macaulay) over $\mathrm{Spec}(\mathbf{Z})$. Since all the fibers of $Z \to \mathrm{Spec}(\mathbf{Z})$ would be equidimensional, miracle flatness would imply that Z is flat over $\mathrm{Spec}(\mathbf{Z})$, as desired.

Assume that R is a field; we need to show that $\dim(Z_{\mathrm{cl}}) = \dim(G)$. By definition, Z_{cl} is the variety of triples $(x, \check{B}_1, \check{B}_2)$ where $\check{B}_1$ and $\check{B}_2$ are Borel subgroups of $\check{G}$ and $x \in \mathfrak{b}_1 \cap \mathfrak{b}_2$. For $w \in W$, let $Z_w \subseteq Z_{\mathrm{cl}}$ denote the locally closed stratum of those triples $(x, \check{B}_1, \check{B}_2)$ for which $\check{B}_1$ and $\check{B}_2$ are in relative position w. If $X_w \subseteq \check{G}/\check{B} \times \check{G}/\check{B}$ denotes the $\check{G}$-orbit corresponding to $w \in W$, then X_w is smooth and irreducible of dimension $\ell(w) + |\Phi^+|$. The restriction of the projection $Z_{\mathrm{cl}} \to \check{G}/\check{B} \times \check{G}/\check{B}$ to X_w is an $\check{G}$-equivariant vector bundle, and its fibers are isomorphic to $\mathfrak{b} \cap \mathrm{Ad}_w \mathfrak{b} \cong \mathfrak{t} \oplus \bigoplus_{\alpha\in\Phi^+, w^{-1}\alpha\in\Phi^+} \mathfrak{g}_\alpha$. This is an affine space of dimension $\mathrm{rank}(G) + |\Phi^+| - \ell(w)$, so that $\dim(Z_{\mathrm{cl}}|_{X_w}) = \mathrm{rank}(G) + 2|\Phi^+| = \dim(G)$, as needed. $\square$

Proposition 4.3.5. *The composite map*

$$^{2\rho}B_F/^{2\rho}\check{B} \times_{^{2\rho}G_F/^{2\rho}\check{G}} {}^{2\rho}B_F/^{2\rho}\check{B} \to {}^{2\rho}B_F/^{2\rho}\check{B} \times {}^{2\rho}B_F/^{2\rho}\check{B} \to T_F \times T_F$$

admits a factorization through the map $\mathcal{W} \to T_F \times T_F$ *(given by the source and target maps of the Weyl groupoid). The resulting map* $\chi : {}^{2\rho}B_F/^{2\rho}\check{B} \times_{^{2\rho}G_F/^{2\rho}\check{G}} {}^{2\rho}B_F/^{2\rho}\check{B} \to \mathcal{W}$ *is a map of groupoid schemes over the map* ${}^{2\rho}B_F/^{2\rho}\check{B} \to T_F$. *Moreover, the composite*

$$\mathcal{W} \cong T_F \times_{T_F/\mathcal{W}} T_F \xrightarrow{\kappa_F} {}^{2\rho}B_F/^{2\rho}\check{B} \times_{^{2\rho}G_F/^{2\rho}\check{G}} {}^{2\rho}B_F/^{2\rho}\check{B} \xrightarrow{\chi} \mathcal{W}$$

is an isomorphism.

Proof. It suffices to construct the desired map in the universal case, when R is the Lazard ring L and F is the universal formal group with a coordinate. There is an isomorphism

$$^{2\rho}B_F/^{2\rho}\check{B} \times_{^{2\rho}G_F/^{2\rho}\check{G}} {}^{2\rho}B_F/^{2\rho}\check{B} \cong Z_F/^{2\rho}\check{G},$$

where $Z_F = {}^{2\rho}\widetilde{\check{G}}_F \times_{^{2\rho}G_F} {}^{2\rho}\widetilde{\check{G}}_F$. We will show that there is a ${}^{2\rho}\check{G}$-equivariant dotted arrow making the following diagram (${}^{2\rho}\check{G}$-equivariantly) commute:

$$\begin{tikzcd} Z_F \arrow[r, dashed, "\chi"] \arrow[d, shift left] \arrow[d, shift right] & \mathcal{W} \arrow[d, shift left] \arrow[d, shift right] \\ {}^{2\rho}\widetilde{\check{G}}_F \arrow[r, "\chi"'] & T_F. \end{tikzcd}$$

By definition, Z_F (which is isomorphic to its underlying classical scheme by Lemma 4.3.4) is the variety of triples $(x, \check{B}_1, \check{B}_2)$ where $\check{B}_1$ and $\check{B}_2$ are Borel subgroups of $\check{G}$ and $x \in ({}^{2\rho}B_1)_F \cap ({}^{2\rho}B_2)_F$. For $w \in W$, let $Z_w \subseteq Z_F$ denote the locally closed stratum of those triples $(x, \check{B}_1, \check{B}_2)$ for which $\check{B}_1$ and $\check{B}_2$ are in relative position w. Let $\overline{Z_w}$ denote the closure of Z_w. Recall that $(Z_F)_{\mathbf{Q}} \cong ({}^{2\rho}\widetilde{\check{\mathfrak{g}}} \times_{\mathfrak{g}} {}^{2\rho}\widetilde{\check{\mathfrak{g}}})^\wedge_{\mathcal{N}}$. This implies that $H^0((Z_F)_{\mathbf{Q}}; \mathcal{O}_{(Z_F)_{\mathbf{Q}}})$ is reduced, and so since since Z_F is flat over L and L is torsion-free, the map $H^0(Z_F; \mathcal{O}_{Z_F}) \to H^0((Z_F)_{\mathbf{Q}}; \mathcal{O}_{(Z_F)_{\mathbf{Q}}})$ is injective. In particular, $H^0(Z_F; \mathcal{O}_{Z_F})$ is also reduced. Since the irreducible components of Z_F are given by the $\overline{Z_w}$ for $w \in W$, the map $H^0(Z_F; \mathcal{O}_{Z_F}) \to \prod_{w\in W} H^0(\overline{Z_w}; \mathcal{O}_{\overline{Z_w}})$ is injective. Since Z_F is a flat locally complete intersection over L by Lemma 4.3.4,

$\mathrm{H}^0(\mathrm{Z_F}; \mathcal{O}_{\mathrm{Z_F}})$ is the equalizer of the diagram

$$\prod_{w\in \mathrm{W}} \mathrm{H}^0(\overline{\mathrm{Z}_w}; \mathcal{O}_{\overline{\mathrm{Z}_w}}) \rightrightarrows \prod_{w\in \mathrm{W}, \alpha\in\Phi^+} \mathrm{H}^0(\overline{\mathrm{Z}_w}\cap\overline{\mathrm{Z}_{s_\alpha w}}; \mathcal{O}_{\overline{\mathrm{Z}_w}\cap\overline{\mathrm{Z}_{s_\alpha w}}}),$$

where the two maps are given by restriction.

The composite

$$\chi_w : \overline{\mathrm{Z}_w} \hookrightarrow \mathrm{Z_F} \xrightarrow{\mathrm{pr}_1} {}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}} \xrightarrow{\chi} \mathrm{T_F}$$

defines a map $\chi_w^* : \mathcal{O}_{\mathrm{T_F}} \to \mathrm{H}^0(\overline{\mathrm{Z}_w}; \mathcal{O}_{\overline{\mathrm{Z}_w}})$. As w ranges over elements of the Weyl group, we obtain a map $\chi^* : \mathrm{Map}(\mathrm{W}, \mathcal{O}_{\mathrm{T_F}}) \to \prod_{w\in\mathrm{W}} \mathrm{H}^0(\overline{\mathrm{Z}_w}; \mathcal{O}_{\overline{\mathrm{Z}_w}})$. By construction, the map $\mathcal{O}_{\mathcal{W}} \to \mathrm{Map}(\mathrm{W}, \mathcal{O}_{\mathrm{T_F}})$ is injective (here, we use that R is the Lazard ring, so as to appeal to Example 2.2.21), with image consisting of those functions $f : \mathrm{W} \to \mathcal{O}_{\mathrm{T_F}}$ such that $f(w) \equiv f(s_\alpha w) \pmod{e(\alpha)}$. To show that $\chi^*|_{\mathcal{O}_{\mathcal{W}}}$ lands in $\mathrm{H}^0(\mathrm{Z_F}; \mathcal{O}_{\mathrm{Z_F}}) \hookrightarrow \prod_{w\in\mathrm{W}} \mathrm{H}^0(\overline{\mathrm{Z}_w}; \mathcal{O}_{\overline{\mathrm{Z}_w}})$, it suffices to show that for each $\alpha \in \Phi^+$, there is a commutative diagram

$$\begin{tikzcd}
\overline{\mathrm{Z}_w}\cap\overline{\mathrm{Z}_{s_\alpha w}} \arrow[rr] \arrow[d] & & \mathrm{Spec}(\mathrm{R}) \arrow[d, "0"] \\
\overline{\mathrm{Z}_w} \arrow[r, "\chi_w"'] & \mathrm{T_F} \arrow[r, "\alpha"'] & \mathrm{F}.
\end{tikzcd}$$

By definition, $\overline{\mathrm{Z}_w}$ is the moduli of triples $(x, \check{\mathrm{B}}_1, \check{\mathrm{B}}_2)$ for which $\check{\mathrm{B}}_1$ and $\check{\mathrm{B}}_2$ are in relative position $\leq w$. If such a point lifts to $\overline{\mathrm{Z}_w}\cap\overline{\mathrm{Z}_{s_\alpha w}}$, then the image of x in $(\check{\mathrm{B}}_1/[\check{\mathrm{B}}_1, \check{\mathrm{B}}_1])_{\mathrm{F}} \cong \mathrm{T_F}$ must lie inside $(\mathrm{T}_\alpha)_{\mathrm{F}} \hookrightarrow \mathrm{T_F}$, so that $\alpha(\chi(x)) = 0$ as desired.

It is straightforward to verify that the ${}^{2\rho}\check{\mathrm{G}}$-equivariant map $\chi : \mathrm{Z_F} \to \mathcal{W}$ described above is a map of groupoid schemes over the map $\chi : {}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}} \to \mathrm{T_F}$, and moreover that the composite

$$\mathcal{W} \cong \mathrm{T_F} \times_{\mathrm{T_F}/\mathcal{W}} \mathrm{T_F} \xrightarrow{\kappa_{\mathrm{F}}} \mathrm{Z_F}/{}^{2\rho}\check{\mathrm{G}} \xrightarrow{\chi} \mathcal{W}$$

is an isomorphism. □

Theorem 4.3.6. *The composite map*

$${}^{2\rho}\mathrm{B_F}/{}^{2\rho}\check{\mathrm{B}} \to \mathrm{T_F} \to \mathrm{T_F}/\mathcal{W}$$

admits a factorization through a map $\chi : {}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}} \to \mathrm{T_F}/\mathcal{W}$. *The resulting map*

$$\mathrm{T_F}/\mathcal{W} \xrightarrow{\kappa_{\mathrm{F}}} {}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}} \xrightarrow{\chi} \mathrm{T_F}/\mathcal{W}$$

is an isomorphism.

Suppose that the bad primes (see Table 1) of G *are inverted in* R. *Then the natural map* $q : {}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}} \to {}^{2\rho}\mathrm{G_F} \times_{\mathrm{T_F}/\mathcal{W}} \mathrm{T_F}$ *exhibits* ${}^{2\rho}\mathrm{G_F} \times_{\mathrm{T_F}/\mathcal{W}} \mathrm{T_F}$ *as the affinization of* ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}$.

Proof. Proposition 4.3.5 produces a ${}^{2\rho}\check{\mathrm{G}}$-equivariant map $\chi : \mathrm{Z_F} \to \mathcal{W}$ of groupoid schemes over the map ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}} \to \mathrm{T_F}$. This induces a ${}^{2\rho}\check{\mathrm{G}}$-equivariant map $\chi : {}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}/\mathrm{Z_F} \to \mathrm{T_F}/\mathcal{W}$. Let $\mu : {}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}} \cong {}^{2\rho}\check{\mathrm{G}} \times^{{}^{2\rho}\check{\mathrm{B}}} {}^{2\rho}\mathrm{B_F} \to {}^{2\rho}\mathrm{G_F}$ denote the natural map. We will show that the map ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}/\mathrm{Z_F} \to {}^{2\rho}\mathrm{G_F}$ is an isomorphism. Filtering F by powers of the augmentation ideal $\ker(\mathcal{O}_{\mathrm{F}} \to \mathrm{R})$ (so that the associated graded is $\hat{\mathbf{G}}_a(2)$) defines a filtered lift of the map ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}/\mathrm{Z_F} \to {}^{2\rho}\mathrm{G_F}$ whose associated graded is the completion of the map ${}^{2\rho}\widetilde{\check{\mathfrak{g}}}/\mathrm{Z} \to {}^{2\rho}\mathfrak{g}$, where $\mathrm{Z} = {}^{2\rho}\widetilde{\check{\mathfrak{g}}} \times_{{}^{2\rho}\mathfrak{g}} {}^{2\rho}\widetilde{\check{\mathfrak{g}}}$. Since the filtrations are complete, it suffices to show that the latter map is an isomorphism. Because $\hat{\mathbf{G}}_a(2)$ over R is base-changed along the unit map $\mathbf{Z} \to \mathrm{R}$, we may assume that $\mathrm{R} = \mathbf{Z}$ concentrated in weight 0. Since Z is the groupoid ${}^{2\rho}\widetilde{\check{\mathfrak{g}}} \times_{{}^{2\rho}\mathfrak{g}} {}^{2\rho}\widetilde{\check{\mathfrak{g}}}$ over ${}^{2\rho}\widetilde{\check{\mathfrak{g}}}$, the map ${}^{2\rho}\widetilde{\check{\mathfrak{g}}}/\mathrm{Z} \to {}^{2\rho}\mathfrak{g}$ is an isomorphism if the map ${}^{2\rho}\check{\mathrm{G}} \times^{{}^{2\rho}\check{\mathrm{B}}} {}^{2\rho}\mathfrak{b} \cong {}^{2\rho}\widetilde{\check{\mathfrak{g}}} \to \mathfrak{g}$ is surjective. This in turn follows from the observation that $\mathfrak{g}$ is the union of conjugates of $\mathfrak{b}$.

The preceding paragraph produces the desired map $\chi : {}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}} \to \mathrm{T}_\mathrm{F}/\mathcal{W}$. That the composite

$$\mathrm{T}_\mathrm{F}/\mathcal{W} \xrightarrow{\kappa_\mathrm{F}} {}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}} \xrightarrow{\chi} \mathrm{T}_\mathrm{F}/\mathcal{W}$$

is an isomorphism follows from Proposition 4.3.5. It remains only to show that the map $q : {}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F} \to {}^{2\rho}\mathrm{G}_\mathrm{F} \times_{\mathrm{T}_\mathrm{F}/\mathcal{W}} \mathrm{T}_\mathrm{F}$ exhibits ${}^{2\rho}\mathrm{G}_\mathrm{F} \times_{\mathrm{T}_\mathrm{F}/\mathcal{W}} \mathrm{T}_\mathrm{F}$ as the affinization of ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}$. In other words, q defines a map $\mathcal{O}_{{}^{2\rho}\mathrm{G}_\mathrm{F} \times_{\mathrm{T}_\mathrm{F}/\mathcal{W}} \mathrm{T}_\mathrm{F}} \to \mathrm{R}\Gamma({}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}; \mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}})$, which we need to show identifies the source as the connective cover of the target.

We first show that $\mathrm{R}\Gamma({}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}; \mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}})$ is concentrated in degree zero, where it is flat over R. As before, filtering $\mathcal{O}_\mathrm{F}$ by powers of the augmentation ideal $\ker(\mathcal{O}_\mathrm{F} \to \mathrm{R})$ (so that the associated graded is $\mathcal{O}_{\hat{\mathbf{G}}_a(2)}$) induces a complete filtration on ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}$ whose associated graded is ${}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2)^\wedge_\mathcal{N}$. To show that $\mathrm{R}\Gamma({}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}; \mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}})$ is concentrated in degree zero, where it is flat over R, it suffices to prove that $\mathrm{R}\Gamma({}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2)^\wedge_\mathcal{N}; \mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2)^\wedge_\mathcal{N}})$ is concentrated in degree zero, where it is flat over R. In turn, it suffices to show that $\mathrm{R}\Gamma({}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2); \mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2)})$ is concentrated in degree zero, where it is flat over R.

To prove this, we may ignore the natural gradings. Let $p : \widetilde{\check{\mathfrak{g}}} \to \check{\mathrm{G}}/\check{\mathrm{B}} =: \mathcal{B}$ denote the natural quotient map. Then p is affine, so that $\mathrm{R}\Gamma(\widetilde{\check{\mathfrak{g}}}; \mathcal{O}_{\widetilde{\check{\mathfrak{g}}}}) \cong \mathrm{R}\Gamma(\mathcal{B}; p_*\mathcal{O}_{\widetilde{\check{\mathfrak{g}}}})$. We will show that this is concentrated in degree zero, where it is isomorphic to $\mathcal{O}_{\mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}}$. We may assume that R is an algebraically closed field. Since $\widetilde{\check{\mathfrak{g}}} \cong \check{\mathrm{G}} \times^{\check{\mathrm{B}}} \mathfrak{b}$, and there is an isomorphism $\mathfrak{b} \cong \check{\mathfrak{u}}^\perp$ (see Remark 4.2.2), $p_*\mathcal{O}_{\widetilde{\check{\mathfrak{g}}}}$ is the quasicoherent sheaf on $\mathcal{B}$ whose fiber over $\check{\mathrm{B}}' \in \mathcal{B}$ (with unipotent radical $\check{\mathrm{U}}'$) is isomorphic to $\mathrm{Sym}(\check{\mathfrak{g}}/\check{\mathfrak{u}}')$. Since $\mathrm{Sym}(\check{\mathfrak{g}}/\check{\mathfrak{u}}') \cong \mathrm{Sym}(\check{\mathfrak{g}}) \otimes_{\mathrm{Sym}(\check{\mathfrak{u}}')} \mathrm{R}$, there is a (Koszul) filtration on $\mathrm{Sym}(\check{\mathfrak{g}}/\check{\mathfrak{u}}')$ with associated graded given by $\mathrm{Sym}(\check{\mathfrak{g}}) \otimes \mathrm{Sym}(\check{\mathfrak{u}}'[1]) \cong \bigoplus_{i \geq 0} \mathrm{Sym}(\check{\mathfrak{g}}) \otimes \wedge^i(\check{\mathfrak{u}}')[i]$. If $\underline{\check{\mathfrak{u}}}$ denotes the sheaf over $\mathcal{B}$ whose fiber over $\check{\mathrm{B}}'$ is $\check{\mathfrak{u}}'$, then $\mathrm{R}\Gamma(\mathcal{B}; \mathrm{Sym}(\check{\mathfrak{g}}) \otimes \wedge^i(\underline{\check{\mathfrak{u}}})) \cong \mathrm{Sym}(\check{\mathfrak{g}}) \otimes_\mathrm{R} \mathrm{R}\Gamma(\mathcal{B}; \wedge^i(\underline{\check{\mathfrak{u}}}))$. To prove the desired vanishing, it therefore suffices to show that $\mathrm{R}\Gamma(\mathcal{B}; \wedge^i(\underline{\check{\mathfrak{u}}}))$ is concentrated in cohomological degree i, where it is a flat R-module. The claim follows from [AJ84, Lemma 3.4(a)] (see also [Jan03, Section II.6.18]), which in fact moreover implies that $\mathrm{R}\Gamma(\mathcal{B}; \wedge^i(\underline{\check{\mathfrak{u}}})[i]) \cong \mathrm{H}^i(\mathcal{B}; \Omega^i_\mathcal{B})$ is a flat R-module whose base-change to any field has dimension $\#\{w \in \mathrm{W} | \ell(w) = i\}$.

Returning to our goal, we have a map $\mathcal{O}_{{}^{2\rho}\mathrm{G}_\mathrm{F} \times_{\mathrm{T}_\mathrm{F}/\mathcal{W}} \mathrm{T}_\mathrm{F}} \to \mathrm{R}\Gamma({}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}; \mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}}) \cong \mathrm{H}^0({}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}; \mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}})$ of flat R-modules which we need to show is an isomorphism. (Note that $\mathrm{T}_\mathrm{F} \to \mathrm{T}_\mathrm{F}/\mathcal{W}$ is flat, so ${}^{2\rho}\mathrm{G}_\mathrm{F} \times_{\mathrm{T}_\mathrm{F}/\mathcal{W}} \mathrm{T}_\mathrm{F} \to {}^{2\rho}\mathrm{G}_\mathrm{F}$ is flat; since ${}^{2\rho}\mathrm{G}_\mathrm{F}$ is flat over R, it follows that ${}^{2\rho}\mathrm{G}_\mathrm{F} \times_{\mathrm{T}_\mathrm{F}/\mathcal{W}} \mathrm{T}_\mathrm{F}$ is also flat over R.) As usual, we reduce to the universal case when R is the Lazard ring L. By flatness over L and completeness along the kernel of $\mathrm{L} \to \mathbf{Z}$, we are further reduced to checking that the map $\mathcal{O}_{{}^{2\rho}\mathfrak{g}(2) \times_{\mathfrak{t}(2)/\mathcal{W}} \mathfrak{t}(2)} \to \mathrm{H}^0(\widetilde{\check{\mathfrak{g}}}(2); \mathcal{O}_{\widetilde{\check{\mathfrak{g}}}(2)})$ is an isomorphism, in other words that the map $\widetilde{\check{\mathfrak{g}}} \to \mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}$ is the affinization map.

It suffices to show that $\mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}$ is normal, and that the map $\widetilde{\check{\mathfrak{g}}} \to \mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}$ is proper and birational. We show the latter first.

(1) The map $\widetilde{\check{\mathfrak{g}}} \to \mathfrak{g}$ is proper, since it is the composite $\widetilde{\check{\mathfrak{g}}} \hookrightarrow \mathfrak{g} \times \mathcal{B} \xrightarrow{\mathrm{pr}} \mathfrak{g}$ (the first map is a closed immersion, and the second is proper because $\mathcal{B}$ is proper). Since the map $\mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t} \to \mathfrak{g}$ is separated (because $\mathfrak{t} \to \mathfrak{t}/\mathcal{W}$ is an affine morphism), it follows that the map $\widetilde{\check{\mathfrak{g}}} \to \mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}$ is proper.

(2) For birationality of the map $\widetilde{\check{\mathfrak{g}}} \to \mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}$, let $\mathfrak{t}_\emptyset = \mathfrak{t} - \bigcup_{\alpha \in \Phi^+} \mathfrak{t}_\alpha$ denote the complement of the root hyperplanes (so W acts freely on $\mathfrak{t}_\emptyset$). Then $\mathfrak{t}_\emptyset/\mathcal{W} \cong \mathfrak{t}_\emptyset/\mathrm{W}$, and the restriction of $\mathfrak{t} \to \mathfrak{t}/\mathcal{W}$ to $\mathfrak{t}_\emptyset/\mathcal{W}$ is the quotient map $\mathfrak{t}_\emptyset \to \mathfrak{t}_\emptyset/\mathrm{W}$. Let $\mathfrak{g}^{\mathrm{rs}}$ denote $\mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}_\emptyset/\mathrm{W}$, and similarly let $\widetilde{\check{\mathfrak{g}}}^{\mathrm{rs}} = \widetilde{\check{\mathfrak{g}}} \times_{\mathfrak{t}} \mathfrak{t}_\emptyset$. It then suffices to show that there is a free W-action on $\widetilde{\check{\mathfrak{g}}}^{\mathrm{rs}}$ such that the map $\widetilde{\check{\mathfrak{g}}}^{\mathrm{rs}} \to \mathfrak{t}_\emptyset$ is W-equivariant, and such that the map $\widetilde{\check{\mathfrak{g}}}^{\mathrm{rs}} \to \mathfrak{g}^{\mathrm{rs}}$ is a principal W-bundle. By Proposition 4.2.8 (applied to $\mathrm{R} = \mathbf{Z}$, so that $e(\alpha) = \alpha \in \mathcal{O}_{\mathfrak{t}} = \mathrm{Sym}(\mathfrak{t}^*)$ is a nontrivial nonzerodivisor for each $\alpha \in \Phi^+$), there is an isomorphism $\widetilde{\check{\mathfrak{g}}}^{\mathrm{rs}} \cong \check{\mathrm{G}} \times^{\check{\mathrm{T}}} \mathfrak{t}_\emptyset \cong \check{\mathrm{G}}/\check{\mathrm{T}} \times \mathfrak{t}_\emptyset$. The Weyl group W acts diagonally, via the natural $\mathrm{W} = \mathrm{N}_{\check{\mathrm{G}}}(\check{\mathrm{T}})/\check{\mathrm{T}}$-action on $\check{\mathrm{G}}/\check{\mathrm{T}}$ and the W-action on $\mathfrak{t}_\emptyset$. To show that the map $\widetilde{\check{\mathfrak{g}}}^{\mathrm{rs}} \to \mathfrak{g}^{\mathrm{rs}}$ is a principal W-bundle, note that if $(g\check{\mathrm{T}}, x), (h\check{\mathrm{T}}, y) \in \check{\mathrm{G}}/\check{\mathrm{T}} \times \mathfrak{t}_\emptyset$ are sent to the same point of $\mathfrak{g}^{\mathrm{rs}}$, so that $\mathrm{Ad}_g(x) = \mathrm{Ad}_h(y)$, then $\mathrm{Ad}_{h^{-1}g}(x) = y$. Because x and y are regular semisimple, $\mathrm{Z}_{\check{\mathrm{G}}}(x)$ and $\mathrm{Z}_{\check{\mathrm{G}}}(y)$ are isomorphic to the maximal torus $\check{\mathrm{T}}$, so that $h^{-1}g \in \mathrm{N}_{\check{\mathrm{G}}}(\check{\mathrm{T}})$. (See also [KW76, Lemma 3.1], and well as [Tot18, Theorem 9.1].) If $w \in \mathrm{W}$ denotes the image of $h^{-1}g$, then $w \cdot (g\check{\mathrm{T}}, x) = (h\check{\mathrm{T}}, y)$, as desired.

(3) It remains to show that $\mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}$ is normal. Note that $\mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t} \to \mathfrak{g}$ is flat, since the map $\mathfrak{t} \to \mathfrak{t}/\mathcal{W}$ is flat. This implies that $\mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}$ is a complete intersection. Normality therefore follows from showing that $\mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}$ is smooth in codimension 1. For this, let $\mathfrak{g}^{\mathrm{reg}}$ denote the open subscheme of $\mathfrak{g}$ of those points whose stabilizer under $\check{\mathrm{G}}$ has dimension equal to rank(G). We claim that $\mathfrak{g}^{\mathrm{reg}} \hookrightarrow \mathfrak{g}$ has complement of codimension 2, and moreover that the map $\mathfrak{g}^{\mathrm{reg}}/\check{\mathrm{G}} \to \mathfrak{t}/\mathcal{W}$ is smooth and surjective. The latter implies that the map $\mathfrak{g}^{\mathrm{reg}} \to \mathfrak{t}/\mathcal{W}$ is smooth and surjective, so that $\mathfrak{g}^{\mathrm{reg}} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}$ is smooth; since $\mathfrak{g}^{\mathrm{reg}} \hookrightarrow \mathfrak{g}$ has complement of codimension 2, it follows that $\mathfrak{g} \times_{\mathfrak{t}/\mathcal{W}} \mathfrak{t}$ is smooth in codimension 1, as desired.

   (a) Let us show that $\mathfrak{g}^{\mathrm{reg}} \hookrightarrow \mathfrak{g}$ has complement of codimension 2. Let $\mathfrak{g}^{\mathrm{irreg}}$ denote this complement. Similarly, let $\mathfrak{b}^{\mathrm{reg}}$ denote the open subvariety of $\mathfrak{b}$ of those $x \in \mathfrak{b}$ whose centralizer has dimension equal to rank(G); this was denoted by $\mathfrak{b}^{\mathrm{reg}'}$ in the proof of Proposition 4.1.14. Let $\mathfrak{b}^{\mathrm{irreg}}$ denote the complement of $\mathfrak{b}^{\mathrm{reg}} \hookrightarrow \mathfrak{b}$. Using this as the definition of $\mathfrak{b}^{\mathrm{reg}}$. By Item 2 in the proof of Proposition 4.1.14, $\mathfrak{b}^{\mathrm{irreg}} \hookrightarrow \mathfrak{b}$ has codimension 2. The map $\pi : \widetilde{\check{\mathfrak{g}}} \to \mathfrak{g}$ is proper and surjective, and moreover the image of $\widetilde{\check{\mathfrak{g}}}^{\mathrm{irreg}} := \check{\mathrm{G}} \times^{\check{\mathrm{B}}} \mathfrak{b}^{\mathrm{irreg}}$ under π is $\mathfrak{g}^{\mathrm{irreg}}$. Therefore, $\dim(\mathfrak{g}^{\mathrm{irreg}}) \leq \dim(\widetilde{\check{\mathfrak{g}}}^{\mathrm{irreg}})$; but this is $\dim(\widetilde{\check{\mathfrak{g}}}) - 2 = \dim(\mathfrak{g}) - 2$, so $\mathfrak{g}^{\mathrm{irreg}} \hookrightarrow \mathfrak{g}$ has codimension 2.

   (b) We now show that the map $\mathfrak{g}^{\mathrm{reg}}/\check{\mathrm{G}} \to \mathfrak{t}/\mathcal{W}$ is smooth and surjective. Recall that from Item 3 in the proof of Proposition 4.1.14 that the map $\mathfrak{t} \to \mathfrak{b}^{\mathrm{reg}}/\check{\mathrm{B}} \cong \widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}}/\check{\mathrm{G}}$ is flat and surjective (namely, it is the quotient by the group scheme $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$, which is flat over $\mathfrak{t}$ by Proposition 4.2.7). We will momentarily show that $\widetilde{\check{\mathfrak{g}}} \times_{\mathfrak{g}} \widetilde{\check{\mathfrak{g}}}|_{\widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}}} = \widetilde{\check{\mathfrak{g}}} \times_{\mathfrak{g}} \widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}}$ is flat over $\widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}}$. Quotienting by the action of this groupoid on $\widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}}$ shows that the map $\widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}} \to \mathfrak{g}^{\mathrm{reg}}$ is flat and surjective. It follows that the composite $\mathfrak{t} \to \widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}}/\check{\mathrm{G}} \to \mathfrak{g}^{\mathrm{reg}}/\check{\mathrm{G}}$ is flat and surjective (since each map is so, by our discussion above). Therefore, the composite $\mathfrak{t} \to \mathfrak{t}/\mathcal{W} \to \mathfrak{g}^{\mathrm{reg}}/\check{\mathrm{G}}$ is flat and surjective. Since $\mathcal{W}$ is flat over $\mathfrak{t}$, it follows that the map $\mathfrak{t}/\mathcal{W} \to \mathfrak{g}^{\mathrm{reg}}/\check{\mathrm{G}}$ is flat and surjective. Because the composite $\mathfrak{t}/\mathcal{W} \to \mathfrak{g}^{\mathrm{reg}}/\check{\mathrm{G}} \to \mathfrak{t}/\mathcal{W}$ is an isomorphism (hence in particular smooth), [SPa, Lemma 05B5] implies that the map $\mathfrak{g}^{\mathrm{reg}}/\check{\mathrm{G}} \to \mathfrak{t}/\mathcal{W}$ is smooth and surjective.

We now show that $\widetilde{\check{\mathfrak{g}}} \times_{\mathfrak{g}} \widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}}$ is flat over $\widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}}$. The scheme $\widetilde{\check{\mathfrak{g}}} \times_{\mathfrak{g}} \widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}}$ is open inside $\widetilde{\check{\mathfrak{g}}} \times_{\mathfrak{g}} \widetilde{\check{\mathfrak{g}}}$, which is Cohen-Macaulay by Lemma 4.3.4. Since $\widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}}$ is smooth, miracle flatness reduces us to checking that the fibers of the projection map $\widetilde{\check{\mathfrak{g}}} \times_{\mathfrak{g}} \widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}} \to \widetilde{\check{\mathfrak{g}}}^{\mathrm{reg}}$, or equivalently of the projection map $\widetilde{\check{\mathfrak{g}}} \times_{\mathfrak{g}} \mathfrak{b}^{\mathrm{reg}} \to \mathfrak{b}^{\mathrm{reg}}$, have the same dimension. In fact, we will show that the fibers are all zero-dimensional. By Item 3 in Proposition 4.1.14, every $y \in \mathfrak{b}^{\mathrm{reg}}$ is conjugate to $f_0 + x$ for $x \in \mathfrak{t}$ and $f_0 = \sum_{\alpha \in \Delta} f_\alpha$. Since all Borel subgroups are conjugate, it further suffices to show that the fibers of the projection map $\widetilde{\check{\mathfrak{g}}} \times_{\mathfrak{g}} \mathfrak{t} \to \mathfrak{t}$ are zero-dimensional, where the map $\mathfrak{t} \to \mathfrak{g}$ sends $x \mapsto f_0 + x$. Unwinding definitions, we need to show that if R is an algebraically field and $x \in \mathfrak{t}$, then the subvariety $\mathcal{B}_{f_0+x}$ of $\mathcal{B} = \check{\mathrm{G}}/\check{\mathrm{B}} \cong \mathrm{G}/\mathrm{B}$ of those Borel subgroups $\mathrm{B}' \subseteq \mathrm{G}$ such that $f_0 + x \in \mathfrak{b}'$ is zero-dimensional. Using the contracting $\mathbf{G}_m$-action on $\mathfrak{t}$ and upper semicontinuity, it suffices to show that the subvariety $\mathcal{B}_{f_0}$ of $\mathcal{B}$ of those Borel subgroups $\mathrm{B}' \subseteq \mathrm{G}$ such that $f_0 \in \mathfrak{b}'$ is zero-dimensional. Observe that if $f_0 \in \mathfrak{b}'$, then $\mathfrak{b}'$ must contain the root space $\mathfrak{g}_\alpha$ for each $\alpha \in \Delta$, and so in fact $\mathfrak{b}' = \mathfrak{b}$. In other words, there is a unique Borel subgroup $\mathrm{B}' \subseteq \mathrm{G}$ such that $f_0 \in \mathfrak{b}'$, as desired. (This shows that the underlying reduced scheme of $\mathcal{B}_{f_0}$ is a point, but $\mathcal{B}_{f_0}$ itself is non-reduced.) □

Before proceeding, we will need some results about the case $\mathrm{F} = \hat{\mathbf{G}}_a(2)$, the goal being to prove Proposition 4.3.9. I am grateful to Akshay Venkatesh for discussions regarding this result. Until otherwise indicated, G is allowed to be multiply-laced.

Lemma 4.3.7. *Assume* G *is possibly multiply-laced, and as in Proposition 4.2.15, let* ψ_0 *be the map* ${}^{2\rho}\check{\mathfrak{u}}_-(2) \to \mathbf{A}^1$ *given by*

$$ {}^{2\rho}\check{\mathfrak{u}}_-(2) \to {}^{2\rho}\check{\mathfrak{u}}_-(2)/[{}^{2\rho}\check{\mathfrak{u}}_-(2), {}^{2\rho}\check{\mathfrak{u}}_-(2)] \cong \prod_{\alpha\in\Delta} \check{\mathfrak{g}}^*_{-\alpha} \xrightarrow{\sum} \mathbf{A}^1. $$

Then the map $\check{\mathfrak{t}}^*(2) \to (\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-$ *sending* $x \mapsto \psi_0 + x$ *is faithfully flat.*

Proof. It is equivalent to showing that the action map ${}^{2\rho}\check{\mathrm{U}}^- \times \check{\mathfrak{t}}^*(2) \to \psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2)$ sending $(g, x) \mapsto \mathrm{Ad}_g(\psi_0 + x)$ is faithfully flat. By miracle flatness, it suffices to show that this map is surjective with zero-dimensional fibers. Since the above map is one of varieties with contracting $\mathbf{G}_m$-action, where the $\mathbf{G}_m$-action on $\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp$ contracts it to ψ_0, it suffices to show that the fiber over ψ_0 is finite. Since the map

$\check{\mathfrak{t}}^* \to \psi_0 + \check{\mathfrak{u}}_-^\perp$ is a closed immersion, the fiber over ψ_0 is isomorphic to the subvariety $\mathrm{X} \subseteq \check{\mathrm{U}}^-$ of those $g \in \check{\mathrm{U}}^-$ such that $\mathrm{Ad}_g(\psi_0) - \psi_0 \in \check{\mathfrak{t}}^*$.

The map $\mathrm{X} \to \check{\mathfrak{t}}^*$ sending $g \mapsto \mathrm{Ad}_g(\psi_0) - \psi_0$ factors through a closed immersion $\mathrm{X}/\mathrm{Z}_{\check{\mathrm{U}}^-}(\psi_0) \to \check{\mathfrak{t}}^*$. Suppose that $\mathrm{Ad}_g(\psi_0) = \psi_0 + x$ with $x \in \check{\mathfrak{t}}^*$; then, any $\check{\mathrm{G}}$-invariant polynomial on $\check{\mathfrak{g}}^*$ with vanishing constant term is zero on $\psi_0 + x$, and hence also on x. That is, $x \in \check{\mathfrak{t}}^* \times_{\mathrm{Spec}(\mathcal{O}_{\check{\mathfrak{g}}^*})^{\check{\mathrm{G}}}} \{0\}$, where this is an underived fiber product. It therefore suffices to show that $\check{\mathfrak{t}}^* \times_{\mathrm{Spec}(\mathcal{O}_{\check{\mathfrak{g}}^*})^{\check{\mathrm{G}}}} \{0\}$ is finite. It suffices to check this over $\mathbf{Z}$, and in turn it suffices to check this case over $\mathbf{F}_p$ and $\mathbf{Q}$. Over $\mathbf{Q}$, Chevalley's theorem identifies $\mathrm{Spec}(\mathcal{O}_{\check{\mathfrak{g}}^*})^{\check{\mathrm{G}}} \cong \check{\mathfrak{t}}^*/\!\!/\mathrm{W}$, so the claim follows from the observation that the (underived) fiber product $\check{\mathfrak{t}}^* \times_{\check{\mathfrak{t}}^*/\!\!/\mathrm{W}} \{0\}$ is finite. Over $\mathbf{F}_p$, if $p > 2$ or $p = 2$ and $\check{\mathrm{G}}$ does not have a simple factor isomorphic to SO_{2n+1}, then [KW76, Theorem 4(i)][9] shows that one still has $\mathrm{Spec}(\mathcal{O}_{\check{\mathfrak{g}}^*})^{\check{\mathrm{G}}} \cong \check{\mathfrak{t}}^*/\!\!/\mathrm{W}$, so we conclude again from the claim that $\check{\mathfrak{t}}^* \times_{\check{\mathfrak{t}}^*/\!\!/\mathrm{W}} \{0\}$ is finite.

It remains only to check the case $p = 2$ and $\check{\mathrm{G}} = \mathrm{SO}_{2n+1}$. Recall that there is a surjective homomorphism $\pi : \mathrm{SO}_{2n+1} \twoheadrightarrow \mathrm{Sp}_{2n}$ over $\mathbf{F}_2$. Let $\mathrm{K} \subseteq \mathrm{SO}_{2n+1}$ denote the preimage of $\mathrm{Sp}_2^n \subseteq \mathrm{Sp}_{2n}$ under π, so that $\mathrm{K} \cong \mathrm{PGL}_2^n$. Write $\check{\mathfrak{g}}^* = \check{\mathfrak{t}}^* \oplus \bigoplus_{\alpha\in\Phi} \check{\mathfrak{g}}_\alpha^*$, and let $\mathrm{V} \subseteq \check{\mathfrak{g}}^*$ denote the K-stable subspace $\check{\mathfrak{t}}^* \oplus \bigoplus_{\alpha\in\Phi_l} \check{\mathfrak{g}}_\alpha^*$ where Φ_l ranges over the long roots. Then the map $\mathcal{O}_{\check{\mathfrak{g}}^*}^{\check{\mathrm{G}}} \to \mathrm{Sym}(\check{\mathfrak{t}})$ factors through the map $\mathcal{O}_\mathrm{V}^\mathrm{K} \to \mathrm{Sym}(\check{\mathfrak{t}})$. Note that $\mathcal{O}_\mathrm{V}^\mathrm{K} = \mathcal{O}_{(\mathfrak{pgl}_2^*)^n}^{\mathrm{PGL}_2^n}$ is isomorphic to $\mathcal{O}_{\mathfrak{sl}_2^n}^{\mathrm{PGL}_2^n} \cong \mathcal{O}_{\mathfrak{sl}_2^n}^{\mathrm{SL}_2^n}$; the restriction map $\mathcal{O}_\mathrm{V}^\mathrm{K} \to \mathcal{O}_{\check{\mathfrak{t}}^*}$ is injective with image isomorphic to the Frobenius twist $\mathcal{O}_{(\check{\mathfrak{t}}^*)^{(1)}}$ (see [CR10, Proposition 6.1.2]). Arguing as in [CR10, Theorem 6.2.2] shows that the map $\mathcal{O}_{\check{\mathfrak{g}}^*}^{\check{\mathrm{G}}} \to \mathcal{O}_\mathrm{V}^\mathrm{K}$ is injective with image $(\mathcal{O}_\mathrm{V}^\mathrm{K})^{\Sigma_n}$, where Σ_n acts by permuting the SO_3 factors in K. It follows that the map $\mathcal{O}_{\check{\mathfrak{g}}^*}^{\check{\mathrm{G}}} \to \mathcal{O}_{\check{\mathfrak{t}}^*}$ is injective with image isomorphic to to the subring of $\mathcal{O}_{\check{\mathfrak{t}}^*} = \mathbf{F}_2[x_1, \cdots, x_n]$ generated by the elementary symmetric polynomials in the squares of $x_1, \cdots, x_n$. This implies that $\check{\mathfrak{t}}^* \times_{\mathrm{Spec}(\mathcal{O}_{\check{\mathfrak{g}}^*}^{\check{\mathrm{G}}})} \{0\}$ is isomorphic to $\mathrm{Spec}\, \mathbf{F}_2[x_1, \cdots, x_n]/(e_j(x_1^2, \cdots, x_n^2))_{1\leq j\leq n}$, which is indeed finite over $\mathbf{F}_2$.

The discussion above implies that the quotient $\mathrm{X}/\mathrm{Z}_{\check{\mathrm{U}}^-}(\psi_0)$ is finite, so to finish, it suffices to show that $\mathrm{Z}_{\check{\mathrm{U}}^-}(\psi_0)$ is finite. Since $\check{\mathrm{B}}$ is the unique Borel whose Lie algebra is annihilated by ψ_0, any $g \in \mathrm{Z}_{\check{\mathrm{U}}^-}(\psi_0)$ must normalize $\check{\mathrm{B}}$. Because $\mathrm{N}_{\check{\mathrm{G}}}(\check{\mathrm{B}}) = \check{\mathrm{B}}$, this implies that for any algebraically closed field L, we have $g \in \check{\mathrm{B}}(\mathrm{L}) \cap \check{\mathrm{U}}^-(\mathrm{L}) = 1$, and so $\mathrm{Z}_{\check{\mathrm{U}}^-}(\psi_0)(\mathrm{L}) = \{1\}$. This means that $\mathrm{Z}_{\check{\mathrm{U}}^-}(\psi_0)$ has trivial reduced subscheme, which (since it is Noetherian, being a closed subscheme of $\check{\mathrm{U}}^-$) implies that it is finite. □

Lemma 4.3.8. *Fix notation as in Lemma 4.3.7, so in particular,* G *is possibly multiply-laced. Then there is a Cartesian square*

$$\begin{tikzcd} \check{\mathfrak{t}}^*(2) \arrow[r, "\kappa_\mathrm{F}"] \arrow[d] & {}^{2\rho}\check{\mathfrak{u}}^\perp(2)/{}^{2\rho}\check{\mathrm{B}} \arrow[d] \\ (\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_- \arrow[r] & {}^{2\rho}\check{\mathfrak{g}}^*(2)/{}^{2\rho}\check{\mathrm{G}}; \end{tikzcd}$$

[9] T.-H. Chen directed me to [ST26], which says that [KW76, Theorem 4(iv)] is unfortunately incorrect; but we only use part (i) of this result, which continues to remain valid.

equivalently, the Whittaker reduction of the Grothendieck-Springer resolution ${}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2)$ *is* $\check{\mathfrak{t}}^*(2)$ *via* κ_{F}.

Proof. For notational simplicity, we will drop the gradings below. Let $\widetilde{\mathfrak{g}}(2) = \check{\mathrm{G}} \times^{\check{\mathrm{B}}} \check{\mathfrak{u}}^\perp$ be the Grothendieck-Springer resolution, and define Y by the Cartesian square

$$\begin{tikzcd} \mathrm{Y} \arrow[r] \arrow[d] & \widetilde{\mathfrak{g}} \arrow[d, "\mu"] \\ \psi_0 + \check{\mathfrak{u}}_-^\perp \arrow[r] & \check{\mathfrak{g}}^*, \end{tikzcd}$$

so that Y is the moduli of pairs $(\check{\mathfrak{b}}', x)$ of a Borel subalgebra of $\check{\mathfrak{g}}$ and $x \in (\psi_0 + \check{\mathfrak{u}}_-^\perp) \cap \check{\mathfrak{u}}'^\perp$, where $\check{\mathfrak{u}}' = [\check{\mathfrak{b}}', \check{\mathfrak{b}}']$ is the nilpotent radical of $\check{\mathfrak{b}}'$. There is a map $\pi : \mathrm{Y} \to \check{\mathfrak{t}}^*$ sending $(\check{\mathfrak{b}}', x) \mapsto x \pmod{\check{\mathfrak{b}}'^\perp}$. There is an action of $\check{\mathrm{U}}_-$ on Y by conjugation, and it suffices to show that π is an $\check{\mathrm{U}}_-$-torsor. If $\mathrm{Y}_0 = \pi^{-1}(0)$, it suffices to show that the $\check{\mathrm{U}}_-$ action on $(\check{\mathfrak{b}}, \psi_0) \in \mathrm{Y}_0$ defines an isomorphism $\check{\mathrm{U}}_- \xrightarrow{\cong} \mathrm{Y}_0$. By definition, the scheme Y_0 is the moduli of pairs $(\check{\mathfrak{b}}', x)$ with $x \in (\psi_0 + \check{\mathfrak{u}}_-^\perp) \cap \check{\mathfrak{b}}'^\perp$.

We will show momentarily that $\check{\mathfrak{b}}'$ and $\check{\mathfrak{b}}_-$ are in opposite position; let us call this claim $(\star)$. Since the variety of Borels in opposite position to $\check{\mathfrak{b}}_-$ forms the open $\check{\mathrm{U}}_-$-orbit in the flag variety $\check{\mathrm{G}}/\check{\mathrm{B}}$, there is some $u \in \check{\mathrm{U}}_-$ such that $\mathrm{Ad}_u(\check{\mathfrak{b}}) = \check{\mathfrak{b}}'$. This implies that there is a *unique* $x \in (\psi_0 + \check{\mathfrak{u}}_-^\perp) \cap \check{\mathfrak{b}}'^\perp$, so the action map $\check{\mathrm{U}}_- \to \mathrm{Y}_0$ sending $g \mapsto \mathrm{Ad}_g(\check{\mathfrak{b}}, \psi_0)$ is an isomorphism, as desired.

We now discuss $(\star)$. Consider the 2ρ-grading $\check{\mathfrak{g}}^* = \bigoplus_{i \in \mathbf{Z}} \check{\mathfrak{g}}^*(i)$ and similarly on $\check{\mathfrak{g}}$, and let $\check{\mathfrak{g}}^{*,\geq j} = \bigoplus_{i \geq j} \check{\mathfrak{g}}^*(i)$, and similarly for $\check{\mathfrak{g}}^{\geq j}$. Then $\check{\mathfrak{b}}'$ and $\check{\mathfrak{b}}'^\perp$ acquire filtrations, and we claim that $\mathrm{gr}(\check{\mathfrak{b}}') = \check{\mathfrak{b}}$; let us call this claim (†). This is equivalent to $(\star)$, because $\check{\mathfrak{b}}' \cap \check{\mathfrak{b}}_-$ is the subspace of elements of $\check{\mathfrak{b}}'$ of filtration ≤ 0, but the only such elements will lie in $\check{\mathfrak{b}}' \cap \check{\mathfrak{g}}(0) = \check{\mathfrak{t}}$.

It remains to check (†). It will be convenient to identify $\mathrm{gr}(\check{\mathfrak{g}}^*) = \check{\mathfrak{g}}^*$. If $x \in \check{\mathfrak{g}}^*$, let $\overline{x}$ denote its associated graded. Since $\check{\mathfrak{u}}_-^\perp = \check{\mathfrak{g}}^{*,\geq 0}$ and $\psi_0 \in \check{\mathfrak{g}}^*(-2)$, any $x \in \psi_0 + \check{\mathfrak{u}}_-^\perp$ has $\overline{x} = \overline{\psi_0}$ (which is just ψ_0 under the identification $\mathrm{gr}(\check{\mathfrak{g}}^*) = \check{\mathfrak{g}}^*$). Therefore, $\mathrm{gr}(\check{\mathfrak{b}}'^\perp) = \mathrm{gr}(\check{\mathfrak{b}}')^\perp$ contains ψ_0. Since $\mathrm{gr}(\check{\mathfrak{b}}')$ is a maximal solvable subalgebra of $\mathrm{gr}(\check{\mathfrak{g}})$, it is a Borel subalgebra of $\mathrm{gr}(\check{\mathfrak{g}}) \cong \check{\mathfrak{g}}$. But now $\mathrm{gr}(\check{\mathfrak{b}}') = \check{\mathfrak{b}}_-$ because $\check{\mathfrak{b}}_-$ is the unique Borel subalgebra annihilated by ψ_0. □

Proposition 4.3.9. *Fix notation as in Lemma 4.3.7, so in particular,* G *is possibly multiply-laced. Then there is an isomorphism* $\check{\mathfrak{t}}^*(2)/\mathcal{W} \xrightarrow{\cong} (\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-$ *such that the following diagram commutes:*

$$\begin{tikzcd} \check{\mathfrak{t}}^*(2) \arrow[r] \arrow[dr, "x \mapsto \psi_0 + x"'] & \check{\mathfrak{t}}^*(2)/\mathcal{W} \arrow[r, "\kappa_{\mathrm{F}}"] \arrow[d, "\cong"] & {}^{2\rho}\check{\mathfrak{g}}^*(2)/{}^{2\rho}\check{\mathrm{G}}. \\ & (\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_- \arrow[ur] & \end{tikzcd}$$

Proof. Let $\mathcal{W}' = \check{\mathfrak{t}}^*(2) \times_{(\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-} \check{\mathfrak{t}}^*(2)$. By Lemma 4.3.7, it suffices to show that there is an isomorphism $\mathcal{W} \cong \mathcal{W}'$ of groupoid schemes over $\check{\mathfrak{t}}^*(2)$. For this, we will work over $\mathbf{Z}$. Lemma 4.3.8 defines an isomorphism

$$\mathcal{W}' \cong \check{\mathfrak{t}}^*(2) \times_{{}^{2\rho}\check{\mathfrak{g}}^*/{}^{2\rho}\check{\mathrm{G}}} {}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2)/{}^{2\rho}\check{\mathrm{G}} \cong \check{\mathfrak{t}}^*(2) \times_{{}^{2\rho}\check{\mathfrak{g}}^*} {}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2),$$

where the map $\check{\mathfrak{t}}^*(2) \to {}^{2\rho}\check{\mathfrak{g}}^*$ sends $x \mapsto \psi_0 + x$, so that $\mathcal{W}'$ is isomorphic to the variety of pairs $(\check{\mathfrak{b}}, x)$ with $x \in \check{\mathfrak{t}}^*$ where $\psi_0 + x$ annihilates $\check{\mathfrak{u}} = [\check{\mathfrak{b}}, \check{\mathfrak{b}}]$. The scheme $\mathcal{W}'$ can be rewritten as

$$\mathcal{W}' \cong \check{\mathfrak{t}}^*(2) \times_{{}^{2\rho}\check{\mathfrak{u}}^\perp(2)} {}^{2\rho}\check{\mathfrak{u}}^\perp(2) \times_{{}^{2\rho}\check{\mathfrak{g}}^*} {}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2).$$

Note that since $\mathcal{W}'$ is finite flat over $\check{\mathfrak{t}}^*$, there is an isomorphism

$$\mathcal{O}_{\mathcal{W}'} \cong \bigcap_{\alpha \in \Phi^+} \mathcal{O}_{\mathcal{W}'}[e(\Phi - \alpha)^{-1}] \subseteq \mathcal{O}_{\mathcal{W}'}[e(\Phi)^{-1}],$$

where as usual for a subset $\mathrm{J} \subseteq \Phi$, we write $e(\mathrm{J}) = \prod_{\beta \in \mathrm{J}} \beta$. Let $\mathrm{I} \subseteq \Phi^+$ be a subset, and let $\check{\mathfrak{t}}^*_{\mathrm{I}} := \check{\mathfrak{t}}^* - \bigcup_{\alpha \in \Phi^+ \setminus \mathrm{I}} \check{\mathfrak{t}}^*_\alpha \hookrightarrow \check{\mathfrak{t}}^*$, where $\check{\mathfrak{t}}^*_\alpha$ is the hyperplane in $\check{\mathfrak{t}}^*$ corresponding to the root α. Let ${}^{2\rho}\check{\mathrm{U}}_{\mathrm{I}} \subseteq {}^{2\rho}\check{\mathrm{U}}$ be a closed subgroup such that the root subgroup ${}^{2\rho}\check{\mathrm{U}}_\alpha \subseteq {}^{2\rho}\check{\mathrm{U}}_{\mathrm{I}}$ for each $\alpha \in \mathrm{I}$, and such that ${}^{2\rho}\check{\mathrm{U}}_{\mathrm{I}}$ is closed under conjugation by $\check{\mathrm{T}}$. The argument of Proposition 4.2.8 shows that the adjoint action map ${}^{2\rho}\check{\mathrm{U}} \times^{{}^{2\rho}\check{\mathrm{U}}_{\mathrm{I}}} {}^{2\rho}\check{\mathfrak{u}}_{\mathrm{I}}^\perp(2) \to {}^{2\rho}\check{\mathfrak{u}}^\perp(2)$ is an isomorphism upon pulling back along $\check{\mathfrak{t}}^*_{\mathrm{I}} \to \check{\mathfrak{t}}^*$. When $\mathrm{I} = \emptyset$, we may choose ${}^{2\rho}\check{\mathrm{U}}_{\mathrm{I}} = \{1\}$, so that the map $\kappa_{\mathrm{F}} : \check{\mathfrak{t}}^*_\emptyset \to {}^{2\rho}\check{\mathfrak{u}}^\perp(2)|_{\check{\mathfrak{t}}^*_\emptyset}$ sending $x \mapsto \psi_0 + x$ is isomorphic to the map $\check{\mathfrak{t}}^*_\emptyset \to {}^{2\rho}\check{\mathrm{U}} \times \check{\mathfrak{t}}^*_\emptyset$ sending $x \mapsto (1, x)$. It follows that

$$\mathcal{W}'|_{\check{\mathfrak{t}}^*_\emptyset} \cong \check{\mathfrak{t}}^*_\emptyset(2) \times_{\check{\mathfrak{t}}^*_\emptyset(2) \times {}^{2\rho}\check{\mathrm{U}}} (\check{\mathfrak{t}}^*_\emptyset(2) \times {}^{2\rho}\check{\mathrm{U}}) \times_{{}^{2\rho}\check{\mathfrak{g}}^*(2)} {}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2) \cong \check{\mathfrak{t}}^*_\emptyset(2) \times_{{}^{2\rho}\check{\mathfrak{g}}^*(2)} {}^{2\rho}\widetilde{\check{\mathfrak{g}}}(2),$$

where the map $\check{\mathfrak{t}}^*_\emptyset(2) \to {}^{2\rho}\check{\mathfrak{g}}^*(2)$ is just the inclusion. Since the composite $\check{\mathfrak{t}}^* \to {}^{2\rho}\check{\mathfrak{g}}^*(2) \to \check{\mathfrak{t}}^*(2)/\mathcal{W}$ is the quotient map, it follows that the map $\check{\mathfrak{t}}^*_\emptyset(2) \to {}^{2\rho}\check{\mathfrak{g}}^*(2)$ lifts to ${}^{2\rho}\check{\mathfrak{g}}^*_{\mathrm{rs}}(2) = {}^{2\rho}\check{\mathfrak{g}}^*(2) \times_{\check{\mathfrak{t}}^*(2)/\mathcal{W}} \check{\mathfrak{t}}^*_\emptyset(2)/\mathcal{W}$. In particular, $\mathcal{W}'|_{\check{\mathfrak{t}}^*_\emptyset}$ is isomorphic to $\check{\mathfrak{t}}^*_\emptyset(2) \times_{{}^{2\rho}\check{\mathfrak{g}}^*_{\mathrm{rs}}(2)} {}^{2\rho}\widetilde{\check{\mathfrak{g}}}_{\mathrm{rs}}(2)$. Arguing as in Item 2 in the proof of Theorem 4.3.6 shows that the map ${}^{2\rho}\widetilde{\check{\mathfrak{g}}}_{\mathrm{rs}} \to {}^{2\rho}\check{\mathfrak{g}}^*_{\mathrm{rs}}$ is a principal W-bundle. It follows that $\mathcal{W}'|_{\check{\mathfrak{t}}^*_\emptyset} \cong \underline{\mathrm{W}}$, and in fact this is an isomorphism of groupoid schemes over $\check{\mathfrak{t}}^*_\emptyset(2)$. In other words, $\mathcal{O}_{\mathcal{W}'}[e(\Phi)^{-1}] \cong \mathrm{Map}(\mathrm{W}, \mathcal{O}_{\check{\mathfrak{t}}^*(2)})[e(\Phi)^{-1}]$, which is in turn isomorphic to $\mathcal{O}_{\mathcal{W}}[e(\Phi)^{-1}]$.

It therefore suffices to show that for each $\alpha \in \Phi^+$, there is an isomorphism $\mathcal{O}_{\mathcal{W}'}[e(\Phi - \alpha)^{-1}] \cong \mathcal{O}_{\mathcal{W}}[e(\Phi - \alpha)^{-1}]$ of subalgebras of $\mathcal{O}_{\mathcal{W}'}[e(\Phi)^{-1}] \cong \mathcal{O}_{\mathcal{W}}[e(\Phi)^{-1}]$. Taking $\mathrm{I} = \{\alpha\}$ and ${}^{2\rho}\check{\mathrm{U}}_{\mathrm{I}} = {}^{2\rho}\check{\mathrm{U}}_\alpha$, we find that the action map ${}^{2\rho}\check{\mathrm{U}} \times^{{}^{2\rho}\check{\mathrm{U}}_\alpha} {}^{2\rho}\check{\mathfrak{u}}_\alpha^\perp(2) \to {}^{2\rho}\check{\mathfrak{u}}^\perp(2)$ is an isomorphism upon pulling back along $\check{\mathfrak{t}}^*_\alpha \to \check{\mathfrak{t}}^*$. This implies that the map $\kappa_{\mathrm{F}} : \check{\mathfrak{t}}^*_\alpha \to {}^{2\rho}\check{\mathfrak{u}}^\perp(2)|_{\check{\mathfrak{t}}^*_\alpha}$ sending $x \mapsto \psi_0 + x$ is isomorphic to the map $\check{\mathfrak{t}}^*_\alpha \to {}^{2\rho}\check{\mathrm{U}} \times^{{}^{2\rho}\check{\mathrm{U}}_\alpha} {}^{2\rho}\check{\mathfrak{u}}_\alpha^\perp(2)$ sending $x \mapsto (1, f_\alpha + x)$. To show the isomorphism $\mathcal{O}_{\mathcal{W}'}[e(\Phi - \alpha)^{-1}] \cong \mathcal{O}_{\mathcal{W}}[e(\Phi - \alpha)^{-1}]$, we may therefore replace ${}^{2\rho}\check{\mathrm{G}}$ with the Levi subgroup generated by $\check{\mathrm{T}}$ and ${}^{2\rho}\check{\mathrm{U}}_{\pm\alpha}$, and hence reduce to the case when G has semisimple rank 1. In fact, by replacing ${}^{2\rho}\check{\mathrm{G}}$ with its derived subgroup, we are reduced to showing that there is an isomorphism $\mathcal{W} \cong \mathcal{W}'$ of groupoid schemes over $\check{\mathfrak{t}}^*(2)$, compatibly with the homomorphisms from $\underline{\mathrm{W}}$, when G is semisimple of rank 1.

When $\mathrm{G} = \mathrm{SL}_2$, so $\check{\mathrm{G}} = \mathrm{PGL}_2$, we may identify $\check{\mathfrak{g}}^* \cong \mathfrak{sl}_2$, so that $\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2) \cong \left(\begin{smallmatrix} x & y \\ 1 & -x \end{smallmatrix}\right)$. The map $\kappa : \mathbf{A}^1 \cong \check{\mathfrak{t}}^*(2) \to \psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2)$ sends $x \mapsto \left(\begin{smallmatrix} x & 0 \\ 1 & -x \end{smallmatrix}\right)$, and $\mathcal{W}'$ is isomorphic to the closed subscheme of $\check{\mathfrak{t}}^*(2) \times \check{\mathfrak{t}}^*(2) \times {}^{2\rho}\check{\mathrm{U}}_- = \mathbf{A}^1(2) \times \mathbf{A}^1(2) \times \mathbf{G}_a(2)$ of those (x, x', a) such that $\mathrm{Ad}_a\left(\begin{smallmatrix} x & 0 \\ 1 & -x \end{smallmatrix}\right) = \left(\begin{smallmatrix} x' & 0 \\ 1 & -x' \end{smallmatrix}\right)$. But $\mathrm{Ad}_a\left(\begin{smallmatrix} x & 0 \\ 1 & -x \end{smallmatrix}\right) = \left(\begin{smallmatrix} a+x & -a(a+2x) \\ 1 & -(a+x) \end{smallmatrix}\right)$, so $\mathcal{W}' \cong \mathrm{Spec}\, \mathbf{Z}[a, x]/(a^2 + 2xa)$, and the two maps to $\mathbf{A}^1$ are given by $(x, a) \mapsto x, x' = x + a$. By Example 2.2.24, it is therefore isomorphic to $\mathcal{W}$ as a groupoid scheme over $\mathbf{A}^1$. The computation for $\mathrm{G} = \mathrm{PGL}_2$ is similar. $\square$

Definition 2.2.28 and Proposition 4.3.9 imply the following; note that the proof is classification-free.

Theorem 4.3.10. *If k is an even $\mathbf{Z}$-algebra and G is possibly multiply-laced, there are isomorphisms of graded derived commutative $\pi_*(k)$-algebras*

$$\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}}) \cong \mathrm{R}\Gamma(\hat{\check{\mathfrak{t}}}^*(2)/\mathcal{W}; \mathcal{O}) \cong \mathrm{R}\Gamma((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O}).$$

Remark 4.3.11. There are no assumptions in Theorem 4.3.10 on the primes invertible in $\pi_0(k)$. The failure of $\mathrm{gr}^\star_{\mathrm{G}}(k^{h\mathrm{G}})$ to be concentrated in degree 0 (which is in turn implied by $k^{h\mathrm{G}}$ not being even) is therefore equivalent to $(\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-$ having higher coherent cohomology, i.e., to the ${}^{2\rho}\check{\mathrm{U}}_-$-action on $\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2)$ not being free. Theorem 4.3.10 and (A.2.1) give a spectral sequence

$$\mathrm{E}_1 = \mathrm{H}^*((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O}) \Rightarrow \mathrm{H}^{-*}(\mathrm{BG}; k), \tag{4.3.1}$$

which can be viewed as analogous to (but much simpler than) the Adams-Novikov spectral sequence coming from the identification $\mathrm{gr}^\star_{\mathrm{ev}}(\mathbb{S}) \cong \mathrm{R}\Gamma(\mathcal{M}_{\mathrm{FG}}^{\omega\text{-triv}}; \mathcal{O})$. A class in weight i and cohomological degree j on the E_∞-page of (4.3.1) represents a class in $\mathrm{H}^{-(i-j)}(\mathrm{BG}; k)$. In Appendix C, we illustrate Theorem 4.3.10 for $\mathrm{G} = \mathrm{PGL}_2$ and PGL_3 in characteristics 2 and 3 respectively; here, the ${}^{2\rho}\check{\mathrm{U}}_-$-action on $\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2)$ is not free. Conjecture A.2.13 says that the spectral sequence (4.3.1) degenerates multiplicatively at the E_1-page.

Before proceeding, we observe a few consequences of Theorem 4.3.10.

Lemma 4.3.12. *Assume G is possibly multiply-laced. There is an action of the finite group scheme $\mathrm{Z}_{{}^{2\rho}\check{\mathrm{U}}_-}(\psi_0)$ on $\mathrm{H}^{-*}(\mathcal{B}; k)$ such that the image of the natural map $\pi_*(k^{h\mathrm{T}}) \to \mathrm{H}^{-*}(\mathcal{B}; k)$ is the invariant subalgebra $\mathrm{H}^{-*}(\mathcal{B}; k)^{\mathrm{Z}_{{}^{2\rho}\check{\mathrm{U}}_-}(\psi_0)}$. Moreover, the quotient of $\mathrm{H}^{-*}(\mathcal{B}; k)$ by the image of $\ker(\pi_*(k^{h\mathrm{T}}) \to \pi_*(k))$ is isomorphic to $\mathcal{O}_{\mathrm{Z}_{{}^{2\rho}\check{\mathrm{U}}_-}(\psi_0)}$, so that the map $\pi_*(k^{h\mathrm{T}}) \to \mathrm{H}^{-*}(\mathcal{B}; k)$ is surjective if and only if $\mathrm{Z}_{{}^{2\rho}\check{\mathrm{U}}_-}(\psi_0)$ is trivial.*

Proof. Since $\pi_*(k^{h\mathrm{T}}) \otimes_{\mathrm{gr}^\star_{\mathrm{G}}(k^{h\mathrm{G}})} \pi_*(k^{h\mathrm{T}}) \cong \mathrm{H}^{-*}(\mathcal{B}; k)$, Theorem 4.3.10 implies that there is an isomorphism $\mathrm{H}^{-*}(\mathcal{B}; k)$ is isomorphic to the global sections of $\check{\mathfrak{t}}^*(2) \times_{(\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-} \{0\}$, where the map $\{0\} \to (\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-$ is the inclusion of ψ_0. This fiber product is isomorphic to the subvariety of ${}^{2\rho}\check{\mathrm{U}}_- \times \check{\mathfrak{t}}^*(2)$ of those (g, x) such that $\mathrm{Ad}_g(\psi_0) = \psi_0 + x$; this is the variety denoted X in the proof of Lemma 4.3.7. It follows that $\mathrm{X} \cong \mathrm{Spec}(\mathrm{H}^{-*}(\mathcal{B}; k))$. The natural map $\mathrm{X}/\mathrm{Z}_{{}^{2\rho}\check{\mathrm{U}}_-}(\psi_0) \to \check{\mathfrak{t}}^*(2)$ given by projection onto the second factor is a closed immersion. Since $\mathrm{X}/\mathrm{Z}_{{}^{2\rho}\check{\mathrm{U}}_-}(\psi_0)$ is an affine scheme, there is no higher cohomology of the $\mathrm{Z}_{{}^{2\rho}\check{\mathrm{U}}_-}(\psi_0)$-action on $\mathrm{H}^{-*}(\mathcal{B}; k)$, and there is a surjection $\mathcal{O}_{\check{\mathfrak{t}}^*(2)} \twoheadrightarrow \mathrm{H}^{-*}(\mathcal{B}; k)^{\mathrm{Z}_{{}^{2\rho}\check{\mathrm{U}}_-}(\psi_0)}$ factoring the natural map $\mathcal{O}_{\check{\mathfrak{t}}^*(2)} \to \mathrm{H}^{-*}(\mathcal{B}; k)$. The quotient of $\mathrm{H}^{-*}(\mathcal{B}; k)$ by the image of $\ker(\pi_*(k^{h\mathrm{T}}) \to \pi_*(k))$ is therefore isomorphic to the ring of functions on the *underived* pullback $\mathrm{X} \times_{\check{\mathfrak{t}}^*(2)} \{0\}$; but this is just $\mathrm{Z}_{{}^{2\rho}\check{\mathrm{U}}_-}(\psi_0)$, as desired. $\square$

Remark 4.3.13. When $k = \mathbf{F}_p$, the surjectivity of the natural map $\pi_*(k^{h\mathrm{T}}) \to \mathrm{H}^{-*}(\mathcal{B}; k)$ is one of the equivalent criteria for $\mathrm{H}^{-*}(\mathrm{BG}; \mathbf{Z})$ to be p-torsionfree, see

(1) $\Leftrightarrow$ (5) in [Bor83, Item 141.1, pages 775-776]. Lemma 4.3.12 implies that the odd-degree and torsion classes in $\mathrm{H}^{-*}(\mathrm{BG};\mathbf{Z})$ stem from the group cohomology of the finite group scheme $\mathrm{Z}_{\check{\mathrm{U}}^-}(\psi_0)_{\mathbf{Z}}$.

Note that according to Grothendieck [Gro58], the quotient of $\mathrm{H}^{-*}(\mathcal{B};\mathbf{Z})$ by the image of $\ker(\pi_*(\mathbf{Z}^{h\mathrm{T}}) \to \mathbf{Z})$ is the Chow ring $\mathrm{CH}^{-*}(\mathrm{G};\mathbf{Z})$. (The Chow ring $\mathrm{CH}^{-*}(\mathrm{G};\mathbf{F}_p) = \mathrm{CH}^{-*}(\mathrm{G};\mathbf{Z}) \otimes_{\mathbf{Z}} \mathbf{F}_p$ is the subalgebra of $\mathrm{H}^{-*}(\mathrm{G};\mathbf{F}_p)$ generated by the even-degree indecomposables for $p > 2$, and by the squares of the (necessarily odd-degree) indecomposables for $p = 2$.) Lemma 4.3.12 therefore implies that $\operatorname{Spec} \mathrm{CH}^{-*}(\mathrm{G};\mathbf{Z}) \cong \mathrm{Z}_{2\rho\check{\mathrm{U}}_-}(\psi_0)_{\mathbf{Z}}$, where the Chow grading on the left-hand side is given by halving the natural grading on $\mathrm{Z}_{2\rho\check{\mathrm{U}}_-}(\psi_0)$. The Chow ring $\mathrm{CH}^{-*}(\mathrm{G};\mathbf{Z})$ can be explicitly computed for every simple algebraic group (see, e.g., [Kac85] and [KN07]). For instance, when G is the simply-connected form of E_6, then

$$\mathrm{Z}_{2\rho\check{\mathrm{U}}_-}(\psi_0) \cong \operatorname{Spec} \mathrm{CH}^{-*}(\mathrm{E}_6;\mathbf{Z}) \cong \operatorname{Spec} \mathbf{Z}[x,y]/(2x, x^2, 3y, y^2)$$

where x and y live in weights -6 and -8 respectively.

For the following, assume G is the Chevalley split form over $\mathbf{Z}$ of a possibly multiply-laced connected reductive group. We will continue to write BG to denote $\mathrm{BG}(\mathbf{C})$. Let ℓ be a prime, and let p be another prime distinct from ℓ. Let σ be a Steinberg endomorphism of $\mathrm{G}_{\overline{\mathbf{F}_p}}$, i.e., an endomorphism such that $\sigma^n = \mathrm{Frob}_p^m$ for some $n, m \gg 0$, where Frob_p is the Frobenius endomorphism of $\mathrm{G}_{\overline{\mathbf{F}_p}}$. Let k be an ℓ-complete even $\mathbf{Z}$-algebra. The homotopy equivalence $\mathrm{BG}(\overline{\mathbf{F}_p})^\wedge_\ell \xrightarrow{\simeq} \mathrm{BG}^\wedge_\ell$ from [FM84] defines an isomorphism $k^{h\mathrm{G}(\overline{\mathbf{F}_p})} \xrightarrow{\simeq} k^{h\mathrm{G}}$, so that σ defines an endomorphism σ^* of $k^{h\mathrm{G}}$. If $\mathrm{H} = \mathrm{G}(\overline{\mathbf{F}_p})^\sigma$, there is also a homotopy equivalence $\mathrm{BH}^\wedge_\ell \xrightarrow{\cong} (\mathrm{BG}^\wedge_\ell)^{h\mathbf{Z}_{\geq 0}}$, where $\mathbf{Z}_{\geq 0}$ by σ (see [GL20, Equation 1.2]). There is a pullback square

$$\begin{tikzcd} \mathrm{BH}^\wedge_\ell \arrow[r] \arrow[d] & \mathrm{BG}^\wedge_\ell \arrow[d, "\Delta"] \\ \mathrm{BG}^\wedge_\ell \arrow[r, "(1,\sigma)"'] & \mathrm{BG}^\wedge_\ell \times \mathrm{BG}^\wedge_\ell, \end{tikzcd}$$

which induces (by [MNN17, Proposition 7.33]) a pushout square in the category of ℓ-complete $\mathbf{E}_\infty$-rings

$$\begin{tikzcd} k^{h\mathrm{G}} \hat{\otimes}_k k^{h\mathrm{G}} \arrow[r, "\sigma^*\otimes \mathrm{id}"] \arrow[d] & k^{h\mathrm{G}} \arrow[d] \\ k^{h\mathrm{G}} \arrow[r] & k^{h\mathrm{H}}. \end{tikzcd}$$

Here, the tensor product is in the ℓ-complete sense. In other words, $k^{h\mathrm{H}}$ is the homotopy orbits $(k^{h\mathrm{G}})_{h\mathbf{Z}_{\geq 0}}$ in the category of ℓ-complete $\mathbf{E}_\infty$-k-algebras.

Corollary 4.3.14. *Assume for simplicity that $\sigma = \mathrm{Frob}_p^r$, so that $\mathrm{H} = \mathrm{G}(\mathbf{F}_q)$ with $q = p^r$. Then there is a filtration on $k^{h\mathrm{G}(\mathbf{F}_q)}$, given by the homotopy orbits $(\mathrm{fil}^\star_{\mathrm{G\text{-}ev}}(k^{h\mathrm{G}}))_{h\mathbf{Z}_{\geq 0}}$ in the category of filtered ℓ-complete $\mathbf{E}_\infty$-$\mathrm{fil}^\star_{\mathrm{ev}}(k)$-algebras, with associated graded isomorphic to the derived ring of functions on the stack*

$$\mathcal{L}_{\mathrm{Frob}_q}((\psi_0 + \check{\mathfrak{u}}_-)/\check{\mathrm{U}}_-) \cong \{(g,x) \in \check{\mathrm{U}}_- \times (\psi_0 + \check{\mathfrak{u}}_-) | \mathrm{Ad}_g(x) = qx\}/\check{\mathrm{U}}_-. \tag{4.3.2}$$

Proof sketch. By construction, the associated graded of the filtration on $k^{h\mathrm{H}}$ is isomorphic to the pushout of ℓ-complete $\mathbf{E}_\infty$-rings

$$\begin{tikzcd} \mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})\hat{\otimes}_{\mathrm{gr}^\star_{\mathrm{ev}}(k)}\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}}) \arrow[r, "\sigma^*\otimes\mathrm{id}"] \arrow[d] & \mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}}) \arrow[d] \\ \mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}}) \arrow[r] & \mathrm{gr}(k^{h\mathrm{G}(\mathbf{F}_q)}), \end{tikzcd}$$

Under the isomorphism of Theorem 4.3.10, the endomorphism σ^* of $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$ identifies with the endomorphism of $(\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_- \cong \hat{\mathfrak{t}}(2)/\mathcal{W}$ given by multiplication by q^n in weight n. This follows from the analogous claims for $\pi_*(k^{h\mathrm{T}})$ and $\mathrm{H}^{-*}_\mathrm{T}(\mathcal{B};k)$; for instance, the isomorphism $\pi_*(k^{h\mathrm{T}}) \cong \mathcal{O}_{\hat{\mathfrak{t}}(2)}$ intertwines σ^* with multiplication by q on $\hat{\mathfrak{t}}(2)$. This implies that $\mathrm{gr}(k^{h\mathrm{G}(\mathbf{F}_q)})$ is isomorphic to the derived ring of functions on the pullback

$$\begin{tikzcd} \mathcal{L}_{\mathrm{Frob}_q}((\psi_0 + \check{\mathfrak{u}}_-)/\check{\mathrm{U}}_-) \arrow[r] \arrow[d] & (\psi_0 + \check{\mathfrak{u}}_-)/\check{\mathrm{U}}_- \arrow[d, "\mathrm{id}\times\sigma^*"] \\ (\psi_0 + \check{\mathfrak{u}}_-)/\check{\mathrm{U}}_- \arrow[r, "\Delta"'] & (\psi_0 + \check{\mathfrak{u}}_-)/\check{\mathrm{U}}_- \times (\psi_0 + \check{\mathfrak{u}}_-)/\check{\mathrm{U}}_-. \end{tikzcd}$$

The isomorphism (4.3.2) follows by unwinding the definition of the fiber product. $\square$

Remark 4.3.15. Note that the only assumption on p and ℓ in Corollary 4.3.14 is that $p \neq \ell$. For instance, if $k = \mathbf{F}_\ell$ and $q \equiv 1 \pmod{\ell}$, then Corollary 4.3.14 identifies $\mathrm{gr}(\mathbf{F}_\ell^{h\mathrm{G}(\mathbf{F}_q)})$ with the Hochschild homology $\mathrm{HH}((\psi_0 + \check{\mathfrak{u}}_-^\perp(2))/\check{\mathrm{U}}_-)$ relative to $\mathbf{F}_\ell$. The effect of other Steinberg endomorphisms under the isomorphism of Theorem 4.3.10 is more interesting and will be explored in future work; it is related to the ideas in [TV16].

We warn the reader that even though the the spectral sequence (4.3.1) is expected to degenerate multiplicatively at the E_1-page by Conjecture A.2.13, the spectral sequence resulting from the filtration on $k^{h\mathrm{G}(\mathbf{F}_q)}$ does *not* degenerate multiplicatively. For instance, if q is odd and $\mathrm{G} = \mathbf{G}_m$, then $\pi_*\mathrm{gr}(\mathbf{F}_2^{h\mathbf{F}_q^\times}) \cong \mathbf{F}_2[x] \otimes \Lambda(w)$, where Λ denotes the exterior algebra. Here, x lives in weight -2 and degree 0, and w lives in weight -2 and degree 1. On the other hand, if $q \equiv 3 \pmod 4$, then $\mathrm{H}^{-*}(\mathrm{B}\mathbf{F}_q^\times; \mathbf{F}_2) \cong \mathbf{F}_2[w]$, where w lives in degree -1; so the relevant spectral sequence collapses, but there is a multiplicative extension on the E_∞-page enforcing the relation $w^2 = x$. However, I believe this is a quirk of 2-torsion coefficients: just like in Conjecture A.2.13, I expect that if k is an ℓ-complete even $\mathbf{E}_\infty$-$\mathbf{Z}_\ell$-algebra such that $2 \in \pi_0(k)$ is a nonzerodivisor (this hypothesis is relevant only when $\ell = 2$), the spectral sequence

$$\mathrm{E}_1 \cong \pi_*\mathrm{gr}(k^{h\mathrm{G}(\mathbf{F}_q)}) \cong \mathrm{H}^{-*}(\mathcal{L}_{\mathrm{Frob}_q}((\psi_0 + \check{\mathfrak{u}}_-)/\check{\mathrm{U}}_-); \mathcal{O}) \Rightarrow \pi_*(k^{h\mathrm{G}(\mathbf{F}_q)})$$

degenerates multiplicatively at the E_1-page. A class in $\mathrm{H}^{-j}(\mathcal{L}_{\mathrm{Frob}_q}((\psi_0 + \check{\mathfrak{u}}_-)/\check{\mathrm{U}}_-); \mathcal{O})$ in weight i represents a class in $\mathrm{H}^{-(i-j)}(\mathrm{BG}(\mathbf{F}_q); k)$.

We now resume our standing assumption that G has simply-laced derived subgroup.

Corollary 4.3.16. *Let $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ be the map from Proposition 4.3.2. Then the resulting commutative diagram*

$$\begin{tikzcd} \mathrm{T}_{\mathrm{F}} \arrow[r, "\kappa_{\mathrm{F}}"] \arrow[d] & {}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}} \arrow[d] \\ \mathrm{T}_{\mathrm{F}}/\mathcal{W} \arrow[r, "\kappa_{\mathrm{F}}"'] & {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}} \end{tikzcd} \tag{4.3.3}$$

is a Cartesian square.

Proof. It remains to show that the square (4.3.3) is Cartesian. Since the square commutes by construction, there is a map $g : \mathrm{T}_{\mathrm{F}} \to \mathrm{T}_{\mathrm{F}}/\mathcal{W} \times_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}} {}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}}$. There is a complete filtration on $\mathcal{O}_{\mathrm{F}}$ given by powers of the augmentation ideal $\ker(\mathcal{O}_{\mathrm{F}} \to \mathrm{R})$, with associated graded $\mathcal{O}_{\hat{\mathbf{G}}_a(2)}$. This in turn induces complete filtrations on T_{F}, $\mathrm{T}_{\mathrm{F}}/\mathcal{W}$, ${}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$, ${}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}}$, and the map g whose associated graded are the respective objects for $\mathrm{F} = \hat{\mathbf{G}}_a(2)$. To check that g is an isomorphism, it suffices to check that the associated graded map $\hat{\mathfrak{t}}(2) \to \hat{\mathfrak{t}}(2)/\mathcal{W} \times_{{}^{2\rho}\mathfrak{g}(2)^{\wedge}_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{G}}} {}^{2\rho}\mathfrak{b}(2)^{\wedge}_0/{}^{2\rho}\check{\mathrm{B}}$ is an isomorphism. In turn, it suffices to check that the uncompleted and ungraded map $\mathfrak{t} \to \mathfrak{t}/\mathcal{W} \times_{\mathfrak{g}/\check{\mathrm{G}}} \mathfrak{b}/\check{\mathrm{B}}$ is an isomorphism. (Over a $\mathbf{Q}$-algebra, this follows from work of Kostant [Kos78].) By Proposition 4.3.9, the map $\mathfrak{t}/\mathcal{W} \to \mathfrak{g}/\check{\mathrm{G}}$ is isomorphic to the map $(f_0 + \mathfrak{b}_-)/\check{\mathrm{U}}_- \to \mathfrak{g}/\check{\mathrm{G}}$. This implies that $\mathfrak{t}/\mathcal{W} \times_{\mathfrak{g}/\check{\mathrm{G}}} \mathfrak{b}/\check{\mathrm{B}} \cong (f_0 + \mathfrak{b}_-)/\check{\mathrm{U}}_- \times_{\mathfrak{g}/\check{\mathrm{G}}} \mathfrak{b}/\check{\mathrm{B}}$. Since $\mathfrak{t} \cong \check{\mathfrak{t}}^*$ and $f_0 + \mathfrak{b}_- \cong \psi_0 + \check{\mathfrak{u}}_-$, the fiber product in question is isomorphic to $\mathfrak{t}$ by Lemma 4.3.8. $\square$

Corollary 4.3.17. *Suppose that k is an even $\mathbf{E}_\infty$-ring, and that the bad primes of* G *(see Table 1) are invertible in $\pi_0(k)$. Continue to assume that* G *is the base-change to $\pi_*(k)$ of the Chevalley split form of a connected reductive group whose derived subgroup is simply-laced. Then there is an isomorphism*

$$\mathrm{Spec}(\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))/\mathcal{W} \cong \mathrm{T}_{\mathrm{F}}/\mathcal{W} \times_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}} \mathrm{T}_{\mathrm{F}}/\mathcal{W} =: \mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}$$

of groupoids over the stack $\mathrm{T}_{\mathrm{F}}/\mathcal{W}$. Moreover, the map $\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}} \to \mathrm{T}_{\mathrm{F}}/\mathcal{W}$ (induced by the map $\chi : {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}} \to \mathrm{T}_{\mathrm{F}}/\mathcal{W}$) is flat. Taking global sections and using Definition 2.2.28, it follows that there is an isomorphism $\mathrm{gr}^\star_{\mathrm{G}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \cong \mathrm{R}\Gamma(\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}; \mathcal{O})$ of Hopf algebroids over $\mathrm{gr}^\star_{\mathrm{G\text{-}ev}}(k^{h\mathrm{G}}) \cong \mathrm{R}\Gamma(\mathrm{T}_{\mathrm{F}}/\mathcal{W}; \mathcal{O})$.

Proof. Corollary 4.3.16 gives a $\mathcal{W}$-equivariant isomorphism

$$\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} = \mathrm{T}_{\mathrm{F}} \times_{{}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}}} \mathrm{T}_{\mathrm{F}} \cong \mathrm{J}_{\check{\mathrm{G}},\mathrm{F}} \times_{\mathrm{T}_{\mathrm{F}}/\mathcal{W}} \mathrm{T}_{\mathrm{F}},$$

so that $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}/\mathcal{W} \cong \mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}$. The desired isomorphism, and the claims about it, now follows from Theorem 4.2.12. $\square$

The group scheme $\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}$ over $\mathrm{T}_{\mathrm{F}}/\mathcal{W}$ will be called the *group scheme of regular* F*-centralizers.* Note that $\mathrm{T}_{\mathrm{F}}/\mathcal{W}$ might be a stack and not a scheme, and the same is true of $\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}$. The structure map $\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}} \to \mathrm{T}_{\mathrm{F}}/\mathcal{W}$ is relatively flat and affine. Since Proposition 4.3.2 only gives a map $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$, and not a map $\mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}$, there is not necessarily a homomorphism $\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}} \to {}^{2\rho}\check{\mathrm{G}} \times \mathrm{T}_{\mathrm{F}}/\mathcal{W}$.

Remark 4.3.18. It follows from Proposition 4.2.15 and Theorem 4.3.10 that if k is an even $\mathbf{E}_\infty$-$\mathbf{Z}$-algebra and G is the Chevalley split form of a connected reductive group (not necessarily with simply-laced derived subgroup), there are isomorphisms

$$\begin{aligned}\mathrm{Spec}(\mathrm{gr}^\star_{\check{\mathrm{G}}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}};k))) &\cong \mathrm{Spec}(\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}};k))/\mathcal{W} \\ &\cong (\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_- \times_{{}^{2\rho}\check{\mathfrak{g}}^*(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{G}}} (\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-\end{aligned}$$

of groupoids over the stack $(\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-$. This groupoid stack will be denoted $\mathrm{J}_{\check{\mathrm{G}},\hat{\mathbf{G}}_a(2)}$, so that

$$\mathrm{J}_{\check{\mathrm{G}},\hat{\mathbf{G}}_a(2)} \cong \hat{\mathrm{T}}^*(2)(({}^{2\rho}\check{\mathrm{U}}_-,\psi)\backslash{}^{2\rho}\check{\mathrm{G}}/({}^{2\rho}\check{\mathrm{U}}_-,\psi)),$$

where the right-hand side is the bi-Whittaker quotient. (See also [Tel14, Theorem 6.3].) It follows that there is an isomorphism $\mathrm{gr}^\star_{\check{\mathrm{G}}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)) \cong \mathrm{R}\Gamma(\mathrm{J}_{\check{\mathrm{G}},\hat{\mathbf{G}}_a(2)};\mathcal{O})$ of Hopf algebroids over $\mathrm{gr}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}}) \cong \mathrm{R}\Gamma((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-;\mathcal{O})$. Note, again, that $(\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-$ might be a stack and not a scheme, and the same is true of $\mathrm{J}_{\check{\mathrm{G}},\hat{\mathbf{G}}_a(2)}$.

For instance, if $\mathrm{G} = \mathrm{PGL}_2$ and $k = \mathbf{F}_2$, then $\pi_*\mathrm{gr}^\star_{\mathrm{G}\text{-ev}}(k^{h\mathrm{G}}) \cong \mathbf{F}_2[x,\tau_0]$ where x is in degree 0 and weight -2, and τ_0 is in homological degree -1 and weight -2 (see Example C.1). Recall the isomorphisms $\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathbf{Z}) \cong \mathbf{Z}[x, a^{\pm 1}, \frac{a^2-1}{x}]$ of Proposition 4.2.15 and $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{B};\mathbf{Z}) \cong \mathbf{Z}[x,b]/(b^2-bx)$ of Example 2.2.24. If we write $y = \frac{a^2-1}{x}$ for simplicity, the coaction map $\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathbf{Z}) \to \mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathbf{Z}) \otimes_{\mathbf{Z}[x]} \mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{B};\mathbf{Z})$ is given by the map $\mathbf{Z}[x, a^{\pm 1}, \frac{a^2-1}{x}] \to \mathbf{Z}[x, a^{\pm 1}, \frac{a^2-1}{x}, b]/(b^2-bx)$ given by the formulas

$$x \mapsto x - 2b,\ y \mapsto y - ba^{-2}y^2,\ a \mapsto a - ba^{-1}y.$$

This formula follows from equivariant localization and the fact that the map $\mathbf{Z}[a^{\pm 1}] \to \mathbf{Z}[a^{\pm 1}, c]/(c^2-c)$ encoding the $\mathbf{Z}/2$-action on $\mathbf{Z}[a^{\pm 1}]$ by inverting a sends $a \mapsto a(1-c) + a^{-1}c$. As in Example 2.2.24 (or Example C.1), one finds that there is an isomorphism

$$\mathrm{H}^{-*}(\mathrm{J}_{\check{\mathrm{G}},\hat{\mathbf{G}}_a(2)};\mathcal{O}) \cong \pi_*\mathrm{gr}^\star_{\check{\mathrm{G}}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathbf{F}_2)) \cong \mathbf{F}_2[x,\tau_0,a^{-1}y]/\tau_0 a^{-1}y.$$

We remind that x is in degree 0 and weight -2, τ_0 is in homological degree -1 and weight -2, and $a^{-1}y$ is in degree 0 and weight 2. Note that the invariant element $a + a^{-1}$ is equal to $xa^{-1}y = a - a^{-1}$, since we are working over $\mathbf{F}_2$. The relation $\tau_0 a^{-1}y$ comes from the fact that the coaction on a is $a - ba^{-1}y$, so that the differential on a is $ba^{-1}y$.

Similarly, when $\mathrm{G} = \mathrm{SL}_2$ and $k = \mathbf{F}_2$, there is an isomorphism

$$\mathrm{H}^{-*}(\mathrm{J}_{\check{\mathrm{G}},\hat{\mathbf{G}}_a(2)};\mathcal{O}) \cong \pi_*\mathrm{gr}^\star_{\check{\mathrm{G}}}(\mathrm{C}^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathbf{F}_2)) \cong \mathbf{F}_2[x^2,y^2,y(1+xy)],$$

where $y = \frac{a-1}{2x}$. We remind that x^2 is in degree 0 and weight -4, y^2 is in degree 0 and weight 4, and $y(1+xy)$ is in degree 0 and weight 2. (Note that $a-1 = 2xy = 0$, so $a = 1$, and therefore the invariant element $a + a^{-1} = 2 = 0$ vanishes.) Unlike the case of $\mathrm{G} = \mathrm{PGL}_2$, it is concentrated in cohomological degree 0.

4.4. **Shifted Poisson structures.** The following is a variant of the main definition of [PTVV13], where one allows the symplectic form to be twisted by a nontrivial line bundle.

Definition 4.4.1. Let S be a scheme, which we assume to be affine for simplicity, and let $\mathcal{L}$ be a line bundle over S. Let $\mathrm{X} \to \mathrm{S}$ be an fpqc stack over S. Let

$\mathrm{fil}^\star_{\mathrm{Hodge}}\mathrm{dR}_{\mathrm{X/S}}$ denote the Hodge-filtered derived de Rham complex of X relative to S, so that $\mathrm{gr}^\star_{\mathrm{Hodge}}\mathrm{dR}_{\mathrm{X/S}} \cong \wedge^\star_{\mathrm{X}}(\mathrm{L}_{\mathrm{X/S}})[-\star]$, where $\mathrm{L}_{\mathrm{X/S}}$ is the cotangent complex. An $\mathcal{L}$-*twisted closed j-form of degree n* on X is a global section of $\mathcal{L}^\vee \otimes \mathrm{fil}^j_{\mathrm{Hodge}}\mathrm{dR}_{\mathrm{X/S}}[j+n]$. An $\mathcal{L}$-twisted closed 2-form ω of degree n is called an *$\mathcal{L}$-twisted n-shifted symplectic form* on X (relative to S) if the section of $\mathcal{L}^\vee \otimes (\wedge^2\mathrm{L}_{\mathrm{X/S}})[n]$ induced by the map $\mathrm{fil}^2_{\mathrm{Hodge}}\mathrm{dR}_{\mathrm{X/S}}[n+2] \to (\wedge^2\mathrm{L}_{\mathrm{X/S}})[n]$ induces an isomorphism $\mathrm{T}_{\mathrm{X/S}} \xrightarrow{\cong} \mathrm{L}_{\mathrm{X/S}} \otimes \mathcal{L}^\vee[n]$. When the twisted symplectic form in question is clear from context, we will typically say that X is an $\mathcal{L}$-twisted n-shifted symplectic stack. If X is an $\mathcal{L}$-twisted n-shifted symplectic stack, $\overline{\mathrm{X}}$ will denote the $\mathcal{L}$-twisted n-shifted symplectic stack with twisted shifted symplectic form $-\omega$.

Let $f : \mathrm{Y} \to \mathrm{X}$ be a morphism of fpqc stacks over S. An *$\mathcal{L}$-twisted n-shifted isotropic structure* on f is the data of a nullhomotopy of the composite

$$\mathrm{T}_{\mathrm{Y/S}} \to f^*\mathrm{T}_{\mathrm{X/S}} \xrightarrow{\cong} f^*\mathrm{L}_{\mathrm{X/S}} \otimes \mathcal{L}^\vee[n] \to f^*\mathrm{L}_{\mathrm{Y/S}} \otimes \mathcal{L}^\vee[n].$$

An $\mathcal{L}$-twisted n-shifted isotropic structure is called *Lagrangian* if the above composite is a cofiber sequence.

With these definitions, many foundational results in the theory of shifted symplectic geometry hold nearly verbatim with the same proofs; one only needs to insert the twisting $\mathcal{L}$ appropriately. Let us collect some of these results:

Proposition 4.4.2. (1) *An $\mathcal{L}$-twisted n-shifted Lagrangian structure on the map* $\mathrm{X} \to \mathrm{S}$ *is the data of an $\mathcal{L}$-twisted $(n-1)$-shifted symplectic structure on* X *relative to* S*.*
(2) *Suppose* X_1*,* X_2*, and* X_3 *are $\mathcal{L}$-twisted n-shifted symplectic stacks relative to* S*. If* $\mathrm{Y}_1 \to \mathrm{X}_1 \times \overline{\mathrm{X}_2}$ *and* $\mathrm{Y}_2 \to \mathrm{X}_2 \times \overline{\mathrm{X}_3}$ *are $\mathcal{L}$-twisted n-shifted Lagrangians, then* $\mathrm{Y}_1 \times_{\mathrm{X}_2} \mathrm{Y}_2 \to \mathrm{X}_1 \times \overline{\mathrm{X}_3}$ *is an $\mathcal{L}$-twisted n-shifted Lagrangian. This is proved as in* [Saf16, Theorem 1.2].

Setup 4.4.3. Fix a graded commutative ring R, and let F be a 1-dimensional graded formal group over R, *without* a choice of coordinate. Let $\mathcal{O}\{1\}$ denote the dual of the Lie algebra of F, so that $\mathcal{O}\{1\}$ is an invertible R-module. (A choice of coordinate on F defines an isomorphism $\mathcal{O}\{1\} = \mathrm{R}$.) If $\mathcal{F}$ is a quasicoherent sheaf on a graded R-scheme and $n \in \mathbf{Z}$, we will write $\mathcal{F}\{n\}$ to denote $\mathcal{F} \otimes \mathcal{O}\{1\}^{\otimes n}$. Let G be the base-change to R of the Chevalley split form of a connected reductive group whose derived subgroup is not necessarily simply-laced; assume that the lacing number of the simple factors of the derived subgroup of G are invertible in R.

When $\mathrm{S} = \mathrm{Spec(R)}$ and $\mathcal{L} = \mathcal{O}\{1\}$, we will use the phrase "Tate-twisted" in place of "$\mathcal{O}\{1\}$-twisted" when discussed twisted shifted symplectic structures.

Proposition 4.4.4. *If* G *is simply-laced, the minimal bilinear form on* G *defines a Tate-twisted* 1*-shifted symplectic structure on* ${}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$*; similarly, it defines a Tate-twisted* 0*-shifted symplectic structure on* ${}^{2\rho}\check{\mathrm{G}} \times {}^{2\rho}\mathrm{G}_{\mathrm{F}}$*. If* G *is multiply-laced (so that, by assumption, the lacing number of the simple factors of the derived subgroup of* G *are invertible in* R*), the minimal bilinear form on* G *defines a Tate-twisted* 1*-shifted symplectic structure on* ${}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$*; similarly, it defines a Tate-twisted* 0*-shifted symplectic structure on* ${}^{2\rho}\check{\mathrm{G}} \times {}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}$*.*

Proof. Assume first that G is simply-laced. If V is a graded ${}^{2\rho}\check{\mathrm{G}}$-representation, let $\underline{\mathrm{V}}$ denote the associated vector bundle over ${}^{2\rho}\mathrm{G}_{\mathrm{F}}$. The tangent complex of

${}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ is the ${}^{2\rho}\check{\mathrm{G}}$-equivariant complex $[\underline{{}^{2\rho}\check{\mathfrak{g}}} \to \underline{{}^{2\rho}\mathfrak{g}}\{-1\}]$ over ${}^{2\rho}\mathrm{G}_{\mathrm{F}}$, where the differential is given by the derivative of the ${}^{2\rho}\check{\mathrm{G}}$-action on ${}^{2\rho}\mathrm{G}_{\mathrm{F}}$. By Remark 4.2.2, the minimal bilinear form gives an isomorphism $\mathfrak{g} \cong \check{\mathfrak{g}}^*$, which defines an isomorphism $\mathrm{T}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}} \cong \mathrm{L}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}}\{-1\}[1]$. The same argument proves the claim about ${}^{2\rho}\check{\mathrm{G}} \times {}^{2\rho}\mathrm{G}_{\mathrm{F}}$. The multiply-laced case is proved similarly, using the isomorphism $\check{\mathfrak{g}} \cong \check{\mathfrak{g}}^*$ from Remark 4.2.2 (afforded by the lacing numbers being invertible). $\square$

Example 4.4.5. If G is simply-laced, the map ${}^{2\rho}\mathrm{G}_{\mathrm{F}} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ admits a Tate-twisted 1-shifted Lagrangian structure, and the Tate-twisted 0-shifted symplectic structure on ${}^{2\rho}\mathrm{G}_{\mathrm{F}} \times_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}} {}^{2\rho}\mathrm{G}_{\mathrm{F}} \cong {}^{2\rho}\mathrm{G}_{\mathrm{F}} \times {}^{2\rho}\check{\mathrm{G}}$ resulting from Proposition 4.4.2 is isomorphic to the Tate-twisted 0-shifted symplectic structure from Proposition 4.4.4. Similarly, if G is multiply-laced (so that the lacing number of the simple factors of the derived subgroup of G are invertible in R), the map ${}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}} \to {}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ admits a Tate-twisted 1-shifted Lagrangian structure, and the Tate-twisted 0-shifted symplectic structure on ${}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}} \times_{{}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}} {}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}} \cong {}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}} \times {}^{2\rho}\check{\mathrm{G}}$ resulting from Proposition 4.4.2 is isomorphic to the Tate-twisted 0-shifted symplectic structure from Proposition 4.4.4.

Example 4.4.6. If G has simply-laced derived subgroup, the map $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ from Proposition 4.3.2 admits a Tate-twisted 1-shifted Lagrangian structure. Note that the tangent fibers of $\mathrm{T}_{\mathrm{F}}/\mathcal{W}$ are isomorphic to the tangent fibers of $\mathrm{T}_{\hat{\mathbf{G}}_a(2)}/\mathcal{W} \cong \hat{\mathfrak{t}}(2)/\mathcal{W}$, and similarly the tangent fibers of ${}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ are isomorphic to the tangent fibers of ${}^{2\rho}\mathfrak{g}(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{G}}$. The stack $\hat{\mathfrak{t}}(2)/\mathcal{W}$ is isomorphic to $(f_0 + {}^{2\rho}\hat{\mathfrak{b}}(2))/{}^{2\rho}\check{\mathrm{U}}_-$ by Proposition 4.3.9. The map induced on tangent bundles by the map $(f_0 + {}^{2\rho}\hat{\mathfrak{b}}(2))/{}^{2\rho}\check{\mathrm{U}}_- \to {}^{2\rho}\mathfrak{g}(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{G}}$ is given by the map $[{}^{2\rho}\check{\mathfrak{u}}_- \to {}^{2\rho}\mathfrak{b}\{-1\}] \to [{}^{2\rho}\check{\mathfrak{g}} \to {}^{2\rho}\mathfrak{g}\{-1\}]$ of complexes, whose cofiber is the complex $[{}^{2\rho}\check{\mathfrak{b}} \to {}^{2\rho}\mathfrak{u}\{-1\}]$ which is isomorphic to $[{}^{2\rho}\check{\mathfrak{u}}_- \to {}^{2\rho}\mathfrak{b}\{-1\}]^*\{-1\}[1]$ as desired. Proposition 4.4.2 therefore defines a Tate-twisted 0-shifted symplectic structure on $\mathrm{T}_{\mathrm{F}}/\mathcal{W} \times_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}} \mathrm{T}_{\mathrm{F}}/\mathcal{W} = \mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}$. Similarly, if G is multiply-laced (so that the lacing number of the simple factors of the derived subgroup of G are invertible in R), the map $\kappa_{\mathrm{F}} : \check{\mathrm{T}}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ from Proposition 4.3.2 admits a Tate-twisted 1-shifted Lagrangian structure.

Remark 4.4.7. When G is a torus, the maps $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ and $\mathrm{T}_{\mathrm{F}} \to \mathrm{T}_{\mathrm{F}}/\check{\mathrm{T}}$ agree, and the Tate-twisted 0-shifted symplectic structure on $\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}} = \mathrm{T}_{\mathrm{F}} \times \check{\mathrm{T}}$ from Example 4.4.6 is just that of Proposition 4.4.4.

Let us assume from now that F is equipped with a choice of coordinate, so that a Tate-twisted shifted symplectic structure is just a shifted symplectic structure where the symplectic form has weight 2. Example 4.4.6 defines a 0-shifted symplectic structure on $\mathrm{T}_{\mathrm{F}}/\mathcal{W} \times_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}} \mathrm{T}_{\mathrm{F}}/\mathcal{W} = \mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}$, which defines a Poisson structure on $\mathrm{H}^0(\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}; \mathcal{O})$ of weight 2. If the bad primes of G are inverted in R, then Theorem 4.2.12 gives an isomorphism $\mathrm{H}^0(\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}; \mathcal{O}) \cong (\mathcal{A}_{\mathrm{G}})^{\mathrm{W}}$, where $\mathcal{A}_{\mathrm{G}}$ is as in Definition 3.1.10.

Proposition 4.4.8. *If the bad primes of* G *are inverted in* R*, the Poisson bracket on* $\mathrm{H}^0(\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}; \mathcal{O}) \cong (\mathcal{A}_{\mathrm{G}})^{\mathrm{W}}$ *defined above is isomorphic to the Poisson bracket from Proposition 3.3.3.*

Proof. As in the proof of Theorem 3.3.7, there is an injection $(\mathcal{A}_{\mathrm{G}})^{\mathrm{W}} \hookrightarrow \mathcal{A}_{\mathrm{G}}[e(\Phi)^{-1}] \xleftarrow{\cong} \mathcal{A}_{\mathrm{T}}[e(\Phi)^{-1}]$. This is a map of Poisson algebras, so to verify that the two Poisson brackets on $(\mathcal{A}_{\mathrm{G}})^{\mathrm{W}}$ agree, it suffices to show that the two Poisson brackets on $\mathcal{A}_{\mathrm{T}}$ agree. Since $\mathcal{A}_{\mathrm{T}} = \mathcal{O}_{\mathrm{T}_{\mathrm{F}} \times \check{\mathrm{T}}}$, this follows from Remark 4.4.7 and Construction 3.3.1. □

In particular, Proposition 4.4.8 and Theorem 3.3.7 imply:

Corollary 4.4.9. *If k is an even $\mathbf{E}_\infty$-ring which is good for* G*, and the bad primes of* G *are invertible in $\pi_0(k)$, then the isomorphism $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}_*^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \cong \mathrm{H}^0(\mathrm{J}_{\check{\mathrm{G}},\mathrm{F}}; \mathcal{O})$ is one of graded Poisson $\pi_*(k)$-algebras, where Construction 3.3.6 provides the Poisson structure on $\mathrm{gr}^0_{\mathrm{G}}(\mathrm{C}_*^{\mathrm{G}_{\mathcal{O}},\mathrm{BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$.*

5. Equivalences of categories

5.1. **The simply-laced case.** As usual, let k be an even $\mathbf{E}_\infty$-ring. Let G be the base-change to $\pi_*(k)$ of the Chevalley split form of a connected *simply-laced semisimple* group. (It is easy to extend the results below to the case when G is a connected reductive group whose derived subgroup is simply-laced. Since keeping track of the torus $\mathrm{G}/[\mathrm{G}, \mathrm{G}]$ will just burden the exposition, I will assume for simplicity that G is semisimple.) Then Lemma 4.2.1 defines an action of ${}^{2\rho}\check{\mathrm{G}}$ on ${}^{2\rho}\mathrm{G}$, and hence on ${}^{2\rho}\mathrm{G}_{\mathrm{F}}$, by conjugation. Let G^{ad} denote the adjoint quotient of G (this agrees with $\check{\mathrm{G}}^{\mathrm{ad}}$ thanks to Lemma 4.2.1), so that its dual group is $\check{\mathrm{G}}^{\mathrm{sc}}$.

Remark 5.1.1. Throughout this section, we will often assume that the bad primes of G are invertible in $\pi_0(k)$. This stems from our reliance on this assumption in Theorem 4.2.12; however, as mentioned there, this assumption is used only to appeal to Proposition 4.2.7 and Proposition 5.1.15. In the former, the assumption is used only in a very minor way and is likely unnecessary, while in the latter, the assumption is used more seriously to appeal to [AJ84, Proposition 4.4]. It seems likely that more careful arguments will eliminate the need to invert the bad primes of G for the results below to hold (like in Proposition 4.2.15 and Theorem 4.3.10), but we do not discuss this further here.

Definition 5.1.2. For each dominant minuscule coweight μ of G^{ad}, let $i_\mu : \mathcal{G}\mathrm{r}^\mu_{\mathrm{G}^{\mathrm{ad}}} \cong \mathcal{G}\mathrm{r}^{\leq\mu}_{\mathrm{G}^{\mathrm{ad}}} \hookrightarrow \mathcal{G}\mathrm{r}_{\mathrm{G}^{\mathrm{ad}}}$ denote the closed inclusion. Let $\mathcal{G}\mathrm{r}^\mu_{\mathrm{G}}$ denote the preimage of $\mathcal{G}\mathrm{r}^\mu_{\mathrm{G}^{\mathrm{ad}}}$ under the map $\mathcal{G}\mathrm{r}_{\mathrm{G}} \to \mathcal{G}\mathrm{r}_{\mathrm{G}^{\mathrm{ad}}}$, and let $i_\mu : \mathcal{G}\mathrm{r}^\mu_{\mathrm{G}} \hookrightarrow \mathcal{G}\mathrm{r}_{\mathrm{G}}$ denote the resulting inclusion (note that $\mathcal{G}\mathrm{r}^\mu_{\mathrm{G}}$ is empty if μ is not a coweight of G). Define $\mathrm{IC}_\mu \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ to be the pushforward $i_{\mu,*}\underline{k}_{\mathcal{G}\mathrm{r}^\mu_{\mathrm{G}}}$. (If $\mathcal{G}\mathrm{r}^\mu_{\mathrm{G}}$ is empty, then this is 0.) More generally, if $\mu_\bullet = (\mu_1, \cdots, \mu_n)$ is a sequence of dominant minuscule coweights of G^{ad}, let $\mathrm{IC}_{\mu_\bullet} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ denote the restriction to $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ of the pushforward of the constant sheaf along the map $\mathcal{G}\mathrm{r}^{\leq\mu_\bullet}_{\mathrm{G}^{\mathrm{ad}}} \to \mathcal{G}\mathrm{r}^{\leq|\mu_\bullet|}_{\mathrm{G}^{\mathrm{ad}}} \subseteq \mathcal{G}\mathrm{r}_{\mathrm{G}^{\mathrm{ad}}}$. Let $\mathcal{G}\mathrm{r}^{\leq\mu_\bullet}_{\mathrm{G}}$ denote the preimage of $\mathcal{G}\mathrm{r}^{\leq\mu_\bullet}_{\mathrm{G}^{\mathrm{ad}}}$ under the map $\mathcal{G}\mathrm{r}_{\mathrm{G}} \to \mathcal{G}\mathrm{r}_{\mathrm{G}^{\mathrm{ad}}}$, so that $\mathrm{IC}_{\mu_\bullet}$ is the pushforward of the constant sheaf along the map $i_{\mu_\bullet} : \mathcal{G}\mathrm{r}^{\leq\mu_\bullet}_{\mathrm{G}} \to \mathcal{G}\mathrm{r}^{\leq|\mu_\bullet|}_{\mathrm{G}} \subseteq \mathcal{G}\mathrm{r}_{\mathrm{G}}$.[10] (If $|\mu_\bullet|$ is not a coweight of G, then $\mathcal{G}\mathrm{r}^{\leq\mu_\bullet}_{\mathrm{G}}$ is empty and $\mathrm{IC}_{\mu_\bullet} = 0$ in $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$.)

Similarly, if μ is a dominant minuscule weight of $\check{\mathrm{G}}^{\mathrm{sc}}$, let $\mathrm{V}_\mu \in \mathrm{Perf}(\mathrm{B}\check{\mathrm{G}}^{\mathrm{sc}})$ denote the Weyl module of $\check{\mathrm{G}}^{\mathrm{sc}}$ with highest weight μ (viewed as a projective $\pi_*(k)$-module via base-change from $\mathbf{Z}$), and more generally if $\mu_\bullet = (\mu_1, \cdots, \mu_n)$ is a sequence of dominant minuscule weights of $\check{\mathrm{G}}^{\mathrm{sc}}$, let $\mathrm{V}_{\mu_\bullet} \cong \bigotimes_{i=1}^n \mathrm{V}_{\mu_i}$. We will use the same

[10] We do not use perverse normalizations here!

notation to denote the projection (i.e., pushforward[11]) of $\mathrm{V}_{\mu_\bullet}$ to $\mathrm{Perf}(\mathrm{B}\check{\mathrm{G}})$. (If $|\mu_\bullet|$ is not a dominant weight of $\check{\mathrm{G}}$, then this is zero.) Note that V_μ canonically lifts to an object of $\mathrm{Perf}^{\mathrm{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$, via the action of $2\rho : \mathbf{G}_m \to \check{\mathrm{G}}$. We will continue to denote this lift by V_μ, and similarly for $\mathrm{V}_{\mu_\bullet}$. We will also write $\mathrm{V}_{\mu_\bullet}(2\rho, |\mu_\bullet|)$ to denote the shift of $\mathrm{V}_{\mu_\bullet}$ in weight by $(2\rho, |\mu_\bullet|)$. We will write $\underline{\mathrm{V}}_{\mu_\bullet}$ to denote $\mathrm{V}_{\mu_\bullet}(2\rho, |\mu_\bullet|) \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T}_\mathrm{F}}$, so that it is a free graded $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-module.

Construction 5.1.3. Let μ be a dominant minuscule coweight of G^{ad}, and let $\mathrm{W}_\mu \subseteq \mathrm{W}$ be the stabilizer of the coweight μ. Let Φ_μ^+ denote the set of positive roots α with $(\alpha, \mu) > 0$, let $\mathrm{E}_\mu = \mathrm{W}/\mathrm{W}_\mu \times \Phi_\mu^+$, and let Γ denote the diagram of finite sets $\mathrm{E}_\mu \rightrightarrows \mathrm{W}/\mathrm{W}_\mu$, where the two maps are given by $(w, \alpha) \mapsto w, s_\alpha(w)$. Let $\mathrm{E}_\mu \to \mathcal{O}_{\mathrm{T}_\mathrm{F}}$ be the map sending $(w, \alpha) \mapsto e(\alpha)$. If M_Γ is the $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-algebra as in Definition 2.2.11, there is an abstract isomorphism $\underline{\mathrm{V}}_\mu \cong \mathrm{M}_\Gamma$ of graded $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-modules. Indeed, since both are free $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-modules of rank $|\mathrm{W}/\mathrm{W}_\mu|$, they are abstractly isomorphic to $\prod_{\mathrm{W}/\mathrm{W}_\mu} \mathcal{O}_{\mathrm{T}_\mathrm{F}}$ with the grading given by shifting the summand corresponding to $w \in \mathrm{W}/\mathrm{W}_\mu$ by $2\ell(w)$. It now follows from Construction 2.2.29 that there is a $\mathcal{W}$-module structure on $\underline{\mathrm{V}}_\mu$ such that upon inverting $e(\Phi_\mu^+)$ and using the isomorphisms $\underline{\mathrm{V}}_\mu[e(\Phi_\mu^+)^{-1}] \cong \mathrm{M}_\Gamma[e(\Phi_\mu^+)^{-1}] \cong \mathrm{Map}(\mathrm{W}/\mathrm{W}_\mu, \mathcal{O}_{\mathrm{T}_\mathrm{F}}[e(\Phi_\mu^+)^{-1}])$, the $\mathcal{O}_{\mathcal{W}}[e(\Phi_\mu^+)^{-1}] \cong \mathrm{Map}(\mathrm{W}, \mathcal{O}_{\mathrm{T}_\mathrm{F}}[e(\Phi_\mu^+)^{-1}])$-coaction on $\underline{\mathrm{V}}_\mu[e(\Phi_\mu^+)^{-1}]$ identifies with the natural W-action on $\mathrm{Map}(\mathrm{W}/\mathrm{W}_\mu, \mathcal{O}_{\mathrm{T}_\mathrm{F}}[e(\Phi_\mu^+)^{-1}])$. Taking tensor products, this defines a graded $\mathcal{W}$-module structure on $\underline{\mathrm{V}}_{\mu_\bullet} := \mathrm{V}_{\mu_\bullet}(2\rho, |\mu_\bullet|) \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T}_\mathrm{F}}$.

Proposition 5.1.4. *Suppose that the bad primes (see Table 1) of* G *are inverted in* $\pi_0(k)$. *Under the isomorphism* $\mathrm{Spec}(\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)) \cong \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ *of Theorem 4.2.12, the* $\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$*-comodule* $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathrm{IC}_{\mu_\bullet})$ *is* $\mathcal{W}$*-equivariantly isomorphic to the* $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$*-module* $\mathrm{Res}^{\check{\mathrm{G}}}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}(\underline{\mathrm{V}}_{\mu_\bullet})$.

Proof. It suffices to prove the claim when $k = \mathrm{MU}$, so we will assume this from now. Using Corollary 3.1.12, we may further reduce to the case when $\mu_\bullet = \mu$. Let us write $\underline{\mathrm{V}}_\mu^{\mathrm{geom}}$ to denote $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathrm{IC}_\mu) \cong \mathrm{H}_{\mathrm{T},c}^{-*}(\mathrm{G}/\mathrm{P}_\mu; k)$. Note that $\mathrm{G}/\mathrm{P}_\mu$ is a k-GKM T-space (using Lemma 2.2.20). If M denotes either of $\underline{\mathrm{V}}_\mu^{\mathrm{geom}}$, $\underline{\mathrm{V}}_\mu$, or $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$, then M is flat over $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$: for $\underline{\mathrm{V}}_\mu^{\mathrm{geom}}$, this follows from the Bruhat decomposition for $\mathrm{G}/\mathrm{P}_\mu$; and for $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$, this is Proposition 4.2.7. Moreover,

$$\mathrm{M} \cong \bigcap_{\alpha \in \Phi} \mathrm{M}[e(\Phi - \{\alpha\})^{-1}] \subseteq \mathrm{M}[e(\Phi)^{-1}];$$

for $\underline{\mathrm{V}}_\mu^{\mathrm{geom}}$, this follows from Theorem 2.2.12; and for $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$, this follows from Theorem 4.2.12. Recall from Corollary 4.2.9 that $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}[e(\Phi)^{-1}] \cong \mathcal{O}_{\check{\mathrm{T}} \times \mathrm{T}_\mathrm{F}}[e(\Phi)^{-1}]$. It therefore suffices to check that there is a $\check{\mathrm{T}}$-equivariant isomorphism $\underline{\mathrm{V}}_\mu[e(\Phi)^{-1}] \cong \underline{\mathrm{V}}_\mu^{\mathrm{geom}}[e(\Phi)^{-1}]$, and that for each root α, the submodules $\underline{\mathrm{V}}_\mu[e(\Phi - \{\alpha\})^{-1}]$ and $\underline{\mathrm{V}}_\mu^{\mathrm{geom}}[e(\Phi - \{\alpha\})^{-1}]$ are $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}[e(\Phi - \{\alpha\})^{-1}]$-equivariantly isomorphic (compatibly with their $\mathcal{W}$-actions).

[11]Since the kernel of the map $\check{\mathrm{G}}^{\mathrm{sc}} \to \check{\mathrm{G}}$ is diagonalizable, the derived pushforward does not have any higher cohomology. In other words, $\mathrm{Perf}(\mathrm{B}\check{\mathrm{G}}^{\mathrm{sc}})$ is a direct sum of categories indexed by $\mathbb{X}^*(\mathrm{Z}(\check{\mathrm{G}}^{\mathrm{sc}}))$, and the pushforward map is just induced by projection to the direct sum of the categories indexed by $\mathbb{X}^*(\mathrm{Z}(\check{\mathrm{G}}))$.

By Proposition 2.2.4, $\underline{\mathrm{V}}_\mu^{\mathrm{geom}}[e(\Phi)^{-1}] \xrightarrow{\cong} \mathrm{Map}(\mathrm{W}/\mathrm{W}_\mu, \mathcal{O}_{\mathrm{T_F}}[e(\Phi)^{-1}])$, and the latter is isomorphic to $\underline{\mathrm{V}}_\mu[e(\Phi)^{-1}]$ as in Construction 5.1.3. Note that this can be rewritten as $\mathrm{Map}(\mathrm{W}/\mathrm{W}_\mu, \mathbf{Z}) \otimes \mathcal{O}_{\mathrm{T_F}}[e(\Phi)^{-1}]$. Under this identification, the $\check{\mathrm{T}}$-action on $\underline{\mathrm{V}}_\mu[e(\Phi)^{-1}]$ is given by the embedding $\mathrm{W}/\mathrm{W}_\mu \hookrightarrow \mathbb{X}_*$. It therefore suffices to show that the $\mathbf{Z}[\mathbb{X}_*] \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T_F}} \cong \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r_T}; k)$-comodule structure on $\underline{\mathrm{V}}_\mu^{\mathrm{geom}}[e(\Phi)^{-1}]$ admits the same description. There is a T-equivariant commutative diagram

$$\begin{tikzcd} \mathrm{W}/\mathrm{W}_\mu \arrow[r] \arrow[d, "i_\mu^{\mathrm{T}}"'] & \mathrm{G}/\mathrm{P}_\mu \arrow[d, "i_\mu"] \\ \mathbb{X}_* \cong \mathcal{G}\mathrm{r_T} \arrow[r] & \mathcal{G}\mathrm{r_G}, \end{tikzcd}$$

where the left vertical map is the T-fixed locus of the right vertical map. The map $i_\mu^{\mathrm{T}} : \mathrm{W}/\mathrm{W}_\mu \to \mathbb{X}_*$ is the inclusion of the W-orbit of $\mu \in \mathbb{X}_*$. The naturality of the homology coaction implies that under the above isomorphisms, the $\mathbf{Z}[\mathbb{X}_*] \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T_F}} \cong \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r_T}; k)$-coaction identifies with the base-change along $\mathbf{Z} \to \mathcal{O}_{\mathrm{T_F}}[e(\Phi)^{-1}]$ of the map

$$\mathrm{Map}(\mathrm{W}/\mathrm{W}_\mu, \mathbf{Z}) \to \mathrm{Map}(\mathrm{W}/\mathrm{W}_\mu, \mathbf{Z}) \otimes \mathbf{Z}[\mathbb{X}_*]$$

which is induced by the tautological coaction of the coalgebra $\mathbf{Z}[\mathrm{W}/\mathrm{W}_\mu]$ on $\mathrm{Map}(\mathrm{W}/\mathrm{W}_\mu, \mathbf{Z})$ and composition with the coalgebra map $\mathbf{Z}[\mathrm{W}/\mathrm{W}_\mu] \to \mathbf{Z}[\mathbb{X}_*]$. This is exactly the $\check{\mathrm{T}}$-action on $\mathrm{Map}(\mathrm{W}/\mathrm{W}_\mu, \mathbf{Z})$ given by the embedding $\mathrm{W}/\mathrm{W}_\mu \hookrightarrow \mathbb{X}_*$, so that that there is a $\check{\mathrm{T}}$-equivariant isomorphism $\underline{\mathrm{V}}_\mu[e(\Phi)^{-1}] \cong \underline{\mathrm{V}}_\mu^{\mathrm{geom}}[e(\Phi)^{-1}]$ compatibly with their $\mathcal{W}[e(\Phi)^{-1}] \cong \underline{\mathrm{W}}$-actions. (Note that Example 2.2.30 already implies that $\underline{\mathrm{V}}_\mu$ and $\underline{\mathrm{V}}_\mu^{\mathrm{geom}}$ are $\mathcal{W}$-equivariantly isomorphic.)

Recall that $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \cong \mathcal{A}_{\mathrm{G}}$ (where $\mathcal{A}_{\mathrm{G}}$ is as in Definition 3.1.10). Fix a root α, and let L_α denote the Levi subgroup of G generated by the root spaces $\mathrm{U}_{\pm\alpha}$, so that L_α has semisimple rank 1. Then $\mathcal{A}_{\mathrm{G}}[e(\Phi - \{\alpha\})^{-1}] \cong \mathcal{A}_{\mathrm{L}_\alpha}[e(\Phi - \{\alpha\})^{-1}]$ as Hopf subalgebras of $\mathcal{A}_{\mathrm{G}}[e(\Phi)^{-1}] \cong \mathcal{A}_{\mathrm{T}}[e(\Phi)^{-1}]$. Moreover, if T_α denotes the kernel of α, then Proposition 2.2.4 gives an isomorphism $\underline{\mathrm{V}}_\mu^{\mathrm{geom}}[e(\Phi - \{\alpha\})^{-1}] \cong \mathrm{H}_{\mathrm{T},c}^{-*}((\mathrm{G}/\mathrm{P}_\mu)^{\mathrm{T}_\alpha}; k)[e(\Phi - \{\alpha\})^{-1}]$ of subalgebras in $\underline{\mathrm{V}}_\mu^{\mathrm{geom}}[e(\Phi)^{-1}] \cong \mathrm{Map}(\mathrm{W}/\mathrm{W}_\mu, \mathcal{O}_{\mathrm{T_F}}[e(\Phi)^{-1}])$. There is an isomorphism $(\mathrm{G}/\mathrm{P}_\mu)^{\mathrm{T}_\alpha} \cong \coprod_{w \in \langle s_\alpha \rangle \backslash \mathrm{W}/\mathrm{W}_\mu} \mathrm{L}_\alpha/(\mathrm{L}_\alpha \cap \mathrm{Ad}_w(\mathrm{P}_\mu))$. Since L_α has semisimple rank 1, each connected component of $(\mathrm{G}/\mathrm{P}_\mu)^{\mathrm{T}_\alpha}$ is an isolated point or isomorphic to $\mathbf{P}^1$ (depending on whether $w^{-1}(\alpha)$ is a root of the Levi quotient of P_μ). Replacing G by L_α, we are reduced in this way to checking that there is a $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$-equivariant isomorphism $\underline{\mathrm{V}}_\mu \cong \underline{\mathrm{V}}_\mu^{\mathrm{geom}}$ when G is (connected) reductive with semisimple rank 1.

For this, we may replace G by its derived subgroup, in which case G is isomorphic either to SL_2 or PGL_2. When $\mathrm{G} = \mathrm{SL}_2$, there is no minuscule coweight, so $\mathrm{G}/\mathrm{P}_\mu = \emptyset$ and $\mathrm{V}_\mu \cong 0$. We may therefore assume $\mathrm{G} = \mathrm{PGL}_2$. In this case, the map $\mathrm{G}/\mathrm{P}_\mu \to \mathcal{G}\mathrm{r_G}$ is isomorphic to the composite $\mathbf{P}^1 \to \Omega_{\mathrm{poly}}(\mathrm{GL}_2) \to \Omega_{\mathrm{poly}}(\mathrm{PGL}_2)$. Using Theorem 2.2.9 (or rather, the special case Example 2.2.14), the equivariant homology $\mathrm{H}_*^{\mathrm{T,BM}}(\mathbf{P}^1; k)$ is the free $\mathcal{O}_{\mathrm{F}}$-module $\mathcal{O}_{\mathrm{F}}\{[0], [\mathbf{P}^1]\}$ where $[0]$ lives in weight 0 (and is the fundamental class of the fixed point $0 \in \mathbf{P}^1$), and $[\mathbf{P}^1]$ lives in weight 2. The fundamental class of $\infty \in \mathbf{P}^1$ is $[\infty] = x[\mathbf{P}^1] - [0]$, where x is a coordinate on F. The map

$$\mathrm{H}_*^{\mathrm{T,BM}}(\mathbf{P}^1; k) \to \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{PGL}_2}; k) \cong \mathcal{O}_{\mathrm{F}}[a^{\pm 1}, b]/(bx = a - a^{-1})$$

sends $[0] \mapsto a$ and $[\infty] \mapsto a^{-1}$ (since the map $\mathbf{P}^1 \to \mathcal{G}\mathrm{r}_{\mathrm{PGL}_2}$ is given on T-fixed points by the inclusion $\{1,-1\} \to \mathbf{Z}$), and so x-torsionfreeness forces $[\mathbf{P}^1] \mapsto b$. Let us therefore write $a = [0]$ and $b = [\mathbf{P}^1]$ in $\mathrm{H}_*^{\mathrm{T,BM}}(\mathbf{P}^1; k)$, so that the map $\mathrm{H}_*^{\mathrm{T,BM}}(\mathbf{P}^1; k) \to \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{PGL}_2}; k)$ is given by the inclusion $\mathcal{O}_\mathrm{F}\{a, b\} \to \mathcal{O}_\mathrm{F}[a^{\pm 1}, b]/(bx = a - a^{-1})$. Note that this is a map of coalgebras, where the coproduct is given by $\Delta(a) = a \otimes a$ and $\Delta(b) = b \otimes a + a \otimes b - xb \otimes b$. The multiplication $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k) \otimes_{\mathcal{O}_\mathrm{F}} \mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k) \to \mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k)$ is adjoint to a map $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k) \to \mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k) \otimes_{\mathcal{O}_\mathrm{F}} \mathrm{H}_*^{\mathrm{T,BM}}(\mathbf{P}^1; k)$ which gives a $\mathrm{H}_*^{\mathrm{T,BM}}(\mathbf{P}^1; k)$-coaction on $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k)$. If we write $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k) \cong \mathcal{O}_\mathrm{F}\{v_0, v_1\}$ with v_0 in weight 0 and v_1 in weight -2, then the cup product is given by $v_0 \otimes v_0 \mapsto v_0$, $v_0 \otimes v_1 \mapsto v_1$, $v_1 \otimes v_0 \mapsto v_1$, $v_1 \otimes v_1 \mapsto -xv_1$ (see Example 2.2.14, where v_0 is the unit element 1 and v_1 is denoted b). The $\mathrm{H}_*^{\mathrm{T,BM}}(\mathbf{P}^1; k)$-coaction, and hence the $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{PGL}_2}; k)$-coaction, on $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k)$ is therefore given by

$$v_0 \mapsto v_0 \otimes a + v_1 \otimes b,\ v_1 \mapsto v_1 \otimes a - xv_1 \otimes b = v_1 \otimes (a - xb) = v_1 \otimes a^{-1}.$$

The map $\mathcal{O}_{{}^{2\rho}\check{\mathrm{B}} \times \mathrm{F}} \to \mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{PGL}_2}; k)$ is induced by the inclusion $\widetilde{\mathrm{J}}_{\mathrm{PGL}_2,\mathrm{F}} \to {}^{2\rho}\check{\mathrm{B}} \times \mathrm{F}$ of invertible matrices of the form $\left(\begin{smallmatrix} a & b \\ 0 & a^{-1} \end{smallmatrix}\right)$, and it is clear from the above formulas that the $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_{\mathrm{PGL}_2}; k)$-coaction on $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k)$ is just given by restricting the ${}^{2\rho}\check{\mathrm{B}}$-action on $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k) \cong \mathcal{O}_\mathrm{F}\{v_0, v_1\}$ via the standard representation. That is, there is a $\widetilde{\mathrm{J}}_{\mathrm{PGL}_2,\mathrm{F}}$-equivariant isomorphism $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k) \cong \underline{\mathrm{V}}_\mu$ compatible with the $\check{\mathrm{T}}$-equivariant isomorphism $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}^1; k)[x^{-1}] \cong \underline{\mathrm{V}}_\mu[x^{-1}]$ upon localization. □

Remark 5.1.5. Proposition 5.1.4 gives a $\mathcal{W}$-equivariant isomorphism $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathrm{IC}_{\mu_\bullet}) \cong \underline{\mathrm{V}}_{\mu_\bullet}$ for the $\mathcal{W}$-action on $\underline{\mathrm{V}}_{\mu_\bullet}$ from Construction 5.1.3. The $\mathcal{W}$-action (and in fact, even the W-action) on $\underline{\mathrm{V}}_{\mu_\bullet}$ typically cannot be "straightened out" so that under the isomorphism $\underline{\mathrm{V}}_{\mu_\bullet} \cong \mathrm{V}_{\mu_\bullet}(2\rho, |\mu_\bullet|) \otimes_\mathbf{Z} \mathcal{O}_{\mathrm{T}_\mathrm{F}}$, the $\mathcal{W}$-action is inherited from the tensor factor $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$. In other words, $\underline{\mathrm{V}}_{\mu_\bullet}$ may not admit a $\mathcal{W}$-invariant basis as a $\mathcal{W}$-equivariant $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-module. For instance, if $\mathrm{G} = \mathrm{PGL}_2$ and k is a discrete $\mathbf{Z}$-algebra (for simplicity), then $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathcal{G}\mathrm{r}_{\mathrm{PGL}_2}; \mathrm{IC}_{\mu_\bullet}) \cong k[x]\{v_0, v_1\}$, and the $\mathrm{W} \cong \mathbf{Z}/2$-action sends $x \mapsto -x$, $v_0 \mapsto v_0$, and $v_1 \mapsto v_1 - x$. Since the $\mathbf{Z}/2$-invariants are $k[x^2]\{v_0, x - 2v_1\}$, we find that a $\mathbf{Z}/2$-invariant $k[x]$-module basis for $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathcal{G}\mathrm{r}_{\mathrm{PGL}_2}; \mathrm{IC}_{\mu_\bullet})$ exists (and is given by $\{v_0, x - 2v_1\}$) if and only if $2 \in k^\times$. (This can be viewed as stemming from the failure of $\pi_*(k^{h\mathrm{T}})$ to be a free $\pi_*\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{PGL}_2})$-module, which in turn comes from the 2-torsion in $\pi_*\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{PGL}_2})$.)

We now analyze the quasi-minuscule case.

Definition 5.1.6. Let $\theta \in \mathbb{X}_*^+$ denote the highest short root of $\check{\mathrm{G}}^{\mathrm{sc}}$. (Since we are assuming that $\check{\mathrm{G}}$ is simply-laced, all root lengths are the same, so θ is the highest weight of the adjoint representation of $\check{\mathrm{G}}^{\mathrm{sc}}$.) Then the quasi-minuscule Schubert variety $\mathcal{G}\mathrm{r}_\mathrm{G}^{\leq\theta}$ admits a stratification $\mathcal{G}\mathrm{r}_\mathrm{G}^\theta \sqcup \mathcal{G}\mathrm{r}_\mathrm{G}^0$, where $\mathcal{G}\mathrm{r}_\mathrm{G}^0$ is the basepoint of $\mathcal{G}\mathrm{r}_\mathrm{G}$. As discussed in [NP01, Section 7] and [Zhu17, Lemma 2.1.14], there is a smooth resolution of $\mathcal{G}\mathrm{r}_\mathrm{G}^{\leq\theta}$ defined as follows. Let $\check{\theta}$ denote the highest root of G_{ad}. Then P_θ stabilizes the line $\mathfrak{g}_{\check{\theta}}$, and we write $\mathcal{L}_\theta = \mathrm{G} \times^{\mathrm{P}_\theta} \mathfrak{g}_{\check{\theta}}$ to denote the associated line bundle over $\mathrm{G}/\mathrm{P}_\theta$. Then $\mathcal{G}\mathrm{r}_\mathrm{G}^\theta$ is isomorphic to the total space of $\mathcal{L}_\theta$. Moreover, $\mathcal{G}\mathrm{r}_\mathrm{G}^{\leq\theta}$ is the projective cone of the embedding of $\mathrm{G}/\mathrm{P}_\theta$ by $\mathcal{L}_\theta$ (i.e., it is the Thom space of the line bundle $\mathcal{L}_\theta$ over $\mathrm{G}/\mathrm{P}_\theta$), and the map $f : \mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}) \to \mathcal{G}\mathrm{r}_\mathrm{G}^{\leq\theta}$ is a resolution of singularities which contracts $\mathrm{G}/\mathrm{P}_\theta$ to the basepoint $\mathcal{G}\mathrm{r}^0_{\mathrm{G}}$. We will write $\widetilde{\mathrm{IC}}_\theta \in \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ to denote the pushforward $f_* \underline{k}_{\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})}$.

In geometric Satake, one typically considers the intersection cohomology sheaf IC_θ associated to the constant local system on $\mathcal{G}\mathrm{r}^{\leq\theta}_{\mathrm{G}}$. However, for a non-discrete $\mathbf{E}_\infty$-ring k, it is not clear how one might define the analogous object. The sheaf $\widetilde{\mathrm{IC}}_\theta$ is supposed to be a workaround: if k is a $\mathbf{Q}$-algebra, then the decomposition theorem says that $\widetilde{\mathrm{IC}}_\theta$ splits as $\mathrm{IC}_\theta \oplus \mathcal{C}$, where $\mathcal{C}$ is a certain k-module supported on $\mathcal{G}\mathrm{r}^0_{\mathrm{G}}$ (see [NP01, Corollaire 8.2]). For general even $\mathbf{E}_\infty$-rings k, we use $\widetilde{\mathrm{IC}}_\theta$ as a substitute for IC_θ.

We now study the analogue of Proposition 5.1.4 for $\widetilde{\mathrm{IC}}_\theta$; for this, we may (and will) assume, as in the proof of Proposition 5.1.4, that $k = \mathrm{MU}$ (or perhaps a localization thereof by a nonzero element of $\pi_0(k)$). Note that $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})$ is not a T-equivariant k-GKM space, but it is nevertheless a $\mathbf{P}^1$-bundle over $\mathrm{G}/\mathrm{P}_\theta$. The cohomology $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta) \cong \mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k)$ is therefore a finite free $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-module equipped with a $\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$-comodule structure. There is therefore an isomorphism

$$\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k) \cong \bigcap_{\alpha \in \Phi} \mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k) \left[e(\Phi - \{\alpha\})^{-1}\right] \subseteq \mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k)[e(\Phi)^{-1}];$$

moreover, this is a $\mathcal{W}$-equivariant isomorphism of $\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$-comodules.

Lemma 5.1.7. *There is a $\mathcal{W}$-equivariant isomorphism*

$$\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k)[e(\Phi)^{-1}] \cong \mathrm{Map}(\check{\Phi} \times \{0, \infty\}, \mathbf{Z}) \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T}_\mathrm{F}}[e(\Phi)^{-1}]$$

of $\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)[e(\Phi)^{-1}] \cong \mathcal{O}_{\mathrm{T}_\mathrm{F}}[\mathbb{X}_*][e(\Phi)^{-1}]$*-comodules, where the* $\mathcal{O}_{\mathrm{T}_\mathrm{F}}[\mathbb{X}_*]$*-comodule structure (i.e.,* $\check{\mathrm{T}}$*-action) on* $\mathrm{Map}(\check{\Phi} \times \{0, \infty\}, \mathbf{Z})$ *is induced by the map* $\check{\Phi} \times \{0, \infty\} \xrightarrow{\mathrm{pr}_0} \check{\Phi} \subseteq \mathbb{X}_*$*, and the* $\mathcal{W}$*-action on the right-hand side is determined by the natural* W*-action on* $\check{\Phi}$ *and* $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$ *via the isomorphism* $\mathcal{O}_{\mathcal{W}}[e(\Phi)^{-1}] \xrightarrow{\cong} \mathcal{O}_{\underline{\mathrm{W}}}[e(\Phi)^{-1}]$.

In other words, under the isomorphism $\mathcal{O}_{\mathrm{T}_\mathrm{F}}[\mathbb{X}_*] \cong \mathcal{O}_{\check{\mathrm{T}} \times \mathrm{T}_\mathrm{F}}$, $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k)[e(\Phi)^{-1}]$ is isomorphic to the free $\mathcal{O}_{\mathrm{T}_\mathrm{F}}[e(\Phi)^{-1}]$-module given by base-changing the $\check{\mathrm{T}}$-representation $\mathbf{Z}^{|\Phi|} \oplus \bigoplus_{\check{\alpha} \in \check{\Phi}} (\check{\mathfrak{g}}_{\check{\alpha}})_{\mathbf{Z}} \cong \check{\mathfrak{g}}_{\mathbf{Z}} \oplus \mathbf{Z}^{|\Phi| - \mathrm{rank}(\check{\mathrm{G}})}$ along the unit map $\mathbf{Z} \to \mathcal{O}_{\mathrm{T}_\mathrm{F}}[e(\Phi)^{-1}]$.

Proof. Using Proposition 2.2.4, it follow that $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k)[e(\Phi)^{-1}] \cong \mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}}; k)[e(\Phi)^{-1}]$. The $\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$-comodule structure on $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k)$ arises via the map $f : \mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}) \to \mathcal{G}\mathrm{r}_{\mathrm{G}}$. Therefore, the $\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}}; k)$-comodule structure on $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}}; k)$ arises via the map $f^{\mathrm{T}} : \mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}} \to \mathbb{X}_* \cong \mathcal{G}\mathrm{r}_{\mathrm{T}}$. Note that there is an isomorphism $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}} \cong \check{\Phi} \times \{0, \infty\}$: indeed, this follows from the analogous isomorphism $(\mathrm{G}/\mathrm{P}_\theta)^{\mathrm{T}} \cong \check{\Phi}$, which in turn is a consequence of the observation that $\mathrm{W}/\mathrm{W}_\theta \cong \check{\Phi}$. (This uses that G is simply-laced, so that all roots have the same length.) This map is just the projection onto $\check{\Phi}$ (via $0 \in \{0, \infty\}$) composed with the inclusion of $\check{\Phi}$ into $\mathbb{X}_*$. □

Recall that a $\check{\mathrm{G}}$-representation on a projective R-module V has a *good filtration* (see [Don81]) if it admits an exhaustive filtration

$$0 = \mathrm{V}_0 \subseteq \mathrm{V}_1 \subseteq \cdots \subseteq \mathrm{V}$$

such that $\mathrm{V}_i/\mathrm{V}_{i-1}$ is isomorphic to a direct sum of dual Weyl modules. Dually, a $\check{\mathrm{G}}$-representation on a projective R-module V which admits a filtration whose subquotients are isomorphic to direct sums of Weyl modules will be said to have a *Weyl filtration.* A *tilting module* is a $\check{\mathrm{G}}$-representation on a projective R-module which admits both a good filtration and a Weyl filtration.

Construction 5.1.8. Let T_θ denote the tilting module (defined over $\mathbf{Z}$; see [Jan03, Section E.19]) with highest weight θ. Let V_0 be a free (graded) $\mathbf{Z}$-module such that $\widetilde{\mathrm{T}}_\theta := \mathrm{V}_0 \oplus \mathrm{T}_\theta$ is a free $\mathbf{Z}$-module of rank $2|\check{\Phi}|$, and view $\widetilde{\mathrm{T}}_\theta$ as a $\check{\mathrm{G}}$-representation via the trivial action on V_0. Then, there is a grading on V_0 such that $\underline{\widetilde{\mathrm{T}}}_\theta = \widetilde{\mathrm{T}}_\theta \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T}_\mathrm{F}}$ admits a graded $\mathcal{W}$-module structure defined as follows. Let $\mathrm{E}_f = \check{\Phi}$. Let Φ_θ^+ denote the set of roots of the unipotent radical of P_θ, so it is the set of positive roots α with $(\alpha, \theta) > 0$, and let $\mathrm{E}_b = \check{\Phi} \times \Phi_\theta^+ \times \{0, \infty\}$. Finally, set $\mathrm{E} = \mathrm{E}_f \sqcup \mathrm{E}_b$. There is a diagram $\mathrm{E}_f \rightrightarrows \check{\Phi} \times \{0, \infty\}$ sending $\check{\alpha} \mapsto (\check{\alpha}, 0), (\check{\alpha}, \infty)$. Similarly, there is a diagram $\mathrm{E}_b \rightrightarrows \check{\Phi} \times \{0, \infty\}$ given by the product of the diagram $\check{\Phi} \times \Phi_\theta^+ \rightrightarrows \check{\Phi}$ (sending $(\check{\alpha}, \beta) \mapsto \check{\alpha}, s_\beta(\check{\alpha})$) with $\{0, \infty\}$. Together, this defines a diagram $\mathrm{E} \rightrightarrows \check{\Phi} \times \{0, \infty\}$ which will be denoted by Γ. There is a map $\mathrm{E} \to \mathcal{O}_{\mathrm{T}_\mathrm{F} \times \mathrm{F}}$ which is given on E_f by the map $\check{\Phi} \ni \check{\alpha} \mapsto e(\alpha) +_\mathrm{F} x$ (where x is the coordinate on F), and on E_b by $\check{\Phi} \times \Phi_\theta^+ \times \{0, \infty\} \ni (\check{\alpha}, \beta, \epsilon) \mapsto e(\beta)$.

Let M_Γ be the resulting $\mathcal{O}_{\mathrm{T}_\mathrm{F} \times \mathrm{F}}$-algebra as in Definition 2.2.11, and let $\mathrm{M}_\Gamma^0 = \mathrm{M}_\Gamma \otimes_{\mathcal{O}_\mathrm{F}} \pi_*(k)$, where the map $\mathcal{O}_\mathrm{F} \to \pi_*(k)$ is the augmentation. There is a grading on V_0 so that there is an abstract isomorphism $\underline{\widetilde{\mathrm{T}}}_\theta \cong \mathrm{M}_\Gamma^0$ of graded $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-modules. Indeed, both are free $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-modules of rank $2|\check{\Phi}|$, so they are both abstractly isomorphic to $\prod_{\check{\Phi} \times \{0,\infty\}} \mathcal{O}_{\mathrm{T}_\mathrm{F}}$, so V_0 can be chosen so that the grading on $\prod_{\check{\Phi} \times \{0,\infty\}} \mathcal{O}_{\mathrm{T}_\mathrm{F}}$ is given by shifting the summand corresponding to $(\check{\alpha}, \epsilon) \in \check{\Phi} \times \{0, \infty\}$ by $(2\rho, \theta + \alpha) + 2\delta_\epsilon$. Here, $\delta_0 = 0$ and $\delta_\infty = 1$. It follows from Construction 2.2.29 that there is a $\mathcal{W}$-module structure on $\underline{\widetilde{\mathrm{T}}}_\theta$ such that upon inverting $e(\Phi_\theta^+)$ and using the isomorphisms $\underline{\widetilde{\mathrm{T}}}_\theta[e(\Phi_\theta^+)^{-1}] \cong \mathrm{M}_\Gamma^0[e(\Phi_\theta^+)^{-1}] \cong \mathrm{Map}(\check{\Phi} \times \{0, \infty\}, \mathcal{O}_{\mathrm{T}_\mathrm{F}}[e(\Phi_\theta^+)^{-1}])$, the $\mathcal{O}_\mathcal{W}[e(\Phi_\theta^+)^{-1}] \cong \mathrm{Map}(\mathrm{W}, \mathcal{O}_{\mathrm{T}_\mathrm{F}}[e(\Phi_\theta^+)^{-1}])$-coaction on $\underline{\widetilde{\mathrm{T}}}_\theta[e(\Phi_\theta^+)^{-1}]$ identifies with the natural W-action on $\mathrm{Map}(\check{\Phi} \times \{0, \infty\}, \mathcal{O}_{\mathrm{T}_\mathrm{F}}[e(\Phi_\theta^+)^{-1}])$.

Proposition 5.1.9. *Suppose that the bad primes (see Table 1) of* G *are inverted in* $\pi_0(k)$. *The* $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$*-comodule* $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathcal{G}\mathrm{r}_\mathrm{G}; \widetilde{\mathrm{IC}}_\theta)$ *is* $\mathcal{W}$*-equivariantly isomorphic to the* $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$*-module* $\mathrm{Res}^{\check{\mathrm{G}}}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}(\underline{\widetilde{\mathrm{T}}}_\theta)$ *under the isomorphism* $\mathrm{Spec}(\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)) \cong \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ *of Theorem 4.2.12.*

Proof. As usual, we may assume k is the localization of MU at the bad primes over G. Let $\underline{\mathrm{V}}^{\mathrm{geom}}$ denote $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathcal{G}\mathrm{r}_\mathrm{G}; \widetilde{\mathrm{IC}}_\theta) \cong \mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k)$. Although $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})$ is not a T-equivariant k-GKM space, it is a $\mathrm{T} \times \mathbf{C}^\times$-equivariant k-GKM space where $\mathbf{C}^\times$ acts on the $\mathbf{P}^1$-fibers by scaling. A direct computation (by an argument similar to Lemma 5.1.7 and the calculations below) shows that there is an isomorphism $\mathrm{H}_{\mathrm{T} \times \mathbf{C}^\times, c}^{-*}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k) \cong \mathrm{M}_\Gamma$ of $\mathcal{O}_{\mathrm{T}_\mathrm{F} \times \mathrm{F}}$-algebras, where M_Γ is as in Construction 5.1.8. This, along with Example 2.2.30, implies that there is a $\mathcal{W}$-equivariant isomorphism $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k) \cong \mathrm{M}_\Gamma^0$ of $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-algebras; the latter is $\mathcal{W}$-equivariantly isomorphic to $\underline{\widetilde{\mathrm{T}}}_\theta$ as a $\mathcal{O}_{\mathrm{T}_\mathrm{F}}$-module (by definition).

We need to check that the isomorphism $\mathrm{H}_{\mathrm{T},c}^{-*}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}); k) \cong \underline{\widetilde{\mathrm{T}}}_\theta$ is one of $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$-modules. Arguing as in Proposition 5.1.4, it suffices to check that there is a

$\check{\mathrm{T}}$-equivariant isomorphism $\widetilde{\underline{\mathrm{T}}}_\theta[e(\Phi)^{-1}] \cong \underline{\mathrm{V}}^{\mathrm{geom}}[e(\Phi)^{-1}]$, and that for each root α, the submodules $\widetilde{\underline{\mathrm{T}}}_\theta[e(\Phi-\{\alpha\})^{-1}]$ and $\underline{\mathrm{V}}^{\mathrm{geom}}[e(\Phi-\{\alpha\})^{-1}]$ are $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}[e(\Phi-\{\alpha\})^{-1}]$-equivariantly isomorphic (compatibly with $\mathcal{W}$-actions). The first part was already checked in Lemma 5.1.7. As in the proof of Proposition 5.1.4, fix a root α, and let L_α denote the Levi subgroup of G generated by the root spaces $\mathrm{U}_{\pm\alpha}$, so that L_α has semisimple rank 1. Then $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}[e(\Phi-\{\alpha\})^{-1}] \cong \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{L}}_\alpha,\mathrm{F}}}[e(\Phi-\{\alpha\})^{-1}]$ as Hopf subalgebras of $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}[e(\Phi)^{-1}] \cong \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{T}},\mathrm{F}}}[e(\Phi)^{-1}]$. (See, e.g., Corollary 4.2.9; alternatively, this follows from the isomorphism $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \cong \mathcal{A}_{\mathrm{G}}$ from Theorem 4.2.12 where $\mathcal{A}_{\mathrm{G}}$ is as in Definition 3.1.10.) We are reduced in this way to checking that there is an $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{L}}_\alpha,\mathrm{F}}}[e(\Phi-\{\alpha\})^{-1}]$-equivariant isomorphism $\widetilde{\underline{\mathrm{T}}}_\theta[e(\Phi-\{\alpha\})^{-1}] \cong \underline{\mathrm{V}}^{\mathrm{geom}}[e(\Phi-\{\alpha\})^{-1}]$, compatible with the $\check{\mathrm{T}}$-equivariant isomorphism $\widetilde{\underline{\mathrm{T}}}_\theta[e(\Phi)^{-1}] \cong \underline{\mathrm{V}}^{\mathrm{geom}}[e(\Phi)^{-1}]$ of Lemma 5.1.7.

If T_α denotes the kernel of α, then Proposition 2.2.4 gives an isomorphism $\underline{\mathrm{V}}^{\mathrm{geom}}[e(\Phi-\{\alpha\})^{-1}] \cong \mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}; k)[e(\Phi-\{\alpha\})^{-1}]$ of subalgebras in $\underline{\mathrm{V}}^{\mathrm{geom}}[e(\Phi)^{-1}] \cong \mathrm{Map}(\check{\Phi} \times \{0,\infty\}, \mathcal{O}_{\mathrm{T}_\mathrm{F}}[e(\Phi)^{-1}])$. Using this, we are reduced to showing that for each positive root α, there is a $\widetilde{\mathrm{J}}_{\check{\mathrm{L}}_\alpha,\mathrm{F}}$-equivariant isomorphism $\mathrm{Res}^{\check{\mathrm{G}}}_{\widetilde{\mathrm{J}}_{\check{\mathrm{L}}_\alpha,\mathrm{F}}}(\widetilde{\underline{\mathrm{T}}}_\theta) \cong \mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}; k)$, compatible with the $\check{\mathrm{T}}$-equivariant isomorphism $\widetilde{\underline{\mathrm{T}}}_\theta[e(\Phi)^{-1}] \cong \underline{\mathrm{V}}^{\mathrm{geom}}[e(\Phi)^{-1}]$. Here, $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}; k)$ is viewed as a $\mathrm{H}^{\mathrm{T},\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{L}_\alpha}; k) \cong \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{L}}_\alpha,\mathrm{F}}}$-comodule via the map $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha} \to \mathcal{G}\mathrm{r}_{\mathrm{L}_\alpha}$ induced by taking T_α-fixed points of the map $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}) \to \mathcal{G}\mathrm{r}_{\mathrm{G}}$.

We now compute $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}$. Recall that $(\mathrm{G}/\mathrm{P}_\theta)^{\mathrm{T}_\alpha} \cong \coprod_{w\in\langle s_\alpha\rangle\backslash \mathrm{W}/\mathrm{W}_\theta} \mathrm{L}_\alpha/(\mathrm{L}_\alpha \cap \mathrm{Ad}_w(\mathrm{P}_\theta))$ is a disjoint union of isolated fixed points (when $w^{-1}(\alpha)$ is a root of the Levi quotient of P_θ) and $\mathbf{P}^1$.

(1) The fiber of $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}$ over an isolated fixed point in $(\mathrm{G}/\mathrm{P}_\theta)^{\mathrm{T}_\alpha}$ (indexed by $w \in \langle s_\alpha\rangle\backslash \mathrm{W}/\mathrm{W}_\theta$) is isomorphic to $\mathbf{P}((\mathcal{L}_\theta)_w \oplus \mathcal{O})^{\mathrm{T}_\alpha}$, where T_α acts on $\mathbf{P}((\mathcal{L}_\theta)_w \oplus \mathcal{O}) \cong \mathbf{P}^1$ through the inclusion $\mathrm{T}_\alpha \hookrightarrow \mathrm{T}$ and the (w-shifted) character $w\theta : \mathrm{T} \to \mathbf{C}^\times$. Since w indexes an isolated fixed point, $(w^{-1}(\alpha), \theta) = 0$, so that $w\theta \neq \pm\alpha$, and therefore $w\theta|_{\mathrm{T}_\alpha}$ is nontrivial. The fixed locus $\mathbf{P}((\mathcal{L}_\theta)_w \oplus \mathcal{O})^{\mathrm{T}_\alpha}$ is therefore $\{0_w, \infty_w\}$.
(2) The fiber of $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}$ over a component $\mathbf{P}^1 \subseteq (\mathrm{G}/\mathrm{P}_\theta)^{\mathrm{T}_\alpha}$ (indexed by $w \in \langle s_\alpha\rangle\backslash \mathrm{W}/\mathrm{W}_\theta$) is isomorphic to $\mathbf{P}((\mathcal{L}_\theta)_w \oplus \mathcal{O})^{\mathrm{T}_\alpha}$. Since w indexes a $\mathbf{P}^1$-component, $(w^{-1}(\alpha), \theta) \neq 0$. Suppose that $w\theta \neq \pm\alpha$. Then again $w\theta|_{\mathrm{T}_\alpha}$ is nontrivial, and the fixed locus $\mathbf{P}((\mathcal{L}_\theta)_w \oplus \mathcal{O})^{\mathrm{T}_\alpha}$ is $\{0_w, \infty_w\} \times \mathbf{P}^1 \cong \mathbf{P}^1 \sqcup \mathbf{P}^1$.
(3) The fiber of $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}$ over a component $\mathbf{P}^1 \subseteq (\mathrm{G}/\mathrm{P}_\theta)^{\mathrm{T}_\alpha}$ (indexed by $w \in \langle s_\alpha\rangle\backslash \mathrm{W}/\mathrm{W}_\theta$) is isomorphic to $\mathbf{P}((\mathcal{L}_\theta)_w \oplus \mathcal{O})^{\mathrm{T}_\alpha}$. Since w indexes a $\mathbf{P}^1$-component, $(w^{-1}(\alpha), \theta) \neq 0$. Suppose that $w\theta = \pm\alpha$. Then $w\theta|_{\mathrm{T}_\alpha}$ is trivial, and T_α fixes all of $\mathbf{P}((\mathcal{L}_\theta)_w \oplus \mathcal{O})$. It follows from this that $\mathbf{P}((\mathcal{L}_\theta)_w \oplus \mathcal{O})^{\mathrm{T}_\alpha}$ is a $\mathbf{P}^1$-bundle over the base $\mathbf{P}^1$ component. Because $\langle w\theta, \alpha\rangle = \pm 2$, the pullback $\mathcal{L}_\theta|_{\mathbf{P}^1}$ is $\mathcal{O}(2)$, so that $\mathbf{P}((\mathcal{L}_\theta)_w \oplus \mathcal{O})^{\mathrm{T}_\alpha}$ is the Hirzebruch surface $\mathbf{P}(\mathcal{O}(2) \oplus \mathcal{O})$.

It follows that the fixed locus $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}$ is a disjoint union of isolated points (which come in pairs), projective lines $\mathbf{P}^1$ (which come in pairs), and Hirzebruch surfaces $\mathbf{P}(\mathcal{O}(2) \oplus \mathcal{O})$. In fact, the Hirzebruch surface appears with multiplicity 1 in this disjoint union. These components are indexed by $w \in \langle s_\alpha\rangle\backslash \mathrm{W}/\mathrm{W}_\theta \cong \langle s_\alpha\rangle\backslash\check{\Phi}$, and correspond to the cases $(\alpha, w\theta) = 0$, $(\alpha, w\theta) \neq 0$ and $w\theta \neq \pm\alpha$ (so that

$(\alpha, w\theta) = \pm 1)$, and $w\theta = \pm\alpha$, respectively. In particular, $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}; k)$ is isomorphic to a direct sum of copies of $\mathrm{H}^{-*}_{\mathrm{T},c}(\{0,\infty\}; k) \cong \mathcal{O}^{\oplus 2}_{\mathrm{T_F}}$ and $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}^1 \sqcup \mathbf{P}^1; k) \cong \mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}^1; k)^{\oplus 2}$, and a single copy of $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{O}(2) \oplus \mathcal{O}); k)$. Moreover, this direct sum decomposition is one of $\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}r_{\mathrm{L}_\alpha}; k)$-comodules.

Recall that we need to show that for each positive root α, there is a $\widetilde{\mathrm{J}}_{\check{\mathrm{L}}_\alpha,\mathrm{F}}$-equivariant isomorphism $\mathrm{Res}^{\check{\mathrm{G}}}_{\widetilde{\mathrm{J}}_{\check{\mathrm{L}}_\alpha,\mathrm{F}}}(\underline{\widetilde{\mathrm{T}}}_\theta) \cong \mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}; k)$ compatible with the $\check{\mathrm{T}}$-equivariant isomorphism $\underline{\widetilde{\mathrm{T}}}_\theta[e(\alpha)^{-1}] \cong \underline{\mathrm{V}}^{\mathrm{geom}}[e(\alpha)^{-1}]$. Since $\underline{\widetilde{\mathrm{T}}}_\theta$ is a tilting $\check{\mathrm{G}}$-module, $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{L}}_\alpha}(\underline{\widetilde{\mathrm{T}}}_\theta)$ is also a tilting $\check{\mathrm{L}}_\alpha$-module (see [Don85, Mat90]). It follows (see [Jan03, Section E]) that $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{L}}_\alpha}(\underline{\widetilde{\mathrm{T}}}_\theta)$ is a direct sum of indecomposable tilting modules for the semisimple rank 1 group $\check{\mathrm{L}}_\alpha$. To describe it, we may replace $\check{\mathrm{L}}_\alpha$ by its derived subgroup, which is isomorphic to SL_2 or PGL_2. Since restriction of representations along $\mathrm{SL}_2 \to \mathrm{PGL}_2$ is fully faithful, it suffices to describe $\mathrm{Res}^{\check{\mathrm{G}}}_{\mathrm{SL}_2}(\underline{\widetilde{\mathrm{T}}}_\theta) \cong \mathrm{V}_0 \oplus \mathrm{Res}^{\check{\mathrm{G}}}_{\mathrm{SL}_2}(\mathrm{T}_\theta)$. Since T_θ is an extension of $\check{\mathfrak{g}}^{\mathrm{sc}}$ by a trivial representation, and the lengths of root strings are at most 2 in the simply-laced case, $\mathrm{Res}^{\check{\mathrm{G}}}_{\mathrm{SL}_2}(\mathrm{T}_\theta)$ is isomorphic to a direct sum of trivial representations (containing those root spaces $\check{\mathfrak{g}}^{\mathrm{sc}}_\beta$ with $(\beta,\alpha) = 0$), the tilting representation $\mathrm{V}^{\oplus 2}_{\varpi_1}$ (containing those root spaces $\check{\mathfrak{g}}^{\mathrm{sc}}_\beta$ with $(\beta,\alpha) = \pm 1$), and a single copy of the tilting representation $\mathrm{V}^{\otimes 2}_{\varpi_1}$ (containing the root spaces $\check{\mathfrak{g}}^{\mathrm{sc}}_{\pm\alpha}$).

It is easy to check that the number of summands of each type (trivial, V_{ϖ_1}, and $\mathrm{V}^{\otimes 2}_{\varpi_1}$; and $\mathrm{H}^{-*}_{\mathrm{T},c}(*; k) \cong \mathcal{O}_{\mathrm{T_F}}$, $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}^1; k)$, and $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{O}(2) \oplus \mathcal{O}); k))$ in $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{L}}_\alpha}(\underline{\widetilde{\mathrm{T}}}_\theta)$ and $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}; k)$ agree, so it suffices to show that there are isomorphisms $\mathcal{O}_{\mathrm{T_F}} \cong \underline{\mathrm{triv}}$, $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}^1; k) \cong \mathrm{Res}^{\mathrm{SL}_2}_{\widetilde{\mathrm{J}}_{\mathrm{SL}_2,\mathrm{F}}}(\underline{\mathrm{V}}_{\varpi_1})$, and $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{O}(2) \oplus \mathcal{O}); k) \cong \mathrm{Res}^{\mathrm{SL}_2}_{\widetilde{\mathrm{J}}_{\mathrm{SL}_2,\mathrm{F}}}(\underline{\mathrm{V}}^{\otimes 2}_{\varpi_1})$ of $\mathcal{W}$-equivariant $\widetilde{\mathrm{J}}_{\mathrm{SL}_2,\mathrm{F}}$-representations. Here, triv denotes the trivial 1-dimensional representation of $\check{\mathrm{G}}$. The isomorphism $\mathcal{O}_{\mathrm{T_F}} \cong \underline{\mathrm{triv}}$ of $\widetilde{\mathrm{J}}_{\mathrm{SL}_2,\mathrm{F}}$-representations is clear. We have already established in Proposition 5.1.4 that $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}^1; k) \cong \mathrm{Res}^{\mathrm{SL}_2}_{\widetilde{\mathrm{J}}_{\mathrm{SL}_2,\mathrm{F}}}(\underline{\mathrm{V}}_{\varpi_1})$ as $\widetilde{\mathrm{J}}_{\mathrm{SL}_2,\mathrm{F}}$-representations. Finally, note that there is an isomorphism $\mathbf{P}(\mathcal{O}(2) \oplus \mathcal{O}) \cong \mathbf{P}^1 \widetilde{\times} \mathbf{P}^1$ such that the map $\mathbf{P}(\mathcal{O}(2) \oplus \mathcal{O}) \to \mathcal{G}r^{\leq\theta}_{\mathrm{PGL}_2} \hookrightarrow \mathcal{G}r_{\mathrm{PGL}_2}$ is the convolution map $\mathbf{P}^1 \widetilde{\times} \mathbf{P}^1 \to \mathcal{G}r_{\mathrm{PGL}_2}$. Therefore, Proposition 5.1.4 implies that $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathbf{P}(\mathcal{O}(2) \oplus \mathcal{O}); k) \cong \mathrm{Res}^{\mathrm{SL}_2}_{\widetilde{\mathrm{J}}_{\mathrm{SL}_2,\mathrm{F}}}(\underline{\mathrm{V}}^{\otimes 2}_{\varpi_1})$, as desired. □

Notation 5.1.10. Say that a dominant coweight $\lambda \in \mathbb{X}^+_*$ is *(quasi-)minuscule* if it is 0, minuscule, or quasi-minuscule. (This definition makes sense even if G is not simply-laced; but if G is assumed to be simply-laced, then the quasi-minuscule coweight is the highest root of $\check{\mathrm{G}}^{\mathrm{sc}}$.) Whenever we discuss sequences $\mu_\bullet = (\mu_1, \cdots, \mu_n)$ of dominant (quasi-)minuscule weights of $\check{\mathrm{G}}^{\mathrm{sc}}$, we will always assume that at most one of the μ_i is 0. If $\mu_\bullet = (\mu_1, \cdots, \mu_n)$ is a sequence of dominant (quasi-)minuscule weights of $\check{\mathrm{G}}^{\mathrm{sc}}$, then (extending Definition 5.1.2) let $\widetilde{\mathrm{IC}}_{\mu_\bullet}$ denote the convolution $\widetilde{\mathrm{IC}}_{\mu_1} \star \cdots \star \widetilde{\mathrm{IC}}_{\mu_n}$, where $\widetilde{\mathrm{IC}}_\mu = \mathrm{IC}_\mu$ if μ is minuscule. Similarly, if μ is (quasi-)minuscule, let T_μ denote the corresponding tilting $\check{\mathrm{G}}$-module (defined over $\mathbf{Z}$), and let $\widetilde{\mathrm{T}}_\mu$ denote T_μ if μ is minuscule, and $\mathrm{T}_\mu \oplus \mathrm{V}_0$ if μ is quasi-minuscule (i.e., the highest root of $\check{\mathrm{G}}^{\mathrm{sc}}$) where V_0 is as in Construction 5.1.8. If $\mu_\bullet = (\mu_1, \cdots, \mu_n)$ is a sequence of dominant (quasi-)minuscule weights of $\check{\mathrm{G}}^{\mathrm{sc}}$, let $\mathrm{T}_{\mu_\bullet} \cong \bigotimes_{i=1}^n \mathrm{T}_{\mu_i}$. As usual, let $\mathrm{T}_{\mu_\bullet}(2\rho, |\mu_\bullet|)$ denote the shift of $\mathrm{T}_{\mu_\bullet}$ in weight by

$(2\rho, |\mu_\bullet|)$, and write $\underline{\mathrm{T}}_{\mu_\bullet}$ to denote $\mathrm{T}_{\mu_\bullet}(2\rho, |\mu_\bullet|) \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T_F}}$, so that it is a free graded $\mathcal{O}_{\mathrm{T_F}}$-module. Similarly for $\widetilde{\mathrm{T}}_{\mu_\bullet}$ and $\underline{\widetilde{\mathrm{T}}}_{\mu_\bullet}$.

Convolving Proposition 5.1.9 with Proposition 5.1.4, and using Corollary 3.1.12, it follows that:

Corollary 5.1.11. *Suppose that the bad primes (see Table 1) of* G *are inverted in* $\pi_0(k)$*, and let* $\mu_\bullet$ *be a sequence of dominant (quasi-)minuscule weights of* $\check{\mathrm{G}}^{\mathrm{sc}}$*. Then the* $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$*-comodule* $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{G}\mathrm{r}_\mathrm{G}; \widetilde{\mathrm{IC}}_{\mu_\bullet})$ *is* $\mathcal{W}$*-equivariantly isomorphic to the* $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$*-module* $\mathrm{Res}^{\check{\mathrm{G}}}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}(\underline{\widetilde{\mathrm{T}}_{\mu_\bullet}})$ *under the isomorphism* $\mathrm{Spec}(\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)) \cong \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ *of Theorem 4.2.12.*

Proposition 5.1.12. *Let* $\mu_\bullet$ *be a sequence of dominant (quasi-)minuscule coweights of* G^{ad}*. Assume Conjecture 3.1.14. Then* $\widetilde{\mathrm{IC}}_{\mu_\bullet}$ *is even (in the sense of Definition 2.3.1).*

Proof. Let $\lambda \in \mathbb{X}_*$, and let $j_\lambda : \mathrm{I}t^\lambda \hookrightarrow \mathcal{G}\mathrm{r}_\mathrm{G}$ denote the inclusion of the corresponding I-orbit. For $? \in \{!, *\}$, an object $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ is ?-even if $j_\lambda^?(\mathcal{F})$ is perfect even in $\mathrm{Shv}_\mathrm{I}(\mathrm{I}t^\lambda; k)$ (in the sense of Definition 2.3.1) for each $\lambda \in \mathbb{X}_*$. As mentioned in Recollection 3.1.1, $\mathrm{I}t^\lambda$ is a vector space on which T acts linearly. In particular, $\mathrm{Shv}_\mathrm{I}(\mathrm{I}t^\lambda; k) \simeq \mathrm{Mod}_k^{h\mathrm{T}}$ (via restriction to the basepoint $t^\lambda \in \mathrm{I}t^\lambda$). It follows that if $i_\lambda : t^\lambda \hookrightarrow \mathcal{G}\mathrm{r}_\mathrm{G}$ denotes the inclusion of the T-fixed point, then $\mathcal{F}$ is ?-even if $i_\lambda^?(\mathcal{F})$ is perfect even in $\mathrm{Mod}_k^{h\mathrm{T}}$ for each $\lambda \in \mathbb{X}_*$.

By Proposition 3.1.16, it suffices to show that $\widetilde{\mathrm{IC}}_\theta$ is even and that IC_μ is even for each minuscule μ. Let $? \in \{!, *\}$. For IC_μ with minuscule μ, it suffices to show that if $\lambda \in \mathbb{X}_*$ is in the W-orbit of μ and $i_\lambda : t^\lambda \hookrightarrow \mathcal{G}\mathrm{r}_\mathrm{G}^\mu \cong \mathrm{G}/\mathrm{P}_\mu$ is the inclusion of the corresponding T-fixed point, then $i_\lambda^?(\mathrm{IC}_\mu)$ is perfect even. This is clear for $? = *$. Note that $i_\lambda^!(\mathrm{IC}_\mu) \cong \Sigma^\mathrm{V} k$ where $\mathrm{V} = \mathrm{T}_{t^\lambda}(\mathrm{G}/\mathrm{P}_\mu)$ is the tangent space at t^λ. This is perfect even since V is a complex T-representation.

To show that $\widetilde{\mathrm{IC}}_\theta$ is even, it suffices to show that if $\lambda \in \mathbb{X}_*$ is zero or a coroot of G, and $i_\lambda : t^\lambda \hookrightarrow \mathcal{G}\mathrm{r}_\mathrm{G}^{\leq\theta}$ is the inclusion of the corresponding T-fixed point, then $i_\lambda^?(\widetilde{\mathrm{IC}}_\theta)$ is perfect even. Since the map $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}) \to \mathcal{G}\mathrm{r}_\mathrm{G}^{\leq\theta}$ is an isomorphism over $\mathcal{G}\mathrm{r}_\mathrm{G}^\theta$, the same argument as above for minuscule μ implies that $i_\lambda^?(\widetilde{\mathrm{IC}}_\theta)$ is perfect even if λ is in the W-orbit of θ. For the case $\lambda = 0$, note that the fiber of the map $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O}) \to \mathcal{G}\mathrm{r}_\mathrm{G}^{\leq\theta}$ over t^0 is isomorphic to $\mathrm{G}/\mathrm{P}_\theta$. If we continue to write i_λ to denote the inclusion $\mathrm{G}/\mathrm{P}_\theta \hookrightarrow \mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})$, then $i_\lambda^?(\widetilde{\mathrm{IC}}_\theta) \cong \mathrm{R}\Gamma_\mathrm{T}(\mathrm{G}/\mathrm{P}_\theta; i_\lambda^?(\underline{k}_{\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})}))$. Let $\mathbf{C}^{\ell(w)}$ denote the Bruhat cell in $\mathrm{G}/\mathrm{P}_\theta$ (corresponding to $w \in \mathrm{W}/\mathrm{W}_\theta$), let $\dot{w}$ denote the T-fixed point to which $\mathbf{C}^{\ell(w)}$ contracts, and let $i_{\lambda,w}$ denote the inclusion $\mathbf{C}^{\ell(w)} \hookrightarrow \mathrm{G}/\mathrm{P}_\theta \hookrightarrow \mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})$. By recollement, it suffices to show that $i_{\lambda,w}^?(\underline{k}_{\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})})$ is perfect even. This is clear for $? = *$. Note that $i_{\lambda,w}^!(\underline{k}_{\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})}) \cong \Sigma^\mathrm{V} k$ where $\mathrm{V} = \mathrm{T}_{\dot{w}}\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})$ is the tangent space at $\dot{w}$. This is perfect even since V is a complex T-representation. □

Definition 5.1.13. Let λ be a dominant coweight of G, and let $\mathcal{P}_{\leq\lambda} = \bigoplus_{|\mu_\bullet| \leq \lambda} \widetilde{\mathrm{IC}}_{\mu_\bullet}$, where the direct sum ranges over sequences $\mu_\bullet$ of dominant (quasi-)minuscule coweights with $|\mu_\bullet| \leq \lambda$. Let $\mathcal{P} \in \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ denote $\mathrm{colim}_\lambda \mathcal{P}_{\leq\lambda}$, and define $\langle\mathcal{P}\rangle$, $\langle\mathcal{P}\rangle_{\mathrm{fil}}$, and $\langle\mathcal{P}\rangle_{\mathrm{gr}}$ as in Construction 2.4.11. These categories will be denoted $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$, $\mathrm{Shv}^{\text{q-min,fil}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$, and $\mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$. Note that

$\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ is a monoidal subcategory of $\mathrm{Shv}^{\text{toral}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ under convolution (recall the notation "toral" from Remark 2.4.12). There is a cellularization functor $\mathrm{Shv}^{\text{toral}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k) \to \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ which is right adjoint to the inclusion $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k) \hookrightarrow \mathrm{Shv}^{\text{toral}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$. The cellularization functor will be denoted $\mathcal{F} \mapsto \mathcal{F}^{\text{q-min}}$; since it is a right adjoint to a monoidal functor, it is automatically lax monoidal.

Let $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)^{!\text{-ev}}$ denote the full subcategory of $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ spanned by those sheaves whose restriction to $\mathrm{Shv}_{\mathrm{I}}(\mathcal{G}r_{\mathrm{G}}; k)$ is !-even with respect to the I-orbit stratification on $\mathcal{G}r_{\mathrm{G}}$, i.e., the pullback $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k) \times_{\mathrm{Shv}^{\text{toral}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)} \mathrm{Shv}^{\text{toral}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)^{!\text{-ev}}$. If Conjecture 3.1.14 is true, then Proposition 3.1.16 implies that $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)^{!\text{-ev}}$ is a monoidal subcategory of $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ under convolution.

Definition 5.1.14. If λ is a dominant coweight of G, let $\mathrm{Perf}^{\text{q-min,gr}}_{\leq\lambda}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ denote the full subcategory of $\mathrm{Perf}^{\text{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ generated by $\mathrm{T}_{\leq\lambda} = \bigoplus_{|\mu_\bullet|\leq\lambda} \mathrm{T}_{\mu_\bullet}(2\rho, |\mu_\bullet|)$, where the (finite!) direct sum ranges over sequences $\mu_\bullet$ of dominant (quasi-)minuscule coweights of G_{ad} with $|\mu_\bullet| \leq \lambda$ and at most one copy of $0 \in \mu_\bullet$. (Note that $\mathrm{T}_{\leq\lambda}$ may be replaced by $\widetilde{\mathrm{T}}_{\leq\lambda} = \bigoplus_{|\mu_\bullet|\leq\lambda} \widetilde{\mathrm{T}}_{\mu_\bullet}(2\rho, |\mu_\bullet|)$, and the resulting category $\mathrm{Perf}^{\text{q-min,gr}}_{\leq\lambda}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ would be the same.) In other words, $\mathrm{Perf}^{\text{q-min,gr}}_{\leq\lambda}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ is the category of bounded complexes of the objects $\mathrm{T}_{\mu_\bullet}(2\rho, |\mu_\bullet|)$ for sequences $\mu_\bullet$ of dominant (quasi-)minuscule coweights of G with $|\mu_\bullet| \leq \lambda$. Let $\mathrm{Perf}^{\text{q-min,gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}}) = \mathrm{colim}_\lambda \mathrm{Perf}^{\text{q-min,gr}}_{\leq\lambda}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$.

Let $\check{\mathrm{X}}$ be a graded affine scheme over $\pi_*(k)$ equipped with an ${}^{2\rho}\check{\mathrm{G}}$-action. Let $\mathrm{IndPerf}^{\text{q-min,gr}}_{\leq\lambda}(\check{\mathrm{X}}/{}^{2\rho}\check{\mathrm{G}})$ denote the full subcategory of $\mathrm{IndPerf}^{\text{gr}}(\check{\mathrm{X}}/{}^{2\rho}\check{\mathrm{G}})$ generated by the essential image of $\mathrm{Perf}^{\text{q-min,gr}}_{\leq\lambda}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ under pullback along $\check{\mathrm{X}}/{}^{2\rho}\check{\mathrm{G}} \to \mathrm{B}^{2\rho}\check{\mathrm{G}}$. Similarly, let $\mathrm{IndPerf}^{\text{q-min,gr}}(\check{\mathrm{X}}/{}^{2\rho}\check{\mathrm{G}}) = \mathrm{colim}_\lambda \mathrm{IndPerf}^{\text{q-min,gr}}_{\leq\lambda}(\check{\mathrm{X}}/{}^{2\rho}\check{\mathrm{G}})$. We will also write $\mathrm{IndPerf}^{\text{all,gr}}_{\leq\lambda}(\check{\mathrm{X}}/{}^{2\rho}\check{\mathrm{G}})$ denote the full subcategory of $\mathrm{QCoh}^{\text{gr}}(\check{\mathrm{X}}/{}^{2\rho}\check{\mathrm{G}})$ generated by the essential image of $\mathrm{Perf}^{\text{gr}}_{\leq\lambda}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ under pullback along $\check{\mathrm{X}}/{}^{2\rho}\check{\mathrm{G}} \to \mathrm{B}^{2\rho}\check{\mathrm{G}}$. Here, the direct sum ranges over all dominant coweights of G_{ad} with $\mu \leq \lambda$. Let $\mathrm{IndPerf}^{\text{all,gr}}(\check{\mathrm{X}}/{}^{2\rho}\check{\mathrm{G}}) = \mathrm{colim}_\lambda \mathrm{IndPerf}^{\text{all,gr}}_{\leq\lambda}(\check{\mathrm{X}}/{}^{2\rho}\check{\mathrm{G}})$.

Note that Definition 5.1.14 makes sense even if $\check{\mathrm{G}}$ is not simply-laced.

We now review a cohomological vanishing statement which will be useful below.

Proposition 5.1.15. *Suppose the bad primes of* G *are invertible in* R*. Then for any* $\check{\mathrm{G}}$*-module* V *with a good filtration, the derived global sections* $\mathrm{R}\Gamma({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}; \mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \otimes_{\mathrm{R}} \mathrm{V})$ *is concentrated in degree* 0*, where it is isomorphic to* $(\mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \otimes_{\mathrm{R}} \mathrm{V})^{{}^{2\rho}\check{\mathrm{G}}}$*. In particular, for any (finitely generated) tilting* $\check{\mathrm{G}}$*-module* V*,* $\mathrm{REnd}_{\mathrm{Perf}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \otimes_{\mathrm{R}} \mathrm{V})$ *is concentrated in degree zero, where it is isomorphic to* $\mathrm{End}_{\mathrm{Perf}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})^{\heartsuit}}(\mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \otimes_{\mathrm{R}} \mathrm{V})$*.*

Proof. The second claim follows from the first, because $\mathrm{REnd}_{\mathrm{Perf}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \otimes_{\mathrm{R}} \mathrm{V})$ is isomorphic to $\mathrm{R}\Gamma({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}; \mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \otimes_{\mathrm{R}} \mathrm{V} \otimes_{\mathrm{R}} \mathrm{V}^*)$. Since V is a tilting module, the same is true of V^*. In particular, $\mathrm{V} \otimes_{\mathrm{R}} \mathrm{V}^*$ is itself a tilting module (see [Don85, Mat90]). Let us now show that $\mathrm{R}\Gamma({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}; \mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \otimes_{\mathrm{R}} \mathrm{V})$ is concentrated in degree 0 if V has a good filtration. Since ${}^{2\rho}\mathrm{G}_{\mathrm{F}}$ is affine, $\mathrm{R}\Gamma({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}; \mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \otimes_{\mathrm{R}} \mathrm{V}) \cong \mathrm{R}\Gamma(\mathrm{B}^{2\rho}\check{\mathrm{G}}; \mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \otimes_{\mathrm{R}} \mathrm{V})$. The canonical degeneration from F to $\hat{\mathbf{G}}_a$ defines a complete exhaustive filtration on $\mathcal{O}_{\mathrm{G_F}}$ whose associated graded is isomorphic to $\mathcal{O}_{\mathrm{G}_{\hat{\mathbf{G}}_a}}$. Recall from Lemma 4.1.2 that $\mathrm{G}_{\hat{\mathbf{G}}_a} \cong \mathfrak{g}^\wedge_{\mathcal{N}}$. Therefore, there is a spectral sequence converging to $\mathrm{H}^*(\mathrm{B}^{2\rho}\check{\mathrm{G}}; \mathcal{O}_{^{2\rho}\mathrm{G_F}} \otimes_{\mathrm{R}} \mathrm{V})$ with E_1-page $\mathrm{H}^*(\mathrm{B}^{2\rho}\check{\mathrm{G}}; \mathcal{O}_{\mathfrak{g}^\wedge_{\mathcal{N}}} \otimes_{\mathrm{R}} \mathrm{V})$; so it suffices to show that the latter is concentrated in cohomological degree 0. Recall from [AJ84, Proposition 4.4] that $\mathcal{O}_{\mathfrak{g}} = \mathrm{Sym}(\mathfrak{g}^*)$ admits a good filtration as a G-representation (because the bad primes of G are invertible). The desired vanishing follows from the observation that $\mathrm{R}\Gamma(\mathrm{B}\check{\mathrm{G}}; \mathrm{V}')$ vanishes in nonzero degrees when V' admits a good filtration: this reduces to the case when V' is a dual Weyl module, in which case it follows from Kempf vanishing (which holds even over $\mathbf{Z}$; see the proof of [Jan03, Proposition II.4.5]). □

Theorem 5.1.16. *Suppose that the bad primes (see Table 1) of* G *are inverted in* $\pi_0(k)$.[12] *Assume Conjecture 3.1.14 for every sequence of dominant (quasi-)minuscule coweights of* G^{ad}. *Then there is a* $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$*-linear equivalence*

$$\mathcal{S} : \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \xrightarrow{\simeq} \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}}),$$

where the right-hand side is $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$*-linear via the map* ${}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}} \to \mathrm{T_F}/\mathcal{W}$ *of Theorem 4.3.6 and the isomorphism* $\mathrm{R}\Gamma(\mathrm{T_F}/\mathcal{W}; \mathcal{O}) \cong \mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$ *of Definition 2.2.28.*

The functor $\underline{\mathcal{T}}_\star : \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \to \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}})$ *sends* $\widetilde{\mathrm{IC}}_{\mu_\bullet} \mapsto \widetilde{\mathrm{T}}_{\mu_\bullet} \otimes_{\mathbf{Z}} \mathcal{O}_{^{2\rho}\mathrm{G_F}}$ *for every sequence* $\mu_\bullet$ *of dominant (quasi-)minuscule coweights of* G^{ad}. *In particular,* $\underline{\mathcal{T}}_\star$ *sends* IC_0 *(the skyscraper sheaf at the basepoint of* $\mathcal{G}\mathrm{r}_{\mathrm{G}}$*) to the structure sheaf* $\mathcal{O}_{^{2\rho}\mathrm{G_F}/^{2\rho}\check{\mathrm{G}}}$. *Moreover, the restriction of* $\underline{\mathcal{T}}_\star$ *to* $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{\text{!-ev}}$ *is monoidal for the convolution monoidal structure on* $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{\text{!-ev}}$ *and the pointwise tensor product on* $\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}})$. *There is also a commutative diagram*

$$\begin{tikzcd}
\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{\text{!-ev}} \arrow[rr, "\underline{\mathcal{T}}_\star"] \arrow[d, "\mathrm{gr}^\star_{\text{G-ev}}\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; -)"'] & & \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \arrow[d, "\mathcal{S}"] \\
\mathrm{Mod}^{\mathrm{gr}}_{\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})} & \mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{T_F}/\mathcal{W}) \arrow[l, "\mathrm{R}\Gamma(\mathrm{T_F}/\mathcal{W}; -)"] & \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}}) \arrow[l, "\kappa^*_{\mathrm{F}}"]
\end{tikzcd}$$

where $\kappa_{\mathrm{F}} : \mathrm{T_F}/\mathcal{W} \to {}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}}$ *is as in Proposition 4.3.2.*

Proof. Recall from Theorem 4.2.12 that there is an isomorphism of group schemes $\mathrm{Spec}(\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)) \cong \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ over $\pi_*(k^{h\mathrm{T}}) \cong \mathcal{O}_{\mathrm{T_F}}$. By Corollary 5.1.11, there is a $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$-equivariant isomorphism $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{P}_{\leq\lambda}) \cong \underline{\widetilde{\mathrm{T}}}_{\leq\lambda}$. For notational simplicity, let us write $\mathcal{X} = {}^{2\rho}\mathrm{G_F} \times_{\mathrm{T_F}/\mathcal{W}} \mathrm{T_F}$, $\mathcal{C} = \mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T_F}}\widetilde{\widetilde{\mathrm{J}}}_{\check{\mathrm{G}},\mathrm{F}})^\heartsuit$, and $\mathcal{A}_{\leq\lambda} = \mathrm{End}^\star_{\mathcal{C}}(\underline{\widetilde{\mathrm{T}}}_{\leq\lambda})$. Proposition 5.1.12 and Proposition 2.4.14 give an equivalence

$$\mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \simeq \mathrm{colim}_\lambda(\mathrm{RMod}^{\mathrm{gr}}_{\mathcal{A}_{\leq\lambda}})^{h\mathcal{W}},$$

and Corollary 5.1.11 implies that the functor $\underline{\mathcal{T}}_\star$ sends the !-even sheaf $\widetilde{\mathrm{IC}}_{\mu_\bullet}$ to $\mathrm{Hom}^\star_{\mathcal{C}}(\underline{\widetilde{\mathrm{T}}}_{\leq\lambda}, \underline{\widetilde{\mathrm{T}}}_{\mu_\bullet})$.

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[12] I expect that this is unnecessary; see Remark 5.1.1.

There is an isomorphism

(5.1.1)
$$\mathcal{A}_{\leq\lambda} \cong \mathrm{End}^{\star}_{\mathrm{Perf}({}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}})^{\mathrm{gr},\heartsuit}}(\mathcal{O}_{{}^{2\rho}\mathrm{B}_{\mathrm{F}}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}_{\leq\lambda}) \cong \mathrm{End}^{\star}_{\mathrm{Perf}({}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})^{\mathrm{gr},\heartsuit}}(\mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}_{\leq\lambda}),$$

where the first isomorphism is via Corollary 4.1.16, and the second isomorphism comes from the definitional isomorphism ${}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\mathrm{B} \cong {}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$. There is an isomorphism

$$\mathcal{A}_{\leq\lambda} \cong \mathrm{End}^{\star}_{\mathrm{Perf}(\mathcal{X}/{}^{2\rho}\check{\mathrm{G}})^{\mathrm{gr},\heartsuit}}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}_{\leq\lambda}) \cong \mathrm{REnd}^{\star}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/{}^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}_{\leq\lambda}),$$

where the first isomorphism is via (5.1.1) and Theorem 4.3.6, and the second isomorphism is via Proposition 5.1.15. The category of graded modules over this algebra is $\mathrm{IndPerf}^{\text{q-min,gr}}_{\leq\lambda}(\mathcal{X}/{}^{2\rho}\check{\mathrm{G}})$. It follows that $(\mathrm{RMod}^{\mathrm{gr}}_{\mathcal{A}_{\leq\lambda}})^{h\mathcal{W}}$ is equivalent to $\mathrm{IndPerf}^{\text{q-min,gr}}_{\leq\lambda}(\mathcal{X}/{}^{2\rho}\check{\mathrm{G}})^{h\mathcal{W}}$, which is in turn $\mathrm{IndPerf}^{\text{q-min,gr}}_{\leq\lambda}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$. Taking the colimit over λ gives the claimed equivalence.

In the same way, there is an isomorphism

$$\mathrm{Hom}^{\star}_{\mathcal{C}}(\underline{\widetilde{\mathrm{T}}}_{\leq\lambda}, \underline{\widetilde{\mathrm{T}}}_{\mu_\bullet}) \cong \mathrm{RHom}^{\star}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/{}^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}_{\leq\lambda}, \mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}_{\mu_\bullet}),$$

so that the equivalence $\mathcal{S}$ sends $\widetilde{\mathrm{IC}}_{\mu_\bullet}$ to $(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}_{\mu_\bullet})^{h\mathcal{W}} \cong \mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}_{\mu_\bullet}$. We now show that the functor $\underline{\mathcal{T}}_\star : \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{!\text{-ev}} \to \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$ is monoidal for the convolution monoidal structure on $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{!\text{-ev}}$ and the pointwise tensor on $\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$. By the construction of the equivalence $\mathcal{S}$ described above, it suffices to show that the functor $\underline{\mathcal{T}}_\star : \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{!\text{-ev}} \to \mathrm{IndPerf}(\mathrm{B}_{\mathrm{T}_{\mathrm{F}}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})^{\heartsuit} = \mathcal{C}$ is monoidal. Recall from Proposition 2.4.14 that $\underline{\mathcal{T}}_\star$ sends $\mathcal{F} \in \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{!\text{-ev}}$ with support contained in $\mathcal{G}\mathrm{r}^{\leq\lambda}_{\mathrm{G}}$ to $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F}) \in \mathcal{C}$ equipped with its $\mathcal{W}$-action from Definition 2.2.28. The desired monoidality therefore follows from (the T-equivariant analogue of) the isomorphism (3.1.1).

Note that it follows from Proposition 2.4.14, Corollary 4.1.16, and Theorem 4.3.6 that $\underline{\mathcal{T}}_\star : \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \to \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$ sends $\mathcal{F} \in \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^{!\text{-ev}}$ to the image under the localization $\mathrm{IndPerf}^{\mathrm{gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}) \to \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$ of the ${}^{2\rho}\check{\mathrm{G}}$-equivariant $\mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}}$-module $((\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}}\otimes_{\pi_*(k)}\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F}))^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}})^{h\mathcal{W}}$. (Here, we use the isomorphism $\mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \cong ((\mathcal{O}_{\mathrm{T}_{\mathrm{F}}\times{}^{2\rho}\check{\mathrm{G}}})^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}})^{h\mathcal{W}}$ coming from Remark 4.2.6 and Theorem 4.3.6 to equip $((\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}}\otimes_{\pi_*(k)}\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F}))^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}})^{h\mathcal{W}}$ with the structure of a $\mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}}$-module.) Now, $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F})^{h\mathcal{W}}$ is the global sections of the pullback of the ${}^{2\rho}\check{\mathrm{G}}$-equivariant $\mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}}$-module $((\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}}\otimes_{\pi_*(k)}\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F}))^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}})^{h\mathcal{W}}$ along the map $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ from Proposition 4.3.2. The desired commutative diagram follows since $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F})^{h\mathcal{W}} = \mathrm{gr}^{\star}_{\text{G-ev}}(\mathrm{R}\Gamma_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F}))$ by Corollary 2.2.31. □

As mentioned at the beginning of this section, Theorem 5.1.16 continues to hold verbatim if G is only assumed to be connected reductive with simply-laced derived subgroup.

Base-changing Theorem 5.1.16 along the augmentation $\mathrm{gr}^{\star}_{\text{G-ev}}(k^{h\mathrm{G}}) \to \pi_*(k)$ yields the following, whose special case $k = \mathbf{Q}$ was proved in [Gin95]:

Corollary 5.1.17. *There is a $\pi_*(k)$-linear equivalence*

$$\mathcal{S}_0 : \pi_*(k) \otimes_{\mathrm{gr}^{\star}_{\text{G-ev}}(k^{h\mathrm{G}})} \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \xrightarrow{\simeq} \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathcal{N}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}), \tag{5.1.2}$$

where ${}^{2\rho}\mathcal{N}_{\mathrm{F}}$ is defined as the derived fiber of the map ${}^{2\rho}\mathrm{G}_{\mathrm{F}} \to \mathrm{T}_{\mathrm{F}}/\mathcal{W}$ of Theorem 4.3.6 along the map $\mathrm{Spec}(\pi_*(k)) \xrightarrow{0} \mathrm{T}_{\mathrm{F}} \to \mathrm{T}_{\mathrm{F}}/\mathcal{W}$, and where the category $\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathcal{N}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$ is defined as in Definition 5.1.14. Let $\kappa_{\mathrm{F}} : \mathrm{Spec}(\pi_*(k)) \to {}^{2\rho}\mathcal{N}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ denote the restriction of $\kappa_{\mathrm{F}} : \mathrm{T}_{\mathrm{F}}/\mathcal{W} \to {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$.

The functor $\mathcal{T}_\star : \mathrm{Shv}^{\text{q-min}}_{(\mathrm{G}_{\mathcal{O}})}(\mathcal{G}r_{\mathrm{G}}; k) \to \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathcal{N}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$ sends $\widetilde{\mathrm{IC}}_{\mu_\bullet} \mapsto \widetilde{\mathrm{T}}_{\mu_\bullet} \otimes_{\mathbf{Z}} \mathcal{O}_{{}^{2\rho}\mathcal{N}_{\mathrm{F}}}$ for every sequence $\mu_\bullet$ of dominant (quasi-)minuscule coweights of G^{ad}. In particular, $\mathcal{T}_\star$ sends IC_0 (the skyscraper sheaf at the basepoint of $\mathcal{G}r_{\mathrm{G}}$) to the structure sheaf $\mathcal{O}_{{}^{2\rho}\mathcal{N}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}}$. There is also a commutative diagram

$$\begin{tikzcd}
\mathrm{Shv}^{\text{q-min}}_{(\mathrm{G}_{\mathcal{O}})}(\mathcal{G}r_{\mathrm{G}}; k)^{!\text{-ev}} \arrow[r, "\mathcal{T}_\star"] \arrow[d, "\mathrm{R}\Gamma(\mathcal{G}r_{\mathrm{G}};-)"'] & \pi_*(k) \otimes_{\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})} \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k) \arrow[d, "\mathcal{S}"] \\
\mathrm{Mod}^{\mathrm{gr}}_{\pi_*(k)} & \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathcal{N}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}). \arrow[l, "\kappa_{\mathrm{F}}^*"]
\end{tikzcd}$$

The left-hand side of (5.1.2) is the associated graded of a filtered lift of $\mathrm{Shv}^{\text{q-min}}_{(\mathrm{G}_{\mathcal{O}})}(\mathcal{G}r_{\mathrm{G}}; k)$, which may be viewed as a full subcategory of those sheaves in $\mathrm{Shv}(\mathcal{G}r_{\mathrm{G}}; k)$ which are constructible along the $\mathrm{G}_{\mathcal{O}}$-orbit stratification.

Remark 5.1.18. The (*a priori* derived) scheme ${}^{2\rho}\mathcal{N}_{\mathrm{F}}$ is in fact flat over $\pi_*(k)$ (and hence is underived), and it will be called the F-*nilpotent cone* of G. If $\mathrm{F} = \hat{\mathbf{G}}_a(2)$, then ${}^{2\rho}\mathcal{N}_{\mathrm{F}}$ is just the usual nilpotent cone ${}^{2\rho}\mathcal{N} \subseteq {}^{2\rho}\mathfrak{g}(2)^\wedge_{\mathcal{N}}$, and if $\mathrm{F} = \hat{\mathbf{G}}_m$, then ${}^{2\rho}\mathcal{N}_{\mathrm{F}}$ is the unipotent cone ${}^{2\rho}\mathcal{U} \subseteq {}^{2\rho}\mathrm{G}^\wedge_{\mathcal{U}}$. In general, Theorem 4.3.6 implies that ${}^{2\rho}\mathcal{N}_{\mathrm{F}}$ is the ${}^{2\rho}\check{\mathrm{G}}$-orbit of ${}^{2\rho}\mathrm{U}_{\mathrm{F}} \hookrightarrow {}^{2\rho}\mathrm{G}_{\mathrm{F}}$. It follows from Proposition 4.1.14 that the ${}^{2\rho}\check{\mathrm{B}}$-orbit of $\mathrm{Spec}(\pi_*(k)) \xrightarrow{0} \mathrm{T}_{\mathrm{F}} \xrightarrow{\kappa_{\mathrm{F}}} {}^{2\rho}\mathrm{B}_{\mathrm{F}}$ is precisely ${}^{2\rho}\mathrm{U}_{\mathrm{F}} \hookrightarrow {}^{2\rho}\mathrm{B}_{\mathrm{F}}$; in fact, it also shows that ${}^{2\rho}\mathrm{U}_{\mathrm{F}}$ is the affinization of $({}^{2\rho}\check{\mathrm{B}} \times \mathrm{Spec}(\pi_*(k)))/\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}|_0$. Moreover, Theorem 4.2.12 gives an isomorphism $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}|_0 \cong \mathrm{Spec}(\mathrm{H}^{\mathrm{BM}}_*(\mathcal{G}r_{\mathrm{G}}; k))$ of graded group schemes over $\pi_*(k)$, so that ${}^{2\rho}\mathrm{U}_{\mathrm{F}}$ is the affinization of $({}^{2\rho}\check{\mathrm{B}} \times \mathrm{Spec}(\pi_*(k)))/\mathrm{Spec}(\mathrm{H}^{\mathrm{BM}}_*(\mathcal{G}r_{\mathrm{G}}; k))$ and ${}^{2\rho}\mathcal{N}_{\mathrm{F}}$ is the affinization of $({}^{2\rho}\check{\mathrm{G}} \times \mathrm{Spec}(\pi_*(k)))/\mathrm{Spec}(\mathrm{H}^{\mathrm{BM}}_*(\mathcal{G}r_{\mathrm{G}}; k))$.

Example 5.1.19. Assume Conjecture 3.1.14 for every sequence of dominant (quasi-)minuscule coweights of G^{ad}. Suppose $k = \mathbf{Z}'$ is the localization of $\mathbf{Z}$ at the bad primes of G. Then $\mathrm{F} = \hat{\mathbf{G}}_a(2)$ over $\pi_*(k)$. By Lemma 4.1.2, ${}^{2\rho}\mathrm{G}_{\mathrm{F}} \cong {}^{2\rho}\mathfrak{g}(2)^\wedge_{\mathcal{N}}$, and Remark 4.2.2 gives a ${}^{2\rho}\check{\mathrm{G}}$-equivariant isomorphism ${}^{2\rho}\mathfrak{g}(2)^\wedge_{\mathcal{N}} \cong {}^{2\rho}\check{\mathfrak{g}}^*(2)^\wedge_{\mathcal{N}}$ (via the minimal bilinear form). Theorem 5.1.16 gives a $\mathrm{gr}^\star_{\text{G-ev}}(\mathbf{Z}'^{h\mathrm{G}})$-linear equivalence

$$\mathcal{S} : \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; \mathbf{Z}') \xrightarrow{\simeq} \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\check{\mathfrak{g}}^*(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{G}})$$

such that the functor $\mathcal{T}_\star : \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; \mathbf{Z}') \to \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\check{\mathfrak{g}}^*(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{G}})$ sends $\widetilde{\mathrm{IC}}_{\mu_\bullet} \mapsto \widetilde{\mathrm{T}}_{\mu_\bullet} \otimes_{\mathbf{Z}} \mathcal{O}_{{}^{2\rho}\check{\mathfrak{g}}^*(2)}$. Moreover, the restriction of $\mathcal{T}_\star$ to $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; \mathbf{Z}')^{!\text{-ev}}$ is monoidal for the convolution monoidal structure on $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; \mathbf{Z}')$ and the pointwise tensor product on $\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\check{\mathfrak{g}}^*(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{G}})$.

Example 5.1.20. Assume Conjecture 3.1.14 for every sequence of dominant (quasi-)minuscule coweights of G^{ad}. Suppose $k = \mathrm{KU}'$ is the localization of KU at the bad primes of G. By Example 4.1.4, ${}^{2\rho}\mathrm{G}_{\mathrm{F}} \cong {}^{2\rho}\mathrm{G}^\wedge_{\mathcal{U}}$, and Theorem 5.1.16 gives a $\mathrm{gr}^\star_{\text{G-ev}}(\mathrm{KU}'^{h\mathrm{G}})$-linear equivalence

$$\mathcal{S} : \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; \mathrm{KU}') \xrightarrow{\simeq} \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}^\wedge_{\mathcal{U}}/{}^{2\rho}\check{\mathrm{G}})$$

such that the functor $\underline{\mathcal{T}}_\star : \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_\mathrm{G}; \mathrm{KU}') \to \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}^\wedge_\mathcal{U}/{}^{2\rho}\check{\mathrm{G}})$ sends $\widetilde{\mathrm{IC}}_{\mu_\bullet} \mapsto \widetilde{\mathrm{T}}_{\mu_\bullet} \otimes_\mathbf{Z} \mathcal{O}_{{}^{2\rho}\mathrm{G}^\wedge_\mathcal{U}}$. Moreover, the restriction of $\underline{\mathcal{T}}_\star$ to $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_\mathrm{G}; \mathrm{KU}')^{\text{!-ev}}$ is monoidal for the convolution monoidal structure on $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_\mathrm{G}; \mathrm{KU}')$ and the pointwise tensor product on $\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}^\wedge_\mathcal{U}/{}^{2\rho}\check{\mathrm{G}})$.

Remark 5.1.21. For future use, we explicitly state the following from the proof of Theorem 5.1.16: the image under the functor $\mathcal{S} \circ \underline{\mathcal{T}}_\star$ (which I will simply denote by $\mathcal{S}$) of $\mathcal{F} \in \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_\mathrm{G}; k)^{\text{!-ev}}$ is the image of the ${}^{2\rho}\check{\mathrm{G}}$-equivariant $\mathcal{O}_{{}^{2\rho}\mathrm{G}_\mathrm{F}}$-module $\mathcal{S}(\mathcal{F}) := ((\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}} \otimes_{\pi_*(k)} \mathrm{H}^{-*}_\mathrm{T}(\mathcal{G}r_\mathrm{G}; \mathcal{F}))^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}})^{h\mathcal{W}}$ under the localization functor $\mathrm{IndPerf}^{\mathrm{gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}}) \to \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}})$. If $\mathcal{F}$ is an $\mathbf{E}_1 \otimes \mathbf{A}_2$-algebra object (for instance, an $\mathbf{E}_2$-algebra object) in $\mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_\mathrm{G}; k)$, then $\mathcal{S}(\mathcal{F})$ is a commutative algebra object of $\mathrm{IndPerf}^{\mathrm{gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}})$, and its relative spectrum over ${}^{2\rho}\mathrm{G}_\mathrm{F}$ is the affinization of the graded stack $({}^{2\rho}\check{\mathrm{G}} \times \mathrm{Spec}(\mathrm{H}^{-*}_\mathrm{T}(\mathcal{G}r_\mathrm{G}; \mathcal{F})))/(\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \rtimes \mathcal{W})$.

Example 5.1.22. Recall from Remark 3.1.6 that $\omega^{\mathrm{ren}}_{\mathcal{G}r_\mathrm{G}}$ is !-even. It follows from Theorem 4.2.12 and Remark 5.1.21 that the corresponding ${}^{2\rho}\check{\mathrm{G}}$-equivariant $\mathcal{O}_{{}^{2\rho}\mathrm{G}_\mathrm{F}}$-module $\mathcal{S}(\omega^{\mathrm{ren}}_{\mathcal{G}r_\mathrm{G}})$ is

$$\mathcal{S}(\omega^{\mathrm{ren}}_{\mathcal{G}r_\mathrm{G}}) \cong ((\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}} \times \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}})^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}})^{h\mathcal{W}} \cong \mathrm{R}\Gamma({}^{2\rho}\check{\mathrm{G}} \times \mathrm{T}_\mathrm{F}/\mathcal{W}; \mathcal{O}).$$

That is, $\mathcal{S}(\omega^{\mathrm{ren}}_{\mathcal{G}r_\mathrm{G}})$ is the pushforward of the structure sheaf along the map κ_F : $\mathrm{T}_\mathrm{F}/\mathcal{W} \to {}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}}$. This isomorphism encodes the commutative diagram of Theorem 5.1.16.

Remark 5.1.23. With the definition of $\mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_\mathrm{G}; k)$ from Definition 2.4.15, Theorem 5.1.16 continues to hold verbatim when k is not necessarily an even $\mathbf{E}_\infty$-ring. The key observation is that the equivalence of Theorem 5.1.16 does not depend on the choice of a coordinate on F, and (therefore) is functorial in the coefficient even $\mathbf{E}_\infty$-ring. The same is true for the remarks concerning Theorem 5.1.16, like Remark 5.1.21: in this generality, it says that the image under the functor $\mathcal{S} \circ \underline{\mathcal{T}}_\star$ (which I will simply denote by $\mathcal{S}$) of $\mathcal{F} \in \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_\mathrm{G}; k)^{\text{!-ev}}$ is the image under the localization $\mathrm{IndPerf}^{\mathrm{gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}}) \to \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}})$ of the ${}^{2\rho}\check{\mathrm{G}}$-equivariant $\mathcal{O}_{{}^{2\rho}\mathrm{G}_\mathrm{F}}$-module

$$\mathcal{S}(\mathcal{F}) = \lim_{k \to \mathrm{A}} ((\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}} \otimes_{\pi_*(\mathrm{A})} \mathrm{H}^{-*}_\mathrm{T}(\mathcal{G}r_\mathrm{G}; \mathcal{F}_\mathrm{A}))^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}})^{h\mathcal{W}}.$$

Here, the limit ranges over all even $\mathbf{E}_\infty$-k-algebras A.

Remark 5.1.24. One can modify Definition 5.1.13 to define an analogue $\mathrm{Shv}^{\text{min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_\mathrm{G}; k)$ of $\mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_\mathrm{G}; k)$ using $\mathcal{P}_{\leq \lambda} = \bigoplus_{|\mu_\bullet| \leq \lambda} \mathrm{IC}_{\mu_\bullet}$, where the direct sum ranges only over sequences $\mu_\bullet$ of dominant *minuscule* coweights with $|\mu_\bullet| \leq \lambda$. Similarly, one can modify Definition 5.1.14 to define an analogue $\mathrm{Perf}^{\text{min,gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ of $\mathrm{Perf}^{\text{q-min,gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ using tensor products of only *minuscule* tilting representations (i.e., of minuscule irreducible representations). Since Conjecture 3.1.14 holds for sequences of dominant minuscule coweights of G by [Hai06, Corollary 1.2], the argument of Theorem 5.1.16 shows that if the bad primes of G are invertible in $\pi_0(k)$, there is (independently of Conjecture 3.1.14) a $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$-linear equivalence

$$\mathcal{S} : \mathrm{Shv}^{\text{min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_\mathrm{G}; k) \xrightarrow{\simeq} \mathrm{IndPerf}^{\text{min,gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}})$$

which satisfies the same properties as the equivalence of Theorem 5.1.16. This continues to hold in the case $\mathrm{G} = \mathrm{E}_8$, except that the statement becomes a triviality: it asserts that the zero category is equivalent to itself.

Remark 5.1.25. Let K be a field of characteristic $p > 2$. Then we may also view $\mathcal{G}\mathrm{r}_\mathrm{G}$ as an ind-scheme over the algebraic closure $\overline{\mathrm{K}}$. The abelian geometric Satake equivalence extends to the setting of perverse étale sheaves on $\mathcal{G}\mathrm{r}_\mathrm{G}$ (see [MV07, Theorem 14.1]), as does the derived geometric Satake equivalence of [BF08] with rational coefficients. Following [BF08, Section 6.5], the main results of the preceding sections similarly hold verbatim in the setting of étale sheaves of spectra. More precisely, let $\ell \neq p$ be a different prime, and let k be an ℓ-complete even $\mathbf{E}_\infty$-ring with a continuous $\mathbf{Z}_\ell^\times$-action by $\mathbf{E}_\infty$-ring maps. We assume for simplicity that k is even-periodic (in the sense that $\pi_*(k) \cong \pi_0(k)[\beta^{\pm 1}]$ with $\beta \in \pi_2(k)$) and that the $\mathbf{Z}_\ell^\times$-action on $\mathrm{H}^0(\mathbf{C}\mathrm{P}^\infty; k) \cong \pi_0(k)[\![x]\!]$ is given by the $\mathbf{Z}_\ell^\times$-action on $\pi_0(k)$ and the cyclotomic character on x. There is a map $\mathrm{Spec}(\mathbf{Z}[1/\ell])_{\mathrm{et}} \to \mathrm{B}\mathbf{Z}_\ell^\times$ of topoi classifying the extension $\mathbf{Z}[1/\ell, \zeta_{\ell^\infty}]$, so precomposition with the unit map $\mathbf{Z}[1/\ell] \to \mathrm{K}$ induces a map $\mathrm{Spec}(\mathrm{K})_{\mathrm{et}} \to \mathrm{B}\mathbf{Z}_\ell^\times$. The $\mathbf{Z}_\ell^\times$-action on k therefore defines an étale sheaf of $\mathbf{E}_\infty$-rings over K, and so one obtains an action of the absolute Galois group Gal_K on $\mathrm{Shv}^{\mathrm{et}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$.

The composition of the homomorphism $\mathrm{Gal}_\mathrm{K} \xrightarrow{\chi} \mathbf{Z}_\ell^\times \to \mathbf{G}_m$ with the map $\rho : \mathbf{G}_m \to \check{\mathrm{G}}^{\mathrm{ad}}$ defines an action of Gal_K on $\check{\mathrm{G}}$ and on G, where these are viewed as group schemes over $\mathbf{Z}_\ell$. By assumption, the Gal_K-action on the 1-dimensional formal group $\mathrm{F} = \mathrm{Spf}\,\mathrm{H}^0(\mathbf{C}\mathrm{P}^\infty; k)$ is via the cyclotomic character. This defines a Gal_K-action on $(\mathrm{G}_{\pi_0(k)})_\mathrm{F}/\check{\mathrm{G}}_{\pi_0(k)} = \mathrm{Hom}(\mathrm{F}^\vee, \mathrm{G}_{\pi_0(k)})/\check{\mathrm{G}}_{\pi_0(k)}$ which is semilinear for the Gal_K-action on $\pi_0(k)$ via the cyclotomic character $\chi : \mathrm{Gal}_\mathrm{K} \to \mathbf{Z}_\ell^\times$ and the $\mathbf{Z}_\ell^\times$-action on k. This defines an action of Gal_K on $\mathrm{IndPerf}^{\text{q-min}}((\mathrm{G}_{\pi_0(k)})_\mathrm{F}/\check{\mathrm{G}}_{\pi_0(k)})$, which is equivalent to $\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}})$ by using the 2-periodicity of k. By tracking the Gal_K-action on the objects in the proof of Theorem 5.1.16, one finds that the equivalence $\mathcal{S}$ of Theorem 5.1.16 is Gal_K-equivariant. (See also [Zhu17, Section 5.5].)

Note that the action of the Frobenius element $\mathrm{Frob} \in \mathrm{Gal}_\mathrm{K}$ on $\mathrm{IndPerf}^{\mathrm{gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}})$ is $\mathbf{Z}_\ell$-linear. Recall from [GKRV22, Section 3] the notion of trace of an endomorphism of a category. The trace of Frob on $\mathrm{IndPerf}^{\mathrm{gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}})$, taken internally to ℓ-complete $\mathbf{Z}_\ell$-linear categories, is linear over the $\mathbf{E}_\infty$-$\mathbf{Z}_\ell$-algebra R given by the ℓ-completion of $\mathrm{HH}(\pi_0(k)/\mathbf{Z}_\ell; {}^{\mathrm{Frob}}\pi_0(k))$, and is isomorphic to the (derived) ring of functions on the stack

$$\mathcal{L}_{\mathrm{Frob}}((\mathrm{G}_{\pi_0(k)})_\mathrm{F}/\check{\mathrm{G}}_{\pi_0(k)}) \cong \{(g,x) \in \check{\mathrm{G}}_\mathrm{R} \times (\mathrm{G}_\mathrm{F})_\mathrm{R} | \mathrm{Ad}_g(x) = [p](x)\}/\check{\mathrm{G}}_\mathrm{R}. \tag{5.1.3}$$

Here, $[p](x)$ is the image of $x \in \mathrm{G}_\mathrm{F}$ under the automorphism of G_F induced by the multiplication by p map $\mathrm{F} \to \mathrm{F}$, and $\check{\mathrm{G}}_\mathrm{R}$ is acting by diagonal conjugation. For instance, if $k = \mathbf{Z}_\ell[u^{\pm 1}]$ where u is in weight 2 and $\mathbf{Z}_\ell^\times$-action given by the cyclotomic character on u, then $\mathrm{R} = \mathbf{Z}_\ell$, and

$$\mathcal{L}_{\mathrm{Frob}}((\mathrm{G}_{\pi_0(k)})_\mathrm{F}/\check{\mathrm{G}}_{\pi_0(k)}) \cong \{(g,x) \in \check{\mathrm{G}}_{\mathbf{Z}_\ell} \times (\check{\mathfrak{g}}^*_{\mathbf{Z}_\ell})^\wedge_\mathcal{N} | \mathrm{Ad}_g(x) = px\}/\check{\mathrm{G}}_{\mathbf{Z}_\ell}.$$

Similarly, when $k = \mathrm{KU}_\ell$ with the $\mathbf{Z}_\ell^\times$-action by Adams operations, then $\mathrm{R} = \mathbf{Z}_\ell$, and

$$\mathcal{L}_{\mathrm{Frob}}((\mathrm{G}_{\pi_0(k)})_\mathrm{F}/\check{\mathrm{G}}_{\pi_0(k)}) \cong \{(g,x) \in \check{\mathrm{G}}_{\mathbf{Z}_\ell} \times (\mathrm{G}_{\mathbf{Z}_\ell})^\wedge_\mathcal{U} | \mathrm{Ad}_g(x) = x^p\}/\check{\mathrm{G}}_{\mathbf{Z}_\ell}.$$

Remark 5.1.26. It is natural to ask how the relationship between the stacks ${}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ and $\mathrm{T}_{\mathrm{F}}/\mathrm{W}$ manifests on the topological side of Theorem 5.1.16. The inclusion $\mathrm{T} \subseteq \mathrm{G}$ induces a span

$$\begin{tikzcd} & \mathrm{T}_{\mathrm{F}}/\mathrm{N}_{{}^{2\rho}\check{\mathrm{G}}}(\check{\mathrm{T}}) \arrow[dl, "q"'] \arrow[dr, "p"] & \\ \mathrm{T}_{\mathrm{F}}/\mathrm{W} & & {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}, \end{tikzcd}$$

where q is induced by the natural map $\mathrm{N}_{{}^{2\rho}\check{\mathrm{G}}}(\check{\mathrm{T}}) \to \mathrm{W}$. Here, $\mathrm{T}_{\mathrm{F}}/\mathrm{W}$ denotes the quotient stack. This diagram induces a functor

$$\mathrm{sgn} \otimes q_* p^* : \mathrm{IndPerf}^{\mathrm{gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}) \to \mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{T}_{\mathrm{F}}/\mathrm{W}), \tag{5.1.4}$$

where sgn denotes the pullback of the sign representation of W to $\mathrm{T}_{\mathrm{F}}/\mathrm{W}$. When $k = \mathbf{C}$ and $\mathrm{F} = \hat{\mathbf{G}}_a(2)$ (so that ${}^{2\rho}\mathrm{G}_{\mathrm{F}} \cong \check{\mathfrak{g}}^*(2)^{\wedge}_{\mathcal{N}}$ and $\mathrm{T}_{\mathrm{F}} \cong \hat{\check{\mathfrak{t}}}^*(2)$), this functor was described in terms of the derived geometric Satake equivalence and Springer theory in [Mat16a], building on [AH13, AHR15].

This picture admits a generalization to arbitrary k. Consider, for instance, the case $\mathrm{G} = \mathrm{GL}_n$; in this case, Theorem 5.1.16 holds verbatim without any primes needing to be inverted. Moreover, a tilting representation of GL_n is "small" in the sense of [Bro95, AH13, AHR15] if and only if it is a direct summand of $(\mathbf{A}^n)^{\otimes n}$ or its dual. Let $\mathcal{N}$ denote the nilpotent cone of $\mathfrak{gl}_n$. Let $\mu = (n, 0, \cdots, 0)$, and let $j : \mathcal{N} \to \mathcal{G}\mathrm{r}^{\leq \mu}_{\mathrm{GL}_n}$ denote the map sending x to the lattice $(t-x)\mathbf{C}[\![t]\!]^n \subseteq \mathbf{C}[\![t]\!]^n$. In [Lus81] (see also [Zhu17, Lemma 2.3.12]), it was shown that j is an open embedding. Moreover, if $\varpi_\bullet$ denotes the tuple $(\varpi_1, \cdots, \varpi_1)$ of minuscule coweights of GL_n, so that $|\varpi_\bullet| = \mu$, there is a Cartesian square

$$\begin{tikzcd} \widetilde{\mathcal{N}} \arrow[r, "j"] \arrow[d, "\mu"'] & \mathcal{G}\mathrm{r}^{\varpi_\bullet}_{\mathrm{G}} \arrow[d, "m"] \\ \mathcal{N} \arrow[r, "j"'] & \mathcal{G}\mathrm{r}^{\leq\mu}_{\mathrm{G}}. \end{tikzcd} \tag{5.1.5}$$

Let $\mathcal{P}^{\mathrm{sm}} = \widetilde{\mathrm{IC}}_{\varpi_\bullet}$, and define $\langle \mathcal{P}^{\mathrm{sm}} \rangle$, $\langle \mathcal{P}^{\mathrm{sm}} \rangle_{\mathrm{fil}}$, and $\langle \mathcal{P}^{\mathrm{sm}} \rangle_{\mathrm{gr}}$ as in Construction 2.4.11. These categories will be denoted $\mathrm{Shv}^{\mathrm{sm}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, $\mathrm{Shv}^{\mathrm{sm,fil}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, and $\mathrm{Shv}^{\mathrm{sm,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$. Theorem 5.1.16 implies that $\mathrm{Shv}^{\mathrm{sm,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ is equivalent to the subcategory of $\mathrm{IndPerf}^{\mathrm{gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$ generated by $\mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \otimes_{\mathbf{Z}} \mathrm{V}^{\otimes n}$, where V is the graded ${}^{2\rho}\mathrm{GL}_n$-representation whose underlying graded $\mathbf{Z}$-module is $\mathbf{Z}^n$ with the grading $(0, 2, \cdots, 2n-2)$. This subcategory will be denoted $\mathrm{IndPerf}^{\mathrm{sm,gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$.

Similarly, let $\mathrm{Spr} = \mu_! \underline{k}_{\widetilde{\mathcal{N}}}$ denote (the k-analogue of) the Springer sheaf, and define $\langle \mathrm{Spr} \rangle$, $\langle \mathrm{Spr} \rangle_{\mathrm{fil}}$, and $\langle \mathrm{Spr} \rangle_{\mathrm{gr}}$ as in Construction 2.4.11. These categories will be denoted $\mathrm{Shv}^{\mathrm{Spr}}_{\mathrm{G}}(\mathcal{N}; k)$, $\mathrm{Shv}^{\mathrm{Spr,fil}}_{\mathrm{G}}(\mathcal{N}; k)$, and $\mathrm{Shv}^{\mathrm{Spr,gr}}_{\mathrm{G}}(\mathcal{N}; k)$. There is an analogue of [CG10, Theorem 7.2.2], which gives an isomorphism

$$\pi_* \mathrm{End}_{\mathrm{Shv}_{\mathrm{G}}(\mathcal{N};k)}(\mathrm{Spr}) \cong \mathcal{O}_{\mathrm{T}_{\mathrm{F}}} \rtimes \mathbf{Z}[\mathrm{W}], \tag{5.1.6}$$

where the right-hand side denotes the smash product of the group algebra of the Weyl group with $\mathcal{O}_{\mathrm{T}_{\mathrm{F}}}$. Here, W acts on $\mathcal{O}_{\mathrm{T}_{\mathrm{F}}}$ in the obvious way. The isomorphism (5.1.6) follows from an isomorphism between $\pi_* \mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathcal{N};k)}(\mathrm{Spr})$ and $\mathcal{O}_{\mathcal{W}} \rtimes \mathbf{Z}[\mathrm{W}]$, which can in turn be proved by identifying $\pi_* \mathrm{End}_{\mathrm{Shv}_{\mathrm{T}}(\mathcal{N};k)}(\mathrm{Spr})$ with

$\mathrm{H}^{\mathrm{T,BM}}_{*+4\dim_{\mathbf{C}}(\mathrm{G/B})}(\mathrm{Z}_0; k)$, where Z_0 is the Steinberg variety $\widetilde{\mathcal{N}} \times_{\mathcal{N}} \widetilde{\mathcal{N}}$. The desired algebra isomorphism $\mathrm{H}^{\mathrm{T,BM}}_{*+4\dim_{\mathbf{C}}(\mathrm{G/B})}(\mathrm{Z}_0; k) \cong \mathcal{O}_{\mathcal{W}} \rtimes \mathbf{Z}[\mathrm{W}]$ now follows from Theorem 2.2.9 by a description of the GKM graph of Z_0, the algebra map

$$\mathrm{H}^{\mathrm{T,BM}}_{*+4\dim_{\mathbf{C}}(\mathrm{G/B})}(\mathrm{Z}_0; k) \hookrightarrow \mathrm{H}^{\mathrm{T,BM}}_{*+4\dim_{\mathbf{C}}(\mathrm{G/B})}(\mathrm{Z}_0; k)[e(\Phi)^{-1}] \cong \mathrm{H}^{\mathrm{T,BM}}_{*+4\dim_{\mathbf{C}}(\mathrm{G/B})}(\mathrm{Z}_0^{\mathrm{T}}; k)[e(\Phi)^{-1}],$$

and the algebra isomorphism $\mathrm{H}^{\mathrm{T,BM}}_{*+4\dim_{\mathbf{C}}(\mathrm{G/B})}(\mathrm{Z}_0^{\mathrm{T}}; k) \cong \mathcal{O}_{\mathrm{T_F}}[\mathrm{W}] \rtimes \mathbf{Z}[\mathrm{W}]$. (This is just as in the proof of Theorem 3.1.11.)

The isomorphism (5.1.6) defines an equivalence

$$\mathrm{Shv}^{\mathrm{Spr,gr}}_{\mathrm{G}}(\mathcal{N}; k) \simeq \mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{T_F/W}).$$

The Cartesian square (5.1.5) identifies $j^*\mathcal{P}^{\mathrm{sm}} \cong \mathrm{Spr}$. This gives a functor

$$j^* : \mathrm{Shv}^{\mathrm{sm,fil}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k) \to \mathrm{Shv}^{\mathrm{Spr,fil}}_{\mathrm{G}}(\mathcal{N}; k),$$

whose induced functor on associated graded is isomorphic, under the equivalences between $\mathrm{Shv}^{\mathrm{sm,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ and $\mathrm{IndPerf}^{\mathrm{sm,gr}}({}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}})$ and between $\mathrm{Shv}^{\mathrm{Spr,gr}}_{\mathrm{G}}(\mathcal{N}; k)$ and $\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{T_F/W})$ discussed above, to the composite

$$\mathrm{IndPerf}^{\mathrm{sm,gr}}({}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}}) \subseteq \mathrm{IndPerf}^{\mathrm{gr}}({}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}}) \xrightarrow{(5.1.4)} \mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{T_F/W}).$$

For the sake of notational simplicity, let us write $\mathcal{X} = {}^{2\rho}\mathrm{G_F} \times_{\mathrm{T_F}/\mathcal{W}} \mathrm{T_F}$ and $\widetilde{\mathcal{X}} = {}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}$ in the following, which says that the right adjoint of the fully faithful functor $\mathrm{Shv}^{\mathrm{q\text{-}min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k) \to \mathrm{Shv}^{\mathrm{gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ is not very destructive.

Proposition 5.1.27. *Suppose that the bad primes of* G *are inverted in* $\pi_0(k)$*, and that the fully faithful functor* $\mathrm{Perf}^{\mathrm{q\text{-}min}}(\mathrm{B}\check{\mathrm{G}}) \to \mathrm{Perf}(\mathrm{B}\check{\mathrm{G}})$ *is an equivalence (see Proposition 5.1.29 for some criteria). Assume Conjecture 3.1.14 for every sequence of dominant (quasi-)minuscule coweights of* G^{ad}*. Let* $\mathcal{F} \in \mathrm{Shv}^{\mathrm{toral}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)^{!\text{-ev}}$*, and let* $\mathcal{S}'(\mathcal{F}) = (\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}} \otimes_{\pi_*(k)} \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}}; \mathcal{F}))^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$*, so that* $\mathcal{S}'(\mathcal{F})$ *is a* ${}^{2\rho}\check{\mathrm{G}}$*-equivariant* $\mathcal{O}_{\mathcal{X}}$*-module (via the isomorphism* $\mathcal{O}_{\mathcal{X}} \cong (\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}} \times \mathrm{T_F}})^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$ *from Remark 4.2.6 and Theorem 4.3.6).*

If $\mathcal{S}'(\mathcal{F})$ *admits a good filtration as a* ${}^{2\rho}\check{\mathrm{G}}$*-module, then the natural map* $\mathcal{F}^{\mathrm{q\text{-}min}} \to \mathcal{F}$ *induces an isomorphism on global sections. Moreover, if* $\mu_\bullet$ *is a sequence of dominant (quasi-)minuscule coweights of* G^{ad}*, there is an isomorphism*

(5.1.7)

$$\mathrm{gr}^{\star}_{\mathrm{G\text{-}ev}}(\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}};k)}(\widetilde{\mathrm{IC}}_{\mu_\bullet}, \mathcal{F})) \cong \mathrm{RHom}_{\mathrm{Perf}^{\mathrm{all,gr}}({}^{2\rho}\mathrm{G_F}/{}^{2\rho}\check{\mathrm{G}})}(\widetilde{\mathrm{T}}_{\mu_\bullet} \otimes_{\mathbf{Z}} \mathcal{O}_{{}^{2\rho}\mathrm{G_F}}, \mathcal{S}(\mathcal{F})),$$

where $\mathcal{S}(\mathcal{F}) = (\mathcal{S}'(\mathcal{F}))^{h\mathcal{W}}$*.*

Proof. For notational simplicity, let $\mathcal{C} = \mathrm{Shv}_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}}; k)$. Suppose $\mathcal{F} \in \mathcal{C}$ is supported on $\mathcal{G}r^{\leq\mu}_{\mathrm{G}}$, and let us for simplicity write $\mathcal{P}$ to denote $\mathcal{P}_{\leq\mu}$. The cellularization $\mathcal{F}^{\mathrm{q\text{-}min}}$ is given by

$$\mathcal{F}^{\mathrm{q\text{-}min}} \cong \underline{\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}};k)}(\mathcal{P}, \mathcal{F})} \otimes_{\underline{\mathrm{End}_{\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}};k)}(\mathcal{P})}} \mathcal{P}.$$

Here, the underlined objects are viewed as constant sheaves on $\mathcal{G}r_{\mathrm{G}}$. The natural map $\mathcal{F}^{\mathrm{q\text{-}min}} \to \mathcal{F}$ is given by "evaluation". Note that

$$\mathrm{H}^*_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}}; \mathcal{F}^{\mathrm{q\text{-}min}}) \cong \pi_*(\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}};k)}(\mathcal{P}, \mathcal{F})) \otimes_{\pi_*(\mathrm{End}_{\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}};k)}(\mathcal{P}))} \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}}; \mathcal{P}).$$

The filtrations $\mathrm{fil}^\star_{\mathrm{G}}(\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}};k)}(\mathcal{P},\mathcal{F}))$ and $\mathrm{fil}^\star_{\mathrm{G}}(\mathrm{End}_{\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}};k)}(\mathcal{P}))$ have associated graded isomorphic to $(\pi_\star \mathrm{Hom}_{\mathcal{C}}(\mathcal{P},\mathcal{F}))^{h\mathcal{W}}$ and $(\pi_\star \mathrm{End}_{\mathcal{C}}(\mathcal{P}))^{h\mathcal{W}}$, respectively. This defines a spectral sequence

$$\mathrm{E}_2 \cong (\pi_\star \mathrm{Hom}_{\mathcal{C}}(\mathcal{P},\mathcal{F}))^{h\mathcal{W}} \otimes_{(\pi_\star \mathrm{End}_{\mathcal{C}}(\mathcal{P}))^{h\mathcal{W}}} \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};\mathcal{P}) \Rightarrow \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};\mathcal{F}^{\text{q-min}}), \tag{5.1.8}$$

where the tensor product is taken in the derived sense. Theorem 2.3.2, Theorem 4.2.12, Corollary 5.1.11, and Proposition 5.1.12 (which uses Conjecture 3.1.14) give ($\mathcal{W}$-equivariant) isomorphisms

$$\begin{aligned}\pi_\star \mathrm{Hom}_{\mathcal{C}}(\mathcal{P},\mathcal{F}) &\cong \mathrm{Hom}^\star_{\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T_F}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})^\heartsuit}(\underline{\widetilde{\mathrm{T}}}, \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};\mathcal{F})),\\ \pi_\star \mathrm{End}_{\mathcal{C}}(\mathcal{P}) &\cong \mathrm{End}^\star_{\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T_F}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})^\heartsuit}(\underline{\widetilde{\mathrm{T}}}),\end{aligned}$$

Here, we have written $\widetilde{\mathrm{T}}$ to denote $\widetilde{\mathrm{T}}_{\leq\mu}$ (just like with $\mathcal{P}$), so that $\mathrm{H}^*_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};\mathcal{P}) \cong \underline{\widetilde{\mathrm{T}}}$, where we recall that $\underline{\widetilde{\mathrm{T}}}_{\mu_\bullet}$ is as defined in Notation 5.1.10.

Write $\widetilde{\mathcal{S}}(\mathcal{F})$ to denote the object of $\mathrm{Perf}^{\mathrm{gr}}(\widetilde{\mathcal{X}}/^{2\rho}\check{\mathrm{G}}) \simeq \mathrm{Perf}^{\mathrm{gr}}(^{2\rho}\mathrm{B_F}/^{2\rho}\check{\mathrm{B}})$ given by $(\mathcal{O}_{^{2\rho}\check{\mathrm{B}}} \otimes_{\pi_*(k)} \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};\mathcal{F}))^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$. Note that $\widetilde{\mathcal{S}}(\mathcal{F})$ is a $^{2\rho}\check{\mathrm{B}}$-equivariant $\mathcal{O}_{^{2\rho}\mathrm{B_F}}$-module (via the isomorphism $\mathcal{O}_{^{2\rho}\mathrm{B_F}} \cong (\mathcal{O}_{^{2\rho}\check{\mathrm{B}}\times\mathrm{T_F}})^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$ from Remark 4.2.6). Let us continue to write $\widetilde{\mathcal{S}}(\mathcal{F})$ to denote the pullback of $\widetilde{\mathcal{S}}(\mathcal{F})$ along the map $\widetilde{\mathcal{X}} \to \widetilde{\mathcal{X}}/^{2\rho}\check{\mathrm{G}}$. Note that $\mathrm{H}^0(\widetilde{\mathcal{X}};\widetilde{\mathcal{S}}(\mathcal{F})) = \mathcal{S}'(\mathcal{F})$. Corollary 4.1.16 implies that

$$\begin{aligned}\mathrm{Hom}^\star_{\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T_F}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})^\heartsuit}(\underline{\widetilde{\mathrm{T}}}, \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};\mathcal{F})) &\cong \mathrm{H}^0(\widetilde{\mathcal{X}}/^{2\rho}\check{\mathrm{G}};\widetilde{\mathcal{S}}(\mathcal{F})\otimes_{\mathbf{Z}} \widetilde{\mathrm{T}}^\vee)\\ &\cong \pi_0\mathrm{R}\Gamma(\mathrm{B}^{2\rho}\check{\mathrm{G}};\mathcal{S}'(\mathcal{F})\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}^\vee),\\ \mathrm{End}^\star_{\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T_F}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})^\heartsuit}(\underline{\widetilde{\mathrm{T}}}) &\cong \mathrm{H}^0(\widetilde{\mathcal{X}}/^{2\rho}\check{\mathrm{G}};\mathcal{O}_{\widetilde{\mathcal{X}}/^{2\rho}\check{\mathrm{G}}}\otimes_{\mathbf{Z}} \mathrm{End}_{\mathbf{Z}}(\widetilde{\mathrm{T}}))\\ &\cong \pi_0\mathrm{R}\Gamma(\mathrm{B}^{2\rho}\check{\mathrm{G}};\mathrm{H}^0(\widetilde{\mathcal{X}};\mathcal{O})\otimes_{\mathbf{Z}}\mathrm{End}_{\mathbf{Z}}(\widetilde{\mathrm{T}})).\end{aligned}$$

Since $\widetilde{\mathrm{T}}$ and $\mathrm{H}^0(\widetilde{\mathcal{X}};\mathcal{O}) \cong \mathcal{O}_{\mathcal{X}}$ admit good filtrations (the latter by Theorem 4.3.6 and Proposition 5.1.15), it follows from Proposition 5.1.15 that $\mathrm{R}\Gamma(\mathrm{B}^{2\rho}\check{\mathrm{G}};\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}} \mathrm{End}_{\mathbf{Z}}(\widetilde{\mathrm{T}}))$ is concentrated in degree zero. Similarly, since $\mathrm{H}^0(\widetilde{\mathcal{X}};\widetilde{\mathcal{S}}(\mathcal{F})) = \mathcal{S}'(\mathcal{F})$ is assumed to have a good filtration, $\mathrm{R}\Gamma(\mathrm{B}^{2\rho}\check{\mathrm{G}};\mathcal{S}'(\mathcal{F})\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}^\vee)$ is also concentrated in degree zero. We conclude that

$$\begin{aligned}\mathrm{Hom}^\star_{\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T_F}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})^\heartsuit}(\underline{\widetilde{\mathrm{T}}}, \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};\mathcal{F})) &\cong \mathrm{RHom}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}},\mathcal{S}'(\mathcal{F}))\\ \mathrm{End}^\star_{\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T_F}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})^\heartsuit}(\underline{\widetilde{\mathrm{T}}}) &\cong \mathrm{REnd}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}).\end{aligned}$$

Combining the isomorphisms discussed above, the E_2-page of (5.1.8) simplifies to

$$\begin{aligned}\mathrm{E}_2 &\cong \mathrm{RHom}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}},\mathcal{S}'(\mathcal{F}))^{h\mathcal{W}} \otimes_{\mathrm{REnd}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}})^{h\mathcal{W}}} \underline{\widetilde{\mathrm{T}}}\\ &\cong \mathrm{RHom}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}},\mathcal{S}'(\mathcal{F})) \otimes_{\mathrm{REnd}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}})} \underline{\widetilde{\mathrm{T}}}\\ &\cong \kappa_{\mathrm{F}}^*(\mathrm{RHom}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}},\mathcal{S}'(\mathcal{F})) \otimes_{\mathrm{REnd}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}})} (\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\underline{\widetilde{\mathrm{T}}})),\end{aligned}$$

where $\kappa_{\mathrm{F}} : \mathrm{T_F} \to \widetilde{\mathcal{X}} \to \mathcal{X}$ is the map from Construction 4.1.12 composed with the map $\widetilde{\mathcal{X}} \to \mathcal{X}$ from Theorem 4.3.6. Let $\mathcal{N} \mapsto \mathcal{N}^{\text{q-min}}$ denote the right adjoint to the fully faithful functor $\mathrm{Perf}^{\text{q-min},\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}}) \hookrightarrow \mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})$. Then there is an isomorphism

$$\mathcal{S}'(\mathcal{F})^{\text{q-min}} \cong \mathrm{RHom}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}}, \mathcal{S}'(\mathcal{F}))\otimes_{\mathrm{REnd}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\widetilde{\mathrm{T}})}(\mathcal{O}_{\mathcal{X}}\otimes_{\mathbf{Z}}\underline{\widetilde{\mathrm{T}}}).$$

By assumption, the fully faithful functor $\mathrm{Perf}^{\text{q-min}}(\mathrm{B}\check{\mathrm{G}}) \to \mathrm{Perf}(\mathrm{B}\check{\mathrm{G}})$ is an equivalence. This implies that the functor $\mathrm{Perf}^{\text{q-min,gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}}) \hookrightarrow \mathrm{Perf}^{\text{all,gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})$ is an equivalence. In particular, the natural map $\mathcal{S}'(\mathcal{F})^{\text{q-min}} \to \mathcal{S}'(\mathcal{F})$ is an isomorphism. We conclude that the E_2-page of (5.1.8) is isomorphic to $\kappa_{\mathrm{F}}^*(\mathcal{S}'(\mathcal{F})) \cong \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{F})$. Since the E_2-page is concentrated in homological degree zero, the spectral sequence (5.1.8) collapses, and we conclude that $\mathrm{H}^*_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{F}^{\text{q-min}}) \cong \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{F})$, as desired.

For the final part of Proposition 5.1.27, note that as above, combining Theorem 2.3.2, Theorem 4.2.12, Corollary 5.1.11, and Proposition 5.1.12 (which uses Conjecture 3.1.14) yields a ($\mathcal{W}$-equivariant) isomorphism

$$\pi_* \mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)}(\widetilde{\mathrm{IC}}_{\mu_\bullet}, \mathcal{F}) \cong \mathrm{Hom}^\star_{\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T}_{\mathrm{F}}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})^\heartsuit}(\widetilde{\mathrm{T}}_{\mu_\bullet} \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T}_{\mathrm{F}}}, \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{F})).$$

Corollary 4.1.16 and Theorem 4.3.6 imply that

$$\mathrm{Hom}^\star_{\mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T}_{\mathrm{F}}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})^\heartsuit}(\widetilde{\mathrm{T}}_{\mu_\bullet}\otimes_{\mathbf{Z}}\mathcal{O}_{\mathrm{T}_{\mathrm{F}}}, \mathrm{H}^*_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{F})) \cong \mathrm{Hom}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})^\heartsuit}(\widetilde{\mathrm{T}}_{\mu_\bullet}\otimes_{\mathbf{Z}}\mathcal{O}_{\mathcal{X}}, \mathcal{S}'(\mathcal{F})).$$

Arguing with good filtrations as above shows that this is in turn isomorphic to $\mathrm{RHom}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\widetilde{\mathrm{T}}_{\mu_\bullet} \otimes_{\mathbf{Z}} \mathcal{O}_{\mathcal{X}}, \mathcal{S}'(\mathcal{F}))$, so that

$$\pi_* \mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)}(\widetilde{\mathrm{IC}}_{\mu_\bullet}, \mathcal{F}) \cong \mathrm{RHom}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/^{2\rho}\check{\mathrm{G}})}(\widetilde{\mathrm{T}}_{\mu_\bullet} \otimes_{\mathbf{Z}} \mathcal{O}_{\mathcal{X}}, \mathcal{S}'(\mathcal{F})).$$

Taking $\mathcal{W}$-invariants produces the isomorphism (5.1.7). □

Remark 5.1.28. Recall from Remark 5.1.21 that the image of $\mathcal{F} \in \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)^{!\text{-ev}}$ under the functor $\mathcal{S}\circ\underline{\tau}_\star$ (which I will denote by $\mathcal{S}$) is the image under the localization $\mathrm{IndPerf}^{\mathrm{gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}) \to \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$ of the ${}^{2\rho}\check{\mathrm{G}}$-equivariant $\mathcal{O}_{{}^{2\rho}\mathrm{G}_{\mathrm{F}}}$-module $\mathcal{S}(\mathcal{F}) := ((\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}} \otimes_{\pi_*(k)} \mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{F}))^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}})^{h\mathcal{W}}$. Assume the hypotheses of Proposition 5.1.27, so that the fully faithful functor $\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}) \to \mathrm{IndPerf}^{\mathrm{gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$ is an equivalence. If $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)^{!\text{-ev}}$, then Proposition 5.1.27 implies that there is an isomorphism

$$\mathcal{S}(\mathcal{F}^{\text{q-min}}) \cong ((\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}} \otimes_{\pi_*(k)} \mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{F}))^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}})^{h\mathcal{W}}.$$

That is, the image of $\mathcal{F}^{\text{q-min}}$ under $\mathcal{S}$ can be computed directly from $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};\mathcal{F})$ with its $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \rtimes \mathcal{W}$-action.

The inclusion $\mathrm{Perf}^{\text{q-min,gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}}) \hookrightarrow \mathrm{Perf}^{\mathrm{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ is an equivalence upon localization at certain primes. The following is essentially [JMW16, Proposition 3.8]. Note that the bounds below are likely not the best possible (this being particularly egregious for E_8)! We include the non-simply-laced cases below for completeness.

Proposition 5.1.29. *Suppose that* $\check{\mathrm{G}}$ *is the split form of a semisimple (not necessarily simply-laced) algebraic group over* R. *If* p *is not one of the primes in Table 2 for the simple adjoint factors of* $\check{\mathrm{G}}$, *then the fully faithful functor* $\mathrm{Perf}^{\text{q-min,gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}}) \hookrightarrow \mathrm{Perf}^{\mathrm{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ *is an equivalence upon* p*-localization.*

Proof. Since $\mathrm{Perf}^{\mathrm{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ is a sum of blocks of $\mathrm{Perf}^{\mathrm{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}})$, we may assume without loss of generality that $\check{\mathrm{G}}$ is simple and simply-connected. It suffices to check that the fully faithful functor $\mathrm{Perf}^{\text{q-min,gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}}) \hookrightarrow \mathrm{Perf}^{\mathrm{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ is an equivalence

TABLE 2. Primes not allowed in Proposition 5.1.29. Note, in particular, that every bad prime (see Table 1) is not allowed in Proposition 5.1.29.

Type	A_n	B_n	C_n	D_n	E_6	E_7	E_8	F_4	G_2
Disallowed p	$\emptyset$	2	$2 \leq p \leq n$	2	2	2, 3	$2 \leq p \leq 37$	2, 3	2

upon base-change to any residue field of R of characteristic $\neq p$ (where p is one of the primes in Table 2 for the simple adjoint factors of $\check{\mathrm{G}}$). For this, it suffices to show that for any algebraically closed field K of characteristic $p \geq 0$ with a map $\mathrm{R} \to \mathrm{K}$ where p is not one of the primes in Table 2 for the simple adjoint factors of $\check{\mathrm{G}}$, the fully faithful functor $\mathrm{Perf}^{\text{q-min,gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}}) \otimes_{\mathrm{R}} \mathrm{K} \hookrightarrow \mathrm{Perf}^{\mathrm{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}}) \otimes_{\mathrm{R}} \mathrm{K}$ is an equivalence.

Note that $\mathrm{T}_\mu \cong \mathrm{V}_\mu$ when μ is minuscule. Also, T_θ admits a filtration whose associated graded is a direct sum of trivial representations and V_θ. It follows that $\mathrm{V}_\theta \in \mathrm{Perf}^{\text{q-min,gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$. The category $\mathrm{Perf}(\mathrm{B}\check{\mathrm{G}}) \otimes_{\mathrm{R}} \mathrm{K}$ is generated by Weyl modules, so it suffices to show that all Weyl modules lie in the subcategory $\mathrm{Perf}^{\text{q-min}}(\mathrm{B}\check{\mathrm{G}}) \otimes_{\mathrm{R}} \mathrm{K}$ of $\mathrm{Perf}(\mathrm{B}\check{\mathrm{G}}) \otimes_{\mathrm{R}} \mathrm{K}$ generated by tensor powers of (quasi-)minuscule tilting modules. Recall that the tensor product of Weyl modules admits a filtration by Weyl modules (see [Don85, Mat90]). Since any dominant weight λ is a $\mathbf{Z}_{\geq 0}$-linear combination $\sum_{i \geq 0} n_i \varpi_i$ of fundamental weights ϖ_i, it follows by inducting on the Weyl filtration on $\bigotimes_{i \geq 0} \mathrm{V}_{\varpi_i}^{\otimes n_i}$ that V_λ lies in the subcategory of $\mathrm{Perf}(\mathrm{B}\check{\mathrm{G}}) \otimes_{\mathrm{R}} \mathrm{K}$ which is generated by $\bigotimes_{i \geq 0} \mathrm{V}_{\varpi_i}^{\otimes n_i}$. It therefore suffices to show that if the primes in Table 2 are invertible in R, then $\mathrm{V}_\varpi \in \mathrm{Perf}^{\text{q-min}}(\mathrm{B}\check{\mathrm{G}})$ for each fundamental weight ϖ of $\check{\mathrm{G}}$. We now go case-by-case, using notation as in [Bou68]. (The decompositions used below were computed using Sage.)

(1) When $\check{\mathrm{G}} = \mathrm{SL}_{n+1}$, all fundamental weights are minuscule, so each V_{ϖ_i} lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BSL}_{n+1})$ by definition.
(2) When $\check{\mathrm{G}} = \mathrm{Spin}_{2n+1}$, the spin representation $\mathrm{S} = \mathrm{V}_{\varpi_n}$ is minuscule. The tensor square $\mathrm{S} \otimes \mathrm{S}$ is isomorphic to the even part of the Clifford algebra of V_{ϖ_1}. The tensor length filtration defines a Spin_{2n+1}-equivariant filtration on $\mathrm{S} \otimes \mathrm{S}$ with associated graded pieces isomorphic to $\wedge^{2i}(\mathrm{V}_{\varpi_1}) \cong \wedge^{2(n-i)+1}(\mathrm{V}_{\varpi_1})$. This filtration admits a Spin_{2n+1}-equivariant splitting if $2 \in \mathrm{R}^\times$.[13] It follows that each $\wedge^{2i}(\mathrm{V}_{\varpi_1})$ is a retract of $\mathrm{S} \otimes \mathrm{S}$, so all the fundamental representations V_{ϖ_i} lie in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BSpin}_{2n+1})$ if $2 \in \mathrm{R}^\times$.
(3) When $\check{\mathrm{G}} = \mathrm{Sp}_{2n}$, the standard representation V_{ϖ_1} is minuscule. If $i! \in \mathrm{R}^\times$, then $\wedge^i \mathrm{V}_{\varpi_1}$ is a direct summand of $\mathrm{V}_{\varpi_1}^{\otimes i}$. The symplectic form defines a map $\wedge^{i-2}\mathrm{V}_{\varpi_1} \to \wedge^i \mathrm{V}_{\varpi_1}$ whose cofiber is V_{ϖ_i}, so all the fundamental representations V_{ϖ_i} lie in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BSp}_{2n})$ if $n! \in \mathrm{R}^\times$.
(4) When $\check{\mathrm{G}} = \mathrm{Spin}_{2n}$, the standard representation V_{ϖ_1} and the half-spin representations $\mathrm{S}^+ = \mathrm{V}_{\varpi_n}$ and $\mathrm{S}^- = \mathrm{V}_{\varpi_{n-1}}$ are minuscule. In any characteristic, $\mathrm{End}(\mathrm{S}^+) \oplus \mathrm{End}(\mathrm{S}^-)$ (resp. $\mathrm{Hom}(\mathrm{S}^+, \mathrm{S}^-) \oplus \mathrm{Hom}(\mathrm{S}^-, \mathrm{S}^+)$) is isomorphic to the even (resp. odd) part of the Clifford algebra of V_{ϖ_1}. The tensor length

[13] Indeed, the map $\mathrm{V}_{\varpi_1} \to \mathrm{End}(\wedge^*(\mathrm{V}_{\varpi_1}))$ sending v to $\mathrm{L}_v : w \mapsto v \wedge w + \langle v, w \rangle$ (where the latter denotes contraction with respect to the bilinear form associated to the quadratic form on V_{ϖ_1}) satisfies $\mathrm{L}_v^2 = q(v)$, so we obtain a map $\mathrm{Cl}(\mathrm{V}_{\varpi_1}) \to \mathrm{End}(\wedge^*(\mathrm{V}_{\varpi_1}))$. Evaluating on $1 \in \wedge^0(\mathrm{V}_{\varpi_1})$ defines the desired isomorphism $\mathrm{Cl}(\mathrm{V}_{\varpi_1}) \to \wedge^*(\mathrm{V}_{\varpi_1})$.

filtration defines a Spin_{2n}-equivariant filtration on $\mathrm{End}(\mathrm{S}^+) \oplus \mathrm{End}(\mathrm{S}^-)$ (resp. $\mathrm{Hom}(\mathrm{S}^+, \mathrm{S}^-) \oplus \mathrm{Hom}(\mathrm{S}^-, \mathrm{S}^+)$) with associated graded pieces isomorphic to $\wedge^{2i}(\mathrm{V}_{\varpi_1}) \cong \wedge^{2(n-i)}(\mathrm{V}_{\varpi_1})$ (resp. $\wedge^{2i+1}(\mathrm{V}_{\varpi_1}) \cong \wedge^{2(n-i)-1}(\mathrm{V}_{\varpi_1})$). As in the case of Spin_{2n+1}, these filtrations admits Spin_{2n}-equivariant splittings if $2 \in \mathrm{R}^\times$, so that each fundamental representation V_{ϖ_i} lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BSpin}_{2n})$ if $2 \in \mathrm{R}^\times$.

(5) When $\check{\mathrm{G}} = \mathrm{E}_6$, we will show that all the fundamental representations V_{ϖ_i} lie in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_6)$ if $2 \in \mathrm{R}^\times$.
  (a) The standard 27-dimensional representations V_{ϖ_1} and $\mathrm{V}_{\varpi_6} \cong \mathrm{V}_{\varpi_1}^\vee$ are minuscule.
  (b) If $2 \in \mathrm{R}^\times$, then $\wedge^2(\mathrm{V}_{\varpi_1})$ is a retract of $\mathrm{V}_{\varpi_1}^{\otimes 2}$ (and similarly for V_{ϖ_6}), so $\wedge^2(\mathrm{V}_{\varpi_1}), \wedge^2(\mathrm{V}_{\varpi_6}) \in \mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_6)$. The 351-dimensional fundamental representation V_{ϖ_3} (resp. V_{ϖ_5}) is isomorphic to $\wedge^2(\mathrm{V}_{\varpi_1})$ (resp. $\wedge^2(\mathrm{V}_{\varpi_6})$).
  (c) The adjoint representation $\mathrm{V}_{\varpi_2} = \mathfrak{e}_6$ always lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_6)$, since $\mathrm{T}_\theta \in \mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_6)$ by definition and T_θ admits a filtration whose associated graded is a direct sum of trivial representations (which lie in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_6)$) and V_{ϖ_2}.
  (d) If $2 \in \mathrm{R}^\times$, then $\wedge^2(\mathrm{V}_{\varpi_2})$ is a retract of $\mathrm{V}_{\varpi_2}^{\otimes 2}$, so $\wedge^2(\mathrm{V}_{\varpi_2}) \in \mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_6)$. The Lie bracket defines a map $\wedge^2 \mathrm{V}_{\varpi_2} \to \mathrm{V}_{\varpi_2}$ whose fiber is the 2925-dimensional representation V_{ϖ_4}, so that V_{ϖ_4} lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_6)$.

(6) When $\check{\mathrm{G}} = \mathrm{E}_7$, we will show that all the fundamental representations V_{ϖ_i} lie in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_7)$ if $6 \in \mathrm{R}^\times$.
  (a) The standard 56-dimensional representation V_{ϖ_7} is minuscule. It is a symplectic representation of E_7, and in particular is self-dual.
  (b) When $p \neq 2$, $\wedge^2(\mathrm{V}_{\varpi_7})$ is a retract of $\mathrm{V}_{\varpi_7}^{\otimes 2}$, and hence lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_7)$. The symplectic form defines a map $\wedge^2(\mathrm{V}_{\varpi_7}) \to \mathrm{R}$ whose fiber is the 1539-dimensional representation V_{ϖ_6}.
  (c) If $p \neq 2, 3$, then $\wedge^3(\mathrm{V}_{\varpi_7})$ is a retract of $\mathrm{V}_{\varpi_7}^{\otimes 3}$, and hence lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_7)$. The symplectic form defines a map $\wedge^3(\mathrm{V}_{\varpi_7}) \to \mathrm{V}_{\varpi_7}$ whose fiber is the 27664-dimensional representation V_{ϖ_5}.
  (d) If $p \neq 2, 3$, then $\wedge^4(\mathrm{V}_{\varpi_7})$ is a retract of $\mathrm{V}_{\varpi_7}^{\otimes 4}$, and hence lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_7)$. The symplectic form defines a map $\wedge^4(\mathrm{V}_{\varpi_7}) \to \wedge^2(\mathrm{V}_{\varpi_7})$ whose fiber is the 365750-dimensional representation V_{ϖ_4}.
  (e) The adjoint representation $\mathrm{V}_{\varpi_1} = \mathfrak{e}_7$ always lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_7)$, since $\mathrm{T}_\theta \in \mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_7)$ by definition and T_θ admits a filtration whose associated graded is a direct sum of trivial representations (which lie in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_7)$) and V_{ϖ_1}.
  (f) When $p \neq 2$, $\wedge^2(\mathrm{V}_{\varpi_1})$ is a retract of $\mathrm{V}_{\varpi_1}^{\otimes 2}$, so it lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_7)$. The Lie bracket defines a map $\wedge^2(\mathrm{V}_{\varpi_1}) \to \mathrm{V}_{\varpi_1}$ whose fiber is the 8645-dimensional representation V_{ϖ_3}, so V_{ϖ_3} lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_7)$.
  (g) The adjoint action of V_{ϖ_1} on V_{ϖ_7} defines a map $\mathrm{V}_{\varpi_1} \otimes \mathrm{V}_{\varpi_7} \to \mathrm{V}_{\varpi_7}$. The fiber of this map is isomorphic in characteristic zero to $\mathrm{V}_{\varpi_1 + \varpi_7} \oplus \mathrm{V}_{\varpi_2}$. Computing the Casimir eigenvalues shows that this splitting exists away from characteristics 2 and 3, so that the 912-dimensional representation V_{ϖ_2} is a retract of $\mathrm{fib}(\mathrm{V}_{\varpi_1} \otimes \mathrm{V}_{\varpi_7} \to \mathrm{V}_{\varpi_7})$ and hence lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_7)$.

(7) When $\check{\mathrm{G}} = \mathrm{E}_8$, we will show that all the fundamental representations V_{ϖ_i} lie in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_8)$ if $p \in \mathrm{R}^\times$ for all $2 \leq p \leq 37$.

(a) The 248-dimensional adjoint representation V_{ϖ_8} is quasi-minuscule.

(b) If $2 \in \mathrm{R}^\times$, then $\wedge^2(\mathrm{V}_{\varpi_8})$ is a retract of $\mathrm{V}_{\varpi_8}^{\otimes 2}$, so $\wedge^2(\mathrm{V}_{\varpi_8}) \in \mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_8)$. The fiber of the Lie bracket $\wedge^2(\mathrm{V}_{\varpi_8}) \to \mathrm{V}_{\varpi_8}$ gives the 30380-dimensional representation V_{ϖ_7}, so $\mathrm{V}_{\varpi_7} \in \mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_8)$.

(c) Again if $2 \in \mathrm{R}^\times$, then $\mathrm{Sym}^2(\mathrm{V}_{\varpi_8})$ is a retract of $\mathrm{V}_{\varpi_8}^{\otimes 2}$, so $\mathrm{Sym}^2(\mathrm{V}_{\varpi_8}) \in \mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_8)$. The minimal bilinear form defines a map $\mathrm{K} \to \mathrm{Sym}^2(\mathrm{V}_{\varpi_8})$ whose cofiber admits a filtration with associated graded $\mathrm{V}_{2\varpi_8} \oplus \mathrm{V}_{\varpi_1}$. Computing Casimir eigenvalues shows that this filtration splits away from characteristics 2 and 7, so that the 3875-dimensional representation V_{ϖ_1} is a retract of $\mathrm{Sym}^2(\mathrm{V}_{\varpi_8})$, and hence lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_8)$.

(d) If $6 \in \mathrm{R}^\times$, then $\wedge^3(\mathrm{V}_{\varpi_8})$ is a retract of $\mathrm{V}_{\varpi_8}^{\otimes 3}$, so $\wedge^3(\mathrm{V}_{\varpi_8}) \in \mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_8)$. The invariant cubic form defines a map $\mathrm{R} \to \wedge^3(\mathrm{V}_{\varpi_8})$ whose cofiber admits a filtration with associated graded isomorphic to $\mathrm{V}_{\varpi_1} \oplus \mathrm{V}_{2\varpi_8} \oplus \mathrm{V}_{\varpi_7} \oplus \mathrm{V}_{\varpi_6}$. Computing Casimir eigenvalues shows that this filtration splits away from characteristics $2, 3, 5, 7$, so that the 2450240-dimensional representation V_{ϖ_6} is a retract of $\wedge^3(\mathrm{V}_{\varpi_8})$, and hence lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_8)$.

(e) The tensor product $\mathrm{V}_{\varpi_7} \otimes \mathrm{V}_{\varpi_8}$ admits a filtration with associated graded isomorphic to $\mathrm{V}_{\varpi_2} \oplus \mathrm{V}_{\varpi_6} \oplus \mathrm{V}_{\varpi_7} \oplus \mathrm{V}_{\varpi_7+\varpi_8} \oplus \mathrm{V}_{\varpi_1+\varpi_8} \oplus \mathrm{V}_{\varpi_8} \oplus \mathrm{V}_{\varpi_1} \oplus \mathrm{V}_{2\varpi_8}$. Computing Casimir eigenvalues shows that this filtration splits away from characteristics $2, 3, 5, 7, 13, 31$, so that in this case the 147250-dimensional representation V_{ϖ_2} is a retract of $\mathrm{V}_{\varpi_7} \otimes \mathrm{V}_{\varpi_8}$, and hence lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_8)$.

(f) The tensor product $\mathrm{V}_{\varpi_6} \otimes \mathrm{V}_{\varpi_8}$ admits a filtration with associated graded isomorphic to $\mathrm{V}_{\varpi_6+\varpi_8} \oplus \mathrm{V}_{\varpi_7+\varpi_8} \oplus \mathrm{V}_{\varpi_2+\varpi_8} \oplus \mathrm{V}_{\varpi_1+\varpi_8} \oplus \mathrm{V}_{\varpi_1+\varpi_7} \oplus \mathrm{V}_{\varpi_7} \oplus \mathrm{V}_{\varpi_6} \oplus \mathrm{V}_{\varpi_5} \oplus \mathrm{V}_{\varpi_3} \oplus \mathrm{V}_{\varpi_2}$. Computing Casimir eigenvalues shows that this filtration splits away from characteristics $p \leq 31$ (as well as $p = 29$), so that in this case the 6696000-dimensional representation V_{ϖ_3} and the 146325270-dimensional representation V_{ϖ_5} are retracts of $\mathrm{V}_{\varpi_6} \otimes \mathrm{V}_{\varpi_8}$, and hence lie in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_8)$.

(g) The tensor product $\mathrm{V}_{\varpi_5} \otimes \mathrm{V}_{\varpi_8}$ admits a filtration with associated graded isomorphic to $\mathrm{V}_{\varpi_5+\varpi_8} \oplus \mathrm{V}_{\varpi_1+\varpi_6} \oplus \mathrm{V}_{\varpi_2+\varpi_8} \oplus \mathrm{V}_{\varpi_6+\varpi_8} \oplus \mathrm{V}_{\varpi_1+\varpi_2} \oplus \mathrm{V}_{\varpi_3+\varpi_8} \oplus \mathrm{V}_{\varpi_2+\varpi_7} \oplus \mathrm{V}_{\varpi_7} \oplus \mathrm{V}_{\varpi_6} \oplus \mathrm{V}_{\varpi_5} \oplus \mathrm{V}_{\varpi_4} \oplus \mathrm{V}_{\varpi_3}$. Computing Casimir eigenvalues shows that this filtration splits away from characteristic $p \leq 37$ (as well as $p = 23$), so that in this case the 6899079264-dimensional representation V_{ϖ_4} is a retract of $\mathrm{V}_{\varpi_5} \otimes \mathrm{V}_{\varpi_8}$, and hence lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BE}_8)$.

(8) When $\check{\mathrm{G}} = \mathrm{F}_4$, we will show that all the fundamental representations V_{ϖ_i} lie in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BF}_4)$ if $6 \in \mathrm{R}^\times$.

(a) The standard 26-dimensional representation V_{ϖ_4} is quasi-minuscule. If $\mathcal{J}$ denotes the 27-dimensional exceptional Jordan algebra (defined already over $\mathbf{Z}$), then V_{ϖ_4} is the subspace of trace zero elements of $\mathcal{J}$.

(b) If $2 \in \mathrm{R}^\times$, then $\wedge^2(\mathrm{V}_{\varpi_4})$ is a retract of $\mathrm{V}_{\varpi_4}^{\otimes 2}$, so $\wedge^2(\mathrm{V}_{\varpi_4}) \in \mathrm{Perf}^{\text{q-min}}(\mathrm{BF}_4)$. The commutator of left multiplication on the Jordan algebra $\mathcal{J}$ defines a map $\wedge^2(\mathrm{V}_{\varpi_4}) \to \mathrm{Der}(\mathcal{J})$, where $\mathrm{Der}(\mathcal{J})$ is the algebra of derivations of $\mathcal{J}$. The 52-dimensional adjoint representation V_{ϖ_1} is $\mathrm{Der}(\mathcal{J})$ (by definition of F_4), and the fiber of the map $\wedge^2(\mathrm{V}_{\varpi_4}) \to \mathrm{V}_{\varpi_1}$ is the 273-dimensional representation V_{ϖ_3}. Computing Casimir eigenvalues for V_{ϖ_1} and V_{ϖ_3} shows that the map $\wedge^2(\mathrm{V}_{\varpi_4}) \to \mathrm{V}_{\varpi_1}$ admits a splitting if K has characteristic > 3. It follows that in this case, $\mathrm{V}_{\varpi_1}, \mathrm{V}_{\varpi_3} \in \mathrm{Perf}^{\text{q-min}}(\mathrm{BF}_4)$.

(c) The adjoint action of V_{ϖ_1} on V_{ϖ_4} defines a map $\mathrm{V}_{\varpi_1} \otimes \mathrm{V}_{\varpi_4} \to \mathrm{V}_{\varpi_4}$. The fiber of this map admits a filtration with associated graded $\mathrm{V}_{\varpi_1} \oplus \mathrm{V}_{\varpi_2}$, and computing Casimir eigenvalues shows that this filtration splits if K has characteristic > 3. It follows that in this case, the 1274-dimensional representation V_{ϖ_2} lies in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BF}_4)$.

(9) When $\check{\mathrm{G}} = \mathrm{G}_2$, we will show that both the fundamental representations V_{ϖ_1} and V_{ϖ_2} lie in $\mathrm{Perf}^{\text{q-min}}(\mathrm{BG}_2)$ if $2 \in \mathrm{R}^\times$.
   (a) The standard 7-dimensional representation V_{ϖ_1} is quasi-minuscule.
   (b) If $2 \in \mathrm{R}^\times$, then $\wedge^2(\mathrm{V}_{\varpi_1})$ is a retract of $\mathrm{V}_{\varpi_1}^{\otimes 2}$, so $\wedge^2(\mathrm{V}_{\varpi_1}) \in \mathrm{Perf}^{\text{q-min}}(\mathrm{BG}_2)$. The 7-dimensional cross product defines a map $\wedge^2\mathrm{V}_{\varpi_1} \to \mathrm{V}_{\varpi_1}$ whose fiber is the adjoint representation V_{ϖ_2}. □

Remark 5.1.30. In fact, if no simple adjoint factor of $\check{\mathrm{G}}$ is isomorphic to E_8, F_4, or G_2, then the conclusion of Proposition 5.1.29 continues to hold for the fully faithful functor $\mathrm{Perf}^{\min,\mathrm{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}}) \hookrightarrow \mathrm{Perf}^{\mathrm{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}})$ (as long as one also prohibits $p = 3$ for E_6). Analyzing the proof of Proposition 5.1.29 (and the table on [Jan91, Page 299]), shows that it suffices to prove that if $6 \in \mathrm{R}^\times$, then the adjoint representation $\mathfrak{e}_6 \in \mathrm{Perf}^{\min}(\mathrm{BE}_6)$ (resp. $\mathfrak{e}_7 \in \mathrm{Perf}^{\min}(\mathrm{BE}_7)$). This can be argued as follows.

The action of $\mathrm{E}_6^{\mathrm{sc}}$ on V_{ϖ_1} defines a map $\mathfrak{e}_6^{\mathrm{sc}} \to \mathrm{End}(\mathrm{V}_{\varpi_1}) \cong \mathrm{V}_{\varpi_1} \otimes \mathrm{V}_{\varpi_6}$ (even over $\mathbf{Z}$). The linear dual gives a map $\mathrm{V}_{\varpi_1} \otimes \mathrm{V}_{\varpi_6} \to (\mathfrak{e}_6^{\mathrm{sc}})^*$, and by Remark 4.2.2, the minimal bilinear form defines an isomorphism $(\mathfrak{e}_6^{\mathrm{sc}})^* \cong \mathfrak{e}_6^{\mathrm{sc}}$ if $p \neq 3$. Assuming $p \neq 3$ and writing $\mathfrak{e}_6$ for $\mathfrak{e}_6^{\mathrm{sc}}$, we obtain a map $\mathfrak{e}_6 \to \mathrm{End}(\mathrm{V}_{\varpi_1}) \to \mathfrak{e}_6$ in $\mathrm{Perf}(\mathrm{BE}_6)$. When $p \neq 3$, the adjoint representation V_{ϖ_2} is irreducible (see [Jan91, Page 299]), and this composite is just multiplication by the image in R of an element of $\mathbf{Z}[1/3]$. This element is the ratio of the "trace form" of V_{ϖ_1} to the minimal bilinear form on $\mathfrak{e}_6$, which by definition is the Dynkin index $d_{\mathrm{V}_{\varpi_1}} \in \mathbf{Z}[1/3]$ of V_{ϖ_1}. By [LS97, Proposition 2.6], $d_{\mathrm{V}_{\varpi_1}} = 6$, so that $\mathfrak{e}_6$ is an E_6-equivariant retract of $\mathrm{V}_{\varpi_1} \otimes \mathrm{V}_{\varpi_6}$ if $6 \in \mathrm{R}^\times$.

One can argue in the same way for E_7. The adjoint representation V_{ϖ_1} is irreducible if $p \neq 2$ (see [Jan91, Page 299]). Using the Dynkin index argument as in the case of E_6 and the calculation $d_{\mathrm{V}_{\varpi_7}} = 12 \in \mathbf{Z}[1/2]$ from [LS97, Proposition 2.6], it follows that V_{ϖ_1} is a retract of $\mathrm{End}(\mathrm{V}_{\varpi_7}) \cong \mathrm{V}_{\varpi_7}^{\otimes 2}$ if $p \nmid 12$. Assuming $p \neq 2, 3$, it follows that $\mathrm{V}_{\varpi_1} \cong \mathfrak{e}_7 \in \mathrm{Perf}^{\min}(\mathrm{BE}_7)$.

Remark 5.1.31. Assume Conjecture 3.1.14 for every sequence of dominant (quasi-)minuscule coweights of G^{ad}. It follows from Theorem 5.1.16 and Proposition 5.1.29 that if p is not one of the primes in Table 2 for the simple adjoint factors of $\check{\mathrm{G}}$, there is a $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$-linear equivalence

$$\mathrm{Shv}^{\text{q-min},\mathrm{gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_{\mathrm{G}}; k) \simeq \mathrm{IndPerf}^{\mathrm{all},\mathrm{gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$$

after p-localization, such that the restriction of $\underline{\tau}_\star$ to $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)^{\text{!-ev}}$ is monoidal for the convolution monoidal structure on $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ and the pointwise tensor product on $\mathrm{IndPerf}^{\mathrm{all,gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}})$.

Example 5.1.32. Suppose $k = \mathbf{Z}'$ is the localization of $\mathbf{Z}$ at the bad primes of G. Assume Conjecture 3.1.14 for every sequence of dominant (quasi-)minuscule coweights of G^{ad}. In [NP01, Proposition 9.6], it was shown that for every dominant coweight $\lambda \in \mathbb{X}_*^+$, the corresponding IC_λ is an object of $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Z}') \otimes_{\mathbf{Z}'} \mathbf{Q}$, in the sense that it is a direct summand of $\widetilde{\mathrm{IC}}_{\mu_\bullet}$ for some sequence $\mu_\bullet$ of dominant (quasi-)minuscule coweights. It follows that the fully faithful functor $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Z}') \hookrightarrow \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Z}')$ is an equivalence upon rationalization. In particular, Example 5.1.19 implies that there is a filtered lift of $\mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Q})$ whose associated graded category $\mathrm{Shv}^{\mathrm{gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Q})$ is equivalent to $\mathrm{IndPerf}^{\mathrm{all,gr}}({}^{2\rho}\check{\mathfrak{g}}^*(2)^\wedge_\mathcal{N}/{}^{2\rho}\check{\mathrm{G}})$. This is a variant of the (renormalized) derived geometric Satake equivalence of [BF08, AG15]. (The latter references use formality to show that the claimed filtered lift of $\mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Q})$ is in fact the *constant* filtration, which implies that $\mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Q})$ is equivalent to the shearing of $\mathrm{IndPerf}^{\mathrm{all}}(\check{\mathfrak{g}}^*(2)^\wedge_\mathcal{N}/\check{\mathrm{G}})$. Note that we did not use perverse normalizations in our discussion above, so one has to be careful in comparing with [BF08, AG15].)

5.2. **The multiply-laced case.** Variants of the results of Section 4.2 and Section 5.1 continue to hold even in the multiply-laced case, as suggested by Proposition 4.2.15. We will make these explicit in this section. Many of the proofs will carry through verbatim, so we will only sketch arguments when they might differ from their analogues in Section 4.2 and Section 5.1. As usual, let $\check{\mathrm{G}}^{\mathrm{sc}}$ denote the simply-connected cover of the Langlands dual group of G. There is a simply-laced group H and a diagram automorphism σ of H such that the fixed subgroup $\mathrm{H}^\sigma \cong \check{\mathrm{G}}^{\mathrm{sc}}$ (see Table 3). Note that $\check{\mathrm{G}}$ is the quotient of H^σ by the subgroup $\pi_1^{\mathrm{alg}}(\check{\mathrm{G}}) \subseteq \mathrm{Z}(\mathrm{H}^\sigma)$. The order of σ will be denoted ℓ; when $\check{\mathrm{G}}$ is simple, this is the lacing number of $\check{\mathrm{G}}$. The data of H along with its diagram automorphism σ determines another simply-connected semisimple group $\mathrm{G} := \mathrm{H}_\sigma$, called the "coinvariant group". If $\mathrm{T} \subseteq \mathrm{H}$ is a σ-stable maximal torus, the coinvariant torus T_σ, defined as the quotient $\mathrm{T}/(1-\sigma)$, is a maximal torus of G. Note that $\mathbb{X}^*(\mathrm{T}_\sigma) \cong \mathbb{X}^*(\mathrm{T})^\sigma$. The root system of G is the dual of the root system of $\check{\mathrm{G}}$, so that the weight lattice of G is $\mathbb{X}^*(\mathrm{T})^\sigma$. See [Moh03, Spr06].

The key is the following, which crucially uses that F is a *formal* group scheme:

Lemma 5.2.1. *Fix a graded commutative ring* R*, and let* F *be a* 1*-dimensional graded formal group over* R*. There are isomorphisms* $(\mathrm{T}^\sigma)_\mathrm{F} \cong (\mathrm{T}_\mathrm{F})^\sigma$ *and* $(\mathrm{T}_\sigma)_\mathrm{F} \cong (\mathrm{T}_\mathrm{F})_\sigma$*, where* $(\mathrm{T}_\mathrm{F})_\sigma$ *is defined as the quotient* $\mathrm{T}_\mathrm{F}/(1-\sigma)$*. If* ℓ *is invertible in* R*, the norm map* $(\mathrm{T}_\mathrm{F})_\sigma \to \mathrm{T}_\mathrm{F}^\sigma$ *(which, in a choice of coordinate on* F*, sends* $x \mapsto x +_\mathrm{F} \cdots +_\mathrm{F} \sigma^{\ell-1}(x)$*) is an isomorphism.*

Proof. It is easy to see that $(\mathrm{T}^\sigma)_\mathrm{F} \cong (\mathrm{T}_\mathrm{F})^\sigma$ and $(\mathrm{T}_\sigma)_\mathrm{F} \cong (\mathrm{T}_\mathrm{F})_\sigma$. For the second claim, note that by degenerating F to its Lie algebra, we are reduced to the case $\mathrm{F} = \hat{\mathbf{G}}_a(2)$. In this case, the assertion is that the norm map $\mathfrak{t}_\sigma \to \mathfrak{t}^\sigma$ is an isomorphism. The inverse to this map is the map $\mathfrak{t}^\sigma \to \mathfrak{t}_\sigma$ sending $x \mapsto \frac{x}{\ell}$, which is well-defined since ℓ is invertible in R. □

TABLE 3. Diagram automorphisms σ of H with $\mathrm{G} = \mathrm{H}^\sigma$ simple. Here, q denotes the diagonal matrix with entries $(1, \cdots, 1, -1)$, and n denotes the antidiagonal matrix with entries $(1, -1, 1, \cdots)$. Except in the case of $\mathrm{G} = \mathrm{G}_2$, σ has order 2; and when $\mathrm{G} = \mathrm{G}_2$, σ has order 3. (There is an exceptional automorphism of $\mathrm{H} = \mathrm{SL}_{2n+1}$ of order 4 given by sending $g \mapsto \mathrm{Ad}_n(g^\mathrm{T})^{-1}$, where n is the antidiagonal matrix with entries $(1, -1, 1, \cdots)$. The fixed subgroup H^σ is SO_{2n+1}, and $\mathrm{H}_\sigma = \mathrm{Sp}_{2n}$. However, we will ignore this automorphism in the following discussion.)

H	SL_{2n}	Spin_{2n+2}	E_6	Spin_8
σ	$g \mapsto \mathrm{Ad}_n(g^\mathrm{T})^{-1}$	$g \mapsto \mathrm{Ad}_q(g)$	Duality on Cayley plane	Triality
$\check{\mathrm{G}}^{\mathrm{sc}} = \mathrm{H}^\sigma$	Sp_{2n}	Spin_{2n+1}	F_4	G_2
$\mathrm{G} = \mathrm{H}_\sigma$	Spin_{2n+1}	Sp_{2n}	F_4	G_2

Remark 5.2.2. The inverse of the norm map $(\mathrm{T_F})_\sigma \to \mathrm{T}_\mathrm{F}^\sigma$ is given (in a choice of coordinate on F) by the map $\mathrm{T}_\mathrm{F}^\sigma \to (\mathrm{T_F})_\sigma$ sending $x \mapsto [\frac{1}{\ell}]_\mathrm{F}(x)$. Here, $[\frac{1}{\ell}]_\mathrm{F}(x)$ is the unique power series $f(x)$ which solves $[\ell]_\mathrm{F}(f(x)) = x$; such a power series exists (e.g., using Hensel's lemma) because ℓ is assumed to be a unit in R. For example, if $\mathrm{F} = \hat{\mathbf{G}}_m$, then

$$[\tfrac{1}{\ell}]_\mathrm{F}(x) = (1+x)^{1/\ell} - 1 = \sum_{n\geq 1} \binom{1/\ell}{n} x^n,$$

which is indeed well-defined because $\binom{1/\ell}{n} \in \mathbf{Z}[\ell^{-1}]$.

Assume from now that ℓ is invertible in R. One can then make the same definitions as in Section 4.2 and Section 5.1:

Definition 5.2.3. Fix a σ-stable Borel $\mathrm{B} \subseteq \mathrm{H}$, and let B_σ denote the corresponding Borel subgroup of G_σ. Since the fixed point subgroup B^σ is a Borel subgroup of $\check{\mathrm{G}}^{\mathrm{sc}} = \mathrm{H}^\sigma$, we will denote its image in $\check{\mathrm{G}}$ by $\check{\mathrm{B}}$. Let $\kappa'_\mathrm{F} : (\mathrm{T_F})_\sigma \to {}^{2\rho}\check{\mathrm{B}}_\mathrm{F}$ denote the composite

$$(\mathrm{T_F})_\sigma \xrightarrow[\cong]{\mathrm{Nm}} \mathrm{T}_\mathrm{F}^\sigma \xrightarrow{\kappa_\mathrm{F}} {}^{2\rho}\check{\mathrm{B}}_\mathrm{F},$$

where κ_F is as in Construction 4.1.12. Using the action of ${}^{2\rho}\check{\mathrm{B}}$ on ${}^{2\rho}\check{\mathrm{B}}_\mathrm{F}$, we follow Definition 4.2.4 and define $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ denote the groupoid scheme over $(\mathrm{T_F})_\sigma$ given by the (*a priori* derived) fiber product

$$\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} := (\mathrm{T_F})_\sigma \times_{{}^{2\rho}\check{\mathrm{B}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{B}}} (\mathrm{T_F})_\sigma,$$

where both maps $(\mathrm{T_F})_\sigma \to {}^{2\rho}\check{\mathrm{B}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{B}}$ are given by composing $\kappa'_\mathrm{F} : (\mathrm{T_F})_\sigma \to {}^{2\rho}\check{\mathrm{B}}_\mathrm{F}$ with the quotient map ${}^{2\rho}\check{\mathrm{B}}_\mathrm{F} \to {}^{2\rho}\check{\mathrm{B}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{B}}$.

With Definition 5.2.3 in place of Construction 4.1.12 and Definition 4.2.4, the results of Section 4.2 go through unchanged. Let us state the most important ones (which are analogues of Proposition 4.1.14, Theorem 4.2.12, Proposition 4.3.2, and Theorem 4.3.6):

Proposition 5.2.4. *Let* R *be a graded commutative ring, and let* F *be a* 1*-dimensional graded formal group over* R *equipped with a coordinate. Suppose that the bad primes* *(in particular, the lacing number ℓ) of* G *are inverted in* R*. If* V *is a graded* ${}^{2\rho}\check{\mathrm{B}}$*-representation in finitely generated projective* R*-modules, then the* $\mathcal{O}_{(\mathrm{T_F})_\sigma}$*-linear map* $(\mathrm{V}\otimes_{\mathrm{R}}\mathcal{O}_{{}^{2\rho}\check{\mathrm{B}}_{\mathrm{F}}})^{{}^{2\rho}\check{\mathrm{B}}}\to(\mathrm{V}\otimes_{\mathrm{R}}\mathcal{O}_{(\mathrm{T_F})_\sigma})^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$ *on invariants is an isomorphism.*

Proposition 5.2.5. *Suppose that k is an even $\mathbf{E}_\infty$-ring, and that the bad primes of* G *are invertible in $\pi_0(k)$. Let $\mathcal{W}_\sigma$ denote the groupoid scheme* $\mathrm{Spf}\,\mathrm{H}^{-*}_{\mathrm{T}_\sigma,c}(\mathrm{G}/\mathrm{B}_\sigma;k)$ *over* $\pi_*(k^{h\mathrm{T}_\sigma})\cong\mathcal{O}_{(\mathrm{T}_\sigma)_\mathrm{F}}$*. Then there is a $\mathcal{W}_\sigma$-equivariant isomorphism* $\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}\cong\mathrm{H}^{\mathrm{T}_\sigma,\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G};k)$ *of Hopf* $\pi_*(k^{h\mathrm{T}_\sigma})\cong\mathcal{O}_{(\mathrm{T}_\sigma)_\mathrm{F}}$*-algebras such that the following diagram commutes:*

$$\begin{tikzcd}
\mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{T}},\mathrm{F}}} \arrow[r, hook] \arrow[d, "\cong"] & \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}} \arrow[r, hook] \arrow[d, "\cong"] & \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{T}},\mathrm{F}}}[e(\Phi)^{-1}] \arrow[d, "\cong"] \\
\mathrm{H}^{\mathrm{T}_\sigma,\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}_\sigma};k) \arrow[r, hook] & \mathrm{H}^{\mathrm{T}_\sigma,\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G};k) \arrow[r, hook] & \mathrm{H}^{\mathrm{T}_\sigma,\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}_\sigma};k)[e(\Phi)^{-1}].
\end{tikzcd}$$

Proposition 5.2.6. *Let* R *be a graded commutative ring, and let* F *be a 1-dimensional graded formal group over* R *equipped with a coordinate. Let* ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}={}^{2\rho}\check{\mathrm{G}}\times^{{}^{2\rho}\check{\mathrm{B}}}{}^{2\rho}\check{\mathrm{B}}_\mathrm{F}$*. Suppose that the bad primes of* G *are inverted in* R*. Then the composite map* ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}}\to(\mathrm{T_F})_\sigma\to(\mathrm{T_F})_\sigma/\mathcal{W}_\sigma$ *factors through the natural map* ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}}\to{}^{2\rho}\check{\mathrm{G}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}}$*, and the resulting map* $q:{}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}\to{}^{2\rho}\check{\mathrm{G}}_\mathrm{F}\times_{(\mathrm{T_F})_\sigma/\mathcal{W}}(\mathrm{T_F})_\sigma$ *exhibits* ${}^{2\rho}\check{\mathrm{G}}_\mathrm{F}\times_{(\mathrm{T_F})_\sigma/\mathcal{W}_\sigma}(\mathrm{T_F})_\sigma$ *as the affinization of* ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_\mathrm{F}$*. Moreover, the composite map*

$$(\mathrm{T_F})_\sigma\xrightarrow{\kappa'_\mathrm{F}}{}^{2\rho}\check{\mathrm{B}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{B}}\to{}^{2\rho}\check{\mathrm{G}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}}$$

admits a factorization through the stacky quotient $(\mathrm{T_F})_\sigma/\mathcal{W}_\sigma$*, and the composite*

$$(\mathrm{T_F})_\sigma/\mathcal{W}_\sigma\xrightarrow{\kappa'_\mathrm{F}}{}^{2\rho}\check{\mathrm{G}}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}}\to(\mathrm{T_F})_\sigma/\mathcal{W}_\sigma$$

is an isomorphism.

For any sequence $\mu_\bullet$ of minuscule dominant coweights of G_{ad}, one can define $\mathrm{IC}_{\mu_\bullet}\in\mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G};k)$ and $\mathrm{V}_{\mu_\bullet}\in\mathrm{Perf}^{\mathrm{gr}}(\mathrm{B}^{2\rho}\check{\mathrm{G}}^{\mathrm{sc}})$ exactly as in Definition 5.1.2, and Proposition 5.1.4 continues to hold:

Proposition 5.2.7. *Suppose that k is an even $\mathbf{E}_\infty$-ring, and that the bad primes of* G *are invertible in $\pi_0(k)$. Under the isomorphism* $\mathrm{Spec}(\mathrm{H}^{\mathrm{T}_\sigma,\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G};k))\cong\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ *of Proposition 5.2.5, the* $\mathrm{H}^{\mathrm{T}_\sigma,\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G};k)$*-comodule* $\mathrm{H}^{-*}_{\mathrm{T}_\sigma,c}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathrm{IC}_{\mu_\bullet})$ *is $\mathcal{W}_\sigma$-equivariantly isomorphic to the* $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$*-module* $\mathrm{Res}^{\check{\mathrm{G}}}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}(\underline{\mathrm{V}}_{\mu_\bullet})$*.*

We now discuss an analogue of Proposition 5.1.9. Here, some care is required because our analysis in Proposition 5.1.9 used the fact that in the simply-laced case, the quasi-minuscule dominant weight is the highest root of $\check{\mathfrak{g}}$. This statement fails when G is multiply-laced. Let $\theta\in\mathbb{X}^+_*$ denote the quasi-minuscule dominant weight of $\check{\mathrm{G}}^{\mathrm{sc}}$, i.e., the highest short coroot of G_{ad}, and like in Definition 5.1.6, let $\widetilde{\mathrm{IC}}_\theta\in\mathrm{Shv}_{(\mathrm{G}_{\mathrm{ad}})_\mathcal{O}}(\mathcal{G}\mathrm{r}_{\mathrm{G}_{\mathrm{ad}}};k)$ denote the pushforward $f_*\underline{k}_{\mathbf{P}(\mathcal{L}_\theta\oplus\mathcal{O})}$ where $f:\mathbf{P}(\mathcal{L}_\theta\oplus\mathcal{O})\to\mathcal{G}\mathrm{r}^{\leq\theta}_{\mathrm{G}_{\mathrm{ad}}}$ is a resolution of singularities which contracts $\mathrm{G}/\mathrm{P}_\theta$ to the basepoint $\mathcal{G}\mathrm{r}^0_{\mathrm{G}_{\mathrm{ad}}}$ (see [NP01] and [Zhu17, Lemma 2.1.14]). As in Section 5.1, we will write $\widetilde{\mathrm{IC}}_\theta$ for the pullback of $\widetilde{\mathrm{IC}}_\theta\in\mathrm{Shv}_{(\mathrm{G}_{\mathrm{ad}})_\mathcal{O}}(\mathcal{G}\mathrm{r}_{\mathrm{G}_{\mathrm{ad}}};k)$ along the map $\mathcal{G}\mathrm{r}_\mathrm{G}\to\mathcal{G}\mathrm{r}_{\mathrm{G}_{\mathrm{ad}}}$ (note that this pullback may be zero). As in Construction 5.1.8, let T_θ denote the tilting module (defined over $\mathbf{Z}$; see [Jan03, Section E.19]) with highest weight θ. Let $\check{\Phi}_{\mathrm{short}}$ denote the set of short coroots. Let V_0 be a free (graded) $\mathbf{Z}$-module such that $\widetilde{\mathrm{T}}_\theta := \mathrm{V}_0 \oplus \mathrm{T}_\theta$ is a free $\mathbf{Z}$-module of rank $2|\check{\Phi}_{\mathrm{short}}|$, and view $\widetilde{\mathrm{T}}_\theta$ as a $\check{\mathrm{G}}^{\mathrm{sc}}$-representation via the trivial action on V_0. Then there is a $\mathcal{W}_\theta$-action on $\widetilde{\mathrm{T}}_\theta$ defined as in Construction 5.1.8, using $\mathrm{E}_f = \check{\Phi}_{\mathrm{short}}$, $\mathrm{E}_b = \check{\Phi}_{\mathrm{short}} \times \Phi_\theta^+ \times \{0, \infty\}$, and the diagram $\mathrm{E}_f \sqcup \mathrm{E}_b \rightrightarrows \check{\Phi}_{\mathrm{short}} \times \{0,\infty\}$.

Proposition 5.2.8. *Suppose that k is an even $\mathbf{E}_\infty$-ring, and that the bad primes of* G *are invertible in $\pi_0(k)$. Then the $\mathrm{H}_*^{\mathrm{T}_\sigma,\mathrm{BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$-comodule $\mathrm{H}^{-*}_{\mathrm{T}_\sigma,c}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta)$ is $\mathcal{W}_\theta$-equivariantly isomorphic to the $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$-module $\mathrm{Res}^{\check{\mathrm{G}}}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}(\underline{\widetilde{\mathrm{T}}}_\theta)$ under the isomorphism of Proposition 5.2.5.*

Proof sketch. The argument of Proposition 5.1.9 goes through unchanged, except that one needs to analyze the indecomposable tilting summands of $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{L}}_\alpha}(\widetilde{\mathrm{T}}_\theta)$, where α is a root of G and L_α is the Levi subgroup of G generated by the root spaces $\mathrm{U}_{\pm\alpha}$. In particular, we need to show that if $(\mathrm{SL}_2)_\alpha \to \check{\mathrm{L}}_\alpha$ is the map given by the simply-connected cover of the derived subgroup of $\check{\mathrm{L}}_\alpha$, then the restriction $\mathrm{Res}^{\check{\mathrm{G}}}_{(\mathrm{SL}_2)_\alpha}(\widetilde{\mathrm{T}}_\theta)$ is equivariantly isomorphic to a direct sum of copies of the trivial representation (which appear in pairs), copies of the standard representation V_{ϖ_1} (which appear in pairs), and a single copy of $\mathrm{V}_{\varpi_1}^{\otimes 2}$. Moreover, these must appear with the same multiplicities as the (unions of) components $\{0,\infty\}$, $\mathbf{P}^1 \sqcup \mathbf{P}^1$, and $\mathbf{P}(\mathcal{O}(2) \oplus \mathcal{O})$ of the fixed subspace $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{(\mathrm{T}_\sigma)^\alpha}$. As in Proposition 5.1.9, this will then give the $\widetilde{\mathrm{J}}_{\check{\mathrm{L}}_\alpha,\mathrm{F}}$-equivariant isomorphism $\mathrm{Res}^{\check{\mathrm{G}}}_{\widetilde{\mathrm{J}}_{\check{\mathrm{L}}_\alpha}}(\underline{\widetilde{\mathrm{T}}}_\theta) \cong \mathrm{H}^{-*}_{\mathrm{T}_\sigma,c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{(\mathrm{T}_\sigma)^\alpha}; k)$.

To do this, we need to analyze $\mathrm{Res}^{\check{\mathrm{G}}}_{(\mathrm{SL}_2)_\alpha}(\mathrm{T}_\theta)$. Note that the only weights appearing in T_θ are 0 and the short roots of $\check{\mathrm{G}}$. If μ is a short root and α is a long root, then $\langle \mu, \check{\alpha}\rangle \in \{-1,0,1\}$, so $\mathrm{Res}^{\check{\mathrm{G}}}_{(\mathrm{SL}_2)_\alpha}(\mathrm{T}_\theta)$ is a direct sum of copies of the trivial representation and the standard representation V_{ϖ_1} of SL_2 (corresponding to whether $\langle \mu, \check{\alpha}\rangle$ is 0 or 1). If, however, α is a short root, then still $\langle \mu, \check{\alpha}\rangle \in \{-1,0,1\}$ unless $\mu = \pm\alpha$ (in which case $\langle \mu, \check{\alpha}\rangle = \pm 2$). It follows that $\mathrm{Res}^{\check{\mathrm{G}}}_{(\mathrm{SL}_2)_\alpha}(\mathrm{T}_\theta)$ is a direct sum of copies of the trivial representation and the standard representation V_{ϖ_1} of SL_2, and a single copy of the tilting representation $\mathrm{V}_{\varpi_1}^{\otimes 2}$ for SL_2. Counting the multiplicities of these indecomposable summands shows that they appear with the same multiplicities as the (unions of) components $\{0,\infty\}$, $\mathbf{P}^1 \sqcup \mathbf{P}^1$, and $\mathbf{P}(\mathcal{O}(2) \oplus \mathcal{O})$ of the fixed subspace $\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{(\mathrm{T}_\sigma)^\alpha}$, as desired. □

With notation as in Definition 5.1.13 and Definition 5.1.14, the argument of Theorem 5.1.16 leads to the following:

Theorem 5.2.9. *Suppose that k is an even $\mathbf{E}_\infty$-ring, and that the bad primes of* G *are invertible in $\pi_0(k)$. Assume Conjecture 3.1.14 for every sequence of dominant (quasi-)minuscule coweights of G^{ad}. Then there is a $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$-linear equivalence*

$$\mathcal{S} : \mathrm{Shv}^{\text{q-min},\mathrm{gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \xrightarrow{\simeq} \mathrm{IndPerf}^{\text{q-min},\mathrm{gr}}({}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}),$$

where the right-hand side is $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$-linear via the map ${}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}} \to (\mathrm{T}_{\mathrm{F}})_\sigma/\mathcal{W}_\sigma$ of Proposition 5.2.6 and the isomorphism $\mathrm{R}\Gamma((\mathrm{T}_{\mathrm{F}})_\sigma/\mathcal{W}_\sigma; \mathcal{O}) \cong \mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$ of Definition 2.2.28.

The functor $\underline{\mathcal{T}}_\star : \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \to \mathrm{IndPerf}^{\text{q-min},\mathrm{gr}}({}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$ sends $\widetilde{\mathrm{IC}}_{\mu_\bullet} \mapsto \widetilde{\mathrm{T}}_{\mu_\bullet} \otimes_{\mathbf{Z}} \mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}}$ for every sequence $\mu_\bullet$ of dominant (quasi-)minuscule coweights of

G^{ad}. *In particular,* $\underline{\mathcal{T}}_\star$ *sends* IC_0 *(the skyscraper sheaf at the basepoint of* $\mathcal{G}\mathrm{r}_\mathrm{G}$*) to the structure sheaf* $\mathcal{O}_{^{2\rho}\check{\mathrm{G}}_\mathrm{F}/^{2\rho}\check{\mathrm{G}}}$. *Moreover, the restriction of* $\underline{\mathcal{T}}_\star$ *to* $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)^{!\text{-ev}}$ *is monoidal for the convolution monoidal structure on* $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)^{!\text{-ev}}$ *and the pointwise tensor product on* $\mathrm{IndPerf}^{\text{q-min,gr}}(^{2\rho}\check{\mathrm{G}}_\mathrm{F}/^{2\rho}\check{\mathrm{G}})$. *There is also a commutative diagram*

$$\begin{tikzcd}
\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)^{!\text{-ev}} \arrow[rr, "\underline{\mathcal{T}}_\star"] \arrow[d, "\mathrm{gr}^\star_\mathrm{G} \mathrm{R}\Gamma_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; -)"] & & \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \arrow[d, "\mathcal{S}"] \\
\mathrm{Mod}^{\mathrm{gr}}_{\mathrm{gr}^\star_{\check{\mathrm{G}}\text{-ev}}(k^{h\mathrm{G}})} & \mathrm{IndPerf}^{\mathrm{gr}}((\mathrm{T}_\mathrm{F})_\sigma/\mathcal{W}_\sigma) \arrow[l, "\mathrm{R}\Gamma((\mathrm{T}_\mathrm{F})_\sigma/\mathcal{W}_\sigma; -)"'] & \mathrm{IndPerf}^{\text{q-min,gr}}(^{2\rho}\check{\mathrm{G}}_\mathrm{F}/^{2\rho}\check{\mathrm{G}}) \arrow[l, "\kappa_\mathrm{F}^*"]
\end{tikzcd}$$

where $\kappa_\mathrm{F} : (\mathrm{T}_\mathrm{F})_\sigma/\mathcal{W}_\sigma \to {}^{2\rho}\check{\mathrm{G}}_\mathrm{F}/^{2\rho}\check{\mathrm{G}}$ *is as in Proposition 5.2.6.*

Remark 5.2.10. Just like Remark 5.1.21, the image under the functor $\mathcal{S}\circ\underline{\mathcal{T}}_\star$ (which I will simply denote by $\mathcal{S}$) of $\mathcal{F} \in \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)^{!\text{-ev}}$ is the image under the localization $\mathrm{IndPerf}^{\mathrm{gr}}(^{2\rho}\mathrm{G}_\mathrm{F}/^{2\rho}\check{\mathrm{G}}) \to \mathrm{IndPerf}^{\text{q-min,gr}}(^{2\rho}\check{\mathrm{G}}_\mathrm{F}/^{2\rho}\check{\mathrm{G}})$ of the $^{2\rho}\check{\mathrm{G}}$-equivariant $\mathcal{O}_{^{2\rho}\check{\mathrm{G}}_\mathrm{F}}$-module $\mathcal{S}(\mathcal{F}) := ((\mathcal{O}_{^{2\rho}\check{\mathrm{G}}} \otimes_{\pi_*(k)} \mathrm{H}^{-*}_{\mathrm{T}_\sigma}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathcal{F}))^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}})^{h\mathcal{W}_\sigma}$. We will often write $\mathcal{S}(\mathcal{F})$ to denote the above object of $\mathrm{IndPerf}^{\mathrm{gr}}(^{2\rho}\check{\mathrm{G}}_\mathrm{F}/^{2\rho}\check{\mathrm{G}})$.

Similarly, just like with Remark 5.1.23, with the definition of $\mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ from Definition 2.4.15, Theorem 5.2.9 continues to hold verbatim when k is not necessarily an even $\mathbf{E}_\infty$-ring. The key observation is that the equivalence of Theorem 5.1.16 does not depend on the choice of a coordinate on F, and (therefore) is functorial in the coefficient even $\mathbf{E}_\infty$-ring.

The obvious analogue of Proposition 5.1.27 also continues to hold: let $\mathcal{F} \in \mathrm{Shv}^{\text{toral}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)^{!\text{-ev}}$, and let $\mathcal{S}'(\mathcal{F}) = (\mathcal{O}_{^{2\rho}\check{\mathrm{G}}} \otimes_{\pi_*(k)} \mathrm{H}^*_{\mathrm{T}_\sigma}(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathcal{F}))^{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$. If $\mathcal{S}'(\mathcal{F})$ admits a pro-good filtration as a $^{2\rho}\check{\mathrm{G}}$-module, then the natural map $\mathcal{F}^{\text{q-min}} \to \mathcal{F}$ induces an isomorphism on global sections. Moreover, if $\mu_\bullet$ is a sequence of dominant (quasi-)minuscule coweights of G^{ad}, there is an isomorphism

$$\mathrm{gr}^\star_\mathrm{G}(\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)}(\widetilde{\mathrm{IC}}_{\mu_\bullet}, \mathcal{F})) \cong \mathrm{RHom}_{\mathrm{Perf}^{\mathrm{all,gr}}(^{2\rho}\check{\mathrm{G}}_\mathrm{F}/^{2\rho}\check{\mathrm{G}})}(\widetilde{\mathrm{T}}_{\mu_\bullet} \otimes_\mathbf{Z} \mathcal{O}_{^{2\rho}\check{\mathrm{G}}_\mathrm{F}}, \mathcal{S}(\mathcal{F})),$$

where $\mathcal{S}(\mathcal{F}) = (\mathcal{S}'(\mathcal{F}))^{h\mathcal{W}_\sigma}$.

Finally, Remark 5.1.25 also holds with the obvious modifications.

Remark 5.2.11. Like Corollary 5.1.17, there is a $\pi_*(k)$-linear equivalence

$$(5.2.1)\quad \mathcal{S}_0 : \pi_*(k) \otimes_{\mathrm{gr}^\star_{\check{\mathrm{G}}\text{-ev}}(k^{h\mathrm{G}})} \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \xrightarrow{\simeq} \mathrm{IndPerf}^{\text{q-min,gr}}(^{2\rho}\mathcal{N}_\mathrm{F}/^{2\rho}\check{\mathrm{G}}),$$

where $^{2\rho}\mathcal{N}_\mathrm{F}$ is defined as the derived fiber of the map $^{2\rho}\check{\mathrm{G}}_\mathrm{F} \to \check{\mathrm{T}}_\mathrm{F}/\mathcal{W}$ of Theorem 4.3.6 along the map $\mathrm{Spec}(\pi_*(k)) \xrightarrow{0} \check{\mathrm{T}}_\mathrm{F} \to \check{\mathrm{T}}_\mathrm{F}/\mathcal{W}$. As in Remark 5.1.18, the (*a priori* derived) scheme $^{2\rho}\mathcal{N}_\mathrm{F}$ is in fact flat over $\pi_*(k)$ (and hence is underived). It is the $^{2\rho}\check{\mathrm{G}}$-orbit of $^{2\rho}\mathrm{U}_\mathrm{F} \hookrightarrow {}^{2\rho}\mathrm{G}_\mathrm{F}$, $^{2\rho}\mathrm{U}_\mathrm{F}$ is the affinization of $(^{2\rho}\check{\mathrm{B}} \times \mathrm{Spec}(\pi_*(k)))/\mathrm{Spec}(\mathrm{H}^{\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G}; k))$, and $^{2\rho}\mathcal{N}_\mathrm{F}$ is the affinization of the quotient stack $(^{2\rho}\check{\mathrm{G}} \times \mathrm{Spec}(\pi_*(k)))/\mathrm{Spec}(\mathrm{H}^{\mathrm{BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G}; k))$.

Remark 5.2.12. Theorem 5.2.9 is somewhat unsatisfying because we used Lemma 5.2.1. When k is an even $\mathbf{Z}$-algebra, for example, the right-hand side of the equivalence of Theorem 5.2.9 is $\mathrm{IndPerf}^{\text{q-min,gr}}$ of $^{2\rho}\check{\mathfrak{g}}(2)^\wedge_{\mathcal{N}}/^{2\rho}\check{\mathrm{G}}$. This stack is, indeed, isomorphic to $^{2\rho}\check{\mathfrak{g}}^*(2)^\wedge_{\check{\mathcal{N}}}/^{2\rho}\check{\mathrm{G}}$: since we have inverted the lacing number, Remark 4.2.2 does identify $^{2\rho}\check{\mathfrak{g}}^* \cong {}^{2\rho}\check{\mathfrak{g}}$ via the minimal bilinear form.

However, unlike with Theorem 5.1.16 (see Footnote 12), one should not expect Theorem 5.2.9 to hold if the bad primes of G are not inverted in $\pi_0(k)$, or if $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ is replaced by a *genuine* equivariant analogue of the category of sheaves. For instance, already when $k = \mathbf{F}_\ell$ where ℓ is the lacing number of G (which we assume to be multiply-laced, so $\ell \in \{2,3\}$), Proposition 4.2.15 suggests that if the stack ${}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}} \cong {}^{2\rho}\check{\mathfrak{g}}(2)^{\wedge}_{\check{\mathcal{N}}}/{}^{2\rho}\check{\mathrm{G}}$ in Theorem 5.2.9 is replaced by the non-isomorphic stack ${}^{2\rho}\check{\mathfrak{g}}^*(2)^{\wedge}_{\check{\mathcal{N}}}/{}^{2\rho}\check{\mathrm{G}}$, then Theorem 5.2.9 might continue to hold with coefficients in k (and more generally with coefficients in an arbitrary commutative ring). This discussion suggests that Theorem 5.2.9 might continue to hold with arbitrary coefficient $\mathbf{E}_\infty$-rings (and perhaps also with genuine equivariance) if the stack ${}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ is replaced by an appropriate variant, although at the moment I do not have a clean definition of this object.

For instance, let $\mathrm{KU}' = \mathrm{KU}[\ell^{-1}]$, and consider a genuine equivariant analogue of $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathrm{KU}')$, which I will denote $\mathrm{Shv}^{\mathrm{gen}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathrm{KU}')$. (This does not seem to exist in the literature, but is expected to be constructed in forthcoming work of Konovalov-Perunov-Prikhodko and Cnossen-Maegawa-Volpe.) In the case $k = \mathrm{KU}'$, genuine equivariance has the effect of replacing the formal group $\mathrm{F} \cong \hat{\mathbf{G}}_m$ over $\pi_*(\mathrm{KU}') \cong \mathbf{Z}[\ell^{-1}, \beta^{\pm 1}]$ by the honest multiplicative group $\mathbf{G}_m$. Taking $\mathrm{F} = \mathbf{G}_m$, it is easy to see that $\mathrm{G}_{\mathrm{F}} \cong \mathrm{G}$ (as opposed to $\mathrm{G}^{\wedge}_{\mathcal{U}}$ for $\mathrm{F} = \hat{\mathbf{G}}_m$). In this case, it is easy to see that ${}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}} \cong {}^{2\rho}\check{\mathrm{G}}/{}^{2\rho}\check{\mathrm{G}}$ *cannot* be the correct Langlands object describing $\mathrm{Shv}^{\mathrm{gen,q\text{-}min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathrm{KU}')$. However, using the works [Spr06, Moh03] and expected properties of the theory of genuine equivariant sheaves, it can be shown with similar methods that if G is simply-connected (for simplicity), there is an equivalence

$$\mathcal{S} : \mathrm{Shv}^{\mathrm{gen,q\text{-}min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathrm{KU}') \xrightarrow{\simeq} \mathrm{IndPerf}^{\mathrm{q\text{-}min,gr}}(\mathrm{H}/_{\sigma}\mathrm{H}'),$$

where $\mathrm{H}' = \mathrm{H}/\mathrm{Z}(\mathrm{H}^\sigma)$ and $\mathrm{H}/_{\sigma}\mathrm{H}'$ denotes the stacky quotient of H by the σ-twisted adjoint action of H'. Note that $\mathrm{H}/_{\sigma}\mathrm{H} \cong \mathrm{Map}(\mathrm{BF}^\vee, \mathrm{BH})^{\mathbf{Z}/\ell}$, where $\mathbf{Z}/\ell$ acts on $\mathrm{BF}^\vee \cong \mathrm{B}\mathbf{Z} = \mathrm{S}^1$ by rotation, and on BH by σ. Similarly, suppose $\mathrm{ku}' = \mathrm{ku}[\ell^{-1}]$, and let $\mathfrak{H}_\beta$ denotes the deformation to the normal cone of the embedding $\mathrm{H}/\mathrm{H}^\sigma \hookrightarrow \mathrm{H}$ sending $x \mapsto x\sigma(x)^{-1}$ viewed as a scheme over $\pi_*\mathrm{ku}' = \mathbf{Z}[\ell^{-1}, \beta]$. Again using expected properties of the theory of genuine equivariant sheaves, it can be shown with similar methods that there is an equivalence

$$\mathcal{S} : \mathrm{Shv}^{\mathrm{gen,q\text{-}min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathrm{ku}') \xrightarrow{\simeq} \mathrm{IndPerf}^{\mathrm{q\text{-}min,gr}}(\mathfrak{H}_\beta/_{\sigma}\mathrm{H}').$$

Upon inverting β, this recovers the equivalence with KU'-coefficients; when $\beta = 0$ (which corresponds to $\mathbf{Z}[\ell^{-1}]$-coefficients), $\mathfrak{H}_\beta$ specializes to $\mathrm{H} \times^{\mathrm{H}^\sigma} \mathfrak{h}^\sigma(2)$, so that the $\beta = 0$ fiber of $\mathfrak{H}_\beta/_{\sigma}\mathrm{H}'$ is isomorphic to $\check{\mathfrak{g}}(2)/\check{\mathrm{G}}$. Moreover, Borel-completing these equivalences recovers Theorem 5.2.9.

5.3. **Loop-rotation equivariance for tori.** When G is a torus T and k is an even $\mathbf{E}_\infty$-ring, Theorem 5.1.16 (or rather, its analogue for connected reductive groups with simply-laced derived subgroup) states that there is a $\pi_*(k^{h\mathrm{T}})$-linear equivalence

$$\mathcal{S} : \mathrm{Shv}^{\mathrm{gr}}_{\mathrm{T}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; k) \xrightarrow{\simeq} \mathrm{IndPerf}^{\mathrm{q\text{-}min,gr}}(\mathrm{T}_{\mathrm{F}}/\check{\mathrm{T}}), \tag{5.3.1}$$

where $\mathrm{Shv}^{\mathrm{gr}}_{\mathrm{T}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; k) = \langle \mathcal{P} \rangle_{\mathrm{gr}}$ is defined as in Construction 2.4.2 with $\mathcal{P}$ being the constant sheaf on $\mathcal{G}\mathrm{r}_{\mathrm{T}}$. In this section, we will explain the analogue of the above equivalence for the category $\mathrm{Shv}_{\mathrm{T}_{\mathcal{O}} \rtimes \mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; k)$; here, $\mathbf{T} \cong \mathbf{C}^\times$, acting by loop-rotation. The case of nonabelian G will be treated in future work. For notational simplicity, I will write $\widetilde{\mathrm{T}}$ below to denote $\mathrm{T}\times\mathbf{T}$. We will allow k to have a nontrivial $\mathbf{T}$-action, so that there is a class $\hbar\in\pi_{-2}(k^{h\mathbf{T}})$ such that $\pi_*(k^{h\mathbf{T}})/\hbar\cong\pi_*(k)$. Note, however, that the natural map $k^{h\mathbf{T}}\to k$ need not admit a splitting unless the $\mathbf{T}$-action on k is trivializable. Let us write $\mathbf{T}_{\mathrm{F}}$ to denote $\operatorname{Spf}\pi_*(k^{h\mathbf{T}})$, so that $\hbar$ defines a divisor $\operatorname{Spec}(\pi_*(k))\hookrightarrow\mathbf{T}_{\mathrm{F}}$. In the following, we will write $\widetilde{\mathrm{T}}_{\mathrm{F}}$ to denote $\operatorname{Spf}\pi_*((k^{h\mathrm{T}_{\mathcal{O}}})^{h\mathbf{T}})\cong\operatorname{Spf}\pi_*((k^{h\mathbf{T}})^{h\mathrm{T}})$, so that $\widetilde{\mathrm{T}}_{\mathrm{F}}$ is a formal group scheme over $\mathbf{T}_{\mathrm{F}}$ of relative dimension equal to $\operatorname{rank}(\mathrm{T})$.

The following is taken from [Dev26] (see also [DM23]):

Definition 5.3.1. Let $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$ denote the graded $\mathcal{O}_{\mathbf{T}_{\mathrm{F}}}$-algebra defined by $\mathcal{O}_{\widetilde{\mathrm{T}}_{\mathrm{F}}}\{x_\lambda\}_{\lambda\in\mathbb{X}^*(\check{\mathrm{T}})}$ subject to the relations

$$x_\lambda x_\mu=x_{\lambda+\mu},\ \ yx_\lambda=x_\lambda(y+_{\mathrm{F}}\lambda^*(y)).$$

Here, $y\in\mathcal{O}_{\mathrm{T}_{\mathrm{F}}}$, and $\lambda^*(y)$ is the pullback of y under the map $\mathbf{T}_{\mathrm{F}}\to\mathrm{T}_{\mathrm{F}}$ induced by λ. This will be called the algebra of F-*differential operators* on $\check{\mathrm{T}}$; see Example 5.3.5 for some examples. Let $\mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}})$ denote the category of graded $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$-modules. The diagonal map on $\mathbb{X}_*\cong\mathbb{X}^*(\check{\mathrm{T}})$ equips $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$ with the structure of a Hopf algebroid over $\mathcal{O}_{\widetilde{\mathrm{T}}_{\mathrm{F}}}$, as well as a left and right action of $\check{\mathrm{T}}$ on $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$. Note that $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}\otimes_{\mathcal{O}_{\mathbf{T}_{\mathrm{F}}}}\pi_*(k)\cong\mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}}$, so that $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$ may be viewed as a "deformation quantization" of $\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}$ equipped with the Poisson structure of Construction 3.3.1.

There is an action of $\check{\mathrm{T}}\times\check{\mathrm{T}}$ on $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$ by left and right multiplication, and we will write $\mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}})^{(\check{\mathrm{T}}\times\check{\mathrm{T}},\mathrm{wk})}$ to denote $\mathrm{LMod}_{\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}}(\mathrm{IndPerf}(\mathrm{B}\check{\mathrm{T}}\times\mathrm{B}\check{\mathrm{T}}))$. Similarly, if we consider only the left or right $\check{\mathrm{T}}$-action (which are the same, because $\check{\mathrm{T}}$ is commutative), we define $\mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}})^{(\check{\mathrm{T}},\mathrm{wk})}$ to be $\mathrm{LMod}_{\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}}(\mathrm{IndPerf}(\mathrm{B}\check{\mathrm{T}}))$. Note that $\mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}})^{(\check{\mathrm{T}},\mathrm{wk})}\simeq\mathrm{IndPerf}^{\mathrm{gr}}(\widetilde{\mathrm{T}}_{\mathrm{F}})$, and similarly $\mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}})^{(\check{\mathrm{T}}\times\check{\mathrm{T}},\mathrm{wk})}\simeq\mathrm{IndPerf}^{\mathrm{gr}}(\widetilde{\mathrm{T}}_{\mathrm{F}}\times\mathrm{B}\check{\mathrm{T}})$.

Note that $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$ does not depend on a choice of Euler class $\hbar\in\pi_{-2}(k^{h\mathbf{T}})$. The key calculation we need below is the following analogue of Theorem 4.2.12:

Proposition 5.3.2. *There is an isomorphism $\mathrm{H}^{\widetilde{\mathrm{T}}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)\cong\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$ of cocommutative Hopf $\pi_*(k^{h\mathbf{T}})$-algebroids.*

Proof. There is an isomorphism $\mathrm{H}^{\widetilde{\mathrm{T}}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)\cong\mathcal{O}_{\widetilde{\mathrm{T}}_{\mathrm{F}}}[\mathbb{X}_*]$ of cocommutative $\pi_*(k^{h\mathbf{T}})$-coalgebroids, because $\mathcal{G}\mathrm{r}_{\mathrm{T}}\cong\mathbb{X}_*$ and the coproduct on $\mathrm{H}^{\widetilde{\mathrm{T}}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)$ is given by the diagonal on $\mathbb{X}_*$. It therefore suffices to check that this isomorphism respects the algebra structure. For $\lambda\in\mathbb{X}_*$, the map $\mathbb{X}_*\to\mathbb{X}_*$ given by multiplication by λ is $\widetilde{\mathrm{T}}$-equivariant for the homomorphism $\widetilde{\mathrm{T}}\to\widetilde{\mathrm{T}}$ sending $(t,\theta)\mapsto(t\lambda(\theta),\theta)$. The induced map $\pi_*(k^{h\widetilde{\mathrm{T}}})\to\pi_*(k^{h\widetilde{\mathrm{T}}})$ sends $(y,\hbar)\mapsto(y,\hbar+_{\mathrm{F}}\lambda^*(y))$. This implies that $x_\lambda^{-1}yx_\lambda$ is equal to $y+_{\mathrm{F}}\lambda^*(y)$, as desired. $\square$

Theorem 5.3.3. *There is a monoidal $\pi_*(k^{h\mathbf{T}})$-linear equivalence*

$$\mathcal{S}:\mathrm{Shv}^{\mathrm{gr}}_{\mathrm{T}_{\mathcal{O}}\rtimes\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)\xrightarrow{\simeq}\mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}})^{(\check{\mathrm{T}}\times\check{\mathrm{T}},\mathrm{wk})}.$$

If $i_\mu:\{t^\mu\}\to\mathcal{G}\mathrm{r}_{\mathrm{T}}$ is the inclusion of the point corresponding to $\mu\in\mathbb{X}_*$, then $\mathcal{S}$ sends $(i_\mu)_*\underline{k}_{t^\mu}$ to $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}\otimes\mathbf{Z}_{0,\mu}$, where $\mathbf{Z}_{0,\mu}$ denotes the $\check{\mathrm{T}}\times\check{\mathrm{T}}$-representation with character $(0,\mu)$. Moreover, this equivalence recovers (5.3.1) *upon base-change along the map $\pi_*(k^{h\mathbf{T}})\to\pi_*(k)$.*

Proof. The category $\mathrm{Shv}_{\mathrm{T}_{\mathcal{O}}\rtimes\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)$ is generated by the objects $\underline{k}_\mu := (i_\mu)_*\underline{k}_{t^\mu}$. Theorem 2.3.2 (see Remark 2.3.6) implies that

$$\pi_* \mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}_{\mathcal{O}}\rtimes\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)}(\underline{k}_\mu, \underline{k}_\lambda) \cong \mathrm{Hom}^\star_{\mathcal{C}^\heartsuit}(\mathrm{H}^*_{\widetilde{\mathrm{T}}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; \underline{k}_\mu), \mathrm{H}^*_{\widetilde{\mathrm{T}}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; \underline{k}_\lambda)),$$

where $\mathcal{C} = \mathrm{coMod}_{\mathrm{H}^{\widetilde{\mathrm{T}}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}};k)}(\mathrm{Mod}^{\mathrm{gr}}_{\mathrm{H}^*_{\widetilde{\mathrm{T}}}(*;k)})$. It is easy to see that

$$\mathrm{Hom}^\star_{\mathcal{C}^\heartsuit}(\mathrm{H}^*_{\widetilde{\mathrm{T}}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; \underline{k}_\mu), \mathrm{H}^*_{\widetilde{\mathrm{T}}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; \underline{k}_\lambda)) \cong \mathrm{RHom}^\star_{\mathcal{C}}(\mathrm{H}^*_{\widetilde{\mathrm{T}}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; \underline{k}_\mu), \mathrm{H}^*_{\widetilde{\mathrm{T}}}(\mathcal{G}\mathrm{r}_{\mathrm{T}}; \underline{k}_\lambda)).$$

The isomorphism $\mathrm{H}^{\widetilde{\mathrm{T}}}_*(\mathcal{G}\mathrm{r}_{\mathrm{T}};k) \cong \mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$ of Proposition 5.3.2 implies that $\mathcal{C} = \mathrm{coMod}_{\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}}(\mathrm{IndPerf}^{\mathrm{gr}}(\widetilde{\mathrm{T}}_{\mathrm{F}}))$. The natural functor $\mathcal{C} \to \mathrm{IndPerf}^{\mathrm{gr}}(\widetilde{\mathrm{T}}_{\mathrm{F}})$ defines an equivalence

$$\mathrm{IndPerf}^{\mathrm{gr}}(\widetilde{\mathrm{T}}_{\mathrm{F}}) \otimes_{\mathcal{C}} \mathrm{IndPerf}^{\mathrm{gr}}(\widetilde{\mathrm{T}}_{\mathrm{F}}) \simeq \mathrm{IndPerf}^{\mathrm{gr}}(\widetilde{\mathrm{T}}_{\mathrm{F}}) \otimes_{\pi_*(k)} \mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}}),$$

which defines an equivalence

$$\mathcal{C} \xrightarrow{\simeq} \mathrm{Tot}\left(\mathrm{IndPerf}^{\mathrm{gr}}(\widetilde{\mathrm{T}}_{\mathrm{F}}) \rightrightarrows \mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}}) \cdots\right).$$

The equivalence $\mathrm{IndPerf}^{\mathrm{gr}}(\widetilde{\mathrm{T}}_{\mathrm{F}}) \simeq \mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}})^{\check{\mathrm{T}},\mathrm{wk}}$ now defines the claimed equivalence $\mathcal{C} \xrightarrow{\simeq} \mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}})^{(\check{\mathrm{T}}\times\check{\mathrm{T}},\mathrm{wk})}$. Unwinding shows that $\underline{k}_\mu$ is sent to $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}} \otimes \mathbf{Z}_{0,\mu}$, as desired. □

Remark 5.3.4. Using Definition 2.4.15, Theorem 5.3.3 extends to the case when k is not necessarily even: it now states that there is a monoidal $\mathrm{gr}^\star_{\mathbf{T}\text{-ev}}(k^{h\mathbf{T}})$-linear equivalence

$$\mathcal{S} : \mathrm{Shv}^{\mathrm{gr}}_{\mathrm{T}_{\mathcal{O}}\rtimes\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};k) \xrightarrow{\simeq} \mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}})^{(\check{\mathrm{T}}\times\check{\mathrm{T}},\mathrm{wk})}$$

satisfying the same properties as the equivalence of Theorem 5.3.3.

Example 5.3.5. Suppose k is an ordinary commutative ring with trivial $\mathbf{T}$-action. Then Theorem 5.3.3 states that there is a monoidal $k[\hbar]$-linear equivalence

$$\mathcal{S} : \mathrm{Shv}^{\mathrm{gr}}_{\mathrm{T}_{\mathcal{O}}\rtimes\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};k) \xrightarrow{\simeq} \mathrm{D\hat{M}od}^{\mathrm{gr}}(\check{\mathrm{T}})^{(\check{\mathrm{T}}\times\check{\mathrm{T}},\mathrm{wk})}.$$

Here, $\mathrm{D\hat{M}od}$ denotes the full subcategory of $\mathrm{DMod}(\check{\mathrm{T}})$ of those D-modules on which the Euler vector fields of $\check{\mathrm{T}}$ act locally nilpotently.

Similarly, suppose $k = \mathrm{ku}$ with trivial $\mathbf{T}$-action. Then $\pi_*(\mathrm{ku}^{h\mathbf{T}}) \cong \mathbf{Z}[\![q-1]\!][\beta,\hbar]/(\beta\hbar = q-1)$ with β in weight 2 and $\hbar$ in weight -2, and

$$\mathrm{D}^{\mathrm{F}}_{\mathbf{G}_m} \cong \pi_*(\mathrm{ku}^{h\mathbf{T}})\{x^{\pm 1}, \theta\}^\wedge_{\beta\theta}/(\theta x = x(q\theta + \hbar)).$$

The commutation relation above implies that $\theta = \hbar(x\partial^q_x)$, where ∂^q_x is the q-derivative sending $f(x) \mapsto \frac{f(qx)-f(x)}{(q-1)x}$. It follows, for instance, that $\mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}})[\hbar^{-1}]$ is equivalent to the category of q-D-modules on $\check{\mathrm{T}}$, so that there is a $\mathbf{Z}[\![q-1]\!]$-linear equivalence

$$\mathcal{S} : \mathrm{Shv}^{\mathrm{gr}}_{\mathrm{T}_{\mathcal{O}}\rtimes\mathbf{T}}(\mathcal{G}\mathrm{r}_{\mathrm{T}};\mathrm{ku})[\hbar^{-1}] \xrightarrow{\simeq} \mathrm{D\hat{M}od}_q(\check{\mathrm{T}})^{(\check{\mathrm{T}}\times\check{\mathrm{T}},\mathrm{wk})}.$$

Here, $\mathrm{D\hat{M}od}_q$ denotes the full subcategory of $\mathrm{DMod}_q(\check{\mathrm{T}})$ of those q-D-modules on which the q-Euler vector fields act locally nilpotently.

Example 5.3.6. Theorem 5.3.3 is very interesting when k admits a nontrivial $\mathbf{T}$-action. For instance, suppose $k = \mathrm{THH}(\mathbf{F}_p)$ with its natural $\mathbf{T}$-action. There is an equivalence $\mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}}) \simeq \mathrm{IndPerf}(\check{\mathrm{T}}^{\mathrm{Nyg}}_{\mathbf{F}_p})$, where $\check{\mathrm{T}}^{\mathrm{Nyg}}_{\mathbf{F}_p}$ denotes the *Nygaard stack* of $\check{\mathrm{T}}_{\mathbf{F}_p}$ in the sense of [Dri24, Bha24]. (In this case, $\pi_*(k^{h\mathbf{T}}) \cong \mathbf{Z}_p[\hbar,\mu]/(\mu\hbar = p)$, which is $\mathrm{R}\Gamma(\mathbf{F}_p^{\mathrm{Nyg}}; \mathcal{O}\{*\})$ by definition.) This can be deduced from the observation that there is an equivalence $\hat{\mathrm{DMod}}^{\mathrm{gr}}(\check{\mathrm{T}}) \simeq \mathrm{IndPerf}(\check{\mathrm{T}}^{\mathrm{dR}}_{\mathbf{Z}_p})$ when $k = \mathbf{Z}_p$ with the trivial $\mathbf{T}$-action (see Example 5.3.5), and if $\mathbf{Z}_p \to \mathrm{THH}(\mathbf{F}_p)$ denotes the natural S^1-equivariant map (see [NS18, Corollary IV.4.16]), then $\mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}})$ (resp. $\mathrm{IndPerf}(\check{\mathrm{T}}^{\mathrm{Nyg}}_{\mathbf{F}_p})$) is the base-change of $\hat{\mathrm{DMod}}^{\mathrm{gr}}(\check{\mathrm{T}})$ (resp. $\mathrm{IndPerf}(\check{\mathrm{T}}^{\mathrm{dR}}_{\mathbf{Z}_p})$) along the map $\pi_*(\mathbf{Z}_p^{h\mathbf{T}}) \to \pi_*(\mathrm{THH}(\mathbf{F}_p)^{h\mathbf{T}})$ given by $\mathbf{Z}_p[\hbar] \to \mathbf{Z}_p[\hbar, \mu]/(\mu\hbar = p)$.

It follows from Theorem 5.3.3 that there is a monoidal $\mathrm{R}\Gamma(\mathbf{F}_p^{\mathrm{Nyg}}; \mathcal{O}\{*\})$-linear equivalence

$$\mathcal{S} : \mathrm{Shv}^{\mathrm{gr}}_{\mathrm{T}_{\mathcal{O}} \rtimes \mathbf{T}}(\mathcal{G}r_{\mathrm{T}}; \mathrm{THH}(\mathbf{F}_p)) \xrightarrow{\simeq} \mathrm{IndPerf}(\check{\mathrm{T}} \backslash \check{\mathrm{T}}^{\mathrm{Nyg}}_{\mathbf{F}_p} / \check{\mathrm{T}}).$$

More generally, suppose $k = \mathrm{THH}(\mathbf{Z}_p)$ with its natural $\mathbf{T}$-action (this is an $\mathbf{E}_\infty$-$\mathbf{Z}$-algebra, but not $\mathbf{T}$-equivariantly). Then $\mathrm{gr}^\star_{\mathbf{T}\text{-ev}}(k^{h\mathbf{T}}) \cong \mathrm{R}\Gamma(\mathbf{Z}_p^{\mathrm{Nyg}}; \mathcal{O}\{*\})$ (see [BL22, Theorem 6.2.8] and [HRW25]), and similarly $\mathrm{DMod}^{\mathrm{gr}}_{\mathrm{F}}(\check{\mathrm{T}}) \simeq \mathrm{IndPerf}(\check{\mathrm{T}}^{\mathrm{Nyg}}_{\mathbf{Z}_p})$. This can be proved similarly to the $\mathbf{F}_p$ case above. Remark 5.3.4 says that there is a monoidal $\mathrm{R}\Gamma(\mathbf{Z}_p^{\mathrm{Nyg}}; \mathcal{O}\{*\})$-linear equivalence

$$\mathcal{S} : \mathrm{Shv}^{\mathrm{gr}}_{\mathrm{T}_{\mathcal{O}} \rtimes \mathbf{T}}(\mathcal{G}r_{\mathrm{T}}; \mathrm{THH}(\mathbf{Z}_p)) \xrightarrow{\simeq} \mathrm{IndPerf}(\check{\mathrm{T}} \backslash \check{\mathrm{T}}^{\mathrm{Nyg}}_{\mathbf{Z}_p} / \check{\mathrm{T}}).$$

In future work, we aim to prove analogues of the above equivalences with T replaced by a general connected complex reductive group G.

6. Complements

6.1. **Levi compatibility.** Throughout this section, let G be connected, *simply-laced*, and semisimple; modifying arguments appropriately as in Section 5.2 extends the discussion below to the case when G is multiply-laced. One can also relax the semisimplicity condition to reductivity. Fix an even $\mathbf{E}_\infty$-ring k such that the bad primes of G are invertible in $\pi_0(k)$. Fix a Levi subgroup $\mathrm{L} \subseteq \mathrm{G}$ containing a chosen maximal torus $\mathrm{T} \subseteq \mathrm{G}$, let $\mathrm{P} \subseteq \mathrm{G}$ be a Levi subgroup with Levi quotient L, and let $i : \mathcal{G}r_{\mathrm{L}} \to \mathcal{G}r_{\mathrm{G}}$ denote the closed embedding. Note that since G is simply-laced, the same is true of L. In the remainder of this section, we will assume Conjecture 3.1.14 for every sequence of dominant (quasi-)minuscule coweights of G^{ad} and of L^{ad}. We will also add subscripts or superscripts to various objects associated to L, like $\mathcal{W}_{\mathrm{L}}$ to denote the Weyl groupoid for L.

Construction 6.1.1. Then there is a functor $i_* : \mathrm{Shv}_{\mathrm{L}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{L}}; k) \to \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$, which (as in Remark 2.4.5) defines a functor $i_*^{\mathrm{fil}} : \mathrm{Shv}^{\text{q-min,fil}}_{\mathrm{L}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{L}}; k) \to \mathrm{Shv}^{\text{q-min,fil}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$. By Theorem 5.1.16, $\mathcal{S} \circ i_*^{\mathrm{gr}} \circ \mathcal{S}_{\mathrm{L}}^{-1}$ defines a functor $\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{L}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{L}}) \to \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$, which will continue to be denoted i_*^{gr}.

Let $\mathrm{T}^*_{\mathrm{F}}({}^{2\rho}\check{\mathrm{G}}/{}^{2\rho}\check{\mathrm{U}}_{\check{\mathrm{P}}}) = {}^{2\rho}\mathrm{P}_{\mathrm{F}} \times^{{}^{2\rho}\check{\mathrm{P}}} ({}^{2\rho}\check{\mathrm{G}} \times {}^{2\rho}\check{\mathrm{L}})$, so that there is a map $\mathrm{T}^*_{\mathrm{F}}({}^{2\rho}\check{\mathrm{G}}/{}^{2\rho}\check{\mathrm{U}}_{\check{\mathrm{P}}}) \to {}^{2\rho}\mathrm{L}_{\mathrm{F}}$. Note that the isomorphism of Remark 4.2.2 identifies $\mathrm{T}^*_{\hat{\mathbf{G}}_a(2)}({}^{2\rho}\check{\mathrm{G}}/{}^{2\rho}\check{\mathrm{U}}_{\check{\mathrm{P}}})$ with the completion ${}^{2\rho}\check{\mathfrak{u}}^{\perp}_{\check{\mathrm{P}}}(2)^\wedge_{\mathcal{N}} \times^{{}^{2\rho}\check{\mathrm{U}}_{\check{\mathrm{P}}}} {}^{2\rho}\check{\mathrm{G}}$ of the usual cotangent bundle $\mathrm{T}^*(2)({}^{2\rho}\check{\mathrm{G}}/{}^{2\rho}\check{\mathrm{U}}_{\check{\mathrm{P}}})$.

Theorem 6.1.2. *Assume that the ring of global sections* $\mathrm{H}^0(\mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{U}}_{\check{\mathrm{P}}}); \mathcal{O})$ *admits a good filtration as a* $\check{\mathrm{G}} \times \check{\mathrm{L}}$*-representation. Then the functor* $i_*^{\mathrm{gr}} : \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{L}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{L}}) \to$

$\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$ *is induced by pull-push along the correspondence*

$$\begin{tikzcd}[column sep=small] & {}^{2\rho}\check{\mathrm{G}}\backslash \mathrm{T}^*_{\mathrm{F}}({}^{2\rho}\check{\mathrm{G}}/{}^{2\rho}\check{\mathrm{U}}_{\check{\mathrm{P}}})^{\mathrm{aff}}/{}^{2\rho}\check{\mathrm{L}} \arrow[dl] \arrow[dr] & \\ {}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}} & & {}^{2\rho}\mathrm{L}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{L}}. \end{tikzcd} \tag{6.1.1}$$

Proof. Fix a dominant coweight λ (resp. λ') of G (resp. L). Recall from the proof of Theorem 5.1.16 that if $\mathcal{C} = \mathrm{IndPerf}^{\mathrm{gr}}(\mathrm{B}_{\mathrm{T}_{\mathrm{F}}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}})^{\heartsuit}$ and $\mathcal{A}_{\leq\lambda} = \mathrm{End}^{\star}_{\mathcal{C}}(\underline{\widetilde{\mathrm{T}}}_{\leq\lambda})$, then $\mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; k)$ is equivalent to $\mathrm{colim}_{\lambda}(\mathrm{RMod}^{\mathrm{gr}}_{\mathcal{A}_{\leq\lambda}})^{h\mathcal{W}}$. Under this equivalence, Remark 2.4.5 says that the functor $i^{\mathrm{gr}}_* : \mathrm{colim}_{\lambda}(\mathrm{RMod}^{\mathrm{gr}}_{\mathcal{A}_{\leq\lambda}})^{h\mathcal{W}} \to \mathrm{colim}_{\lambda}(\mathrm{RMod}^{\mathrm{gr}}_{\mathcal{A}^{\mathrm{L}}_{\leq\lambda'}})^{h\mathcal{W}_{\mathrm{L}}}$ is induced by the $\mathcal{W}\times_{\mathrm{T}_{\mathrm{F}}}\mathcal{W}_{\mathrm{L}}$-equivariant $\mathcal{A}_{\leq\lambda}$-$\mathcal{A}^{\mathrm{L}}_{\leq\lambda'}$-bimodule $\mathrm{Ext}^{\star}_{\mathrm{Shv}_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};k)}(\mathcal{P}_{\leq\lambda}, i_*\mathcal{P}^{\mathrm{L}}_{\leq\lambda'})$. The Hopf algebra map $i_* : \mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}r_{\mathrm{L}}; k) \to \mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}r_{\mathrm{G}}; k)$ defines a homomorphism $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \to \widetilde{\mathrm{J}}_{\check{\mathrm{L}},\mathrm{F}}$ of group schemes over T_{F}, which will be denoted $\check{i}$; note that if M is a $\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}r_{\mathrm{L}}; k)$-comodule in $\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\mathrm{T}_{\mathrm{F}}}$, then $\check{i}^*\mathrm{M}$ is the $\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}r_{\mathrm{G}}; k)$-comodule given by composing the coaction map with the Hopf algebra map i_*.

It follows from [Gin25, Lemma 6.3.1] that the assumptions of Theorem 2.3.2 are satisfied for the map $i : \mathcal{G}r_{\mathrm{L}} \subseteq \mathcal{G}r_{\mathrm{G}}$, so that Corollary 5.1.11 gives an isomorphism

$$\mathrm{Ext}^{\star}_{\mathrm{Shv}_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};k)}(\mathcal{P}_{\leq\lambda}, i_*\mathcal{P}^{\mathrm{L}}_{\leq\lambda'}) \xrightarrow{\cong} \mathrm{Hom}^{\star}_{\mathcal{C}}(\underline{\widetilde{\mathrm{T}}}_{\leq\lambda}, \check{i}^*\underline{\widetilde{\mathrm{T}}}^{\mathrm{L}}_{\leq\lambda'}).$$

For notational simplicity, let us write $\mathrm{V} = \widetilde{\mathrm{T}}_{\leq\lambda}$ and $\mathrm{V}' = \widetilde{\mathrm{T}}^{\mathrm{L}}_{\leq\lambda'}$, so that V (resp. V') is a $\check{\mathrm{G}}$-representation (resp. $\check{\mathrm{L}}$-representation) in free abelian groups. Then $\underline{\widetilde{\mathrm{T}}}_{\leq\lambda}$ is isomorphic to $\mathrm{Res}^{{}^{2\rho}\check{\mathrm{B}}}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}(\mathcal{O}_{\mathrm{T}_{\mathrm{F}}} \otimes_{\mathbf{Z}} \mathrm{V})$, and similarly $\underline{\widetilde{\mathrm{T}}}^{\mathrm{L}}_{\leq\lambda'}$ is isomorphic to $\mathrm{Res}^{{}^{2\rho}\check{\mathrm{B}}_{\mathrm{L}}}_{\widetilde{\mathrm{J}}_{\check{\mathrm{L}},\mathrm{F}}}(\mathcal{O}_{\mathrm{T}_{\mathrm{F}}} \otimes_{\mathbf{Z}} \mathrm{V}')$. As in Definition 4.2.4, the homomorphism $\check{i} : \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \to \widetilde{\mathrm{J}}_{\check{\mathrm{L}},\mathrm{F}}$ fits into a commutative diagram

$$\begin{tikzcd} \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \arrow[r] \arrow[d, "\check{i}"'] & \mathrm{T}_{\mathrm{F}} \times {}^{2\rho}\check{\mathrm{B}} \arrow[d, "\check{i}"] \\ \widetilde{\mathrm{J}}_{\check{\mathrm{L}},\mathrm{F}} \arrow[r] & \mathrm{T}_{\mathrm{F}} \times {}^{2\rho}\check{\mathrm{B}}_{\check{\mathrm{L}}} \end{tikzcd}$$

of affine group schemes over T_{F}, where the right vertical map is induced by the quotient map $\check{i} : {}^{2\rho}\check{\mathrm{B}} \to {}^{2\rho}\check{\mathrm{B}}_{\check{\mathrm{L}}}$ coming from the semidirect product ${}^{2\rho}\check{\mathrm{B}} \cong {}^{2\rho}\check{\mathrm{B}}_{\check{\mathrm{L}}} \ltimes \check{\mathrm{U}}_{\check{\mathrm{P}}}$. (This follows, for instance, from the naturality of the isomorphism of Theorem 4.2.12.) This defines a commutative diagram

$$\begin{tikzcd} \mathrm{B}_{\mathrm{T}_{\mathrm{F}}}\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}} \arrow[r] \arrow[d, "\check{i}"'] & {}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}} \arrow[d, "\check{i}"] \\ \mathrm{B}_{\mathrm{T}_{\mathrm{F}}}\widetilde{\mathrm{J}}_{\check{\mathrm{L}},\mathrm{F}} \arrow[r] & {}^{2\rho}\mathrm{B}_{\mathrm{L},\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}}_{\check{\mathrm{L}}}. \end{tikzcd}$$

Using the isomorphisms ${}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}} \cong {}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}}$ and ${}^{2\rho}\mathrm{B}_{\mathrm{L},\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}}_{\check{\mathrm{L}}} \cong {}^{2\rho}\widetilde{\check{\mathrm{L}}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{L}}$, we may identify $\check{i}$ with a map ${}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}} \to {}^{2\rho}\widetilde{\check{\mathrm{L}}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{L}}$ (which we will continue to denote by $\check{i}$). Note that $\check{i}^*\underline{\widetilde{\mathrm{T}}}^{\mathrm{L}}_{\leq\lambda'}$ is isomorphic to $\mathrm{Res}^{{}^{2\rho}\check{\mathrm{B}}}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}(\mathcal{O}_{\mathrm{T}_{\mathrm{F}}} \otimes_{\mathbf{Z}} \check{i}^*\mathrm{V}')$. Corollary 4.1.16

gives isomorphisms

$$\begin{aligned}\mathrm{Ext}^{\star}_{\mathrm{Shv}_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};k)}(\mathcal{P}_{\leq\lambda}, i_*\mathcal{P}^{\mathrm{L}}_{\leq\lambda'}) &\xrightarrow{\cong} \mathrm{Hom}_{\mathrm{Perf}({}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}})^{\mathrm{gr},\heartsuit}}(\mathcal{O}_{{}^{2\rho}\mathrm{B}_{\mathrm{F}}}\otimes_{\mathbf{Z}}\mathrm{V}, \check{i}^*(\mathcal{O}_{{}^{2\rho}\mathrm{B}_{\mathrm{L,F}}}\otimes_{\mathbf{Z}}\mathrm{V}'))\\ &\cong \mathrm{Hom}_{\mathrm{Perf}({}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})^{\mathrm{gr},\heartsuit}}(\mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}}\otimes_{\mathbf{Z}}\mathrm{V}, \check{i}^*(\mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{L}}}_{\mathrm{F}}}\otimes_{\mathbf{Z}}\mathrm{V}'))\\ &\cong \mathrm{Hom}_{\mathrm{Perf}({}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})^{\mathrm{gr},\heartsuit}}(\mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}}\otimes_{\mathbf{Z}}\mathrm{V}, \mathcal{O}_{{}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}}\otimes_{\mathbf{Z}}\check{i}^*(\mathrm{V}')).\end{aligned}$$

There is a homomorphism $\check{\mathrm{B}}\to\check{\mathrm{L}}$ given by the composite $\check{\mathrm{B}}\to\check{\mathrm{P}}\to\check{\mathrm{L}}$, which alternatively factors as $\check{\mathrm{B}}\to\check{\mathrm{B}}_{\check{\mathrm{L}}}\to\check{\mathrm{L}}$. Define

$$\mathrm{Y}_{\mathrm{F}} = ({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}})\times^{{}^{2\rho}\check{\mathrm{B}}}{}^{2\rho}\mathrm{B}_{\mathrm{F}} \cong \mathrm{T}^*_{\mathrm{F}}({}^{2\rho}\check{\mathrm{G}}/{}^{2\rho}\check{\mathrm{U}}_{\check{\mathrm{P}}})\times_{{}^{2\rho}\mathrm{L}_{\mathrm{F}}}{}^{2\rho}\widetilde{\check{\mathrm{L}}}_{\mathrm{F}}.$$

so that there is a ${}^{2\rho}\check{\mathrm{G}}$-equivariant isomorphism $\mathrm{Y}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{L}}\cong{}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\mathrm{F}}$, and there is a Cartesian square

$$\begin{tikzcd} \mathrm{Y}_{\mathrm{F}}/({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}})\cong{}^{2\rho}\mathrm{B}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{B}} \arrow[r] \arrow[d] & {}^{2\rho}\widetilde{\check{\mathrm{L}}}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{L}} \arrow[d] \\ \mathrm{T}^*_{\mathrm{F}}({}^{2\rho}\check{\mathrm{G}}/{}^{2\rho}\check{\mathrm{U}}_{\check{\mathrm{P}}})/({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}})\cong{}^{2\rho}\widetilde{\check{\mathrm{G}}}_{\check{\mathrm{P}}}/{}^{2\rho}\check{\mathrm{G}} \arrow[r] & {}^{2\rho}\mathrm{L}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{L}}. \end{tikzcd}$$

The above calculation gives isomorphisms

$$\begin{aligned}\mathrm{Ext}^{\star}_{\mathrm{Shv}_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};k)}(\mathcal{P}_{\leq\lambda}, i_*\mathcal{P}^{\mathrm{L}}_{\leq\lambda'}) &\xrightarrow{\cong} \mathrm{H}^0(\mathrm{Y}_{\mathrm{F}}/({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}}); \mathcal{O}_{\mathrm{Y}_{\mathrm{F}}}\otimes_{\mathbf{Z}}(\mathrm{V}^\vee\otimes_{\mathbf{Z}}\mathrm{V}'))\\ &\cong \mathrm{R}\Gamma(\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}; \mathcal{O}_{\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}}\otimes_{\mathbf{Z}}(\mathrm{V}^\vee\otimes_{\mathbf{Z}}\mathrm{V}'))^{{}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}}}.\end{aligned}$$

We claim that this is isomorphic to the derived global sections over $\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}/({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}})$ of $\mathcal{O}_{\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}}\otimes_{\mathbf{Z}}(\mathrm{V}^\vee\otimes_{\mathbf{Z}}\mathrm{V}'))$. As in the proof of Proposition 5.1.15, the degeneration of F to $\hat{\mathbf{G}}_a(2)$ defines a spectral sequence converging to $\mathrm{H}^*(\mathrm{B}({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}}); \mathcal{O}_{\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}}\otimes_{\mathbf{Z}}(\mathrm{V}^\vee\otimes_{\mathbf{Z}}\mathrm{V}'))$ with E_1-page isomorphic to $\mathrm{H}^*(\mathrm{B}({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}}); \mathcal{O}_{\mathrm{Y}^{\mathrm{aff}}_{\hat{\mathbf{G}}_a(2)}}\otimes_{\mathbf{Z}}(\mathrm{V}^\vee\otimes_{\mathbf{Z}}\mathrm{V}'))$. It therefore suffices to show that the latter is concentrated in degree 0, for which (as in the proof of Proposition 5.1.15) it suffices to show that $\mathcal{O}_{\mathrm{Y}^{\mathrm{aff}}_{\hat{\mathbf{G}}_a(2)}}$ admits a good filtration as a ${}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}}$-representation. For readability, we will ignore the implicit gradings in the following. It follows from the definition of Y_{F} that $\mathrm{Y}^{\mathrm{aff}}_{\hat{\mathbf{G}}_a}$ is isomorphic to a completion of $\mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{U}}_{\check{\mathrm{P}}})^{\mathrm{aff}}\times_{\mathfrak{l}}\widetilde{\check{\mathfrak{l}}}^{\mathrm{aff}}$. By Theorem 4.3.6, $\widetilde{\check{\mathfrak{l}}}^{\mathrm{aff}}\cong\mathfrak{l}\times_{\mathfrak{t}/\mathcal{W}_{\mathrm{L}}}\mathfrak{t}$, so that $\mathrm{Y}^{\mathrm{aff}}_{\hat{\mathbf{G}}_a}\cong\mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{U}}_{\check{\mathrm{P}}})^{\mathrm{aff}}\times_{\mathfrak{t}/\mathcal{W}_{\mathrm{L}}}\mathfrak{t}$. The $\check{\mathrm{G}}\times\check{\mathrm{L}}$ action on $\mathrm{Y}^{\mathrm{aff}}_{\hat{\mathbf{G}}_a}$ is through the natural action on $\mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{U}}_{\check{\mathrm{P}}})^{\mathrm{aff}}$, so the claim follows from our assumption that the ring of functions on $\mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{U}}_{\check{\mathrm{P}}})^{\mathrm{aff}}$ admits a good filtration as a $\check{\mathrm{G}}\times\check{\mathrm{L}}$-representation.

As in the proof of Theorem 5.1.16, let $\mathcal{X}=\mathrm{T}_{\mathrm{F}}\times_{\mathrm{T}_{\mathrm{F}}/\mathcal{W}}{}^{2\rho}\mathrm{G}_{\mathrm{F}}$ and similarly $\mathcal{X}_{\mathrm{L}}=\mathrm{T}_{\mathrm{F}}\times_{\mathrm{T}_{\mathrm{F}}/\mathcal{W}_{\mathrm{L}}}{}^{2\rho}\mathrm{L}_{\mathrm{F}}$. There is a correspondence

(6.1.2)
$$\begin{tikzcd} & \mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}/({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}}) \arrow[dl, "p"'] \arrow[dr, "q"] & \\ \mathcal{X}/{}^{2\rho}\check{\mathrm{G}} & & \mathcal{X}_{\mathrm{L}}/{}^{2\rho}\check{\mathrm{L}}. \end{tikzcd}$$

We conclude from the above discussion that there is an isomorphism

$$\begin{aligned}\mathrm{Ext}^\star_{\mathrm{Shv}_{\mathrm{T}}(\mathcal{G}r_{\mathrm{G}};k)}(\mathcal{P}_{\leq\lambda}, i_*\mathcal{P}^{\mathrm{L}}_{\leq\lambda'}) &\xrightarrow{\cong} \mathrm{R}\Gamma(\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}/({}^{2\rho}\check{\mathrm{G}} \times {}^{2\rho}\check{\mathrm{L}}); \mathcal{O}_{\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}} \otimes_{\mathbf{Z}} (\mathrm{V}^\vee \otimes_{\mathbf{Z}} \mathrm{V}'))\\ &\cong \mathrm{RHom}_{\mathrm{Perf}^{\mathrm{gr}}(\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}/({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}}))}(\mathcal{O}_{\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}} \otimes_{\mathbf{Z}} \mathrm{V}, \mathcal{O}_{\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}} \otimes_{\mathbf{Z}} \mathrm{V}')\\ &\cong \mathrm{RHom}_{\mathrm{Perf}^{\mathrm{gr}}(\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}/({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{L}}))}(p^*(\mathcal{O}_{\mathcal{X}} \otimes_{\mathbf{Z}} \mathrm{V}), q^*(\mathcal{O}_{\mathcal{X}_{\mathrm{L}}} \otimes_{\mathbf{Z}} \mathrm{V}')).\end{aligned}$$

This is an isomorphism of $\mathcal{W} \times_{\mathrm{T}_{\mathrm{F}}} \mathcal{W}_{\mathrm{L}}$-equivariant $\mathcal{A}_{\leq\lambda}$-$\mathcal{A}^{\mathrm{L}}_{\leq\lambda}$-bimodules, where the right-hand side is viewed as a bimodule via the isomorphisms $\mathcal{A}_{\leq\lambda} \cong \mathrm{REnd}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}/{}^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}} \otimes_{\mathbf{Z}} \mathrm{V})$ and $\mathcal{A}^{\mathrm{L}}_{\leq\lambda} \cong \mathrm{REnd}_{\mathrm{Perf}^{\mathrm{gr}}(\mathcal{X}_{\mathrm{L}}/{}^{2\rho}\check{\mathrm{G}})}(\mathcal{O}_{\mathcal{X}_{\mathrm{L}}} \otimes_{\mathbf{Z}} \mathrm{V}')$. Under the equivalences $\mathrm{RMod}^{\mathrm{gr}}_{\mathcal{A}_{\leq\lambda}} \simeq \mathrm{IndPerf}^{\text{q-min,gr}}_{\leq\lambda}(\mathcal{X}/{}^{2\rho}\check{\mathrm{G}})$ and $\mathrm{RMod}^{\mathrm{gr}}_{\mathcal{A}^{\mathrm{L}}_{\leq\lambda}} \simeq \mathrm{IndPerf}^{\text{q-min,gr}}_{\leq\lambda'}(\mathcal{X}_{\mathrm{L}}/{}^{2\rho}\check{\mathrm{L}})$, this bimodule defines the functor $\mathrm{IndPerf}^{\text{q-min,gr}}_{\leq\lambda'}(\mathcal{X}_{\mathrm{L}}/{}^{2\rho}\check{\mathrm{L}}) \to \mathrm{IndPerf}^{\text{q-min,gr}}_{\leq\lambda}(\mathcal{X}/{}^{2\rho}\check{\mathrm{G}})$ given by pull-push along the correspondence (6.1.2). It follows from the case $\mathrm{F} = \hat{\mathbf{G}}_a(2)$ that there is an isomorphism $\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}} \cong \mathrm{T}^*_{\mathrm{F}}(\check{\mathrm{G}}/\check{\mathrm{U}}_{\check{\mathrm{P}}})^{\mathrm{aff}} \times_{\mathrm{T}_{\mathrm{F}}/\mathcal{W}_{\mathrm{L}}} \mathrm{T}_{\mathrm{F}}$, so that $\mathrm{Y}^{\mathrm{aff}}_{\mathrm{F}}/\mathcal{W}_{\mathrm{L}} \cong \mathrm{T}^*_{\mathrm{F}}(\check{\mathrm{G}}/\check{\mathrm{U}}_{\check{\mathrm{P}}})^{\mathrm{aff}}$. Unwinding, we find that the functor $\check{i}^{\mathrm{gr}}_* : \mathrm{IndPerf}^{\text{q-min,gr}}_{\leq\lambda'}({}^{2\rho}\mathrm{L}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{L}}) \to \mathrm{IndPerf}^{\text{q-min,gr}}_{\leq\lambda}({}^{2\rho}\mathrm{G}_{\mathrm{F}}/{}^{2\rho}\check{\mathrm{G}})$ is given by pull-push along the correspondence (6.1.1). □

Remark 6.1.3. Unfortunately, I do not know in what generality the global sections $\mathrm{H}^0(\mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{U}}_{\check{\mathrm{P}}}); \mathcal{O})$ admit a good filtration as a $\check{\mathrm{G}} \times \check{\mathrm{L}}$-representation (except, of course, over a $\mathbf{Q}$-algebra). However, it is true that $\mathrm{H}^0(\mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{U}}); \mathcal{O})$ admits a good filtration as a $\check{\mathrm{G}} \times \check{\mathrm{T}}$-representation over a commutative ring R over which the bad primes of $\check{\mathrm{G}}$ are inverted. Indeed, this can be checked over the localization of $\mathbf{Z}$ at the bad primes of $\check{\mathrm{G}}$, and moreover by standard arguments one can reduce to the case $\mathrm{R} = \mathbf{F}_p$ where p is not a bad prime for $\check{\mathrm{G}}$. The exact sequence $\check{\mathfrak{t}}^* \to \check{\mathfrak{u}}^\perp \to \check{\mathfrak{b}}^\perp$ defines a filtration on $\mathrm{H}^0(\mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{U}}); \mathcal{O})$ with associated graded $\mathrm{H}^0(\mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{B}}); f^*(\mathrm{Ind}_{\check{\mathrm{B}}}^{\check{\mathrm{G}}}(\mathcal{O}_{\mathrm{T}^*\check{\mathrm{T}}})))$, where $f : \mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{B}}) \to \check{\mathrm{G}}/\check{\mathrm{B}}$ is the canonical map. It therefore suffices to observe that for each $\lambda \in \mathbb{X}^*(\check{\mathrm{T}})$, the global sections $\mathrm{H}^0(\mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{B}}); f^*\mathcal{O}(\lambda))$ admits a good filtration as a $\check{\mathrm{G}}$-module (where $\mathcal{O}(\lambda)$ is the line bundle over $\check{\mathrm{G}}/\check{\mathrm{B}}$ corresponding to λ) by [KLT99, Theorem 7].

The same argument reduces the case of a general parabolic subgroup to showing that $\mathrm{H}^0(\check{\mathrm{G}} \times^{\check{\mathrm{P}}} (\check{\mathfrak{p}}^\perp \times \mathrm{T}^*\check{\mathrm{L}}); \mathcal{O})$ admits a good filtration as an $\check{\mathrm{G}} \times \check{\mathrm{L}}$-module; this would follow from the statement that $\mathrm{H}^0(\check{\mathrm{G}} \times^{\check{\mathrm{U}}_{\check{\mathrm{P}}}} \check{\mathfrak{p}}^\perp; \mathcal{O})$ admits a good filtration as an $\check{\mathrm{L}} \times \check{\mathrm{L}}$-module, which is an analogue of [KLT99, Theorem 7]. We note that $\mathrm{H}^0(\mathrm{T}^*(\check{\mathrm{G}}/\check{\mathrm{U}}_{\check{\mathrm{P}}}); \mathcal{O})$ was studied in [Gan24].

6.2. **Projectively perverse sheaves.** In [BF08], Bezrukavnikov and Finkelberg proved that there is a $\mathbf{Q}$-linear equivalence $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; \mathbf{Q}) \simeq \mathrm{IndPerf}(\check{\mathfrak{g}}^*[2]/\check{\mathrm{G}})$, where $\check{\mathrm{G}}$ lives over $\mathbf{Q}$ and $\check{\mathfrak{g}}^*[2]$ denotes the shifted affine space $\mathrm{Spec}\,\mathrm{Sym}_{\mathbf{Q}}(\check{\mathfrak{g}}[-2])$. There is a Koszul duality equivalence $\mathrm{IndPerf}(\check{\mathfrak{g}}^*[2]/\check{\mathrm{G}}) \simeq \mathrm{IndCoh}(\check{\mathfrak{g}}[-1]/\check{\mathrm{G}})$, under which the standard t-structure on $\mathrm{IndCoh}(\check{\mathfrak{g}}[-1]/\check{\mathrm{G}})$ corresponds to the t-structure on $\mathrm{IndPerf}(\check{\mathfrak{g}}^*[2]/\check{\mathrm{G}})$ where the connective subcategory is the smallest full subcategory which is stable under small colimits and contains $\delta_*(\mathrm{V})$ for all (finite-dimensional) $\check{\mathrm{G}}$-representations V, where $\delta : \mathrm{Spec}(\mathbf{Q}) \to \check{\mathfrak{g}}^*[2]$ is the inclusion of the origin. By [Ber21, Proposition 2.3.4], the derived geometric Satake equivalence is t-exact for the perverse t-structure on $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}r_{\mathrm{G}}; \mathbf{Q})$, the standard t-structure on $\mathrm{IndCoh}(\check{\mathfrak{g}}[-1]/\check{\mathrm{G}})$, and the above t-structure on $\mathrm{IndPerf}(\check{\mathfrak{g}}^*[2]/\check{\mathrm{G}})$. Taking hearts produces the usual abelian geometric Satake equivalence $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathbf{Q})^{\heartsuit} \simeq \mathrm{Rep}(\check{\mathrm{G}})^{\heartsuit}$. (This is not quite correct as stated; see the caveats in [BZSV23, Section 6.5].)

When k is a general $\mathbf{E}_\infty$-ring spectrum, there need not be a t-structure on Mod_k, and in particular on $\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, anymore. This happens already for $k = \mathbf{Q}[u^{\pm 1}]$ with u in homological degree 2. However, the problem is not merely periodicity: already for ring spectra k which are connective (so that Mod_k admits a t-structure) but which are not concentrated in degree 0, the naive extension of the perverse t-structure to sheaves of k-module spectra does not seem to be very well-behaved. Instead, we use a *weight* structure to define "projectively perverse" sheaves (see Definition 6.2.7). To proceed, let us assume for ease of exposition that G is semisimple. Recall that $\mathbb{X}_*(\mathrm{T})$ indexes the semi-infinite orbits in $\mathcal{G}\mathrm{r}_{\mathrm{G}}$. For $\lambda \in \mathbb{X}_*(\mathrm{T})$, let $s_\lambda : \mathrm{S}_\lambda \hookrightarrow \mathcal{G}\mathrm{r}_{\mathrm{G}}$ denote the corresponding semi-infinite orbit. If $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, then the fiber over $\lambda \in \mathcal{G}\mathrm{r}_{\mathrm{T}}$ of $q_! i^*(\mathcal{F})$ is precisely $\mathrm{R}\Gamma_c(\mathrm{S}_\lambda; s_\lambda^* \mathcal{F})$. Here, $i : \mathcal{G}\mathrm{r}_{\mathrm{B}} \to \mathcal{G}\mathrm{r}_{\mathrm{G}}$ and $q : \mathcal{G}\mathrm{r}_{\mathrm{B}} \to \mathcal{G}\mathrm{r}_{\mathrm{T}} = \mathbb{X}_*(\mathrm{T})$.

Notation 6.2.1. Let k be an $\mathbf{E}_\infty$-ring, and let $\check{\mathrm{T}}_k = \mathrm{Spec}\, k[\mathbb{X}_*(\mathrm{T})]$; this is a spectral scheme over k (in the sense of [Lur17]). It is furthermore a group scheme, and we will write $\mathrm{IndPerf}(\mathrm{B}\check{\mathrm{T}}_k)$ to denote its category of representations, so that $\mathrm{IndPerf}(\mathrm{B}\check{\mathrm{T}}_k) = \bigoplus_{\lambda \in \mathbb{X}_*(\mathrm{T})} \mathrm{Mod}_k$.

For $\mathcal{F} \in \mathrm{Shv}_{(\mathrm{G}_{\mathcal{O}})}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, the stratification on $\mathcal{G}\mathrm{r}_{\mathrm{G}}$ by the semi-infinite orbits S_λ defines a $\mathbb{X}_*(\mathrm{T})$-indexed filtration on $\mathrm{R}\Gamma_c(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathcal{F})$ with associated graded $\mathrm{F}_{\mathrm{MV}}(\mathcal{F}) := \bigoplus_{\lambda \in \mathbb{X}_*(\mathrm{T})} \mathrm{R}\Gamma_c(\mathrm{S}_\lambda; s_\lambda^* \mathcal{F}) \in \mathrm{IndPerf}(\mathrm{B}\check{\mathrm{T}}_k)$. This defines a functor $\mathrm{F}_{\mathrm{MV}} : \mathrm{Shv}_{(\mathrm{G}_{\mathcal{O}})}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \to \mathrm{IndPerf}(\mathrm{B}\check{\mathrm{T}}_k)$. Note that $\mathrm{gr}^\star_{\mathrm{ev}/k}(\mathrm{F}_{\mathrm{MV}}(\mathcal{F}))$ acquires a $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$-grading: the $\mathbb{X}_*(\mathrm{T})$-grading is from the $\check{\mathrm{T}}_k$-action on $\mathrm{F}_{\mathrm{MV}}(\mathcal{F})$, and the $\mathbf{Z}$-grading is via the internal grading from $\mathrm{gr}^\star_{\mathrm{ev}}$.

Remark 6.2.2. Let $\alpha \in \mathbb{X}^*(\mathrm{T})$. Then there is an endofunctor of $\mathrm{IndPerf}(\mathrm{B}\check{\mathrm{T}}_k)$ which sends $\bigoplus_{\lambda \in \mathbb{X}_*(\mathrm{T})} \mathrm{M}_\lambda \mapsto \bigoplus_{\lambda \in \mathbb{X}_*(\mathrm{T})} \Sigma^{(\alpha, \lambda)} \mathrm{M}_\lambda$. When $\check{\mathrm{T}}_k = \mathbf{G}_m$, this is the shearing functor of Remark A.1.3. As mentioned there, this shearing functor associated to α *cannot* be equipped with a symmetric monoidal structure if $k = \mathbb{S}$ (but if $\alpha(\mathbb{X}_*(\mathrm{T})) \subseteq 2\mathbf{Z}$, the shearing functor associated to α does acquire a symmetric monoidal structure from an $\mathbf{E}_\infty$-complex orientation of k). The shearing functor associated to $\alpha = 2\rho$ is implicitly used in [MV07], so some care is required in adapting arguments from *loc. cit.* to the present setting.

Proposition 6.2.3. *Let k be an $\mathbf{E}_\infty$-ring. Suppose μ is a dominant minuscule coweight of* G. *Then $\mathrm{R}\Gamma_c(\mathrm{S}_\lambda; s_\lambda^* \mathrm{IC}_\mu)$ is isomorphic to $\Sigma^{-(2\rho, \mu + \lambda)} k$ if $\lambda \in \mathrm{W}\mu$, and is* 0 *otherwise. In particular, $\mathrm{gr}^\star_{\mathrm{ev}/k}(\mathrm{F}_{\mathrm{MV}}(\mathrm{IC}_\mu))$ is isomorphic as a $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$-graded $\mathrm{gr}^\star_{\mathrm{ev}}(k)$-module to $\mathrm{V}_\mu \otimes_{\mathbf{Z}} \mathrm{gr}^\star_{\mathrm{ev}}(k)$, where V_μ is the restriction to $\check{\mathrm{T}}$ of the $\check{\mathrm{G}}$-representation with highest weight μ. Here, the $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$-grading is induced from the natural $\mathbf{Z}$-grading on $\mathrm{gr}^\star_{\mathrm{ev}}(k)$, and the $\mathbf{Z}$-grading on V_μ places the λ-weight space in $\mathbf{Z}$-weight $-(2\rho, \lambda + \mu)$.*

Proof. Base-change gives an isomorphism $\mathrm{R}\Gamma_c(\mathrm{S}_\lambda; \mathrm{IC}_\mu) \cong \mathrm{R}\Gamma_c(\mathrm{S}_\lambda \cap \mathcal{G}\mathrm{r}_{\mathrm{G}}^{\leq \mu}; k)$. It follows from [MV07, Theorem 3.2] that $\mathrm{S}_\lambda \cap \mathcal{G}\mathrm{r}_{\mathrm{G}}^{\leq \mu}$ is empty unless $\lambda \in \mathrm{W}\mu$, in which case it is isomorphic to $\mathbf{C}^{(\rho, \mu + \lambda)}$; this implies the desired isomorphism. □

Example 6.2.4. Although $\mathrm{F}_{\mathrm{MV}}(\mathrm{IC}_\mu) \cong \bigoplus_{\lambda \in \mathrm{W}\mu} \Sigma^{-(2\rho, \mu + \lambda)} k$ is isomorphic to $\mathrm{R}\Gamma_c(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathrm{IC}_\mu)$ if k is even, this isomorphism is not canonical in k. If it were, then by writing $\mathbb{S} = \mathrm{Tot}(\mathrm{MU}^{\otimes \bullet + 1})$, it would follow that if $k = \mathbb{S}$, then $\mathrm{R}\Gamma_c(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathrm{IC}_\mu)$

would also be isomorphic to $\bigoplus_{\lambda\in \mathrm{W}\mu} \Sigma^{-(2\rho,\mu+\lambda)}\mathbb{S}$. This, however, is false: indeed, the filtration on $\mathrm{R}\Gamma_c(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathrm{IC}_\mu) \cong \mathrm{R}\Gamma(\mathcal{G}\mathrm{r}^\mu_{\mathrm{G}}; \mathbb{S})$ is just the cellular filtration, so it does not split because there are (stably) nontrivial attaching maps between the cells of $\mathcal{G}\mathrm{r}^\mu_{\mathrm{G}}$, at least if G has semisimple rank ≥ 2.

Proposition 6.2.5. *Let k be an $\mathbf{E}_\infty$-ring. Suppose θ is a dominant quasi-minuscule coweight of* G. *Then there are isomorphisms*

$$\mathrm{R}\Gamma_c(\mathrm{S}_\lambda; s_\lambda^* \widetilde{\mathrm{IC}}_\theta) \cong \begin{cases} \bigoplus_{w\in \mathrm{W}\theta} \Sigma^{-(2\rho,w\theta+\theta)+\mathrm{sgn}(2\rho,w\theta)} k & \lambda = 0,\\ \Sigma^{-(2\rho,\theta+\lambda)} k & \lambda\in \mathrm{W}\theta,\\ 0 & \textit{else.}\end{cases}$$

Here, $\mathrm{sgn}(2\rho,\alpha)$ *for* $\alpha \in \Phi$ *is equal to 2 if* $\alpha\in\Phi^+$ *and* -2 *if* $\alpha \in \Phi^-$. *In particular,* $\mathrm{gr}^\star_{\mathrm{ev}/k}(\mathrm{F}_{\mathrm{MV}}(\widetilde{\mathrm{IC}}_\theta))$ *is isomorphic as a* $\mathbb{X}_*(\mathrm{T})\times\mathbf{Z}$*-graded* $\mathrm{gr}^\star_{\mathrm{ev}}(k)$*-module to* $(\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{T}'_\theta) \oplus \mathrm{V}^{\mathrm{triv}}_\theta)\otimes_{\mathbf{Z}} \mathrm{gr}^\star_{\mathrm{ev}}(k)$. *Here,* $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{T}'_\theta)$ *is the* $\mathbb{X}_*(\mathrm{T})\times\mathbf{Z}$*-graded free abelian group given by the restriction to* $\check{\mathrm{T}}$ *of the direct sum of a trivial* $\check{\mathrm{G}}$*-representation and the tilting* $\check{\mathrm{G}}$*-module* T_θ *with highest weight* θ*, where the* λ*-weight space lies in* $\mathbf{Z}$*-weight* $-(2\rho,\lambda+\theta)$*; and* $\mathrm{V}^{\mathrm{triv}}_\theta$ *is a* $\mathbf{Z}$*-graded free abelian group which vanishes in weight* $-(2\rho,\theta)$ *and which is placed in zero* $\mathbb{X}_*(\mathrm{T})$*-grading.*

Proof. We may assume that G is simple. Recall from Definition 5.1.6 the resolution $f : \mathbf{P}(\mathcal{L}_\theta\oplus\mathcal{O}) \to \mathcal{G}\mathrm{r}^{\leq\theta}_{\mathrm{G}}$, where $\mathcal{L}_\theta = \mathrm{G}\times^{\mathrm{P}_\theta}\mathfrak{g}_{\check{\theta}}$ is the line bundle over $\mathrm{G}/\mathrm{P}_\theta$ where $\check{\theta}$ is the highest root of G_{ad} and P_θ stabilizes the line $\mathfrak{g}_{\check{\theta}}$. Let $\mathcal{L}_{-\theta}$ be the inverse of $\mathrm{G}/\mathrm{P}_\theta$. Let $\pi_{\pm\theta} : \mathcal{L}_{\pm\theta}\to \mathrm{G}/\mathrm{P}_\theta$ denote the projections, and let $z_{\pm\theta} : \mathrm{G}/\mathrm{P}_\theta \to \mathcal{L}_{\pm\theta}$ denote the zero section.

Let $\widetilde{\mathrm{S}}_\lambda$ denote the preimage of $\mathrm{S}_\lambda \hookrightarrow \mathcal{G}\mathrm{r}_{\mathrm{G}}$ under f, so that base-change gives an isomorphism $\mathrm{R}\Gamma_c(\mathrm{S}_\lambda;\widetilde{\mathrm{IC}}_\theta)\cong \mathrm{R}\Gamma_c(\widetilde{\mathrm{S}}_\lambda; k)$. By [MV07, Theorem 3.2], $\mathrm{S}_\lambda\cap\mathcal{G}\mathrm{r}^{\leq\theta}_{\mathrm{G}}$ is nonempty if and only if $\lambda = 0$ or $\lambda\in \mathrm{W}\theta$; in particular, $\widetilde{\mathrm{S}}_\lambda$ is empty unless $\lambda\in\{0\}\cup \mathrm{W}\theta$. Assume first that $\lambda\neq 0$. Then $\mathrm{S}_\lambda\cap\mathcal{G}\mathrm{r}^{\leq\theta}_{\mathrm{G}} = \mathrm{S}_\lambda\cap\mathcal{G}\mathrm{r}^{\theta}_{\mathrm{G}}$ is isomorphic to $\mathbf{C}^{(\rho,\theta+\lambda)}$. Since the map f is an isomorphism over $\mathcal{G}\mathrm{r}^\theta_{\mathrm{G}}$, it follows that $\widetilde{\mathrm{S}}_\lambda\cong\mathbf{C}^{(\rho,\theta+\lambda)}$ if $\lambda\in \mathrm{W}\theta$. When $\lambda = 0$, [NP01, Corollaire 7.5] gives an isomorphism

$$\widetilde{\mathrm{S}}_0 \cong \bigcup_{w\in \mathrm{W}\theta, w\theta\in\Phi^+} z_{-\theta}(\mathrm{U}w\mathrm{P}_\theta/\mathrm{P}_\theta) \cup \bigcup_{w\in\mathrm{W}\theta, w\theta\in\Phi^-} \pi^{-1}_{-\theta}(\mathrm{U}w\mathrm{P}_\theta/\mathrm{P}_\theta).$$

Here, $\mathrm{U}w\mathrm{P}_\theta/\mathrm{P}_\theta$ are the Bruhat cells in $\mathrm{G}/\mathrm{P}_\theta$. As in the proof of [NP01, Lemme 8.3], $z_{-\theta}(\mathrm{U}w\mathrm{P}_\theta/\mathrm{P}_\theta)$ is isomorphic to $\mathbf{C}^{(\rho,w\theta+\theta)-1}$ and $\pi^{-1}_{-\theta}(\mathrm{U}w\mathrm{P}_\theta/\mathrm{P}_\theta)$ is isomorphic to $\mathbf{C}^{(\rho,w\theta+\theta)+1}$. This implies the desired isomorphism.

The final claim about $\mathrm{gr}^\star_{\mathrm{ev}/k}(\mathrm{F}_{\mathrm{MV}}(\widetilde{\mathrm{IC}}_\theta))$ follows from the observation that for $\lambda\in\mathbb{X}_*(\mathrm{T})$, the λ-weight space of $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{T}_\theta)$ is 0 if $\lambda\notin\{0\}\cup\mathrm{W}\theta$; is $\mathbf{Z}$ if $\lambda\in\mathrm{W}\theta$; and is a $\mathbf{Z}$-submodule of $\mathrm{Map}(\pm\Delta_s,\mathbf{Z})$ if $\lambda = 0$, where $\pm\Delta_s = \{w\in\mathrm{W}\theta | (2\rho,w\theta) = \pm 2\}$ is the set of short simple roots and their negatives. Since the Weyl filtration of T_θ consists of exactly one copy of the Weyl module with highest weight θ and the trivial module with some multiplicity, the only nontrivial claim is that the dimension of the 0-weight space of T_θ is bounded above by $2|\Delta_s|$, i.e., that the multiplicity of the trivial module in T_θ is bounded above by $|\Delta_s|$. This in turn follows from [Kor18, Lemma 4.2 and Remark 4.4], which in fact implies that the multiplicity of the trivial module in T_θ is at most 2. (Alternatively, it also follows from the usual geometric Satake equivalence and the comparison between parity sheaves and tilting representations; see [BGM+19, JMW16].) $\square$

Like in Example 6.2.4, the $\mathbb{X}_*(\mathrm{T})$-indexed filtration on $\mathrm{R\Gamma}_c(\mathcal{G}\mathrm{r}_\mathrm{G}; \widetilde{\mathrm{IC}}_\theta)$ with associated graded $\mathrm{F}_{\mathrm{MV}}(\widetilde{\mathrm{IC}}_\theta)$ generally does not split.

Corollary 6.2.6. *Let k be an $\mathbf{E}_\infty$-ring. As in Notation 5.1.10, let $\mu_\bullet = (\mu_1, \cdots, \mu_n)$ be a sequence of dominant (quasi-)minuscule weights of $\check{\mathrm{G}}^{\mathrm{sc}}$, and let $\widetilde{\mathrm{IC}}_{\mu_\bullet}$ denote the convolution $\widetilde{\mathrm{IC}}_{\mu_1} \star \cdots \star \widetilde{\mathrm{IC}}_{\mu_n}$. Then $\mathrm{R\Gamma}_c(\mathrm{S}_\lambda; s_\lambda^* \widetilde{\mathrm{IC}}_{\mu_\bullet})$ is isomorphic to a direct sum of even shifts of k, and $\mathrm{gr}^\star_{\mathrm{ev}/k}(\mathrm{F}_{\mathrm{MV}}(\widetilde{\mathrm{IC}}_{\mu_\bullet}))$ is isomorphic as a $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$-graded $\mathrm{gr}^\star_{\mathrm{ev}}(k)$-module to $(\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{T}'_{\mu_\bullet}) \oplus \mathrm{V}^{\mathrm{triv}}_{\mu_\bullet}) \otimes_{\mathbf{Z}} \mathrm{gr}^\star_{\mathrm{ev}}(k)$. Here, $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{T}'_{\mu_\bullet})$ is a $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$-graded free abelian group given by the restriction to $\check{\mathrm{T}}$ of the direct sum of a trivial $\check{\mathrm{G}}$-representation and the tensor product $\mathrm{T}_{\mu_\bullet} = \bigotimes_{i=1}^n \mathrm{T}_{\mu_i}$ of the tilting $\check{\mathrm{G}}$-modules with highest weight μ_i, where the λ-weight space lies in $\mathbf{Z}$-weight $-(2\rho, \lambda + |\mu_\bullet|)$; and $\mathrm{V}^{\mathrm{triv}}_{\mu_\bullet}$ is a $\mathbf{Z}$-graded free abelian group which vanishes in weight $-(2\rho, |\mu_\bullet|)$ and is placed in zero $\mathbb{X}_*(\mathrm{T})$-grading.*

Proof. This follows from Proposition 6.2.3 and Proposition 6.2.5 by arguing as in [NP01, Théorème 3.1]. □

Definition 6.2.7. Let k be an $\mathbf{E}_\infty$-ring, and let $\mathcal{F} \in \mathrm{Shv}_{(\mathrm{G}_\mathcal{O})}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$. Say that $\mathcal{F}$ is *projectively perverse* if for each $\lambda \in \mathbb{X}_*(\mathrm{T})$, the weight λ component of $\mathrm{F}_{\mathrm{MV}}(\mathcal{F}) \in \mathrm{IndPerf}(\mathrm{B}\check{\mathrm{T}}_k)$ is isomorphic to a retract of direct sum of copies of $\Sigma^{-(2\rho,\lambda)} k$. (Note that if k is 2-periodic, this is just the condition that $\mathrm{F}_{\mathrm{MV}}(\mathcal{F})$ is a retract of a direct sum of copies of k.) If $\mathcal{F} \in \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$, say that $\mathcal{F}$ is projectively perverse if its image under the functor $\mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \to \mathrm{Shv}_{(\mathrm{G}_\mathcal{O})}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ is projectively perverse. If $\mathcal{F}$ is projectively perverse, then $\mathrm{gr}^\star_{\mathrm{ev}/k}(\mathrm{F}_{\mathrm{MV}}(\mathcal{F}))$ is isomorphic as a $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$-graded $\mathrm{gr}^\star_{\mathrm{ev}}(k)$-module to $\mathrm{M} \otimes_{\mathbf{Z}} \mathrm{gr}^\star_{\mathrm{ev}}(k)$ where the $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$-grading is inherited from the $\mathbf{Z}$-grading of $\mathrm{gr}^\star_{\mathrm{ev}}(k)$ (which is placed in $\mathbb{X}_*(\mathrm{T})$-weight 0) and the $\mathbb{X}_*(\mathrm{T})$-grading of M where the λ-weight space is placed in $\mathbf{Z}$-weight $-(2\rho, \lambda)$.

Example 6.2.8. Let k be an ordinary commutative ring. Then [MV07, Lemma 3.9] says that any projectively perverse object of $\mathrm{Shv}_{(\mathrm{G}_\mathcal{O})}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ is perverse in the usual sense. Moreover, the homotopy category of the additive category of projectively perverse objects of $\mathrm{Shv}_{(\mathrm{G}_\mathcal{O})}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ is equivalent to the additive 1-category of those perverse sheaves $\mathcal{F}$ such that $\pi_* \mathrm{F}_{\mathrm{MV}}(\mathcal{F})$ is a projective k-module. In particular, its abelianization is the usual perverse heart $\mathrm{Shv}_{(\mathrm{G}_\mathcal{O})}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)^\heartsuit$. One important observation in [MV07] is that if k is a noetherian ring of finite global dimension and $\mathcal{F}$ is perverse, then there is a canonical isomorphism $\mathrm{F}_{\mathrm{MV}}(\mathcal{F}) \cong \mathrm{R\Gamma}_c(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathcal{F})$. Moreover, in this case $\pi_{-(2\rho,\lambda)} \mathrm{R\Gamma}_c(\mathrm{S}_\lambda; s_\lambda^* \mathcal{F})$ is isomorphic to the λ-weight space of the $\check{\mathrm{G}}$-representation corresponding to $\mathcal{F}$ under the abelian geometric Satake equivalence of [MV07].

Example 6.2.9. Let k be an $\mathbf{E}_\infty$-ring. If μ is a minuscule dominant coweight of $\check{\mathrm{G}}$, then Proposition 6.2.3 implies that $\mathrm{IC}^\natural_\mu := \Sigma^{(2\rho,\mu)} \mathrm{IC}_\mu$ is projectively perverse.

Construction 6.2.10. Let k be an even $\mathbf{E}_\infty$-ring. Let θ be a quasi-minuscule dominant coweight of G; we will construct a projectively perverse sheaf $\mathrm{IC}^\natural_\theta \in \mathrm{Shv}_{(\mathrm{G}_\mathcal{O})}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$.

Let IC_0 denote the skyscraper sheaf at the basepoint of $\mathcal{G}\mathrm{r}_\mathrm{G}$. For notational simplicity, let $\mathcal{C}$ denote $\mathrm{coMod}_{\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)}(\mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\mathcal{O}_{\mathrm{T_F}}})$. Note that by Theorem 3.1.11, there is an isomorphism $\mathrm{H}_*^{\mathrm{T,BM}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \xrightarrow{\cong} \mathcal{A}_\mathrm{G}$ of commutative and cocommutative

Hopf $\mathcal{O}_{\mathrm{T_F}}$-algebras where $\mathcal{A}_{\mathrm{G}}$ is as in Definition 3.1.10. There is an isomorphism $\mathcal{A}_{\mathrm{T}} = \mathcal{O}_{\mathrm{T_F}}[\mathbb{X}_*(\mathrm{T})]$, so there is a forgetful functor $\mathcal{C} \to \mathrm{IndPerf}(\check{\mathrm{T}})^{\heartsuit} \otimes \mathrm{Mod}^{\mathrm{gr},\heartsuit}_{\mathcal{O}_{\mathrm{T_F}}}$.

Recall that $\mathcal{A}_{\mathrm{G}} = \bigcap_{\alpha\in\Phi^+} \mathcal{A}_{\mathrm{G}}[e(\Phi^+ - \{\alpha\})^{-1}] \subseteq \mathcal{A}_{\mathrm{T}}[e(\Phi^+)^{-1}]$, and $\mathcal{A}_{\mathrm{G}}[e(\Phi^+ - \{\alpha\})^{-1}] \cong \mathcal{A}_{\mathrm{L}_\alpha}[e(\Phi^+ - \{\alpha\})^{-1}]$ where L_α is the Levi subgroup of G generated by the root spaces $\mathrm{U}_{\pm\alpha}$. Note that the proof of Theorem 4.2.12 shows that $\mathcal{A}_{\mathrm{L}_\alpha}$ is isomorphic as a graded Hopf $\mathcal{O}_{\mathrm{T_F}}$-algebra to $\mathcal{O}_{\mathrm{J}_{\check{\mathrm{L}}_\alpha,\mathrm{F}}}$, and therefore in particular admits a surjective map $\mathcal{O}_{\check{\mathrm{L}}_\alpha\times\mathrm{T_F}} \to \mathcal{A}_{\mathrm{L}_\alpha}$ of Hopf $\mathcal{O}_{\mathrm{T_F}}$-algebras (without inverting any primes!).

Let $\widetilde{\mathrm{T}}_\theta$ denote the $\mathbf{Z}$-graded $\check{\mathrm{G}}$-representation in free abelian groups given by $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{T}'_\theta) \oplus \mathrm{V}^{\mathrm{triv}}_\theta$, where T'_θ is the direct sum of a trivial $\check{\mathrm{G}}$-representation with the tilting $\check{\mathrm{G}}$-module with highest weight θ (where the $\mathbf{Z}$-grading places the λ-weight space in $\mathbf{Z}$-weight $(2\rho,\lambda)$), and $\mathrm{V}^{\mathrm{triv}}_\theta$ is a $\mathbf{Z}$-graded free (finitely generated) abelian group which vanishes in weight $-(2\rho,\theta)$ and has trivial $\check{\mathrm{G}}$-action. It follows from Lemma 5.1.7 and Proposition 5.2.8 that there is an isomorphism of $\mathbf{Z}$-graded $\mathcal{A}_{\mathrm{T}}$-comodules (i.e., $\check{\mathrm{T}}$-representations in graded $\mathcal{O}_{\mathrm{T_F}}$-modules)

$$\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta)[e(\Phi^+)^{-1}] \cong \mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\widetilde{\mathrm{T}}_\theta) \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T_F}}[e(\Phi^+)^{-1}]. \tag{6.2.1}$$

For $\alpha\in\Phi^+$, let T_α denote the kernel of α. There is an isomorphism of $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta)[e(\Phi^+ - \{\alpha\})^{-1}]$ with the base-change along $\pi_*(k^{h\mathbf{C}^\times}) \to \pi_*(k^{h\mathrm{T}})[e(\Phi^+ - \{\alpha\})^{-1}]$ of $\mathrm{H}^{-*}_{\mathbf{C}^\times,c}(\mathbf{P}(\mathcal{L}_\theta \oplus \mathcal{O})^{\mathrm{T}_\alpha}; k)$. The analyses of $\mathbf{P}(\mathcal{L}_\theta\oplus\mathcal{O})^{\mathrm{T}_\alpha}$ from Proposition 5.1.9 and Proposition 5.2.8 now show that for each $\alpha\in\Phi^+$, the isomorphism (6.2.1) extends to an isomorphism of $\mathbf{Z}$-graded $\mathcal{A}_{\mathrm{L}_\alpha}$-comodules

$$\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta)[e(\Phi^+ - \{\alpha\})^{-1}] \cong \mathrm{Res}^{\check{\mathrm{L}}_\alpha}_{\mathrm{Spec}(\mathcal{A}_{\mathrm{L}_\alpha})} \mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{L}}_\alpha}(\widetilde{\mathrm{T}}_\theta) \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T_F}}[e(\Phi^+ - \{\alpha\})^{-1}].$$

Moreover, this splitting is compatible (upon inverting $e(\alpha)$) with the splitting (6.2.1) of $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta)[e(\Phi^+)^{-1}]$ as a $\mathbf{Z}$-graded $\check{\mathrm{T}}$-representation.

Since $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta)$ is isomorphic to $\bigcap_{\alpha\in\Phi^+} \mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta)[e(\Phi^+ - \{\alpha\})^{-1}]$ as a submodule of $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta)[e(\Phi^+)^{-1}]$, and similarly for $\mathcal{A}_{\mathrm{G}}$, the preceding paragraph implies that there is a $\mathbf{Z}$-graded free abelian group $\mathrm{V}^{\mathrm{triv}}_\theta$ which vanishes in weight $-(2\rho,\theta)$ and has trivial $\mathcal{A}_{\mathrm{G}}$-coaction such that: $\mathrm{V}^{\mathrm{triv}}_\theta \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T_F}}$ splits off $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta)$; and upon inverting $e(\Phi^+)$, it identifies with the summand $\mathrm{V}^{\mathrm{triv}}_\theta \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T_F}}[e(\Phi^+)^{-1}]$ of $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta)[e(\Phi^+)^{-1}]$ via (6.2.1). Since $\widetilde{\mathrm{IC}}_\theta$ is even by Proposition 5.1.12, Theorem 2.3.2 gives an isomorphism (without inverting any primes!)

$$\pi_* \mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)}(\widetilde{\mathrm{IC}}_\theta, \mathrm{IC}_0) \xrightarrow{\cong} \mathrm{Hom}^*_{\mathcal{C}}(\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta), \mathcal{O}_{\mathrm{T_F}}), \tag{6.2.2}$$

and similarly for $\pi_*\mathrm{Hom}_{\mathrm{Shv}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}};k)}(\mathrm{IC}_0, \widetilde{\mathrm{IC}}_\theta)$. It follows that if $\mathrm{V}^{\mathrm{triv}}_{\theta,\mathbb{S}}$ is a (finite) direct sum of even shifts of $\mathbb{S}$ (none of which are $\Sigma^{(2\rho,\theta)}\mathbb{S}$) such that there is a $\mathbf{Z}$-graded isomorphism $\pi_*(\mathrm{V}^{\mathrm{triv}}_{\theta,\mathbb{S}} \otimes_{\mathbb{S}} \mathbf{Z}) \cong \mathrm{V}^{\mathrm{triv}}_\theta$, then the retraction of $\mathrm{V}^{\mathrm{triv}}_\theta \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{T_F}}$ off $\mathrm{H}^{-*}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \widetilde{\mathrm{IC}}_\theta)$ in $\mathcal{C}$ admits a lifting to a retraction of $\mathrm{V}^{\mathrm{triv}}_{\theta,\mathbb{S}} \otimes_{\mathbb{S}} \mathrm{IC}_0$ off $\widetilde{\mathrm{IC}}_\theta$ in $\mathrm{Shv}_{\mathrm{T}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$. Forgetting T-equivariance, this of course implies that $\mathrm{V}^{\mathrm{triv}}_{\theta,\mathbb{S}} \otimes_{\mathbb{S}} \mathrm{IC}_0$ splits off $\widetilde{\mathrm{IC}}_\theta$ in $\mathrm{Shv}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$; let $\mathrm{IC}^\natural_\theta$ denote the $(2\rho,\theta)$-fold suspension of the complementary summand of $\widetilde{\mathrm{IC}}_\theta$. Since both $\widetilde{\mathrm{IC}}_\theta$ and IC_0 lie in the subcategory $\mathrm{Shv}_{(\mathrm{G}_{\mathcal{O}})}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \subseteq \mathrm{Shv}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$, it follows that $\mathrm{IC}^\natural_\theta \in \mathrm{Shv}_{(\mathrm{G}_{\mathcal{O}})}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$.

Remark 6.2.11. Construction 6.2.10 can be refined to show that $\mathrm{IC}_\theta^\natural \in \mathrm{Shv}_{(\mathrm{G}_\mathcal{O})}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ admits a lift to $\mathrm{Shv}_{\mathrm{G}_\mathcal{O}}^{\mathrm{toral}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ (recall the notation "toral" from Remark 2.4.12). It is also well-defined for a general $\mathbf{E}_\infty$-ring k which admits an eff cover by an even $\mathbf{E}_\infty$-ring.

Lemma 6.2.12. *Let k be an even $\mathbf{E}_\infty$-ring. The sheaf $\mathrm{IC}_\theta^\natural \in \mathrm{Shv}_{(\mathrm{G}_\mathcal{O})}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ from Construction 6.2.10 is projectively perverse, and there are isomorphisms*

$$\mathrm{R}\Gamma_c(\mathrm{S}_\lambda; s_\lambda^* \mathrm{IC}_\theta^\natural) \cong \begin{cases} \bigoplus_{w\in \mathrm{W}\theta, (2\rho, w\theta) = \pm 2} k & \lambda = 0, \\ \Sigma^{-(2\rho,\lambda)} k & \lambda \in \mathrm{W}\theta, \\ 0 & \textit{else}. \end{cases}$$

Moreover, $\pi_*(\mathrm{F}_{\mathrm{MV}}(\mathrm{IC}_\theta^\natural))$ is isomorphic as a $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$-graded $\pi_*(k)$-module to $(\mathrm{Res}_{\check{\mathrm{T}}}^{\check{\mathrm{G}}}(\mathrm{T}_\theta) \oplus \mathrm{V}_{\theta,0}) \otimes_\mathbf{Z} \pi_*(k)$, where $\mathrm{V}_{\theta,0}$ is a trivial $\check{\mathrm{G}}$-representation. Here, the $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$-grading is inherited from the $\mathbf{Z}$-grading of $\pi_*(k)$ (which is placed in $\mathbb{X}_*(\mathrm{T})$-weight 0), and the $\mathbb{X}_*(\mathrm{T})$-grading of $\mathrm{Res}_{\check{\mathrm{T}}}^{\check{\mathrm{G}}}(\mathrm{T}_\theta) \oplus \mathrm{V}_{\theta,0}$ where the λ-weight space is placed in $\mathbf{Z}$-weight $-(2\rho,\lambda)$.

Proof. Recall that there is a $\mathbb{X}_*(\mathrm{T})$-indexed filtration $\mathrm{fil}^\star_{\mathrm{MV}}\mathrm{R}\Gamma_c(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathcal{F})$ on $\mathrm{R}\Gamma_c(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathcal{F})$ whose associated graded is isomorphic to $\mathrm{F}_{\mathrm{MV}}(\mathcal{F})$. Because $\mathrm{V}_{\theta,\mathbb{S}}^{\mathrm{triv}} \otimes_\mathbb{S} \Sigma^{(2\rho,\theta)}\mathrm{IC}_0$ is a retract of $\Sigma^{(2\rho,\theta)}\widetilde{\mathrm{IC}}_\theta$, there is a retraction of $\mathrm{fil}^\star_{\mathrm{MV}}\mathrm{R}\Gamma_c(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathrm{V}_{\theta,\mathbb{S}}^{\mathrm{triv}} \otimes_\mathbb{S} \Sigma^{(2\rho,\theta)}\mathrm{IC}_0)$ off $\mathrm{fil}^\star_{\mathrm{MV}}\mathrm{R}\Gamma_c(\mathcal{G}\mathrm{r}_\mathrm{G}; \Sigma^{(2\rho,\theta)}\widetilde{\mathrm{IC}}_\theta)$. Since k is even and $\mathrm{F}_{\mathrm{MV}}(\widetilde{\mathrm{IC}}_\theta)$ is a direct sum of even shifts of k by Proposition 6.2.5, the filtration $\mathrm{fil}^\star_{\mathrm{MV}}\mathrm{R}\Gamma_c(\mathcal{G}\mathrm{r}_\mathrm{G}; \Sigma^{(2\rho,\theta)}\widetilde{\mathrm{IC}}_\theta)$ admits a splitting (the canonicity of this splitting is not relevant here), so that $\mathrm{R}\Gamma_c(\mathcal{G}\mathrm{r}_\mathrm{G}; \Sigma^{(2\rho,\theta)}\widetilde{\mathrm{IC}}_\theta) \cong \Sigma^{(2\rho,\theta)}\mathrm{F}_{\mathrm{MV}}(\widetilde{\mathrm{IC}}_\theta)$. The filtration $\mathrm{fil}^\star_{\mathrm{MV}}\mathrm{R}\Gamma_c(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathrm{V}_{\theta,\mathbb{S}}^{\mathrm{triv}} \otimes_\mathbb{S} \Sigma^{(2\rho,\theta)}\mathrm{IC}_0)$ therefore also admits a splitting.

By Proposition 6.2.5, $\Sigma^{(2\rho,\theta)}\mathrm{F}_{\mathrm{MV}}(\mathrm{V}_{\theta,\mathbb{S}}^{\mathrm{triv}} \otimes_\mathbb{S} \mathrm{IC}_0)$ is a direct summand of $\bigoplus_{w\in\mathrm{W}\theta} \Sigma^{-(2\rho,w\theta)+\mathrm{sgn}(2\rho,w\theta)}k$ placed in $\mathbb{X}_*(\mathrm{T})$-weight 0. Taking $k = \mathbf{Z}$, we see that $\pi_*(\Sigma^{(2\rho,\theta)}\mathrm{F}_{\mathrm{MV}}(\mathrm{V}_{\theta,\mathbb{S}}^{\mathrm{triv}} \otimes_\mathbb{S} \mathrm{IC}_0))$ is isomorphic to $\mathrm{V}_\theta^{\mathrm{triv}}$ shifted in $\mathbf{Z}$-weight by $(2\rho,\theta)$. Construction 6.2.10 therefore implies that $\Sigma^{(2\rho,\theta)}\mathrm{F}_{\mathrm{MV}}(\mathrm{V}_{\theta,\mathbb{S}}^{\mathrm{triv}} \otimes_\mathbb{S} \mathrm{IC}_0)$ is the direct sum of those $\Sigma^{-(2\rho,w\theta)+\mathrm{sgn}(2\rho,w\theta)}k$ such that $(2\rho, w\theta) \neq \pm 2$. Taking the complementary summand and shifting by $(2\rho,\theta)$, it follows $\mathrm{R}\Gamma_c(\mathrm{S}_0; \mathrm{IC}_\theta^\natural)$ is a direct sum of copies of k indexed by those $w \in \mathrm{W}\theta$ such that $(2\rho, w\theta) = \pm 2$, and $\mathrm{R}\Gamma_c(\mathrm{S}_\lambda; \mathrm{IC}_\theta^\natural) \cong \Sigma^{(2\rho,\theta)}\mathrm{R}\Gamma_c(\mathrm{S}_0; \widetilde{\mathrm{IC}}_\theta)$ for $\lambda \neq 0$. Using Proposition 6.2.5, we see that $\mathrm{IC}_\theta^\natural$ is projectively perverse, as desired. □

Note that if k is an ordinary commutative ring, $\mathrm{IC}_\theta^\natural \in \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ is perverse in the usual sense, but it is not necessarily the IC-sheaf of $\mathcal{G}\mathrm{r}_\mathrm{G}^{\leq\theta}$.

Remark 6.2.13. Since both $\widetilde{\mathrm{IC}}_\theta$ is even (in the sense of Definition 2.3.1) and $\mathrm{IC}_\theta^\natural$ is a retract of an even shift of $\widetilde{\mathrm{IC}}_\theta$, it follows that $\mathrm{IC}_\theta^\natural \in \mathrm{Shv}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ is also even. The image of $\mathrm{IC}_\theta^\natural$ in $\mathrm{IndPerf}^{\mathrm{q\text{-}min,gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}})$ under the functor $\mathcal{S} \circ \underline{\tau}_\star$ of Theorem 5.1.16 and Theorem 5.2.9 is isomorphic to $(\mathrm{Res}_{\check{\mathrm{T}}}^{\check{\mathrm{G}}}(\mathrm{T}_\theta) \oplus \mathrm{V}_{\theta,0}) \otimes_\mathbf{Z} \mathcal{O}_{{}^{2\rho}\mathrm{G}_\mathrm{F}}((2\rho,\theta))$, where $\mathcal{O}_{{}^{2\rho}\mathrm{G}_\mathrm{F}}((2\rho,\theta))$ denotes a shift in $\mathbf{Z}$-weight by $(2\rho,\theta)$.

Proposition 6.2.14. *Let k be an even $\mathbf{E}_\infty$-ring, and let $\mu_\bullet = (\mu_1, \cdots, \mu_n)$ be a sequence of dominant (quasi-)minuscule weights of $\check{\mathrm{G}}^{\mathrm{sc}}$. As in Notation 5.1.10, let $\mathrm{IC}_{\mu_\bullet}^\natural$ denote the convolution $\mathrm{IC}_{\mu_1}^\natural \star \cdots \star \mathrm{IC}_{\mu_n}^\natural$. Then $\mathrm{IC}_{\mu_\bullet}^\natural$ is projectively perverse, and $\pi_*(\mathrm{F}_{\mathrm{MV}}(\mathrm{IC}_{\mu_\bullet}^\natural))$ is isomorphic as a $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$-graded $\pi_*(k)$-module to $(\mathrm{Res}_{\check{\mathrm{T}}}^{\check{\mathrm{G}}}(\mathrm{T}_{\mu_\bullet}) \oplus$*

$\mathrm{V}_{\mu_\bullet,0}) \otimes_{\mathbf{Z}} \pi_*(k)$*, where* $\mathrm{V}_{\mu_\bullet,0}$ *is a trivial* $\check{\mathrm{G}}$*-representation in free abelian groups. Here, the* $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$*-grading is inherited from the* $\mathbf{Z}$*-grading of* $\pi_*(k)$ *(which is placed in* $\mathbb{X}_*(\mathrm{T})$*-weight* 0*), and the* $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$*-grading of* $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{T}_{\mu_\bullet}) \oplus \mathrm{V}_{\mu_\bullet,0}$ *where the* λ*-weight space is placed in* $\mathbf{Z}$*-weight* $-(2\rho,\lambda)$*.*

Proof. As in the proof of [NP01, Théorème 3.1], there is an isomorphism

$$\mathrm{R}\Gamma_c(\mathrm{S}_\lambda; \mathrm{IC}^\natural_{\mu_\bullet}) \cong \bigoplus_{|\lambda_\bullet|=\lambda} \bigotimes_{i=1}^n \mathrm{R}\Gamma_c(\mathrm{S}_{\lambda_i}; \mathrm{IC}^\natural_{\mu_i}).$$

Example 6.2.9 and Lemma 6.2.12 imply that each $\mathrm{R}\Gamma_c(\mathrm{S}_{\lambda_i}; \mathrm{IC}^\natural_{\mu_i})$ is a direct sum of copies of $\Sigma^{-(2\rho,\lambda_i)}k$, so that the entire tensor product is a direct sum of copies of $\Sigma^{-(2\rho,\lambda)}k$ as desired. □

Definition 6.2.15. Let k be an even $\mathbf{E}_\infty$-ring. Let X denote the graded $\pi_*(k)$-scheme given by ${}^{2\rho}\mathrm{G}_{\mathrm{F}}$ if G has simply-laced derived subgroup, and ${}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}$ if G has multiply-laced derived subgroup and ℓ is inverted in $\pi_0(k)$. Let $\mathrm{IndPerf}^{\text{q-min,gr}}(\mathrm{X}/{}^{2\rho}\check{\mathrm{G}})^\natural$ denote the idempotent completion of the full subcategory of $\mathrm{IndPerf}^{\text{q-min,gr}}(\mathrm{X}/{}^{2\rho}\check{\mathrm{G}})$ spanned by the vector bundles $(\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{T}_{\mu_\bullet}) \oplus \mathrm{V}_{\mu_\bullet,0}) \otimes_{\mathbf{Z}} \mathcal{O}_{\mathrm{X}/{}^{2\rho}\check{\mathrm{G}}}((2\rho,|\mu_\bullet|))$, where $\mathcal{O}_{\mathrm{X}/{}^{2\rho}\check{\mathrm{G}}}((2\rho,|\mu_\bullet|))$ denotes a shift in $\mathbf{Z}$-weight by $(2\rho,|\mu_\bullet|)$, and $\mu_\bullet$ ranges over sequences of dominant (quasi-)minuscule weights of $\check{\mathrm{G}}^{\mathrm{sc}}$. Here, notation is as in Proposition 6.2.14. Similarly, let $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_{\mathrm{G}}; k)^\natural$ denote the idempotent completion of the full subcategory of $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_{\mathrm{G}}; k)$ spanned by $\mathrm{IC}^\natural_{\mu_\bullet}$ as $\mu_\bullet$ ranges over sequences of dominant (quasi-)minuscule weights of $\check{\mathrm{G}}^{\mathrm{sc}}$.

Combined with Proposition 6.2.14, Theorem 5.1.16 and Theorem 5.2.9 imply:

Corollary 6.2.16. *Let* k *be an even* $\mathbf{E}_\infty$*-ring. If* $\mathcal{F} \in \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_{\mathrm{G}}; k)^\natural$*, then* $\mathcal{F}$ *is even (in the sense of Definition 2.3.1) and projectively perverse (in the sense of Definition 6.2.7). Suppose that the bad primes (see Table 1) of* G *are inverted in* $\pi_0(k)$*, and assume Conjecture 3.1.14 for every sequence of dominant (quasi-)minuscule coweights of* G^{ad}*. Then the essential image of* $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_{\mathrm{G}}; k)^\natural$ *under the functor* $\mathcal{S} \circ \underline{\mathcal{T}}_\star : \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_{\mathrm{G}}; k)^{\text{!-ev}} \to \mathrm{IndPerf}^{\text{q-min,gr}}(\mathrm{X}/{}^{2\rho}\check{\mathrm{G}})$ *is equivalent to* $\mathrm{IndPerf}^{\text{q-min,gr}}(\mathrm{X}/{}^{2\rho}\check{\mathrm{G}})^\natural$*.*

Moreover, if $\mathcal{F} \in \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_{\mathrm{G}}; k)^\natural$*, there is a* $\check{\mathrm{G}}$*-representation* $\mathrm{V}_\mathcal{F}$ *in free abelian groups and a* $\mathbb{X}_*(\mathrm{T})$*-indexed filtration on the fiber over* $0 \in \mathrm{T}_{\mathrm{F}}/\mathcal{W}$ *of* $\kappa^*_{\mathrm{F}}(\mathcal{S}(\underline{\mathcal{T}}_\star(\mathcal{F})))$*, i.e., on* $\pi_*(k) \otimes_{\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})} \mathrm{gr}^\star_{\text{G-ev}} \mathrm{R}\Gamma_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_{\mathrm{G}}; \mathcal{F})$*, with associated graded* $\pi_* \mathrm{F}_{\mathrm{MV}}(\mathcal{F})$*, which is in turn isomorphic to* $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{V}_\mathcal{F}) \otimes_{\mathbf{Z}} \pi_*(k)$*. The* $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$*-grading on* $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{V}_\mathcal{F}) \otimes_{\mathbf{Z}} \pi_*(k)$ *is inherited from the* $\mathbf{Z}$*-grading of* $\pi_*(k)$ *(which is placed in* $\mathbb{X}_*(\mathrm{T})$*-weight* 0*) and the* $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$*-grading of* $\mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{V}_\mathcal{F})$ *where the* λ*-weight space is placed in* $\mathbf{Z}$*-weight* $-(2\rho,\lambda)$*.*

This implies that there is a free $\mathbb{X}_*(\mathrm{T})$-graded $\mathbb{S}$-module $\mathrm{V}_{\mathcal{F},\mathbb{S}}$ such that there is a $\mathbb{X}_*(\mathrm{T}) \times \mathbf{Z}$-graded isomorphism $\pi_*(\mathrm{V}_{\mathcal{F},\mathbb{S}} \otimes_{\mathbb{S}} \mathbf{Z}) \cong \mathrm{Res}^{\check{\mathrm{G}}}_{\check{\mathrm{T}}}(\mathrm{V}_\mathcal{F})$, and such that there is an isomorphism $\mathrm{F}_{\mathrm{MV}}(\mathcal{F}) \cong \mathrm{V}_{\mathcal{F},\mathbb{S}} \otimes_{\mathbb{S}} k$ of $\mathbb{X}_*(\mathrm{T})$-graded k-modules.

Remark 6.2.17. Let k be an ordinary commutative ring. By Example 6.2.8, the homotopy category of $\mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}r_{\mathrm{G}}; k)^\natural$ is equivalent to the idempotent completion of the full subcategory of the 1-category of $\mathrm{Perv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)$ spanned by convolutions of (quasi-)minuscule IC-sheaves. On the other hand, $\mathrm{X} \cong {}^{2\rho}\check{\mathfrak{g}}^*(2)^\wedge_{\mathcal{N}}$, and the weight 0 component of $\mathcal{O}_{\check{\mathfrak{g}}^*(2)}$ is k. This implies that the $\mathbf{Z}$-weight 0 truncation of $\mathrm{IndPerf}^{\text{q-min,gr}}(\mathrm{X}/{}^{2\rho}\check{\mathrm{G}})^\natural$ is equivalent to the idempotent completion of the full subcategory of the 1-category $\mathrm{IndPerf}(\check{\mathrm{G}})^\heartsuit$ spanned by $\mathrm{T}_{\mu_\bullet}$ as $\mu_\bullet$ ranges over sequences of dominant (quasi-)minuscule weights of $\check{\mathrm{G}}^{\mathrm{sc}}$. Moreover, if $\mathcal{F} \in \mathrm{Shv}^{\text{q-min}}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k)^\natural$, then the $\check{\mathrm{G}}$-representation $\mathrm{V}_{\mathcal{F}}$ is the image of $\mathcal{F}$ under the geometric Satake equivalence of [MV07]. Corollary 6.2.16 can therefore be viewed as a variant of the *abelian* geometric Satake equivalence.

6.3. **Steenrod operations and Springer isomorphism.** We will momentarily review some of the rich theory of power operations in homotopy theory; these force the existence of additional structures on the stacks appearing on the spectral side of Theorem 5.1.16 and Theorem 5.2.9. Our goal in this section is to describe these structures explicitly. This section is motivated by a discussion with David Treumann.

Throughout this section, we will fix a prime p and an even $\mathbf{E}_\infty$-ring k such that the C_p-Tate construction $k^{t\mathrm{C}_p}$ (for the trivial C_p-action on k) is concentrated in even weights. Note that if we fix a complex orientation $\hbar \in \pi_{-2}(k^{h\mathrm{S}^1})$ of k, then $\pi_*(k^{t\mathrm{C}_p}) \cong \pi_*(k)[\hbar]^\wedge[\hbar^{-1}]/\langle p\rangle$ where $\hbar$ is in weight -2, $\langle p\rangle = \frac{[p](\hbar)}{\hbar}$, and the completion is taken in the sense of Remark A.1.6. The ring $\pi_*(k^{t\mathrm{C}_p})$ is therefore concentrated in even weights if and only if $\langle p\rangle \in \pi_0(k^{h\mathrm{S}^1})$ is a nontrivial nonzerodivisor. This is implied by p being a nontrivial nonzerodivisor in $\pi_0(k)$. For notational simplicity, we will simply write $\pi_*(k^{t\mathrm{C}_p}) \cong \pi_*(k)[\hbar^{\pm 1}]^\wedge/\langle p\rangle$.

We will also assume that G is connected, simply-laced, and semisimple; modifying arguments appropriately as in Section 5.2 extends the discussion below to the case when G is multiply-laced. One can also relax the semisimplicity condition to reductivity. We require that the bad primes of G are inverted in $\pi_0(k)$; this does not place any restrictions on p, although it does imply that if p is a bad prime of G, then $k^{t\mathrm{C}_p} = 0$, so our discussion below will be vacuous in this case. We will sometimes allow the case when k is an even $\mathbf{E}_\infty$-$\mathbf{Z}$-algebra (without inverting any primes), and will make this explicit when we do so.

Definition 6.3.1. Recall that there is an $\mathbf{E}_\infty$-map $\varphi_k : k \to k^{t\mathrm{C}_p}$, called the Tate-valued Frobenius, which is defined by the composite of the Tate diagonal $k \to (k^{\otimes p})^{t\mathrm{C}_p}$ with the C_p-Tate construction of the multiplication map $k^{\otimes p} \to k$. See [NS18, Definition IV.1.1] for a modern reference. There is also a canonical (unit) map $k \to k^{t\mathrm{C}_p}$, and the base-change of the graded 1-dimensional group $\mathrm{F} = \mathrm{Spf}(\pi_*(k^{h\mathrm{S}^1}))$ over $\pi_*(k)$ to $\pi_*(k^{t\mathrm{C}_p})$ will continue to be denoted F; any coordinate chosen on F will be base-changed from a choice over $\pi_*(k)$.

Example 6.3.2. If $k = \mathbf{Z}$, then $\mathbf{Z}^{t\mathrm{C}_p} \cong \bigoplus_{n > -\infty} \Sigma^{2n}\mathbf{F}_p$, because $\pi_*(\mathbf{Z}^{t\mathrm{C}_p}) \cong \mathbf{Z}[\hbar^{\pm 1}]/p$. The composite of $\varphi_{\mathbf{Z}}$ with the projection map $\mathbf{Z}^{t\mathrm{C}_p} \to \Sigma^{2n}\mathbf{F}_p$ is zero unless $(p-1)|n$, in which case it is the composite $\mathbf{Z} \to \mathbf{F}_p \xrightarrow{\mathrm{P}^{n/(p-1)}} \Sigma^{2n}\mathbf{F}_p$. Here, $\mathrm{P}^{n/(p-1)}$ is the $n/(p-1)$st Steenrod operation.

Remark 6.3.3. If X is an anima, then the map $\mathrm{C}^*(\mathrm{X}; k) \to \mathrm{C}^*(\mathrm{X}; k^{t\mathrm{C}_p})$ given by applying $\mathrm{C}^*(\mathrm{X}; -)$ to φ_k encodes the action of power operations of k on $\mathrm{C}^*(\mathrm{X}; k)$. Similarly, the map $\mathrm{C}_*(\mathrm{X}; k) \to \mathrm{C}_*(\mathrm{X}; k^{t\mathrm{C}_p})$ given by applying $\mathrm{C}_*(\mathrm{X}; -)$ to φ_k

encodes the action of power operations of k on $\mathrm{C}_*(\mathrm{X}; k)$. See [NS18, Theorem IV.1.21]. For instance, if $k = \mathbf{Z}$, we obtain a map $\mathrm{H}^{-*}(\mathrm{X}; \mathbf{Z}) \to \mathrm{H}^{-*}(\mathrm{X}; \mathbf{Z}^{t\mathrm{C}_p}) \cong \mathrm{H}^{-*}(\mathrm{X}; \mathbf{F}_p)[\hbar^{\pm 1}]$, whose components encode the Steenrod operations on the mod p reductions of classes in $\mathrm{H}^{-*}(\mathrm{X}; \mathbf{Z})$ by Example 6.3.2. We will continue to refer to these as "Steenrod operations", even though they act on classes in $\mathrm{H}^{-*}(\mathrm{X}; \mathbf{Z})$ (instead of classes in $\mathrm{H}^{-*}(\mathrm{X}; \mathbf{F}_p)$, which is the more traditional meaning of Steenrod operations).

Example 6.3.4. Let T be a torus. Then φ_k induces a map $\varphi_k^* : k^{h\mathrm{T}} \to (k^{t\mathrm{C}_p})^{h\mathrm{T}}$, and hence in particular a $k^{t\mathrm{C}_p}$-linear map $k^{h\mathrm{T}} \otimes_k {}^{\varphi} k^{t\mathrm{C}_p} \to (k^{t\mathrm{C}_p})^{h\mathrm{T}}$. Applying $\mathrm{Spf}(\pi_*(-))$, we obtain a map $\varphi_k : \mathrm{T}_\mathrm{F} \to \varphi_k^*(\mathrm{T}_\mathrm{F})$ of group schemes over $\pi_*(k^{t\mathrm{C}_p})$. Note that both T_F and $\varphi_k^*(\mathrm{T}_\mathrm{F})$ are abstractly isomorphic as group schemes over $\pi_*(k^{t\mathrm{C}_p})$ (via φ_k), but φ_k is very nontrivial in chosen coordinates on F. For instance, it follows from [Qui71a, Proposition 3.17] that the map $\varphi_k : \mathrm{T}_\mathrm{F} \to \varphi_k^*(\mathrm{T}_\mathrm{F})$ is given in coordinates on T_F by the universal formula

$$\varphi_k(x) = x \prod_{i=1}^{p-1} \tfrac{x +_\mathrm{F} [i](\hbar)}{\hbar}. \tag{6.3.1}$$

This, of course, is analogous to the Artin-Schreier polynomial, except that it is a power series in x with coefficients in $\pi_*(k^{t\mathrm{C}_p}) \cong \pi_*(k)[\hbar^{\pm 1}]^\wedge/\langle p \rangle$. For instance, if k is an even $\mathbf{Z}$-algebra, then (6.3.1) simply reads $\varphi_k(x) = x - \hbar^{1-p} x^p$, so it is in fact the Artin-Schreier polynomial. Similarly, if $k = \mathrm{ku}$, then there is an isomorphism

$$\pi_*(\mathrm{ku}^{t\mathrm{C}_p}) \cong \mathbf{Z}[\beta][\hbar^{\pm 1}]^\wedge/\langle p \rangle \cong \mathbf{Z}[\![q-1]\!][\hbar^{\pm 1}]/[p]_q \cong \mathbf{Z}_p[\zeta_p][\hbar^{\pm 1}],$$

where $\beta\hbar = q - 1$ and $\langle p \rangle = [p]_q = \frac{q^p - 1}{q - 1}$. The final isomorphism sends $q \mapsto \zeta_p$, where ζ_p is a primitive pth root of unity. Note that the image of β in this ring is $\frac{\zeta_p - 1}{\hbar}$. Write $y = \frac{x}{\hbar}$; then $\frac{x +_\mathrm{F} [i](\hbar)}{\hbar} = q^i y + [i]_q$, so the formula (6.3.1) reads

$$\varphi_{\mathrm{ku}}(x) = \hbar y \prod_{i=1}^{p-1} (q^i y + [i]_q) = (-1)^{p-1} \hbar \tfrac{(1 + (q-1)y)^p - 1}{(q-1)^p} = (-1)^{p-1} \sum_{i=1}^{p} \tfrac{\hbar}{(q-1)^{p-i}} \binom{p}{i} y^i.$$

This is well-defined because $\binom{p}{i}$ is divisible by p for $i \neq p$, and p is a unit multiple of $(q-1)^{p-1}$ in $\mathbf{Z}_p[\zeta_p]$.

Construction 6.3.5. The homomorphism $\varphi_k : \mathrm{F} \to \varphi_k^*(\mathrm{F})$ over $\pi_*(k^{t\mathrm{C}_p})$ induces a homomorphism $\varphi_k^*(\mathrm{F}^\vee) \to \mathrm{F}^\vee$. If H is a graded group scheme over $\pi_*(k^{t\mathrm{C}_p})$, we obtain a map $\varphi_k : \mathrm{H}_\mathrm{F} \to \mathrm{H}_{\varphi_k^*(\mathrm{F})}$ over $\pi_*(k^{t\mathrm{C}_p})$, where $\mathrm{H}_\mathrm{F} = \mathrm{Hom}(\mathrm{F}^\vee, \mathrm{H})$ is as in Definition 4.1.1. The map φ_k is H-equivariant for the action of H on H_F by conjugation, so it induces a map $\mathrm{H}_\mathrm{F}/\mathrm{H} \to \mathrm{H}_{\varphi_k^*(\mathrm{F})}/\mathrm{H}$ of quotient stacks.

Example 6.3.6. Let k be an even $\mathbf{Z}$-algebra, so that $\mathrm{F} = \hat{\mathbf{G}}_a(2)$. Under the isomorphism $\mathrm{H}_\mathrm{F} \cong \mathfrak{h}(2)^\wedge_\mathcal{N}$ of Lemma 4.1.2, the $\mathbf{F}_p[\hbar^{\pm 1}]$-linear map $\varphi_k : \mathrm{H}_\mathrm{F} \to \mathrm{H}_{\varphi_k^*(\mathrm{F})}$ is given by the map $x \mapsto x - \hbar^{1-p} x^{[p]}$.

Example 6.3.7. Let $k = \mathrm{ku}$, so that $\mathrm{F} = \hat{\mathbf{G}}_\beta$. Under the identification from Remark 4.1.5 of $(\mathrm{SL}_n)_\mathrm{F}$ with the subscheme of $\mathrm{SL}_n \times \mathrm{Spec}(\mathbf{Z}[\beta])$ of matrices A such that $\frac{\det(\mathrm{I}_n + \beta\mathrm{A}) - 1}{\beta} = 0$, the $\mathbf{Z}_p[\zeta_p][\hbar^{\pm 1}]$-linear map $\varphi_k : (\mathrm{SL}_n)_\mathrm{F} \to (\mathrm{SL}_n)_{\varphi_k^*(\mathrm{F})}$ is given by the map

$$x \mapsto \tfrac{(\mathrm{I}_n + \beta x)^p - \mathrm{I}_n}{p\beta} = \tfrac{(\hbar\mathrm{I}_n + (q-1)x)^p - \hbar^p \mathrm{I}_n}{p\hbar^{p-1}(q-1)}.$$

The following is a consequence of the naturality of Definition 2.2.23 in the coefficient $\mathbf{E}_\infty$-ring:

Proposition 6.3.8. *Define* $\varphi_{\mathbf{Z}} : \mathcal{O}_{\mathcal{W}} \to \mathcal{O}_{\mathcal{W}}[\hbar^{\pm 1}]/p$ *via the action of* $\varphi_{\mathbf{Z}}$ *on* F *and the isomorphism* $\mathcal{O}_{\mathcal{W}} \cong \mathrm{M}_{\Gamma_{\mathcal{B}}}$ *from Example 2.2.21. Then the isomorphism* $\mathrm{H}^{-*}_{\mathrm{T},c}(\mathcal{B}; k) \cong \mathcal{O}_{\mathcal{W}}$ *respects the action of* $\varphi_{\mathbf{Z}}$.

Using Definition 2.2.28, Proposition 6.3.8 in turn defines an action of $\varphi_{\mathbf{Z}}$ on $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$. When $k = \mathbf{Z}[1/p]$ and p is not a bad prime for G, the action of $\varphi_{\mathbf{Z}}$ on $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}}) \cong \mathrm{H}^{-*}(\mathrm{BG}; \mathbf{Z}[1/p])$ was studied by Borel and Serre in [BS53] and [BS51] (among many others).

Like with Proposition 6.3.8, the following is a consequence of the naturality of Theorem 5.1.16 in the coefficient $\mathbf{E}_\infty$-ring:

Theorem 6.3.9. *Under the* $\mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}})$*-linear equivalence*

$$\mathcal{S} : \mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \xrightarrow{\simeq} \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}})$$

of Theorem 5.1.16, there is a commutative diagram

$$\begin{tikzcd}
\mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \otimes_{\pi_*(k)} \pi_*(k^{t\mathrm{C}_p}) \arrow[r, "\mathcal{S}"] \arrow[d, "\varphi_k"] & \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_{\varphi_k^*(\mathrm{F})}/{}^{2\rho}\check{\mathrm{G}}) \arrow[d, "\varphi_k^*"] \\
\mathrm{Shv}^{\text{q-min,gr}}_{\mathrm{G}_\mathcal{O}}(\mathcal{G}\mathrm{r}_\mathrm{G}; k^{t\mathrm{C}_p}) \arrow[r, "\mathcal{S}"] & \mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\mathrm{G}_\mathrm{F}/{}^{2\rho}\check{\mathrm{G}})
\end{tikzcd}$$

Theorem 6.3.9, or rather the ingredients in the proof of Theorem 5.1.16, can be used to describe the action of φ_k on $\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$. Namely:

Proposition 6.3.10. *Under the isomorphism* $\mathrm{Spec}\,\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \cong \widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ *from Theorem 4.2.12, the map* $\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \otimes_{\pi_*(k)} \pi_*(k^{t\mathrm{C}_p}) \to \mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G}; k^{t\mathrm{C}_p})$ *identifies with the endomorphism of* $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}$ *coming from the commutative diagram*

$$\begin{tikzcd}
\mathrm{T}_\mathrm{F} \arrow[r, "\varphi_k"] \arrow[d, "\kappa_\mathrm{F}"'] & \mathrm{T}_{\varphi_k^*(\mathrm{F})} \arrow[d, "\kappa_\mathrm{F}"] \\
{}^{2\rho}\mathrm{B}_\mathrm{F}/{}^{2\rho}\check{\mathrm{B}} \arrow[r, "\varphi_k"'] & {}^{2\rho}\mathrm{B}_\mathrm{F}/{}^{2\rho}\check{\mathrm{B}}.
\end{tikzcd}$$

Proof. By Theorem 4.2.12, both $\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G}; k) \cong \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$ injects into its $e(\Phi)$-localization, which is isomorphic to $\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{T}; k)[e(\Phi)^{-1}] \cong \mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_\mathrm{F}}[e(\Phi)^{-1}]$. Furthermore, the maps $\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{T}; k) \to \mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G}; k)$ and $\mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_\mathrm{F}} \to \mathcal{O}_{\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}}$ are φ_k-equivariant, so we are reduced to the case when G is a torus. The claim in this case follows from Example 6.3.4 and the observation that under the isomorphism $\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{T}; k) \cong \mathcal{O}_{\mathrm{T}_\mathrm{F}}[\mathbb{X}_*(\mathrm{T})]$, the power operation φ_k acts trivially on the Laurent polynomial generators from $\mathbb{X}_*(\mathrm{T})$. $\square$

Remark 6.3.11. Recall from Proposition 4.2.15 that if G is the Chevalley split form of a connected reductive group (*not* necessarily with simply-laced derived subgroup), there is an integral isomorphism

$$\mathrm{Spec}\,\mathrm{H}^{\mathrm{T,BM}}_*(\mathcal{G}\mathrm{r}_\mathrm{G}; \mathbf{Z}) \cong \hat{\check{\mathfrak{t}}}^*(2) \times_{{}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{B}}} \hat{\check{\mathfrak{t}}}^*(2)$$

without inverting any primes. Let us denote the right-hand side by $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\hat{\mathbf{G}}_a}$. Proposition 6.3.10 continues to hold in this generality (with the same proof): under the isomorphism (4.2.2), the map $\mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{Z}) \otimes_{\mathbf{Z}} \pi_*(\mathbf{Z}^{t\mathrm{C}_p}) \to \mathrm{H}_*^{\mathrm{T},\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{Z}^{t\mathrm{C}_p})$ identifies with the endomorphism of $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\hat{\mathbf{G}}_a}$ coming from the commutative diagram

$$\begin{tikzcd} \hat{\check{\mathfrak{t}}}^*(2) \arrow[r, "\varphi_{\mathbf{Z}}"] \arrow[d, "\kappa_\mathrm{F}"'] & \hat{\check{\mathfrak{t}}}^*(2) \arrow[d, "\kappa_\mathrm{F}"] \\ {}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{B}} \arrow[r, "\varphi_{\mathbf{Z}}"'] & {}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}/{}^{2\rho}\check{\mathrm{B}}. \end{tikzcd}$$

Here, the map $\varphi_{\mathbf{Z}} : {}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}} \to {}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}$ over $\pi_*(\mathbf{Z}^{t\mathrm{C}_p}) \cong \mathbf{F}_p[\hbar^{\pm 1}]$ is given by $x \mapsto x - \hbar^{1-p}x^{[p]}$ (see Example 6.3.6).

This implies the following calculation of the total Steenrod operation $\varphi_{\mathbf{Z}}$ on the *non*equivariant homology $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{Z})$; note that this is an operation

$$\varphi_{\mathbf{Z}} : \mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{Z}) \to \mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{Z}^{t\mathrm{C}_p}) \cong \mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{F}_p)[\hbar^{\pm 1}].$$

The action of $\varphi_{\mathbf{Z}}$ on ${}^{2\rho}\check{\mathfrak{u}}^\perp(2)^\wedge_{\mathcal{N}}$ sends ψ_0 to some $\varphi_{\mathbf{Z}}(\psi_0)$. Because $\varphi_{\mathbf{Z}}(\psi_0)$ has no poles in $\hbar^{-1}$ and is congruent to ψ_0 modulo $\hbar^{-1}$, there is some $b \in {}^{2\rho}\check{\mathrm{B}}$ such that $\mathrm{Ad}_b(\varphi_{\mathbf{Z}}(\psi_0)) = \psi_0$. Conjugation by b^{-1} defines a homomorphism $\varphi_{\mathbf{Z}} : \mathrm{Z}_{^{2\rho}\check{\mathrm{B}}}(\psi_0) \to \mathrm{Z}_{^{2\rho}\check{\mathrm{B}}}(\psi_0)$, which is independent of the choice of b because b is unique up to left-translation by $\mathrm{Z}_{^{2\rho}\check{\mathrm{B}}}(\psi_0)$ and $\mathrm{Z}_{^{2\rho}\check{\mathrm{B}}}(\psi_0)$ is commutative. Under the isomorphism $\mathrm{Spec}\,\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{Z}) \cong \mathrm{Z}_{^{2\rho}\check{\mathrm{B}}}(\psi_0)$ coming from (4.2.2), the action of $\varphi_{\mathbf{Z}}$ on $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{Z})$ identifies with the above endomorphism $\varphi_{\mathbf{Z}}$ of $\mathrm{Z}_{^{2\rho}\check{\mathrm{B}}}(\psi_0)$. (Dualizing, this also computes the Steenrod operations on $\mathrm{H}^{-*}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{Z})$.)

Using Remark 6.3.11, one obtains a proof of the following; it is an elementary observation, but we present it anyway as a proof of concept.

Proposition 6.3.12 ([Jan86, Section 4.1])**.** *Suppose $\check{\mathrm{G}}$ is the Chevalley split form of a connected simple group (not necessarily simply-laced), and let h be the Coxeter number of $\check{\mathrm{G}}$. Then the restricted pth operation on the nilpotent cone $\mathcal{N} \subseteq \check{\mathfrak{g}}^*_{\mathbf{F}_p}$ vanishes if $p \geq h$.*

Proof. Since $\mathcal{N} \subseteq \check{\mathfrak{g}}^*_{\mathbf{F}_p}$ is the $\check{\mathrm{G}}$-orbit of $\check{\mathfrak{b}}^\perp \subseteq \check{\mathfrak{g}}^*$, it suffices to show that the restricted pth operation on $\check{\mathfrak{b}}^\perp$ vanishes if $p \geq h$. Note that if $p \geq h$, then in particular p is not a bad prime for G. The $\check{\mathrm{B}}$-orbit $\check{\mathrm{B}}/\mathrm{Z}_{\check{\mathrm{B}}}(\psi_0) \subseteq \check{\mathfrak{b}}^\perp$ is open and has complement of codimension 2 (the latter follows from the proof of Proposition 4.1.14). The operation $\varphi_{\mathbf{Z}}$ on $\check{\mathfrak{b}}^\perp$ is induced by the homomorphism $\varphi_{\mathbf{Z}} : \mathrm{Z}_{\check{\mathrm{B}}}(\psi_0) \to \mathrm{Z}_{\check{\mathrm{B}}}(\psi_0)$ from Proposition 4.2.15, so it suffices to show that $\varphi_{\mathbf{Z}}$ is the identity on $\mathrm{Z}_{\check{\mathrm{B}}}(\psi_0)$ if $p \geq h$. Since the action of $\varphi_{\mathbf{Z}}$ on $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{Z})$ identifies with the endomorphism $\varphi_{\mathbf{Z}}$ of $\mathrm{Z}_{^{2\rho}\check{\mathrm{B}}}(\psi_0)$ under the isomorphism $\mathrm{Spec}\,\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{Z}) \cong \mathrm{Z}_{^{2\rho}\check{\mathrm{B}}}(\psi_0)$ coming from (4.2.2), it suffices to show that $\varphi_{\mathbf{Z}}$ is the just the mod p reduction map $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{Z}) \to \mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};\mathbf{F}_p)[\hbar^{\pm 1}]$.

This can be proved using the generating complexes from [Bot58], as elaborated upon in [LM11]. Namely, recall that if X is a homotopy commutative H-space and $f : \mathrm{Y} \to \mathrm{X}$ is a map from a CW-complex into X, then f is said to exhibit Y as a generating complex for X if f induces a surjection $\mathrm{Sym}(\mathrm{H}_*(\mathrm{Y};\mathbf{Z})) \twoheadrightarrow \mathrm{H}_*(\mathrm{X};\mathbf{Z})$. In [LM11], it was shown that if θ denotes the highest short coroot of G, then the Schubert variety $\mathcal{G}\mathrm{r}_\mathrm{G}^{\leq\theta}$ is a generating complex for $\mathcal{G}\mathrm{r}_\mathrm{G}$. Since $\mathrm{H}_*(\mathcal{G}\mathrm{r}_\mathrm{G}^{\leq\theta};\mathbf{Z})$ generates

$\mathrm{H}_*(\mathcal{G}r_{\mathrm{G}}; \mathbf{Z})$ as a ring, and $\varphi_{\mathbf{Z}}$ is a ring map, it suffices to show that $\varphi_{\mathbf{Z}}$ is the identity on $\mathrm{H}_*(\mathcal{G}r_{\mathrm{G}}^{\leq\theta}; \mathbf{Z})$. By Example 6.3.2, it equivalently suffices to show that all Steenrod operations P^i act trivially on $\mathrm{H}_*(\mathcal{G}r_{\mathrm{G}}^{\leq\theta}; \mathbf{Z})$ for $i > 0$.

To see this, observe that the dimension of $\mathcal{G}r_{\mathrm{G}}^{\leq\theta}$ is given by $2(h-1)$, where h is the Coxeter number of $\check{\mathrm{G}}$. The operation P^i sends a class in $\mathrm{H}_*(\mathcal{G}r_{\mathrm{G}}^{\leq\theta}; \mathbf{Z})$ in weight j to a class in $\mathrm{H}_*(\mathcal{G}r_{\mathrm{G}}^{\leq\theta}; \mathbf{F}_p)$ in weight $j-2i(p-1)$. Since $\mathrm{H}_*(\mathcal{G}r_{\mathrm{G}}^{\leq\theta}; \mathbf{F}_p)$ is concentrated in nonnegative weights and $p \geq h$, we see that P^i could only possibly act nontrivially when $p = h$ and $i = 1$, and that too only on classes in $\mathrm{H}_{2(h-1)}(\mathcal{G}r_{\mathrm{G}}^{\leq\theta}; \mathbf{Z})$. However, P^1 applied to such a class would land in $\mathrm{H}_0(\mathcal{G}r_{\mathrm{G}}^{\leq\theta}; \mathbf{F}_p)$. This implies that it is zero: any degree-shifting Steenrod operation landing in $\mathrm{H}_0(\mathrm{X}; \mathbf{F}_p)$ necessarily vanishes if X is a connected space. □

Remark 6.3.13. In Proposition 6.3.12, we showed that $\varphi_{\mathbf{Z}}$ is just the mod p reduction map on $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}r_{\mathrm{G}}; \mathbf{Z})$ for $p \geq h$. This can be proved alternatively as follows. By [Ser53, Kum65], it is shown that there is a map $\prod_{i=1}^{\mathrm{rank(G)}} \mathrm{S}^{2d_i-1} \to \mathrm{G}$ (where the d_i are the fundamental degrees of G) which induces an isomorphism in mod p homology if and only if $p \geq h$. This implies that the map $\prod_{i=1}^{\mathrm{rank(G)}} \Omega\mathrm{S}^{2d_i-1} \to \Omega\mathrm{G}$ induces an isomorphism in mod p homology iff $p \geq h$; this is the key homotopical fact underlying why $p \geq h$ is much simpler than smaller p. The action of $\varphi_{\mathbf{Z}}$ on $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}r_{\mathrm{G}}; \mathbf{Z}) \cong \mathrm{H}_*(\Omega\mathrm{G}; \mathbf{Z})$ is therefore determined by the action of $\varphi_{\mathbf{Z}}$ on $\mathrm{H}_*(\Omega\mathrm{S}^{2d_i-1}; \mathbf{Z}) \cong \mathbf{Z}[x_{2d_i-2}]$. For degree reasons, and the fact that any degree-shifting Steenrod operation landing in $\mathrm{H}_0(\mathrm{X}; \mathbf{F}_p)$ necessarily vanishes if X is a connected space, it follows that $\varphi_{\mathbf{Z}}$ is just the mod p reduction map, as desired.

Example 6.3.14. Let $\mathrm{G} = \mathrm{GL}_n$, so that there is an isomorphism $\mathfrak{gl}_n^* \cong \mathfrak{gl}_n$. Under this isomorphism, ψ_0 is sent to the maximal Jordan block (with 1 on the subdiagonal and 0 elsewhere). The calculation of Remark 6.3.11 leads to the following description of $\varphi_{\mathbf{Z}}$ on $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}r_{\mathrm{GL}_n}; \mathbf{Z}) \cong \mathbf{Z}[a_0^{\pm 1}, a_1, \cdots, a_{n-1}]$ (where a_i lives in weight $2i$). First, $\varphi_{\mathbf{Z}}(a_0) = a_0$, and for $0 < i < n$, one has

$$\varphi_{\mathbf{Z}}(a_i) = \sum_{j=0}^{\lfloor i/p \rfloor} (-1)^{i-jp} \binom{i-j(p-1)}{j} \hbar^{-j(p-1)} a_{i-j(p-1)}. \tag{6.3.2}$$

This can alternatively be deduced as follows: there is a map $\mathbf{CP}_+^{n-1} \to \mathcal{G}r_{\mathrm{GL}_n}$ which is a generating complex (in the sense of [Bot58], as reviewed in Proposition 6.3.12), and so the action of $\varphi_{\mathbf{Z}}$ on $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}r_{\mathrm{GL}_n}; \mathbf{Z})$ is determined by its action on $\mathrm{H}_*(\mathbf{CP}^{n-1}; \mathbf{Z})$. To describe this, it is convenient to take $n = \infty$, i.e., to describe the action of $\varphi_{\mathbf{Z}}$ on $\mathrm{H}_*(\mathbf{CP}^\infty; \mathbf{Z})$. Write $\mathrm{H}_*(\mathbf{CP}^\infty; \mathbf{Z}) \cong \Gamma_{\mathbf{Z}}(a)$, so that the classes a_i described above are $\gamma_i(a)$. The homology $\mathrm{H}_*(\mathbf{CP}^\infty; \mathbf{Z})$ is $\mathbf{Z}$-linearly dual to the cohomology $\mathrm{H}^{-*}(\mathbf{CP}^\infty; \mathbf{Z}) \cong \mathbf{Z}[x]$, on which $\varphi_{\mathbf{Z}}$ acts by $x \mapsto x - \hbar^{1-p} x^p$. Consider the generating series $\exp(ax) := \sum_{i \geq 0} \gamma_i(a) x^i$. Then the action of $\varphi_{\mathbf{Z}}$ on $\mathrm{H}_*(\mathbf{CP}^\infty; \mathbf{Z})$ is obtained by extracting the coefficients of x^i in $\exp(a\varphi_{\mathbf{Z}}(x)) = \exp(a(x - \hbar^{1-p} x^p))$, which yields (6.3.2).

This can be rephrased alternatively as follows. If we view $\operatorname{Spec} \mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}r_{\mathrm{GL}_n}; \mathbf{Z})$ as $\mathrm{Z}_{2\rho_{\check{\mathrm{B}}}}(\psi_0) \cong \mathbf{W}_n^\times$, where $\mathbf{W}_n$ is the big Witt vectors of length n, so that its functor of points is given by $\mathrm{R} \mapsto (\mathrm{R}[x]/x^n)^\times$, then $\varphi_{\mathbf{Z}}$ is given on R-points by the effect on unit groups of the map $\mathrm{R}[x]/x^n \to \mathrm{R}[\hbar^{\pm 1}, x]/(p, x^n)$ sending $x \mapsto x - \hbar^{1-p} x^p$.

Remark 6.3.15. More generally, if k is an even $\mathbf{E}_\infty$-ring, there is an isomorphism $\operatorname{Spec} \mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_{\mathrm{GL}_n}; k) \cong \mathbf{W}_n^\times$, where $\mathbf{W}_n$ is the big Witt vectors of length n over $\pi_*(k)$. If R is a graded $\pi_*(k)$-algebra, then (6.3.1) implies that φ_k is given on R-points by the effect on unit groups of the map $\mathrm{R}[x]/x^n \to \mathrm{R}[\hbar^{\pm 1}]^\wedge[x]/(\langle p\rangle, x^n)$ sending $x \mapsto x \prod_{i=1}^{p-1} \frac{x +_{\mathrm{F}} [i](\hbar)}{\hbar}$.

Example 6.3.16. Continuing Example 4.2.16, recall that there is an isomorphism

$$\mathrm{H}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}_2}; \mathbf{Z}) \cong \mathbf{Z}[u, v, w]/(2v = u^2),$$

where u, v, and w live in weights 2, 4, and 10. As discussed in Example 4.2.16, $\operatorname{Spec} \mathrm{H}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}_2}; \mathbf{Z}) \cong \mathrm{Z}_{^{2\rho}\check{\mathrm{B}}}(\psi_0)$ where $\psi_0 \in \check{\mathfrak{u}}^\perp \subseteq \mathfrak{g}_2^*$. Using the calculations of [Ken87] and Remark 6.3.11, we find that the action of $\varphi_{\mathbf{Z}}$ on $\mathrm{H}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}_2}; \mathbf{Z})$ can be described as follows. For degree reasons, $\varphi_{\mathbf{Z}}$ is the mod p reduction map if $p > 5$ (see also Proposition 6.3.12), so we only need to describe the cases $p = 2, 3, 5$. In this case, one finds:

$$\varphi_{\mathbf{Z}}(u) = u,$$
$$\varphi_{\mathbf{Z}}(v) = \begin{cases} v + \hbar^{-1} u & p = 2 \\ v & p > 2, \end{cases}$$
$$\varphi_{\mathbf{Z}}(w) = \begin{cases} w + \hbar^{-4} u & p = 5 \\ w & p \neq 5. \end{cases}$$

Of course, this could have also been deduced by direct calculation with $\mathrm{H}_*(\mathcal{G}\mathrm{r}_{\mathrm{G}_2}; \mathbf{Z})$.

Recall from Remark 5.1.18 that if R is a graded commutative ring in which the bad primes of G are invertible, F is a 1-dimensional graded formal group law over R, and ${}^{2\rho}\mathcal{N}_{\mathrm{F}} = {}^{2\rho}\mathrm{G}_{\mathrm{F}} \times_{\mathrm{T}_{\mathrm{F}}/\mathcal{W}} \operatorname{Spec}(\mathrm{R})$ is the F-nilpotent cone, then ${}^{2\rho}\mathcal{N}_{\mathrm{F}}$ is the ${}^{2\rho}\check{\mathrm{G}}$-orbit of ${}^{2\rho}\mathrm{U}_{\mathrm{F}} \hookrightarrow {}^{2\rho}\mathrm{G}_{\mathrm{F}}$. Moreover, ${}^{2\rho}\mathrm{U}_{\mathrm{F}} \cong ({}^{2\rho}\check{\mathrm{B}} \times \operatorname{Spec}(\mathrm{R}))/\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}|_0$, where $\widetilde{\mathrm{J}}_{\check{\mathrm{G}},\mathrm{F}}|_0 \cong \operatorname{Spec}(\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \mathrm{MU})) \otimes_{\pi_*(\mathrm{MU})} \mathrm{R}$. When $\mathrm{F} = \hat{\mathbf{G}}_m$, a version of the following (with less restrictive condition on primes) is proved in [Spr69], using [SS70, Chapter III, Section 3.11].

Proposition 6.3.17. *Let* G *be the Chevalley split form over* $\mathbf{Z}$ *of a connected semisimple group (not necessarily simply-laced). Suppose that* $p \geq h$*, and that* p *does not divide the order of the center of* G *(this condition is only meaningful if* G *has type* A*). Assume that* R *is a* p*-complete graded commutative ring, and let* F *be a* 1*-dimensional formal group law over* R*. Then there is a* ${}^{2\rho}\check{\mathrm{G}}$*-equivariant isomorphism* ${}^{2\rho}\mathcal{N}_{\mathrm{F}} \cong {}^{2\rho}\mathcal{N}$ *over* $\operatorname{Spec}(\mathrm{R})$*, where* ${}^{2\rho}\mathcal{N} \hookrightarrow {}^{2\rho}\check{\mathfrak{g}}(2)$ *denotes the usual nilpotent cone.*

Proof. We may work in the universal case and assume R is the p-completion of the Lazard ring, so that $\mathrm{R} = \pi_*(k)$ where $k = \mathrm{MU}_p^\wedge$. Because p does not divide the order of the center of G, an argument similar to Lemma 5.2.1 shows that ${}^{2\rho}\mathrm{G}_{\mathrm{F}} \cong {}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}$, so ${}^{2\rho}\mathcal{N}_{\mathrm{F}} = {}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}} \times_{\check{\mathrm{T}}_{\mathrm{F}}/\mathcal{W}} \operatorname{Spf}(\pi_*(k))$. Moreover, ${}^{2\rho}\mathcal{N}_{\mathrm{F}}$ is the ${}^{2\rho}\check{\mathrm{G}}$-orbit of ${}^{2\rho}\check{\mathrm{U}}_{\mathrm{F}} \hookrightarrow {}^{2\rho}\check{\mathrm{G}}_{\mathrm{F}}$ and ${}^{2\rho}\check{\mathrm{U}}_{\mathrm{F}} \cong ({}^{2\rho}\check{\mathrm{B}} \times \operatorname{Spec}(\pi_*(k)))/\operatorname{Spec}(\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k))$. It therefore suffices to show that there is an isomorphism $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \cong \mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \pi_*(k))$ of cocommutative Hopf algebras over $\pi_*(k)$. First, observe that the Whitehead filtration on k defines a Hopf algebra filtration on $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \cong \mathrm{H}_*(\Omega\mathrm{G}; k)$ with associated graded $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; \pi_*(k)) \cong \mathrm{H}_*(\Omega\mathrm{G}; \pi_*(k))$. (This is of course just the Atiyah-Hirzebruch filtration.) Observe that this filtration on $\mathrm{H}_*(\Omega\mathrm{G}; k)$ splits as $\pi_*(k)$-algebras. For this, it suffices to show that $\mathrm{H}_*(\Omega\mathrm{G};\pi_*(k))$ is a polynomial ring, which in turn follows from showing that the localization $\mathrm{H}_*(\Omega\mathrm{G};\mathbf{Z}')$ of $\mathrm{H}_*(\Omega\mathrm{G};\mathbf{Z})$ at the bad primes of G is a polynomial ring. This in turn follows from Proposition 4.2.15, which gives an isomorphism $\operatorname{Spec}\mathrm{H}_*(\Omega\mathrm{G};\mathbf{Z}') \cong \mathrm{Z}_{\check{\mathrm{G}}}(f_0)$ of group schemes, and the fact that $\mathrm{Z}_{\check{\mathrm{G}}}(f_0)$ is a smooth commutative unipotent group scheme (the smoothness is proved in [Spr66, Theorem 5.9(a)]).

It remains only to show that for some choice of splitting of the Atiyah-Hirzebruch filtration, the resulting algebra isomorphism $\mathrm{H}_*(\Omega\mathrm{G};k) \cong \mathrm{H}_*(\Omega\mathrm{G};\pi_*(k))$ is one of coalgebras. As reviewed in Remark 6.3.13, it is shown in [Ser53, Kum65] the loop space map $f : \prod_{i=1}^{\mathrm{rank}(\mathrm{G})} \Omega\mathrm{S}^{2d_i-1} \to \Omega\mathrm{G}$ induces an isomorphism in mod p homology iff $p \geq h$, and hence f is in fact a p-complete equivalence of loop spaces. Because $\mathrm{H}_*^{\mathrm{BM}}(\mathcal{G}\mathrm{r}_\mathrm{G};k) \cong \mathrm{H}_*(\Omega\mathrm{G};k)$, the Künneth formula implies that it suffices to show that there is an isomorphism $\mathrm{H}_*(\Omega\mathrm{S}^{2d_i-1};k) \cong \mathrm{H}_*(\Omega\mathrm{S}^{2d_i-1};\pi_*(k))$ of cocommutative Hopf algebras over $\pi_*(k)$. This, however, is clear: since G is semisimple, $d_i > 1$, and so both are polynomial algebras over $\pi_*(k)$ on a primitive generator in weight $2d_i - 2$. □

Remark 6.3.18. In [Spr69], it is shown that when $\mathrm{F} = \hat{\mathbf{G}}_m$, there is an isomorphism ${}^{2\rho}\mathcal{N}_\mathrm{F} \cong {}^{2\rho}\mathcal{N}$ over Spec(R) under the much weaker assumption that p is not a bad prime for G and that p does not divide the order of the center of G. I do not know if Proposition 6.3.17 holds in this generality. However, one strategy to prove it in this generality is as follows. Recall that it suffices to show that there is a Hopf algebra isomorphism $\mathrm{H}_*(\Omega\mathrm{G};k) \cong \mathrm{H}_*(\Omega\mathrm{G};\pi_*(k))$, where k is the localization of MU at the bad primes of G and the order of the center of G. For this, recall from [Bot58] that there is a finite CW-complex X with a map $f : \mathrm{X} \to \Omega\mathrm{G}$ such that the induced map $\mathrm{Sym}_{\pi_*(k)}(\mathrm{H}_*(\mathrm{X};\pi_*(k))) \to \mathrm{H}_*(\Omega\mathrm{G};\pi_*(k))$ (and hence also the induced map $\mathrm{Sym}_{\pi_*(k)}(\mathrm{H}_*(\mathrm{X};k)) \to \mathrm{H}_*(\Omega\mathrm{G};k)$) is surjective. Since both of these maps are Hopf algebra maps, where the coproduct on $\mathrm{Sym}_{\pi_*(k)}(\mathrm{H}_*(\mathrm{X};\pi_*(k)))$ and $\mathrm{Sym}_{\pi_*(k)}(\mathrm{H}_*(\mathrm{X};k))$ are induced by the cocommutative coalgebra structures on $\mathrm{H}_*(\mathrm{X};\pi_*(k))$ and $\mathrm{H}_*(\mathrm{X};k)$, it suffices to show that there is an isomorphism $\mathrm{H}_*(\mathrm{X};\pi_*(k)) \cong \mathrm{H}_*(\mathrm{X};k)$ of coalgebras over $\pi_*(k)$. Since X can be chosen to have an even cell decomposition, $\mathrm{H}_*(\mathrm{X};\pi_*(k))$ and $\mathrm{H}_*(\mathrm{X};k)$ are isomorphic finite free $\pi_*(k)$-modules with duals $\mathrm{H}^{-*}(\mathrm{X};\pi_*(k))$ and $\mathrm{H}^{-*}(\mathrm{X};k)$. It therefore suffices to show that there is a choice of X and an isomorphism $\mathrm{H}^{-*}(\mathrm{X};\pi_*(k)) \cong \mathrm{H}^{-*}(\mathrm{X};k)$ of $\pi_*(k)$-*algebras*. Unfortunately, I do not know if this is possible for each simple G; however, we have the following partial extension of Proposition 6.3.17 to the classical groups and the "honorary" classical group G_2.

Proposition 6.3.19. *Suppose* G *is a classical group (i.e., of type* A, B, C, *or* D*) or of type* G_2*, and suppose that the bad primes of* G *and the order of the center of* G *are invertible in* R*. Then there is a* ${}^{2\rho}\check{\mathrm{G}}$*-equivariant isomorphism* ${}^{2\rho}\mathcal{N}_\mathrm{F} \cong {}^{2\rho}\mathcal{N}$ *over* Spec(R)*, and this isomorphism restricts to a* ${}^{2\rho}\check{\mathrm{B}}$*-equivariant isomorphism* ${}^{2\rho}\mathrm{U}_\mathrm{F} \cong {}^{2\rho}\mathfrak{u}(2)$*.*

Proof. By Remark 6.3.18, it suffices to show that there is a choice of generating variety X with even cells and an isomorphism $\mathrm{H}^{-*}(\mathrm{X};\pi_*(k)) \cong \mathrm{H}^{-*}(\mathrm{X};k)$ of $\pi_*(k)$-algebras. Here, k is the localization of MU at the bad primes of G and the order of the center of G.

- Type A: in this case, one can take $\mathrm{X} = \mathbf{C}\mathrm{P}^{n-1}$, and then
$$\mathrm{H}^{-*}(\mathrm{X}; \pi_*(k)) \cong \mathrm{H}^{-*}(\mathrm{X}; k) \cong \pi_*(k)[x]/x^n.$$
- Type B: in this case, one can take $\mathrm{X} = \mathrm{Q}_{2n-1}$ to be a quadric in $\mathbf{C}\mathrm{P}^{2n}$. Studying the Gysin map for the inclusion $\mathrm{X} \subseteq \mathbf{C}\mathrm{P}^{2n}$ shows that
$$\mathrm{H}^{-*}(\mathrm{X}; k) \cong \pi_*(k)[x, y]/(x^n = y\tfrac{[2](x)}{x}, y^2),$$
where x and y live in weights -2 and $-2n$, respectively. Here, $[2](x)$ denotes the 2-series of x in the formal group law F over $\pi_*(k)$; note that the fraction $\frac{[2](x)}{x}$ is well-defined, and is a power series in x with constant term 2. The element x is the restriction of the Chern class of the line bundle $\mathcal{O}(1)$ on $\mathbf{C}\mathrm{P}^{2n}$, and y comes from the linear subspace $\mathbf{C}\mathrm{P}^n \subseteq \mathrm{Q}_{2n-1} \subseteq \mathbf{C}\mathrm{P}^{2n}$. If the bad prime 2 is invertible in $\pi_0(k)$, then $\frac{[2](x)}{2x}$ is well-defined and invertible in $\pi_*(k)[x]^\wedge$ (where the completion is taken in the sense of Remark A.1.6); note that it is invertible because its constant term is 1. It follows that there is a ring isomorphism $\mathrm{H}^{-*}(\mathrm{X}; \pi_*(k)) \xrightarrow{\cong} \mathrm{H}^{-*}(\mathrm{X}; k)$ by sending $(x', y') \mapsto (x, y\frac{[2](x)}{2x})$.
- Type C: in this case, one can take $\mathrm{X} = \mathrm{Sp}(n)/\mathrm{U}(n)$. There is an isomorphism
$$\mathrm{H}^{-*}(\mathrm{X}; k) \cong \pi_*(k)[c_1, \cdots, c_n]/(p_{1,k}, \cdots, p_{n,k}),$$
where c_i is the ith elementary symmetric polynomial in $x_1, \cdots, x_n$, and $p_{i,k}$ is the ith elementary symmetric polynomial in $-x_1\overline{x_1}, \cdots, -x_n\overline{x_n}$. Here, $\overline{x}$ denotes the inverse of x in the formal group law F over $\pi_*(k)$. Since $\overline{x} = -x \pmod{x^2}$, there is a power series $\iota(x)$ with constant term 1 such that $-\overline{x} = x\iota(x)$. If 2 is a unit in $\pi_0(k)$, $\iota(x)$ admits a square root $\iota(x)^{1/2}$. There is an isomorphism $\mathrm{H}^{-*}(\mathrm{X}; k) \cong \mathrm{H}^{-*}(\mathrm{X}; \pi_*(k))$ given by sending c_i to the ith elementary symmetric polynomial in $y_1 = x_1\iota(x_1)^{1/2}, \cdots, y_n = x_n\iota(x_n)^{1/2}$. Note that $-x\overline{x} = (x\iota(x)^{1/2})^2$, so $p_{i,k}$ is the ith elementary symmetric polynomial in $y_1^2, \cdots, y_n^2$, as needed.
- Type D: in this case, one can take $\mathrm{X} = \mathrm{Q}_{2n}$ to be a quadric in $\mathbf{C}\mathrm{P}^{2n+1}$. Studying the Gysin map for the inclusion $\mathrm{X} \subseteq \mathbf{C}\mathrm{P}^{2n+1}$ shows that
$$\mathrm{H}^{-*}(\mathrm{X}; k) \cong \pi_*(k)[x, y]/(x^{n+1} = y[2](x), y^2 = yf),$$
where x and y live in weights -2 and $-2n$, respectively. The element x is the restriction of the Chern class of the line bundle $\mathcal{O}(1)$ on $\mathbf{C}\mathrm{P}^{2n+1}$, y comes from one of the two subspaces $\mathbf{C}\mathrm{P}^n \subseteq \mathrm{Q}_{2n} \subseteq \mathbf{C}\mathrm{P}^{2n+1}$, and f is the power series in x given by the image under the map $\mathrm{H}^{-*}(\mathbf{C}\mathrm{P}^{2n+1}; k) \to \mathrm{H}^{-*}(\mathrm{X}; k)$ of the Euler class of the virtual bundle $\mathcal{O}(1)^{\oplus(n+1)} - \mathcal{O}(2)$. The relation $y^2 = yf$ comes from the observation that the normal bundle to one of the maximal linear subspaces $\mathbf{C}\mathrm{P}^n \subseteq \mathrm{Q}_{2n}$ is represented by $\nu_{\mathbf{C}\mathrm{P}^n/\mathbf{C}\mathrm{P}^{2n+1}} - \nu_{\mathrm{Q}_{2n}/\mathbf{C}\mathrm{P}^{2n+1}}|_{\mathbf{C}\mathrm{P}^n} = \mathcal{O}(1)^{\oplus(n+1)} - \mathcal{O}(2)$ in the Grothendieck group of complex vector bundles on Q_{2n}.

Note that the image of f in $\mathrm{H}^{-*}(\mathrm{X}; \pi_0(k))$ under the map $k \to \pi_0(k)$ is the Euler class of $\mathcal{O}(1)^{\oplus(n+1)} - \mathcal{O}(2)$, which is $x^n(1+(-1)^n)/2$. In particular, if n is odd, then $f \mapsto 0$ in $\mathrm{H}^{-*}(\mathrm{X}; \pi_0(k))$, so that the leading term of f is 0, i.e., $f = x^{n+1}g$ for some power series $g(x)$. It follows that
$$f = x^{n+1}g = y[2](x)g,$$

so that

$$y^2 - yf = y^2(1 - [2](x)g) = 0.$$

But $1 - [2](x)g$ is a unit, so that $y^2 = 0$. Arguing as in the case of type B above, we find that if 2 is a unit in $\pi_0(k)$, then there is a ring isomorphism $\mathrm{H}^{-*}(\mathrm{X}; \pi_*(k)) \xrightarrow{\cong} \mathrm{H}^{-*}(\mathrm{X}; k)$ by sending $(x', y') \mapsto (x, y\frac{[2](x)}{2x})$.

Now suppose n is even. Then $f \mapsto x^n$ in $\mathrm{H}^{-*}(\mathrm{X}; \pi_0(k))$, so that

$$\mathrm{H}^{-*}(\mathrm{X}; \pi_*(k)) \cong \pi_*(k)[x', y']/(x'^{n+1} = 2x'y', y'^2 = x'^n y'),$$

Let us write $f = x^n g$ for some power series $g(x)$ with constant term 1. Observe that $g(x)\frac{[2](x)}{x} - 1$ is a power series in x with constant term $2-1 = 1$, so it is invertible. Moreover, the square root $h(x) := \left(g(x)\frac{[2](x)}{x} - 1\right)^{-1/2}$ is well-defined as soon as 2 is a unit in $\pi_0(k)$. Then there is a ring isomorphism $\mathrm{H}^{-*}(\mathrm{X}; \pi_*(k)) \xrightarrow{\cong} \mathrm{H}^{-*}(\mathrm{X}; k)$ by sending

$$(x', y') \mapsto (x, \tfrac{x^n}{2} + h(x)(y\tfrac{[2](x)}{2x} - \tfrac{x^n}{2})).$$

Since $h(0) = 1$, it follows that the map is invertible. To finish, we only need to see that the necessary relations hold; I leave this as an exercise to the interested reader.

- Type G_2: in this case, one can take $\mathrm{X} = \mathrm{G}_2/\mathrm{U}(2)^{\mathrm{short}}$, using the short root embedding $\mathrm{U}(2)^{\mathrm{short}} \hookrightarrow \mathrm{G}_2$. In this case, there is an isomorphism

$$\mathrm{H}^{-*}(\mathrm{G}_2/\mathrm{U}(2)^{\mathrm{short}}; \mathbf{Z}) \cong \mathbf{Z}\{1, x, \tfrac{x^2}{3}, \tfrac{x^3}{6}, \tfrac{x^4}{18}, \tfrac{x^5}{18}\},$$

where x lives in weight -2. (Note that there is a small typo in [Bot58, Page 60].) One can also compute $\mathrm{H}^{-*}(\mathrm{G}_2/\mathrm{U}(2)^{\mathrm{short}}; k)$ using geometric methods like in the previous examples, but we give a homotopy-theoretic argument. (Such an argument would have worked for the preceding examples, too.) Note that upon inverting 3 (in the sense of [Sul05]), both embeddings $\mathrm{U}(2)^{\mathrm{short}}, \mathrm{U}(2)^{\mathrm{long}} \hookrightarrow \mathrm{G}_2$ are homotopic as $\mathbf{E}_1$-maps. Therefore, $\mathrm{G}_2/\mathrm{U}(2)^{\mathrm{short}}$ and $\mathrm{G}_2/\mathrm{U}(2)^{\mathrm{long}}$ are homotopy equivalent upon inverting 3. The quotient $\mathrm{G}_2/\mathrm{SU}(2)^{\mathrm{long}}$ by the long root $\mathrm{SU}(2)$ is isomorphic to the Stiefel variety $\mathrm{SO}_7/\mathrm{SO}_5 \cong \mathrm{V}_2(\mathbf{R}^7)$. Upon inverting 2 (in the sense of [Sul05]), it is homotopy equivalent to S^{11}. It follows that $\mathrm{G}_2/\mathrm{U}(2)^{\mathrm{long}}$ is homotopy equivalent to $\mathbf{C}\mathrm{P}^5$ after inverting 2, so that $\mathrm{G}_2/\mathrm{U}(2)^{\mathrm{short}}$ is homotopy equivalent to $\mathbf{C}\mathrm{P}^5$ after inverting 6. (This is clear in integral homology from the description of $\mathrm{H}^{-*}(\mathrm{G}_2/\mathrm{U}(2)^{\mathrm{short}}; \mathbf{Z})$ above.) As we have already seen in type A, $\mathrm{H}^{-*}(\mathbf{C}\mathrm{P}^5; \pi_*(k)) \cong \mathrm{H}^{-*}(\mathbf{C}\mathrm{P}^5; k)$ as $\pi_*(k)$-algebras, so that $\mathrm{H}^{-*}(\mathrm{X}; \pi_*(k)) \cong \mathrm{H}^{-*}(\mathrm{X}; k)$ as desired. □

Appendix A. Even $\mathbf{E}_\infty$-rings

A.1. **Gradings and filtrations.** Throughout this document, we will often work in graded algebraic geometry, so let us make our conventions about this precise. Even though our algebro-geometric constructions will take place only in graded algebraic geometry, we will sometimes need to work in the setting of filtered categories. A rather convenient foundation is provided by [HP25, HP24], which develops algebraic geometry for graded rings with the Koszul sign rule.

Recollection A.1.1. Let $\mathbf{Z}_{\geq}$ denote the integers viewed as a poset via the standard ordering. The category $\mathrm{Sp}^{\mathrm{fil}}$ of filtered spectra is the functor category $\mathrm{Fun}(\mathbf{Z}_{\geq}, \mathrm{Sp})$ equipped with the Day convolution symmetric monoidal structure, and similarly the category $\mathrm{Sp}^{\mathrm{gr}}$ of graded spectra is the functor category $\mathrm{Fun}(\mathbf{Z}, \mathrm{Sp})$ where $\mathbf{Z}$ is viewed a discrete category. The Beilinson t-structure on $\mathrm{Sp}^{\mathrm{gr}}$ is given by declaring $\mathrm{X}^\star$ to be connective if $\mathrm{X}^n \in \mathrm{Sp}_{\geq n}$ for all $n \in \mathbf{Z}$. Let $\mathrm{Ab}^{\mathrm{gr}}$ denote the symmetric monoidal category of $\mathbf{Z}$-graded abelian groups, where the symmetry isomorphism in the tensor product has the Koszul sign rule (so that $a \otimes b = (-1)^{\mathrm{wt}(a)\mathrm{wt}(b)} b \otimes a$). The heart $(\mathrm{Sp}^{\mathrm{gr}})^\heartsuit$ is equivalent to $\mathrm{Ab}^{\mathrm{gr}}$ (via $\mathrm{X}^n \mapsto \pi_n(\mathrm{X}^n)$). We will write $\mathrm{X}^\star(n)$ to denote the shift of $\mathrm{X}^\star$ in weight by n.

According to [HP25], a "Dirac ring" is just a commutative algebra object in $\mathrm{Ab}^{\mathrm{gr}}$, and one defines the notion of faithfulness and flatness exactly as in classical algebra. By [HP25, Example 2.2], the homotopy groups of any homotopy commutative ring spectrum form a Dirac ring. One can also define a "Dirac stack" [HP24, Definition 2.6], which is a sheaf on Dirac rings with the faithfully flat topology. When we discuss Dirac rings in the main body of this document, we will always (except in the generality of Construction 2.4.2) assume they are *evenly graded*, and then simply refer to them as *graded commutative rings*; and similarly for *graded stacks*.

Let $\mathbf{Z}(0)$ denote the unit object in $\mathrm{Ab}^{\mathrm{gr}}$; then we will write $\mathrm{B}\mathbf{G}_m$ to denote $\mathrm{Spec}_{\mathrm{Dir}}(\mathbf{Z}(0))$, where the spectrum is taken in the sense of [HP24]. Similarly, if x is in weight 2, and $\mathbf{Z}[x^{\pm 1}]$ denotes the associated Dirac ring, we will write $\mathrm{Spec}(\mathbf{Z})$ to denote $\mathrm{Spec}_{\mathrm{Dir}}(\mathbf{Z}[x^{\pm 1}])$. In the body of this document, we will only be concerned with Dirac stacks Y which are colimits of affine Dirac stacks $\mathrm{Spec}_{\mathrm{Dir}}(\mathrm{R})$ with R being evenly graded. In this case, we will simply view the datum of the Dirac stack Y as the data of $\mathrm{X} = \mathrm{Y} \times_{\mathrm{B}\mathbf{G}_m} \mathrm{Spec}(\mathbf{Z})$ with the $\mathbf{G}_m$-action which keeps track of the grading. In this setting, we will also write $\mathrm{QCoh}^{\mathrm{gr}}(\mathrm{X})$ to denote $\mathrm{QCoh}(\mathrm{Y})$.

If R is a graded commutative ring, we will write $\mathrm{Mod}_{\mathrm{R}}^{\mathrm{gr}}$ to denote the derived category of graded R-modules (in $\mathrm{Ab}^{\mathrm{gr}}$). Equivalently (by [HP24, Proposition 3.4]), this is the category of R-modules in $\mathrm{Sp}^{\mathrm{gr}}$, where R is placed in homological degree 0.

Remark A.1.2. If $\mathrm{R} \in \mathrm{CAlg}(\mathrm{Sp}^{\mathrm{gr}})$, we will write $\mathrm{Mod}_{\mathrm{R}}^{\mathrm{gr}} = \mathrm{Mod}_{\mathrm{R}}(\mathrm{Sp}^{\mathrm{gr}})$, and refer to a $\mathrm{Mod}_{\mathrm{R}}^{\mathrm{gr}}$-module category in $\mathrm{Pr}^{\mathrm{L,st}}$ as a *graded* R-*linear category*. Similarly for filtered R-linear categories. If $\mathcal{C}^{\mathrm{fil}} \in \mathrm{Mod}_{\mathrm{Sp}^{\mathrm{fil}}}(\mathrm{Pr}^{\mathrm{L,st}})$ is a filtered category, it has an associated graded category given by $\mathcal{C}^{\mathrm{fil}} \otimes_{\mathrm{Sp}^{\mathrm{fil}}} \mathrm{Sp}^{\mathrm{gr}}$ (where $\mathrm{Sp}^{\mathrm{gr}}$ is an $\mathrm{Sp}^{\mathrm{fil}}$-module category via $\mathrm{gr} : \mathrm{Sp}^{\mathrm{fil}} \to \mathrm{Sp}^{\mathrm{gr}}$), as well as an underlying category given by $\mathcal{C} = \mathcal{C}^{\mathrm{fil}} \otimes_{\mathrm{Sp}^{\mathrm{fil}}} \mathrm{Sp}$ (where Sp is an $\mathrm{Sp}^{\mathrm{fil}}$-module category via $\mathrm{colim} : \mathrm{Sp}^{\mathrm{fil}} \to \mathrm{Sp}$). We will refer to $\mathcal{C}^{\mathrm{fil}}$ as a "filtered lift" of $\mathcal{C}$.

Remark A.1.3. There is an $\mathbf{E}_{2,\mathrm{BS}^1}$-monoidal functor $\mathrm{sh} : \mathrm{Sp}^{\mathrm{gr}} \to \mathrm{Sp}^{\mathrm{gr}}$ sending $\mathrm{X}^\star \mapsto \mathrm{X}^\star[2\star]$ called *shearing* (see [DHL+23]). As shown in [Rak20, Proposition 3.3.4], this functor admits a canonical symmetric monoidal structure on $\mathrm{Mod}_{\mathbf{Z}}^{\mathrm{gr}}$ (and, in fact, already on $\mathrm{Mod}_{\mathrm{MU}}^{\mathrm{gr}}$). However, [DHL+23, Remark 3.11] shows that the shearing endofunctor of $\mathrm{Sp}^{\mathrm{gr}}$ *cannot* be equipped with an $\mathbf{E}_3$-monoidal structure.

If $\mathcal{C}^{\mathrm{fil}}$ is a filtered category with associated graded $\mathcal{C}^{\mathrm{fil}} \otimes_{\mathrm{Sp}^{\mathrm{fil}}} \mathrm{Sp}^{\mathrm{gr}}$, we will call the tensor product $(\mathcal{C}^{\mathrm{fil}} \otimes_{\mathrm{Sp}^{\mathrm{fil}}} \mathrm{Sp}^{\mathrm{gr}}) \otimes_{\mathrm{Sp}^{\mathrm{gr}}} {}^{\mathrm{sh}^{-1}}\mathrm{Sp}^{\mathrm{gr}}$ the *shearing* of the associated graded of $\mathcal{C}^{\mathrm{fil}}$, and denote it simply by $\mathcal{C}^{\mathrm{gr}}$ (see also [BZSV23, Section 6.3]). Note that if $\mathcal{C}^{\mathrm{fil}}$ is symmetric monoidal, then $\mathcal{C}^{\mathrm{gr}}$ is $\mathbf{E}_{2,\mathrm{BS}^1}$-monoidal, but if $\mathcal{C}^{\mathrm{fil}}$ is an $\mathbf{E}_\infty$-algebra in $\mathrm{Mod}_{\mathbf{Z}}^{\mathrm{fil}}$-module categories, then $\mathcal{C}^{\mathrm{gr}}$ is in fact an $\mathbf{E}_\infty$-algebra in $\mathrm{Mod}_{\mathbf{Z}}^{\mathrm{gr}}$-module categories. We will sometimes denote this setup by $\mathcal{C} \rightsquigarrow \mathcal{C}^{\mathrm{gr}}$, and refer to it as a 1-parameter degeneration of $\mathcal{C}$ to $\mathcal{C}^{\mathrm{gr}}$. Note that there is a functor $\mathcal{C}^{\mathrm{fil}} \to \mathcal{C}^{\mathrm{gr}}$ given by tensoring along the functor $\mathrm{Sp}^{\mathrm{fil}} \to \mathrm{Sp}^{\mathrm{gr}}$ sending $\mathrm{fil}^\star \mathrm{X} \mapsto \mathrm{sh}^{-1}(\mathrm{gr}^\star \mathrm{X})$.

Our use of terminology is rather abusive, but it is intended to reflect our focus on homotopy groups and not the associated graded of the Whitehead filtration. If R is an $\mathbf{E}_\infty$-ring, then $\tau_{\geq\star}(\mathrm{R}) \in \mathrm{Sp}^{\mathrm{fil}}$ is an $\mathbf{E}_\infty$-algebra in $\mathrm{Sp}^{\mathrm{fil}}$, and if $\mathcal{C}^{\mathrm{fil}} = \mathrm{Mod}^{\mathrm{fil}}_{\tau_{\geq\star}\mathrm{R}}$, then $\mathcal{C}^{\mathrm{gr}}$ in our notation is $\mathrm{Mod}^{\mathrm{gr}}_{\pi_\star(\mathrm{R})}$. On the other hand, $\mathcal{C}^{\mathrm{fil}} \otimes_{\mathrm{Sp}^{\mathrm{fil}}} \mathrm{Sp}^{\mathrm{gr}}$ is $\mathrm{Mod}^{\mathrm{gr}}_{\pi_\star(\mathrm{R})[\star]}$.

One can produce a large collection of graded stacks as follows.

Construction A.1.4. Let $\mathrm{A}^\bullet$ be a cosimplicial diagram of $\mathbf{E}_\infty$-ring spectra. Then there is a cosimplicial diagram of Dirac rings given by $\pi_*(\mathrm{A}^\bullet)$, and $\mathrm{Spec}_{\mathrm{Dir}}(\pi_*(\mathrm{A}^\bullet))$ defines a simplicial Dirac stack. If X is the geometric realization of $\mathrm{Spec}_{\mathrm{Dir}}(\pi_*(\mathrm{A}^\bullet))$ (so that it is a graded stack if each $\mathrm{A}^\bullet$ has even homotopy groups), there is a Tot spectral sequence

$$\mathrm{E}_1 \cong \mathrm{H}^*(\mathrm{X}; \mathcal{O}(n)) \Rightarrow \pi_*(\mathrm{Tot}(\mathrm{A}^\bullet)).$$

We will often be interested in the case that $\mathrm{A}^\bullet$ is the descent diagram for a map $\mathrm{A} \to \mathrm{B}$ of $\mathbf{E}_\infty$-rings (so that $\mathrm{A}^\bullet = \mathrm{B}^{\otimes_{\mathrm{A}} \bullet+1}$). If $\pi_*(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{B})$ is flat over $\pi_*(\mathrm{B})$, then there is an isomorphism $\pi_*(\mathrm{A}^\bullet) \cong \pi_*(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{B})^{\otimes_{\pi_*(\mathrm{B})} \bullet}$. Moreover, the pair $(\pi_*(\mathrm{B}), \pi_*(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{B}))$ forms a Hopf algebroid (see [Rav86, Appendix A]), and $\mathrm{X} = |\,\mathrm{Spec}_{\mathrm{Dir}}(\pi_*(\mathrm{A}^\bullet))|$ is the quotient stack of $\mathrm{Spec}_{\mathrm{Dir}}(\pi_*(\mathrm{B}))$ by the groupoid $\mathrm{Spec}_{\mathrm{Dir}}(\pi_*(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{B}))$. Note that the category of perfect complexes on the stack X is Hovey's category of stable comodules [Hov04]; see [BHV18, Proposition 4.8] and [Pst23b, Remark 3.11].

One of the key objects used in the main body of the article is the following.

Definition A.1.5. Let R be a graded commutative ring. A *graded (1-dimensional) formal group* over R is a 1-dimensional formal group over $\mathrm{Spec}(\mathrm{R})/\mathbf{G}_m$ which is locally isomorphic to $\hat{\mathcal{O}}(2)/\mathbf{G}_m$, where $\hat{\mathcal{O}}(2)$ denotes the formal completion at the origin of the total space of the line bundle $\mathcal{O}(2)$ over $\mathrm{B}\mathbf{G}_m$. Similarly, a *graded (1-dimensional) formal group law* over R is a graded formal group over $\mathrm{Spec}(\mathrm{R})/\mathbf{G}_m$ equipped with a coordinate, i.e., a trivialization of the ideal sheaf of the identity section. Let $\mathcal{M}_{\mathrm{FG}}$ denote the moduli stack of graded formal groups, so that there is a canonical map $\mathcal{M}_{\mathrm{FG}} \to \mathrm{B}\mathbf{G}_m$ which classifies a square root $\mathcal{O}\{\frac{1}{2}\}$ of the line bundle given by the coLie algebra ω of the universal graded formal group.

Remark A.1.6. If R is a graded commutative ring and x is a variable in weight $2m$, let $\mathrm{R}[x]^\wedge$ denote the inverse limit $\lim_n \mathrm{R}[x]/\mathrm{I}_n$, where $\mathrm{I}_n \subseteq \mathrm{R}[x]$ is the ideal generated by homogeneous weight 0 elements of the form $\sum_{i=1}^n r_i x^i$ with $r_i \in \mathrm{R}_{-2mi}$. We similarly define $\mathrm{R}[x_1, \cdots, x_n]^\wedge$ for a sequence $(x_1, \cdots, x_n)$ of variables in weights $2m_1, \cdots, 2m_n$. We will also write $\mathrm{Spf}(\mathrm{R}[x]^\wedge)$ to denote the ind-scheme $\mathrm{colim}_n \mathrm{Spec}(\mathrm{R}[x]/\mathrm{I}_n)$ (which is equipped with a $\mathbf{G}_m$-action); this is somewhat abusive notation, because $\mathrm{Spf}(\mathrm{R}[x]^\wedge)$ need not be a formal scheme.

For example, if R in weight 0, then $\mathrm{R}[x]^\wedge \cong \mathrm{R}[x]$ unless x is in weight 0, in which case $\mathrm{R}[x]^\wedge \cong \mathbf{Z}[\![x]\!]$; more generally, if R is concentrated in nonpositive weights and x is in weight < 0, then $\mathrm{R}[x]^\wedge \cong \mathrm{R}[x]$. If $\mathrm{R} = \mathbf{Z}[\beta]$ with β in weight $2m$, and x is in weight $-2m$, then $\mathrm{R}[x]^\wedge \cong \mathbf{Z}[\beta, x]^\wedge_{(\beta x)}$.

With this notation in hand, a graded formal group law over R is the datum of $\mathrm{F}(x, y) \in \mathrm{R}[x, y]^\wedge$ with x and y in weight -2 which satisfies the usual axioms of a formal group law: $F(x,y) = F(y,x)$, $F(x,0) = x$, and $F(x, F(y,z)) = F(F(x,y),z)$. We will often use F to denote the graded formal group underlying the graded formal group law $F(x,y)$, and refer to x as the chosen *coordinate* of F.

One of the most important and useful results about formal group laws is the following:

Theorem A.1.7 (Lazard, [Haz78, Theorem I.5.3.1]). *The functor from the category of graded commutative rings to the category of sets sending a graded commutative ring* R *to the set of graded formal group laws over* R *is corepresented by the Lazard ring* $\mathrm{L} := \mathbf{Z}[b_1, b_2, \cdots]$ *where* b_i *lives in weight* $2i$.

Remark A.1.8. We will frequently use Quillen's theorem [Qui71a, Qui69], which gives an isomorphism $\mathrm{L} \xrightarrow{\cong} \pi_*(\mathrm{MU})$ via the map which classifies the graded formal group over MU given by $\mathrm{H}^{-*}(\mathbf{C}\mathrm{P}^\infty; \mathrm{MU})$. See [HP24, Definition 5.12] for an exposition in the language of Dirac geometry. This result, along with the isomorphism between $\pi_*(\mathrm{MU} \otimes \mathrm{MU})$ and the ring classifying formal group laws equipped with an automorphism, shows that the Hopf algebroid associated to the cosimplicial diagram $\mathrm{MU}^{\otimes \bullet+1}$ (in the sense of Construction A.1.4) is isomorphic to the moduli stack $\mathcal{M}_{\mathrm{FG}}$.

A.2. (Even) $\mathbf{E}_\infty$-rings.

Definition A.2.1. Let k be an even $\mathbf{E}_\infty$-ring (i.e., $\pi_*(k)$ is concentrated in even weights). Then there is a (1-dimensional) graded formal group defined over $\pi_*(k)$ defined by $\mathrm{Spf}(\pi_*(k^{h\mathrm{S}^1}))/\mathbf{G}_m$. This formal group will be denoted F, or sometimes F_k to denote its dependence on k. The choice of a complex orientation of k defines a class $x \in \pi_{-2}(k^{h\mathrm{S}^1})$, which gives an isomorphism $\pi_*(k^{h\mathrm{S}^1}) \cong \pi_*(k)[x]^\wedge$; here, the completion is as in Remark A.1.6. The Cartier dual $\mathrm{F}^\vee$ of F is given by $\mathrm{Spf}(\pi_*(k[\mathrm{BS}^1]))/\mathbf{G}_m$, i.e., given by the k-homology of $\mathbf{C}\mathrm{P}^\infty$. If T is a torus, then $\mathrm{Spf}(\pi_*(k^{h\mathrm{T}}))/\mathbf{G}_m \cong \mathrm{Hom}(\mathbb{X}^*(\mathrm{T}), \mathrm{F})$, which we will often denote by T_F. Again, Cartier duality gives an isomorphism $\mathrm{T}_\mathrm{F} \cong \mathrm{Hom}(\mathrm{F}^\vee, \mathrm{T})$.

Example A.2.2. Suppose k is an $\mathbf{E}_\infty$-$\mathbf{Q}$-algebra. Then $\mathrm{F} \cong \hat{\mathbf{G}}_a(2)$, and more generally $\mathrm{T}_\mathrm{F} \cong \hat{\mathfrak{t}}(2)$. Similarly, if k is an $\mathbf{E}_\infty$-KU-algebra, then $\mathrm{F} \cong \hat{\mathbf{G}}_m$, and more generally $\mathrm{T}_\mathrm{F} \cong \hat{\mathrm{T}}$.

The following is the main definition of [HRW25] (albeit, importantly, with $\tau_{\geq \star}$ in place of $\tau_{\geq 2\star}$; this in particular has the effect of halving the weights):

Definition A.2.3. Let k be an $\mathbf{E}_\infty$-ring. Define $\mathrm{fil}^\star_{\mathrm{ev}}(k) := \lim_{k \to \mathrm{R}} \tau_{\geq \star}(\mathrm{R})$, where the limit runs over all even $\mathbf{E}_\infty$-k-algebras R. Similarly, if M is a k-module, define $\mathrm{fil}^\star_{\mathrm{ev}/k}(\mathrm{M}) = \lim_{k \to \mathrm{R}} \tau_{\geq \star}(\mathrm{R} \otimes_k \mathrm{M})$, where again the limit runs over all even $\mathbf{E}_\infty$-k-algebras R. We will define $\mathrm{gr}^\star_{\mathrm{ev}}(k)$ (and similarly for $\mathrm{gr}^\star_{\mathrm{ev}/k}(\mathrm{M})$) to be the *shearing* of the associated graded of $\mathrm{fil}^\star_{\mathrm{ev}}(k)$, so that $\mathrm{gr}^\star_{\mathrm{ev}}(k) := \lim_{k \to \mathrm{R}} \pi_\star(\mathrm{R})$, and similarly $\mathrm{gr}^\star_{\mathrm{ev}/k}(\mathrm{M}) = \lim_{k \to \mathrm{R}} \pi_\star(\mathrm{R} \otimes_k \mathrm{M})$ where the limit runs over all even $\mathbf{E}_\infty$-k-algebras R. Note that $\mathrm{gr}^\star_{\mathrm{ev}}(k)$ is always evenly graded (although this need not be true for $\mathrm{gr}^\star_{\mathrm{ev}/k}(\mathrm{M})$). It is often convenient to view $\mathrm{gr}^\star_{\mathrm{ev}}(k)$ as the global sections of the structure sheaf of a graded stack $\mathrm{Spev}(k)$ defined as $\mathrm{colim}_{k \to \mathrm{R}} \mathrm{Spec}(\pi_\star(\mathrm{R}))$, where the colimit runs over all even $\mathbf{E}_\infty$-k-algebras R.

A map $k \to k'$ of $\mathbf{E}_\infty$-rings is called *evenly faithfully flat* (or *eff* for short) if for any map $k \to \mathrm{R}$ of $\mathbf{E}_\infty$-rings with R being an even $\mathbf{E}_\infty$-ring, $\mathrm{R} \otimes_k k'$ is even and the map $\pi_*(\mathrm{R}) \to \pi_*(\mathrm{R} \otimes_k k')$ is faithfully flat. A large class of such maps are in fact *evenly free* (or *evenly projective*), in the sense that the tensor product $\mathrm{R} \otimes_k k'$ is a direct sum of even shifts of R (or a retract thereof). This property is satisfied if k' has an even cell structure as an k-module; in this case, the map $k \to k'$ is called *even cellular.*

Proposition A.2.4 ([HRW25, Corollary 2.2.17]). *Let $k \to k'$ be an eff map of $\mathbf{E}_\infty$-rings. Then $\mathrm{fil}^\star_{\mathrm{ev}}(k) \xrightarrow{\cong} \mathrm{Tot}(\mathrm{fil}^\star_{\mathrm{ev}}(k'^{\otimes_k \bullet+1}))$. Similarly, if M is a k-module, then $\mathrm{fil}^\star_{\mathrm{ev}/k}(\mathrm{M}) \xrightarrow{\cong} \mathrm{Tot}(\mathrm{fil}^\star_{\mathrm{ev}}(\mathrm{M} \otimes_k k'^{\otimes_k \bullet+1}))$. In particular, if $k \to \mathrm{R}$ is an eff cover by an even $\mathbf{E}_\infty$-ring, then $\mathrm{fil}^\star_{\mathrm{ev}}(k) \xrightarrow{\cong} \mathrm{Tot}(\tau_{\geq \star}(\mathrm{R}^{\otimes_k \bullet+1}))$ and $\mathrm{fil}^\star_{\mathrm{ev}/k}(\mathrm{M}) \xrightarrow{\cong} \mathrm{Tot}(\tau_{\geq \star}(\mathrm{M} \otimes_k \mathrm{R}^{\otimes_k \bullet+1}))$.*

Example A.2.5. One very important example (albeit not one used seriously in this document) is the case when $k = \mathbb{S}$. The map $\mathbb{S} \to \mathrm{MU}$ is even cellular, hence in particular is eff. By [Mil60], $\pi_*(\mathrm{MU})$ is concentrated in even degrees, so Proposition A.2.4 implies that $\mathrm{gr}^\star_{\mathrm{ev}}(\mathbb{S}) \cong \mathrm{Tot}(\pi_*(\mathrm{MU}^{\otimes \bullet+1}))$. It follows from Remark A.1.8 that $\mathrm{Spec}(\pi_*(\mathrm{MU}^{\otimes \bullet+1}))/\mathbf{G}_m$ is isomorphic to the moduli stack of graded formal groups equipped with $\bullet$ many coordinates. In particular, $\mathrm{gr}^\star_{\mathrm{ev}}(\mathbb{S}) \cong \mathrm{R}\Gamma(\mathcal{M}_{\mathrm{FG}}; \mathcal{O}\{\frac{\star}{2}\}) = \mathrm{R}\Gamma(\mathcal{M}_{\mathrm{FG}}; \omega^\star)$, and $\mathrm{Spev}(\mathbb{S}) \cong \mathcal{M}_{\mathrm{FG}}$. The resulting spectral sequence assembling $\pi_*(\mathbb{S})$ from $\mathrm{H}^*(\mathcal{M}_{\mathrm{FG}}; \omega^\star)$ is the Adams-Novikov spectral sequence. The natural map $\mathcal{M}_{\mathrm{FG}} \to \mathrm{B}\mathbf{G}_m$ classifies the dual of the Lie algebra of the universal 1-dimensional formal group, and it is a rational isomorphism. An object of $\mathrm{QCoh}(\mathcal{M}_{\mathrm{FG}})$ which descends along the map $\mathcal{M}_{\mathrm{FG}} \to \mathrm{B}\mathbf{G}_m$ behaves as though it "comes from $\mathbf{F}_1$"; for instance, the graded group scheme ${}^{2\rho}\check{\mathrm{G}}$ in Theorem 1.1.1 is pulled back along this map.

Remark A.2.6. For a general $\mathbf{E}_\infty$-ring k, the unit map $\mathbb{S} \to k$ induces a map $\mathrm{Spev}(k) \to \mathrm{Spev}(\mathbb{S}) \cong \mathcal{M}_{\mathrm{FG}}$. The resulting 1-dimensional graded formal group over $\mathrm{Spev}(k)$ is $\mathrm{colim}_n \mathrm{Spev}(\mathrm{C}^*(\mathbf{CP}^n; k))$, which defines a formal group by Definition A.2.1.

Definition A.2.3 needs to be modified in the equivariant setting. (We will only work Borel-equivariantly in this document.) There are several possible extensions of Definition A.2.3, especially in the case of disconnected groups, and we will use the following. There are also variants of this story with genuine equivariance (e.g., [AP25]), which we will not discuss here.

Definition A.2.7. Let G be a compact Lie group. Let $\mathcal{T}$ denote the full subcategory of the orbit ∞-category of G of those homogeneous G-spaces whose isotropy is a finite intersection of maximal tori of G. If p is a prime, let $\mathcal{P}_p$ denote the full subcategory of the orbit ∞-category of G of those homogeneous G-spaces whose isotropy is a finite intersection of maximal p-toral subgroups of G; recall that a subgroup is called p-toral if its neutral component is a torus and its component group is a p-group. Let $k \in \mathrm{CAlg}(\mathrm{Sp}^{\mathrm{BG}})$ be an $\mathbf{E}_\infty$-ring with G-action. A spectrum $\mathrm{M} \in \mathrm{Mod}_k^{h\mathrm{G}}$ will be called G-*even* if $\mathrm{M}^{h\mathrm{T}}$ is even for every maximal *connected* compact abelian subgroup $\mathrm{T} \subseteq \mathrm{G}$, i.e., maximal torus. This notion obviously only depends on the neutral connected component of G; since all maximal tori in a compact connected Lie group are conjugate, it follows that M is G-even if and only if $\mathrm{M}^{h\mathrm{T}}$ is even for *any* maximal torus $\mathrm{T} \subseteq \mathrm{G}$. For instance, any M whose underlying k-module has even homotopy is G-even.

Define

$$\mathrm{fil}^\star_{\mathrm{G\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}}) = \lim_{k\to \mathrm{R}} \lim_{\mathrm{A}\in\mathcal{T}^{\mathrm{op}}} \tau_{\geq\star}((\mathrm{M}\otimes_k \mathrm{R})^{h\mathrm{A}}),$$

where the outer limit is over even G-equivariant $\mathbf{E}_\infty$-k-algebras R. (Here, R is interpreted to be even in the sense described above.) Note that $\mathrm{fil}^\star_{\mathrm{G\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$ is just $\lim_{k\to\mathrm{R}} \mathrm{fil}^\star_{\mathrm{G\text{-}ev}/\mathrm{R}^{h\mathrm{G}}}((\mathrm{M}\otimes_k \mathrm{R})^{h\mathrm{G}})$. Similarly, if p is a fixed prime, let

$$\mathrm{fil}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}}) = \lim_{k\to \mathrm{R}} \lim_{\mathrm{A}\in\mathcal{P}_p^{\mathrm{op}}} \tau_{\geq\star}((\mathrm{M}\otimes_k \mathrm{R})^{h\mathrm{A}}),$$

where the outer limit is over even G-equivariant $\mathbf{E}_\infty$-k-algebras R. This definition is only reasonable upon p-completion. Note that $\mathrm{fil}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$ is just $\lim_{k\to\mathrm{R}} \mathrm{fil}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ev}/\mathrm{R}^{h\mathrm{G}}}((\mathrm{M}\otimes_k \mathrm{R})^{h\mathrm{G}})$.

If $\mathrm{M} = k$, we will denote these simply by $\mathrm{fil}^\star_{\mathrm{G\text{-}ev}}(k^{h\mathrm{G}})$ and $\mathrm{fil}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ev}}(k^{h\mathrm{G}})$. We will also define $\mathrm{gr}^\star_{\mathrm{G\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$ (and similarly for $\mathrm{gr}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$) to be the *shearing* of the associated graded of $\mathrm{fil}^\star_{\mathrm{G\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$ (resp. $\mathrm{fil}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$). Note that $\mathrm{gr}^\star_{\mathrm{ev}}(k^{h\mathrm{G}})$ is always evenly graded (although this need not be true for $\mathrm{gr}^\star_{\mathrm{G\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$, $\mathrm{gr}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$, or $\mathrm{fil}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ev}}(k^{h\mathrm{G}})$). We will only use $\mathrm{fil}^\star_{\mathrm{G\text{-}ev}}$ in the body of this document.

Remark A.2.8. Already when G is a torus T and $k = \mathbf{Z}$ (necessarily with trivial T-action), an object $\mathrm{M} \in \mathrm{Mod}_{\mathbf{Z}}^{h\mathrm{T}}$ can be even in the sense of Definition A.2.7 if its underlying $\mathbf{Z}$-module is not even. For instance, $\mathrm{M} = \mathrm{C}_*(\mathrm{T};\mathbf{Z})$ will satisfy this property.

Remark A.2.9. There are, of course, several variants of Definition A.2.7, especially in the case when G is disconnected. For instance, one could replace $\mathcal{P}_p$ by the full subcategory $\mathcal{A}$ (resp. $\mathcal{A}_p$) of the orbit ∞-category of G of those homogeneous G-spaces whose isotropy is a finite intersection of maximal, possibly disconnected, compact abelian subgroups of G (resp. maximal elementary abelian p-subgroups of G). For $\mathrm{M} \in \mathrm{Mod}_k^{h\mathrm{G}}$, this defines $\mathrm{fil}^\star_{\mathrm{G\text{-}ab\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$ and $\mathrm{fil}^\star_{\mathrm{G\text{-}ab\text{-}ev}}(k^{h\mathrm{G}})$ as well as the graded variants (resp. $\mathrm{fil}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ab\text{-}ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$ and $\mathrm{fil}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ab\text{-}ev}}(k^{h\mathrm{G}})$ as well as the graded variants).

Example A.2.10. Suppose G is a *finite* group and $k \in \mathrm{CAlg}(\mathrm{Sp}^{\mathrm{BG}})$. Then k is even in the sense of Definition A.2.7 if and only if its underlying $\mathbf{E}_\infty$-ring is even, so that $\mathrm{fil}^\star_{\mathrm{G\text{-}ev}}(k^{h\mathrm{G}})$ is just $\mathrm{fil}^\star_{\mathrm{ev}}(k)$ in the sense of Definition A.2.3. However, $\mathrm{fil}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ev}}(k^{h\mathrm{G}})$, and therefore $\mathrm{gr}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ev}}(k^{h\mathrm{G}})$, is more interesting. Assume for simplicity that k is even (this is harmless, because it determines the general case). Since a p-toral subgroup of G is just a p-subgroup of G, a cofinality argument shows that if $\mathcal{O}_p$ is the full subcategory of the orbit category of G of those finite G-sets whose isotropy is a p-group, then $\mathrm{gr}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ev}}(k^{h\mathrm{G}}) = \lim_{\mathrm{A}\in\mathcal{P}_p^{\mathrm{op}}} \mathrm{H}^{-\star}(\mathrm{BA};k)$ is isomorphic to $\lim_{\mathrm{A}\in\mathcal{O}_p^{\mathrm{op}}} \mathrm{H}^{-\star}(\mathrm{BA};k)$. If $k = \mathbf{F}_p$, then this *a priori* derived inverse limit is in fact concentrated in homological degree 0 by [Mis87, Section 2]. Then the inverse limit is isomorphic to $\mathrm{H}^{-*}(\mathrm{BG};\mathbf{F}_p)$ by the Cartan-Eilenberg stable elements formula [CE56, Theorem XII.10.1]. Let us also note that in [Qui71b], Quillen showed that the natural map $\mathrm{H}^{-*}(\mathrm{BG};\mathbf{F}_p) \to \mathrm{gr}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ab\text{-}ev}}(\mathbf{F}_p^{h\mathrm{G}})$ is an F-isomorphism, where $\mathrm{gr}^\star_{\mathrm{G\text{-}}p\mathrm{\text{-}ab\text{-}ev}}$ is defined in Remark A.2.9.

Remark A.2.11. If k is p-complete and A is a finite p-group, one could apply the even filtration of Definition A.2.3 to $k^{h\mathrm{A}}$. Unfortunately, this is typically not very reasonable, because $k^{h\mathrm{A}}$ need not admit an eff cover by an even $\mathbf{E}_\infty$-ring.

Example A.2.12. Let k be an even $\mathbf{E}_\infty$-ring, let G be a connected compact Lie group, and let $\mathrm{M} \in \mathrm{Mod}_k^{h\mathrm{G}}$. If T is a maximal torus of G, then the uniqueness of maximal tori up to conjugacy and a cofinality argument show that $\mathrm{colim}_{\boldsymbol{\Delta}^{\mathrm{op}}} \mathrm{BT}^{\times_{\mathrm{BG}} \bullet+1} \xrightarrow{\simeq} \mathrm{colim}_{\mathrm{A} \in \mathcal{T}} \mathrm{BA}$. This implies that there is an isomorphism $\mathrm{fil}^\star_{\text{G-ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}}) \xrightarrow{\cong} \mathrm{Tot}(\tau_{\geq\star}((\mathrm{C}^*(\mathrm{G}/\mathrm{T}; k) \otimes_k \mathrm{M})^{h\mathrm{T}}))$. Note that there is an isomorphism $\mathrm{fil}^\star_{\text{G-ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}}) \cong \mathrm{fil}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}})$, where the notation $\mathrm{fil}^\star_{\mathrm{G}}$ is from Definition 2.2.28. As such, we will often write $\mathrm{fil}^\star_{\text{G-ev}/k^{h\mathrm{G}}}(\mathrm{M}^{h\mathrm{G}})$ or $\mathrm{fil}^\star_{\mathrm{G}}(\mathrm{M}^{h\mathrm{G}})$ to denote the same object. When k is p-complete, the analogous picture with $\mathrm{fil}^\star_{\text{G-}p\text{-ev}}$ was studied by Jackowski-McClure-Oliver in [JMO94] (in terms of "p-stubborn groups").

There is a spectral sequence

$$\mathrm{E}_1 \cong \pi_* \mathrm{gr}^\star_{\text{G-ev}}(k^{h\mathrm{G}}) \Rightarrow \pi_*(k^{h\mathrm{G}}). \tag{A.2.1}$$

If $\mathrm{M} \in \mathrm{Mod}_k^{h\mathrm{G}}$, there is, of course, an analogous spectral sequence with $\mathrm{E}_1 \cong \pi_* \mathrm{gr}^\star_{\text{G-ev}/k^{h\mathrm{G}}} \mathrm{M}^{h\mathrm{G}}$ abutting to $\pi_*(\mathrm{M}^{h\mathrm{G}})$.

Conjecture A.2.13. *For any connected reductive group* G *over* $\mathbf{C}$ *(or, equivalently, any connected compact Lie group) and any even* $\mathbf{E}_\infty$-$\mathbf{Z}$*-algebra* k*, the spectral sequence* (A.2.1) *degenerates at the* E_1*-page with no multiplicative extensions on the* E_∞*-page.*

Remark A.2.14. A conjecture attributed to Frank Adams (see, e.g., the introduction of [Bro07]) states that if $p > 2$ and G is a connected compact Lie group, the map $\mathrm{H}^{-*}(\mathrm{BG}; \mathbf{F}_p) \to \prod_{\mathrm{A}} \mathrm{H}^{-*}(\mathrm{BA}; \mathbf{F}_p)$ is injective, where the product is taken over representatives of conjugacy classes of maximal elementary abelian p-subgroups of G. However, this conjecture is false for $\mathrm{G} = \mathrm{PGL}_{p^2}$, as shown in [Fan25]. If one views Conjecture A.2.13 as a "remedy" to the analogous failure of injectivity of the map $\mathrm{H}^{-*}(\mathrm{BG}; \mathbf{F}_p) \to \mathrm{H}^{-*}(\mathrm{BT}; \mathbf{F}_p)$ for arbitrary p, one natural modification of Adams' conjecture would be that the spectral sequence

$$\mathrm{E}_1 \cong \pi_* \mathrm{gr}^\star_{\text{G-}p\text{-ab-ev}}(\mathbf{F}_p^{h\mathrm{G}}) \Rightarrow \pi_*(\mathbf{F}_p^{h\mathrm{G}})$$

degenerates at the E_1-page with no multiplicative extensions on the E_∞-page, for arbitrary p and connected compact Lie groups G, where $\mathrm{gr}^\star_{\text{G-}p\text{-ab-ev}}$ is defined in Remark A.2.9. Just as Theorem 4.3.10 gives a "combinatorial" description of $\mathrm{gr}^\star_{\text{G-ev}}(\mathbf{F}_p^{h\mathrm{G}})$, i.e., a description in terms of the Langlands dual group of G, it would be interesting to give a calculation of $\mathrm{gr}^\star_{\text{G-}p\text{-ab-ev}}(\mathbf{F}_p^{h\mathrm{G}})$ using the "finite maximal tori" of [HV14].

If k is an even $\mathbf{E}_\infty$-$\mathbf{Z}$-algebra and G is a connected compact Lie group, we describe $\mathrm{gr}^\star_{\mathrm{G}}(k^{h\mathrm{G}})$ in Theorem 4.3.10 (see Appendix C for examples) as the global sections of the structure sheaf of a certain stack built from the Langlands dual group of G. In Definition 2.2.28, we also describe $\mathrm{gr}^\star_{\mathrm{G}}(k^{h\mathrm{G}})$ for a general even $\mathbf{E}_\infty$-ring as the global sections of the structure sheaf of the classifying stack of the combinatorially-defined Weyl groupoid from Definition 2.2.23. It would be interesting to extend these results to disconnected compact Lie groups; this seems most feasible for $\mathrm{gr}^\star_{\text{G-ab-ev}}(k^{h\mathrm{G}})$ from Remark A.2.9.

Appendix B. Spectral liftings

In this section, we discuss how the main results of this article (Theorem 5.1.16 in the simply-laced case, and Theorem 5.2.9 in the multiply-laced case) would follow from applying the principles of geometric Langlands duality assuming a lifting of the shearing of the graded group scheme ${}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}$ over $\pi_*(k)$ to a group object in spectral schemes over the coefficient $\mathbf{E}_\infty$-ring k. (See also [Lur10, Remark 10.3].) This appendix is motivational and is not used in the proof of any results in the body of the text; as such, almost every statement appearing below are conjectural.

Unfortunately, I do not know how to construct such a lift of ${}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}$ (outside of the easy case of a torus), and in fact we will show that such a lift cannot exist if one insists on sufficient multiplicative structure. Still, it is useful to spell out this story, because it predicts statements which one *can* in fact prove.[14] Some such statements will be discussed in future work.

As usual in this article, we will always use the "arithmetic shearing" normalization of the usual geometric Satake equivalence (see [BZSV23, Section 6.5.3]). With this convention, the geometric Langlands program suggests that if k is an ordinary commutative ring, then there is a k-linear $\mathbf{T}$-equivariant $\mathbf{E}_3$-monoidal equivalence

$$\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \simeq \mathrm{IndCoh}_{\mathcal{N}}(\mathrm{B}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}} \times_{{}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}/{}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}} \mathrm{B}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}), \tag{B.0.1}$$

where the stack appearing on the right-hand side[15] is defined over k, the quotient stack ${}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}/{}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}$ is by the conjugation action of ${}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}$ on itself, and $\mathbf{T} \cong \mathrm{S}^1$ is the loop-rotation circle. When k is a $\mathbf{Q}$-algebra, this follows (without the $\mathbf{E}_3$-monoidal structure) from [BF08, AG15].

Note that the right-hand side of (B.0.1) can be rewritten as

$$\mathrm{IndCoh}_{\mathcal{N}}((1 \times_{{}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}} 1)/{}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}) \simeq \mathrm{IndCoh}_{\mathcal{N}}({}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}} \backslash \mathrm{Spec}(\mathrm{HH}({}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}/k))/{}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}),$$

simply because ${}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}} \backslash \mathrm{Spec}(\mathrm{HH}({}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}/k))$ is isomorphic to $1 \times_{{}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}} 1$. Koszul duality and the definition of nilpotent singular support from [AG15] now allow us to rewrite the right-hand side again, as $\mathrm{IndPerf}({}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}} \backslash \mathrm{Spec}(\mathfrak{Z}_{\mathbf{E}_2}({}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}/k))^{\wedge}_{\mathcal{N}}/{}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}})$. Here, $\mathfrak{Z}_{\mathbf{E}_2}({}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}/k) = \mathrm{REnd}_{\mathrm{HH}({}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}/k)}(\mathcal{O}_{{}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}})$ is the $\mathbf{E}_2$-center of ${}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}$ over k. Despite the notation, it is in fact a $\mathbf{T}$-equivariant $\mathbf{E}_3$-k-algebra, and a (non-$\mathbf{T}$-equivariant) $\mathbf{E}_2$-$\mathcal{O}_{{}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}}$-algebra. Stitching these equivalences together, we obtain a conjectural k-linear $\mathbf{T}$-equivariant $\mathbf{E}_3$-monoidal equivalence

$$\mathrm{Shv}_{\mathrm{G}_{\mathcal{O}}}(\mathcal{G}\mathrm{r}_{\mathrm{G}}; k) \simeq \mathrm{IndPerf}({}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}} \backslash \mathrm{Spec}(\mathfrak{Z}_{\mathbf{E}_2}({}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}/k))^{\wedge}_{\mathcal{N}}/{}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}) \tag{B.0.2}$$

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[14] The situation is in some sense analogous to that of chromatic homotopy theory, where, following foundational work of Quillen [Qui69, Qui71a] and Morava [Mor85], the existence of a spectral lift of the moduli stack $\mathcal{M}_{\mathrm{FG}}$ of formal groups was long suspected. This object has now been constructed [Gre21], building on [Lur18]. But historically, even before it was constructed, the then-hypothetical spectral lift of $\mathcal{M}_{\mathrm{FG}}$ motivated some very important directions in homotopy theory, like the construction of Morava E-theory and the theory of topological modular forms [DFHH14]; and conversely, it suggested several interesting algebro-geometric constructions motivated by homotopy-theoretic calculations, like Rezk's $\mathbf{T}$-monad [Rez14]. I hope that similarly, the hypothesized spectral lift of the shearing of ${}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}$ discussed below will suggest further directions in homotopy theory and geometric representation theory.

[15] A technical point is that the right-hand side of (B.0.1) is not yet defined precisely in the literature, because [AG15] only define ind-coherent sheaves over $\mathbf{Q}$-algebras; an extension of this theory to general bases should certainly exist.

when k is an ordinary commutative ring. Our goal in this section is to describe some consequences of the following (and in particular, its relation to Theorem 5.1.16 and Theorem 5.2.9):

Conjecture B.1. *There is an* $\mathbf{E}_1$*-coalgebra* $\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_{\mathbb{S}}}$ *in* $\mathbf{E}_2^{\mathrm{fr}}$*-algebras in* Sp *such that for every even* $\mathbf{E}_\infty$*-ring* k*,* $\pi_*(\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_k})$ *is isomorphic (as a graded Hopf algebra over* $\pi_*(k)$*) to* $\mathcal{O}_{2\rho\check{G}}$*, where* $\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_k} := \mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_{\mathbb{S}}} \otimes_{\mathbb{S}} k$*. For notational simplicity, write* $\mathrm{Rep}({}^{\mathrm{sh}(2\rho)}\check{G}_{\mathbb{S}})$ *to denote* $\mathrm{coMod}_{\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_{\mathbb{S}}}}$*, and similarly for* $\mathrm{Rep}({}^{\mathrm{sh}(2\rho)}\check{G}_{\mathbb{S}} \times_{\mathrm{Spec}(\mathbb{S})} {}^{\mathrm{sh}(2\rho)}\check{G}_{\mathbb{S}})$*. Then, for every even* $\mathbf{E}_\infty$*-ring* k*, there is a completion* $\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_k}/k)^\wedge_{\mathcal{N}_k}$ *of* $\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_k}/k)$ *such that there is an* k*-linear* $\mathbf{T}$*-equivariant* $\mathbf{E}_3$*-monoidal equivalence (which is natural in* k*)*
(B.0.3)

$$\mathrm{Shv}_{G_{\mathcal{O}}}(\mathcal{G}r_G; k) \simeq \mathrm{IndPerf}_{\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_k}/k)^\wedge_{\mathcal{N}_{\mathbb{S}}}}(\mathrm{Rep}({}^{\mathrm{sh}(2\rho)}\check{G}_k \times_{\mathrm{Spec}(k)} {}^{\mathrm{sh}(2\rho)}\check{G}_k)).$$

This equivalence sends the δ*-sheaf at the basepoint of* $\mathcal{G}r_G$ *to* $\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_k}/k)^\wedge_{\mathcal{N}_k}$*.*[16] *If* k *is an ordinary commutative ring,* (B.0.3) *recovers* (B.0.2)*.*

Remark B.2. Conjecture B.1 is true if G is a torus: in this case, $\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_{\mathbb{S}}}$ is just the spherical group ring $\mathbb{S}[\mathbb{X}_*(\mathrm{T})]$. One can also ask for several variants of Conjecture B.1: for instance, one could ask for an $\mathbf{E}_1$-coalgebra $\mathcal{O}_{2\rho\check{G}_{\mathbb{S}}}$ in $\mathbf{E}_2^{\mathrm{fr}}$-algebras in $\mathrm{Sp}^{\mathrm{gr}}$ whose shearing is $\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_{\mathbb{S}}}$ and such that for every even $\mathbf{E}_\infty$-ring k, $\pi_*(\mathcal{O}_{2\rho\check{G}_k})$ is isomorphic (as a *bi*graded Hopf algebra over $\pi_*(k)$) to $\mathcal{O}_{2\rho\check{G}}$. One may also ask for a lift not just of $\mathcal{O}_{2\rho\check{G}}$ or of $\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}}$, but also of Borel subgroups and their unipotent radicals; these would be necessary for variants of Conjecture B.1 involving the affine flag variety (as in [ABG04, Bez16] over $\mathbf{Q}$).

Fix an even $\mathbf{E}_\infty$-ring k, possibly with $\mathbf{T}$-action. Applying the Whitehead filtration to the right-hand side of (B.0.3) produces a filtered lift of this category, whose associated graded is the graded $\pi_*(k)$-linear category $\mathrm{IndPerf}^{\mathrm{gr}}_{\pi_*\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_k}/k)^\wedge_{\mathcal{N}_k}}(\mathrm{Rep}({}^{2\rho}\check{G} \times {}^{2\rho}\check{G}))$. Note that Conjecture B.1 guarantees that $\tau_{\geq\star}\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_k}$ is still a coalgebra over $\tau_{\geq\star}(k)$.

Expectation B.3. The aforementioned filtered lift of the right-hand side of (B.0.3) is equivalent (as a filtered $\tau_{\geq\star}(k)$-linear category) to a topologically-defined filtered lift of $\mathrm{Shv}_{G_{\mathcal{O}}}(\mathcal{G}r_G; k)$ which contains $\mathrm{Shv}^{\mathrm{q\text{-}min,fil}}_{G_{\mathcal{O}}}(\mathcal{G}r_G; k)$ from Definition 5.1.13 as a full subcategory.

Let us assume for ease of exposition that G is semisimple and simply-laced (in the multiply-laced case, one needs to assume in addition that the lacing number of G is inverted in $\pi_0(k)$, replace ${}^{2\rho}G_F$ below by ${}^{2\rho}\check{G}_F$, and appeal to Theorem 5.2.9 instead of Theorem 5.1.16). Expectation B.3 and Theorem 5.1.16 then imply that if the bad primes of G are inverted in $\pi_0(k)$, there is a fully faithful (monoidal)

[16] There should also be a version of (B.0.3) when $k = \mathbb{S}$ itself. However, the failure of $\mathrm{Sp}^{h\mathrm{T}}$ to be unipotent in the sense of [MNN17] implies that the spectral side cannot just be $\mathrm{IndPerf}_{\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{G}_{\mathbb{S}}}/\mathbb{S})^\wedge_{\mathcal{N}_{\mathbb{S}}}}(\mathrm{Rep}({}^{\mathrm{sh}(2\rho)}\check{G}_{\mathbb{S}} \times_{\mathrm{Spec}(\mathbb{S})} {}^{\mathrm{sh}(2\rho)}\check{G}_{\mathbb{S}}))$; rather, it must be "$\mathrm{IndCoh}_{\mathcal{N}}({}^{\mathrm{sh}(2\rho)}\check{G}_{\mathbb{S}} \backslash \mathrm{Spec}(\mathrm{HH}({}^{\mathrm{sh}(2\rho)}\check{G}_{\mathbb{S}}/\mathbb{S}))/{}^{\mathrm{sh}(2\rho)}\check{G}_{\mathbb{S}})$". I chose not to state Conjecture B.1 this way because I do not know a definition of ind-coherent sheaves for spectral schemes, and also because $\mathrm{Spec}(\mathrm{HH}({}^{\mathrm{sh}(2\rho)}\check{G}_{\mathbb{S}}/\mathbb{S}))$ does not make sense (since ${}^{\mathrm{sh}(2\rho)}\check{G}_{\mathbb{S}}$ is only an $\mathbf{E}_2$-scheme, its Hochschild homology is only an $\mathbf{E}_1$-algebra).

embedding

$$\mathrm{IndPerf}^{\text{q-min,gr}}({}^{2\rho}\check{\mathrm{G}}\backslash({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\mathrm{G}_{\mathrm{F}})/{}^{2\rho}\check{\mathrm{G}}) \hookrightarrow \mathrm{IndPerf}^{\mathrm{gr}}_{\pi_*\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{\mathrm{G}}_k}/k)^\wedge_{\mathcal{N}_k}}(\mathrm{Rep}({}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{G}})),$$

where $\mathrm{F}\cong\mathrm{Spf}(\mathrm{H}^{-*}(\mathbf{C}\mathrm{P}^\infty;k))$ is the 1-dimensional graded formal group over $\pi_*(k)$. Computing endomorphisms of the structure sheaf, this in turn implies that there is a graded $\pi_*(k)$-linear ${}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\check{\mathrm{G}}$-equivariant algebra isomorphism

$$\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\mathrm{G}_{\mathrm{F}}} \cong \pi_*\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{\mathrm{G}}_k}/k)^\wedge_{\mathcal{N}_k}. \tag{B.0.4}$$

In fact, since $\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{\mathrm{G}}_k}/k)$ is an $\mathbf{E}_3$-k-algebra, the Browder bracket equips $\pi_*\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{\mathrm{G}}_k}/k)$ with the structure of a graded Poisson algebra whose Poisson bracket has weight 2 (among a multitude of other structures which we will explore in future work; see [CLM76, Chapter III]). The isomorphism (B.0.4) is moreover an isomorphism of graded Poisson algebras, where the Poisson structure on ${}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\mathrm{G}_{\mathrm{F}}$ comes from its 0-shifted symplectic structure (see Example 4.4.5). The isomorphism (B.0.4) was the primary motivation for Theorem 5.1.16. Let us explicitly verify (B.0.4) in the two cases where ${}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}_k$ is known to exist:

Example B.4. If k is an ordinary commutative ring, then $\mathrm{F}=\hat{\mathbf{G}}_a(2)$, so that ${}^{2\rho}\mathrm{G}_{\mathrm{F}}\cong{}^{2\rho}\mathfrak{g}(2)^\wedge_{\mathcal{N}}$ by Lemma 4.1.2. For (B.0.4), it suffices to show that there is a graded isomorphism

$$\mathcal{O}_{\check{\mathrm{G}}\times\mathfrak{g}(2)^\wedge_{\mathcal{N}}} \cong \pi_*\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\check{\mathrm{G}}}/k)^\wedge_{\mathcal{N}}.$$

This follows from [Dev26, Theorem 2.3], which identifies $\pi_*\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\check{\mathrm{G}}}/k)\cong\mathcal{O}_{\mathrm{T}^*(2)(\check{\mathrm{G}}/k)}$, and Remark 4.2.2, which gives isomorphisms

$$\mathrm{T}^*(2)(\check{\mathrm{G}}/k)\cong\check{\mathrm{G}}\times\check{\mathfrak{g}}^*(2)\cong\check{\mathrm{G}}\times\mathfrak{g}(2)$$

via the minimal bilinear form on $\mathfrak{g}$.

Remark B.5. Conjecture B.1 and Expectation B.3 imply (by taking endomorphisms of the unit objects) that for arbitrary connected reductive groups G and even $\mathbf{E}_\infty$-$\mathbf{Z}$-algebras k, there is a graded isomorphism $\mathrm{gr}^\star_{\mathrm{G}}(k^{h\mathrm{G}})\cong\mathrm{R}\Gamma({}^{2\rho}\check{\mathfrak{g}}^*(2)/{}^{2\rho}\check{\mathrm{G}};\mathcal{O})$. In Theorem 4.3.10, we saw that there is an isomorphism $\mathrm{gr}^\star_{\mathrm{G}}(k^{h\mathrm{G}})\cong\mathrm{R}\Gamma((\psi_0+{}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-;\mathcal{O})$, so it would suffice to show that the natural map $(\psi_0+{}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-\to{}^{2\rho}\check{\mathfrak{g}}^*(2)/{}^{2\rho}\check{\mathrm{G}}$ induces an isomorphism upon applying $\mathrm{R}\Gamma(-;\mathcal{O})$. This is a very reasonable conjecture which is at least a question only about $\check{\mathrm{G}}$ (and not about its relationship to G). The known calculations of $\mathrm{H}^{-*}(\mathrm{BG};k)$ in many cases, the calculations of Appendix C and [Tot18], as well as some calculations by Venkatesh, support this conjecture.

Example B.6. Let k be a general even $\mathbf{E}_\infty$-ring, and assume G is a torus T. The isomorphism (B.0.4) follows from [Dev26, Theorem 3.1], whose proof is just the observation that

$$\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{\mathrm{G}}_k}/k)\cong\mathfrak{Z}_{\mathbf{E}_2}(k[\mathbb{X}_*(\mathrm{T})]/k)\cong k[\mathbb{X}_*(\mathrm{T})]^{h\mathrm{T}},$$

where the final isomorphism is Remark 3.2.6, and that

$$\pi_*(k[\mathbb{X}_*(\mathrm{T})]^{h\mathrm{T}})\cong\pi_*(k)[\mathbb{X}_*(\mathrm{T})]\otimes_{\pi_*(k)}\mathcal{O}_{\mathrm{T}_{\mathrm{F}}}\cong\mathcal{O}_{\check{\mathrm{T}}\times\mathrm{T}_{\mathrm{F}}}.$$

The $\mathbf{T}$-equivariant $\mathbf{E}_3$-k-algebra $\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{\mathrm{G}}_k}/k)$ defines an $\mathbf{E}_1$-$k^{h\mathbf{T}}$-algebra $\mathfrak{Z}_{\mathbf{E}_2}(\mathcal{O}_{\mathrm{sh}(2\rho)\check{\mathrm{G}}_k}/k)^{h\mathbf{T}}$. Taking homotopy groups, (B.0.4) implies that there is an associative $\pi_*(k^{h\mathbf{T}})$-algebra $\mathrm{D}^{\mathrm{F}}_{{}^{2\rho}\check{\mathrm{G}}}$ which is a lift of $\mathcal{O}_{{}^{2\rho}\check{\mathrm{G}}\times{}^{2\rho}\mathrm{G}_{\mathrm{F}}}$ along the divisor $\mathrm{Spec}(\pi_*(k))\hookrightarrow\mathbf{T}_{\mathrm{F}}=\mathrm{Spf}(\pi_*(k^{h\mathbf{T}}))$. In future work, I hope to give a definition of $\mathrm{D}^{\mathrm{F}}_{{}^{2\rho}\check{\mathrm{G}}}$ and use it to extend the results of Section 5.3 to nonabelian G. Again, we can explicitly describe $\mathrm{D}^{\mathrm{F}}_{{}^{2\rho}\check{\mathrm{G}}}$ in the two cases where ${}^{\mathrm{sh}(2\rho)}\check{\mathrm{G}}_k$ is known to exist:

Example B.7. If k is an ordinary commutative ring, then $\mathrm{D}^{\mathrm{F}}_{{}^{2\rho}\check{\mathrm{G}}}$ is just the completion of the Rees construction of the usual algebra $\mathrm{D}_{{}^{2\rho}\check{\mathrm{G}}}$ of (crystalline) differential operators of ${}^{2\rho}\check{\mathrm{G}}$ with respect to the order filtration. This again follows from [Dev26, Theorem 2.3].

Example B.8. If k is a general even $\mathbf{E}_\infty$-ring and G is a torus T, then the associative $\pi_*(k^{h\mathbf{T}})$-algebra $\mathrm{D}^{\mathrm{F}}_{\check{\mathrm{T}}}$ was described in Definition 5.3.1. The survey [Dev26] describes of some of its fascinating properties.

When $\mathrm{G} = \mathrm{PGL}_2$, Conjecture B.1 predicts the existence of an $\mathbf{E}_1$-coalgebra $\mathcal{O}_{{}^{\mathrm{sh}(2\rho)}\mathrm{SL}_{2\mathrm{KU}}}$ in $\mathbf{E}_2^{\mathrm{fr}}$-algebras in $\mathrm{Mod}_{\mathrm{KU}}$ such that $\pi_*(\mathcal{O}_{{}^{\mathrm{sh}(2\rho)}\mathrm{SL}_{2\mathrm{KU}}})$ is isomorphic (as a graded Hopf algebra over $\pi_*(\mathrm{KU}) \cong \mathbf{Z}[\beta^{\pm 1}]$) to $\mathcal{O}_{{}^{2\rho}\mathrm{SL}_2}[\beta^{\pm 1}]$. Since ${}^{2\rho}\mathrm{SL}_2$ is evenly graded and β lives in weight 2, there is an isomorphism $\mathcal{O}_{{}^{2\rho}\mathrm{SL}_2}[\beta^{\pm 1}] \cong \mathcal{O}_{\mathrm{SL}_2}[\beta^{\pm 1}]$ of graded Hopf algebras over $\mathbf{Z}[\beta^{\pm 1}]$. We note that one cannot even construct a natural $\mathbf{E}_4$-KU-algebra $\mathcal{O}_{{}^{\mathrm{sh}(2\rho)}\mathrm{SL}_{2\mathrm{KU}}}$, ignoring the coalgebra structure entirely:

Proposition B.9. *There is no $\mathbf{E}_4$-KU-algebra $\mathcal{O}_{{}^{\mathrm{sh}(2\rho)}\mathrm{SL}_{2\mathrm{KU}}}$ equipped with an $\mathbf{E}_4$-KU-algebra map $\mathrm{KU}[a,b,c,d] \to \mathcal{O}_{{}^{\mathrm{sh}(2\rho)}\mathrm{SL}_{2\mathrm{KU}}}$ from the flat polynomial KU-algebra on generators a,b,c,d such that the induced map on π_0 is quotienting by the ideal generated by $ad - bc - 1$.*

Proof. If p is a prime and A is an $\mathbf{E}_{2n+1}$-KU-algebra, then there is an operation $\delta : \pi_0(\mathrm{A}) \to \pi_0(\mathrm{A}/p^n)$, which is defined exactly as the power operation θ from [Hop14]; see the proof of [HL24, Proposition 4.22] for some discussion. If A further refines to an $\mathbf{E}_{2n+2}$-KU-algebra, then this operation satisfies the usual relations of a δ-ring (I am grateful to Ishan Levy for a discussion about this point). Namely, there are identities

$$\delta(x+y) = \delta(x) + \delta(y) - \tfrac{(x+y)^p - x^p - y^p}{p},$$
$$\delta(xy) = \delta(x)y^p + x^p\delta(y) + p\delta(x)\delta(y).$$

In particular, if A is an $\mathbf{E}_4$-KU-algebra, there is a map $\delta : \pi_0(\mathrm{A}) \to \pi_0(\mathrm{A}/p)$ satisfying the above identities. The operation δ is zero on $a,b,c,d \in \pi_0\mathrm{KU}[a,b,c,d]$, so the same must be true of the images of these elements in $\pi_0\mathcal{O}_{{}^{\mathrm{sh}(2\rho)}\mathrm{SL}_{2\mathrm{KU}}} = \mathcal{O}_{\mathrm{SL}_2}$. Since $ad - bc = 1$ in $\mathcal{O}_{\mathrm{SL}_2}$, and $\delta(1) = 0$, it follows that $\delta(ad - bc) = 0$. If $p = 2$, then

$$\delta(ad - bc) = \delta(ad) + \delta(-bc) + adbc = \delta(ad) - \delta(bc) - b^2c^2 + adbc = bc,$$

because $ad - bc = 1$ and $\delta(ad) = \delta(bc) = 0$. Since $bc \neq 0$, this is a contradiction. Note that the issue persists after p-localization at odd p: for the same reason, $\delta(ad - bc) = \frac{(1+bc)^p - b^p c^p - 1}{p}$, which does not vanish in $\mathcal{O}_{\mathrm{SL}_2}$. $\square$

Appendix C. Mod 3 cohomology of PGL_3

Let G be a connected reductive group over $\mathbf{C}$, and let k be an ordinary commutative ring. In Theorem 4.3.10, we saw that there is an isomorphism $\mathrm{gr}^\star_{\mathrm{G}}(k^{h\mathrm{G}}) \cong \mathrm{R}\Gamma((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O})$. This defines the spectral sequence (4.3.1) (coming from (A.2.1)), which we will illustrate with $k = \mathbf{F}_p$ for $(\mathrm{G}, p) = (\mathrm{PGL}_p, p)$ with $p = 2, 3$.

Example C.1. When $(\mathrm{G}, p) = (\mathrm{PGL}_2, 2)$, ${}^{2\rho}\check{\mathrm{U}}_- = \mathbf{G}_a(2)$ and $\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2) \subseteq {}^{2\rho}\check{\mathfrak{g}}^* \cong {}^{2\rho}\mathfrak{pgl}_2(2)$ is the affine subspace $\left(\begin{smallmatrix} x & y \\ 1 & 0 \end{smallmatrix}\right)$. Here, x lives in weight -2 and y lives in weight -4. The action of $a \in \mathbf{G}_a$ (which lives in weight -2) is given by

$$\begin{aligned} x &\mapsto x + 2a \equiv x \pmod 2, \\ y &\mapsto y - a^2 - ax \equiv y + a^2 + ax \pmod 2. \end{aligned}$$

Note that the homomorphism $\alpha_2 \to \mathrm{Z}_{\check{\mathrm{U}}^-}(\psi_0)$ sending $a \mapsto \left(\begin{smallmatrix} 1 & 0 \\ a & 1 \end{smallmatrix}\right)$ is an isomorphism; the interesting cohomology we see below stems from the group cohomology of α_2. Let $\underline{\mathbf{G}_a}$ denote the constant group scheme $\operatorname{Spec} \mathbf{F}_2[a, x]$ over $\operatorname{Spec} \mathbf{F}_2[x]$. Then the action of $\mathbf{G}_a$ on $\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2)$ can be described as follows: it is the composite of the homomorphism $\varphi : \underline{\mathbf{G}_a} \to \underline{\mathbf{G}_a}$ sending $a \mapsto a^2 + ax$ with the translation action of $\underline{\mathbf{G}_a}$ on $\underline{\mathbf{A}}^1 = \operatorname{Spec} \mathbf{F}_2[y, x]$. By Shapiro's lemma, $\mathrm{R}\Gamma((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O}) \cong \mathrm{R}\Gamma(\mathrm{B}_{\mathbf{F}_2[x]} \ker(\varphi); \mathcal{O})$. This is easy to compute: recall that

$$\mathrm{H}^*(\mathrm{B}_{\mathbf{F}_2[x]}\underline{\mathbf{G}_a}; \mathcal{O}) \cong \mathbf{F}_2[x, \tau_0, \tau_1, \cdots],$$

where τ_i lives in cohomological degree 1 and weight -2^{i+1}, and is given by the cocycle $a \mapsto a^{2^i}$. Since $(a^2 + ax)^{2^i} = a^{2^{i+1}} + a^{2^i} x^{2^i}$, the map φ sends $\tau_i \mapsto \tau_{i+1} + x^{2^i}\tau_i$, so that

$$\mathrm{H}^*(\mathrm{B}_{\mathbf{F}_2[x]} \ker(\varphi); \mathcal{O}) \cong \mathbf{F}_2[x, \tau_0, \tau_1, \cdots]/(\tau_{i+1} = x^{2^i}\tau_i)_{i \geq 0} \cong \mathbf{F}_2[x, \tau_0].$$

The spectral sequence (4.3.1) collapses at the E_1-page; under the isomorphism $\mathrm{H}^{-*}(\mathrm{BPGL}_2; \mathbf{F}_2) \cong \mathbf{F}_2[w_2, w_3]$, the Stiefel-Whitney class w_2 (resp. w_3) is represented by x (resp. τ_0). See also Example 2.2.24.

Remark C.2. When $k = \mathbf{Z}$ and $\mathrm{G} = \mathrm{PGL}_2$, there is an isomorphism

$$\mathrm{H}^*((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O}) \cong \mathbf{Z}[x^2 + 4y, \tau_0]/2\tau_0,$$

where $x^2 + 4y$ lives in weight -4 and τ_0 lives in weight -2 and cohomological degree 1. This is isomorphic to $\mathrm{H}^{-*}(\mathrm{BPGL}_2; \mathbf{Z})$. Note that $(\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-$ is isomorphic to the moduli stack of curves $t = u^2 + xu + y$ modulo change of coordinate $u \mapsto u + a$, which is in turn isomorphic to the even stack of ko (see [DFHH14, Chapter 9, Section 3.2]).

A somewhat more interesting example is the case $(\mathrm{G}, p) = (\mathrm{PGL}_3, 3)$, which will occupy the remainder of this section. Then ${}^{2\rho}\check{\mathrm{U}}_-$ is the group of upper-triangular matrices $\left(\begin{smallmatrix} 1 & a_1 & b \\ 0 & 1 & a_2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$; let $\mathrm{Z} = \mathbf{G}_a(4)$ denote its center. Similarly, $\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2) \subseteq {}^{2\rho}\check{\mathfrak{g}}^* \cong {}^{2\rho}\mathfrak{pgl}_3(2)$ is the affine subspace $\left(\begin{smallmatrix} x_1 & y_1 & z \\ 1 & x_2 & y_2 \\ 0 & 1 & 0 \end{smallmatrix}\right)$. The coordinates x_1, x_2, a_1, and a_2 have weights -2; the coordinates y_1, y_2, and b have weight -4; and z has weight -6.

The adjoint action sends

$$\begin{aligned} x_1 &\mapsto x_1 + a_1 + a_2, \\ x_2 &\mapsto x_2 - a_1 + 2a_2 \equiv x_2 - (a_1 + a_2) \pmod 3, \\ y_1 &\mapsto y_1 - a_1(x_1 - x_2 + a_1) + b, \\ y_2 &\mapsto y_2 + a_2(a_1 - a_2 - x_2) - b, \\ z &\mapsto z + a_1 y_2 - a_2 y_1 + a_1 a_2 (x_1 - x_2 + a_1) - b(x_1 + a_2 + a_1). \end{aligned}$$

Note that the homomorphism $\alpha_3 \to \mathrm{Z}_{\check{\mathrm{U}}^-}(\psi_0)$ sending $a \mapsto \left(\begin{smallmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ a^2 & 2a & 1 \end{smallmatrix}\right)$ is an isomorphism; the interesting cohomology we see below stems from the group cohomology of α_3. Let (a_1, a_2) denote the coordinates on ${}^{2\rho}\check{\mathrm{U}}^-/\mathrm{Z} \cong \mathbf{G}_a(2)^2$, and define $(\gamma_1, \gamma_2) = (a_1 + a_2, a_2)$. Let $\mathrm{V} \subseteq {}^{2\rho}\check{\mathrm{U}}_-$ denote the preimage of $\mathbf{G}_a(2)_{\gamma_1} \subseteq {}^{2\rho}\check{\mathrm{U}}^-/\mathrm{Z}$ inside ${}^{2\rho}\check{\mathrm{U}}_-$. Then it is easy to verify that

$$\mathrm{H}^*((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/\mathrm{V}; \mathcal{O}) \cong \mathrm{H}^0((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/\mathrm{V}; \mathcal{O}) \cong \mathbf{F}_3[c, v, w],$$

where

$$\begin{aligned} c &= x_1 + x_2, \\ v &= y_1 + y_2 - x_2 x_1, \\ w &= z - y_2 x_1. \end{aligned}$$

These are just the coefficients of the characteristic polynomial of $\left(\begin{smallmatrix} x_1 & y_1 & z \\ 1 & x_2 & y_2 \\ 0 & 1 & 0 \end{smallmatrix}\right)$.

Remark C.3. It is true even over $\mathbf{Z}$ that $\mathrm{H}^*((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/\mathrm{V}; \mathcal{O}) \cong \mathrm{H}^0((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/\mathrm{V}; \mathcal{O}) \cong \mathbf{Z}[c, v, w]$. The residual action of ${}^{2\rho}\check{\mathrm{U}}_-/\mathrm{V} \cong \mathbf{G}_a(2)_{\gamma_2}$ is given by

$$\begin{aligned} c &\mapsto c + 3\gamma_2, \\ v &\mapsto v - 2\gamma_2 c - 3\gamma_2^2, \\ w &\mapsto w - \gamma_2 v + \gamma_2^2 c + \gamma_2^3. \end{aligned}$$

Note that this is exactly the transformation law for cubic curves $t^2 = u^3 + cu^2 - vu + w$ modulo change of coordinate $u \mapsto u + \gamma_2$, so that $(\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2))/{}^{2\rho}\check{\mathrm{U}}_-$ is isomorphic to the moduli stack of such curves; after localizing at 3, this is in turn isomorphic to the moduli stack of cubic curves (which is also the even stack of $\mathrm{tmf}_{(3)}$; see [DFHH14, Chapter 9, Section 3.2] and [Mat16c])).

If $\mathrm{J} = v^3 + c^2 v^2 - c^3 w$, then the $\mathbf{G}_a(2)_{\gamma_2}$-invariant subring of $\mathbf{F}_3[c, v, w]$ is $\mathbf{F}_3[c, \mathrm{J}]$, so $\mathrm{H}^*(\mathbf{G}_a(2)_{\gamma_2}; \mathbf{F}_3[c, v, w])$ is a graded $\mathbf{F}_3[c, \mathrm{J}]$-algebra. Note that in terms of the original coordinates on $\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^\perp(2)$, we have $c = x_1 + x_2$ and

$$\mathrm{J} = (x_1 + x_2)^2 (y_1 + y_2 - x_1 x_2)^2 + (y_1 + y_2 - x_1 x_2)^3 + (x_1 + x_2)^3 (x_1 y_2 - z).$$

Recall that there is an isomorphism

$$\mathrm{H}^*(\mathrm{B}\mathbf{G}_a(2)_{\gamma_2}; \mathbf{F}_3) \cong \Lambda(\tau_0, \tau_1, \cdots) \otimes \mathbf{F}_3[\zeta_1, \zeta_2, \cdots],$$

where τ_i lives in cohomological degree 1 and weight $-2p^i$ and is given by the 1-cocycle $\gamma_2 \mapsto \gamma_2^{p^i}$; and ζ_i lives in cohomological degree 2 and weight $-2p^i$, and is given by the Bockstein of τ_{i-1}. In other words, ζ_i is the 2-cocycle $(\gamma_2, \gamma_2') \mapsto \gamma_2^{2p^{i-1}} \gamma_2'^{p^{i-1}} + \gamma_2^{p^{i-1}} \gamma_2'^{2p^{i-1}}$. The cocycles τ_0 and τ_1 define classes $\tau_0, \tau_1 \in \mathrm{H}^1(\mathbf{G}_a(2)_{\gamma_2}; \mathbf{F}_3[c, v, w])$. Since $d(v) = \gamma_2 c$, it follows that $c\tau_0 = 0$. Moreover, $\tau_0^2 = 0$. Similarly, $d(w) = \gamma_2^3 + \gamma_2^2 c - \gamma_2 v$ and $d(v^2) = c^2\gamma_2^2 - c\gamma_2 v$, so that $d(cw - v^2) = \gamma_2^3 c$. It follows that $c\tau_1 = 0$ and $\tau_1^2 = 0$, too. The 2-cocycle ζ_1 sends $(\gamma_2, \gamma_2') \mapsto \gamma_2^2 \gamma_2' + \gamma_2 \gamma_2'^2$.

Lemma C.4. *There is an equality $\tau_0 \tau_1 + c\zeta_1 = 0$ in $\mathrm{H}^2(\mathbf{G}_a(2)_{\gamma_2}; \mathbf{F}_3[c, v, w])$.*

Proof. Consider the 1-cochain $\kappa(\gamma_2) = \gamma_2^2 v + \gamma_2 w$. We begin by calculating $d(\kappa)$; recall that

$$d(\kappa) : (\gamma_2, \gamma_2') \mapsto \gamma_2 \kappa(\gamma_2') - \kappa(\gamma_2 + \gamma_2') + \kappa(\gamma_2).$$

We compute each term. First,

$$\gamma_2\kappa(\gamma_2') = \gamma_2'^2 v + \gamma_2\gamma_2'^2 c + \gamma_2' w - \gamma_2\gamma_2' v + \gamma_2^2\gamma_2' c + \gamma_2^3\gamma_2'.$$

Next,

$$\kappa(\gamma_2 + \gamma_2') = \gamma_2^2 v + \gamma_2'^2 v - \gamma_2\gamma_2' v + \gamma_2 w + \gamma_2' w.$$

It follows that $d(\kappa)$ sends

$$d(\kappa) : (\gamma_2, \gamma_2') \mapsto c(\gamma_2\gamma_2'^2 + \gamma_2^2\gamma_2') + \gamma_2^3\gamma_2'.$$

The cup product $\tau_1\tau_0$ is the 2-cocycle sending $(\gamma_2, \gamma_2') \mapsto \gamma_2^3\gamma_2'$, so we find that $d(\kappa) = c\zeta_1 + \tau_0\tau_1$, as desired. □

Lemma C.5. *The cohomology ring* $\mathrm{H}^*(\mathbf{G}_a(2)_{\gamma_2}; \mathbf{F}_3[c, v, w])$ *is generated by* c*,* J*,* τ_0*,* τ_1*, and* ζ_1*.*

Proof. Consider the cohomology $\mathrm{H}^*(\mathbf{G}_a(2)_{\gamma_2}; \mathbf{F}_3[v, w])$. The action of $\mathbf{G}_a(2)_{\gamma_2}$ on $\mathbf{F}_3[v, w]$ is just given by $v \mapsto v$ and $w \mapsto w - \gamma_2 v + \gamma_2^3$. As in Example C.1, let $\underline{\mathbf{G}_a}$ denote the constant group scheme $\mathrm{Spec}\,\mathbf{F}_3[\gamma_2, v]$ over $\mathrm{Spec}\,\mathbf{F}_3[v]$. Then the action of $\mathbf{G}_a$ on $\mathbf{F}_3[v, w]$ can be described as follows: it is the composite of the homomorphism $\varphi : \underline{\mathbf{G}_a} \to \underline{\mathbf{G}_a}$ sending $\gamma_2 \mapsto \gamma_2^3 - \gamma_2 v$ with the translation action of $\underline{\mathbf{G}_a}$ on $\underline{\mathbf{A}^1} = \mathrm{Spec}\,\mathbf{F}_3[v, w]$. By Shapiro's lemma, $\mathrm{H}^*(\mathbf{G}_a(2)_{\gamma_2}; \mathbf{F}_3[v, w]) \cong \mathrm{H}^*(\mathrm{B}_{\mathbf{F}_3[v]} \ker(\varphi); \mathcal{O})$. As in Example C.1, this is isomorphic to $\Lambda(\overline{\tau}_0) \otimes \mathbf{F}_3[v, \overline{\zeta}_1]$, where $\overline{\tau}_0$ lives in cohomological degree 1 and $\overline{\zeta}_1$ lives in cohomological degree 2. Now consider the c-Bockstein spectral sequence, given by

$$\mathrm{E}_1 = \mathrm{H}^*(\mathbf{G}_a(2)_{\gamma_2}; \mathbf{F}_3[v, w])[\overline{c}] \cong \mathbf{F}_3[v, \overline{\zeta}_1, \overline{c}] \otimes \Lambda(\overline{\tau}_0) \Rightarrow \mathrm{H}^*(\mathbf{G}_a(2)_{\gamma_2}; \mathbf{F}_3[c, v, w]).$$

The classes c, J, ζ_1, τ_0, and τ_1 discussed above are detected on the E_1-page by $\overline{c}$, v^3, $\overline{\zeta}_1$, $\overline{\tau}_0$, and $v\overline{\tau}_0$. It follows that $\mathrm{H}^*(\mathbf{G}_a(2)_{\gamma_2}; \mathbf{F}_3[c, v, w])$ is indeed generated by c, J, ζ_1, τ_0, and τ_1, as desired. □

Piecing these calculations together, we conclude:

Proposition C.6. *There is an isomorphism*

$$\mathrm{H}^*((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O}) \cong \mathbf{F}_3[c, \mathrm{J}, \zeta_1, \tau_0, \tau_1]/(\tau_0^2, c\tau_0, \tau_1^2, c\tau_1, \tau_0\tau_1 + c\zeta_1),$$

where c *lives in weight* -2*,* J *lives in weight* -12*,* ζ_1 *lives in weight* -6 *and cohomological degree* 2 *(hence in total degree* -8*),* τ_0 *lives in weight* -2 *and cohomological degree* 1 *(hence in total degree* -3*), and* τ_1 *lives in weight* -6 *and cohomological degree* 1 *(hence in total degree* -7*).*

It follows from [KMS75, Theorem 4.11] that the spectral sequence (4.3.1) degenerates at the E_1-page with no multiplicative extensions on the E_∞-page, and there is an isomorphism

$$\mathrm{H}^*((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O}) \cong \mathrm{H}^{-*}(\mathrm{BPGL}_3; \mathbf{F}_3).$$

Note that although ζ_1 lives in even total degree, it corresponds to a class in $\mathrm{H}^2((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O})$.

Remark C.7. We mentioned in Remark C.3 that the stack $(\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-$ is isomorphic to the moduli stack $\mathcal{M}$ of cubic curves $t^2 = u^3 + cu^2 - vu + w$ modulo change of coordinate $u \mapsto u + \gamma_2$. It follows that over $\mathbf{Z}$, $\mathrm{H}^*((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O})$ is essentially the cohomology of the moduli stack of cubic curves; the reason for the "essentially" is that $\mathcal{M}$ is only isomorphic to the moduli stack of cubic curves after localizing at 3. In any case, by replicating the calculations above, one obtains an isomorphism

$$\mathrm{H}^*((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O}) \cong \mathbf{Z}[c_2, c_3, \Delta, \tau_0, \zeta_1]/\mathrm{I},$$

where I is the ideal generated by

$$\tau_0^2, 3\tau_0, 3\zeta_1, c_2\tau_0, c_2\zeta_1, c_3\tau_0, c_3\zeta_1, 27\Delta - c_3^2 - 4c_2^3.$$

Here, c_2, c_3, and Δ live in cohomological degree 0 and weights -4, -6, and -12 respectively, and τ_0 and ζ_1 live in the same degrees as in Proposition C.6. It follows from [Vis07, Theorem 3.7] that the spectral sequence (4.3.1) also degenerates over $\mathbf{Z}$ at the E_1-page with no multiplicative extensions on the E_∞-page. Note that after localizing at 3, $\mathrm{H}^0((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O})$ is precisely the ring of modular forms over $\mathbf{Z}_{(3)}$ (as expected from Remark C.3).

Remark C.8. A similar calculation, albeit a more tedious one, shows that if $\mathrm{G} = \mathrm{G}_2$ and $k = \mathbf{F}_2$, then there is an isomorphism $\mathrm{H}^*((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O}) \cong \mathbf{F}_2[c, v, \tau_0]$ with c and v in cohomological degree 0 and weights -4 and -6 respectively, and τ_0 in cohomological degree 1 and weight -6. This matches with the calculation of $\mathrm{H}^{-*}(\mathrm{BG}_2; \mathbf{F}_2)$, so the spectral sequence (4.3.1) again degenerates at the E_1-page with no multiplicative extensions on the E_∞-page.

This degeneration occurs also for $\mathrm{G} = \mathrm{SO}_{2n}, \mathrm{SO}_{2n+1}$ and $k = \mathbf{F}_2$. If $\mathrm{G} = \mathrm{SO}_{2n+1}$, for instance, one finds that there is an isomorphism

$$\mathrm{H}^*((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O}) \cong \mathbf{F}_2[\tau_0, c_1, \cdots, \tau_{n-1}, c_n, \tau_n],$$

where c_i lives in cohomological degree 0 and weight $-2i$, and τ_i lives in cohomological degree 1 and weight $-2i$. The classes τ_i (resp. c_i) detect the odd (resp. even) Stiefel-Whitney classes in $\mathrm{H}^{-*}(\mathrm{BG}; \mathbf{F}_2)$.

These calculations motivate Conjecture A.2.13. For general p, the stack $(\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-$ for $\mathrm{G} = \mathrm{PGL}_p$ is isomorphic (even over $\mathbf{Z}$) to the moduli stack of curves $t^{p-1} = u^p + c_1 u^{p-1} + \cdots + c_p$ (where c_i lives in weight $-2i$) modulo change of coordinate $u \mapsto u + \alpha$ with α in weight -2, just like in Remark C.2 and Remark C.3. The cohomology of this stack was studied for $p = 5$ in [Hil08, Theorem 6.2]. If Conjecture A.2.13 holds, then [Hil08, Theorem 6.2] can be viewed as providing a computation of $\mathrm{H}^*(\mathrm{BPGL}_5; \mathbf{Z}_{(5)})$, and (as expected) the resulting algebra is very complicated.

Question C.9. Is there an analogue of the isomorphism $\mathrm{gr}^\star_{\mathrm{G}}(k^{h\mathrm{G}}) \cong \mathrm{R}\Gamma((\psi_0 + {}^{2\rho}\check{\mathfrak{u}}_-^{\perp}(2))/{}^{2\rho}\check{\mathrm{U}}_-; \mathcal{O})$ from Theorem 4.3.10, and hence of the spectral sequence (4.3.1), for p-complete commutative rings k and p-compact groups G (in the sense of [DW94, AGMV08])? Does Conjecture A.2.13 hold for this putative spectral sequence?

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